UNION PUBLIC SERVICE COMMISSION (UPSC)

COMBINED DEFENCE SERVICES (CDS)

ELEMENTARY MATHEMATICS PREVIOUS YEAR QUESTIONS (PYQ) COMPLETE ARCHIVE

ALL 2,799 QUESTIONS EXTRACTED CHAPTER-WISE (2009–2024)

Target Examinations:	IMA, INA, AFA (Combined Defence Services Examination)
Subject / Paper:	Paper 3: Elementary Mathematics (100 Questions / 100 Marks)
Source Question Archive:	UPSC CDS Papers (2009 – 2024) [36 Complete Exam Booklets / 1,340 Pages]
Total Extracted Questions:	**2,799 Authentic Questions** Organized into 13 Subject Chapters
Feature Highlights:	Syllabus Breakdown • Topic Weightage Matrix • Exam Tagged Practice Drill

MASTER QUESTION REPOSITORY FOR CDS MATHEMATICS

This exhaustive volume contains all **2,799 questions** extracted from the official UPSC CDS Elementary Mathematics archive. Every question is organized strictly chapter-wise and indexed with its examination year for targeted topic-by-topic drill and preparation.

TABLE OF CONTENTS & CHAPTER REPOSITORY

Chapter / Section	Exam Weightage & Description	Total Questions
Part I: Syllabus & Strategy	Exam Blueprint, 13-Chapter Matrix & 15-Year Trends	Analysis
Chapter 01: Number System & Divisibility	15 - 20 Questions (15 - 20 Marks)	**480 Qs**
Chapter 02: HCF, LCM & Logarithms	5 - 8 Questions (5 - 8 Marks)	**65 Qs**
Chapter 03: Percentages, Profit, Loss & Discount	6 - 10 Questions (6 - 10 Marks)	**158 Qs**
Chapter 04: Simple & Compound Interest, Ratio & Mixtures	6 - 10 Questions (6 - 10 Marks)	**156 Qs**
Chapter 05: Time, Speed & Distance, Time & Work	10 - 15 Questions (10 - 15 Marks)	**224 Qs**
Chapter 06: Algebra, Polynomials & Linear Equations	10 - 15 Questions (10 - 15 Marks)	**559 Qs**
Chapter 07: Quadratic Equations, Roots & Progressions	5 - 8 Questions (5 - 8 Marks)	**51 Qs**
Chapter 08: Trigonometry & Heights and Distances	12 - 16 Questions (12 - 16 Marks)	**264 Qs**
Chapter 09: Geometry: Lines, Angles, Triangles & Polygons	10 - 15 Questions (10 - 15 Marks)	**287 Qs**
Chapter 10: Geometry: Circles, Chords & Tangents	8 - 12 Questions (8 - 12 Marks)	**129 Qs**
Chapter 11: Mensuration 2D: Areas & Perimeters	6 - 10 Questions (6 - 10 Marks)	**148 Qs**
Chapter 12: Mensuration 3D: Surface Areas & Volumes	8 - 12 Questions (8 - 12 Marks)	**156 Qs**
Chapter 13: Statistics & Data Interpretation	8 - 12 Questions (8 - 12 Marks)	**122 Qs**
TOTAL REPOSITORY ARCHIVE	**13 Core Mathematics Chapters (2009–2024 Archive)**	**2,799 Qs**

Part I: CDS Elementary Mathematics Syllabus & Exam Architecture

The Elementary Mathematics paper in the UPSC Combined Defence Services (CDS) examination is taken by candidates competing for admission to the **Indian Military Academy (IMA)**, **Indian Naval Academy (INA)**, and **Air Force Academy (AFA)**. (Officers Training Academy candidates are exempt from this paper). The standard of the paper corresponds to the matriculation level (Class X CBSE/State Board), but questions test deep conceptual understanding, speed, and numerical agility.

1. Examination Blueprint & Marking Scheme

Parameter	Details / Specifications	Strategic Implications
Total Questions	100 Multiple Choice Questions (MCQs)	Requires high speed and selective attempt strategy
Maximum Marks	100 Marks (1.0 Mark per correct response)	Clean 1:1 mark allocation per question
Duration	2 Hours (120 Minutes)	Exact 72 seconds per question including OMR
Negative Marking	-0.33 Marks per incorrect response (1/3rd penalty)	Strict avoidance of blind guessing is required
Qualifying Cutoff	Minimum 20% sectional cutoff (20 Marks)	Target score for top merit: 65 - 80+ Marks
Medium of Paper	Bilingual: English & Hindi side-by-side	English questions printed on alternating pages

2. Chapter Weightage Matrix & Question Archive Distribution

No.	Chapter / Module	Exam Qs	Exam Marks	Frequency	Archive Total
01	Number System & Divisibility	15 Qs	15 M	100% of Exams	**480 Qs**
02	HCF, LCM & Logarithms	5 Qs	5 M	100% of Exams	**65 Qs**
03	Percentages, Profit, Loss & Discount	6 Qs	6 M	100% of Exams	**158 Qs**
04	Simple & Compound Interest, Ratio & Mixtures	6 Qs	6 M	100% of Exams	**156 Qs**
05	Time, Speed & Distance, Time & Work	10 Qs	10 M	100% of Exams	**224 Qs**
06	Algebra, Polynomials & Linear Equations	10 Qs	10 M	100% of Exams	**559 Qs**
07	Quadratic Equations, Roots & Progressions	5 Qs	5 M	100% of Exams	**51 Qs**
08	Trigonometry & Heights and Distances	12 Qs	12 M	100% of Exams	**264 Qs**
09	Geometry: Lines, Angles, Triangles & Polygons	10 Qs	10 M	100% of Exams	**287 Qs**
10	Geometry: Circles, Chords & Tangents	8 Qs	8 M	100% of Exams	**129 Qs**
11	Mensuration 2D: Areas & Perimeters	6 Qs	6 M	100% of Exams	**148 Qs**
12	Mensuration 3D: Surface Areas & Volumes	8 Qs	8 M	100% of Exams	**156 Qs**
13	Statistics & Data Interpretation	8 Qs	8 M	100% of Exams	**122 Qs**
—	**TOTAL QUESTIONS IN REPOSITORY**	**100 / Exam**	**100 Marks**	**36 Exam Sets**	**2,799 Qs**

3. 15-Year Evolution Trend Analysis (2009 – 2024)

• **Dominance of Advanced Geometry & Mensuration (35–40% Weightage):**
Unlike SSC or banking exams that prioritize commercial arithmetic, UPSC CDS tests heavy classical Euclidean geometry (circle theorems, similarity, triangle centers) and multi-step 3D mensuration (frustums, melted spheres, inscribed solids).

• **Trigonometry & Heights and Distances (12–16 Questions):**
Trigonometry questions consistently reward algebraic substitutions (e.g. setting theta = 0, 30, or 45 degrees to eliminate options) alongside conceptual multi-angle applications.

• **Algebra & Number Theory Precision:**
Questions on divisibility, remainder theorem for polynomials, cyclic unit digits, and quadratic discriminants require mastery of fundamental theorems.

4. The 5 Golden Formulas for CDS Math Mastery

Rule 1: Focus on the 'Big Three' (Geometry, Mensuration, Trigonometry)	These three areas constitute 55 to 65 questions (over 60% of the entire paper!). Mastering these guarantees a score above 60+ marks.
Rule 2: Option Elimination via Value Substitution	In Algebra and Trigonometry, substitute simple values (x = 1, y = 2, theta = 45°) to test options directly. This solves problems in 20 seconds without lengthy derivations.
Rule 3: Master Remainder & Factor Theorems	Understand polynomial division: P(a) is remainder when P(x) is divided by (x - a). For numbers, use Fermat's and Euler's remainder concepts.
Rule 4: Formula Memorization for Statistics	8-10 questions in Statistics are direct formula applications: Mode = 3 Median - 2 Mean, grouped median/mean formulas, and standard deviation invariance under origin shifts.
Rule 5: The Three-Pass Exam Strategy	Pass 1 (0-50 min): Solve all direct/short questions (target 40-45 Qs). Pass 2 (50-100 min): Moderate 2-step calculations (target 25-30 Qs). Pass 3 (100-120 min): Verification and difficult geometry proofs.

CHAPTER 01

Total Archive Questions: 480 Questions

NUMBER SYSTEM & DIVISIBILITY

Typical Exam Weightage: 15 - 20 Questions (15 - 20 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q3] What is the remainder when 2n is divided by 4 ? What is the last digit in the expansion of **(a)** I only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 What is x
2. [CDS (I) Paper 1 Q4] If log, 6 m and log, 3 log, (r/2) equal to ? = n, then what is Consider the following statements : A number is divisible by 9 if a, + a2 + + + is divisible by **(a)** I only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 What is x
3. [CDS (I) Paper 1 Q9] 2. a, — + a, —a, + as is divisible by **(a)** I only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 What is x
4. [CDS (I) Paper 1 Q8] 9. 2 If the remainder Of the polynomial + atx + + — + when divided by (x — 1) is 1, then which one of the following is correct ? When (x' — 2x2 + px — q) is divided by (x' — 2x — 3), the remainder is (x — 6). What are the values Of p, q respectively ? What are the values of c when the HCF Of x' + cx2 — x + 2c and + cx — 2 over the rationals is a linear polynomial ? If (x + 2) is the HCF of x2 + ax + b and x2 + cx d (a c and b d), then which one Of the following is correct ? **(b)** 2a + b 2c + d **(c)** b + 2c = 2a + d **(d)** b-2c-2a-d What is the LCM of (x2 — Y2 — z2 — 2yz), (x' — + + 2xz) (x' + ya — z2 — ? **(a) (b) (c) (d)** (Cond)
5. [CDS (I) Paper 1 Q16] A person bought 5 tickets from a station P to a station Q and IO tickets from the station p to a station R. He paid Rs. **(a)** I only **(b)** 2 only **(c)** O or I If x —I + •Æ, then what is the value of x' — 4x3 + ? What is the magnitude of difference Of the roots of x2 — ax + b = O ? a2 —4b What is the solution Of the equations x -y— and ll **(d)** x = and y
6. [CDS (I) Paper 1 Q12] What is the larger number ? **(a)** I only **(b)** 2 only **(c)** O or I If x —I + •Æ, then what is the value of x' — 4x3 + ? What is the magnitude of difference Of the roots of x2 — ax + b = O ? a2 —4b What is the solution Of the equations x -y— and ll **(d)** x = and y
7. [CDS (I) Paper 1 Q22] 23. 24. log(5/2) ? **(b)** If x cos 600 + y cos 00 3 and 4x sin 300 — y cot 450 = 2, then what is the value of x ? If the unit Of weight is 15/4 kg, what number will 3/2 quintal represent ? **(a)** 25 **(c)** 1/9 **(d)** None of the above A dishonest dealer professes to sell his goods at cost price, but uses a false weight and thus gains 20%. For a kilogram he uses a weight of **(a)** 700 g **(b)** 750 g **(c)** 800 g **(d)** 850 g If we divide a positive integer by another positive integer, what is the resulting number ? **(a) (b) (c) (d)** It is always a natural number It is always an integer It is a rational number It is an irrational number is 6
8. [CDS (I) Paper 1 Q27] Nine numbers are Written in ascending Order. The middle number is the average Of the nine numbers. The average of the first five larger numbers is 68 and thit of five smaller numbers is **(a)** If x cos 600 + y cos 00 3 and 4x sin 300 — y cot 450 = 2, then what is the value of x ? If the unit Of weight is 15/4 kg, what number will 3/2 quintal represent ? **(b)** 25 **(c)** 1/9 **(d)** None of the above A dishonest dealer professes to sell his goods at cost price, but uses a false weight and thus gains 20%. For a kilogram he uses a weight of
9. [CDS (I) Paper 1 Q39] What is the value Of 13th observation ? **(a)** 20 **(b)** 23 **(c)** 32 **(d)** 40 26
10. [CDS (I) Paper 2 Q41] 42. What is the value Of k for which the System of equations x + 2y — 3 O and 5x + ky + 7 = O has no solution ? -3/14 **(b)** —14/3 1/10 **(d)** 10 For two natural numbers m and n, let g denote the greatest common factor or m and n. Consider the following in respect of three natural numbers k, m and n . gm(nk) 2. grnngnk gmk Which Of the above Ware correct ? **(a) (b) (c) (d)** I only 2 Only Both I and 2 Neither I nor 2 What is equal to ? **(a) (b) (d)** —3/4 1/4 3/4 Jf the roots of the equation — are equal, then what is the value Of m ? **(a) (b) (c)** 1/2 0
11. [CDS (I) Paper 2 Q47] If x a + a + b 3 3 then what is the value of xa + Sbx — 2a ? **(a) (b) (c)** 2a3 -2aa What least value must be given to @ so that the number 84705@2 is divisible by 9 ? If p is an integer, then every Square integer is Of the form **(a)** 2p Or (4p — I) **(b)** 4p or (4p— I) **(c)** 3p or (3p + I) **(d)** 4p or (4p + I) The paint in a certain container is sumcient to paint an area equal to 5875 m2. HOW many 51 bricks Of dimensions 12•5 cm x IO cm x cm can be painted out of this container ? **(b) (c)** 225 180 150 100 The ratio of volumes Of two cones is 4 : 5 and the ratio or the radii of their bases is 2 :
12. [CDS (I) Paper 2 Q63] 'hat the lual the **(a) (b) (c) (d)** From a cylindrical log whose height is equal to its diameter, the greatest possible sphere has been taken Out. What is the fraction of the original log which is eut away ? In thc figure given above, PT is a tangent to a circle Of radius 6 cm. If P is at a distance Of IO cm from the centre O and PB = 5 cm, then what is the length of the chord BC ?
13. [CDS (I) Paper 2 Q68] Consider the following statements : **(a) (b) (c) (d)** 900 +

14. [CDS (I) Paper 2 Q98] 99. is m 26 -A) The median of three positive integers, no two of which are equal, is **(a) (b) (c) (d)** 1/2 2 loglo 13

15. [CDS (I) Paper 2 Q99] 100. The median of three positive integers, no two of which are equal, is **(a) (b) (c) (d)** 1/2 2 loglo 13

16. [CDS (I) Paper 3 Q4] Consider those numbers between 300 and 400 such that when each number is divi&d by 6, 9 and 12. it leaves 4 as remainder in each case. What is the sum of the numbers ? **(a)** 692 **(b)** 764 **(c)** 1080 **(d)** 1092 T.DETA.K-FN - A 2

17. [CDS (I) Paper 3 Q5] Consider the following assumption and two statements : Assumption : A number 'ABCDE is divisible by **(a)** Last digit is 4 **(b)** Last digit is 8 **(c)** Last digit is 2 the expansion is a positive S. Consider the following assumption and two statements : A number •ABCDE' is Assumption divisible by 11. E-D+C-B+A is Statement — I divisible by 11. • E-D+C-B+A=O. Statement — , Which one Of the following is corect ? **(d)** Last two digits are zero 2. What is the locus Of centres of circles which touch a given line at a given point ?

18. [CDS (I) Paper 3 Q11] Statement—I : E —D + C —B + A is divisible by **(a)** Last digit is 4 **(b)** Last digit is 8 **(c)** Last digit is 2 the expansion is a positive S. Consider the following assumption and two statements : A number •ABCDE' is Assumption divisible by 11. E-D+C-B+A is Statement — I divisible by 11. • E-D+C-B+A=O. Statement — , Which one Of the following is corect ? **(d)** Last two digits are zero 2. What is the locus Of centres of circles which touch a given line at a given point ?

19. [CDS (I) Paper 3 Q1] What can said about Of 212" —64", where n integer ? **(a)** Last digit is 4 **(b)** Last digit is 8 **(c)** Last digit is 2 the expansion is a positive S. Consider the following assumption and two statements : A number •ABCDE' is Assumption divisible by 11. E-D+C-B+A is Statement — I divisible by 11. • E-D+C-B+A=O. Statement — , Which one Of the following is corect ? **(d)** Last two digits are zero 2. What is the locus Of centres of circles which touch a given line at a given point ?

20. [CDS (I) Paper 3 Q9] A three digit nurntrr is divisible by 11 and has its digit in the unit's place equal to l. The number is 297 more than tlr number obtained by reversing the digits. What is the number ? **(a)** 121 **(b)** 231 **(c)** 561 **(d)** 451 IO. If u. , v, w are real such that u3 ISuvw, which one of - the following is correct ? **(c)** u — 2 v = 3w **(d)** 11+2v=-3w

21. [CDS (I) Paper 3 Q12] Which one of the following is a corect statement ? 13, What is the least integral value of k for which the equation has both the roots positive ? **(a)** 3 years **(b)** 4 years **(c)** 5 years **(d)** 6 years 8. What is the median of the ages of minor children ?

22. [CDS (I) Paper 3 Q10] If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? **(d)** u +2v=-3w **(a)** 3 years **(b)** 4 years **(c)** 5 years **(d)** 6 years 8. What is the median of the ages of minor children ?

23. [CDS (I) Paper 3 Q13] What is the least integral value of k for which the equation has both the roots positive ? **(a)** 3 years **(b)** 4 years **(c)** 5 years **(d)** 6 years 8. What is the median of the ages of minor children ?

24. [CDS (I) Paper 3 Q17] If the highest common factor of two positive integers is 24 then their least Common multiple cannot **(a)** 72 **(b)** 216 **(c)** 372 **(d)** 600

25. [CDS (I) Paper 3 Q18] If r and s are any real numbers such that Oss r + then what is the maximum value Of the product rs ? 19. r is a non-zero real nurntwr such that r75 > PO. This is possible only when **(a)** 72 **(b)** 216 **(c)** 372 **(d)** 600 18. If r and s are any real, numbers such that 1, then what is the maximum value Of the product rs ? 19. r is a non•zero real Humber such that r75 > PO. This is possible only when 20. The median BD of Che triangle ABC meets AC at D. If then which one of the following is correct ?

26. [CDS (I) Paper 3 Q21] The remainder on dividing given integers a and b by 7 are resrrctively 5 and **(a)** 72 **(b)** 216 **(c)** 372 **(d)** 600 18. If r and s are any real, numbers such that 1, then what is the maximum value Of the product rs ? 19. r is a non•zero real Humber such that r75 > PO. This is possible only when 20. The median BD of Che triangle ABC meets AC at D. If then which one of the following is correct ?

27. [CDS (I) Paper 3 Q4] What is the remainder when ab is divided by7? **(a)** 72 **(b)** 216 **(c)** 372 **(d)** 600 18. If r and s are any real, numbers such that 1, then what is the maximum value Of the product rs ? 19. r is a non•zero real Humber such that r75 > PO. This is possible only when 20. The median BD of Che triangle ABC meets AC at D. If then which one of the following is correct ?

28. [CDS (I) Paper 3 Q36] For any integer n, what is the HCF Of integers m = + I and k 9n +4 ? **(a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** days 33. If of

29. [CDS (I) Paper 3 Q38] Which one of the following is an infinite set ? **(a)** {x: x is a whole number less than or equal to 1000} **(b)** {x: x is a natural number less than 1000} **(c)** {x: x is a positive integer less than or equal to 1000} **(d)** {x: x is an integer and less than 1000}

30. [CDS (I) Paper 3 Q48] Three lines intersect each other in pairs. What is the påmber of angles so formed ? **(d)** 12 **(a)** 12 49. ABCD is a square. p. Q, R, S are points on the sidesAB, BC, CD, DA respectively such that CR=DS. What is ZSPQ equal ? **(b)** 300 **(c)** 450 **(d)** 600

31. [CDS (I) Paper 3 Q51] A ladder of 17 ft. length reaches a window Which is 15 ft. above the ground on one side of the street. Keeping its foot at the same point the ladder is turned to the other side of the street and now it reaches a window 8 ft. high. What is the Width of the street ? **(a)** 23 ft **(b)** 15 ft. **(c)** 25 ft. **(d)** 30ft.

32. [CDS (I) Paper 3 Q73] What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? **(d)** 4-5 **(a)** 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? **(b)** 9th **(c)** 15m **(d)** I •5m

33. [CDS (I) Paper 3 Q78] A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? **(a)** 100 **(b)** 90 **(c)** 80 **(d)** 75

34. [CDS (I) Paper 3 Q82] With the help of histograrn one can prepare **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

35. [CDS (I) Paper 3 Q99] What is the value of c ? **(a)** IO cm 5 cm **(b)** cm **(c)** 56 94. When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the 2 2 97. IA there three simultaneous linear equations in two unknowns. What can be the number of solutions **(d)** One or infinite

36. [CDS (I) Paper 4 Q2] What is the last digit expansion Of 34798 ? **(a)** x—2y of and third side in cm? **(b)** g, 2v'fi **(c) (d)**

37. [CDS (I) Paper 4 Q3] What is in the equal to? **(a)** x—2y of and third side in cm? **(b)** g, 2v'fi **(c) (d)**

38. [CDS (I) Paper 4 Q6] What is the sum Of the digits Of the least number which, when divided by 52, leaves 33 as remainder; when divided by 78, leaves 59 and when divided by 117, leaves 98 as remainder? **(a)** 17 **(b)** 18 **(c)** 19 **(d)** 21 If f(x) and g(x) are two polynomials with integral coefficients which vanish at x 112, then what is the factor of HCF of f (x) and dx)? **(c)** 2x —1 **(d)** 2x+1 If the expression (px3 + x2 —2x— is divisible by (x —1) and (x + 1), what are the values of p and q respectively? **(a)** 2-1 **(b)** -21 **(d)** 21 TWO sides acute •angled triangle are 6 cm and 2 cm respectively. Which one of. the following represents ,the correct range of the

39. [CDS (I) Paper 4 Q11] What is the smallest positive integer which when divided by 4, 5, 8, 9 leaves remainder's 3, 4, 7, 8 respectively? **(b)** 319 **(a)** 20 years **(b)** 22 years **(c)** 24 years **(d)** 25 years IO. A mixture contains niilk and water in the ratio 5 : On Adding 5 litres Of water, the ratio Of' milk and water becomes 5 : 2. Wh#t is the quantity Of milk in the original mixture?

40. [CDS (I) Paper 4 Q12] What is the value Of x for which x, x + I, x + 3 are all' prime numbers? **(a)** 20 years **(b)** 22 years **(c)** 24 years **(d)** 25 years IO. A mixture contains niilk and water in the ratio 5 : On Adding 5 litres Of water, the ratio Of' milk and water becomes 5 : 2. Wh#t is the quantity Of milk in the original mixture?

41. [CDS (I) Paper 4 Q22] What is the solution of the equation 2 23. **(a)** 72 s 68 s 18, If **(b) (c) (d)** Myz 2xyz 3xyz —3xyz 21. If the 14th term Of an arithmetic series is 6 and 6th term is 14, then What is the 95th term? @ -75

42. [CDS (I) Paper 4 Q24] If k is any even positive integer, then (k 2 + 2k) is **(a)** divisible by 24 **(b)** d ivisible 8 but may no t be divisible 24 **(c)** divisible by 4 but may not be divisible by 8 **(d)** divisible by 2 but may not be divisible by 4

43. [CDS (I) Paper 4 Q28] Under what condition do the equations and have a unique solution? **(d)** kgo **(a)** divisible by 24 **(b)** d ivisible 8 but may no t be divisible 24 **(c)** divisible by 4 but may not be divisible by 8 **(d)** divisible by 2 but may not be divisible by 4 25. If salary of X is more than salary Y, then by how much percentöge is salary Of Y less than X?

44. [CDS (I) Paper 4 Q31] If a+b+czO, then what is the value a2 b2 fc2 Of _ + ? bc ab If I is added eo the denominator of 32. a fraction, it pecomes 1/2 and if I is added to the numerator, the fraction becomes **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

45. [CDS (I) Paper 4 Q1] What is the fraction? **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

46. [CDS (I) Paper 4 Q32] 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction? **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

47. [CDS (I) Paper 4 Q94] For a set Of pdsitive numbers, consider the following statements : **(a)** + Crn 2 **(b)** 44 **(c)** cm2 Which of the above statements is are correct? **(d)**

48. [CDS (I) Paper 4 Q100] What is ZAEB equal to? **(d)** None Of the above **(a)** + Crn 2 **(b)** 44 **(c)** cm2 Which of the above statements is are correct? **(d)**

49. [CDS (I) Paper 5 Q31] What is the square root Of 0-324 x 0-64 x 129-6 0-729 x 1-024 x 36 **(a) (b)** 4 2 **(c) (d)** 3 -2 Consider the following statements : What is the number of circles passing through all the vertices Of a given triangle ?

50. [CDS (I) Paper 5 Q46] If thræe sides Of a right-angled triangle are 50 integers in their lowest form, then One Of its sides is always divisible by **(a) (b) (c)** A certain amount of money has to be divided between two persons P and Q in the ratio 3 : 5. But it Was divided in the ratio Of 2 : 3 and thereby Q loses IO. What was the amount ? None Of the above Consider the following statements . If p is a prime such that p + 2 is algo a prime, **(d)**

51. [CDS (I) Paper 5 Q2] Which Of the above statements Ware correct ? **(a) (b)** I only 2 only Both and 2 Neither nor 2 When a positive integer n is divided by 5, the remainder is 2. is the remainder When the number 3n is divided by 5 ? **(a) (b) (c) (d)** 2 4 Of x rupees is equal to t times q% Of y rupees. What is the ratio of x to y ? **(a) (b) (c)** p q : qt e

52. [CDS (I) Paper 5 Q5] But it Was divided in the ratio Of 2 : 3 and thereby Q loses IO. What was the amount ? **(a) (b) (c)** 250 soo 350 400 What is the value Of [loglo (5 loglo ? **(a) (b) (c)** 4 2 Which one of the following three digit numbers divides 9238 and 7091 with the same remainder in each case ? **(a) (b) (c) (d)** 113 209 317 191 If a and b are positive integers, x and y are non-negative integers and a = bx y, then which One of the following is correct ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

53. [CDS (I) Paper 5 Q48] 49. q% Of (14 -A) If thræe in tegers its sides sides Of a right-angled triangle are 50. in their lowest form, then One Of is always divisible by **(a) (b) (c)** A certain amount of money has to be divided between two persons P and Q in the ratio 3 : 5. But it Was divided in the ratio Of 2 : 3 and thereby Q loses IO. What was the amount ? None Of the above Consider the following statements . If p is a prime such that p + 2 is algo a prime, **(d)**

54. [CDS (I) Paper 5 Q53] What is the value Of [loglo (5 loglo ? **(a) (b) (a) (b) (c)** 4 2 I only 2 only Both and 2 Neither nor 2 When a positive integer n is divided by 5, the remainder is 2. is the remainder Which one of the following three digit numbers divides 9238 and 7091 with the same remainder in each case ? When the number **(a) (b) (c) (d)** 2 4 3n is divided by 5 is equal to t times the ratio of x to y **(a) (b) (c) (d)** 113 209 317 191 Of x rupees y rupees. What is **(a) (b) (c)** p q : qt If a and b are positive integers, x and y are non-negative integers and a = bx y, then which One of the following is correct ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

55. [CDS (I) Paper 5 Q67] The HCF of two polynomials p(x) And q(x) is 2x (x + 2) and LCM is 24x (x (x — 2). If + 32x2 + 32x, then what is q(x) equal to ? **(a)** 4xa — 16x **(b)** 6x3 - 24x **(c)** 12x3 + 24x **(d)** 12x3 - 24K What is the condition that the equation ax2 bx + c O, where a O, has both the roots positive ? **(a) (b) (c) (d)** a, b and c are Of same sign a and b are of same sign b and c have the same sign opposite to that of a a and c have the same sign opposite to that of b What is the volume (in cm) of a spherical shell With 8 cm and IO em as its internal and external diameters respectively ? 3 12k 2441 3 2501 ABC is a triangle and AD is perpendicular to BC. It is given that the lengths Of AB, BC, CA are all rational numbers. Which one Of the following is correct ? **(a) (b) (c) (d)** AD and BD must be rational AD must be rational but BD need not be rational BD must be rational but AD need not be rational Neither AD nor BD need be rational What is the missing frequency x ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

56. [CDS (I) Paper 5 Q66] 67. The HCF of two polynomials p(x) And q(x) is 2x (x + 2) and LCM is 24x (x (x — 2). If + 32x2 + 32x, then what is q(x) equal to ? **(a)** 4xa — 16x **(b)** 6x3 - 24x **(c)** 12x3 + 24x **(d)** 12x3 - 24K What is the condition that the equation ax2 bx + c O, where a O, has both the roots positive ? a, b and c are Of same sign a and b are of same sign b and c have the same sign opposite to that of a a and c have the same sign opposite to that of b **(a) (c)** 31 27 24 14 What is the missing frequency y ? **(a) (b) (c)** 27 24 11 What is the volume (in cm) of a spherical shell With 8 cm and IO em as its internal and external diameters respectively ? 3 12k 2441 3 2501 ABC is a triangle and AD is perpendicular to BC. It is given that the lengths Of AB, BC, CA are all rational numbers. Which one Of the following is correct ? If one root of the equation 2x2 + 3x + c 0-5, then what is the value of c ? **(b)** z O is (18 **(a) (b) (c) (d)** AD and BD must be rational AD must be rational but BD need not be rational BD must be rational but AD need not be rational Neither AD nor BD need be rational P-PQFA-L-RZ

57. [CDS (I) Paper 5 Q88] I APB Cpl) i A, B, C **(a)** 121-7 121-75 **(b)** 121-75 ABC t, X Y AX.AYur **(c)** 16 2 **(d)** 20 cm 2

58. [CDS (I) Paper 5 Q84] 88. I APB Cpl) i A, B, C **(a)** 121-7 121-75 **(b)** 121-75 ABC t, X Y AX.AYur **(c)** 16 2 **(d)** 20 cm 2

59. [CDS (I) Paper 6 Q23] What is the second number ? **(a)** 423 **(b)** 843 **(c)** 1683 **(d)** 2523 Two numbers are in the ratio 3 : 5. If 9 is subtracted from each number, then they 2. 3. 4. Rs. 240 Rs. 480 Rs. 800 are in the ratio of 12 : 23. second number ?

60. [CDS (I) Paper 6 Q54] 55. A number when divided by 2, 3 or 5 gives remainder **(a)** 53/9 days **(b)** 34/7 days **(c)** 85/13 days **(d)** 53/10 days 49.. The radii of the circular ends Of a bucket Of height 40 em are of lengths 35 cm and 14 cm. What is the volume Of the bucket ?

61. [CDS (I) Paper 6 Q63] Consider the following statements 64. **(a)** 36 **(b)** 44 **(c)** 52 64. **(d)** 150

62. [CDS (I) Paper 6 Q3] Set of points of a given line is a finite set. Intelligent students in a class is a Good books in a school library is a Which of the above statements is/are not correct ? **(a)** 1 only **(b)** 2 and 3 Only **(c)** I and 2 only **(d)** 1, 2 and 3 If a and ß are the roots of the equation x2 — 6x + 6 = O, what is a3 + ß3 + + + a + ß equal to ? **(d)** 124 — 24, then what is value of m ? **(b)** 1/3 If n is a natural number, then is **(a)** always a natural number **(b)** always a rational number **(c)** always an irrational number **(d)** either a natural nurnERr or an irrational number 18

63. [CDS (I) Paper 6 Q66] 20 : 27 9:10 27 : 20 Consider the following statements **(a)** 36 **(b)** 44 **(c)** 52 64. **(d)** 150

64. [CDS (I) Paper 6 Q2] 3. Set of points of a given line is a finite set. Intelligent students in a class is a Good books in a school library is a x frequency 23 4 2 7 14 22 6 30 What is the cumulative frequency of variate Which of the above statements is/are not correct ? **(a)** 1 only **(b)** 2 and 3 Only **(c)** I and 2 only **(d)** 1, 2 and 3 If a and ß are the roots of the equation x2 — 6x + 6 = O, what is a3 + ß3 + + + a + ß equal to ? **(c)** 45 **(d)** 124 21 — 24, then what as value of m ? **(b)** 1/3 If n is a natural number, then **(a)** always a natural number **(b)** always a rational number **(c)** always an irrational number **(d)** either a natural nurnERr or an irrational number Consider the following statements in respect of the set S = {l, 2, 3, (n + l)f2 is the median of the numbers in S. 2. n is the mode of the numbers in S. Which Of the above statements is/are correct ? **(a)** 1 only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither nor 2 The radii of two cylinders are in the ratio 2 : 3 and their curved surface areas are in the ratio 5 :

65. [CDS (I) Paper 6 Q76] The product of two numbers is 6912 and their GCD is **(a)** 286 **(b)** 296 73. The areas of two circles are in the ratio I : 2. If the two circles are bent in the form Of squares, what is the ratio of their areas ? The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? **(c)** 3 units **(d)** 5 units

66. [CDS (I) Paper 6 Q84] Consider the following statements : **(a)** u = 2•25v **(b)** u = 3v **(c)** 3v < u < 4v 80. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? **(d)** 256

67. [CDS (I) Paper 6 Q2] The product of any three consecutive integers is divisible by **(a)** u = 2•25v **(b)** u = 3v **(c)** 3v < u < 4v 80. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? **(d)** 256

68. [CDS (I) Paper 6 Q6] Any integer can be expressed in one of the three forms 3k, 3k + l, 3k + 2 where is an integer. Which of the above statements is/are correct ? **(a)** I Only **(b)** 2 Only **(c)** Both 1 and 2 **(d)** Neither I nor 2 The fourth. proportional to 7, 11, **(a)** 16 **(b)** 18 **(c)** 20 **(d)** 22 14 is ABC is an isosceles triangle such that AB BC = 8 cm and Z ABC -

69. [CDS (I) Paper 6 Q91] 88. Which one of the following is a prime **(a)** 280 **(b)** 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ? **(c)** r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca • cosec ß sin cosß equal to ? **(d)** None of the abo ve Consider the following statements : 88. 89. 90. 91. Which one of the following is a prime number ?

70. [CDS (I) Paper 6 Q92] In the figure given above, LM is parallel to QR. If LM divides the triangle PQR such that area of trapezium LMRQ is two times the area of triangle PLM, then what 93. **(d)** 24 **(a)** **(b)** ¯ equal to ? 2 The largest integer that divides product of any four consecutive integers is **(a)** **(c)** 4 12 94. c D lat

71. [CDS (I) Paper 7 Q13] Consider the fogowing statements ; I. Eyery composite nun!ber is a natural number **(a)** For size Of readymade garments, mode is the best suitable measure **(b)** For average rate Of increase when the rate of population growth is given, geometric is best suitable **(c)** For average rate of speed when different distances are covered by different rates ofspeed, harmonic mean is best suitable Z **(d)** For average level of iniélligence of students in a class, arithmetic mean is the best suitable If the surface area of a sphere is 616 square cm, what is its volume ?

72. [CDS (I) Paper 7 Q75] What is the sum Of their reciprocals ? **(a)** 64 minutes **(b)** 80 minutes **(c)** 96 minutes **(d)** 120 minutes The pair of numbers which are relatively prime to each other is :

73. [CDS (I) Paper 7 Q99] If A = {x : x is an even natural number}, B = {x : x is a natural number and multiple of 5} and C = {x : x is a natural number and multiple of 10}, then what is A n (B u C) equal to ? **(a)** **(b)** **(c)** **(d)** {10, 20, 30, . {5, 10, 15,20,...} {20, 40, 60,

74. [CDS (I) Paper 8 Q11] 12. Consider the following Statements I. Every integer is a rational n umber. **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 The product of HCF and LCM of 18 and 15 is :

75. [CDS (I) Paper 8 Q2] Every rational number is a real number. Which of the above statements is/are correct ? **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 The product of HCF and LCM of 18 and 15 is : **(a)** 120 **(b)** 150 **(c)** 175 **(d)** 270 What is 3 x 03 x 0-03 x 0003 x 00003 x 30 equal to ? **(b)** (0-009)3 **(c)** (0-0009)3 **(d)** None Of the above

76. [CDS (I) Paper 8 Q15] 16. What is 0.242424.... **(a)** 23/99 **(b)** 8/33 **(c)** 7/33 **(d)** 47/198 equal to ? 4

77. [CDS (I) Paper 8 Q18] Which one of the following has least number of divisors ? **(a)** 88 **(b)** 91 **(c)** 96 **(d)** 99

78. [CDS (I) Paper 8 Q25] The price of an article is Rs. **(a)** x- 10 **(b)** x = 11 **(c)** x = 11.5 If 0.764 y = 1.236 x, Y—x value Of **(d)** 0.764

79. [CDS (I) Paper 8 Q27] What is the value of **(a)** x- 10 **(b)** x = 11 **(c)** x = 11.5 If 0.764 y = 1.236 x, Y—x value Of **(d)** 0.764

80. [CDS (I) Paper 8 Q31] What is the greatest number that divides 13850 and 17030 and leaves a remainder 17? **(a)** 477 **(b)** 159 **(c)** 107 **(d)** 87 If the sum of an integer and its reciprocal is 10/3, then the integer is equal to **(c)** 12 **(d)** 13 Let x e {2, 3, 4} and y {4, 6, 9, 10}. Let A be the set Of all order pairs (x, y) such that x is a factor of y. How many elements does the set A contain ? **(a)** 12 **(b)** 10

81. [CDS (I) Paper 8 Q32] Which one of the following is a null set ? 8 O { x is a real number : x > I and = { x is a real number : x 2 1 and (Cond) **(a)** x- 10 **(b)** x = 11 **(c)** x = 11.5 If 0.764 y = 1.236 x, Y—x value Of **(d)** 0.764

82. [CDS (I) Paper 8 Q35] 36. 37. = 100- lox = — 5x2 If (x - 6) is the HCF ofx2 - 2x - 24 and k? **(a) (c)** kx — 6, then what is the value Of 3 6 If the expression x3 + 4x2 + 4x + k has (x + 4) as a factor, then what is the value a) -16 **(b)** 16 **(c)** 32 - 32 If = 4, then what is x equal **(a)** 150 **(b)** 216 **(c)** 316 **(d)** 450 Which one of the following is neither prime number nor composite number ?

83. [CDS (I) Paper 8 Q2] If n is a prime number greater than 5, then n' — 1 is divisible by 2400. Every square number is of the form 5n or (5n — 1) or (5n + l) where n is a whole number. Which of the above Statements is/are correct ? **(a)** 1 only **(b)** 2 only **(c)** Both land 2 **(d)** Neither 1 nor 2 33. kx — 6, then what is the value Of

84. [CDS (I) Paper 8 Q34] 36. 37. = 100- lox = — 5x2 If (x - 6) is the HCF ofx2 - 2x - 24 and k? **(a) (b)** 3 6 If the expression x3 + 4x2 + 4x + k has **(c)** 16 **(d)** 32 - 32 35. If

85. [CDS (I) Paper 8 Q43] 44. What is the value Of k which will make the expression 4k2 + 12k + k a perfect square ? If one root of the equation — + —+ — O is reciprocal of the other, then which one of the following is correct ? **(c)** ac 1 If f(x) is a polynomial with constant term IO having a factor (x — k) where k is an integer, then what is the possible value of k? **(a)** -20 **(b)** 20 If a —24 •J: , then what is the value of **(a)** 12 **(b)** 14 **(c)** 16 **(d)** 18 12 4 47 48

86. [CDS (I) Paper 8 Q46] There are some benches in a class room having the number of rows 4 more than the number of columns. If each bench is seated with 5 students, there are two seats vacant in a class Of 158 students. The number of rows is **(d)** 10 **(a)** ac I If f **(b)** -20 **(c)** 20 **(d)** 12

87. [CDS (I) Paper 8 Q65] Consider the following statements **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 If the diagonals of a quadrilateral are equal and bisect each other at right angles, then the quadrilateral is a

88. [CDS (I) Paper 9 Q2] There iS Of rational numbers any rationctl There is number Of between 'There iS a number numbers •beiween any rational n umberg. Which Of the above terpenes ia/üre eorrect ? L only 2 only 3 only 2 and 3 Representation Of — . Where p, are integers, q is 781 -3330 1171 2341 seh00i are teachers Who teech M thematieg or Phy.ics. Of these teacher. 20 teach Mathematics and Ifi teach Physics, 5 both Mathematics and Physics. The of teachers i.eat:hing only emetics is **(a)** days 15 18 d eye if a body B at the rate Of x and gnother distance at or y km/hr, thon the average 'Deed in krn'hr is XN-Y 2 A sails a distance 48 elong the of a river in 8 it takes 12 to return the distance, then the epeed of the Lhe river is O•ti Y. k'n/br A a in hours, Arter working for 6 hours, he look the help of boy finished goather 5 The time that the alone ecanplete Gbe both Mathematics and Physics. The of teachers i.eat:hing only emetics is 12 -A **(b) (c)** 30 45 60 hrs 5. 6. sly 'X' comp **(d)** days 15 18 d eye if a body B at the rate Of x and gnother distance at or y km/hr, thon the average 'Deed in krn'hr is XN-Y 2 A sails a distance 48 elong the of a river in 8 it takes 12 to return the distance, then the epeed of the Lhe river is O•ti Y. k'n/br A a in hours, Arter working for 6 hours, he look the help of boy finished goather 5 The time that the alone ecanplete Gbe

89. [CDS (I) Paper 9 Q8] Which a simple intere•t of 55 obtained after 9 rate Of S — •S per is 1.000 2,000 a natural number an integer and a number a not •n integer a number but not a The Of numbers is 2376 While their HCF 33, If one Of the is 297, other is 216 264 trgvel• L distanee at 3 knybr reaehæ 15 minutes late. he travel' at 15 minute earlier. The he has to travel 72 km O-DETA The digit number, which gum of the digi±8 and product Of the digit* the remainde•• Of quotients is one, is 32 Tile short and long hands Of a cm 6 Cyr. long resp«tiveiy. Theo distance. by Of 'bort hand in days long hand in 3 day* i' vahze Of 725 x 725 x 371 1960 1096 **(d)** 1016 N. (N 2). 41 the Of possn:Ae solutions N are . Of above **(a)** 1/2 **(b)** 1/√2 **(c)** √3/2 **(d)** 1

90. [CDS (I) Paper 9 Q26] Consider the i" Of A and B **(a)** 32 36 40 **(b)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only **(c)** 32 36 40 30. If 3 men 4 buys can a piece Of work in 8 days, then 6 men and B eon in 2 as•ys 4 days e days i 6 days X. Y and had taken dinner together. cost Of the meal Of Z 2W more than that Of Y and the east the meal Of X was 5/6 aB much as the cost Of the of Z. If paid then what W•øg the that the three Of them had paid ? 285 300 35b None of the above 26. Consider the i" Of A and B 2. Which Of the above are ? and 2 only Z and 3 only 3 •ancy iB the HCF of the poiynomiaiz **(d)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only

91. [CDS (I) Paper 9 Q3] 4. **(a)** 32 36 40 **(b)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only **(c)** 32 36 40 30. If 3 men 4 buys can a piece Of work in 8 days, then 6 men and B eon in 2 as•ys 4 days e days i 6 days X. Y and had taken dinner together. cost Of the meal Of Z 2W more than that Of Y and the east the meal Of X was 5/6 aB much as the cost Of the of Z. If paid then what W•øg the that the three Of them had paid ? 285 300 35b None of the above 26. Consider the i" Of A and B 2. Which Of the above are ? and 2 only Z and 3 only 3 •ancy iB the HCF of the poiynomiaiz **(d)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only

92. [CDS (I) Paper 9 Q36] 87. 88. in equation — and •mabest prime p) — not a 29 **(a)** a3r03 **(b)** b4/a4 following is 35. tables together cost nd two together 36. 4 chai-rs and 5 cable* proporvion, $7. differenee in Of equation 2x2- 4 3-5 and **(c) (d)** The

93. [CDS (I) Paper 9 Q5] What is cog (A R) to ? **(a) (b) (c)** {b **(d)**

94. [CDS (I) Paper 9 Q56] Direct ü»ng : Consider the foa.ou'ing for the (04) questions that follow : 16 Ai four **(a) (b) (c)** {b **(d)**

95. [CDS (I) Paper 9 Q2] Height of i ndividua measured in cm. **(a) (b)** I only and 4 3 and 4 **(c)** o: 10 10—20 20 -so 30-40 weight Of Of 15 boys and IO girls is S8•4 kg. If average weight Of is 40 kg, then what is the average weight Of the girls ? 366 kg 35 kg 36 856 kg : ' Consi&r the following for the next three question. that a di8t.ribütion having class intervals O- 10, 10-20. 20-30 SO-40S respective frequencies are x, x 8, x — 2 and x —4 and the arithmetic mean Of the Consider the statements I. A freq uenCy ution condenses the data and reveals ite important features. 2. A frequency distribution is an representation of originpl Which of the above statements Ware corect ? **(d)** only 2 onbr Both and 2 Neither nor 2 96. dstribution is 17•8. value Of x ie

96. [CDS (I) Paper 9 Q3] Waiting tim e to Gi lure electric **(a) (b)** I only and 4 3 and 4 **(c)** o: 10 10—20 20 -so 30-40 weight Of Of 15 boys and IO girls is S8•4 kg. If average weight Of is 40 kg, then what is the average weight Of the girls ? 366 kg 35 kg 36 856 kg : ' Consi&r the following for the next three question. that a di8t.ribütion having class intervals O- 10, 10-20. 20-30 SO-40S respective frequencies are x, x 8, x — 2 and x —4 and the arithmetic mean Of the Consider the statements I. A freq uenCy ution condenses the data and reveals ite important features. 2. A frequency distribution is an representation of originpl Which of the above statements Ware corect ? **(d)** only 2 onbr Both and 2 Neither nor 2 96. dstribution is 17•8. value Of x ie

97. [CDS (I) Paper 10 Q2] 3. Consider the following statements . **(a)** 1398 142% **(b)** 1484 4. What iS the HCF of a2b4 5 and **(c)** ab **(d)** a2b3 . What iS the LCM of x2 +2x-8, 2. 3. Consider the following statements . 2. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7. Every composite number less than 121 is divisible by a prime number less than 11. x

98. [CDS (I) Paper 10 Q7] Every composite number less than 121 is divisible by a prime number less than **(a)** 1398 142% **(b)** 1484 4. What iS the HCF of a2b4 5 and **(c)** ab **(d)** a2b3 . What iS the LCM of x2 +2x-8, 2. 3. Consider the following statements . 2. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7. Every composite number less than 121 is divisible by a prime number less than 11. x

99. [CDS (I) Paper 10 Q11] What is equal to? **(d)** 4 -2 If x + and then what is 4x2 + *4xy equal to? **(a) (b) (d)** 75 85 91 100 If the expression x +3x2 +4x+k has a factor x + 5, then what is the value of k ? **(a)** -70 **(b)** 70 **(c)** 48 B-BCRB-M-NBU 48A

100. [CDS (I) Paper 10 Q1] 7710312401 is divisible by **(a)** 15 **(b)** 25 **(c)** 20 **(d)** 30 9. 10. 11. What is equal to?

101. [CDS (I) Paper 10 Q16] 17. 64 74 48 **(d)** 42 **(a)** O only **(b)** 20 only **(c)** both 10 and 20 **(d)** neither IO nor 20 If

102. [CDS (I) Paper 10 Q18] For what values of k will 4x5 + 9x4 —7x3 — 5x2 —4kx+3k2 contain x — 1 as a factor? xty2 -z2)+ y(z2 '2 —X —y2)LS 19. divisible by **(a)** (y —z) only **(b)** (z — x) only **(c)** both (y— z) and (z — x) **(d)** neither (y —z) nor (z —x) B-BCRB-M-NBU / 48A 195 +21 S is divisible by **(a)** O only **(b)** 20 only **(c)** both 10 and 20 **(d)** neither IO nor 20 If (49)2 — (25) 2 = then what is x equal to? 6

103. [CDS (I) Paper 10 Q23] "IXvo 195 +21 S is divisible by **(a)** O only **(b)** 20 only **(c)** both 10 and 20 **(d)** neither IO nor 20 If (49)2 — (25) 2 = then what is x x + Y = 4 and 1 then what is x + y equal to? **(a)** 11 **(b)** 10 A person's salary has increased from e 7,200 toe 8,100. What is the percentage increase in his salary? equal to? **(a)** 64 **(b)** 74 **(c)** 48 **(d)** 42 For what values of k will 4x5 + 9x4 —7x3 — 5x2 —4kx+3k2 contain x — 1 as a factor? **(a)**

104. [CDS (I) Paper 10 Q26] 27. For integers a, b and if HCF@ b) 1 and HCF(g c) = 1, then which one of the following is conect? **(a)** HCF(g bc) = 1 **(b)** HCF(Q **(c)** HCF(a, bc) -b **(d)** None Of the above If b is the largest square divisor Of c and a2 divides C, then which one of the following is correct? (a, b, c are i ntegers) **(b) (d)** b divides a a does not divide b a divides b a and b are co-prime If k is a positive integer, then every Square integer is of the form **(a)** 4k only **(b)** 4k or 4k+3 **(c)** 4k + 1 or 4k + 3 **(d)** 4k or 4k + I Every prime number of the form 3k can be represented in the for-rn 6m + 1 (k, m are integers) when **(a)** k is odd **(b)** k is even **(c)** k can be both odd and even **(d)** No such form is possible B-BCRB-M-NBU / 48A 8

105. [CDS (I) Paper 10 Q32] What is the last digit in 7402 + 3402 ? **(d)** None Of the above **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

106. [CDS (I) Paper 10 Q35] The pair of rational numbers that lies between — and — is 4 4 262 752 **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

107. [CDS (I) Paper 10 Q70] The number Of passengers with which the bus started was **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

108. [CDS (I) Paper 10 Q38] If x is positive even integer and y is negative Odd integer, then xy is **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

109. [CDS (I) Paper 10 Q42] 20 workers working for 5 hours a day complete a work in IO days. If 25 workers are employed to work IO hours a day, what is the time required to complete the work? **(a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** 8 days If m and n are natural numbers, then is **(a)** always irrational **(b)** irrational unless n is the mth power of an integer **(c)** irrational unless m is the nth power of an integer **(d)** irrational unless m and n are

110. **[CDS (I) Paper 10 Q74]** Consider the following statements • **(a) (b) (c)** 1 only 2 only Both 1 and 2 Neither 1 nor 2 75. Consider the following statements • 1. The perpendicular bisector of a chord of a circle does not pass through the centre of the 2. The angle in a semicircle is a right angle. Which of the statements given above is/ are correct? **(d)** I only

111. **[CDS (I) Paper 10 Q75]** Consider the following statements • **(a) (b) (c)** 1 only 2 only Both 1 and 2 Neither 1 nor 2 75. Consider the following statements • 1. The perpendicular bisector of a chord of a circle does not pass through the centre of the 2. The angle in a semicircle is a right angle. Which of the statements given above is/ are correct? **(d)** I only

112. **[CDS (I) Paper 10 Q82]** Consider the following statements . **(a)** 2 only **(b)** Both 1 and 2 **(c)** Neither 1 nor 2 84. 26 If tang + sec8= m, then what sec9 equal to? is 80. If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 **(d)** 2m

113. **[CDS (I) Paper 11 Q4]** 5. 6. **(b) (c) (d)** Let is Odd Or even is never divisible by 2 is always even None of the above a, b be positive integers. What is HCF equal to ? **(a)** (e) HCF (a, b) product Of four consecutive nßtural numbers plus one is **(a)** a non•square **(b)** always sum of two square numbers **(c)** a square **(d)** None of the above For any integers 'a' and with HCF(a, b) what is HCF (a b, a — b) equal to ? (8) It is always 1 **(b)** It is always 2 **(c)** Either I or 2 **(d)** None of th e above The expression 2xa * x2 by **(a) (b) (c)** 2x-i B-ABQA-N-Nau — I is divisible (2-

114. **[CDS (I) Paper 11 Q18]** 19. (4—A) The value Of x2 — **(a)** 8 11 22 4x I I can never be less 17 , then what is equal to ? **(a)** 16 63 95 **(c) (d)** None of the abov e If 3x4— 2x3 3x2 — 2x 3 is divided by + 2), then the remainder is **(b)** 27 25 What should be added to the expression x (x + a) (x + 2a) (x + 3a) so that the sum may a perfect square ? **(b) (c)** 4a2 None Of the above

115. **[CDS (I) Paper 11 Q26]** 27. The sum Of two numbers x and y is 2-5 times their difference. If the product of the numbers is 84, then What is the sum Of the two numbers ? **(b) (c)** 26 24 22 20 If n is a whole number greater than I, then n (n2— I) is always divisible by **(a) (c) (d)** 12 24 48 60 If the roots Of the equation (a2 — be) + 2 (b2 — ac) x + (c2 — ab) = O are equal, where b * O, then which one Of the following is correct ? **(a) (b) (d)** a + b c z abc 3 abe What is the remainder When 41 by 7? is divided **(a) (b) (c)** 2 4 None Of the above

116. **[CDS (I) Paper 11 Q2]** No integer Of the form 4k + g , where k is an integer, can be expressed as the sum Of two squares. Square of an Odd integer can be expressed in the form 8k + I, where k is an integer. Which Of the above Statements Ware correct ? **(a) (b) (c) (d)** 1 only 2 only Both 1 and 2 Neither I nor 2 For any integer n, what is HCF(22n+ 7, 3311 + 10) equal to? **(a) (d)** 11 None of the above

117. **[CDS (I) Paper 11 Q1]** The average of B an d C must be gre ater than 25, **(a) (b) (c)** 9 : 04 am 9 : 08 am 9: 14 am None of the above then **(d)**

118. **[CDS (I) Paper 11 Q2]** The average of A an d D must be I ess than **(a) (b) (c)** 9 : 04 am 9 : 08 am 9: 14 am None of the above then **(d)**

119. **[CDS (I) Paper 11 Q98]** 99. Consider the following : **(a) (b) (c) (d)** 1 only 2 only Both I and 2 Neither I nor 2

120. **[CDS (I) Paper 12 Q16]** HOW many pairs Of positive integers m and n satisw the equation m n 12 where n is an odd integer less than 60? 17 **(d)** . The 7 4 3 sides Of a triangle are in the ratio — — : If its perimeter is 234 52 cm, then What is the length Of the smallest side? **(a)** 9 cm **(b)** I O Crn **(c)** 11 cm **(d)** 12 cm

121. **[CDS (I) Paper 12 Q19]** Consider all those two-digit positive integers less than 50, which when divided by 4 yield unity as remainder. What is their sum? **(a)** 310 **(b)** 314 **(c)** 318 **(d)** 323 If every side Of an equilateral triangle is doubled, then the area Of new triangle becomes k times the area Of the Old One. What is k equal to?

122. **[CDS (I) Paper 12 Q26]** What is the highest common factor Of 2x3 x 2 — x —2 and 3x3 -2x2 + x-2? **(c)** 2X+1 B-ABQB-N-LzS/ IA *itive nd Y Y2. if is **(d)** 184 8 **(a)** 2X+1 B-ABQB-N-LzS/ IA 23. 24. **(b)** 11 25. **(c)** 184 tx + y **(d)**

123. **[CDS (I) Paper 12 Q30]** What is the remainder when (1235x4523x2451) is divided by **(c)** 5 7 What is the remainder when (1723 +2323 + 2923) is divided by 23? 29. p, q and r are prime numbers such that p < q < r < **(a)** 2X+1 B-ABQB-N-LzS/ IA 23. 24. **(b)** 11 25. **(c)** 184 tx + y **(d)**

124. **[CDS (I) Paper 12 Q13]** In how many Cases Would (p + q + r) also be a prime number? 3 None of the above The LCM of two numbers is 90 times their HCF. The sum Of LCM and HCF is 1456. If one of the numbers is 160, then what is the other ber? 120 136 144 HOW many pairs Of X and Y are possible in the number 763X4Y2. if the number is divisible by 9? 8 10 What is the remainder when 41012 is divided by 2 3 **(a)** 2X+1 B-ABQB-N-LzS/ IA 23. 24. **(b)** 11 25. **(c)** 184 tx + y **(d)**

125. **[CDS (I) Paper 12 Q27]** 28. What is the remainder when (1235x4523x2451) is divided by **(a)** 2X+1 B-ABQB-N-LzS/ IA 23. 24. **(b)** 11 25. **(c)** 184 tx + y **(d)**

126. **[CDS (I) Paper 12 Q32]** What is 2b equal to? 4b 3 **(a)** 37 **(b)** 2ab 34. 35. 10 If ax + by — 2 O and axby = I, Where a O, b O, then what is x + b 2 y **(c) (d)**

127. **[CDS (I) Paper 12 Q38]** What is of 360? 12 18 24 the number Of divisors 39. **(b)** 15 **(d)** 18 is **(a)** 11 37. **(b)** 15 **(c)** 18 710 _ 510 is divisible by 10 5 38. What is the number Of divisors of 360? Consider the following statements in respect Of four spheres A, B, C and D having respective radii 6, 8, 10 and 12 cm 2. The surface area Of sphere C is equal to the sum Of areas of spheres A and B The volume Of sphere D is equal to the sum of volumes of spheres A, B and C Which Of the above statements is / are correct? **(d)**

128. **[CDS (I) Paper 12 Q40]** 12 The None Of the above multiplication of a three-digit number W 5 With digit Z yields X215. What is X+Y+Z equal to? 13 17 If the equation xz +2(1 + k)x+ k 2 O has equal roots, then what is the value of k? 2 2 **(c)** 36. **(d)** 11 37. **(b)** 15 **(d)** 18 710 _ 510 is divisible by 10 5 **(a)** 11 37. **(b)** 15 **(c)** 18 710 _ 510 is divisible by 10 5 38. What is the number Of divisors of 360? Consider the following statements in respect Of four spheres A, B, C and D having respective radii 6, 8, 10 and 12 cm 2. The surface area Of sphere C is equal to the sum Of areas of spheres A and B The volume Of sphere D is equal to the sum of volumes of spheres A, B and C Which Of the above statements is / are correct? **(d)**

129. **[CDS (I) Paper 12 Q39]** 40. 12 12 18 24 None Of the above The multiplication of a three-digit number W 5 With digit Z yields X215. What is X+Y+Z equal to? 13 17 If the equation xz +2(1 + k)x+ k 2 O has equal roots, then what is the value of k? 2 2 **(c)** B;-ABQB-N-LZS/ IA **(a)** 11 37. **(b)** 15 **(c)** 18 710 _ 510 is divisible by 10 5 38. What is the number Of divisors of 360? Consider the following statements in respect Of four spheres A, B, C and D having respective radii 6, 8, 10 and 12 cm 2. The surface area Of sphere C is equal to the sum Of areas of spheres A and B The volume Of sphere D is equal to the sum of volumes of spheres A, B and C Which Of the above statements is / are correct? **(d)**

130. **[CDS (I) Paper 12 Q41]** 42. If m and n are the roots Of the equation x + + b = O, and m 2 and are the roots of the equation x — cx+d O, then which of the following is/ are correct? 2b — a2 c Select the correct answer using the code given below. **(a)** 1 only **(b)** 2 only Both 1 and 2 **(d)** Neither 1 nor 2 If N2 -33, N2 -31 and N2 -29 are prime numbers, then what is the number of possible values of N, where N is an integer? **(c)** **(d)** 6 None Of the above IWSBQB-N-LZS/ IA

131. **[CDS (I) Paper 12 Q44]** The HCF Of two natural numbers m and n is 24 and their product is **(a)** 16 42. **(b)** 14a If m and n are the roots Of the equation x + + b = O, and m 2 and are the roots Of the equation x — cx+d O, then which of the following is/ are correct? 43. There are 48 cricket balls, 72 hockey balls and 84 tennis balls, and they have to be arranged in several rows in such a way that every row contains the same number Of balls of one type. Whåt is the minimum number of rows required for this to happen? 2b — a2 c Select the correct answer using the code given below. **(c)** 1 only **(d)** 2 only Both 1 and 2

132. **[CDS (I) Paper 12 Q48]** What is the remainder when x5 — 5x2 + 125 is divided by x +5? **(b)** 125 **(c)** -3125 **(d)** 3125 B-ABQB-N-LZS/ IA 16 •and ab(x2 — —b2)? (a2x2 bx)2 (a2x2 of 45 **(a)** 125 **(b)** -3125 **(c)** 3125 B-ABQB-N-LZS/ IA 46. **(d)** e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

133. **[CDS (I) Paper 12 Q49]** What is the lowest common multiple of ab(x2 + + b2) **(b)** **(d)** (a2x2 SO. A certain number Of two digits is three times the sum Of its digits. if 45 is added to the number, the digits Will be reversed. What is the sum Of the squares Of the two digits of the number? **(c)** 41 53 64 16 If (x + k) is the common factor of x + b and x2 + d, then What is k equal to? **(a)** 125 **(b)** -3125 **(c)** 3125 B-ABQB-N-LZS/ IA 46. **(d)** e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

134. **[CDS (I) Paper 12 Q68]** Consider the following statements **(a)** sin x **(b)** cosx 62. What is **(c)** sin x + cos x **(d)** sin x — cos x 63. Consider the following statements • l. sin > sin I 2. COS < COS Which Of the above Stater-nents 66. What is the number of pairs of perpendicular planes in a cuboid?

135. **[CDS (I) Paper 12 Q69]** 71. Consider the following statements : **(a)** **(b)** **(c)** **(d)** I and 2 only 2 and 3 only 1 and 3 only 1, 2 and 3 Which of the above statements is / are correct?

136. **[CDS (I) Paper 12 Q89]** Consider the following statements . **(a)** The triangle is isosceles **(b)** The triangle is equilateral **(c)** None Of the ab ove For the next three **(d)** 3124

137. **[CDS (I) Paper 12 Q92]** Consider the following statements : **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither nor 2 93. If one Of the roots Of the equation px + qx+r=O is three times the Other, then Which one of the following relations is correct? **(d)** 63 m 2 96 m 2 126 m2 252 m 2

138. **[CDS (I) Paper 12 Q97]** Consider the following data : **(a)** 18 **(b)** 20 **(c)** 24 Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range 98. When we take class intervals On the x-axis and corresponding frequencies on the y-axis, and draw rectangles with the areas proportional to the frequencies Of the respective class intervals, the graph so Obtained is called 2. The cumulative frequency polygon gives an equivalent representation Of frequency distribution table. The cumulative frequency polygon is a closed polygon with one horimntal and one vertical side. The have nan-negative Which of the above is/ are correct? 1 only **(d)** 2 only Both 1 and 2

139. [CDS (I) Paper 13 Q2] 3. 4 3 2 37 13 where x, y, z are natural numbers, then what is z equal to? **(d)** 3 Cannot be determined due to insufficient data For what value Of k is (x — 5) a factor ofx3 -3x2 +kx-10? -8 2 B-EFUA-O-PDW/S2A is 2 **(a)** 3 Cannot be determined due to insufficient data 4. 5. 6. For what value Of k is **(b)** 2 3 B-EFUA-O-PDW/S2A 2 is 2 4. What is equal to? 5. The digit in the units place Of the product 81 x 82 x 83 x 84 is 4 6 6. The square root of **(c)** 2 3 I. What is the remainder when 496 is divided by 6? 2. 3. 4 3 2 37 13 where x, y, z are natural numbers, then what is z equal to? **(d)** 3 Cannot be determined due to insufficient data For what value Of k is

140. [CDS (I) Paper 13 Q4] What is equal to? **(a)** 3 Cannot be determined due to insufficient data 4. 5. 6. For what value Of k is **(b)** 2 3 B-EFUA-O-PDW/S2A 2 is 2 4. What is equal to? 5. The digit in the units place Of the product 81 x 82 x 83 x 84 is 4 6 6. The square root of **(c)** 2 3 I. What is the remainder when 496 is divided by 6? 2. 3. 4 3 2 37 13 where x, y, z are natural numbers, then what is z equal to? **(d)** 3 Cannot be determined due to insufficient data For what value Of k is

141. [CDS (I) Paper 13 Q6] The square root of (0.75)3 +11 1-0.75 **(a)** 3 Cannot be determined due to insufficient data 4. 5. 6. For what value Of k is **(b)** 2 3 B-EFUA-O-PDW/S2A 2 is 2 4. What is equal to? 5. The digit in the units place Of the product 81 x 82 x 83 x 84 is 4 6 6. The square root of **(c)** 2 3 I. What is the remainder when 496 is divided by 6? 2. 3. 4 3 2 37 13 where x, y, z are natural numbers, then what is z equal to? **(d)** 3 Cannot be determined due to insufficient data For what value Of k is

142. [CDS (I) Paper 13 Q7] 9. What is the remainder obtained when 1421 x 1423 x 1425 is divided by 12? **(c)** 2 3 **(a)** 2 3 **(b)** 7 6 None of the above 8. Consider the following statements : 2. The equation 1990x — 173y= I I has no solution in integers for x and y. The equation 3x— 12y= 7 has no solution in integers for x and y. The difference between the squares Of two consecutive odd integers is always divisible by Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only Both I and 2 Neither 1 nor 2 A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was

143. [CDS (I) Paper 13 Q8] Consider the following statements : **(a)** 2 3 **(b)** 7 6 None of the above 8. Consider the following statements : 2. The equation 1990x — 173y= I I has no solution in integers for x and y. The equation 3x— 12y= 7 has no solution in integers for x and y. The difference between the squares Of two consecutive odd integers is always divisible by Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only Both I and 2 Neither 1 nor 2 A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was

144. [CDS (I) Paper 13 Q2] The equation 1990x — 173y= I I has no solution in integers for x and y. The equation 3x— 12y= 7 has no solution in integers for x and y. Which Of the above statements is/ are correct? **(a)** 2 3 **(b)** 7 6 None of the above 8. Consider the following statements : 2. The equation 1990x — 173y= I I has no solution in integers for x and y. The equation 3x— 12y= 7 has no solution in integers for x and y. The difference between the squares Of two consecutive odd integers is always divisible by Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only Both I and 2 Neither 1 nor 2 A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was

145. [CDS (I) Paper 13 Q11] 12. What is the number of possible pairs Of (P, QI if the number 357P25Q is divisible by both 3 and 5? **(b)** 7 6 None of the above The difference between the squares Of two consecutive odd integers is always divisible by **(c)** 3 8 16 The expression x —x3pt+4x2pt—4x q is divisible by + 3xq — 3xpt **(a)** 2 3 **(b)** 7 6 None of the above 8. Consider the following statements : 2. The equation 1990x — 173y= I I has no solution in integers for x and y. The equation 3x— 12y= 7 has no solution in integers for x and y. The difference between the squares Of two consecutive odd integers is always divisible by Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only Both I and 2 Neither 1 nor 2 A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was

146. [CDS (I) Paper 13 Q28] A runs I — times as fast as B. If A gives B a start of 80 m, how far If must the winning post from the starting point be so that A and B might reach it at the same time? **(a)** 200 m **(b)** 300 m **(c)** 270 m **(d)** 160 m

147. [CDS (I) Paper 13 Q36] Let A and B be finite non-empty sets With the number of elements in A m and number Of elements in B z n. Let m > n. If for some integer k l, the number Of nan-empty subsets of A 2k the number Of non•empty subsets Of B, then which one Of the following is correct? **(b) (d)** m n + p for some Odd prime number p m n + t for some composite number t If n is a natural number and n = where m, P2, P3 are distinct prime factors, then the number Of prime factors for n is (C) Ill X 3 +1) **(d)** None Of the above Consider the following statements for the sequence of numbers given below Each number can be expressed in the form (4m 3), where m is a natural number. Some numbers are squares. Which Of the above statements is/ are correct? **(b)** 1 only 2 only Both I and 2 Neither I nor 2 **(a)** 378 **(b)** 384 32. 33. 34. 35. 36. Let A and B be finite non-empty sets With the number of elements in A m and number Of elements in B z n. Let m > n. If for some integer k l, the number Of nan-empty subsets of A 2k the number Of non•empty subsets Of B, then which one Of the following is correct? **(c)** 375 383 **(d)**

148. **[CDS (I) Paper 13 Q38]** Let p be a prime number Other than 2 Or **(a) (b)** 3x4 — 8x3y+IOx2y 3x4 — 8x3y— IOx2y2 +7 xy + 2y 3x4 +7xy3 * 2y 3x4 8x3y— IOX2y +7xy3 + 2y4 38. Let p be a prime number Other than 2 Or 5. One would like to express the vulgar fraction l/ p in the form of a recurring decimal. Then the decimal will be Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only

149. **[CDS (I) Paper 13 Q5]** One would like to express the vulgar fraction l/ p in the form of a recurring decimal. Then the decimal will be **(b) (c) (d)** a pure recurring and its period necessarily (p— I) a mixed recurring and its period necessarily (p— I) a pure recurring and its period Will factor of (p — l) a mixed recurring and its period will factor of (p— I) decimal will be decimal will be decimal be Some decimal be some B-EFUA-O-PD W S2A **(a) (b)** 3x4 — 8x3y+IOx2y 3x4 — 8x3y— IOx2y2 +7 xy + 2y 3x4 +7xy3 * 2y 3x4 8x3y— IOX2y +7xy3 + 2y4 38. Let p be a prime number Other than 2 Or 5. One would like to express the vulgar fraction l/ p in the form of a recurring decimal. Then the decimal will be Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only

150. **[CDS (I) Paper 13 Q39]** 40. be be Consider the following statements **(a) (b)** 3x4 — 8x3y+IOx2y 3x4 — 8x3y— IOx2y2 +7 xy + 2y 3x4 +7xy3 * 2y 3x4 8x3y— IOX2y +7xy3 + 2y4 38. Let p be a prime number Other than 2 Or 5. One would like to express the vulgar fraction l/ p in the form of a recurring decimal. Then the decimal will be Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only

151. **[CDS (I) Paper 13 Q2]** There exists only one prime number p such that (17 p + I) is a square. If x is the product of IO consecutive prime numbers starting from 2, then (x + l) is also a prime number. Which Of the above statements is/ are correct? **(a)** I only **(b)** 2 only **(c)** Both I and 2 Neither 1 nor 2 The sign Of the quadratic polynornial ax2 + c is always positive if **(a) (b) (c) (d)** a is positive and b2 — O a is positive and b2 —4ac20 a can and b2 a can and b2 be any real number —4ac S O be any real number — 4ac2 0

152. **[CDS (I) Paper 13 Q61]** Consider the following Statements . **(a) (b)** , where — , where K < — or K > 3 **(c) (d)** I only 2 only Both I and 2 Neither I nor 2 62. If COSA — tan B, cos B tan C and cos C = tan A, then sin A is equal to JB-I

153. **[CDS (I) Paper 13 Q64]** 22 3— tan2 A 1—3tan2A where K is a real number, then cosecA(3sinA — 4sin3 A) is equal to **(b) (c)** , where — , where K < — or K > 3 The angles Of elevation Of the top of a tower from two points P and Q at distances m 2 and n2 respectively, from the base and in the same straight line with it are complementary. The height of the **(a) (b) (c) (d)** (mn)l/2

154. **[CDS (I) Paper 13 Q63]** 64. 22 3— tan2 A 1—3tan2A where K is a real number, then cosecA(3sinA — 4sin3 A) is equal to **(a) (b)** , where — , where K < — or K > 3 **(c) (d)** I only 2 only Both I and 2 Neither I nor 2 62. If COSA — tan B, cos B tan C and cos C = tan A, then sin A is equal to JB-I

155. **[CDS (I) Paper 13 Q71]** Consider the following statements **(a) (b)** 73. If 71. Consider the following statements 1. There exists at least one value Of x between O and — which satisfies the equation sin x —2sin x 2 _ sin 5 is greater th an cos I - S. Which Of the above statements is/are correct? **(c)** only **(d)** 2 only

156. **[CDS (I) Paper 13 Q82]** The last digit in the expansion Of 17256 9 3 **(a) (b)** 83. 86. If a **(c)** 264 **(d)** 300

157. **[CDS (I) Paper 13 Q86]** If a a and b are positive integers, then a 3 + b 6 is equal to of **(a) (b)** 83. 86. If a **(c)** 264 **(d)** 300

158. **[CDS (I) Paper 13 Q93]** At 8:30, the hour hand and the minute hand Of a clock form an angle of **(b)** 800 750 **(a) (b)** 2 cm Cannot be determined due to insufficient data 92. Let on, OC and OD are rays in the anticlockwise direction such that ZAOB= ZCOD= 1 ocp, ZBOC= 820 and ZAOD= 780. Consider the following statements 89. From a circular piece of cardboard of radius 3 cm, two sectors of4 **(c)** I I square cm **(d)** 22 square cm

159. **[CDS (I) Paper 13 Q94]** The complement angle of 80' is **(b)** — radian **(d)** rad Ian 96. **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither I nor 2 Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 Cm is tangent to the inner circle. The radius Of the inner circle is **(d)** B-EFUA-O-PDW/52A 94. The complement angle of 80' is

160. **[CDS (I) Paper 13 Q95]** Consider the following statements . **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither I nor 2 Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 Cm is tangent to the inner circle. The radius Of the inner circle is **(d)** B-EFUA-O-PDW/52A 94. The complement angle of 80' is

161. **[CDS (I) Paper 14 Q1]** Ihe number of values of x satisfying 100 x + — > 50 , where x is a natural number less than or equal to 100 is **(a)** 51 **(b)** 53 **(c)** 55 **(d)** 57

162. **[CDS (I) Paper 14 Q2]** What is the sum of digits Of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11 respectively as the remainders ? **(a)** 51 **(b)** 53 **(c)** 55 **(d)** 57 2. What is the sum of digits Of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11 respectively as the remainders ? 3. A milkman claims to sell milk at its cost price only. Still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture ?

163. **[CDS (I) Paper 14 Q4]** What is 4+ 4 — equal to ? **(b)** 2 B-ABVB-O-GPU - A 2 S. The largest natural number which divides every natural number of the forn — where n is a natural number greater than 2 is **(b)** 12 **(c)** 24 **(d)** 48 **(a)** 51 **(b)** 53 **(c)** 55 **(d)** 57 2. What is the sum of digits Of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11 respectively as the remainders ? 3. A milkman claims to sell milk at its cost price only. Still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture ?

164. **[CDS (I) Paper 14 Q12]** A number when divided by 7 leaves a remainder 3 and the resulting quotient when divided by 1 1 leaves a remainder **(a)** 35 **(b)** 40 **(c)** 45 **(d)** 50 106

165. **[CDS (I) Paper 14 Q6]** If the same number when divided by 11 leaves a remainder m and the resulting quotient when divided by 7 leaves a remainder n. What are the values Of m and n respectively ? **(a)** I and 4 **(b)** 4 and 1 **(c)** 3 and 6 **(d)** 6 and 3 4

166. **[CDS (I) Paper 14 Q15]** The seven digit .number 876p37q is divisible by **(a)** I only **(b)** 2 only **(c)** I and 2 only l, 2 and Y is/are 14. If **(d)** 2 minutes

167. **[CDS (I) Paper 14 Q18]** Which one of the following is correct ? **(a)** I only **(b)** 2 only **(c)** I and 2 only l, 2 and Y is/are 14. If **(d)** 2 minutes

168. **[CDS (I) Paper 14 Q21]** The number of pairs (x, y) where x, y are integers satisfying the equation 21x+48y=5is **(a)** Zero **(b)** One **(c)** Two **(d)** Infinity

169. **[CDS (I) Paper 14 Q22]** Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct ? 1. (y —x) can be an even integer. 2. can an even integer. 3. , can be an even integer. B-ABVB-O-GPU - A of Select the correct answer using the code given below : **(a)** I and 2 only **(b)** 2 and 3 only **(c)** I and 3 only **(d)** 1, 2 and 3

170. **[CDS (I) Paper 14 Q23]** Let x and y be positive integers such that x > y. The expressions 3x 4 2}' and + 3y when divided by 5 leave remainders 2 and 3 respectively. What is the remainder when (x —y) is divided **(a)** 43-2% **(b)** 56-8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ? **(c)** 5<k$11 21. The number of pairs **(d)** Zero

171. **[CDS (I) Paper 14 Q33]** The sum of first 47 terms of the series 11 4564 56456 **(b)** 6 6 9 20 B-ABVB-O-GPU - A 12 34.. Which one of the fellowing is correct ? **(a)** (x + 2) is a factor of — + — 24.1 + 32 **(b)** (x +2) is a factor Of -12r2+24x-32 **(c)** (x — 2) is a factor Of x' + 32 **(d)** (x — 2) is a factor of x' + 0-1* +24x-32

172. **[CDS (I) Paper 14 Q34]** Which one of the following is correct ? **(a) (b) (c) (d)** (x +2) is a factor of x' -G + 12r2-24x+32 (x +2) is a factor Of x4+6x3-12r2+24x-32 (x —2) is a factor of X4-6X3+ 12r2-24X+ 32 (x— 2) is a factor of x' + G +24x-32

173. **[CDS (I) Paper 14 Q47]** A number consists of two digits, whose sum is **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

174. **[CDS (I) Paper 14 Q27]** Ihe product of digits of the number is **(c)** 10 **(d)** 12 **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

175. **[CDS (I) Paper 14 Q50]** Consider all positive two digit numbers each of which when divided by 7 leavés a remainder **(a)** 3600 **(b)** 6200 **(c)** 6028 **(d)** 6400 50. Consider all positive two digit numbers each of which when divided by 7 leavés a remainder 3. What is their sum ?

176. **[CDS (I) Paper 14 Q3]** What is their sum ? **(a)** 661 **(b)** 666 **(c)** 676 **(d)** 777

177. **[CDS (I) Paper 14 Q66]** The diameter Of a wheel. that makes 452 revolutions to move 2 km and 26 decametre is equal to **(a)** 2-0 **(b)** 2+6 66. The diameter Of a wheel that makes 452 revolutions to move 2 km and 26 decametre is equal to 67. If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow **(c)** I only **(d)** 2 only

178. **[CDS (I) Paper 14 Q11]** From a rectangular sheet of sides 18 cm and 14 cm, a semicircular portion with smaller side as diameter is taken out. Then the area of the remaining sheet will be **(a)** will be 75. Let **(b)** 52% **(c)** 540/0 **(d)** 98 cm 108 cm2

179. **[CDS (I) Paper 15 Q9]** IO. The sum and difference of two expressions are 5x2 — x — 4 and x 2 + 9x — 10 respectively. The HCF of the two expressions Will be **(d)** (2x-3) If then value of b2+C2 will be If the polynomial x + px + qx — x the is divisible by (x 4 — I), then the value Of p 2 + q2 is **(c)** 10 **(d)** 13 Let p and q be non-zero integers. Consider the polynomial A(x) = x2+ px+q It is given that (x —m) and (x — kin) are simple factors Of A (x), where m is a non•zero integer and k is a positive integer, k **(a)** 3 7. 8. 9. The sum and difference of two expressions are 5x2 — x — 4 and x 2 + 9x — 10 respectively. The HCF of the two expressions Will be **(b) (c)** -15 If then value of b2+C2 will be If the polynomial x + px + qx — x the 3. If 2 + g m and _ + n, then is — equal to? **(d)** np — mq 4. what and and is divisible by

180. **[CDS (I) Paper 15 Q14]** Let m be a non•zero integer and n be a positive integer. Let R be the remainder Obtained On dividing the polynomial xn + mn by (x Then **(a)** R is a non-zero even integer **(b)** R is odd, if m is Odd **(c)** R s 2 for some integer s, if n is **(d)** R = t 3 for some integer t, if 3 divides n If4X2Y 128 and 33*32y— 9XY O, then the value Of x + y can be equal to

181. **[CDS (I) Paper 15 Q17]** Consider the following statements in respect Of the expression where n is an integer : I. There are exactly two values of n for which Sn 861. and hence for any integer m, we have two values Of n for which Sn m. Which Of the above statements is/ are co rrect? **(a)** I only **(b)** 2 only **(c)** Both and 2 **(d)** Neither I nor 2

182. **[CDS (I) Paper 15 Q18]** Consider the following statements in respect of two different non-zero integers p and q : I. For (p + q) to be less than (p q must bc negative. **(a)** R is a non-zero even integer **(b)** R is odd, if m is Odd **(c)** R s 2 for some integer s, if n is **(d)** R = t 3 for some integer t, if 3 divides n If4X2Y 128 and 33*32y— 9XY O, then the value Of x + y can be equal to 13. If thc linear factors —

183. **[CDS (I) Paper 15 Q19]** 20. then which Of the following is/ are correct? Select the correct answer using the code given below. **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 710 — 510 is divisible by 10 **(d)** 11

184. **[CDS (I) Paper 15 Q21]** Let a two-digit number be k times the sum of its digits. If thc number formed by interchanging the digits is m times the sum of the digits, then the value Of **(b)** 10-k **(c)** 11-k **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 710 — 510 is divisible by 10

185. **[CDS (I) Paper 15 Q28]** Let S be a Set of first fourteen natural numbers. The possible number of pairs (a, b), Where a, b ES and a such that ab leaves remainder I when divided by 15, is **(c) (d)** 6 None of the above **(a) (b)** 6 None of the above B'E'2+CF'2 Which one Of the following is correct in respect of the above statements? **(c)** Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I **(d)** Both Statement I and Statement 11 are true but Statement 11 is not thc correct explanation of Statement I

186. **[CDS (I) Paper 15 Q32]** What is the maximum value of m, if the number ,V-90x42x324X55 is divisible by 3m ? **(d)** 8 7 6 5 **(a)** I + q + q2 = 3pq **(b)** p 3+ q + q2 3pq 2 3pq 34. 36. Consider the following statements : 2. 3. 4. Every natural number is a real number. Every real number is a rational number. Every integer is a real number. Every rational number is a real n umber. 32. What is the maximum value of m, the number divisible by 3m ? if **(c)** 8 7 6 5 33. A bike consumes 20 mL Of petrol per kilometre, if it is driven at a speed in the range of 25—50 km/hour and consumes 40 mL of petrol per kilometrc at any other speed. How much petrol is consumed by the bike in travelling a distance Of 50 km, if the bike is driven at a speed Of 40 km/hour for the first IO km, at a speed Of 60 km/hour for the next 30 km and at a speed of 30 km/hour for the last IO km? Which Of the above Statements are correc t? **(d)** I, 2 and 3

187. **[CDS (I) Paper 15 Q34]** 36. Consider the following statements : **(a)** I + q + q2 = 3pq **(b)** p 3+ q + q2 3pq 2 3pq 34. 36. Consider the following statements : 2. 3. 4. Every natural number is a real number. Every real number is a rational number. Every integer is a real number. Every rational number is a real n umber. 32. What is the maximum value of m, the number divisible by 3m ? if **(c)** 8 7 6 5 33. A bike consumes 20 mL Of petrol per kilometre, if it is driven at a speed in the range of 25—50 km/hour and consumes 40 mL of petrol per kilometrc at any other speed. How much petrol is consumed by the bike in travelling a distance Of 50 km, if the bike is driven at a speed Of 40 km/hour for the first IO km, at a speed Of 60 km/hour for the next 30 km and at a speed of 30 km/hour for the last IO km? Which Of the above Statements are correc t? **(d)** I, 2 and 3

188. **[CDS (I) Paper 15 Q4]** Every natural number is a real number. Every real number is a rational number. Every integer is a real number. Every rational number is a real n umber. Which Of the above Statements are correc t? **(a)** I, 2 and 3 **(b)** 1, 3 and 4 **(c)** 2 and 3 0 nly **(d)** 3 and 4 only

189. **[CDS (I) Paper 15 Q35]** For what value Of k is (x + I) a factor of **(c)** _2 Consider the following Statements : I. There exists a positive real number m such that cos x z 2m 2. mn m + n for all m, n belonging to set of natural numbers. Which of the above statements is/are correct? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2

190. **[CDS (I) Paper 15 Q41]** Consider thc following in respect of natural numbers a, b and c : **(a)** 38. Which of the points P **(b)** Q and R o nly **(c)** P and R o nly **(d)** P and Q o Illy

191. **[CDS (I) Paper 15 Q68]** Which of the following is correct in respect Of the equation 3 e)? (Given that is a real number.) u13,— u13, **(d)** None of the above **(a)** 2 -Be , thenC+2B **(b)** 0-1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e **(c)** only **(d)** 2 only

192. **[CDS (I) Paper 15 Q75]** The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 61 where x, y and z are natural numbers, then What is z equal to? **(a)** 165% 21 B-ABPA-P-JRW/ 54A 74. Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(b)** 12% 76. If 61 where x, y and z are natural numbers, then What is z equal to? **(c)** 75. The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 77. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times thc arca Of the smaller ring, then n is equal to **(d)** 10 78. Consider the following statements : 2. If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n 3. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is

193. [CDS (I) Paper 15 Q78] Consider the following statements : **(a)** 165% 21 B-ABPA-P-JRW/ 54A 74. Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(b)** 12% 76. If 61 where x, y and z are natural numbers, then What is z equal to? **(c)** 75. The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 77. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times thc arca Of the smaller ring, then n is equal to **(d)** 10 78. Consider the following statements : 2. If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n 3. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is

194. [CDS (I) Paper 15 Q76] If 61 where x, y and z are natural numbers, then What is z equal to? **(a)** 165% 21 B-ABPA-P-JRW/ 54A 74. Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(b)** 12% 76. If 61 where x, y and z are natural numbers, then What is z equal to? **(c)** 75. The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 77. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times thc arca Of the smaller ring, then n is equal to **(d)** 10 78. Consider the following statements : 2. If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n 3. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is

195. [CDS (I) Paper 15 Q79] Consider the following statements : **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In a triangle PQR, point X is on PQ and point Y is on PR such that XP = I •5 units, 6 units, PY = 2 units and YR 8 units. Which Of the following are correct? 1. QR=5XY 2. OR is parallel to XY. 3. Triangle PYX is similar to triangle PRQ. Select the correct answer using the code given below.

196. [CDS (I) Paper 15 Q99] If a and b are negative numbers and c i s a pos itive real num , then Which of the following is/ are c **(a)** AM < Median **(b)** GM Median **(c)** AM = Me dian Suppose for Which one of the corre ct? **(d)** GM>1/6

197. [CDS (I) Paper 16 Q2] If is an integer and a, are the roots of 4x2 — such that I < a < 2 4 and 2 < < 3, then how many values can take ? **(c)** 14 **(d)** 15 62 +72 +82 +92 +102 **(a)** x = 100 **(b)** x = IO **(c)** 330 **(d)** 340

198. [CDS (I) Paper 16 Q6] A is a set of positive integers such that when divided by 2, 3, 4, 5 and the remainder l, 2, and 5 respectively. How many integers between 0 and 100 belong to the set A ? **(a)** No integer **(b)** One **(c)** Two **(d)** Three

199. [CDS (I) Paper 16 Q9] The sum of the squares Of two positive integers is **(a)** 2 and 4 **(b)** 4 and 2 **(c)** 2 and 2 **(d)** 4 and 4 11. A boy saves Rs. 4•65 daily. What is the least number of days in which he will be able to save an exact number of rupees ?

200. [CDS (I) Paper 16 Q13] Which one Of the following rational numbers has non-tenninating and repeat- ing decimal expansion ? 15 1600 23 35 50 17 **(a)** 2 and 4 **(b)** 4 and 2 **(c)** 2 and 2 **(d)** 4 and 4 11. A boy saves Rs. 4•65 daily. What is the least number of days in which he will be able to save an exact number of rupees ?

201. [CDS (I) Paper 16 Q15] What is the remainder when 2100 divided by 101 ? **(b)** 11 **(c)** 99 **(d)** 100 IS **(a)** 2 and 4 **(b)** 4 and 2 **(c)** 2 and 2 **(d)** 4 and 4 11. A boy saves Rs. 4•65 daily. What is the least number of days in which he will be able to save an exact number of rupees ?

202. [CDS (I) Paper 16 Q21] "Ihe cost Of kg rice is Rs. **(a)** p -loniy 20. 6 If p and q are the roots of x2+px+q = 0, then which one of the following is correct ? remaining work ? **(b)** 6 hours **(c)** 12 hours **(d)** hours

203. [CDS (I) Paper 16 Q22] Which one Of the following is correct in respect of the number 1729 ? **(a)** It cannot be written as the sum Of the cubes of two positive integers **(b)** It can be written as the Sum Of the cubes of two positive integers in one way only **(c)** It can be written as the sum of the cubes Of two positive integers in two ways only **(d)** It can be wHtten as the sum of the cubes Of two positive integers in three ways Only

204. [CDS (I) Paper 16 Q2] One number may be prime and the other composite. **(a)** 1 and 2 only **(b)** 2 and 3 Only **(c)** 1 and 3 only **(d)** l, 2 and 3 25. What is 23. A shopkeeper increases the cost price Of an item by 20% and offers a discount Of 10% on this marked price. What is his percentage gain ?

205. [CDS (I) Paper 16 Q27] Consider the following statements in respect of positive Odd integers x and y : l. x2 + Y2 is even integer. 2. is divisible by **(a)** 1 and 2 only **(b)** 2 and 3 Only **(c)** 1 and 3 only **(d)** l, 2 and 3 25. What is 23. A shopkeeper increases the cost price Of an item by 20% and offers a discount Of 10% on this marked price. What is his percentage gain ?

206. **[CDS (I) Paper 16 Q30]** 2122+462+842+464 +2130 is divisible by which one of the following integers ? **(d)** 11 **(a)** 11 31. If2p+3q= 12 and4p2+4pq—3q2 — 126, then what is the value of p + 2q ? 29. There are twelve friends A, B, C, D, E, 33. What is the unit digit of 7139 F, G, H, I, J, K and L who invested money in some business in the ratio of and the duration for which they invested the money is in the ratio of respectively. Who will get the maximum profit at the end of the year ? **(b)** F only **(c)** G only **(d)** Both F and G

207. **[CDS (I) Paper 16 Q33]** What is the unit digit of 7139 ? Rs. 336 Rs. 366 Rs. 633 Rs. 663 then what is the value of x — ax+a **(a)** 11 31. If2p+3q= 12 and4p2+4pq—3q2 — 126, then what is the value of p + 2q ? 29. There are twelve friends A, B, C, D, E, 33. What is the unit digit of 7139 F, G, H, I, J, K and L who invested money in some business in the ratio of and the duration for which they invested the money is in the ratio of respectively. Who will get the maximum profit at the end of the year ? **(b)** F only **(c)** G only **(d)** Both F and G

208. **[CDS (I) Paper 16 Q3]** Which one of the following statements is correct ? **(a)** Mahesh saves more than what Kamal saves **(b)** Savings of both of them are equal **(c)** Kamal saves more than what Mahesh saves **(d)** It is not possible to determine who saves more

209. **[CDS (I) Paper 16 Q41]** The area of four walls Of a room is 120 rn2. The length of the room is twice its breadth. If the height of the room is 4 m, What is area of the floor ? **(a)** 40 m2 **(b)** 50 1112 (C) 60 m2 **(d)** 80 rn2

210. **[CDS (I) Paper 16 Q85]** What is the value of tanl Otan20tan30tan40 . . tan890 ? **(a)** 150 **(b)** 300 **(c)** 450 **(d)** 600 82. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ?

211. **[CDS (I) Paper 16 Q86]** The average score of class X is **(a)** 150 **(b)** 300 **(c)** 450 **(d)** 600 82. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ?

212. **[CDS (I) Paper 16 Q76]** The average score of class Z is **(a)** 150 **(b)** 300 **(c)** 450 **(d)** 600 82. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ?

213. **[CDS (I) Paper 16 Q75]** If the average marks Of the complete class is 72, then what is the average marks Of the girls ? **(a)** 64 **(b)** 65 **(c)** 68 **(d)** 74

214. **[CDS (I) Paper 16 Q94]** Consider the following statements : **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 95. Suppose the class interval 10—15 has frequency 30, then what is the frequency density of this class interval ? 98. 99. 100. 28 W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ?

215. **[CDS (I) Paper 16 Q96]** Let S be a set of first ten natural numbers. What is the possible number Of pairs (a, b) where a, b e S and a b such that the product ab (>12) leaves remainder 4 when divided by 12 ? **(d)** 10 **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 95. Suppose the class interval 10—15 has frequency 30, then what is the frequency density of this class interval ? 98. 99. 100. 28 W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ?

216. **[CDS (I) Paper 16 Q97]** What is the remainder when 135+145+155+165 is divided by 29 ? SDC-S-TRD - A 28 **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 95. Suppose the class interval 10—15 has frequency 30, then what is the frequency density of this class interval ? 98. 99. 100. 28 W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ?

217. **[CDS (I) Paper 16 Q98]** W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? **(a)** 2280 **(b)** 2640 **(c)** 3820 **(d)** 4130

218. **[CDS (I) Paper 17 Q7]** 8. What is the square root Of (035)2 + 0•70+1 **(a) (b) (c) (d)** +019? 225 2 3 4 What is the number Of digits in 240 ? (Given that log102 - 0301) 14 13 12 11 If one'root of (a2 — 5a + 3)x2 + (3a— 1) x + 2 = O is twice the Other, then What is the value of'a'? 2 **(a)** 3 2 **(b)** 3 **(d)** 3 What is the remainder when the number (4444)4444 is divided by 9 ? **(a) (b) (c) (d)** 4 6 7 8 What is the square root Of (035)2 + 0•70+1

219. **[CDS (I) Paper 17 Q22]** 23. The pair Of linear equations + 3y + I and + y + 3 = O intersect each other, if **(a) (b) (c) (d)** The number Of prime numbers • are less than 100 is **(a) (b) (c) (d)** 24 25 26 27 The cost of a diamond varies directly as the square Of its weight. A diamond broke into four pieces with their weights in the ratio of I : 2 : 3 :

220. **[CDS (I) Paper 17 Q43]** 44. What is the value of 12 125 **(a) (b) (c) (d)** 2 20 9 27 9 7 27 9 2 29— 9 **(a) (b) (d)** 2 4 5 2 2—

221. **[CDS (I) Paper 17 Q46]** Consider the following statements : I. Of two consecutive integers , o ne is. even. **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

222. **[CDS (I) Paper 17 Q2]** Square of an Odd integer is Of the form 8n + **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

223. **[CDS (I) Paper 17 Q1]** Which Of the above statements Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

224. **[CDS (I) Paper 17 Q49]** 50. 7:11 **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 The system of equations 2x + 4y = 6 and 4x + By 8is **(a) (b) (c) (d)** Consistent With a unique solution Consistent With infinitely many solutions Inconsistent None Of the — I) is a multiple Of p, if N is prime to p and p is a **(a) (b) (c) (d)** Prime number Rational number Real number Composite number The ratio Of two numbers is I : 5 and their product is

225. **[CDS (I) Paper 17 Q17]** What is the included number ? 52. to p **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

226. **[CDS (I) Paper 17 Q56]** 57. In a pie diagram, there are four slices with angles 1500, 900, 600 and **(a)** The I argest slice has angle 1506 **(b)** The smallest slice has angle 700 **(c)** The largest slice has angle 1800 **(d)** The smallest slice has angle 900 In an asymmetrical distribution, if the mean and median Of the distribution are 270 and 220 respectively, then the mode of the data is

227. **[CDS (I) Paper 17 Q96]** In the figure given below, Z A 800 and Z ABC **(a) (b) (c)** S 600 750 1000 **(d)**

228. **[CDS (I) Paper 18 Q3]** 4. What is the value of u (u O) for which 5. x2 — 5x + a and x2 — 7x + 2u have a common factor ? **(a) (b) (c) (d)** 6 4 3 2 HOW many numbers from I to 1000 are divisible by 2, 3, 4 and 5 ? **(a) (b) (c) (d)** 16 17 32 None Of the above What are the factors of xa + 4x2 **(a) (b) (c) (d)** (x — 2), (x + 3) and (x + 5) (x + 2), (x + 3) and (x — 5) +_2), (x — 3) and (x + (x + 2), (x — 3) and (x — 5) If x 111... 1 (20 digits), y = 333... 3 (10 digits) and z = 222 2 (IO digits), then what is equal to ? G.

229. **[CDS (I) Paper 18 Q9]** If the points p and Q represent the real **(a)** 90 19 90 21 100 56 **(b)** 90 Sudhir purchased a chair with three consecutive discounts of 20%, and 5%. The actual deduction will be If a train crosses a km-stone in 12 seconds, how long will it, take to cross 91 km-stones completely if its Speed is 60 km/hr ? **(c) (d)**

230. **[CDS (I) Paper 18 Q16]** Let fix) aoxn + alxn¯l + a2 xn¯2 + + an_1X+ an, where ao, al, a2, an are real numbers. If (x) is divided by (ax—b), then the remainder is **(b) (a)** 19. 20. 21. 22. The factors of x **(b)** x, **(c) (d)** "

231. **[CDS (I) Paper 18 Q17]** 18. Consider the following numbers : **(a)** 19. 20. 21. 22. The factors of x **(b)** x, **(c) (d)** "

232. **[CDS (I) Paper 18 Q25]** 26. The non-zero solution b b2 ac b2 — ac **(b)** b2 —ac b2 c2 b2 + ac **(d)** If k is an integer, then of the x2+7x-14 k2_! = O has **(a) (b) (c) (d)** 8 Both integral roots At least one integral root No integral root Both positive integral roots equation 27 where

233. **[CDS (I) Paper 18 Q28]** 29. How many numbers between 500 and 1000 are divisible by 13 ? **(a)** b2 —ac b2 c2 b2 + ac **(b)** If k is an integer, then of the equation 27. where A student has to secure of marks to pass an examination. He gets only 45 marks and fails by 5 marks. The maximum marks are **(c) (d)**

234. **[CDS (I) Paper 18 Q36]** O, then what is the value of **(a) (b) (c) (d)** If ab + bc + ca 2 — bc b2 a

235. **[CDS (I) Paper 18 Q39]** What is equal to ? 3 3 3 **(c) (d) (a) (b) (c) (d)** If ab + bc + ca 2 — bc b2 a

236. **[CDS (I) Paper 18 Q2]** 3. If 450 < e < 600, then sec2 B + cosec2 6 = for some real number > **(a) (b) (c) (d)** STS-M-TPT 52. 53.

237. **[CDS (I) Paper 18 Q3]** If 450 < e < 600, then sec2 B + cosec2 6 = for some real number > **(a) (b) (c) (d)** STS-M-TPT 52. 53.

238. **[CDS (I) Paper 19 Q7]** Which one of the following is correct ? **(a) (b) (c) (d)** Decimal expansion of a rational number is terminating. Decimal expansion Of a rational number is non-termin ating. Decimal expansion of an irrational number is terrninating. Decimal expansion of an irrational number is norm-terminating and non•repeating. If the roots of the equation px2 + x + r O are reciprGcal to each other, then which one of the following is correct ? **(a) (b) (c) (d)** p=2r 2p r If and 33K What is xy equal to ? — 65y = I, then **(a) (b)** 2 -3 4, 1 and I , then what is ab + xy equal to ? bx **(a) (b) (c) (d)** 0 517 + 518 + 519 is divisible by

239. **[CDS (I) Paper 19 Q6]** 7. Which one of the following is correct ? **(c) (d)** 9 11 13 **(a) (b) (c) (d)** If a + b = 2c, then what is the value Of Decimal expansion of a rational number is terminating. Decimal expansion Of a rational number is non-termin ating. Decimal expansion of an irrational number is terminating. Decimal expansion of an irrational number is norm-terminating and non-repeating. **(a) (b) (c)** If x If the roots of the equation px2 + x + r O are reciprocal to each other, then which one of the following is correct ? **(a) (b) (c) (d)** p=2r 2p r y = zlå and z = Where x I, y I, z I, then what is the value of abc ? If and 33K What is xy equal to ? — 65y = I, then **(a) (b) (c) (d) (a) (b)** 3 then what is 2 -3 4, 1 and I , If 2b a • c and = xz, xb—e b equal to ? **(a) (c) (d)** then what is ab + xy equal to ? bx **(a) (b) (c) (d)** 0 SPSS-T-XHI

240. **[CDS (I) Paper 19 Q11]** 12. If x + y + z = O, then what is — equal to ? **(a) (b) (c)** 24xyz — 24xyz If (x + 3) is a factor Of x3 + 3x2 + 4x k, then What is the value of k ? **(b) (c) (d)** 12 24 36 72 The smallest integer with 4 digits which is a perfect 41are is **(b) (c) (d)** 1000 1024 1089 None of the abcrve SPSS-T-XHI

241. **[CDS (I) Paper 19 Q14]** 15. There are two numbers which are greater than 21 and their LCM and HCF are and 21 respectively. What is the sum of these numbers ? 24xyz — 24xyz If (x + 3) is a factor Of x3 + 3x2 + 4x k, then What is the value of k ? **(a) (c) (d)** 504 508 514 528 **(b) (c) (d)** 12 24 36 72 The smallest integer with 4 digits which is a perfect 41are is If u and are the roots of the equation ax2 + bx + c = O, then what is the value of the expression (0 + l) (p + I) ? **(a) (c) (b) (c) (d)** 1000 1024 1089 None of the abcrve SPSS-T-XHI

242. **[CDS (I) Paper 19 Q7]** What is the value of **(a) (b)** -14 14 21. **(c) (d)**

243. [CDS (I) Paper 19 Q28] 120 is distributed among A, B and C so that A's share is 20 more than B's and 20 less than C's. What is B's share ? **(a) (b) (c)** : E = 3 : 4, then What is B: E equal to? 28. 29. 30. 120 is distributed among A, B and C so that A's share is 20 more than B's and 20 less than C's. What is B's share ? **(d)**

244. [CDS (I) Paper 19 Q38] Which one Of the following decimal numbers is a rational number with denominator 37 ? Three numbers which are eo-prime to each other, are such that the product Of the first two is 286 and that Of the last two is **(a) (b) (c) (d)** 44 years 45 years 50 years 52 years 36.

245. [CDS (I) Paper 19 Q4] If 20 kg Of alloy A, 28 kg of alloy B and some more pure N)pper are mixed to form a third alloy C which now contains copper and tin in the ratio Of 6 : 7, then what is the quantity of pure copper mixed in the alloy C ? **(a)** 3 kg **(b)** 4 kg **(c)** 5 kg **(d)** 7 kg A quadratic polynomial ax2 + bx + c is such that when it is divided by x, (x — I) and (x + I), the remainders are 3, 6 and 4 respectively. What is the value of (a + b) ? 3 2 **(b) (c)** If the average of 9 consecutive positive integers is 55, then what is the largest integer ? **(a) (d)** 57 58 59 60 The average of the ages of 15 students in a class is 19 years. When 5 new students are admitted to the class, the average age of the class becomes 18-5 years. What is the average age of the 5 newly admitted students ? **(a) (b) (c) (d)** 17 years 175 years 18 years 18•5 years The age Of a woman is a twcvdigit integer. On

246. [CDS (I) Paper 19 Q42] 39. reversing this integer, the new integer is the age of her husband Who is elder to her. The difference between their ages is one-eleventh of their sum. What is the difference between their ages ? **(a) (b) (c)** 8 years 9 years IO years 11 years 43. An alloy A contains two elements, copper and tin in the ratio Of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3 : 4. If 20 kg Of alloy A, 28 kg of alloy B and some more pure N **(d)** 3 kg

247. [CDS (I) Paper 19 Q45] Odd prime numbers upto 110 are multiplied together. What is the unit digit in this product ? The average of the ages of 15 students in a class is 19 years. When 5 new students are admitted to the class, the average age of the class becomes 18-5 years. What is the average age of the 5 newly admitted students ? **(a) (b) (c) (d)** None of the above **(a) (b) (c) (d)** 17 years 175 years 18 years 18' 5 years SPSS-T-XHI

248. [CDS (I) Paper 19 Q83] What is the largest power of IO that divides 87. the product I x 2 x 3 x 4 ..x23x24x25? **(a)** 4:31 SPSS-T-XHI 83. What is the largest power of IO that divides 87. the product I x 2 x 3 x 4 ..x23x24x25? 84. 85. 86. **(b) (c)** 2 4 5 None of the above Walls **(d)**

249. [CDS (I) Paper 19 Q94] 91. cut from a squ are of side units ? **(a) (b) (c) (d)** SPSS-T-XHI What is the area Of the largest circular disc 94. 91. cut from a squ are of side units ? 92. 93.

250. [CDS (I) Paper 19 Q2] 8. 4. **(a) (b) (c) (d)** abc is **(a) (b) (d)** 9180 9360 9630 9840 1 2 = Zb and z = xe, then the value of If x 2 + 22/3 + 21/3, then the value of the expression x3 — 6x2 + 6x will be **(b) (c) (d)** 2 —2 How many five-digit numbers of the form XXYXX Ware divisible by 33 ? **(b) (c) (d)** 3 5 Infinite GTHY-T-XMT (3- is

251. [CDS (I) Paper 19 Q8] A five-digit number XY235 is divisible by 3 where X and Y are digits satisfying X + Y s **(a) (b) (c) (d)** abc is

252. [CDS (I) Paper 19 Q5] What is the number of possible pairs of values **(a) (b) (c) (d)** 5 6 7 9 If x2 — 6x — 27 > O, then which one of the following is correct ? **(a) (b) (c) (d)** x < 9 or x > —3 x < — 3 only The number Of divisors Of the number 38808, exclusive of the divisors 1 and itself, is **(b) (c) (d)** 74 72 70 68 HCF and LCM of two polynomials are (x + 3) and (xa — 9x2 — x + 105) respectively. If one of the two x2 - — 21, then other is **(a) (b) (c) (d)** X2 + — 21 x2 + 2x + 15 x2-2x-15 x2—x— 15 The highest four-digit number which is 5. divisible by each of the numbers 16, 36, 45, 48 A five-digit number XY235 is divisible by 3 where X and Y are digits satisfying X + Y s

253. [CDS (I) Paper 19 Q13] 14. 15. -A) If the sum of the squares of three consecutive natural numbers is 110, then the sum of their cubes is **(a) (b) (c)** 625 654 684 725 **(d)**

254. [CDS (I) Paper 19 Q16] The sum of these two integers is **(a) (b) (c)** 625 654 684 725 **(d)**

255. [CDS (I) Paper 19 Q12] 9. — g and aß — , where 1 < p < q r, p then which one of the following is the **(a) (b) (c)** 625 654 684 725 **(d)**

256. [CDS (I) Paper 19 Q15] The product of two integers p and q, where p > 60 and q > 60, is 7168 and their HCF is **(a) (b) (c)** 625 654 684 725 **(d)**

257. [CDS (I) Paper 19 Q1] The sum of the numbers is not divisible by(X+Y + Z). **(a)** ab 1 **(b)** abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to **(c)** abc **(d)** xyz

258. [CDS (I) Paper 19 Q2] The sum of the numbers is divisible by **(a)** ab 1 **(b)** abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to **(c)** abc **(d)** xyz

259. [CDS (I) Paper 19 Q90] Consider the following statements : **(a)** ab 1 **(b)** abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to **(c)** abc **(d)** xyz

260. [CDS (I) Paper 19 Q3] Consider the following statements : **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11

261. **[CDS (I) Paper 19 Q67]** B) (21 —A) Consider the following statements : **(a)** 4/3 **(b)** -4/3 **(c)** 1/3 , then 2 65. 66. 67. 1. 2. cos 0 cos 0 = 4 , where If 1 — sin 0 1 + sin 6 O < O < 900, then O = 600. If 3 tan O + cot O = 5 cosec O, where O < 9 < 900, then 0 = 600. If 0 < 0 < 900, O < 900 and cos 9 < cos 'i, then which one of the following is correct ? **(d)** 900

262. **[CDS (I) Paper 19 Q91]** 92. 320, 64 640, 64 **(a) (b) (c) (d)** pq rt

263. **[CDS (I) Paper 20 Q2]** 3. **(b) (d)** 106 x 1512 x 2014 x x 2516 ? 65 55 50 45 Ho•€v many pairs (A, B) are possible in the number 479865AB if the number is divisible by 9 and it is given that the last digit of the is Odd ? **(a)** Consider the multiplication 999 x abc def132 in decimal notation, where a, b, c, d, e and f are digits. What are the values of a, b, c, d, e and f respeetively ? **(b)** 8.6.8.6. 7.8 2. What is the largest value Of n such that 100 5. divides the product 25x3ax48x53x67x76x812x99* 106 x 1512 x 2014 x x 2516 ? 3. **(c) (d)** 65 55 50 45

264. **[CDS (I) Paper 20 Q9]** Which of the following statements is not **(a) (b)** I'he difference of two prime numbers, both greater than 2, is divisible by 2. For two different integers m, n and a prime number p, if p divides the product m x n, then p divides either m or n. If a ig Of fm•m Gn — **(c) (d)** 2 Cannot be determined In a hostel the rent per room is increased by 20%. If number of rooms in the hostel is also increased by and the hostel is always full, then what is the percentage change in the total collection at the cash counter ?

265. **[CDS (I) Paper 20 Q2]** For two different integers m, n and a prime number p, if p divides the product m x n, then p divides either m or n. If a ig Of fm•m Gn — (n being a natural number), then it is a prime number. There is • only one set of three prime numi_wrs such that there is a gap of 2 between two a(Uacent prime numbers. For x > O, what is the minimum value of **(a) (b)** I'he difference of two prime numbers, both greater than 2, is divisible by 2. For two different integers m, n and a prime number p, if p divides the product m x n, then p divides either m or n. If a ig Of fm•m Gn — **(c) (d)** 2 Cannot be determined In a hostel the rent per room is increased by 20%. If number of rooms in the hostel is also increased by and the hostel is always full, then what is the percentage change in the total collection at the cash counter ?

266. **[CDS (I) Paper 20 Q25]** 26. A plane is going in circles around an airport. The plane takes 3 minutes to complete one round. The angle Of elevation of the plane from a point P on the ground at time t seconds is equal to that at time (t + 30) seconds. At time (t + x) seconds, the plane flies vertically above the point p. What is x equal to ? **(a) (b) (c)** 3 9 A plane is going in circles around an airport. The plane takes 3 minutes to complete one round. The angle Of elevation Of the plane from a point P on the ground at time t seconds is equal to that at time **(d)**

267. **[CDS (I) Paper 20 Q2]** Both p and q may be composite **(a) (b) (c)** 3 9 A plane is going in circles around an airport. The plane takes 3 minutes to complete one round. The angle Of elevation Of the plane from a point P on the ground at time t seconds is equal to that at time **(d)**

268. **[CDS (I) Paper 20 Q3]** One of p an d q may prime and the Other composite. Which Of the above statements are ? **(a) (b) (c) (d)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 In a class of 100 students, the average weight is 30 kg. If the average weight Of the girls is 24 kg and that of the boys is 32 kg, then what is the number Of girls in the class ? **(b) (c)** 25 26 27 28 For any two real numbers a and b, (a —b)2• is **(a)** always zero Zero **(c)** positive only if a b **(d)** positive if and only if a > b

269. **[CDS (I) Paper 20 Q24]** Of Consider the following statements in respect of two integers p and q (both > I) which are relatively prime : I. Both p and q m ay be prime n umbers. **(a) (b) (c)** 3 9 A plane is going in circles around an airport. The plane takes 3 minutes to complete one round. The angle Of elevation Of the plane from a point P on the ground at time t seconds is equal to that at time **(d)**

270. **[CDS (I) Paper 20 Q29]** 30. If a : b c : d I : 6, then what is the value of 600 60 **(c)** 36 What is • equal t.o ? **(a)** 36 28. 29. 30. **(b) (c)** What is • equal t.o ? 8• 17 kmph 7 •80 kmph 777 kmph , then which one Of **(d)**

271. **[CDS (I) Paper 20 Q32]** 84. the following Statements is correct ? The inequality 3N > Na holds when **(a) (b) (d)** Each fraction is equal to I or —I. Each fraction is equal to — or I. — or —I. Each fraction is equal to — only **(c) (d)** N is any natural number N is a natural number greater than 2 N is a natural number greater than 3 N is a natural number except 3 The 3521 is divided by 8. is remainder ? **(a) (c)** 3 9 Which one of the following is an irrational 231 593 0-45454545..... DFGT-T-HTM A prime number contains the digit X at unit's place. How many such digits Of X are possible ? **(a) (b) (d)** 3 4 5 6

272. **[CDS (I) Paper 20 Q65]** 66. The average of 50 consecutive natural 6 numbers is x. What will be the new average when the next four natural numbers are also included ? **(a) (b) (c) (d)** 44 55 66 If the above data is represented by a percentage bar diagram of height 15 em, then what are the lengths of the two segments of the bar diagram corresponding to education and miscellaneous respectively ? Diagrammatic representation Of data ineludes which of the following ? I.

273. **[CDS (I) Paper 20 Q63]** The average of 50 consecutive natural 67. numbers is x. What will be the new average when the next four natural numbers are also included ? **(a) (b) (c) (d)** 44 55 66 If the above data is represented by a percentage bar diagram of height 15 em, then what are the lengths of the two segments of the bar diagram corresponding to education and miscellaneous respectively ? Diagrammatic representation Of data ineludes which of the following ? I.

274. [CDS (I) Paper 20 Q81] 77. Three consecutive integers form the lengths of a right-angled triangle. How many sets of such three consecutive integers Ware possible ? **(a) (b)** Only one Only two Only three Infirutely many Two circles are drawn with the same centre. The circumference of the smaller circle is 44 cm and that of the bigger circle is double the smaller one. What is the area between these two circles ? 154 Square Square Cm 462 Square Cm Square Cm A rectangular red carpet of size 6 ft x 12 ft has a dark red border 6 inches wide. What is the area of the dark red border ? 82. The of a right-angled triangle is k times the shortest side. If the ratio of the other side to hym»tenuse is 4 : 5, then What is the value of k ? **(c) (d)** A 12 m long wire is cut into two pieces. one Of ' which is bent into a circle and the other into a square enclosing the circle. What is the radius Of the circle ? 12

275. [CDS (I) Paper 21 Q9] What is åe of bikers in mad 2017 ? **(a)** 163 **(b)** 152 **(c)** 141 **(d)** 134

276. [CDS (I) Paper 21 Q19] If qudons whA is vahE Of k ? 10 3 10 5 3 10 **(a)** 26 18 **(b)** 42 mimnes 36 **(c)** 45 minutæ **(d)** 46 mimit— 12 sæonds 2 4 18. WbatistheLOAof,-, —, — ? 9 27 18 27 10 27 20 19. If qudons whA is vahE Of k ? 10 3 10 5 3 10 9. 20. A Int to B at tlr time me to C sne 7% sinple intaut. Aner 4 yeas A 01200 intæt B md C. HOW mueh A to C ?

277. [CDS (I) Paper 21 Q37] The sides AD, BC of a ü•apdum ABCD prallel md the di.N1s AC BD at O. If the of Gangle AOB is 3 cm2 the Of tiangle BDC is 8 then what is åe of aianøe AOD ? **(a)** 33 cm **(b)** 44 **(c)** 66 cm 37. The sides AD, BC of a ü•apdum ABCD prallel md the di.N1s AC BD at O. If the of Gangle AOB is 3 cm2 the Of tiangle BDC is 8 then what is åe of aianøe AOD ? **(d)** 8 cm2

278. [CDS (I) Paper 21 Q42] If I is åe of Of an quilatnl Eiangle. thal what is its 3 2 **(c)** 612 **(d)** 212 **(a)** 612 **(b)** 212 43. A of wire in of a Etor Of a circle Of 20 cm, suM01ding an mgle 1500 at If it is ban ih 6rm Of a circle. what will be its ? **(c)** None Of the P, Q R Üe mid-points Of si&s of a &iangle of 128 cm2. If a &imøe ABC is &awn by joining of side Of riangle POR, what is of timgle ABC? **(d)** 4 cm2 O, 8 cm2

279. [CDS (I) Paper 21 Q41] All Sides of Ze of qual diapnals in the rafo 1 : **(a)** 612 **(b)** 212 43. A of wire in of a Etor Of a circle Of 20 cm, suM01ding an mgle 1500 at If it is ban ih 6rm Of a circle. what will be its ? **(c)** None Of the P, Q R Üe mid-points Of si&s of a &iangle of 128 cm2. If a &imøe ABC is &awn by joining of side Of riangle POR, what is of timgle ABC? **(d)** 4 cm2 O, 8 cm2

280. [CDS (I) Paper 21 Q71] If of a Üid its rec9mcal is , thn numbers **(a)** 1 only **(b)** 96" cm2 **(c)** 2 only **(d)** Boå 1 2 69T cm2

281. [CDS (I) Paper 21 Q2] If p divi&s qr if p divides q, p must divi& r. 2' 2' WNch of the **(a)** 1 only **(b)** 96" cm2 **(c)** 2 only **(d)** Boå 1 2 69T cm2

282. [CDS (I) Paper 21 Q77] What is the numb' in 125, 823 md 920 ? [Givn hg102 = 0-301, log. —&477, 10867 - 0845] **(a)** 21, 20, 19 **(b)** 20, 19, 18 **(c)** 22, 21,20 **(d)** n, 20,21 LA x be integer such that 14 divid« x, is 7; wha 15 x, is

283. [CDS (I) Paper 21 Q5] Which of the 6110wing is ? **(d)** x > 50 X nd y to a tank. If only X is open«i, it drains talk of wata 20 mimB. If X Y are opa14 they drain out the tank of wat« in 15 Ifonly Y is how long it tüe &ain out the tank of wata ? 21 **(a)** 30 minutes **(b)** 45 minutes **(c)** 60 minute **(d)** 90 minute 80. ale ftllowiné : 1. is a rational 4x 7 x that < 8 3. — < 1 Br all values Ofx. 4. can be in wh«e p q Which Of **(b)** 2 and 3 **(d)** 2 and4

284. [CDS (I) Paper 22 Q1] The number 2x3*5x7x11+1 is **(a)** a prime number **(b)** not a prime, but power of a prime **(c)** not a power of a prime, but a composite even number **(d)** not a of a prime, but a composite odd number

285. [CDS (I) Paper 22 Q4] Which one of the following is a set of solutions of the equation x x , if n is a positive integer ? **(a) (b)** {l,cn} **(c) (d)** a competitive examination, 250 students have registered. Out of these, 50 students have registered for Physics, 75 students for Mathematics and 35 students for both Mathematics and Physics. What is the number of students who have registered neither for Physics nor for Mathematics ? **(a)** 90 **(b)** 100 **(c)** 150 **(d)** 160

286. [CDS (I) Paper 22 Q6] If the sum of the digits of a number 10" — 1, where n is a natural number, is equal to 3798, then what is the value of n? **(a)** 421 **(b)** 422 **(c)** 423 **(d)** 424

287. [CDS (I) Paper 22 Q8] If m is the number Of prime numbers between 0 and 50•, and n is the number of prime numbers between 50 and 100, then what is (m — n) equal to ? **(a)** 13 **(b)** 39 **(c)** 117 5 11. A two-digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two-digit number is 8, then what is the number ? **(d)** 81

288. [CDS (I) Paper 22 Q9] If b --2 — 3 and — — , then what is the value of — def 4 3 8 27 4 IO. Which one of the following is the largest divisor of 3x+ + 3x+2, if x is any natural number ? **(b)** 13 **(c)** 39 **(d)** 117 5 5 **(a)** 13 **(b)** 39 **(c)** 117 5 11. A two-digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two-digit number is 8, then what is the number ? **(d)** 81

289. [CDS (I) Paper 22 Q11] A two-digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two-digit number is 8, then what is the number ? **(a)** 81 **(b)** 42 **(c)** 24 **(d)** 18

290. **[CDS (I) Paper 22 Q16]** A number divides 12288, 28200 and 44333 so as to leave the same remainder in each case. What is that number ? 272 **(b)** 232 **(c)** 221 **(d)** 120 **(a)** 1000 **(b)** 100000 **(c)** 1000000 16. A number divides 44333 so as to remainder in each 19. What is **(d)** 2am-n equal to ? a 12288, 28200 and leave the same case. What is that 20. 22. If x = •a, y=VÄ and z=fi, then which one of the following is correct ? number ? 272

291. **[CDS (I) Paper 22 Q18]** If m and n are positive integers such that mn = 1331, then what is the value of (m — l)n-l ? **(b)** 100 **(c)** 121 **(d)** 125 **(a)** 1000 **(b)** 100000 **(c)** 1000000 16. A number divides 44333 so as to remainder in each 19. What is **(d)** 2am-n equal to ? a 12288, 28200 and leave the same case. What is that 20. 22. If x = •a, y=VÄ and z=fi, then which one of the following is correct ? number ? 272

292. **[CDS (I) Paper 22 Q22]** If nx) is divided by _r/mr, then what is the where a ß, then remainder ? A - PIXI-T-MTK **(a)** 1000 **(b)** 100000 **(c)** 1000000 16. A number divides 44333 so as to remainder in each 19. What is **(d)** 2am-n equal to ? a 12288, 28200 and leave the same case. What is that 20. 22. If x = •a, y=VÄ and z=fi, then which one of the following is correct ? number ? 272

293. **[CDS (I) Paper 22 Q25]** If the points p and Q represent real numbers 0-73 and 0-56 on the number line, then what is the distance between P and Q ? 6 16 45 11 90 9 **(a)** 49x2 -f- 3x + 2 **(b)** 49x2 _ 3x + 2 **(c)** 49++2 **(d)** 59++2 24. If x varies asyz, theny varies inversely

294. **[CDS (I) Paper 22 Q5]** What is the number of students in the class ? **(a)** 100 **(b)** 105 **(c)** 110 **(d)** 120

295. **[CDS (I) Paper 22 Q41]** The number of three digit numbers (all digits are different) which are divisible by 7 and also divisible by 7 on reversing the order Of the digits, is **(a)** Six **(b)** Five **(c)** Four **(d)** Three

296. **[CDS (I) Paper 22 Q3]** S is always divisible by 3 and divisible by 9 divisible by 37 15 15 Select the corect answer using the code given below : **(a)** 1 only **(b)** 2 only **(c)** 1 and 2 **(d)** I and 3

297. **[CDS (I) Paper 22 Q47]** What is the least perfect square which is divisible by 3, 4, 5, 6 and 7? **(a)** 1764 **(b)** 17640 **(c)** 44100 **(d)** 176400

298. **[CDS (I) Paper 22 Q100]** Consider the following statements with reference to the given figure : **(a)** 924 square units **(b)** 824 square units **(c)** 624 square units **(d)** 462 square units 99. 6 cm What is the approximate area of the shaded region in the figure given ?

299. **[CDS (I) Paper 22 Q1]** The sum of the areas of AAOD and ABOC is equal to the sum of the areas of MOB and ADOC **(a)** 924 square units **(b)** 824 square units **(c)** 624 square units **(d)** 462 square units 99. 6 cm What is the approximate area of the shaded region in the figure given ?

300. **[CDS (I) Paper 23 Q6]** 7. The number of solutions of the equation x + y + z 12, where each of x, y and z is a m:vsitive integer, is **(b) (c) (d)** 53 54 55 56 + b2 + Where a and b consecutive integers and c = ab, then I is **(a) (b) (c) (d)** an even number and it is not a Square Of an integer an odd number and it is not a square of an integer Of an even integer square Of an Odd integer If the number 23P62971335 is divisible by the smallest odd composite then What is the Of P ? **(a) (b) (c) (d)** 4 5 6 7 What is the remainder When the 15 + 25 + 35 + 46 + 55 is divided by 4 ? **(a) (c) (d)** sum xa + + 16 is e*actly divisible by x, where x is 4• positive integer. The number of all such possible values Of x is The number of solutions of the equation x + y + z 12, where each of x, y and z is a m:vsitive integer, is **(b) (c) (d)** 53 54 55 56 **(d)** 3 4 5 6

301. **[CDS (I) Paper 23 Q5]** 6. 7. + b2 + Where a and b consecutive integers and c = ab, then I is The number of (a, b, c), where a, b, c are positive such that abc 30, is **(a) (b) (d)** 30 27 9 8 **(a) (b) (c) (d)** an even number and it is not a Square Of an integer an odd number and it is not a square of an integer Of an even integer square Of an Odd integer If the number 23P62971335 is divisible by the smallest odd composite then What is the Of P ? If the roots Of the quadratic equation — — loglo N O are real, then what is the minimum value of N ? 10 **(d)** 10000 **(a) (b) (c) (d)** 4 5 6 7 What is the remainder When the 15 + 25 + 35 + 46 + 55 is divided by 4 ? **(a) (c) (d)** sum DZOL-T-LKM

302. **[CDS (I) Paper 23 Q10]** What is the digit in the unit place Of 399 ? **(a) (d)** 7 9 LCM of two numbers is 28 times their HCF. sum of the HCF and the LCM is 1740. If one Of these numbers is 240. then what is the other number ? 420 640 820 1040 (xn — an) is divisible by (x — a), where x a, natural number n **(b) (a) (b) (c) (d)** even natural numiRr n only odd natural number n Only prime number n only DZOL-T-LKM (5- If 1720X) ig divided by 18, then what is the remainder ? **(d)** 16 17 12. the 1 What is the value Of **(a) (c) (d)** If **(a) (d)** 5 10 x , then what is the value of 8 3

303. **[CDS (I) Paper 23 Q19]** 20. Lat d(n) denote the number Of positive divisors of a pysitive integer n. Which of the fillowing are correct ? **(a)** 25% 40% **(b) (c) (d)** 1 and 3 only 22. I and 2 only 2 and 3 only I, 2 and 3 A river 3 m deep and 40 m wide is flowing at the rate Of 2 km/hr and falls into the sea. What is the amount of water in litres that will fall into the æa from this river in a minute ? If An Pn + I, where pn is the product of the first n prime numbers, then consider the following statements • I. An is alw ays a composite number. 2. Au + 2 is alw ays an number. 3. An I is always an even number. Which of the above statements Ware correct ?

304. [CDS (I) Paper 23 Q1] 3. d(5) d(ll) d(55) d(5) + d(ll) - d(16) Select the correct answer using the code given **(a)** 25% 40% **(b) (c) (d)** 1 and 3 only 22. I and 2 only 2 and 3 only I, 2 and 3 A river 3 m deep and 40 m wide is flowing at the rate Of 2 km/hr and falls into the sea. What is the amount of water in litres that will fall into the æa from this river in a minute ? If An Pn + l, where pn is the product of the first n prime numbers, then consider the following statements • I. An is alw ays a composite number. 2. Au + 2 is alw ays an number. 3. An I is always an even number. Which of the above statements Ware correct ?

305. [CDS (I) Paper 23 Q3] An is always a composite number. Au + 2 is always an number. An + I is always an even number. Which of the above statements Ware correct ? **(b)** (e) **(d)** 1 only 2 only 3 only 2 and 3 only DZOL-T-LKM (9. **(a)** 25% 40% **(b) (c) (d)** 1 and 3 only 22. I and 2 only 2 and 3 only I, 2 and 3 A river 3 m deep and 40 m wide is flowing at the rate Of 2 km/hr and falls into the sea. What is the amount of water in litres that will fall into the æa from this river in a minute ? If An Pn + l, where pn is the product of the first n prime numbers, then consider the following statements • I. An is alw ays a composite number. 2. Au + 2 is alw ays an number. 3. An I is always an even number. Which of the above statements Ware correct ?

306. [CDS (I) Paper 23 Q22] I and 2 only 2 and 3 only I, 2 and 3 A river 3 m deep and 40 m wide is flowing at the rate Of 2 km/hr and falls into the sea. What is the amount of water in litres that will fall into the æa from this river in a minute ? If An Pn + l, where pn is the product of the first n prime numbers, then consider the following statements • I. An is alw ays a composite number. **(a)** 25% 40% **(b) (c) (d)** 1 and 3 only 22. I and 2 only 2 and 3 only I, 2 and 3 A river 3 m deep and 40 m wide is flowing at the rate Of 2 km/hr and falls into the sea. What is the amount of water in litres that will fall into the æa from this river in a minute ? If An Pn + l, where pn is the product of the first n prime numbers, then consider the following statements • I. An is alw ays a composite number. 2. Au + 2 is alw ays an number. 3. An I is always an even number. Which of the above statements Ware correct ?

307. [CDS (I) Paper 23 Q38] 39. 40. (15-A) X, Y and Z travel from the same place with uniform 4 km.mr, 5 knvhr and 6 km.•'hr respectively. Y starts 2 hours after X. HOW long after Y must Z start in order that they overtake X at the same instant ? **(b) (c)** — hours — hours 1 — x — xn + x-rwl, where n is a natural number, is divisible by A sold an article e 75 which cost him x. He finds that he realised profit on his outlay. What is x equal to ? **(c) (d)** 25% 100%

308. [CDS (I) Paper 23 Q46] 47. (17. A) The Euclidean algorithm is used to calculate **(a) (b) (c)** square root Of an integer cube of an integer Square Of an HCF Of two integers **(d)**

309. [CDS (I) Paper 23 Q47] If radius of a sphere is rational, then which of the fi•llowing Ware correct ? **(a) (b) (c)** square root Of an integer cube of an integer Square Of an HCF Of two integers **(d)**

310. [CDS (I) Paper 23 Q78] 79. 80. (27 —A) In a ABC, if 2 z A 3 6 Z C, then what is z A + Z C equal to ? 900 1350 1500 If the perimeter of a circle and a are equal, then what is the ratio of the area of the circle to that Of the square ? **(a) (b)** Hexagon Nonagon Decagon 79. A bicycle wheel makes 5000 revolutions in moving 11 km. What is the radius of the wheel ? **(c) (d)**

311. [CDS (I) Paper 24 Q1] If the number 413283P759387 is divisible by 13, then What is the value of P? **(b)** 3 6 7 **(a)** 3 6 7 **(b)** How 2. What is the remainder when 21000000 divided by 7? 2 3 4 5 many terms are there in the **(c) (d)** 2 How many pairs of

312. [CDS (I) Paper 24 Q2] What is the remainder when 21000000 divided by 7? **(a)** 3 6 7 **(b)** How 2. What is the remainder when 21000000 divided by 7? 2 3 4 5 many terms are there in the **(c) (d)** 2 How many pairs of

313. [CDS (I) Paper 24 Q4] Consider the pairs of prime numbers (m, n) between 50 and 100 such that m — n = **(a)** 3 6 7 **(b)** How 2. What is the remainder when 21000000 divided by 7? 2 3 4 5 many terms are there in the **(c) (d)** 2 How many pairs of

314. [CDS (I) Paper 24 Q9] What is the unit digit in the expansion of 6732? 3 IO. What is the value of x, if 2bx z a? b 2 — 2bx P.T.O. **(a)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 3q-2 then what is the value of p2 o XVWS—T-MTK/SOA - 3pq+2? 5 9. What is the unit digit in the expansion of 6732? **(b)** 3 IO. What is the value of x, if 2bx z a? b 2 — 2bx **(c)** P.T.O. 9. What is the unit digit in the expansion of 6732? 3 IO. What is the value of x, if 2bx z a? b 2 — 2bx P.T.O. 7. If a b + c O, then which of the following are correct? a 3 + b 3 c 3 3abC —2 **(d)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 3q-2 then what is the value of p2 o XVWS—T-MTK/SOA - 3pq+2? 5

315. [CDS (I) Paper 24 Q21] 24. For how many real values of k is 2 + 12kx 16 a "uare for every integer x ? **(a)** Zem **(b)** One **(c)** Two **(d)** Four

316. [CDS (I) Paper 24 Q22] If equal to? 195 16 16 then What is If the equation 4x 2 — 2kx + 3k O has equal roots, then What are the values of k? **(a)** Zem **(b)** One **(c)** Two **(d)** Four 22. If equal to? 195 16 16 4x3 + 12x2 — x— 3 is divisible by

317. [CDS (I) Paper 24 Q26] Which one Of the following fractions Will have minimum change in its value if 3 is added to both the numerator and the denominator of an the fractions? **(c)** 5 **(d)** Ikt the average score of a class of boys and girls in an examination be p. The ratio Of boys and girls in the class is 3: I . If the average score Of the boys is (p I), then What is the average score of **(a)** Zem **(b)** One **(c)** Two **(d)** Four 22. If equal to? 195 16 16 4x3 + 12x2 — x— 3 is divisible by

318. [CDS (I) Paper 24 Q27] 11 the **(d)** girls? **(a)** Zem **(b)** One **(c)** Two **(d)** Four 22. If equal to? 195 16 16 4x3 + 12x2 — x— 3 is divisible by

319. [CDS (I) Paper 24 Q24] For how many real values of k is 2 + 12kx 16 a "uare for every integer x ? **(a)** Zem **(b)** One **(c)** Two **(d)** Four

320. [CDS (I) Paper 24 Q72] What is the principal amount? e 6,500 e 6,000 e 5,500 e 5,000 **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

321. [CDS (I) Paper 24 Q41] What is logro 31•25 equal to? 3 — 510g10 2 **(a)** 16 44. I — loglo 2 3 log102 I-loglo 2 5 **(b)** 1-log102 What is the 3sin2e +4cos2 e ? 3 2 least value Of . The sum Of the reciprocals of two What alternate natural numtrrs is is the sum Of the numbers? **(c)** 12 13 14 If n is any natural then 52n — 1 is always divisible by how many natural numbers? 47. If sinecose k, where os 9 s then which one of the following is correct? **(d)**

322. [CDS (I) Paper 24 Q42] What is the square root of 15 — 4N'Tä? 43 **(d)** 16 44. . The sum Of the reciprocals of two What alternate natural numtrrs is is the sum Of the numbers? **(b)** 12 13 14 If n is any natural then 52n — 1 is always divisible by how many natural numbers? **(a)** 16 44. I — loglo 2 3 log102 I-loglo 2 5 **(b)** 1-log102 What is the 3sin2e +4cos2 e ? 3 2 least value Of . The sum Of the reciprocals of two What alternate natural numtrrs is is the sum Of the numbers? **(c)** 12 13 14 If n is any natural then 52n — 1 is always divisible by how many natural numbers? 47. If sinecose k, where os 9 s then which one of the following is correct? **(d)**

323. [CDS (I) Paper 24 Q44] I — loglo 2 3 log102 I-loglo 2 5 **(d)** 1-log102 What is the 3sin2e +4cos2 e ? 3 2 least value Of . The sum Of the reciprocals of two What alternate natural numtrrs is is the sum Of the numbers? **(b)** 12 13 14 If n is any natural then 52n — 1 is always divisible by how many natural numbers? **(a)** 16 44. I — loglo 2 3 log102 I-loglo 2 5 **(b)** 1-log102 What is the 3sin2e +4cos2 e ? 3 2 least value Of . The sum Of the reciprocals of two What alternate natural numtrrs is is the sum Of the numbers? **(c)** 12 13 14 If n is any natural then 52n — 1 is always divisible by how many natural numbers? 47. If sinecose k, where os 9 s then which one of the following is correct? **(d)**

324. [CDS (I) Paper 24 Q60] 60• 45' B t, cote **(a)** 11 **(b)** 22 **(c)** 77 **(d)** 99 k sece 59. 60' 60. 60• 45'

325. [CDS (I) Paper 24 Q79] ABC is a trianøe angled at A and AD is perpendicular to BC If BD 8 cm and DC 12 5 cm. then What is AD equal to? **(a)** 3850 m **(b)** 7700 m 15400 m 76. In a trapezium ABCD, AB is parallel to DC. The diagonals AC and BD intersect at P. If AP : PC 4 : Bx — 4 **(c)** 11:23 **(d)** 23:25

326. [CDS (I) Paper 24 Q32] Later on 5 grace marks are added to each student. What are the average marks of the revised marks Of the students? 30 31 32 1 P.T.O. **(a)** In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? **(b)** and E **(c)** A, C and D only **(d)** C D and E

327. [CDS (I) Paper 25 Q1] What are distinct prime mctors of the number 26381 ? **(a)** 29, 17, 37 **(b)** 31, 17, 47 **(c)** 19, 37, 13 **(d)** 23, 31, 37

328. [CDS (I) Paper 25 Q2] Which one of the following is a factor of the **(a)** 29, 17, 37 **(b)** 31, 17, 47 **(c)** 19, 37, 13 **(d)** 23, 31, 37 2. Which one of the following is a factor of the

329. [CDS (I) Paper 25 Q3] What is the square root of 23 —4•v'fiS ? **(b)** 7-3-6 **(a)** 29, 17, 37 **(b)** 31, 17, 47 **(c)** 19, 37, 13 **(d)** 23, 31, 37 2. Which one of the following is a factor of the

330. [CDS (I) Paper 25 Q7] If x is a negative real number, then which of the following are not correct ? l. There is some natural number k such that > 0 2. + x > O always 4. x2 is always a rational number Select the correct answer using the code given below : **(a)** 1, 2 and 3 **(b)** 1, 2 and 4 **(c)** and 4 **(d)** 2, 3 and 4

331. [CDS (I) Paper 25 Q23] What is **(d)** **(a)** 30 **(b)** 45 **(c)** 60 **(d)** 75 equal to ?

332. [CDS (I) Paper 25 Q32] A two-digit number is such that the product of the digits is **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 only when Ware correct ? 28. Let work done by

333. [CDS (I) Paper 25 Q8] If 63 is added to this number, the digits interchange their places. What is the sum of the digits in the number ? **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 only when Ware correct ? 28. Let work done by

334. [CDS (I) Paper 25 Q35] The sum of numerator and denominator of a fraction is **(a)** 36 years **(b)** 40 years **(c)** 42 years **(d)** 45 years 35. The sum of numerator and denominator of a fraction is 10. If the numerator is increased by 3 and denominator is decreased by 1 , the fraction becomes I. What is the difference between numera- tor and denominator of the fraction ? 36. If the system of equations 7x + ky 27 and kx + 7y = 19 have unique solution, then which one of the following is correct ?

335. [CDS (I) Paper 25 Q10] If the numerator is increased by 3 and denominator is decreased by 1 , the fraction becomes I. What is the difference between numera- tor and denominator of the fraction ? **(a)** 36 years **(b)** 40 years **(c)** 42 years **(d)** 45 years 35. The sum of numerator and denominator of a fraction is 10. If the numerator is increased by 3 and denominator is decreased by 1 , the fraction becomes I. What is the difference between numera- tor and denominator of the fraction ? 36. If the system of equations 7x + ky 27 and kx + 7y = 19 have unique solution, then which one of the following is correct ?

336. **[CDS (I) Paper 25 Q89]** What is Z BAP equal to ? **(a)** 300 **(b)** 400 **(c)** 450 **(d)** 600

337. **[CDS (I) Paper 25 Q98]** In one Of the yean the increase in production of cars was minimum as compared to its previous year. What was the minimum percen- tage ? **(a)** 100% **(b)** 150% **(c)** 200% 96. Consider the production of each company separately. In how many instances was the percentage increase in production of cars over the previous year's production greater than 20% ? 97. In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? **(d)** 2015

338. **[CDS (I) Paper 25 Q99]** How many such sequences of five consecutive integers are possible ? **(a)** One **(b)** Two **(c)** Three **(d)** Five

339. **[CDS (I) Paper 26 Q2]** 4143 4343 is divisible by **(a)** 80 **(c)** 86 **(b)** 84 **(d)** 88

340. **[CDS (I) Paper 26 Q4]** 461 +462 +463 +464 is divisible by **(c)** 11 **(d)** 17 **(a)** 2 7. A man walks at an average speed of 3 km/hr from his residence and reaches Office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence 3 2. 4143 4343 is divisible by and office ? **(b)** 6 km **(c)** IO km **(d)** 8 km

341. **[CDS (I) Paper 26 Q14]** Let a, b, G d be positive integers. If 17 then What is the product Of b, G d ? **(a)** 24 **(c)** 68 **(b)** 51 **(d)** 102

342. **[CDS (I) Paper 26 Q20]** What is the smallest natural number from the following Which must be subtracted from 9410 to make the remaining number a perfect square? 5 1 P.T.O. **(a)** 6 units **(b)** 8 units **(c)** IO units **(d)** 12 units 13. If43Xx47Y then what is the value of the following? 4xy+ x + y 2xy-x-y 16. 17. 18. 20. 5 What is the least value of n if 194480* nzm4, where m and n are natural numbers? If + yv'Q ¯ then What is the value Of ?

343. **[CDS (I) Paper 26 Q21]** 22. 27 S + 313 is divisible by **(c)** 12 **(b)** 10 **(d)** 21 2n.2 + m and q •2 — m Let p 2 (where n is even natural number). What should be the least value of m such that p as well as q is divisible by 5? **(a)** 12 **(b)** 10 **(c)** 21 26. 28. If -2 then what is the value Of as + b3? 2n.2 + m and q •2 — m Let p 2 **(d)**

344. **[CDS (I) Paper 26 Q23]** If squaring a positive real number x is same as adding 12, then what is x equal to? **(a)** 12 **(b)** 10 **(c)** 21 26. 28. If -2 then what is the value Of as + b3? 2n.2 + m and q •2 — m Let p 2 **(d)**

345. **[CDS (I) Paper 26 Q32]** 34. A shopkeeper marks the price Of an article at ? **(a)** 15 **(b)** 17 **(c)** 16 **(d)** 18 32. 34. A shopkeeper marks the price Of an article at ? 200. After allowing a discount Of 10%, he still gains 20% on the cost price. What is the cost price 37. 38. 39 40. 9 A piece of cleth costs 10,000. If a 2 m longer piece Of the same cloth is purchased for the same amount, it would cost 250 less per metre. What is the original length of the piece Of the article?

346. **[CDS (I) Paper 26 Q33]** A person borrowed e 9,000 at e 12,000 at 8% and e 15,000 at 9% simple interest per annum. He had to pay e 50,700 at the end Of n years. What is the value Of n ? Consider the following statements : l. The sum of the cutxs Of three consecutive natural numbers is divisible by **(a)** 15 **(b)** 17 **(c)** 16 **(d)** 18 32. 34. A shopkeeper marks the price Of an article at ? 200. After allowing a discount Of 10%, he still gains 20% on the cost price. What is the cost price 37. 38. 39 40. 9 A piece of cleth costs 10,000. If a 2 m longer piece Of the same cloth is purchased for the same amount, it would cost 250 less per metre. What is the original length of the piece Of the article?

347. **[CDS (I) Paper 26 Q9]** 2. Every even power of every Odd number I) when divided by 8 gives I as remainder. Which Of the above statements is/ are correct? **(a)** I only **(b)** 2 only (C) Both I and 2 **(d)** Neither I nor 2

348. **[CDS (I) Paper 26 Q43]** Consider a question and two statements : Question : Is a2 + b2 + c2 are distinct real numbers) always positive? Statement—I : a > b > c Statement—II : a + b + c 0 Which one of the following is correct in respect Of the question and the statemen ts ? **(b) (d)** Statement—I alone is required to answer the question Statement—II alone is required to answer the question Both Statement—I and Statement—ll are required to answer the question Neither Statement—I nor State- ment—ll is required to answer the question **(a) (b) (c)** Statement—I alone is sufficient to answer the question Statement—II alone is sufficient to answer the question Both Statement—I and Statement—II are together sufficient to answer the question Both Statement—I and Statement—II are not sufficient to answer the question **(d)**

349. **[CDS (I) Paper 26 Q61]** Consider the following statements : **(a)** cos29 **(b)** 2 sin B **(c)** 2cOsB 59. If O, where then what is 'the value Of 12cot2 B + 3cosece? 61. Considér the following statements : 2. If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. **(d)**

350. **[CDS (I) Paper 26 Q3]** EDC Select the correct answer using the code given below. **(a)** 2x-90' **(b)** 2x-18 **(c)** 18m-2x ABC is a triangle right angled at B with AC-2BC If ZA-x, then what is ZC equal to? 78. 80. Let D, E and F be the midpoints Of the sides BC, CA and AB respectively Of a triangle ABC Triangle DEF is congruent to which of the following 1. AEF 3. EDC Select the correct answer using the code given below. 81. 84. In a triangle ABC, AB— 16 cm , 12 cm and AD is the bisector Of ZA If BD —4 cm, then what is CD equal to? **(d)** 2-5 cm 82. An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct?

351. [CDS (I) Paper 27 Q8] If three times the greater Of two numbers is 1 divided by the smaller number, the quotient Will be 6 and the remainder Will be **(a)** FDGT-T-EMT 8. If three times the greater Of two numbers is 11• divided by the smaller number, the quotient Will be 6 and the remainder Will be 6. If five times the smaller number is divided by the greater number, the quotient Will be 2 and the remainder Will 3. What is the difference between the numbers ? 9. 10. **(b) (c) (d)** Consider the following statements : l. x'

352. [CDS (I) Paper 27 Q6] If five times the smaller number is divided by the greater number, the quotient Will be 2 and the remainder Will **(a)** FDGT-T-EMT 8. If three times the greater Of two numbers is 11• divided by the smaller number, the quotient Will be 6 and the remainder Will be 6. If five times the smaller number is divided by the greater number, the quotient Will be 2 and the remainder Will 3. What is the difference between the numbers ? 9. 10. **(b) (c) (d)** Consider the following statements : l. x'

353. [CDS (I) Paper 27 Q3] What is the difference between the numbers ? **(a)** FDGT-T-EMT 8. If three times the greater Of two numbers is 11• divided by the smaller number, the quotient Will be 6 and the remainder Will be 6. If five times the smaller number is divided by the greater number, the quotient Will be 2 and the remainder Will 3. What is the difference between the numbers ? 9. 10. **(b) (c) (d)** Consider the following statements : l. x'

354. [CDS (I) Paper 27 Q9] 10. (5—A **(a) (c) (d)** 10 12 Consider the following statements : I. (ab + bc + ca) is a factor a2 (b — + b2 (c — + c2(a — bp. (a + b + c) is a factor + b2(c — + b)3. Of Of Which Of the statements given above Ware **(a) (b) (c) (d)** Only I Only 2 Both I and 2 Neither I nor 2 -y) is

355. [CDS (I) Paper 27 Q20] If x is the HCF and y is the LCM Of ' 3 6 9 27 then which one Of the 5' 25' 20' 50' following is correct ? **(a)** y = 180* y 270* y = 360x the one 21. LCM of Of the Consider the number N = 126 x 38 x 53. Which Of the following statements Ware correct ? I. The number Of Odd factors of N is 60. 2. The number Of even factors of N is 720. Select the correct answer using the code' given below : **(b) (c) (d)**

356. [CDS (I) Paper 27 Q3] What is the remainder when (x — y) is divided by 6 ? **(a)** y = 180* y 270* y = 360x the one 21. LCM of Of the Consider the number N = 126 x 38 x 53. Which Of the following statements Ware correct ? I. The number Of Odd factors of N is 60. 2. The number Of even factors of N is 720. Select the correct answer using the code' given below : **(b) (c) (d)**

357. [CDS (I) Paper 27 Q53] Which Of the following statements Ware correct ? I. The number Of Odd factors of N is **(a)** y = 180* y 270* y = 360x the one 21. LCM of Of the Consider the number N = 126 x 38 x 53. Which Of the following statements Ware correct ? I. The number Of Odd factors of N is 60. 2. The number Of even factors of N is 720. Select the correct answer using the code' given below : **(b) (c) (d)**

358. [CDS (I) Paper 27 Q60] 2. The number Of even factors of N is **(a)** y = 180* y 270* y = 360x the one 21. LCM of Of the Consider the number N = 126 x 38 x 53. Which Of the following statements Ware correct ? I. The number Of Odd factors of N is 60. 2. The number Of even factors of N is 720. Select the correct answer using the code' given below : **(b) (c) (d)**

359. [CDS (I) Paper 27 Q27] 28. 11 -A) Consider the question and two statements given below, : A 2•digit number is added to the number formed by reversing the digits of the 2-digit number. Questitm : What is the 2-digit number ? Statement-I The sum is divisible by **(a) (b)** 12 11 10 9 HOW many of the following values Of x would satisfy the equation qx2 — 2px + q O ? Consider the question and two statements given below, : A 2•digit number is added to the number formed by reversing the digits of the 2-digit number. Questitm : What is the 2-digit number ? Statement-I The sum is divisible by 9. Statement-2 : The sum is divisible by 2. Which one of the following is correct in Of the question and the statements ? 2. , Where p > q , wher•e p > q , where p > q **(c) (d)**

360. [CDS (I) Paper 27 Q9] Statement-2 : The sum is divisible by **(a) (b)** 12 11 10 9 HOW many of the following values Of x would satisfy the equation qx2 — 2px + q O ? Consider the question and two statements given below, : A 2•digit number is added to the number formed by reversing the digits of the 2-digit number. Questitm : What is the 2-digit number ? Statement-I The sum is divisible by 9. Statement-2 : The sum is divisible by 2. Which one of the following is correct in Of the question and the statements ? 2. , Where p > q , wher•e p > q , where p > q **(c) (d)**

361. [CDS (I) Paper 27 Q25] 26. (11 -A) **(a) (b)** 12 11 10 9 HOW many of the following values Of x would satisfy the equation qx2 — 2px + q O ? Consider the question and two statements given below, : A 2•digit number is added to the number formed by reversing the digits of the 2-digit number. Questitm : What is the 2-digit number ? Statement-I The sum is divisible by 9. Statement-2 : The sum is divisible by 2. Which one of the following is correct in Of the question and the statements ? 2. , Where p > q , wher•e p > q , where p > q **(c) (d)**

362. **[CDS (I) Paper 27 Q29]** 30. Consider the question and two statements given below : Question : Is (xn * yn) divisibl e by (x + y) ? Statement- I : n is a natural number. Statement-2 : n is an even natural number. Which one of the following is correct in respect Of the question and the statements ? **(a) (b) (c)** Statement-I alone is sufficient to answer the question Statement-2 alone is suffcient to answer the question Both Statement-I and Statement-2 are sufficient to answer the question Both Statement-I and Statement-2 are not sufficient to answer the question Consider the question and two statements given : Let x and y be two real numbers. Question : Is xy > O ? Statement-I : x8y9 < O. Statement-2 : x9yIO < O. Which one Of the following is correct in respect Of the question and the statements ? **(a) (b) (c) (d)** Statement-I alone is sufficient to answer the question Statement-2 alone is sufficient to answer the question Both Statement-I and Statement-2 are sufficient to answer the question Both Statement-I and Statement-2 are not sufficient to answer the question F DGT-T-EMT (134 Its

363. **[CDS (I) Paper 28 Q5]** 235 — land 2 1-1? **(a)** 34 **(b)** 90 **(c)** 127 **(d)** 129 What is the largest power Of IO that divides the product 29 x 28 x 27 x 2 x I ?

364. **[CDS (I) Paper 28 Q2]** 3. 4. **(a) (b) (d)** 5 6 What is the remainder when 65 is divided by 11 ? **(a) (c) (d)** 10

365. **[CDS (I) Paper 28 Q1]** 437 797 1073 How many of the above numbers are prime ? **(a) (b) (c) (d)** Only one Only two All three None What is the largest number which divides both

366. **[CDS (I) Paper 28 Q16]** If the positive square of (5 + (5 - 30) is a, then what is the positive square root of 8 + 2a ? **(d)** When every even of every odd integer (greater than I) is divided by 8, what is the remainder ? **(a) (b) (c) (d)** 3 2 The remainder is not necessarily I. Consider the following statements : I.

367. **[CDS (I) Paper 28 Q2]** If n is a natural number, then the n(n2 + 2) number is algo a natural number. If m is an Odd integer, then the number m +4m + 11 is an integer. 16 Which Of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 JNBY-D-MTK of

368. **[CDS (I) Paper 28 Q19]** 20. It is given that 5 does not divide n — I, n and n + I, where n is a positive integer. Which one Of the following is correct ? **(a) (b) (c) (d)** 5 divides (n2 + l) 5 divides (n2 — l) 5 divides (n2 + n) 5 divides (n2 — n) What is the largest 5-digit number, which leaves remainder 7, when divided by 18 as well as by 11 ? **(a) (c)** 99981 99988 99997 99999 In a business dealing, A owes B e payable after 5 years, whereas B owes A 12,000 payable after 4 years. They want to settle it now at the rate of 5% simple interest. Who gives how much money in this settlement ? **(a) (b) (c) (d)** Both are at par B gives 6,000 to A A gives e 6,000 to B A gives e 4,000 to B Average marks in Mathematics of Section A comprising 30 students is 65 and that of Section B comprising 35 students is

369. **[CDS (I) Paper 28 Q20]** What is the largest 5-digit number, which leaves remainder 7, when divided by 18 as well as by 11 ? **(a) (c)** 99981 99988 99997 99999 **(a) (b) (c) (d)** 3 2 The remainder is not necessarily I. Consider the following statements : In a business dealing, A owes B e payable after 5 years, whereas B owes A 12,000 payable after 4 years. They want to settle it now at the rate of 5% simple interest. Who gives how much money in this settlement ? I.

370. **[CDS (I) Paper 28 Q24]** 2. The sum Of all factors is 1170. Which Of the above Statements Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 Consider a 6-digit number Of the XYXYXY. The number is divisible by : form 27. **(a) (b) (c) (d)** 3 and 7 only 7 and 13 only 3, 13 and 37 only 3, 7, 13 and 37 JNBY.D-MTK n

371. **[CDS (I) Paper 28 Q25]** 26. 27. —A) **(c)** Of x ? **(a) (b) (c) (d)** , then What is the value 4 8 16 let m and n be natural numbers. What is the minimum value Of (m + n) such that 33m + 22n is divisible by 121 ? **(a) (b) (c)** 4 5 10 The product Of two numbers is 2160 and their HCF is

372. **[CDS (I) Paper 28 Q26]** 27. let m and n be natural numbers. What is the minimum value Of (m + n) such that 33m + 22n is divisible by 121 ? **(a) (b) (c)** 4 5 10 Consider a 6-digit number Of the XYXYXY. The number is divisible by : form **(a) (b) (c) (d)** 3 and 7 only 7 and 13 only 3, 13 and 37 only 3, 7, 13 and 37 The product Of two numbers is 2160 and their HCF is

373. **[CDS (I) Paper 28 Q28]** 29. The age of Q exceeds the age Of P by 3 years. SI The age of R is twice the age Of P and the age of Q is twice the age Of S. Further, the age difference of R and S is 30 years. What is the sum of the ages Of P and Q ? **(a) (b) (c) (d)** 35 years 38 years 39 years 45 years If a, b and c are the sides of a triangle ABC, then + — is always : **(a) (b) (c) (d)** Negative Positive Non-negative Non-positive There are four bells which ring at an interval Of 15 minutes, 25 minutes, 35 minutes and 45 minutes respectively. If all Of them ring at 9 A.M., how many more times will they ring together in the next 72 hours ? **(a) (c)** 2 JNBY-D-MTK Let a, b, c and d be four integers such that a + b + c + d =

374. **[CDS (I) Paper 28 Q32]** 11 -A) The number 9730 — 1480 is divisible by : **(b) (c) (d)** 37 but not 83 83 but not 37 Both 37 and 83 Neither 37 nor 83 Consider the following statements : I. loglo 50 is a rational number. 2. log 100 10 is an irrational number. Which of the statements given above Ware **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 The age of Q exceeds the age Of P by 3 years. Sl.

375. **[CDS (I) Paper 28 Q37]** 13-A) Consider the following statements : l. n — n is divisible by **(a) (b) (c) (d)** Effciency Of 13 women = Effciency Of 18 men = Efficiency Of Effciency Of 11 women men Emciency of Efficiency Of 13 women 17 men Effciency Of I I women = Efficiency Of

376. **[CDS (I) Paper 28 Q6]** 2. n — n is divisible by **(a) (b) (c) (d)** Effciency Of 13 women = Effciency Of 18 men = Efficiency Of Effciency Of 11 women men Emciency of Efficiency Of 13 women 17 men Effciency Of I I women = Efficiency Of

377. **[CDS (I) Paper 28 Q35]** 86. (13-A) Consider the following statements : l. n — n is divisible by **(a) (b) (c) (d)** Effciency Of 13 women = Effciency Of 18 men = Efficiency Of Effciency Of 11 women men Emciency of Efficiency Of 13 women 17 men Effciency Of I I women = Efficiency Of

378. **[CDS (I) Paper 28 Q45]** 15-A) Consider the following statements in respect of the polynomial I — x — xn + xn +1, Where n is a natural number : I. It is divisible by I — 2x + **(a) (b)** Consider the following statements in respect of the polynomial I — x — xn + xn +1, Where n is a natural number : I. It is divisible by I — 2x + 2. It is divisible by I — xn. Which of the statements given above Ware correct ? e 842 e 756 e 641 e 542 **(c) (d)**

379. **[CDS (I) Paper 28 Q2]** It is divisible by I — xn. Which of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 For What values of m, is mx2 + mx + + 9 a square ? **(c)** 9, 16 4, 16 and Y = 4a 4b then what is the value Of (x — a '2 — (y — ? **(a) (b) (c)** What is the difference between simple 43• 41. interest and compound interest on 10,000 for two years at 20% per annum compounded half-yearly ? 42. (15-A) **(a) (c)** Consider the following statements in respect of the polynomial I — x — xn + xn +1, Where n is a natural number : I. It is divisible by I — 2x +

380. **[CDS (I) Paper 28 Q73]** 74. A number 277XY5 (where X, Y are digits) is divisible by **(a) (b) (c) (d)** Choose this option if the Question can be answered by one of the Statements alone but not by the Other. Choose this option if the Question can be answered by either Statement alone. Choose this option if the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. Choose this option if the Question cannot be answered even by using both Statements together. Question : Is m > n, where m, n are nori-zero numbers ? Statement I : Statement 11 : Question : Can a circle be drawn through the points A, B and C ? Statement : Statement 11 : AB=5cm, BC AB=3cm, BC 4 em, A number 277XY5

381. **[CDS (I) Paper 28 Q25]** Question : What is the value Of X ? Statement I : Statement : JNBY-D-MTK The given number divisible by **(a) (b) (c) (d)** Choose this option if the Question can be answered by one of the Statements alone but not by the Other. Choose this option if the Question can be answered by either Statement alone. Choose this option if the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. Choose this option if the Question cannot be answered even by using both Statements together. Question : Is m > n, where m, n are nori-zero numbers ? Statement I : Statement 11 : Question : Can a circle be drawn through the points A, B and C ? Statement : Statement 11 : AB=5cm, BC AB=3cm, BC 4 em, A number 277XY5

382. **[CDS (I) Paper 28 Q75]** Question : What are the unique values Of a, b and c if 2 is a root of the equation ax2 + bx + c = O ? Statement : Statement 11 : Ratio of c to a is I. Ratio of b to a is 5/2). **(a) (b) (c) (d)** Choose this option if the Question can be answered by one of the Statements alone but not by the Other. Choose this option if the Question can be answered by either Statement alone. Choose this option if the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. Choose this option if the Question cannot be answered even by using both Statements together. Question : Is m > n, where m, n are nori-zero numbers ? Statement I : Statement 11 : Question : Can a circle be drawn through the points A, B and C ? Statement : Statement 11 : AB=5cm, BC AB=3cm, BC 4 em, A number 277XY5 (where X, Y are digits) is divisible by

383. **[CDS (I) Paper 28 Q78]** Question : Is p2 + q2 + q Odd, where p , q are positive integers ? **(a)** 24 **(b)** 36 **(c)** 48 **(d)** 60

384. **[CDS (I) Paper 29 Q4]** If a2 — bc = u, b2 — ac = B, c2 — ab = y, then what is equal to ? **(a) (b) (c) (d)** If (x — IP is a factor Of x4 + + ßx2 + — then the other factor will be : **(a) (b) (c)** A 2-digit number is such that the sum of the number and the number obtained by reversing the order Of the digits Of the number is

385. **[CDS (I) Paper 29 Q17]** If3X-1.33-x= equal to ? 6, then what is ¯ 1 + 23 ¯ x 18. **(a) (c)** If 4 x + y = 2aa, then What is x — y equal to ? - 2b3 **(a) (b)** — 2ab3 2b3 **(c)** 2ab3 **(d)** Which one of the following is a factor of + 2v'iy3 — 18xy+ ? **(c) (d)** 3x2 + 2y2 — — 3x2 * 2y2 + + Jiiy — v'Gxy + 6 JNUP-T-EMT

386. **[CDS (I) Paper 29 Q20]** What is number Of digits in the expansion of 125100? 2=0-301) **(a) (b) (c)** 69 70 209 210 What is the HCF of acxa bcx2 adx2 + acdx + bdx * bcd and adxa + acx2 + + bex + aedx + bed if **(a) (b) (c) (d)** bx + c cx + d ax+ b If xn — pyn + qzn is divisible by + abyz — bzx — axy, then what is — — — bn a **(a) (b) (c)** 0

387. **[CDS (I) Paper 29 Q28]** 24. Consider the following statements : **(a) (b) (c)** I only 2 only Both I and 2 Neither I nor 2 **(d)**

388. **[CDS (I) Paper 29 Q50]** 12 15 20 Consider the following statements : **(a)** 1, ' **(b) (c)** 10 50. 12 15 20 Consider the following statements : 2. One of the roots Of the equation is always less than I if a, b and c are all positive. One Of the roots of the equation is always negative if a, b and c are all negative. What is the median **(d)**

389. **[CDS (I) Paper 29 Q5]** Moreover, none of them is more than 40 kg. **(a)** 15 **(b)** 16 **(c)** 18 JNUP-T-EMT In a triangle ABC, angle B = 900 and p is the 87. 84. length Of the perpendicular from B to AC. If BC IO cm and AC = 12 em, then what is the value Of p ? 5Jfi **(d)** 1 oJii

390. **[CDS (I) Paper 29 Q95]** 96. Question : What is the 3-digit number which is divisible by IO ? If the digits in hundred's place and ten's place Of the number are interchanged, the resulting number is diminished by **(a)** if the question can be answered by using one Of the statements alone, but cannot be answered using the other statement alone. Mark option **(b)** if the question can answered by using either statement alone. Mark option **(c)** if the question can be answered by using both the statements together, but cannot be answered using either statement alone. Mark option **(d)** if the question Cannot be 94. answered even by using both the statements

391. **[CDS (I) Paper 29 Q96]** Question : What is the 3-digit number which is divisible by IO ? If the digits in hundred's place and ten's place Of the numi:ær are interchanged, the resulting number is diminished by **(a)** if the question can be answered by using one Of the statements alone, but cannot be answered using the other statement alone. Mark option **(b)** if the question can answered by using either statement alone. Mark option **(c)** if the question can be answered by using both the statements together, but cannot be answered using either statement alone. Mark option **(d)** if the question Cannot be 94. answered even by using both the statements

392. **[CDS (I) Paper 29 Q97]** 98. Question : Is average Of the largest and the smallest of 4 given numbers greater than the average Of the 4 numbers ? The difference between the largest and the second largest numbers is less than the difference between the second smallest and the smallest Of the numbers. Statement •11 : The difference between the largest and the IO smallest numbers is greater than the difference between the second largest and the second smallest Of the numbers. Question : Is (a — b + c) > (a + b — c), where a, b and c are real numbers ? Statement-I : b is negative. c is negative. JNUP-T.EMT (35-A e **(a)** 11 **(b)** 13 **(c)** 17 **(d)** 19

393. **[CDS (I) Paper 29 Q99]** Question : What is the cost of 15 pens, 21 pencils and 18 note books ? The cost of 7 pens, 6 pencils and 5 note books is **(a)** 15° **(b)** 30° **(c)** 45° **(d)** 60°

394. **[CDS Paper 30 Q17]** Consider the following data : 110, 41, 43, 95, 127, 99, 61, 92, 71, 93, 110, **(a) (b)** 4 6 3. old median is I. The difference between new Old mean is less than 0-1. The difference between new old mode is zero. mean mode above Which of the statements given correct ? and and are 19. When the number **(c) (d)**

395. **[CDS Paper 30 Q18]** 19. 20. -A) If n is a natural number, then What is the sum Of all distinct remainders Of 4n + 6n 90 + 11 n when divided by IO for various values Of n ? **(a) (b) (d)** 4 6 When the number (12345678910111213 99100) is divided by 16, what will be the remainder ? **(b) (d)** 15 12 4 3 A, B, C, D can complete a work in 3, 6, 9, 12 hours respectively. Further, only one can work at a time in each hour and nobody can work for two consecutive hours. It is not necessary to engage all. What is the minimum number of hours that they Will take to finish the work ? **(c)** 36/25 12/5 4 2

396. **[CDS Paper 30 Q19]** When the number (12345678910111213 99100) is divided by 16, what will be the remainder ? **(a) (b) (c) (d)** I and 2 only 2 and 3 only I and 3 only I, 2 and 3 What is the digit at the 100th place Of number (225)40 ? **(b) (d)** 15 12 4 3 20. **(b) (c)** 6 5 4 If a, b, c, d are natural numbers, then how many possible remainders are there when Ia + 2b + 3c + 4d is divided by IO ? A, B, C, D can complete a work in 3, 6, 9, 12 hours respectively. Further, only one person can work at a time in each hour and nobody can work for two consecutive hours. It is not necessary to engage all. What is the minimum number of hours that they Will take to finish the work ? **(a) (b) (c)** 4 5 **(c)** 36/25 12/5 4 2 SRSU-T-EMT

397. **[CDS Paper 30 Q27]** If HCF o/ 768 and x6y2 is 32xy for natural numbers x 2 2, y 2, then What is the value of **(a) (b) (c)** 2a **(d)**

398. **[CDS Paper 30 Q2]** 3 37 Select the correct answer using the code given **(a) (b) (c) (d)** I and 2 only 2 and 3 only I and 3 only I, 2 and 3 Consider the following statements : I. If (3m3 2m2 + 5m + n) / m is not an integer, where m and n are integers, then n is not divisible by m.

399. **[CDS Paper 30 Q30]** The sum of two positive numbers is **(a) (b)** 12 18 24 28 **(c) (d)**

400. **[CDS Paper 30 Q63]** Let a, b, c and d be positive integers. Question : Statement -I : Statement-II : Which one Of a, b, e, d is closest to the product abcd ? c is not the smallest. Let mn = k, where m and n are prime numbers and k is an even number. Question : Statement •I : Statement-II : Question : Statement •I : Statement •11 : What is the value Of mn — n + I ? One of the numbers is **(a) (b) (c) (d)** If the Question can be answered by one Of the Statements alone, but not by the 65. If the Question can be answered by either Statement alone. If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. If the Question cannot be answered even by using both Statements together. 66. Statement-II : y z Let a, b and c be the sides of a triangle ABC. Question : Statement •I : Statement-II Is the triangle equilateral ? a2 + b2 + c2

401. **[CDS Paper 30 Q67]** One of the numbers is **(a) (b) (c) (d)** If the Question can be answered by one Of the Statements alone, but not by the 65. If the Question can be answered by either Statement alone. If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. If the Question cannot be answered even by using both Statements together. 66. Statement-II : y z Let a, b and c be the sides of a triangle ABC. Question : Statement •I : Statement-II Is the triangle equilateral ? a2 + b2 + c2

402. **[CDS Paper 30 Q3]** 01 o 02 What is AB BC equal to ? **(a)** IO cm **(b)** 12 cm **(c)** 13 cm **(d)** 14 cm

403. [CDS (I) Paper 31 Q1] A real number x is such that the sum of the number and four times its square is the least. What is that number ? **(a) (b) (c)** 100 -0-125 **(d)** 0125 2. The difference of the square of two natural numbers m and n

404. [CDS (I) Paper 31 Q2] The difference of the square of two natural numbers m and n (m > n) is **(a) (b) (c)** 100 -0-125 **(d)** 0125 2. The difference of the square of two natural numbers m and n

405. [CDS (I) Paper 31 Q72] How many pairs of natural numbers will satisfr• ? **(a) (b) (c)** 100 -0-125 **(d)** 0125 2. The difference of the square of two natural numbers m and n

406. [CDS (I) Paper 31 Q3] Let N be a 5-digit number. When N is divided by 6, 12, 15, 24 it leaves respectively 2, 8, 11, 20 as remainders. What is the greatest value of N ? **(a)** 99960 **(b)** 99956 **(c)** 99950 **(d)** 99946

407. [CDS (I) Paper 31 Q4] What is the remainder when 111222 + 222333 + 333444 is divided by 5 ? -3 S. What are the last digits in the multiplication of 4321012345 x 98766789 ? 6. p varies directly as (x-2 + Y2 + zz). When 3, thenp—70. What is the value of p when x —1, y = 1, **(a)** 100 **(b)** 125 **(c)** 135 **(d)** 140

408. [CDS (I) Paper 31 Q7] Let N be the least positive multiple of 11 that leaves a remainder of 5 when divided by 6, 12, 15, **(a) (b) (c)** 100 -0-125 **(d)** 0125 2. The difference of the square of two natural numbers m and n

409. [CDS (I) Paper 31 Q24] What is the sum of the largest and the smallest 4-digit numbers made by using single digit prime numbers (without repetition) ? **(a)** 7887 **(b)** 7997 **(c)** 8998 **(d)** 9889

410. [CDS (I) Paper 31 Q25] What is the remainder when 3255 is divided by 28 ? **(b)** 11 **(c)** 24 **(d)** 27 **(a)** nM — k **(b)** n **(c)** 81 **(d)** 105

411. [CDS (I) Paper 31 Q26] What is the value of x 8) if (10097 + 10054 + x + 1) leaves a remainder O when divided by 9 ? A - RAKU-T-EMT **(a)** nM — k **(b)** n **(c)** 81 **(d)** 105

412. [CDS (I) Paper 31 Q28] Let N 12345678AB be a Io-digit number, where A, B are digits. If N is divisible by 9, then which of the following statements is/are correct ? I. (A + B) is divisible by 9 **(a)** 16 **(b) (c) (d)**

413. [CDS (I) Paper 31 Q30] If I lx + 5y is a prime number where x, y are natural numbers then what is the minimum value of (x + y) ? **(a)** 16 **(b) (c) (d)**

414. [CDS (I) Paper 31 Q34] In a class containing 200 students, n students prefer both tea and coffee; 2n students prefer coffee, 3n students prefer tea; 4n students prefer neither tea nor coffee. What is the value of n ? **(a)** 20 **(b)** 25 (C) 30 **(d)** 35

415. [CDS (I) Paper 31 Q36] Let n be a natural number. The HCF of n, IO is **(a)** 20 **(b)** 25 **(c)** 30 **(d)** 35 35. 36. Let ABC be a triangle with area 36 square cm. If AB=9 cm, BC' 12 cm and LABC = O, then what is cos9 equal to ? 4 3 2 Let n be a natural number. The HCF of n, n +10 is 10. If the LCM is x

416. [CDS (I) Paper 31 Q47] Let k be a positive integer. What is the quotient when xu•3 + xu•6 + xu•9 + xu+12 is divided by (1 + + ? **(c)** xu•2 **(a)** xu•2 48. A square is drawn inside a square of side 14 cm in such a way that the corners of the inner square coincide with the mid points of the sides of the outer square. What is the area lying between the two squares ? **(b)** 98 square cm **(c)** 56 square cm **(d)** 49 square cm

417. [CDS (I) Paper 31 Q51] What is tane equal to ? 2 3 **(a)** xu•2 48. A square is drawn inside a square of side 14 cm in such a way that the corners of the inner square coincide with the mid points of the sides of the outer square. What is the area lying between the two squares ? **(b)** 98 square cm **(c)** 56 square cm **(d)** 49 square cm

418. [CDS (I) Paper 31 Q53] Consider the following statements : I. The line segment AP divides the area of the triangle ABC into two equal parts **(a)** I and 11 only **(b)** 11 and Ill only **(c)** I and Ill only **(d)** 1, 11 and 111 Consider the following for the next two

419. [CDS (I) Paper 31 Q58] What is OQ equal to ? **(a)** 5 cm **(b)** 6 cm 59. What is PM equal to ? **(c)** IO cm **(d)** 12 cm

420. [CDS (I) Paper 31 Q59] What is PM equal to ? **(a)** 5 cm **(b)** 6 cm 59. What is PM equal to ? **(c)** IO cm **(d)** 12 cm

421. [CDS (I) Paper 31 Q11] Consider the Question and the Statements and mark the correct option. : 1s327"+ 173" divisible Question by 500? : n is odd natural num- Statement-I n is a positive integer Statement-II : Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using any of the 25 25

422. [CDS (I) Paper 31 Q2] What is the average of these integers ? • The sum of all the 5 integers is a multiple of 5 : The number of odd integers is odd 27 Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone **(b)** •Ihe Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements A - RAKU-T-EMT

423. **[CDS (I) Paper 31 Q11]** Consider the Question and the Statements and mark the correct option. nere are three dif- Question : ferent weights. All the weights are integers and their sum is a prime number. What are the weights ? : One of the weights is Statement-I twice the another weight One of the weights is Statement-II : thrice the another weight Which one of the following is correct in respect of the above Question and the Statements ? **(a) (b) (c) (d)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question cannot be answered even by using both the Statements together 29

424. **[CDS (I) Paper 31 Q73]** What is the remainder if we divide 310 by 7? **(a) (b) (c)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question cannot be answered even by using both the Statements together **(d)** 800

425. **[CDS (I) Paper 31 Q75]** The difference of 1031 — 5 and 1030 +p is divisible by 3 where p is a digit. How many values of p are possible ? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 77. The average of the temperatures recorded at noontime from Monday to Sunday is 31 oc. Ifthe lowest tempera- ture recorded is 300C, then what is the maximum of temperature that is possible to record at noontime on any one of the days

426. **[CDS (I) Paper 31 Q76]** Consider the following statements : I. 61 divides 107100 -76100 **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 77. The average of the temperatures recorded at noontime from Monday to Sunday is 31 oc. Ifthe lowest tempera- ture recorded is 300C, then what is the maximum of temperature that is possible to record at noontime on any one of the days

427. **[CDS (I) Paper 31 Q11]** 100 divides 675 + 335 Which of the statements given above is/ are correct ? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

428. **[CDS (I) Paper 31 Q79]** Consider the following statements in respect of where n is a natural number : I. p is always (Xid 11. p is a perfect square Which of the statements given above is/ are correct ? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 A - RAKU-T-EMT

429. **[CDS (I) Paper 31 Q80]** What is the difference between the average of first 50 even natural numbers and the average of first 50 odd natural numbers ? **(b) (a)** 81. Three amounts x, y, z are such that y is the compound interest on x; and z is the compound interest on y. The rate of interest per annum and the time period in years are same. Which one of the following is correct ? **(b)** Y2 = zx **(c)** z2 — xy 82. There are n concentric squares. The area of the innermost square is 1 unit and the distance between corres- ponding comers of any two consecu- tive squares is 1 unit. Consider the following statements : I. The diagonal of the nth square is 11. The area included between nth square and **(d)** I only

430. **[CDS (I) Paper 31 Q88]** If + px2 + qx + r is an integer for all integral values of x, then consider the following statements : I. p must be an integer 11. q must be an integer Ill. r must be an integer Which of the statements given above is/are correct ? **(a)** I and 11 only **(b)** Ill only **(c)** I, Iland Ill **(d)** None of the statements is correct A - RAKU-T-EMT

431. **[CDS (I) Paper 31 Q96]** The number 199 can be written as m2 — n2, where m, n are natural numbers (m > n). What is the value of mn ? **(a)** 9900 **(b)** 9800 **(c)** 9701 **(d)** Cannot be uniquely determined

432. **[CDS (I) Paper 31 Q99]** Let S — 7b + IIC +13d, where a, b, c and d are natural numbers. What is the number of distinct remainders of S when it is divided by 10 ? **(d)** More than 5 **(a)** 1100 **(b)** 1200 **(c)** 1300 **(d)** 1400 96. The number 199 can be written as m2 — n2, where m, n are natural numbers

433. **[CDS (I) Paper 32 Q9]** What is the correct sequence Of the following 11. salts in ascending Orcier according to their presence in the sea water ? I. Calcium Carbonate **(a) (b) (c)** Which of the following statements With reference to Humus Ware correct ? I. It is a dark coloured amorphous 12. substance. 2. Being colloidal in nature , it senes as a reservoir of nutrients. Select the answer using the code given below : Which Of the following statements With reference to Easter Island, a UNESCO World Heritage Site, Ware correct ? I. It is part Of Polynesia group Of is lands. 2. It is a territory of the South American nation Chile. Select the answer using the code given below : **(d)**

434. **[CDS (I) Paper 32 Q2]** Calcium Sulphate **(a) (b) (c)** Which of the following statements With reference to Humus Ware correct ? I. It is a dark coloured amorphous 12. substance. 2. Being colloidal in nature , it senes as a reservoir of nutrients. Select the answer using the code given below : Which Of the following statements With reference to Easter Island, a UNESCO World Heritage Site, Ware correct ? I. It is part Of Polynesia group Of is lands. 2. It is a territory of the South American nation Chile. Select the answer using the code given below : **(d)**

435. **[CDS (I) Paper 32 Q3]** Magnesium Chloride Select the answer using the code given below : **(a) (b) (c)** Which of the following statements With reference to Humus Ware correct ? I. It is a dark coloured amorphous 12. substance. 2. Being colloidal in nature , it senes as a reservoir of nutrients. Select the answer using the code given below : Which Of the following statements With reference to Easter Island, a UNESCO World Heritage Site, Ware correct ? I. It is part Of Polynesia group Of is lands. 2. It is a territory of the South American nation Chile. Select the answer using the code given below : **(d)**

436. **[CDS (I) Paper 32 Q18]** Barrier Reef from the South Pacific. **(a) (b) (c) (d)** Bangkok Hong Kong Macau Singapore Which of the following statements With reference to saline soil in India Ware correct ? I. They are rich in nitrogen. 2. They

437. **[CDS (I) Paper 32 Q32]** Which one among the following motifs does not appear on Harappan seals ? A light bulb rated as 60 W at 220 V has a potential difference of 110 V across its ends. The B)wer dissipated in this light bulb is : **(a) (b) (c) (d)** Brahmacharya Samyag-darshana Sunrita Asteya

438. **[CDS (I) Paper 32 Q38]** 35. the Ajivikas is not correct ? **(a) (b) (c) (d)** They did not practise discrimination on the basis of caste or class. They had a corporate organisation and canonical texts. They practised severe asceticism, Often eating very little food. They formed an alliance With Jainas against Buddhists. 39.

439. **[CDS (I) Paper 32 Q45]** 46. Which one of the following statements about the Buddhist Goddess Hariti is correct ? **(a) (b) (c) (d)** Ayurveda Vyakarana Ganita Jyotisha

440. **[CDS (I) Paper 32 Q51]** 47. rulers Of Kashmir, in the time of Zainul Abidin ? **(a) (b) (c) (d)** Bhattavatara Jonaraja Shrivara 52. Vachaspati

441. **[CDS (I) Paper 32 Q87]** Which, amongst the following, is the latest 90. language to be included in the Eighth Schedule Of the Constitution Of India ? **(a) (b) (c) (d)** Maithili N epali Konkani Manipuri Which of the following statements Ware correct as the Constitution of India ? The Constitution

442. **[CDS (I) Paper 32 Q98]** 101. (30 —A) 99. ? **(a) (b)** 7å1,7tt2 100. 103. **(c)** $ZüZ **(d)**

443. **[CDS (I) Paper 32 Q100]** 103. (firewall) **(a)** $ZüZ **(d) (a)** (Spreadsheet) **(b)** (AutoCAD) **(c)** (MS PowerPoint) (d) (Linker) (Malicious software) ? **(b)** (Password Cracking) **(c)** (Trojan Horse) **(d)** (Firewall)

444. **[CDS (I) Paper 32 Q100]** Consider the following statements With regard to the setting up of the Backward Class **(a) (b) (c) (d)** Poland Ro mania Lithuania Belarus

445. **[CDS (I) Paper 32 Q4]** Shri Gukesh D Shri Harmanpreet Singh Shri Praveen Kumar Ms. Manu Bhaker **(a) (b) (c) (d)** The college students studying in the undergraduate programme in the age group Of 19 — 25 years. The research scholars in recognized research laboratories Of national importance. The college teachers of the country. The school children studying in classes 6 — IO in the age group Of IO — 15 years. 'The Order Of Mubarak Al-Kabeer' is the highest national award of :

446. **[CDS (I) Paper 33 Q7]** Let x each How and y natural numbers, less than 20, such that x , y, and x — y are prime numbers. many such combinations Of x + y, x — y) are possible? One Two Three **(d)** None If (x + — l, then What is the value of p? Cannot be determined If then what is — (2 — equal to? **(b)** 0-5 **(d)** 1-5 If — + _ — what is 5 6 equal to? **(a)** 31/216 **(b)** 35/216 **(c)** 37/216 **(d)** 41/216 13 P.T.O. I. Let p(x) a polynomial. When Mx) is divided by (x — I), it leaves 2 as the remainder. When p(x) is divided by (x — 2b it leaves as the remainder. What is the remainder when nx) is divided by (X — — ? **(d)** 3-2* Consider the following in Of a 2. positive real number x

447. **[CDS (I) Paper 33 Q8]** 9. 10. **(b)** 106 **(c)** 107 **(d)** 108 What is the remainder when x6 divided by x 2 + ? is **(b)** 0 (x is a factor of which one of the following? **(a) (b) (c) (d)** 4 — 3x3 + 8x2 — 3x3 + 8x2 +4x4 —4x4 + 3x X X +8x2 4 — 3x3 - 12 -14x+12 - 12 +14X+12

448. **[CDS (I) Paper 33 Q17]** Let x = n (n + where n is an even natural number. Which of the following statements is/ are correct? I. x is always divisible by **(a) (b)** Either p O or p None Of the above 15. If n is natural numirr less than 7, then What is the number Of values Of n for which **(c) (d)** ANFB-T-EMT/4SA 7 17. Let x = n

449. **[CDS (I) Paper 33 Q48]** 11. x 2 is always divisible by **(a) (b)** Either p O or p None Of the above 15. If n is natural numirr less than 7, then What is the number Of values Of n for which **(c) (d)** ANFB-T-EMT/4SA 7 17. Let x = n

450. **[CDS (I) Paper 33 Q18]** 19. What is the LCM of x4.x2y2+Y4, x3y•ty4 and x4y2 —x3y3? **(a)** x3y3(x6 — y6) **(b)** X3y2(x6 — y6) **(c)** x3y(x6 — y6) **(d)** .g(x6 — y6) Let be a 3-digit number. Let D be the difference between XYZ and ZYX. What is the remainder when D is divided by 99? 7 9 P.T.O.

451. **[CDS (I) Paper 33 Q19]** Let XW be a 3-digit number. Let D be the difference between XYZ and ZYX What is the remainder when D is divided by 99? 7 9 P.T.O. **(a) (b)** Either p O or p None Of the above 15. If n is natural numirr less than 7, then What is the number Of values Of n for which **(c) (d)** ANFB-T-EMT/4SA 7 17. Let x = n

452. **[CDS (I) Paper 33 Q20]** Let p and q be two natural numbers such that (p + is divisible by **(a)** 12 21. What is c2 + bc—a2 —ab equal to? **(b)** -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ? **(c)** 2p **(d)** then what is then what is x 22. If ab ba , then what is equal to?

453. **[CDS (I) Paper 33 Q30]** 4, 6, 7 or 9, it leaves 3 as remainder. What is the smallest 4-digit number that satisfies this property? **(a)** 40 If **(b)** 18 **(c)** 322 **(d)** 72VS ANFB-T-EMT A number N is such that when divided by 4, 6, 7 or 9, it leaves 3 as remainder. What is the smallest 4-digit number that satisfies this property?

454. [CDS (I) Paper 33 Q32] If 4, Where O then what is cosece equal to? **(b)** 3/2 5/3 **(a)** 3/2 5/3 33. If 2 tane=sec29 —2, where R 12, then what is cote equal to? **(b)** v'ä-l **(c)** 0+2 I — sin 34. What is **(d)** mane

455. [CDS (I) Paper 33 Q39] Which Of the following is/ are the factor(s) of(3x+ y) 2 + (3x + 5y) + 5y)2? **(a)** 11 only **(b)** Both I and 11 **(c)** Neither I nor 11 39. Which Of the is/ are the Of **(d)** Both I and 11

456. [CDS (I) Paper 33 Q41] What is tanß equal to? p.T.O. **(a)** 11 only **(b)** Both I and 11 **(c)** Neither I nor 11 39. Which Of the is/ are the Of **(d)** Both I and 11

457. [CDS (I) Paper 33 Q47] What is Wc e equal to? **(c)** 1 P.T.O. **(a)** mn — I O For the following two **(b)** 2/3 For the following tvo **(c)** 1 P.T.O. For the following two **(d)** 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

458. [CDS (I) Paper 33 Q70] Consider the following angles : l. 40 50 **(a) (b)** 6-4 m 67. ABC is a triangle right anøed at B. Further , **(c) (d)** 75 units 100 square units 125 square units 150 units 68. TWO poles are situated 24 m apart and their heights differ by I O m. What is the distance between their tips?

459. [CDS (I) Paper 33 Q11] Consider the Question and the Statements. Question : What is the remainder when — y2n I is divided by xn + yn , where n is a natural number? Statement—I : n is odd. Statement—II : n is even. Which one Of the following is correct in respect Of the above Question and the S tatements? **(b) (c) (d)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any Of the Statements ANFB-T-EMT/45A 31 **(a) (b) (c)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any Of the Statements ANFB-T-EMT/45A **(d)**

460. [CDS (I) Paper 33 Q11] Consider the Question and the Statements. Question : The last digit in the expansion of the number is I. What is the value Of the digit D ? D is a multiple Of **(a)** The Question can be answered by using One of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be an swered by using tx'th the Statements together, but cannot be answered using either Statement alone **(d)** The Question Can be answered even without using any of the ANFB-T-EMT/45A 83. ZB- ZC 84. A Question is given followed by Statements I and 11. Consider Question and the Statements. In a triangle ABC, ZA Is angle A acute? two the Statement—I ABC is Statement—II : not an Obtu se-angled Angle C is acute. Which One of the following is correct in respect of the above Question and the Statements?

461. [CDS (I) Paper 33 Q13] Statement—II : The length Of BD is an even integer. Which one Of the following is correct in respect Of the above Question and the Statements ? **(a)** The Question can be answered by using one of the Statements alone, but cannot be ansqered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using any of the Statements P.T.O.

462. [CDS (I) Paper 33 Q97] What is the value of u ? **(c)** 3 **(a)** 20 **(b)** 25 For the following tvo **(c)** 3 98. What is the value of v? 5 8 ANFB-T-EMT/4SA 99. What is the value of k ? 22 30 100. What is the mean Of the distribution? 21-625 22.225 **(d)** 23 225 24 • 625 P.T.O. For the following tvo

463. [CDS (I) Paper 33 Q99] What is the value Of k ? 22 30 **(a)** 20 **(b)** 25 For the following tvo **(c)** 3 98. What is the value of v? 5 8 ANFB-T-EMT/4SA 99. What is the value of k ? 22 30 100. What is the mean Of the distribution? 21-625 22.225 **(d)** 23 225 24 • 625 P.T.O. For the following tvo

464. [CDS (I) Paper 34 Q11] Seven times a two-digit positive number is equal to four times the number obtained by reversing the order of the digits. How many such two-digit numbers are there ? **(a)** x4 — 2a2x2 + a4 **(b)** x4 + 2a2x2 + a4 **(c)** xa — a2x2 4 a4 8. Which of the following expressions can divide both the polynomials x3+2x2—5x+2 and x3 +4x2+x—6 exactly ? 11. X+l 111. X+2 Select the correct answer using the code given below : **(d)** I and 11 only

465. [CDS (I) Paper 34 Q21] 2727-940-379 is divisible by how many natural numbers less than 10 ? **(d)** More than 3 A - KPRS-T-MTE **(a)** 10 minutes **(b)** 12 minutes **(c)** 1 S minutes **(d)** 20 minutes 18. A person goes downstream at x km/h and upstream at y km/h by the same boat. What is the ratio of speed of boat in still water to speed of •water ? Y 19. Let Q=x2 + bx + c. If the sum of the roots is equal to product Of the roots of the equation Q = 0, then which of the following statements is/are correct ? 20. 21. 11. Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below :

466. [CDS (I) Paper 34 Q23] The HCF and LCM of two numbers p and q are 44 and 4620 respectively. When p is divided by 55, the quotient is 4 and remainder is **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 23. The HCF and LCM of two numbers p and q are 44 and 4620 respectively. When p is divided by 55, the quotient is 4 and remainder is 0. What is the value of q ?

467. [CDS (I) Paper 34 Q24] Let N be the greatest number that will divide 600, 631 and 724, leaving the same remainder. What is the value of N? **(a)** 41 **(b)** 37 **(c)** 33 **(d)** 31 11

468. [CDS (I) Paper 34 Q28] Consider the following statements : I. Two consecutive natural numbers are always co-prime. **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11 29. If and then what is log10

469. [CDS (I) Paper 34 Q31] Let P = N2 where N is an odd integer. What is the remainder when P is divided by 8 ? **(d)** Cannot be determined due to insufficient data **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11 29. If and then what is log10

470. [CDS (I) Paper 34 Q60] The minute hand of a clock is 15 cm long and sweeps an area of 15" cm on the dial of the clock. How much angle does it describe during this period ? **(a)** 120 **(b)** 180 **(c)** 240 **(d)** 300

471. [CDS (I) Paper 34 Q11] Consider the Question and the Statements and mark the correct option. Question : Statement-I Statement-II : Is (p2 + q2) always composite number, where p and q are different prime num- bers ? (p —q) is an odd integer (p + q) is an odd integer Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements 35 35

472. [CDS (I) Paper 34 Q83] What is 25 16 16 19 4 4 **(a)** 487 **(b)** 483 **(c)** 481 **(d)** 480 85. What

473. [CDS (I) Paper 34 Q85] What (X2 — Y2) equal to ? **(d)** 8800 **(a)** 487 **(b)** 483 **(c)** 481 **(d)** 480 85. What

474. [CDS (I) Paper 34 Q93] What is tang equal to ? (C) pq **(a)** 15° **(b)** 30° **(c)** 45° **(d)** 60°

475. [CDS (I) Paper 34 Q95] What is tan2a equal to ? **(b) (a)** 96. What is cot2ß equal to ? p For the next two **(b)** 10 **(c)** 13 For the next two **(d)** 2pq — —

476. [CDS (I) Paper 35 Q2] 3. = k, then what is q3 + r3 83 equal to ? **(a) (b)** p q 2. 3. 4. 5. 6. Suppose p and q are positive integers where q is not a perfect square. If is a root of a quadratic equation having integer coemcients, then which Of the following statements Ware correct ? l. The sum of the roots is 2p times the product Of the roots. IL The sum Of the reciprocals Of the roots is equal to 2p. Select the answer using the code given below : data q = k, then what is **(c) (d)**

477. [CDS (I) Paper 35 Q12] 13. If , then What x + k x +2k x +5k ¯ k is the solution of the equation ? **(a)** 3 k 2k Consider the polynomial p(x) x5 + 2x4 + x3 —x2 — 2x I : l. x2 + 2x + I is a factor of p(x). 11. + x 1 divides Which of the statements given above Ware correct ? **(a) (b) (c) (d)** I only only Both 11 Neither I nor 11 If 5 is a root of the equation x2 + px — 55 = O and the quadratic equation x2 + px + k = O has equal roots, then what is the value of k ? **(a) (b) (c) (d)** 6 9 12

478. [CDS (I) Paper 35 Q15] 16. 17. (7-A **(a) (b) (c) (d)** 2 •5 knvhr 2 • 75 km.'•hr 3 knvhr 3 •25 km/hr If the roots of the equation mx n = O are increased by the same quantity k, then they become the Of the equation x2 + nx + m O. What is the value Of (m + n) ? **(c) (d)** 2 4 If a and are the rt»ts of the equation 2x2 — + I)x + (112 + n + I) O, then What is (2a3 + 2ß3) equal to ? **(b) (c) (d)** - (2na + 1) The product Of two numbers is 1050. The quotient when the larger number is divided by the smaller number is 4 and the remainder is IO. What is the sum Of the two numbers ? **(a) (b) (d)** 70 75 80 85 BFVS-T-TME

479. [CDS (I) Paper 35 Q11] The set Of all re al numbers between O and IO is a finite set. Which Of the statements given above Ware correct ? **(a) (b) (c)** 6% 8% 31. 32. 33. Consider the follmving statements : I. The set Of all birds living on the Earth is an infinite set. 11. The set Of all re al numbers between O and IO is a finite set. Which Of the statements given above Ware correct ? **(d)**

480. [CDS (I) Paper 35 Q98] Consider the following discrete grouped data : **(a)** 1369 13-39 13-19 1309 and Environment Transport and C ommunication Rural Developme nt Industry and Minerals Agricultural Services 3,000 1,500 3,800 2,500 Consider the following incomplete frequency No. 2 4 Which of the following statements Ware distribution with a total frequency Of 100 : correct ? Class Frequency 10 20 l. 11. The percentage investment in the 'Science, Technol **(b) (c) (d)**

CHAPTER 02

Total Archive Questions: 65 Questions

HCF, LCM & LOGARITHMS

Typical Exam Weightage: 5 - 8 Questions (5 - 8 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 2 Q25] 26. Students of a class are made to stand in rows. If one student is extra in a row, there would be two rows less. If one student is less in a row, there would be three rows more. What is the number of students in the class ? **(a) (b) (c)** 65 55 60 50 What is the LCM of 3 (a3 - b3) 4 _ b4) ? 11 (a and **(a) (b) (c) (d)** 33 (aa — b3) (a2 + b2) (a + b) 33 (a —b) (a2 + b2— ab) 33 (a3 - b3) - b4) A number consists Of two digits, whose sum is IO. If 18 is subtracted from the number, digits Of the number are reversed. What is the product of digits in the number ? **(c) (d)** 15 18 24 32 If the HCF of x3 — 27 and + 4x2 + 12x + k is a quadratic polynomial, then what is the value of k ? **(c)** 27 9 3
2. [CDS (I) Paper 3 Q14] What is the value of (4 log,0125-2 log,04+log1032+ log,ot) 15. .14+4" is divisible by which One Of the following ? **(a) (b) (c) (d)** 4 (P + + 2y2) (xz — 2y2) None of above (For the next two questions to follow) : •me average age of 06 persons living in a. house is 23-5 years. Three of them are majors and their average age is 42 years. The difference in ages of the three minor children IS same.
3. [CDS (I) Paper 4 Q8] 2 What •is the HCF 4x3 + 3x2y—9xy2 uy3 **(a)** x—2y of and third side in cm? **(b)** g, 2v'fi **(c) (d)**
4. [CDS (I) Paper 4 Q7] 8. What •is the HCF 4x3 + 3x2y—9xy2 uy3 **(a)** x—2y of and third side in cm? **(b)** g, 2v'fi) **(c)** (442 **(d)** (44ä, 21,110)
5. [CDS (I) Paper 5 Q43] 80 60 50 If the HCF of three numbers 144, x and 192 is 12, then the number x cannot be 38 litres of milk was poured into a tub and the tub was found to be 5% empty. To completely fill the tub, what amount of additional milk must be poured ? **(a) (b) (c)** In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% candidates failed in both English and Mathematics, what percentage Of candidates passed in both the subjects ? 300 g 250 g 200 g None Of the above **(d)**
6. [CDS (I) Paper 6 Q4] Rs. 240 Rs. 480 Rs. 800 are in the ratio of 12 : 23. second number ? **(a)** 44 **(b)** 55 **(c)** 66 **(d)** 77 What is —I? the HCF x3 _ of the 3x2 + **(d)** None of the above -7Y+10 3x What is the polynomials 1 and Ilx +4y —10 9
7. [CDS (I) Paper 6 Q78] What is the LCM Of 9 15 7 14 3 2 3 **(a)** u = 2•25v **(b)** u = 3v **(c)** 3v < u < 4v 80. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? **(d)** 256
8. [CDS (I) Paper 6 Q89] 90. +log.o — + **(b)** 280 **(d)** 52 number ? **(a)** 161 **(b)** 171 **(c)** 173 **(d)** 221 What is **(a)** 16 What is log,o — + log,o ¯ up to 8 terms equal to ? equal to ? log105 None of the above 380 c In the figure given above, Z B = 380, AC BC and AD = CD. What is Z D equal to ? **(a) (c)** 260 380
9. [CDS (I) Paper 6 Q90] 91. Which one of the following is a prime number ? **(a)** 161 **(b)** 171 **(c)** 173 **(d)** 221 What is **(a)** 16 equal to ? What is log,o — + log,o ¯ + log.o — + up to 8 terms equal to ? **(a) (c) (d)**
10. [CDS (I) Paper 7 Q5] 6. Which one among the following statements is correct ? **(a)** Simple bar- diagrams are those diagrams which show two characteristics of the data **(b)** In pie diagrams all the items are converted into angles **(c)** A bar diagram is one in which data are shown in terms Of bars **(d)** Bar diagrams present data through length and breadth Ihe following characteristics relate to the. persons participating in athletic events : I. Height Of the person
11. [CDS (I) Paper 7 Q41] 42. equal to ? **(a)** 5/4 **(b)** 6/5 **(c)** 25/31 **(d)** 31/42 The sum of two numbers is 232 and their HCF is
12. [CDS (I) Paper 8 Q8] If 18 is added to the number, the digits are reversed. The number is equal to **(a)** 26 **(b)** 35 **(c)** 53 **(d)** 62 If log, o a = p and loglo b = q, then what is loglo (aP bq) equal to ? **(a)** P2 + q2 **(c)** p2q2 q
13. [CDS (I) Paper 9 Q2] Which Of the above are ? and 2 only Z and 3 only 3 •ancy iB the HCF of the poiynomiaiz (e **(a)** 32 36 40 **(b)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only **(c)** 32 36 40 30. If 3 men 4 buys can a piece Of work in 8 days, then 6 men and B eon in 2 as•ys 4 days e days i 6 days X. Y and had taken dinner together. cost Of the meal Of Z 2W more than that Of Y and the east the meal Of X was 5/6 aB much as the cost Of the of Z. If paid then what W•øg the that the three Of them had paid ? 285 300 35b None of the above 26. Consider the i" Of A and B 2. Which Of the above are ? and 2 only Z and 3 only 3 •ancy iB the HCF of the poiynomiaiz **(d)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only

14. **[CDS (I) Paper 10 Q1]** The HCF of two numbers is 98 and their LCM iS 2352. The sum of the numbers may be 1372 **(b)** 1398 142% **(d)** 1484 **(a)** 1398 142% **(b)** 1484 4. What iS the HCF of a2b4 5 and **(c)** ab **(d)** a2b3 . What iS the LCM of x2 +2x-8, 2. 3. Consider the following statements . 2. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7. Every composite number less than 121 is divisible by a prime number less than 11. x
15. **[CDS (I) Paper 10 Q4]** 5 2 What iS the HCF of a2b4 + 2a2b2 and (ab)7 —4a2b9 ? **(a)** 1398 142% **(b)** 1484 4. What iS the HCF of a2b4 5 and **(c)** ab **(d)** a2b3 . What iS the LCM of x2 +2x-8, 2. 3. Consider the following statements . 2. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7. Every composite number less than 121 is divisible by a prime number less than 11. x
16. **[CDS (I) Paper 12 Q31]** The LCM of two integers is 1237. What is their HCF? **(a)** 37 **(b) (c) (d)** 19 Cannot be determined
17. **[CDS (I) Paper 14 Q14]** If (x+l) is the HCF of Ax2+Bx+C and Bx2 + Ax + C where A B, then the value of C is **(a)** I only **(b)** 2 only **(c)** I and 2 only l, 2 and Y is/are 14. If **(d)** 2 minutes
18. **[CDS (I) Paper 14 Q86]** The LCM of two numbers is 12 times their HCF. The sum Of HCF and LCM is **(a)** 20 square t **(b)** 30 square unit **(c)** 40 square unit **(d)** None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m
19. **[CDS (I) Paper 15 Q1]** LCM(ab, = a LCM(b, c) **(a)** 38. Which of the points P **(b)** Q and R o nly **(c)** P and R o nly **(d)** P and Q o Illy
20. **[CDS (I) Paper 15 Q2]** HCF(ab, HCF(b, c) **(a)** 38. Which of the points P **(b)** Q and R o nly **(c)** P and R o nly **(d)** P and Q o Illy
21. **[CDS (I) Paper 15 Q3]** HCF(a, b) < LCM(a, b) **(a)** 38. Which of the points P **(b)** Q and R o nly **(c)** P and R o nly **(d)** P and Q o Illy
22. **[CDS (I) Paper 15 Q4]** HCF(a, b) divides LCM (a, b). Which of the above are correct? **(a)** I and 2 Only **(b)** 3 and 4 only **(c)** , 2 and 4 only **(d)** 1, 2, 3 and 4
23. **[CDS (I) Paper 16 Q24]** There are two numbers p and q such that their HCF is **(a)** 1 and 2 only **(b)** 2 and 3 Only **(c)** 1 and 3 only **(d)** l, 2 and 3 25. What is 23. A shopkeeper increases the cost price Of an item by 20% and offers a discount Of 10% on this marked price. What is his percentage gain ?
24. **[CDS (I) Paper 16 Q28]** What are the roots of the equation **(a) (b) (c) (d)** 21 q;) 2 2, 1-10 3 log 3 -2,1- log 2 log 2 -2,1- log 3 given below : **(a)** 1 and 2 only **(b)** 2 and 3 Only **(c)** 1 and 3 only **(d)** l, 2 and 3
25. **[CDS (I) Paper 17 Q4]** 0064 x 625 What is equal to ? 0081 x 4-84 10 **(a)** 99 100 **(b)** 99 9 **(d)** 99 (x + 4) is a factor of which one of the fo expressions ? **(a)** 12-7*+44 **(b)** x2 + 7x — 44 **(c)** x2-7x-44 **(d)** x2 + 7x + 44 If and ß are the roots Of the' quadrauc equation 9x2 k = where k e O, what is the maximum value of — + — ? **(a) (c) (d)** 2 9 -9 Consider the following statements : If a-bc with HCF(c, HCF (c, d). If a = with HCF (b, c) = 1, then LCM (a, d) LCM (c, bd). Which Of the above statements Ware correct ? **(a)**, **(b) (c) (d)** BLS-s-FDE 1 only 2 only Both I and 2 Neither I nor 2 **(a) (b) (c) (d)** iratic
26. **[CDS (I) Paper 17 Q42]** Sunil wants to spend 200 on two types Of sweets, costing 7 and IO respectively. What is the maximum number Of sweets he can get so that no money is left over ? **(b) (c) (d)** 25 26 27 28 What is the LCM Of xa + 8, x2 + 5K + 6 and x3 + 4x2 + 4x ? **(b)** x (x — (x — 3) (x2 + 2x + 4) **(c)** (x +2)2 (x + 3) (x2 — 2x + 4) The HCF Of two expressions p and q is I. What is the reciprocal of their LCM ? **(a)** P+q **(c)** pq **(d)** (pq)-l BLS-S-FDE (12- are
27. **[CDS (I) Paper 20 Q15]** It is given that log, 0 2 = 0-301 and log, O a 0477. How many digits are there in ite **(c)** 19 20 21 22 The sum of three prime numbers is **(a)** 19 20 21 22 The sum of three prime numbers is 100. If one Of them exceeds another by 36, then one Of the numbers is Proportion of WSpgp« 11 Both I and 11 Both 11 and 111 Both 1 and 111 All the three 1% **(b) (c)** 17 16. 29 43 Nono of the above 17. **(d)**
28. **[CDS (I) Paper 21 Q12]** Wych me Of xver LCM ? **(a)** be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? **(b)** 80 **(c)** 60 36 • **(d)** 24 14. WEch Of
29. **[CDS (I) Paper 21 Q31]** What is value of bgIdsin9) + log10(tnØ) + log I o(cote) + loglßw9) + ? **(c)** 0-5 **(a)** Its lia ixuide O **(b)** Its lies on the triülgle **(c)** It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? **(d)** AB<RX
30. **[CDS (I) Paper 21 Q57]** The Sida Of a re 30 cm, 28 an md 16 cm resp«tively. In to deer-mine its na, the logarithm of which of ql.Ätitie ? **(a)** 37, 11, 28, 16 **(b)** 37, 21, 11, 9 58. If — 3•3000, what is value of **(c) (d)** Q. IA XYZ be an Äiangle in 61. 21 which XY 7 cm. If A der»ta üe of the tinge, what is value of logwA4?
31. **[CDS (I) Paper 21 Q58]** If — 3•3000, what is value of (0-001995)8 ? **(a)** 21 **(c)** Q. IA XYZ be an Äiangle in which XY 7 cm. If A der»ta åe of tinge, what is value of log,oA4 ? (Giva 10801050 3-0212 108035 - 1-5441) **(a)** 5-3070 **(b)** 5-3700 **(c)** 5-5635 **(d)** 5-6535
32. **[CDS (I) Paper 22 Q2]** Two unequal pairs of numbers the following conditions : (i) The product of the two numbers in each pair is 2160 (ii) The HCF of the two numbers in each pair is **(a)** a prime number **(b)** not a prime, but power of a prime **(c)** not a power of a prime, but a composite even number **(d)** not a of a prime, but a composite odd number 2. Two unequal pairs of numbers satisfy the following conditions :

33. [CDS (I) Paper 22 Q3] How many digits are there in (54)10 ? (Given that log102 — 0-301 and log,03 0477) **(a)** 16 **(b)** 18 **(c)** 19 **(d)** 27 3 **(b) (c)** 3

34. [CDS (I) Paper 22 Q14] If (X+k) is the HCF of x2+5X+6 and x2 + 8x + 15, then what is the value of k ? A - PIXI-T-MTK **(a)** 13 **(b)** 39 **(c)** 117 5 11. A two-digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two-digit number is 8, then what is the number ? **(d)** 81

35. [CDS (I) Paper 23 Q31] 32. 33. (13—A) What is the HCF of the polynomials x6 — 3x4 + — I and xa + 3x + I ? **(a) (b)** (e) **(d)** The HCF and the LCM of two polynomials are 3x + I and 30x3 + 7x2 IOX — 3 respectively. If one is 6x2 + I, then What is the other polynomial ? **(a) (b) (c) (d)** 15x2 + 3 15x2 + — 3 15x2 + 3 15x2 — — 3 If (p and (p — 2) (2q — I) 2pq — IO, then what is pq equal to ? **(c) (d)** 10 Mahesh is 60 years old. Ram is 5 years 28E younger to Mahesh and 4 years elder to Babu is a younger brother Of Raju and he is 6 years younger. What is the age difference between Mahesh and Babu ? 29. (13—A) **(a) (b) (c)** 18 years 15 years 13 years

36. [CDS (I) Paper 23 Q32] The number Of items in a t»oklet is N. In the first year there is an increase of in this number and in the subsequent year there is a decrease of At the end of the two years, what will be the number of items in the booklet ? What is the HCF of the polynomials x6 — 3x4 + — I and xa + 3x + I ? **(a) (b)** (e) **(d)** The HCF and the LCM of two polynomials are 3x + I and 30x3 + 7x2 IOX — 3 respectively. If one is 6x2 + I, then What is the other polynomial ? **(b) (c) (d)** Less than N Equal to N More than N It depends on the value of N **(a) (b) (c) (d)** 15x2 + 3 15x2 + — 3 15x2 + 3 15x2 — — 3 ab xy — Xb O and bc + yz — cy O, then 33. equal ? what is — — **(a) (c)** DZOL-T-LKM If (p and (p — 2) (2q — I) 2pq — IO, then what is pq equal to ? **(c) (d)** 10

37. [CDS (I) Paper 24 Q16] What is the HCF of x 3 — 30 and -5x+6? (x +41x-3) **(c) (d) (a)** Which Of the above statements is/are correct? **(b)** I only **(c)** 2 only **(d)** Both I and 2

38. [CDS (I) Paper 24 Q90] A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? **(a)** 14 cm **(b)** 13 cm **(c)** 12 cm **(d)** I lcm

39. [CDS (I) Paper 25 Q4] What is the remainder aner dividing the number by 9 ? S. The sum of LCM and HCF of two numbers is 536 and the difference between LCM and HCF is **(a)** 29, 17, 37 **(b)** 31, 17, 47 **(c)** 19, 37, 13 **(d)** 23, 31, 37 2. Which one of the following is a factor of the

40. [CDS (I) Paper 25 Q13] If loglox+logIox2 = 2 log,ox+I, then what is the value ofx ? **(d)** 10 5 5 **(a)** only **(b)** k>2v5 only **(c)** k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is **(d)** I—aßy

41. [CDS (I) Paper 26 Q5] Suppose p and q are the LCM and HCF respectively of two positive numbers. If p : q = 14 I and pq 1134, then what is the difference between the two numbers? **(a)** 27 **(b)** 35 **(c)** 45 **(d)** Cannot be determined due insufficient data

42. [CDS (I) Paper 27 Q4] 5. 6. —A) If , then what is the value of ? **(a) (b) (c) (d)** 9 10 12 What is the HCF (x7 — ya + x5y2_x2y5)? **(b)** (x3 — ya _ x2y + **(c)** (xa + x2y — xy2) **(d)** of and If the Sum of the squares of the rtxjts Of the equation x2 — 14x + k = O is 100, then What is the value Of k ? **(a) (b) (c) (d)** 42 48 52 56 If (x + k) is the HCF of x2 px + q and x2 + qx • p, where p q, then What is the value of k ? **(b) (c) (d)** 2

43. [CDS (I) Paper 27 Q21] LCM of Of the Consider the number N = 126 x 38 x **(a)** y = 180* y 270* y = 360x the one 21. LCM of Of the Consider the number N = 126 x 38 x 53. Which Of the following statements Ware correct ? I. The number Of Odd factors of N is 60. 2. The number Of even factors of N is 720. Select the correct answer using the code' given below : **(b) (c) (d)**

44. [CDS (I) Paper 27 Q24] 25. 26. is the sum Of the rt»ts of the equation ? **(a) (b)** 12 11 10 9 HOW many of the following values Of x would satisfy the equation qx2 — 2px + q O ? 2. , Where p > q , wher•e p > q , where p > q Select the correct answer using the code given **(a) (b) (c) (d)** Only one value Only two values All three values None Let p(x) x4 + x2 + I, q(x) = x4 — 2x3 + 3x2 — 2x + I. If GCD Of p(x) and q(x) is x2 — x + I, then what is their LCM ? **(a) (b) (c) (d)** FDGT-T-EMT 'hat

45. [CDS (I) Paper 27 Q2] Which one of the following is correct in Of the question and the statements ? **(a) (b) (c) (d)** Statement-I alone is sufficient to answer the question Statement-2 alone is Sufficient to answer the question Both Statement-I and Statement-2 are sufficient to answer the question Both Statement-I and Statement-2 are not sufficient to answer the question Consider the question and two statements given below : LCM Of two numbers x and y is 481 where Question : What is the value of (3x — 2y) ? Statement-I : y > I. Statement-2 : HCF of x and y is I. Which one Of the following is correct in Of the question and the statements ? **(a) (b) (c)** Statement-I alone is sufficient to answer the question Statement-2 alone is sufficient to answer the question Both Statement- I and Statement-2 are sufficient to answer the question Both Statement-I and Statement-2 are not sumcient to answer the question

46. [CDS (I) Paper 29 Q13] 14. -A) How many real roots does the **(a)** Onle one Only two **(b)** Only three **(c) (d)** None Ifx= 97 +56" **(a)** 7 6 **(c)** 5 4 , then What is the value Of Let L be the LCM and H be the HCF of two given numbers. L and H are in the ratio 3 :

47. [CDS (I) Paper 29 Q20] 125100 ? (mT loglo 2 = 0•301) **(c)** 69 70 209 210 acx3 + bcx2 + adx2 aedx + bdx + bed adx3 + acx2 + bdx2 + bcx + acdx + bcd HCF (c, d) = 1, c d ? **(a) (b) (c) (c)** cx+ d ax + b xn — pyn + qzn, + abyz — bzx — axy bn

48. [CDS (I) Paper 29 Q18] 19. 125100 ? (mT loglo 2 = 0•301) **(a) (b) (b) (c) (d)** t? **(a) (b) (c) (d)** 4 3 2 - 2b3 — 2ab3 2bg 2ab3 **(c)** 69 70 209 210 acx3 + bex2 + adx2 aedx + bdx + bed adx3 + acx2 + bdx2 + bcx + acdx + bcd HCF (c, d) = 1, c d ? + - + 20. *lix • v'5y — **(a) (b) (c) (c)** cx+ d ax + b xn — pyn + qzn, + abyz — bzx — axy bn 3x2 + 2y2 — v"iÄx — 3x2 + 2y2 + + — •'Gxy + 6 — OGxy + 6 JNUP-T-EMT

49. [CDS (I) Paper 29 Q18] 19. What is number Of digits in the expansion of 125100? 2=0-301) **(a) (c)** If 4 **(a) (b) (c)** 69 70 209 210 What is the HCF of acxa bcx2 adx2 + acdx + bdx * bcd and adxa + acx2 + + bex + aedx + bed if x + y = 2aa, then What is x — y equal to ? - 2b3 **(a) (b)** — 2ab3 2b3 **(c)** 2ab3 **(d)**

50. [CDS (I) Paper 31 Q91] What is the HCF of 236-1 and 245-1 ? **(a)** 1023 **(b)** 512 **(c)** 511 **(d)** 255

51. [CDS (I) Paper 32 Q2] 3. Identify the Mineral on the basis Of the following characteristics relating to it : l. It is usually a greenish crystal. **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

52. [CDS (I) Paper 32 Q5] 6. Identify the cloud on the basis of the following characteristics relating to it : L They 100k like cotton wool _ **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

53. [CDS (I) Paper 33 Q11] What is the remainder when (1725 1925) is divided by 18? **(c)** . The 3 9 HCF of x and y is H. Consider 12 13. the following statements in respect of the HCF of p — **(a)** 106 **(b)** 107 **(c)** 108 What is the remainder when x6 divided by x 2 + ? is 11. What is the remainder when **(d)** 0 9 HCF of x and y is H. Consider

54. [CDS (I) Paper 33 Q16] What is the HCF of x3 + ya 3xy — I and -1? **(b) (c)** ANFB-T-EMT/4SA 1 7 **(a) (b)** Either p O or p None Of the above 15. If n is natural numirr less than 7, then What is the number Of values Of n for which **(c) (d)** ANFB-T-EMT/4SA 7 17. Let x = n

55. [CDS (I) Paper 34 Q6] If (x— 5) is the HCF of —x —p and x2 — qx— IO, then what is the value of (P+q)? **(a)** -23 **(b)** -17 **(c)** 17 **(d)** 23 A - KPRS-T-MTE

56. [CDS (I) Paper 34 Q7] The LCM and HCF of two polyno- mials p(x) and q(x) are (x + — a3) and (x2 —ax a2), respectively. If p(x) + a2x2 + a4, then what is q(x) equal to ? **(a)** x4 — 2a2x2 + a4 **(b)** x4 + 2a2x2 + a4 **(c)** xa — a2x2 4 a4 8. Which of the following expressions can divide both the polynomials x3+2x2—5x+2 and x3 +4x2+x—6 exactly ? 11. X+l 111. X+2 Select the correct answer using the code given below : **(d)** I and 11 only

57. [CDS (I) Paper 34 Q22] Ifthe HCF of p and is G, then which of the following statements is/are correct ? l. HCF Of p and (p + q) is G **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 23. The HCF and LCM of two numbers p and q are 44 and 4620 respectively. When p is divided by 55, the quotient is 4 and remainder is 0. What is the value of q ?

58. [CDS (I) Paper 34 Q11] HCF ofp, (p-q) is G Select the answer using the code given **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

59. [CDS (I) Paper 34 Q25] What is the LCM of 3-6, 0009, 027 ? **(b)** 10-8 **(c)** 16-2 **(d) (a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 23. The HCF and LCM of two numbers p and q are 44 and 4620 respectively. When p is divided by 55, the quotient is 4 and remainder is 0. What is the value of q ?

60. [CDS (I) Paper 34 Q11] Consider the Question and the Statements and mark the correct option. LCM of three numbers Question : p, q and > q > r > O) is 30 and their HCF is S. What is the product of the three numbers ? : 15 Statement-I Statement-II : Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using both the

61. [CDS (I) Paper 35 Q19] If the LCM Of the two numbers is 1615, then what is (x • y) equal to ? 36 180 360 Cannot be determined due to insufficient **(c) (d)** p q data q **(a) (b)** p q 2. 3. 4. 5. 6. Suppose p and q are positive integers where q is not a perfect square. If is a root of a quadratic equation having integer coemcients, then which Of the following statements Ware correct ? l. The sum of the roots is 2p times the product Of the roots. IL The sum Of the reciprocals Of the roots is equal to 2p. Select the answer using the code given below : data q = k, then what is **(c) (d)**

62. [CDS (I) Paper 35 Q9] A number is formed by three digits, each less by unity than the digit that follows it. If 15 is added to the number, then the sum is 30 times the sum of the digits of the number. What is the product of the digits of the number ? **(b) (c)** 6 24 60 120 Consider the following statements in respect of common logarithms : l. The logarithm of a number greater than 100 but less than 1000 lies between 2 and **(a) (b)** 6 24 60 120 Consider the following statements in respect of common logarithms : l. The logarithm of a number greater than 100 but less than 1000 lies between 2 and 3. 11. The logarithm of a positive n umber less than unity is negative. Which of the statements given above Ware correct ? 12. 13. If , then What x + k x +2k x +5k ¯ k is the solution of the equation ? **(c)** 3 k 2k Consider the polynomial p **(d)**

63. **[CDS (I) Paper 35 Q3]** 11. The logarithm of a positive n umber less than unity is negative. Which of the statements given above Ware correct ? **(a) (b)** 6 24 60 120 Consider the following statements in respect of common logarithms : l. The logarithm of a number greater than 100 but less than 1000 lies between 2 and 3. 11. The logarithm of a positive n umber less than unity is negative. Which of the statements given above Ware correct ? 12. 13. If , then What x + k x +2k x +5k ¯ k is the solution of the equation ? **(c)** 3 k 2k Consider the polynomial p **(d)**

64. **[CDS (I) Paper 35 Q20]** What is the HCF ofxs + x2y — 10xy2 + 8ya and x3 x2y — 4xy2 — 4ya ? **(a) (b)** (x — 2y) **(c)** (x + 2y) **(d)** (x + 4y) equal to ? **(a) (b) (c) (d)** 60 40 24 12 The LCM and HCF Of two polynomials p(x) and q(x) are (x + (xz — 7x + 6) and (x + k) respectively. If [xa + (2k — I)x2 + k(k — — k21 is one Of the polynomials, then which Of the following statements Ware correct ? I. The other polynomial is (x + k) (x — 6) where k •g —

65. **[CDS (I) Paper 35 Q6]** 11. (x + k) cannot be the HCF if k — **(a) (b) (c) (d)** 2 •5 knvhr 2 • 75 km.'•hr 3 knvhr 3 •25 km/hr

CHAPTER 03

Total Archive Questions: 158 Questions

PERCENTAGES, PROFIT, LOSS & DISCOUNT

Typical Exam Weightage: 6 - 10 Questions (6 - 10 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q34] 35. What w;uld be the printed price Of a watch purchased at Rs. 380, so that aner giving 5% discount, there is 25% profit ? **(a)** Rs. 400 **(b)** Rs. 450 **(c)** Rs. 500 **(d)** Rs. 600 A person A sells a table costing Rs. 2000 to a person B and earns a profit Of The person B sells it to another person C at a loss of At what price did B sell the table ? **(a)** Rs. 2054 **(b)** Rs. 2050 **(c)** Rs. 2024 **(d)** Rs. 2014 If and what is a : b : c equal to ? **(a)** 12: 8:21 **(b)** 8:21 : 12 **(c)** 8: 12 : 21 **(d)** 21 : 8 : 12 then 8 A bag contains Rs. 114 in the form of 1 rupee, 50 paisa and IO paisa coins in the ratio 3 : 4 : IO. What is the number of 50 paisa coins ? 56

2. [CDS (I) Paper 2 Q4] 5. A person borrowed Rs. 7,500 at 16% compound interest. HOW much does he have to pay at the end Of two years to clear the loan ? **(a) (b) (d)** Rs. 9,900 Rs. 10,092 Rs. 11,000 Rs. 11,052 A trader sens two cycles at Rs. 1,188 eich and gains 10% on the first and loses ! on the second. What is the profit or loss percent on the whole ? **(a)** 1% loss **(b)** 1% gain **(c)** No loss no ga in **(d)** 2% loss What is the least number which when divided by 42, 72 and 84 leaves the remainders 25, 55 and 67 respectively ? **(a) (c) (d)** 521 512 504 487 If a + b 2m2, b + c 6m, a + c = 2, where m is a real number and a b Sc, then which one of the following is correct ? **(a) (c) (d)** P.DETB-J.FN (2- can

3. [CDS (I) Paper 2 Q17] 18. tich If the rate Of interest is 10% per annum and is compounded half yearly, the principal Of Rs. 400 in 3/2 years will amount to **(a) (c) (d)** Rs. 463-00 Rs. 463-05 Rs. 463-15 Rs. 463-20 of the inhabitants of a certain city left that city. Later on 10% Of the remaining inhabitants Of that city again left the city. What is the remaining percentage Of population of that city ? **(a) (c) (d)** 80% 80-4% 80-6% Which one following ? IS the largest among the **(a) (b) (c)** 0-725 0-725 0-725 0-725 If m and n are two integers such that 2 — n, then (m divisible by — 2m) is always **(c)** 9 16 24 48 nvo trains in the same direction at 50 knv'hr and 32 km/hr respectively. A man in the slower train observes that 15 seconds elapse before the faster train completely passes him. What is the length Of the faster train ? If the rate Of interest is 10% per annum and is compounded half yearly, the principal Of Rs. 400 in 3/2 years will amount to **(a) (b) (c)** 75 m 125 m 150 m 625 3 **(a) (c) (d)** Rs. 463-00 Rs. 463-05 Rs. 463-15 Rs. 463-20

4. [CDS (I) Paper 2 Q19] A milk vendor bought 28 litres of milk at the cost of Rs. 8-50 per litre. After adding some water, he sold thc mixture at the same price. If he gains 125%, how much water did he add ? **(a) (b) (c) (d)** litres litres 3-5 litres 2-5 litres

5. [CDS (I) Paper 2 Q34] If ax by cz and abc = xy + ya + zx equal to ? **(a) (c)** XYZ 0 then what is then what is the If - 0037Y 10000, value of — — — ? **(a) (c) (d)** 1/2 1/4 The following table shows the percentage of male and female coffee drinkers and non-coffee drinkers in two Towns A and B Town-A Attributes Male Female Town-B Coffee drinkers Non-coffee drinkers Male 25% 5% 35% Fema I e 30% If the total population of the Towns A and B be 10,000 and 20,000 respectively, then what is the total number Of female coffee drinkers in both towns ? **(a) (b) (c) (d)** 8,000 6,000 3,500 2,500

6. [CDS (I) Paper 2 Q52] 54. If the radius of the base and the height Of a right circular cone are increased by 20%, then what is the approximate percentage increase in volume ? **(a) (b) (c) (d)** 60 68 73 75 PQRS is a diameter of a circle of radius 6 cm as shown in the figure above. The lengths PQ, QR and RS are equal. Semi-circles are drawn on PQ and QS as diameters. What is the perimeter of the shaded region ? **(a) (b) (c) (d)** 127 147t 167t 1811 Hcrw many titres Of water (approximately) can a hemispherical container of radius 21 cm hold ? **(c)** 194 388 194 388 P-DETB.J.FN (14 - Of

7. [CDS (I) Paper 2 Q54] 14 If the radius of the base and the height Of a right circular cone are increased by 20%, then what is the approximate percentage increase in volume ? **(a) (b) (c) (d)** 60 68 73 75

8. [CDS (I) Paper 2 Q2] 3. Information on households collected by an investigator by door-to-door visits. Data on the percentage of sex-wise, for the different districts of a state collected from records of the census of India. General information about families, collected by telephonic interviews. Which one Of the following in respect of the above is correct ? **(a) (b) (c) (d)** 1/2 2 loglo 13

9. [CDS (I) Paper 3 Q30] A man bought a number of oranges at 3 for a rupee and an equal number at 2 for a rurre. At what price per dozen, should he •sell them to make a profit of ? **(c)** Rs. 6 **(d)** Rs.7 **(a)** 26. The shadow of a pole 6 metre high is 15 rnetre long and at the same time the shadow of a tree is 25 metre long. What is the height of the tree ? **(b)** 21 m **(c)** 10m **(d)** 35 m

10. [CDS (I) Paper 3 Q33] If of (x-y) of then what per cent of x is y ? **(a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** days 33. If of

11. [CDS (I) Paper 3 Q46] One saree was purchased for Rs. 564 after getting a discount of 6% and another saree was purchased for Rs. 396 after getting a discount of Taking both the items as a single transaction, what is the percentage of discount ? **(a)** tan2t50 + tan2200 + tan2250 . + tan275Ø **(b)** cos2150 + cos2200 + cos2250 cos2750 **(c)** cot2150 — cot220e + cot2250 — + cot2750 **(d)** sec2150 + sec2200 + sec2250 + sec2750 42. Consider the following data : 5 9 2 The arithmetic mean of the above distribution is 296. What is the missing frequency ? 43. If a is an acute angle and then what is tan equal to ?

12. [CDS (I) Paper 3 Q67] A person invested some amount at the rate of 12% simple interest and the remaining at 10%. He received yearly an interest of Rs. **(a)** Rs. 500 @ 10%, Rs. 800 @ 12% **(b)** Rs. 700 @ 10%, Rs.600 @ **(c)** Rs. 800 @ 10%, Rs. 400 @ **(d)** @ 10%, ROOO @ 68. A man is watching from the top of a tower a boat sl*eding away from the tower. The boat makes an angle Of depression of 450 with the man's eye when at a distance of 60m from the bottom of tower. After 5 seconds, the angle of depression becomes 300. What is the approximate speed of the boat assuming that it is running in still water ? 70. The marks of the students of a class who appeared for a test in English are represented in the following frequency table : 11-20 21-30 31-40 41-50 51.60 What is/are the modal class

13. [CDS (I) Paper 4 Q13] A person spends Of monthly salary on rent, 25% on food, 20% on childreNs education and 12% on electricity, and the balance of Rs 1,040 on the remaining items. What is the monthly salary of the person? **(a) (b) (c) (d)** Rs 8,000 Rs 9,000 Rs 9,600 Rs 10,600 in Of ty 9 8

14. [CDS (I) Paper 4 Q25] If salary of X is more than salary Y, then by how much percentöge is salary Of Y less than X? **(a)** divisible by 24 **(b)** d ivisible 8 but may no t be divisible 24 **(c)** divisible by 4 but may not be divisible by 8 **(d)** divisible by 2 but may not be divisible by 4 25. If salary of X is more than salary Y, then by how much percentöge is salary Of Y less than X?

15. [CDS (I) Paper 4 Q36] On selling an article for Rs 240, a trader loses 4%. In order to gain the 10%, he must sell the article for Rs 275 **(b)** Rs 280 Rs 285 **(d)** RS 300 **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

16. [CDS (I) Paper 4 Q37] In a class, the number Of boys is more than the number Of girls by 12% Of the total strength. What is the ratio of number of boys to that Of girls? 14:11 28 : 25 25 : 28 **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

17. [CDS (I) Paper 4 Q71] If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? **(a)** 100/3 **(b)** 40 **(c)** 50 **(d)** 200/3 Q-DETB-K-FN/26A

18. [CDS (I) Paper 4 Q81] In ant examination, 40% of the candidates wrote their answers in Hindi and the others in English. The laverage marks Of the candidates written in Hindi is 74 and the average marks of the candidates written in English is **(a)** hoo 83. 84. 26 Consider the following statements in respect of histogram 2. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. **(b)** do 81. In ant examination, 40% of the candidates wrote their answers in Hindi and the others in English. The laverage marks Of the candidates written in Hindi is 74 and the average marks of the candidates written in English is 77. What is the average marks Of all the candidates? **(c)** 758 **(d)** 75-8

19. [CDS (I) Paper 4 Q93] If 28 **(a)** The area Of the circle is e qual to that of square **(b)** The area o' the circle is larger than that Of square **(c)** The area of the circle is less than that bf square **(d)** No conclusiion can be drawn 87• If

20. [CDS (I) Paper 4 Q92] ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? **(a)** The area Of the circle is e qual to that of square **(b)** The area o' the circle is larger than that Of square **(c)** The area of the circle is less than that bf square **(d)** No conclusiion can be drawn 87• If

21. [CDS (I) Paper 5 Q18] 19. The set S lx E N 3) is a **(a) (b) (c)** Null set Singleton set Infinite set None Of the above A t,elegraph post gets broken at a point against a storm and its top touches the ground at a distance 20 m from the base of the post making an angle 3co with the ground. What is the height Of the post ? **(b) (c) (d)** 40 m 30 m The sum of two numbers is

22. [CDS (I) Paper 5 Q22] 23. **(a) (b) (c)** Null set Singleton set Infinite set None Of the above 20. 21. 22. 23. **(d)**

23. [CDS (I) Paper 5 Q40] 41. Water contains 14— % Of hydrogen and the rest is oxygen. In 350 g of water, oxygen will be **(a) (b) (c)** 300 g 250 g 200 g None Of the above What is the number Whose 20% is 30% of 40 ? **(a) (b) (d)** 90 80 60 50 38 litres of milk was poured into a tub and the tub was found to be 5% empty. To completely fill the tub, what amount of additional milk must be poured ? **(a) (c) (d)** The litre 2 litres 3 litres 4 litres income Of 'A' is higher than that Of The income Of "B' is 25% less than that of C. What percent less is Ns income from C's income ? **(a) (c)** 7% 12-5% P-PQFA-L-RZ (12 -A 42. ill

24. **[CDS (I) Paper 5 Q45]** In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% candidates failed in both English and Mathematics, what percentage Of candidates passed in both the subjects ? **(a) (b) (c) (d)** 21% 23% If the HCF of three numbers 144, x and 192 is 12, then the number x cannot be **(a) (b) (d)** 180 84 60 48 What is the HCF of 3-0, 1-2 and ? **(a) (b) (c)** 06 0-06 60 6,06 What is the least number Of square tiles required to pave the floor of a room 9 m 99 cm long and 4 m 7 cm broad ? **(a) (b) (c) (d)** 247 277 297 307

25. **[CDS (I) Paper 5 Q42]** Water contains 14— % Of hydrogen and the rest is oxygen. In 350 g of water, oxygen will be **(a) (b) (c)** In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% candidates failed in both English and Mathematics, what percentage Of candidates passed in both the subjects ? 300 g 250 g 200 g None Of the above **(a) (b) (c) (d)** 21% 23% What is the number Whose 20% is 30% of 40 ? **(a) (b) (d)** 90

26. **[CDS (I) Paper 5 Q44]** 45. What is the HCF of 3-0, 1-2 and ? **(a) (c) (d)** The litre 2 litres 3 litres 4 litres income Of 'A' is higher than that **(a) (b) (c)** 06 0-06 60 6,06 Of The income Of "B' is 25% less than that of C. What percent less is Ns income from C's income ? What is the least number Of square tiles required to pave the floor of a room 9 m 99 cm long and 4 m 7 cm broad ? **(a) (c)** 7% 12-5% **(a) (b) (c) (d)** 247 277 297 307 P-PQFA-L-RZ (12 -A)

27. **[CDS (I) Paper 5 Q60]** A trader marks higher than the Cost price. He gives a discount Of 10% on the marked price. In this kind of sales how much percent does the trader gain or lose ? **(c) (d)** 5% gain 2% gain loss 3% loss The speeds of three cars are in the ratio 4 : 3 : **(a) (b) (c) (d)** 112 108 96 84

28. **[CDS (I) Paper 5 Q92]** D 3k E BC 3 ADE 3 cm2 I EC ? **(a)** 16 cm **(b)** 20 em **(c)** 25 cm **(d)** 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

29. **[CDS (I) Paper 5 Q95]** 16Jä -8T 16Jä - 27 -A) **(a)** 16 cm **(b)** 20 em **(c)** 25 cm **(d)** 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ? **(a)** 1-5 C m **(d)** 2-1 cm qx 9,680 10,648 I **(c)** 15%

30. **[CDS (I) Paper 6 Q18]** What is the difference between the digits of the number ? A fruit-seller buys lemons at 2 for a rupee 14. and sells them at 5 for three rupees. What is his gain percent ? **(b)** 15% **(c)** 20% A man sold two watches, each for **(a)** 200 rn2 **(b)** 280 rn2 **(c)** 300 rn2 **(d)** 320 tn2 wide is What is The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance Of I OO m from its base is 450. If the angle Of elevation of the top of complete pillar at the same point is to be 600, then the height of the incomplete pillar is to be increased by

31. **[CDS (I) Paper 6 Q20]** Cm. If its radius is changed so that the area gets reduced by 75%, then the radius becomes **(a) (b)** cm **(c)** cm **(d)** cm 2 From a lighthouse the angles Of depression of two ships on opposite sides of the light- house are observed to be 300 and

32. **[CDS (I) Paper 6 Q27]** An employee is required to contribute 10% of his pay to General Provident Fund. If he gets Rs. 13,500 as net pey in a month, then what is the monthly General Provident Fund contribution (assuming no other deductions) ? **(a)** Rs. 1215 **(b)** Rs. 1350 **(c)** Rs. 1500 **(d)** Rs. 1650 If RS. 8400 is divided among A, B and C in the ratio — . of A? **(a)** Rs. 1800 **(b)** Rs. 3000 **(c)** Rs. 3600 **(d)** Rs. 4000 what is P : Q what is the share 2 5' 5 9 20 3 5 3 5 15 7 4 • R equal to ? 2 7 3 4 7 2 5 4

33. **[CDS (I) Paper 6 Q62]** The angles of elevation of the top of a tower from two points situated at distances 36 m and 64 m from its base and in the same straight line with it are complementary. What is the height of the tower ? **(a)** 50 m **(b)** 48 m **(c)** 25 m **(d)** 24m A grassy field has the shape of an equilateral triangle Of side 6 m. A horse is tied to one of its vertices with a rope of length m. The percentage of the total area Of the field which is available for grazing is best approximated by **(b)** 55% **(c)** 59% **(d)** 62% 00032 What is the value of 0•32 **(a)** 0 0001 **(b) (c) (d)**

34. **[CDS (I) Paper 7 Q5]** If there is 30% loss on the first article, and 20% gain on the second article, what is Overall percentage of loss or gain ? **(a)** 16 In the triangle ABC, the base BC is trisected at D and E. The line through D, parallel to AB, meets AC at F and the line through E parallel to AC meets AB at G Let EG and DF intersect at H. What is the ratio of the sum Of the area of parallelogramÅGEF and the area of the triangle DHE to the area Of the triangle ABC ? **(b)** 1/6 Ifx — 2 is a factor + kx2 what is the value of k ? — 2K —24 then AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number ofpoints which are at a distance of I cm from AB and 5 cm from C ? 18. 19. 20. 21. Ihe angle of elevation of the tip of a tower from a point On the ground is 450. Moving 21 m directly towards the base ofthe tower, the angle ofelevation changes to 600. What is the height Of the tower, to the nearest meter ? **(c)** 48 m **(d)** . 49 m

35. **[CDS (I) Paper 7 Q37]** 38. If the outer and inner diameters of a stone parapet around a well are 112 cm and 7() cm respectively, then what is the area of the parapet ? **(a)** 264 square cm **(b)** 3003 square cm **(c)** 6006 square cm **(d)** 24024 square cm Whatis 0009 +1783+310-0202 equal to? **(a)** 32786638 **(b)** 327,866" **(c)** 327-866ij **(d)** 327-866S A man loses 20% of his money. After spending 25% of the remainder, he has Rs. 480 left. What is the amount of money he originally had ? **(a) (b)** Rs. 720 **(c)** Rs. 800 **(d)** Rs. 840 Assuming that profit of a shopkeeper in a particular commodity is a linear expression of transportation charge (t) and of commodity (q). He earns a profit of Rs. 10,000 by selling 20 units at the transport charge-of Rs.

36. [CDS (I) Paper 7 Q29] What is the number of such pairs of numbers satisfying the above condition ? **(a)** One **(b)** Two **(c)** Four **(d)** Noné of the above A person bought 8 quintal ofrice for certain rupees. After a week, he sold 3 quintal Of rice at 10% profit, 3 quintal of rice with neither profit nor loss and 2 quintal ai 5% loss. In this transaction, what is the profit ? **(b)** 20% **(c)** 25% **(d)** None of the above What is the angle of elevation of the sun when the shadow of a pole is times the length of the pole ? **(a)** 300 **(b)** 450 **(c)** 600 **(d)** None of the above

37. [CDS (I) Paper 7 Q63] Two students A and B solve an equation Of the form ± px + q = O. A starts with a %T0ng value of p and obtains the roots as 2 and **(a)** p **(b)** 3 and —4 **(c)** —3 and—4 **(d)** —3 and 4

38. [CDS (I) Paper 7 Q82] 83. 22 The total surface area Ofa cube is 150 Square cm. What is its volume ? **(a)** 64 cubic cm **(b)** 81 cubic cm **(c)** 125 cubic cm **(d)** 160 cubic cm The HCF and LCM of two natural numbers are 12 and 72 respectively. What is the difference between the two numbers if one of the numbers is 24 ? **(a)** 12 **(b)** 18 **(c)** 21 **(d)** 24 A person sold an article for Rs. 136 and got 15% loss. Had he sold it for Rs. x, he would have got a profit of 15%. Which one Of the following is correct ? Consider the following statements • l. The angular measure in.'radian of a circular arc of fixed length subtending at its centre decreases if the radius of the arc increases. 18000 is equal to 5n radian.

39. [CDS (I) Paper 8 Q25] The product of the two numbers is : **(a)** 45 **(b)** 36 **(c)** 54 **(d)** 63 Age Of X is six times that of Y. After 4 years, X is four times elder to Y. What is the present age Of Y ? **(a)** 4 years **(b)** 5 years **(c)** 6 years **(d)** 7 years A milkman bought 15 kg Of milk and mixed 3 kg Of water in it. If the price per kg Of the mixture becomes Rs. 22, what is cost price of the milk per kg ? **(a)** Rs. 28.00 **(b)** Rs. 26.40 **(c)** Rs. 24.00 **(d)** Rs. 22.60

40. [CDS (I) Paper 8 Q17] What is 5% of of 500 ? **(a)** 12.5 **(b)** 25 **(c)** 1.25 **(d)** 6.25

41. [CDS (I) Paper 8 Q19] A man buys a television set which lists for Rs. 5,000 at 10% discount. He gets an additional 2% discount (after the first discount) for paying cash. What does he actually pay for the set ? **(a)** Rs. 4,410 **(b)** Rs. 4,400 **(c)** Rs. 4,000 **(d)** Rs. 4,500

42. [CDS (I) Paper 9 Q22] 24. -Ai if one Of the Of the — bx + c O the square at the then Which one the following is ? 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf — 5x2 — 81 ia ta) equgi to ? Let A : % is odd integer) and B — : xa — 8x + % — **(a)** 5% **(b)** 8% **(c)** 10% **(d)** 12.5%

43. [CDS (I) Paper 9 Q1] 1%en which one of the ig ? **(b)** (41 ACB sum Of numbers 7 and sum Of 20. their ' is **(a)** Rs. 1,200 **(b)** Rs. 1,440 **(c)** Rs. 1,500 **(d)** Rs. 1,800

44. [CDS (I) Paper 9 Q75] Each *ide of an equitatgret trianøe 6 em. attitude ig What is the quantity of cloth required to gp firm a right circular tent whose base i. of radius 12 m and height 5 m ? **(a)** dC•R **(b)** metres (e) metres **(d)** 156% squ are metres ne CID long. distance end 40 mi is **(c)** tOr/3 em • 311,120 cm 10 em A pool 70 m long, 44 rn Wide and

45. [CDS (I) Paper 10 Q8] A person sold an article for e 3,600 and got a profit of 20%. Had he sold the article for 3,150, how much profit would he have got? **(a)** 15 **(b)** 25 **(c)** 20 **(d)** 30 9. 10. 11. What is equal to?

46. [CDS (I) Paper 10 Q21] A person's salary has increased from e 7,200 toe 8,100. What is the percentage increase in his salary? 23. •Nyo 6 **(c)** 164% **(d)** 124% **(a)** O only **(b)** 20 only **(c)** both 10 and 20 **(d)** neither IO nor 20 If

47. [CDS (I) Paper 10 Q22] X can do a work in 16 days. In how many days will the work be completed by Y if the effciency Of Y is 60% more than that of X? **(a) (b) (c) (d)** 10 days 12 days 25 days 30 days lots of onions with equal quantity, one costing 10 per kg and the other costing 15 per kg, are mixed together and whole lot is sold at 15 per kg. What is the profit Or loss? **(a)** loss **(b)** profit **(c)** 20% profit **(d)** 20% loss

48. [CDS (I) Paper 11 Q22] 23. If the roots of the equation Axa * Bx C O are —I and I, then which one of the following is correct ? **(a) (b) (c) (d)** A and C are both zero A and B are both positive A and C are both negative A and C are of opposite sign If A and B are any two non•empty subsets of a set E, then what is A (A n B) equal to ? AnB AUB **(c) (d)** Consider the following in respect of the numbers u, : I. is the greatest number. 2. is the smalle st num ber. Which of the above statements Ware correct ? **(a) (b) (c)** I only 2 only Both 1 and 2 Neither I nor 2 A water pipe is eut into two pieces. The longer piece is 70% of the length Of the pipe. By how much percentage is the longer piece longer than the shorter piece ? **(a) (d)** 140% None Of the above B.ABQA-N-NBu (6- 0

49. [CDS (I) Paper 11 Q37] 38. In a fire range, 4 shooters are firing at their respective targets. The first, the second, the third and the fourth shooter hit the target once every 5 s, 6 s, 7 s and 8 s respectively. If all Of them hit their target at 9 : 00 am, when will they hit their target together again ? **(b) (c) (d)** 9 : 04 am 9 : 08 am 9: 14 am None of the above If A is a non-empty subset Of a set E, then what is E — (A — e)) equal to ? **(b)** Complement of A then what is A : D equal to ? If the average Of A and B is 30, the average of C and D is 20, then which of the following is/are ? **(a) (b) (c)** 9 : 04 am 9 : 08 am 9: 14 am None of the above then **(d)**

50. **[CDS (I) Paper 12 Q9]** When an article is sold at 20% discount, the selling price is e **(a)** I Q years 12•5 years **(b)** 10-5 years 13. A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(c)** 14 11. 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : 4. What is the ratio Of ages Of A, B and C? **(d)** 6: 15 : 20 9. When an article is sold at 20% discount, the selling price is e 24. What will be the selling price when the discount is 30%?

51. **[CDS (I) Paper 12 Q24]** What will be the selling price when the discount is 30%? **(a)** e 25 **(b)** e 23 **(c)** e 21 **(d)** e 20 A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000 g for 800 g. What is his actual profit p tage? 11. **(b) (d)** The 25% 20% 40% difference between compound interest and simple interest for 2 years at the rate Of 10% over principal amount of e X is e IO. What is the value of X? **(a)** e 100 e 1,000 500 e 5,000 B-ABQB-N-LZS/ LA

52. **[CDS (I) Paper 12 Q14]** If Of m + Of n of (m x n), then what percentage of m is n? 50% **(b)** 75% **(c)** 100% **(d)** Cannot be deterrnined due to in sufficient data If the side of a cube is increased for by 100%, then by what percentage is the surface area Of the cube 10. increased? 1 soon **(b)** 200% 300% 400% 4 **(a)** I Q years 12•5 years **(b)** 10-5 years 13. A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(c)** 14 11. 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : 4. What is the ratio Of ages Of A, B and C? **(d)** 6: 15 : 20 9. When an article is sold at 20% discount, the selling price is e 24. What will be the selling price when the discount is 30%?

53. **[CDS (I) Paper 12 Q13]** A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(b)** 14 **(a)** I Q years 12•5 years **(b)** 10-5 years 13. A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(c)** 14 11. 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : 4. What is the ratio Of ages Of A, B and C? **(d)** 6: 15 : 20 9. When an article is sold at 20% discount, the selling price is e 24. What will be the selling price when the discount is 30%?

54. **[CDS (I) Paper 12 Q46]** 47. A person selling an article for 96 finds that his loss percent is one-fourth Of the amount Of rupees that he paid for the article. What can be the cost price? **(d)** e 160 only e 240 only e 160 or 240 Neither e 160 nor 240 If (x + k) is the common factor of x + b and x2 + d, then What is k equal to? **(a)** 125 **(b)** -3125 **(c)** 3125 B-ABQB-N-LZS/ IA 46. **(d)** e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

55. **[CDS (I) Paper 12 Q47]** 45 A person selling an article for 96 finds that his loss percent is one-fourth Of the amount Of rupees that he paid for the article. What can be the cost price? **(a)** 125 **(b)** -3125 **(c)** 3125 B-ABQB-N-LZS/ IA 46. **(d)** e 160 only e 240 only e 160 or 240 Neither e 160 nor 240

56. **[CDS (I) Paper 12 Q51]** If from the top of a post a string twice the length Of the post is stretched tight to a point on the ground, then what angle will the string make with the post? **(b)** TC 14 Sit/ 12 **(d)** -rt/3 S2. The price of a commodity increased by from 2010 to 2011, from 2011 to 2012 and 77% from 2012 to 2013. What is the average price increase (approximate) from 2010 to 2013? **(a)** TC 14 Sit/ 12 **(b)** -rt/3 S2. The price of a commodity increased by from 2010 to 2011, from 2011 to 2012 and 77% from 2012 to 2013. What is the average price increase **(c)** 26% If then what equal to ? **(d)**

57. **[CDS (I) Paper 12 Q94]** If the radius Of a circle is increased by 6%, then its area Will increase by 95. **(d)** The 6% 1236% class Which has frequency is knou•n as **(b)** median class mean class modal class None Of the above For the next three (3) items that follow : A piece of land is in the form of a parallelogram and the perimeter Of the land is 86 m. The length Of one side exceeds the other by 13 m and one Of the diagonals Which of the is/ are correct? above statements is 41 m **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither nor 2 93. If one Of the roots Of the equation px + qx+r=O is three times the Other, then Which one of the following relations is correct? **(d)** 63 m 2 96 m 2 126 m2 252 m 2

58. **[CDS (I) Paper 13 Q96]** What is the value of R? 8 10 12 Cannot be determined due to insufficient data What is the maximum value of m if the number N- 35x45x 55x is divisible by 5m ? 4 5 7 A milkman claims to sell milk at its cost price only, but he is making a profit of since he has mixed some amount of water in the milk, What is the percentage Of milk in the mixture? **(b) (c)** 250 75% 200 3 B-EFUA-O-PDW/52A **(a) (b)** A person could save of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **(c)** 23 **(d)**

59. **[CDS (I) Paper 13 Q18]** A person could save of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? 1k. 22 _ % 24—0/0 25 _ % 6 **(b)** 23 **(a) (b)** A person could save of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **(c)** 23 **(d)**

60. **[CDS (I) Paper 13 Q32]** 33. **(c)** 375 383 An automobile financer claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest at the rate of 10%, the effective rate Of interest becomes 10-25% 10-5% 10-75% **(c) (d)** What are the possible solutions for x Of the x and — where equation x n are positive integers? **(a) (b) (c)** n, B-EFUA-O-PDW / S2A

61. **[CDS (I) Paper 13 Q52]** A cylindrical vessel of radius 4 Crn contains water. A solid sphere Of radius 3 cm is lowered into the water until it is completely immersed. The water level in the vessel Will rise by **(c)** 2-25 cm **(d)** 4 •5 Crn If the height Of a right circular cone is increased by 200% and the radius Of the base is reduced by 50%, then the volume of the cone **(a) (b) (c)** remains unaltered decreases by 25% increases by 25% increases by The angles of a triangle are in the ratio 4 : I : l. Then the ratio of the largest side to the perimeter is 2 2+0 **(b) (d)**

62. **[CDS (I) Paper 13 Q59]** 60. 20 If the radius of a sphere is increased by 10%. then the volume will be increased by 33-1% **(b)** 30% 10% ABCD is a parallelogram, where AB AD 2 l. One of the angles of the parallelogram is 60'. The tim diagonals are in the ratio **(d)** 7:5 None of the above **(a)** 2r2 3r2 **(b)** 4r2 SS. AD is the diameter of a circle and AB is a chord. If AD 34 cm. AB— 30 cm, the distance Of AB from the centre Of the circle is 58. is equal to — I, then **(c)** 0 **(d)**

63. **[CDS (I) Paper 14 Q3]** A milkman claims to sell milk at its cost price only. Still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture ? **(b)** 75% **(a)** 51 **(b)** 53 **(c)** 55 **(d)** 57 2. What is the sum of digits Of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11 respectively as the remainders ? 3. A milkman claims to sell milk at its cost price only. Still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture ?

64. **[CDS (I) Paper 14 Q19]** The value of a single discount on some amount which is equivalent to a series of discounts of 10%, 20% and on the same amount, is equal to **(a)** 43-2% **(b)** 56-8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ? **(c)** 5<k$11 21. The number of pairs **(d)** Zero

65. **[CDS (I) Paper 14 Q35]** 20% of a number when added to 20 the number itself, then nurntEr is **(a)** 10 s **(b)** 16 s **(c)** 21 s 32. Three athletes run a 4 km race. Their srredsareintheratio 16:d5: 11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is **(d)** 1000m

66. **[CDS (I) Paper 14 Q46]** A's salary was increased by and then decreased by 20%. On the whole A's salary is increased by **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

67. **[CDS (I) Paper 14 Q49]** In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. nere were Only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 votes more than his rival. The number of voters on the voter list is **(a)** 3600 **(b)** 6200 **(c)** 6028 **(d)** 6400

68. **[CDS (I) Paper 14 Q72]** ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by **(b)** 52% **(c)** 540/0 **(d)** 56% **(a)** will be 75. Let **(b)** 52% **(c)** 540/0 **(d)** 98 cm 108 cm2

69. **[CDS (I) Paper 15 Q45]** A cloth merchant buys cloth from a weaver and cheats him by using a scale Which is IO cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but While selling uses a scale which is IO Crn shorter than a normal metre scale. What is his gain? **(c)** 22—0/0 **(d)** 23—0/0 B-ABPA-PÄIRW / 54A **(a)** 9 < xs30 45. A cloth merchant buys cloth from a weaver and cheats him by using a scale Which is IO cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but While selling uses a scale which is IO Crn shorter than a normal metre scale. What is his gain? **(b)** 22—0/0 **(c)** 23—0/0 B-ABPA-PÄIRW / 54A 46. Two pipes A and B can fill a tank in 60 minutes and 75 minutes respectively. There is also an outlet C. If A, B and C are opened together, the tank is full in 50 minutes. How much time will be taken by C to empty the full **(d)**

70. **[CDS (I) Paper 15 Q74]** Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(a)** 165% 21 B-ABPA-P-JRW/ 54A 74. Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(b)** 12% 76. If 61 where x, y and z are natural numbers, then What is z equal to? **(c)** 75. The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 77. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times thc arca Of the smaller ring, then n is equal to **(d)** 10 78. Consider the following statements : 2. If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n 3. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is

71. **[CDS (I) Paper 15 Q2]** 3. The median of the age distribution is 7 years. 70% Of the children are in the age group 6—9 years. The modal age Of the children is 8 years. Which of the above statements are corre ct? **(b) (d)** 32 1 and 2 only 2 and 3 only 1 and 3 only I, 2 and 3 **(a) (b) (c) (d)** 400, — 502, y- 40', y 5x-2y= 10 z- 1050 z-II

72. [CDS (I) Paper 16 Q23] A shopkeeper increases the cost price Of an item by 20% and offers a discount Of 10% on this marked price. What is his percentage gain ? **(a)** 1 and 2 only **(b)** 2 and 3 Only **(c)** 1 and 3 only **(d)** l, 2 and 3 25. What is 23. A shopkeeper increases the cost price Of an item by 20% and offers a discount Of 10% on this marked price. What is his percentage gain ?

73. [CDS (I) Paper 16 Q36] X and Y entered into partnership with Rs. 700' and Rs. 600 respectively. After 3 months X withdrew 2/7 of his stock but after 3 months, he puts back 3/5 of what he had withdrawn. The profit at the end of the year is Rs. **(a)** 11 31. If2p+3q= 12 and4p2+4pq—3q2 — 126, then what is the value of p + 2q ? 29. There are twelve friends A, B, C, D, E, 33. What is the unit digit of 7139 F, G, H, I, J, K and L who invested money in some business in the ratio of and the duration for which they invested the money is in the ratio of respectively. Who will get the maximum profit at the end of the year ? **(b)** F only **(c)** G only **(d)** Both F and G

74. [CDS (I) Paper 16 Q93] A pie chart depicts the classification of total funds Of an organization according to different sources of funds. A particular sector Of pie chart for corporate tax has 1080 angle at' the centre. What is the percentage Of income from corporate tax to total funds ? **(a)** 20% **(c)** 30% **(d)** 35% **(b)** 25% variable takes discrete

75. [CDS (I) Paper 17 Q10] What is the sum ? **(a) (b) (c) (d)** 5,000 e 6,000 6,250 7,500 If Of a + Of b percent of a is b ? = 2% Of ab, then What **(a) (b) (c) (d)** 75% 100% Cannot be determined A passenger train departs from Delhi at 6 p.m.

76. [CDS (I) Paper 17 Q86] 82. increased by p% without increasing its height, then what is the percentage increase in the volume Of the cone ? **(b)** 2p2 100 **(d)** p 2+_p_ 100 A copper Wire when bent in the form Of a **(a)** 2p2 100 **(b)** p 2+_p_ 100 A copper Wire when bent in the form Of a 87. 83. square encloses an area Of 121 cm2. If the same Wire is bent in the form Of a circle, it encloses an area equal to 84. 85. **(c) (d)**

77. [CDS (I) Paper 17 Q88] The radius of a circle is increased so that its circumference increases by 15%. The area Of the circle will increase by If the surface area of a sphere is reduced to one-ninth Of the area, its radius reduces to **(a) (b) (c) (d)** BLS-S-FDE One-fourth One-third One-fifth One-ninth **(a) (b) (c) (d)** 3325%

78. [CDS (I) Paper 18 Q12] IO. 11. (4—A) numbers 0-83 and on the number line, then the distance between P and Q is 21 **(a)** 90 19 90 21 100 56 **(d)** 90 Sudhir purchased a chair with three consecutive discounts of 20%, and 5%. The actual deduction will be **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

79. [CDS (I) Paper 18 Q13] 14. 32% 35% A fruit seller has a certain number of mangoes of which 5% are rotten. He sells 75% of the remainder and he is left with 95 mangoes. HOW many mangoes did he have originally ? 500 450 400 350 **(a)** 90 19 90 21 100 56 **(b)** 90 Sudhir purchased a chair with three consecutive discounts of 20%, and 5%. The actual deduction will be If a train crosses a km-stone in 12 seconds, how long will it, take to cross 91 km-stones completely if its Speed is 60 km/hr ? **(c) (d)**

80. [CDS (I) Paper 18 Q37] In an examination, 35% students failed in Hind; , 45% students failed in English and 20% students failed in both Lhe subjects. What is the percentage of students passing in both the subjects ? **(a) (b) (c) (d)** 20 30 40

81. [CDS (I) Paper 18 Q44] IO _ % 45. -A) **(a) (c) (d)** 6,000 4,800 e 3,600 2,400 When prices rise by 12%, if the expenditure is to be the same, what is the percentage of consumption to be reduced ? **(a) (b) (d)** A man rows down a river 18 km in 4 hours with the stream and returns in 10 hours. Consider the following statements :

82. [CDS (I) Paper 18 Q89] 90. The number of girls studying Statistics is what percent (approximate) of the total number of students studying Chemistry ? **(a) (b) (c) (d)** 58-8 73-5 78•7 80•6 In which subjects is the difference between the number of boys and girls equal ? **(a) (b) (c) (d)** Mathematics and Chemistry Statistics and Chemistry Mathematics and Physics Mathematics and Statistics What is the difference between the number of boys studying Mathematics and the number Of girls studying Physics ? **(a) (b) (c) (d)** 20 30 60 80 What is the ratio ofthe total number of boys to the total number Of girls ? **(a) (b) (c)** STS-M-TPT 67 : 83 17 : 26 27 : 19 189 : 179 (28

83. [CDS (I) Paper 19 Q20] Then the number is **(a) (b) (c) (d)** 20r3 — 5 only If the price Of wheat rises by 25%, then by how much must a man reduce his consumption in order to keep his budget the same as before ? 15% **(b) (c) (d)** 30% — Of the students who registered did not for the examination, — of those who 20 appeared passed. If the number Of registered students is 2000, the number who passed is **(a) (b) (c) (d)** 1920 1056 1020 864 What is the difference between and O •9 ? **(a) (b) (c) (d)** 0-099 0•09 SPSS-T-XHI The remainder when 3x8 + kx2 + — 6 is

84. [CDS (I) Paper 19 Q29] 10,-1, 30 30. -A) **(a) (c)** 10 15 20 25 In the following table of inverse variation, what are the values of A, B and C respectively ? c 60 **(a) (b) (c) (d)** 15 10,-30 1 -30, 10, -1 - 1, -30,10 A person borrowed 5,000 at rate of interest annum and immediately lent it at 5-5%. After two years he collected the amount and settled his loan. What is the amount gained by him in this transaction ? **(b)** e 25 50 100 200

85. [CDS (I) Paper 19 Q37] A merchant with a certain capital and gains annually at the rate of 25%. At the end of 3 years he has 10,000. What is the original amount that the merchant invested ? **(b) (c)** 5,120 5,210 5,350 5,500 Which one Of the following decimal numbers is a rational number with denominator 37 ? **(a) (b) (c) (d)** 44 years 45 years 50 years 52 years 36.

86. **[CDS (I) Paper 19 Q48]** 49. 4—A) A car has an average speed Of 60 km per hour while going from Delhi to Agra and has an average speed Of y km per hour while returning to Delhi from Agra (by travelling the same distance). If the average speed of the car for the whole journey is 48 km per hour, then what is the value Of y ? **(a) (b) (c) (d)** 30 km hour 35 km per hour 40 km per hour 45 km per hour An article is sold at a profit Of 32%. If the cost price is increased by 20% and the sale price remains the same, then the profit percentage becomes **(a) (b) (c) (d)** 12% 15% 20% SPSS-T-XHI A. man can row ae a Speed Of x km/hr in still

87. **[CDS (I) Paper 19 Q49]** An article is sold at a profit Of 32%. If the cost price is increased by 20% and the sale price remains the same, then the profit percentage becomes To know what is the capacity Of the tank, which one of the following is correct ? **(a) (b) (c)** S2 is only sufficient SI, S2 and S3 are necessary Any two out of SI, S2 and sufficient N one Of the above are **(a) (b) (c) (d)** 12% 15% 20%

88. **[CDS (I) Paper 19 Q33]** The sum of ages of a father, a mother, a son Sonu and daughters Savita and Sonia is 96 years. Sonu is the youngest member of the family. The year Sonu was born, the sum of the ages of all the members of the family was 66 years. If the father's age now is 6 times that of Sonu's present age, then 12 years hence, the father's age will be **(b) (c) (d)** 44 45 years 46 years 48 years 'A' is thrice as good a workman as 'B' and takes 10 days less to do a piece of work than 'B' takes. The number of days taken by 'B' alone to finish the work is **(a) (b) (c)** 12 15 20 30 Out of 85 children playing badminton or table tennis or both, the total number of girls in the group is 70% of the total number of boys in the group. The number of boys playing only badminton is 50% of the number of boys and the total number of boys playing badminton is 60% of the total number of boys. The number of children playing only table tennis is 40% of the total number of children and a total of 12 children play badminton and table tennis both. The number of girls playing only badminton is **(a) (c)** 14 16 17 35 GTHY-T-XMT (11 84. of nd 'B'

89. **[CDS (I) Paper 19 Q40]** 41. If 13-A) Two equal amounts were borrowed at 5% and 4% simple interest. The total interest after 4 years amounted to **(a) (b)** 1075 1100 e 1125 1150 **(c)** 75 50 46 e 30 40. 41. Twelve **(d)**

90. **[CDS (I) Paper 19 Q37]** 38. (13-A) Two equal amounts were borrowed at 5% and 4% simple interest. The total interest after 4 years amounted to **(a) (b)** 1075 1100 e 1125 1150 **(c)** 75 50 46 e 30 40. 41. Twelve **(d)**

91. **[CDS (I) Paper 19 Q46]** 42. hand after every 72 minutes of correct time. How much time does the clock lose or gain in a day of normal time ? **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11

92. **[CDS (I) Paper 19 Q53]** 54. What percentage of students passed in all the three subjects ? **(a) (b) (c) (d)** 6 5 4 3 What percentage of students passed in exactly one subject ? **(a) (b) (c) (d)** 21 23 25 27 If the number of students is 360, then how many passed in at least two subjects ? **(a) (b) (c)** 270 263 265 260 What is the ratio of number of students who passed in both Physics and Chemistry to number of students who passed in both Biology and Physics but not Chemistry ? **(a) (b) (c) (d)** 7:10 10: 7

93. **[CDS (I) Paper 19 Q49]** 50. b2 —ac (17 -A) Consider the following for the next 04 (four) It is given that the equations x2 — Y2 = 0 and items that follow : (x — + Y2 = 1 have single solution. For this, the value of 'a' is In an examination of Class XII, 55% students passed in Biology, 62% passed in Physics, 60% passed in Chemistry, 25% passed in Physics and Biology, 30% passed in Physics and Chemistry, 28% passed in Biology and Chemistry. Only 2% failed in all the subjects. **(b) (c) (d)** 2 1

94. **[CDS (I) Paper 20 Q12]** In a hostel the rent per room is increased by 20%. If number of rooms in the hostel is also increased by and the hostel is always full, then what is the percentage change in the total collection at the cash counter ? **(a) (b) (d)** 30% Radha and Hema are neighbours and study in the same school. Both Of them use bicycles to go to the school. Radha's speed is 8 knv'hr whereas Hema's speed is 10 krrv•hr. Hema takes 9 minutes less than Radha to reach the far is the from the Of and Hema ? **(a) (b) (c) (d)** 6 km 6•5 km Which of the following pair of numbers is the solution of the equation + z IO ? **(b) (d)** 0,-2

95. **[CDS (I) Paper 20 Q17]** If a, b and c are positive integers such that , then What is the mean of 23 2 a, b and c ? **(a)** 19 20 21 22 The sum of three prime numbers is 100. If one Of them exceeds another by 36, then one Of the numbers is Proportion of WSpgp« 11 Both I and 11 Both 11 and 111 Both 1 and 111 All the three 1% **(b) (c)** 17 16. 29 43 Nono of the above 17. **(d)**

96. **[CDS (I) Paper 20 Q19]** 20. What is the unit place digit in the expansion of773? **(a) (c) (d)** 3 9 Suppose n is a positive integer such that (n2 + 48) is a perfect square. What is the number of such n ? **(a) (b) (c)** For **(a) (d)** One TWO Three 2 4Jä what is the value of y and x iß of z and y is 40% Of z. If x is p% of y, then what is the value Of p ? **(d)** 45 55 65 75 DFGT-T-HTM

97. **[CDS (I) Paper 20 Q39]** 35. loss of 6%, the seller gets 6 more. What is the cost price of the article ? 18 36 42 e 50 **(a) (b)** 4 and O O and 3 3 and O O and any integer Tushar takes 6 hours to complete a piece of work, while Amar completes the same Work in IO hours. If both Of them work together, then what is the time required to complete the **(c) (d)**

98. [CDS (I) Paper 20 Q74] 75. 3-A) What is the pementage decrease in the area of a triangle if its each side is halved ? **(d)** NO change The volume of a spherical balloon is increased by 700'h What is the percentage increase in its surface area ? **(b)** 300% 400% 450% 500% If the lengths of two parallel chords in a circle of radius IO cm are 12 cm and 16 em, then what is the distance between these two Chords 2 em or 14 em 3 or 21 Cm 4 cm or 28 em Considering two opposite vertices of a square of side 'a' as centres, two circular arcs are drawn Within Square joining the Other two vertices, thus forming two sectors. What is the common area in these two sectors ? **(a) (b)** The Median Mean Geometric mean Mode Graphical location Graphical location Ogive 73. **(c)** NO change following pairs relate to frequency The volume of a spherical balloon is increased by 700'h What is the percentage increase in its surface area ? distribution Of a discrete variable and its frequency polygon. Which one Of the following pairs is not correctly matched ? **(d)** 300% 400% 450% 500%

99. [CDS (I) Paper 21 Q7] WILt the average number of med pa in 2017? 53 (C) 54 **(d)** 56 is ffe Qproxmtate in Ltaliüß during F-iod 2014—17 ? **(a)** 66% **(b)** 68% **(d)** 76%

100. [CDS (I) Paper 21 Q11] Le g nd b be real + = 32 50 + b) 7 **(d)** be 2 one Of the 61%wing ? **(a)** be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? **(b)** 80 **(c)** 60 36 • **(d)** 24 14. WEch Of

101. [CDS (I) Paper 21 Q20] A Int to B at time me to C at sne 7% sinple intaut. 4ner 4 yeas A 01200 int«t B md C. HOW mueh A to C ? **(a) (b)** uSOOO **(c)** 050M) **(d)** e 10000

102. [CDS (I) Paper 21 Q21] A Ells two at se price, ma_king a profit of 3" on me a of on Otha. What is loss pfit on åe üÄEtion ? **(c)** 9% loss **(d)** 9% pin n. Ofl B in the ratio 4 : **(a)** 26 18 **(b)** 42 mimnes 36 **(c)** 45 minutæ **(d)** 46 mimit— 12 sæonds 2 4 18. WbatistheLOAof,-, —, — ? 9 27 18 27 10 27 20 19. If qudons whA is vahE Of k ? 10 3 10 5 3 10 9. 20. A Int to B at tlr time me to C sne 7% sinple intaut. Aner 4 yeas A 01200 intæt B md C. HOW mueh A to C ?

103. [CDS (I) Paper 21 Q3] Euh sava If ffeir expaldituræ ue in the ratio 3 : 2, ofl ? **(a)** 0800 **(d)** 0600 C — SDHY-T-MTH 1'. X, Z strt at point nd me time in the •le direction to run around a circular stadium. X complet— a mmd in 252 swonds, y in 308 *conds and Z in 198 %onds. After what time will they again at poim ? **(a)** 26 18 **(b)** 42 mimnes 36 **(c)** 45 minutæ **(d)** 46 mimit— 12 sæonds 2 4

104. [CDS (I) Paper 21 Q9] 20. A Int to B at tlr time me to C sne 7% sinple intaut. Aner 4 yeas A 01200 intæt B md C. HOW mueh A to C ? **(a) (b)** uSOOO **(c)** 050M) **(d)** e 10000

105. [CDS (I) Paper 21 Q23] The &iin nd bus two is in the ratio 3 4, If the by md bus ftre incras— by 30%, is åe mtio reviEd bus ? 9 **(a)** 17 12 32 19 21 24. N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ? **(b)** 2975 **(c)** 3094 **(d)** 3101

106. [CDS (I) Paper 21 Q3] Whd is ratio of radii ? SO. lagth nd Of a are by 200,4 nd tively. is the in of ? **(a)** 32% **(b)** 30% Sl. If the Of h+ue of a riÖt trian* is, IO cm, is tie of awh a riørt mgld ? **(a)** cm2 **(b)** 50 **(c)** 25 cm2 **(d)** IO cm2 52. squre is drawn such are lying on a circle Of What is the ratio of that Of "uue ? **(a)** 11 **(b)** 7:11 **(c)** 20:19 **(d)** 19:20 iu vatica •us 201 nun. of cinle

107. [CDS (I) Paper 21 Q84] It 11 a km if 120 is by ü•ain md the by cu. It takæ 40 minutes if 200km by trin by What ratio of of cu of ? 29 A x is is muimun. a; x to **(a)** -16 **(b)** -05 **(c)** 0-5 **(d)** 1-5 If divide wlnt is tarwt v*ue of n ? (i) 20 **(b)** 22 **(c)** 23 **(d)** 28 87. diøt in tbe Init's plEe of the by 30 ? The m Of the Of natural is 294. is of åe ? (i) 38 0) 34 (C) 30 **(d)** 26 C - SDHY-T-MTH 11 ze natural is A —n Ells two itans at Rs. one a Of at a of What is Of profit or 10" two ? profit **(c)** No ß0fit **(d)** p-ofit

108. [CDS (I) Paper 22 Q21] The price of an article X increases by 20% every year and price of article Y increases by 10% every year. In the year 2010, the price of article X was C 5000 and price Of article Y was 0000. In which year the difference in their prices exceeded 0000 for the first time ? **(a)** 2012 **(b)** 2013 **(c)** 2014 **(d)** 2015

109. [CDS (I) Paper 22 Q38] Jails with highest and smallest percent- age of trained convicts are respectively **(a)** F and D **(b)** D and F **(c)** C and A **(d)** D and A

110. [CDS (I) Paper 22 Q46] If the annual income of X is 20% more than that of then the income of Y is less than that of X by pO/o. What is the value of p ? **(a)** 10 **(b)** 16i **(c)** 36 **(d)** 27

111. [CDS (I) Paper 23 Q23] A shopkeeper sells his articles at their cost price but uses a faulty balance which reads IO(N) gm for gm. What is the actual profit **(d)** 25% 40% A river 3 m deep and 40 m wide is flowing at the rate Of 2 km/hr and falls into the sea. What is the amount of water in litres that will fall into the æa from this river in a minute ? **(a) (b) (c) (d)** litres ,OOO litres 40,000 litres 4,000 litres If a television set is sold at x, a loss Of 28% would be incurred. If it is sold at e y, a profit of 12% would be incurred. What is the ratio of **(b)** 41:9 31:9 23:9 14:9

112. [CDS (I) Paper 23 Q73] 74. -A) In a triangle, values of all the angles are integers (in degree measure). Which one Of the following cannot be the proportion of their measures ? **(a) (b) (d)** The length of a rectangle is increased by and breadth is decreased by Then the area of the new rectangle is **(a) (b) (c) (d)** neither increased nor decreased increased by 1% decreased by I % decreased by The surface areas of two spheres are in the ratio I **(b) (c) (d)** :

113. **[CDS (I) Paper 24 Q31]** A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(b)** overall discount? 316% **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

114. **[CDS (I) Paper 24 Q36]** The cost price of 100 manoes is equal to the selling price Of 80 manoes. What is the profit percentage? 16% 20% 24% 25% **(a) (b)** 48 minutes 50 minutes 54 minutes 60 minutes **(c) (d)** 15 litres 20 litres 24 litres 25 litres 36. The cost price of 100 manoes is equal to the selling price Of 80 manoes. What is the profit percentage? 16% 20% 24% 25% 37. X sells his goods 25% cheaper than Y and 25% dearer than Z. How much percentage is Z's goods cheaper than Y? 100 200 XVWS-T-MTK/SOA 39. If 20 can clean 20 noors in 20 days, then in how many days can 16 persons clean 16 floors?

115. **[CDS (I) Paper 24 Q37]** X sells his goods 25% cheaper than Y and 25% dearer than Z. How much percentage is Z's goods cheaper than Y? 100 200 XVWS-T-MTK/SOA **(a) (b)** 48 minutes 50 minutes 54 minutes 60 minutes **(c) (d)** 15 litres 20 litres 24 litres 25 litres 36. The cost price of 100 manoes is equal to the selling price Of 80 manoes. What is the profit percentage? 16% 20% 24% 25% 37. X sells his goods 25% cheaper than Y and 25% dearer than Z. How much percentage is Z's goods cheaper than Y? 100 200 XVWS-T-MTK/SOA 39. If 20 can clean 20 noors in 20 days, then in how many days can 16 persons clean 16 floors?

116. **[CDS (I) Paper 24 Q92]** In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? **(a)** and E **(b)** A, C and D only **(c)** C D and E **(d)** A, D and E only The following table shows the marks of 90 students in a test Of 80 marks : 1-10 11-20 21-30 31-40 41-50 51-60 61-70 71-80 Number o students 5 13 18 The Of students who have obtained less than or equal to 50% marks is 40% 45% XVWS-T-MTK/SOA

117. **[CDS (I) Paper 25 Q41]** If p varies directly as q and inversely as square of r, what is the 'rrcentage increase in p due to an increase in q by 20% and a decrease in r by 20% ? **(a)** 36 years **(b)** 40 years **(c)** 42 years **(d)** 45 years 35. The sum of numerator and denominator of a fraction is 10. If the numerator is increased by 3 and denominator is decreased by 1 , the fraction becomes I. What is the difference between numera- tor and denominator of the fraction ? 36. If the system of equations 7x + ky 27 and kx + 7y = 19 have unique solution, then which one of the following is correct ?

118. **[CDS (I) Paper 25 Q43]** A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ? **(a)** *400 **(b)** *450 **(c)** V550 **(d)** e600

119. **[CDS (I) Paper 25 Q90]** What is ZAQP equal to ? **(a)** 300 **(b)** 400 **(c)** 450 **(d)** 600 Consider the following for the next four (04) items that follow : The following Pie-Charts show the per- centage of different categories (A, B, C, D, E and F) of employees in a company in the year 2019 and 2020. The total number of employees was 4000 in 2019 and 5000 in 2020. 22% 20% 27% 2019 22% 25

120. **[CDS (I) Paper 25 Q92]** In which one of the following cate- gories of employees, the percentage change in number of employees in 2020 was maximum as compared to 2019 ? **(a)** 101 •39 cm2 **(b)** 111-39cm2 **(c)** 121 •39cm2 **(d)** 131-39cm2 Consider the following for the next two

121. **[CDS (I) Paper 25 Q93]** What was the percentage increase of category-E employees in 2020 as compared to that in 2019 ? **(a)** 101 •39 cm2 **(b)** 111-39cm2 **(c)** 121 •39cm2 **(d)** 131-39cm2 Consider the following for the next two

122. **[CDS (I) Paper 25 Q94]** There are two categories of employees whose strength increased by same percentage in 2020. What are the two categories ? **(a)** E and F **(b)** C and E **(c)** A and E **(d)** and C A - NYRG-T-MTK

123. **[CDS (I) Paper 25 Q95]** The rrrcentage increase in the total production of cars from 2014 to 2018 **(b)** 100% **(c)** 150% **(d)** 200% **(a)** 100% **(b)** 150% **(c)** 200% 96. Consider the production of each company separately. In how many instances was the percentage increase in production of cars over the previous year's production greater than 20% ? 97. In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? **(d)** 2015

124. **[CDS (I) Paper 25 Q96]** Consider the production of each company separately. In how many instances was the percentage increase in production of cars over the previous year's production greater than 20% ? **(a)** 100% **(b)** 150% **(c)** 200% 96. Consider the production of each company separately. In how many instances was the percentage increase in production of cars over the previous year's production greater than 20% ? 97. In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? **(d)** 2015

125. **[CDS (I) Paper 25 Q97]** In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? 27 **(a)** 2015 **(b)** 2016 **(c)** 2017 **(d)** 2018

126. **[CDS (I) Paper 26 Q15]** If — 17x+yand — x +17% x then What is the value Of ? **(a)** 17 **(c)** 23 GHSA-T-MTK/45A **(b)** 19 **(d)** 27 5

127. **[CDS (I) Paper 26 Q88]** What is the radius of the circle inscribed in a triangle whose sides are 4 cm, 75 cm and 85 cm? 2 cm shower, 5 em Of rain falls. What Direction : Consider the following data for the two (02) items that follow The table below gives the age-wise population **(a)** 21 90. In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? **(b)** 12 cm **(c)** 12•2 cm **(d)** 12•4 cm

128. **[CDS (I) Paper 26 Q89]** In a shower, 5 cm Of rain falls. What is the volume Of water that on Direction : Consider the following data for the two (02) items that follow The table below gives the age-wise population percentage of a city : Age group Below 30 years 30-3499 35-39-99 40—44-99 45_49-99 50 years and above 14•00 29•75 2625 18-50 11•50 2 hectare area Of land? **(a)** 100 cubic metre **(b)** 1000 cu bic metre **(c)** 4000 cubic me tre **(d)** 10000 cubic metre A bicycle wheel Of radius 35 cm makes n revolutions in moving I I km. What is the value Of n ? (Take **(a)** 500 **(b)** 1000 2500 5000 GHSA-•r-MTK/45A The number of persons below the age Of 40 years is 10•5 lakhs.

129. **[CDS (I) Paper 26 Q94]** In Which year, the percentage increase in expenditure is maximum as compared to its previous year? **(a)** 2012 **(c)** 2015 **(b)** 2014 **(d)** 2017 Direction Consider the following for the two (02) items that follow : The budget *lineation R represented in a pie diagram under five different heads A, B, C, D and E are respectively 40%, 18%, 9%, 25% and 8%. The total budget allocation is e300•4 lakhs.

130. **[CDS (I) Paper 26 Q100]** What is the percentage Of candidates who failed in at least two subjects? What is the of candidates Who failed in only one subject? **(a)** 306 under : Year 2011 2012 2013 2014 2015 2016 2017 138 15-4 10-4 13-1 15•8 172 19-4 **(b) (c)** Direction e 36048 lakhs e 300-36 lakhs 04032 lakhs e 18040 lakhs : Consider the following for the 93. HOW many times the increase in expenditure in a year exceeded by more than as compared to previous 94. In Which year, the percentage increase in expenditure is maximum as compared to its previous year? **(d)** 2012

131. **[CDS (I) Paper 26 Q99]** What is the percentage Of candidates Who failed in at least one subject? **(a)** 31% **(c)** 388% **(b)** 354% **(d)** 41-5% HOW many candidates passed in two or more subjects? **(a)** 461 **(c)** 345 **(b)** 405 **(d)** 306 1 P.T.O. Direction : Consider the ft)llowing data for

132. **[CDS (I) Paper 28 Q13]** If a, b, c, d, e and f satisfy 2a = 3b = 6c = 9d = 12e = 18f, then What is the value of (a + + d + e + fi ? **(b) (c) (d)** 4/7 2 5/2 9/2 If a, b, c are non•zero real numbers such that a + b + c = O, then What are the roots Of the equation ax2 + bx + c = O ? **(b) (c) (d)** 2, (da)-l Twelve percent of bananas bought by a fruit vendor got lost during transportation. On selling the remaining bananas, the vendor's overall profit turned Out to be If the vendor had not lost any bananas and had sold them at the price Of the remaining bananas, what would have been his profit percentage ? **(a) (b) (c)** 18 — % None of the above

133. **[CDS (I) Paper 28 Q61]** 25%. If p is the percentage increase in its length, then which one Of the following is **(a) (b) (c) (d)** 16 < p < 18 14 cpc 16 12 < p < 14 62.

134. **[CDS (I) Paper 28 Q84]** 85. 86. —A) If M is the median, then what is the value Of **(b) (c)** The 53 - 60 160 180 plinth Of a house has an area of 200 square metres. It is rectangular in shape and its length and breadth are in the ratio 2 : I. The owner Of the house extends the terrace by I m on each side. What is the percentage Of area that has increased in the terrace relative to the plinth ? **(a) (b) (c) (d)** 32% 15.5% A square sheet Of side length 44 em is rolled along one of its sides to form a cylinder by making opposite edges just to touch each other. What is the volume Of the cylinder ? (Take = **(a) (b)** (e) **(d)** 7 6776 cubic cm 6248 cubic cm 5896 cubic cm 5680 cubic cm

135. **[CDS (I) Paper 29 Q79]** 80. (29-A) 6 7 , x > I, then what is 81• **(a) (b) (c)** 60 m 55 m 675m A l»ught an article and sold it at a profit Of 20%. Had he bought it at less, What would have been the profit percentage if the selling price had been the same ? **(a) (b) (c)** 25% 40% If 2s a + b + c, then what is equal to ? **(a) (b) (c) (d)** ab + bc + ca 2 (ab + bc + ca)

136. **[CDS (I) Paper 29 Q82]** 83. Consider the following statements : The angle in a sector greater than a semi-circle is less than a right angle. If two sides of a pair of opposite sides Of a cyclic quadrilateral are equal, then its diagonals are also equal. Which Of the statements given above Ware correct ? **(a) (b) (c)** 25% 40% **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 If 2s a + b + c, then what is equal to ? If a, b, c, x, y, z are real numbers such that (a • b + e)' — 3 (ab + bc + ca) + 3 (x2 + z2) = O, then which one of the following is correct ? **(a) (b) (c) (d)** ab + bc + ca 2 (ab + bc + ca) **(a) (b) (c) (d)** azbzc, x zyzz*O a x zyzz=O JNUP-T-EMT

137. **[CDS Paper 30 Q7]** A person bought a book at a/4th Of its listed price and sold it at 50% more than its listed price. What is the percentage Of gain in the transaction ? **(a) (b) (c) (d)** 100% If the difference between the interior and exterior angles Of a regular polygon is 1440, then What is the number of sides of the polygon ? **(a) (b) (c) (d)** 12 16 18 20 If the sum and product Of the roots Of a quadratic equation are 2 and — 100 respectively, then which one of the following is correct ? **(a) (b) (c) (d)** There are infinitely many such equations having different roots. 'lhere is only one such equation which is Ther•e is only one such equation which is 100 = o. There is no such equation. Ifa : b: c: d •IS : , then what 5. (—a2 +c2 +d2) is the value Of A person bought a book at a/4th Of its listed price and sold it at 50% more than its listed price. What is the percentage Of gain in the transaction ? **(a) (b) (c) (d)** 6 **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

138. **[CDS Paper 30 Q43]** In a right-angled triangle ABC, AB 15 em, 44• BC 20 em and AC z 25 cm. Further, BP is the perpendicular on AC. What is the difference in the area of triangles PAB and PCB? **(a) (b) (c) (d)** 40 square em 42 square cm 45 square cm 48 square cm Let the positive numbers at, a2, as, agn be in GP. IfPis the GMofa1, as, ...,anandQ is the GM of 1, an an +3, a3n, then What is the GM Of 3n numbers ? p2Q PQ2 pV3 The cost price of y articles is equal to selling price of z articles. If y : z 5 : 4, what is the profit percentage ? **(a) (c) (d)** 25% SRSU-T-EMT (15-

139. **[CDS Paper 30 Q47]** The cost price of y articles is equal to selling price of z articles. If y : z 5 : 4, what is the profit percentage ? **(c)** 5 6 8 If (a + b) : c) a) z 5: 7 ; 6, then what is the value Of (a —b + c) : (a b — c) ? **(a) (b) (c)** Let x the compound interest at the end Of 3 years on a sum of 1000 at the rate Of annually and y be the simple interest at the end of 3 years on a sum Of 1000 at the annual rate of 11%. What is the difference between x and y ? **(a) (c) (d)** 25% **(a) (c) (d)** 16 e 15 SRSU-T-EMT

140. **[CDS Paper 30 Q59]** 60. p number of men can finish a piece Of work in q days. If there are more men, then the work will be finished 12 days earlier. What is the value of q ? **(a) (b) (c)** 100% 99% **(d)** 48 40 36 Cannot be determined insufficient data due to 9400 is distributed among p, Q, R in such a way that if 93, 24, 55 are from their respective shares, then they have money in the ratio 3 : 4 :

141. **[CDS (I) Paper 31 Q43]** A spherical wooden ball of radius r is to be divided into eight identical parts by cutting by planes passing through the same diameter. What is the surface area of each final piece ? 3 2 2m•2 3 4%r2 **(a)** IO mimin **(b)** 5 mimin **(c)** 10 cm/sec **(d)** 5 cm/sec 43. A spherical wooden ball of radius r is to be divided into eight identical parts by cutting by planes passing through the same diameter. What is the surface area of each final piece ? 3 2 2m•2 3 4%r2 A trolley with two wheels one metre 44. apan is moved clockwise on the circular track around a ground with radius 50 m

142. **[CDS (I) Paper 31 Q65]** A question is given followed by two statements I and 11 Consider the Question and the Statements and mark the correct option. : If the price of petrol Question goes up by 20%, by what percentage should the consumption be reduced so that the expenditure remains the same ? price of petrol per litre Statement-I was Rs. 90 Consumption was 24 Statement-II . litre before price hike Which one of the following is correct in respect of the above Question and the Statements ? **(a)** Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using any of the Statemen ts A - RAKU-T-EMT

143. **[CDS (I) Paper 31 Q71]** The total population of an area is 10,000 out of which males and females are equal in number. Out of the total population 30% are Newspaper readers. Out of the total newspaper readers, one-third read English News- paper. Out of the total English News- paper readers, 20% are females. What is the number ofmales who do not read Fnglish Newspaper ? 31 31 **(a)** 800 **(b)** 2100 **(c)** 4200 **(d)** Cannot be detennined due to insufficient data

144. **[CDS (I) Paper 31 Q74]** What is the square root of 64% ? **(a) (b) (c)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question cannot be answered even by using both the Statements together **(d)** 800

145. **[CDS (I) Paper 32 Q36]** 87. (13-A **(a) (b) (c) (d)** They did not practise discrimination on the basis of caste or class. They had a corporate organisation and canonical texts. They practised severe asceticism, Often eating very little food. They formed an alliance With Jainas against Buddhists. Consider the following statements about the of stirrup in India

146. **[CDS (I) Paper 33 Q53]** There are two employees X and Y. X's salary is first increased by 12% and then decreased by 10%, and Y's salary is first increased by 10% and then decreased by 12%. If their salaries at present are equal, then what was the ratio of initial salary of X to initial salary of Y? **(a)** 50: 53 **(b)** 51: 53 **(c)** 121: 126 **(d)** 121: 125 I P.T.O.

147. **[CDS (I) Paper 33 Q56]** Two students X and Y appeared in a test. The score Of X is 20 more than that of Y. If the score Of X is 75% of the sum of the scores of X and Y, then what is the ratio Of score Of X to score Of Y ? ANFB-T-EMT/4SA 21 **(a)** py/x SS. X and Y are two alloys Of copper **(b)** 8/3 **(c)** 2/3 **(d)** 1/3 An amount Of 10,000 is borrowed at 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount Of each installment

148. **[CDS (I) Paper 33 Q54]** In a village consisting Of p persons, x % can read and write. Of the males, only y% can read and write. Of the females, only z% can read and write. If x, y > Z, then what is the number Of males in the village? **(d)** py/x SS. X and Y are two alloys Of copper (Cu) and zinc (Zn) Alloy X is prepared by mixing Cu and Zn in the ratio 5 : 4, and alloy Y is prepared by mixing Cu and Zn in the ratio 5: 13 respectively. If equal quantities of alloys X and Y are melted to form a third alloy Z, then What is the ratio of Cu to Zn in Z ? **(a)** py/x SS. X and Y are two alloys Of copper **(b)** 8/3 **(c)** 2/3 **(d)** 1/3 An amount Of 10,000 is borrowed at 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount Of each installment

149. **[CDS (I) Paper 33 Q60]** A shopkeeper gives three consecutive discounts 10%, and 25% after which he sells the article at a profit Of 8% on the cost price. Had he sold the article after the first discount, how much profit would he have got? **(a)** 20% **(b)** 40% **(c)** None of the above 61. Let AD be the altitude of a triangle ABC If and **(d)** I only

150. **[CDS (I) Paper 34 Q32]** A reduction of 10% in the price of sugar enables a person to buy kg more for *2790. What is the reduced price per kilogram ? **(b)** z48 **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11 29. If and then what is log10

151. **[CDS (I) Paper 34 Q33]** A single discount which is equivalent to a series of discounts p % % is What is the value of p ? **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11 29. If and then what is log10

152. **[CDS (I) Paper 34 Q86]** What is x2—xy+y equal to ? **(a)** 487 **(b)** 483 **(c)** 481 **(d)** 480 For the next two (02) items that 39 follow : X is an alloy of copper and zinc. Y is an alloy containing 80% of copper, 4% of zinc and 16% of tin. A fused mass of X and Y is found to contain 74% of copper, 16% of zinc and 10% of tin.

153. **[CDS (I) Paper 34 Q87]** What is the percentage of copper in **(b)** 64% **(a)** 487 **(b)** 483 **(c)** 481 **(d)** 480 85. What

154. **[CDS (I) Paper 34 Q88]** What is the percentage of zinc in X ? **(a)** 400/0 **(b)** 36% **(c)** 34% **(d)** 30% A - KPRS-T-MTE

155. **[CDS (I) Paper 35 Q23]** A shopkeeper his goods at the cost price. By using false weights, he gained . What weight did he for I kg ? **(a) (b) (c) (d)** 960 gm 950 gm 925 gm 900 gm If the roots of the equation px2 — 40K • 96 = O are even integers, then what is the value Of p? **(a) (d)** 2 3 4 A number N consists Of two digits. The digit in the tens place is 3 times the digit in the units place. The digits are reversed and the resulting number is denoted by R. If N x R = 3627, then what is the product of the digits of the N ? **(b) (c)** 27 18 12 BFVS-T-TME

156. **[CDS (I) Paper 35 Q29]** 30. The ratio Of the cost price Of two articles X 31, and Y is I : **(a) (b) (c)** 6% 8% 31. 32. 33. Consider the follmving statements : I. The set Of all birds living on the Earth is an infinite set. 11. The set Of all re al numbers between O and IO is a finite set. Which Of the statements given above Ware correct ? **(d)**

157. **[CDS (I) Paper 35 Q2]** A businessman earns a profit of by selling X and a loss of by selling Y. If he loses 20% in this transaction, then what is the value of p ? **(a) (c) (d)** 6% 8% A sum Of money at the rate of 5% per annum compounded annually becomes n times in 100 years. What is the value of n ? (Given log102 0-301, log103 z 0-477 and log107 z 0845) **(a) (c) (d)** 98 99 100 More than 100 A variable y is directly proportional to a variable quantity xn. Given that when x z 2, y 208 and when x 3, y = 105•3. Which one Of the following is the constant of proportionality if it is greater than I ? **(a)** 1-1 **(b)** 1-3 **(c)** 2 4 If x z 3 — — 32/3, then what is — 3) (x — 6) equal to ? - 12 0 12 BFVS-T-TME

158. **[CDS (I) Paper 35 Q11]** The percentage investment in the 'Science, Technol(W and Environment' category is 40%. The angle (in degrees) for 'Agricultural Services' in a pie diagram is **(a)** 1369 13-39 13-19 1309 and Environment Transport and C ommunication Rural Developme nt Industry and Minerals Agricultural Services 3,000 1,500 3,800 2,500 Consider the following incomplete frequency No. 2 4 Which of the following statements Ware distribution with a total frequency Of 100 : correct ? Class Frequency 10 20 l. 11. The percentage investment in the 'Science, Technol **(b) (c) (d)**

CHAPTER 04

Total Archive Questions: 156 Questions

SIMPLE & COMPOUND INTEREST, RATIO & MIXTURES

Typical Exam Weightage: 6 - 10 Questions (6 - 10 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q51] The diameter of the moon is approximately one-fourth of that of the earth. What is the (approximate) ratio Of the volume of the moon to that Of earth ? **(d)** 1/64 12 **(a) (b)** (C) 1/16 1/32 1/48

2. [CDS (I) Paper 2 Q21] 22. Rs. 22 Rs. 38 'Rs. 46 **(a) (b) (c) (d)** litres litres 3-5 litres 2-5 litres If the ratio Of x to y ig 25 times the ratio Of y to x, then what is the ratio of x to y ? **(a) (c)** Out 25 of a group of swans, 7/2 times the square root Of the number are swimming in thc pool while the two remaining are playing Outside the pool. What is the total number Of swans ? 12 **(d)** 16 A railway ticket for a child costs half the full fare but the reservation charge is the same on half ticket as much as on full ticket. One reserved first class ticket for a journey between two stations is Rs. 362; 2 one full and one half reserved first class tickets cost Rs.

3. [CDS (I) Paper 2 Q3] What is the ratio Of their vertical heights ? **(a) (b) (c)** 2a3 -2aa **(d)**

4. [CDS (I) Paper 3 Q23] Three circular laminas Of the same radius are cut out from a Irger circular lamina. When the radius of each lamina cut out is ttw largest possible. tfrn what is the ratio (approximate) of the area Of the residual piece of the original lamina to its original total area T **(b)** 0-35 **(c)** 040 **(d)** (b45 **(a)** 72 **(b)** 216 **(c)** 372 **(d)** 600 18. If r and s are any real, numbers such that 1, then what is the maximum value Of the product rs ? 19. r is a non•zero real Humber such that r75 > PO. This is possible only when 20. The median BD of Che triangle ABC meets AC at D. If then which one of the following is correct ?

5. [CDS (I) Paper 3 Q29] The simple interest on a certain sum of money for 3 years at 8% per annum is half the compound interest On Rs. for 2 years at per annum. What is the sum placed On simple interest ? **(a)** 26. The shadow of a pole 6 metre high is 15 rnetre long and at the same time the shadow of a tree is 25 metre long. What is the height of the tree ? **(b)** 21 m **(c)** 10m **(d)** 35 m

6. [CDS (I) Paper 3 Q34] The wages, of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : **(a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** days 33. If of

7. [CDS (I) Paper 3 Q40] For what value of k, will the roots of the equation k.x2 — + 6 O in ratio of 2: 3? value of sin2150 + sin2200 **(a)** tan2t50 + tan2200 + tan2250 . + tan275Ø **(b)** cos2150 + cos2200 + cos2250 cos2750 **(c)** cot2150 — cot220e + cot2250 — + cot2750 **(d)** sec2150 + sec2200 + sec2250 + sec2750 42. Consider the following data : 5 9 2 The arithmetic mean of the above distribution is 296. What is the missing frequency ? 43. If a is an acute angle and then what is tan equal to ?

8. [CDS (I) Paper 3 Q60] Two Similar parallelograms have corres- ponding sides in the ratio I : k. What is the ratio of their areas ? **(a) (b) (c) (d)** 16 1:4k2 1:2k2

9. [CDS (I) Paper 3 Q71] The 10-5-20-5 only 205-30•5 only 10-5-20-5 and 20-5 - 30-5 There is no modal class length, breadth and height Of a rectangular parallelopiped are in ratio : **(a)** Rs. 500 @ 10%, Rs. 800 @ 12% **(b)** Rs. 700 @ 10%, Rs.600 @ **(c)** Rs. 800 @ 10%, Rs. 400 @ **(d)** @ 10%, ROOO @ 68. A man is watching from the top of a tower a boat sl*eding away from the tower. The boat makes an angle Of depression of 450 with the man's eye when at a distance of 60m from the bottom of tower. After 5 seconds, the angle of depression becomes 300. What is the approximate speed of the boat assuming that it is running in still water ? 70. The marks of the students of a class who appeared for a test in English are represented in the following frequency table : 11-20 21-30 31-40 41-50 51.60 What is/are the modal class

10. [CDS (I) Paper 4 Q9] The ratio between the ages of A and Bis 2 : **(a)** 20 years **(b)** 22 years **(c)** 24 years **(d)** 25 years IO. A mixture contains niilk and water in the ratio 5 : On Adding 5 litres Of water, the ratio Of' milk and water becomes 5 : 2. Wh#t is the quantity Of milk in the original mixture?

11. [CDS (I) Paper 4 Q5] After 8 ykars, their ages Will be in the ratio I : **(a)** 20 years **(b)** 22 years **(c)** 24 years **(d)** 25 years IO. A mixture contains niilk and water in the ratio 5 : On Adding 5 litres Of water, the ratio Of' milk and water becomes 5 : 2. Wh#t is the quantity Of milk in the original mixture?

12. [CDS (I) Paper 4 Q2] What is the difference between I their present ages? **(a)** 20 years **(b)** 22 years **(c)** 24 years **(d)** 25 years IO. A mixture contains niilk and water in the ratio 5 : On Adding 5 litres Of water, the ratio Of' milk and water becomes 5 :

13. [CDS (I) Paper 4 Q38] 10 **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

14. [CDS (I) Paper 4 Q89] If two corresponding sides Of two similar triangles, are in the ratio 9 : 4, then What • the ratio Of their areas ? **(c)** 81 16 Q-DETB-K-FN / 26A (d' 27 : 8 28 **(a)** The area Of the circle is e qual to that of square **(b)** The area o' the circle is larger than that Of square **(c)** The area of the circle is less than that bf square **(d)** No conclusiion can be drawn 87• If

15. [CDS (I) Paper 5 Q52] 53. A certain amount of money has to be divided between two persons P and Q in the ratio 3 : **(a) (b) (c)** A certain amount of money has to be divided between two persons P and Q in the ratio 3 : 5. But it Was divided in the ratio Of 2 : 3 and thereby Q loses IO. What was the amount ? None Of the above Consider the following statements . If p is a prime such that p + 2 is algo a prime, **(d)**

16. [CDS (I) Paper 5 Q56] 57. A bag contains 112 in the form of I-rupee, 50-paise and IO•paise coins in the ratio 3 : 8 : IO. What is the number Of 50-paise coins ? **(a) (b) (c) (d)** 112 108 96 84 TWO vessels are full Of milk With milk-water ratio I : 3 and 9 5 respectively. If both are mixed in the ratio 3 : 2, what is the ratio of milk and water in the new mixture ? 59. **(a) (b) (c)** 4: 15 None Of the above What i. least number of complete years in which a sum Of money at compound interest will be more than doubled ? **(a) (b) (c) (d)** 6 4 A piece Of Wire 78 cm long is bent in the form of an isosceles triangle. If the ratio of one Of the equal sides to the base is 5 : 3, then what is the length Of the base ? **(a) (b) (c) (d)** 16 cm 18 Crn 20 cm 30 cm P-PQFA-L-RZ (16 -A f

17. [CDS (I) Paper 5 Q93] A sum Of money On compound interest amounts to 9,680 in 2 years and to 10,648 in 3 years. What is the rate of interest per annum ? **(d)** 20% The eurwed surface Of a cylinder is 1000 cm2. **(a)** 15 cm **(b)** 20 cm **(c)** 25 cm **(d)** 30 cm Let D, E be the points on sides AB and AC respectively Of a triangle ABC such that DE is parallel to BC. Let AD 2 cm, DB = I em, AE = 3 cm and area' Of triangle ADE 3 em What is EC equal to ?

18. [CDS (I) Paper 6 Q5] If 9 is subtracted from each number, then they are in the ratio of 12 : **(a)** 423 **(b)** 843 **(c)** 1683 **(d)** 2523 Two numbers are in the ratio 3 : 5. If 9 is subtracted from each number, then they 2. 3. 4. Rs. 240 Rs. 480 Rs. 800 are in the ratio of 12 : 23. second number ?

19. [CDS (I) Paper 6 Q2] 3. 4. **(a)** 44 **(b)** 55 **(c)** 66 **(d)** 77 What is —I? the HCF of the polynomials 3x2 + 3x — I and **(d)** None of the above 5x-7Y+10 3x+2Y+1 Ilx+4y-10 9 then what is x + y equal to ? A certain sum at simple interest amounts 5. to Rs. 1350 in 5 years and to Rs. 1620 in 8 years. What is the sum ? **(a)** Rs. 700 **(b)** Rs. 800 **(c)** Rs. 900 **(d)** Rs. 1000 What is the number Of circles which pass 6. through three given points not in a straight line ? A cylindrical tank 7 m in diameter, contains 7. water to a depth Of 4 m. What is the total area of wetted surface **(a)** 110m2 **(b)** 126.5 m2 (C) 131.5 tn2 **(d)** 136.5 rn2 A wall iS of the form of a trapezium with 8. height 4 m and parallel sides being 3 m and 5 m. What is the Cost Of painting the wall, if the rate of painting is Rs. 251- per square metre Rs. 240 Rs. 480 Rs. 800 **(a) (b) (c) (d)** 400 The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 is **(a)** 423 **(b)** 843 **(c)** 1683 **(d)** 2523 Two numbers are in the ratio 3 :

20. [CDS (I) Paper 6 Q7] 8. then what is x + y equal to ? A certain sum at simple interest amounts to Rs. 1350 in 5 years and to Rs. 1620 in 8 years. What is the sum ? **(a)** Rs. 700 **(b)** Rs. 800 **(c)** Rs. 900 **(d)** Rs. 1000 What is the number Of circles which pass through three given points not in a straight line ? A cylindrical tank 7 m in diameter, contains water to a depth Of 4 m. What is the total area of wetted surface **(a)** 110m2 **(b)** 126.5 m2 (C) 131.5 tn2 **(d)** 136.5 rn2 A wall iS of the form of a trapezium with height 4 m and parallel sides being 3 m and 5 m. What is the Cost Of painting the wall, if the rate of painting is Rs. 251- per square metre **(a) (b) (c) (d)** 400

21. [CDS (I) Paper 6 Q70] The areas of two circles are in the ratio I : **(a)** 286 **(b)** 296 73. The areas of two circles are in the ratio I : 2. If the two circles are bent in the form Of squares, what is the ratio of their areas ? The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? **(c)** 3 units **(d)** 5 units

22. [CDS (I) Paper 6 Q2] If the two circles are bent in the form Of squares, what is the ratio of their areas ? **(a)** 286 **(b)** 296 73. The areas of two circles are in the ratio I : 2. If the two circles are bent in the form Of squares, what is the ratio of their areas ? The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? **(c)** 3 units **(d)** 5 units

23. [CDS (I) Paper 6 Q73] The Sum which amounts to Rs. 364•80 in 8 years at simple interest per annum **(a)** Rs. 285 **(b)** Rs. 280 Rs. 275 **(c) (d)** Rs. 270

24. [CDS (I) Paper 7 Q3] If 9 is added to each number, they will be in the ratio 3 : **(a)** Both statement-I and statement-ll are true and statement-II is the correct explanation of statement-I **(b)** Both statement-I and statement-II are true but statement-II is not the correct e*iplanation of statement-I **(c)** Statement-I is true, but statement-II is **(d)** Statement-ll is true, but statement-I is false A Square, a circle and an equilateral triangle have same perimeter. Consider the following statements : I. The area of square is greater than the area of the triangle. 2. The area of circle is less than the area oftriangle. Which of the statements Ware correct ?

25. [CDS (I) Paper 7 Q91] 24 The ratio of two complementary angles is I : **(a)** 64 minutes **(b)** 80 minutes **(c)** 96 minutes **(d)** 120 minutes The pair of numbers which are relatively prime to each other is :

26. [CDS (I) Paper 7 Q90] 91. The ratio of two complementary angles is I : **(a)** 64 minutes **(b)** 80 minutes **(c)** 96 minutes **(d)** 120 minutes The pair of numbers which are relatively prime to each other is :

27. [CDS (I) Paper 8 Q7] 8. 2 In a certain school the ratio of boys to girls is 7 : **(a)** 25% **(b)** 334% X can do a piece of work in 25 days. Y is 25% more efficient than X. Ihe number of days taken by Y is • **(c)** 15 days **(d)** 20 days

28. [CDS (I) Paper 8 Q20] Sex ratio is defined as the number of females per 1000 males. In a place the total inhabitants are out of which 935,000 are females. What is the sex ratio for the place ? **(a)** 935 **(b)** 1000 **(c)** 1935 **(d)** 9350 6 2 23 2

29. [CDS (I) Paper 8 Q93] 94. 26 O The circumferences Of two circles are in the ratio 2 : **(a)** 32.25% **(b)** 34.75% If the circumference Of a circle is equal to the perimeter of square, then which one of the following is correct ? **(c)** Area of circle = Area Of square **(d)** Area of circle 2 Area of square

30. [CDS (I) Paper 8 Q3] What is the ratio of their areas ? : 27 The area of a circle inscribed in an equilateral triangle is 154 cm2. What is the perimeter of the triangle ? **(a)** 21 cm **(b)** 426 cm **(d)** 42 cm The lengths of three line segments in cm are given in each of the four cases. Which one of the following cases is not suitable to be the three sides of a triangle ? In histogram the width of the bars is proportional to : **(a)** frequency **(b)** number of classes **(c)** class interval **(d)** None of the above

31. [CDS (I) Paper 9 Q83] ABC ia a right-angled triande such that a point On BC such that ÉD AB. ratio of BD : DC for any value Of a and b given by O.DETA.N+MN 22 **(a)** ABC be an equilateral triangle. If the BC is produced to the point D So Olat 2CT **(b)** 40 em 45 em 50 em 80 em 83. ABC ia a right-angled triande such that a point On BC such that ÉD AB. he ratio of BD : DC for any value Of a and b given by ABC is a tnangle where BC = 2AB, C 30' and A Thc magnitude of the side AC 3 3BC BC of triangle ABC incel in J. What is BJC equal to ? 900 4 soo 2 ABC, - 900 CD is perpendicular AB. if An 4 cm ana 9 em, ehe of DC O.DETA.N+MN 22 84. 85. of Of ig 22 Let side **(c)** ABC be an equilateral triangle. If the BC is produced to the point D So Olat 2CT **(d)** 40 em 45 em 50 em 80 em 83. ABC ia a right-angled triande such that a point On BC such that ÉD AB. ratio of BD : DC for any value Of a and b given by O.DETA.N+MN 22

32. [CDS (I) Paper 12 Q8] The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : **(a)** I Q years 12•5 years **(b)** 10-5 years 13. A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(c)** 14 11. 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : 4. What is the ratio Of ages Of A, B and C? **(d)** 6: 15 : 20 9. When an article is sold at 20% discount, the selling price is e 24. What will be the selling price when the discount is 30%?

33. [CDS (I) Paper 12 Q4] What is the ratio Of ages Of A, B and C? **(a)** I Q years 12•5 years **(b)** 10-5 years 13. A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(c)** 14 11. 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : 4. What is the ratio Of ages Of A, B and C? **(d)** 6: 15 : 20 9. When an article is sold at 20% discount, the selling price is e 24. What will be the selling price when the discount is 30%?

34. [CDS (I) Paper 12 Q12] A sum of money becomes 3 times in S years at simple interest. In how Of many years Will the same Sum become 6 times at the same rate Of simple interest? IO years **(b)** I Q years 12•5 years **(d)** 10-5 years **(a)** I Q years 12•5 years **(b)** 10-5 years 13. A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(c)** 14 11. 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : 4. What is the ratio Of ages Of A, B and C? **(d)** 6: 15 : 20 9. When an article is sold at 20% discount, the selling price is e 24. What will be the selling price when the discount is 30%?

35. [CDS (I) Paper 12 Q11] 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : **(a)** I Q years 12•5 years **(b)** 10-5 years 13. A man buys 200 Oranges for I ,OOO HOW many oranges for e 100 can he sell So that his profit percentage is 25%? 10 16 20 **(c)** 14 11. 8. The ratio Of ages Of A and B is 2 : 5 and the ratio Of ages Of B and C is 3 : 4. What is the ratio Of ages Of A, B and C? **(d)** 6: 15 : 20 9. When an article is sold at 20% discount, the selling price is e 24. What will be the selling price when the discount is 30%?

36. [CDS (I) Paper 13 Q15] The difference bet•ween compound interest and simple interest at the same rate Of interest R percent per annum on an amount Of 15,000 for 2 years is **(a) (b)** A person could save of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **(c)** 23 **(d)**

37. [CDS (I) Paper 13 Q16] There is 60% increase in an amount the in 6 years at simple interest. What will be the compound interest on 12,000 after 3 years at the same rate of Interest? e 2,160 e 3,972 6,240 **(a) (b)** A person could save of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **(c)** 23 **(d)**

38. [CDS (I) Paper 13 Q17] 16 litres of a mixture contains milk and water in thc ratio 5 **(a) (b)** A person could save of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **(c)** 23 **(d)**

39. [CDS (I) Paper 13 Q3] If 4 litres of milk is added to this mixture, the ratio Of milk to water in the new mixture would be **(a) (b)** A person could save of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **(c)** 23 **(d)**

40. [CDS (I) Paper 13 Q19] In how much time would thc simple interest on a principal amount be Ov 125 times the principal amount at per annum? **(b)** years — years years 2 — years **(a)** years — years years 2 — years **(b) (c) (d)** _g_f_ km _æ_ km Pc 20. A man rows downstream 32 km and 14 km upstream, and he takes 6 hours to cover each distance. What is the speed Of the current?

41. [CDS (I) Paper 13 Q30] When a ball bounces, it rises to — 3 of the height from which it fell. If the ball is dropped from a height Of 36 m, how high will it rise at the third bounce? IO— **(d)** The efficiency of p is twice that Of Q, whereas the efficiency Of P and Q together is three times that Of R, If P, Q and R work together on a job, in what ratio should they share their earnings? **(a)** 200 m **(b)** 300 m **(c)** 270 m **(d)** 160 m 29. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of IO km/hr and I I km/hr respec- tively. What is the distance between them after 6 minutes?

42. [CDS (I) Paper 14 Q2] The speeds of A, B and C are in the ratio 50 : 45 : **(a)** 43-2% **(b)** 56-8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ? **(c)** 5<k$11 21. The number of pairs **(d)** Zero

43. [CDS (I) Paper 14 Q32] Three athletes run a 4 km race. Their srredsareintheratio 16:d5: **(a)** 10 s **(b)** 16 s **(c)** 21 s 32. Three athletes run a 4 km race. Their srredsareintheratio 16:d5: 11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is **(d)** 1000m

44. [CDS (I) Paper 14 Q53] The speeds of three buses are in the ratio 2 : 3 : **(a)** 3600 **(b)** 6200 **(c)** 6028 **(d)** 6400 50. Consider all positive two digit numbers each of which when divided by 7 leavés a remainder 3. What is their sum ?

45. [CDS (I) Paper 14 Q4] Ihe time taken by these buses to travel the same distance will be in the ratio **(a)** 3600 **(b)** 6200 **(c)** 6028 **(d)** 6400 50. Consider all positive two digit numbers each of which when divided by 7 leavés a remainder 3. What is their sum ?

46. [CDS (I) Paper 14 Q73] A water tank, open at the top, is hemi- spherical at the bottom and cylindrical above it. Ihe radius is 12m and the capacity is 3312rt m3. The ratio of the surface areas of the spherical and cylindrical portions is B-ABVB-O-GPU - A **(a)** will be 75. Let **(b)** 52% **(c)** 540/0 **(d)** 98 cm 108 cm2

47. [CDS (I) Paper 14 Q78] The area of a trapezium is 336cm2. If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is **(a)** 20 cm **(b)** 22 cm **(c)** 24 cm **(d)** 26 cm

48. [CDS (I) Paper 14 Q88] A sum of Rs. 10,000 is deposited for 1 year at the rate of interest 10% com- pounded half yearly. What will be the interest at the end of one year ? **(a)** Rs. 1000 **(b)** RK1025 **(c)** Rs. 1050 **(d)** RS. 1100 a mixture of milk and water of

49. [CDS (I) Paper 14 Q89] In volume 30 litre, the ratio of milk and water is 7 : **(a)** Rs. 1000 **(b)** RK1025 **(c)** Rs. 1050 **(d)** RS. 1100 a mixture of milk and water of 89. In volume 30 litre, the ratio of milk and water is 7 : 3. The quantity of water to be added to the mixture to make the • ratio of milk and water 1 : 2 is

50. [CDS (I) Paper 14 Q3] The quantity of water to be added to the mixture to make the • ratio of milk and water 1 : 2 is **(a)** 30 **(b)** 32 **(c)** 33 **(d)** 35

51. [CDS (I) Paper 15 Q2] Which one of the following is correct? **(b) (c) (d)** (k + kp2 k2q— (k + l)p2 k2p2 — (kt1)2q I. If the roots or the equation O are in the ratio p: q, then is equal to **(a)** 3 7. 8. 9. The sum and difference of two expressions are 5x2 — x — 4 and x 2 + 9x — 10 respectively. The HCF of the two expressions Will be **(b) (c)** -15 If then value of b2+C2 will be If the polynomial x + px + qx — x the 3. If 2 + g m and _ + n, then is — equal to? **(d)** np — mq 4. what and and is divisible by

52. [CDS (I) Paper 15 Q27] The annual incomes or two persons are in the ratio 9 : 7 and their expenses are in the ratio 4 : **(a) (b)** 6 None of the above B'E'2+CF'2 Which one Of the following is correct in respect of the above statements? **(c)** Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I **(d)** Both Statement I and Statement 11 are true but Statement 11 is not thc correct explanation of Statement I

53. [CDS (I) Paper 16 Q12] TWO men, and B run a 4 km racc on a course km round. If their speeds are in the ratio 5 : 4, how often does the winner pass the other ? **(a)** Once **(b)** Twice **(c)** Thrice **(d)** Four times SDC-S-TRD - A 4

54. [CDS (I) Paper 16 Q16] In an office, one-third of the workers are women, half of the women are married and one-third of the married women have children. If three-fourth of the men are married and one-third of the married men have children, then what is the ratio of married women to married men ? **(a)** 2 and 4 **(b)** 4 and 2 **(c)** 2 and 2 **(d)** 4 and 4 11. A boy saves Rs. 4•65 daily. What is the least number of days in which he will be able to save an exact number of rupees ?

55. [CDS (I) Paper 16 Q29] There are twelve friends A, B, C, D, E, F, G, H, I, J, K and L who invested money in some business in the ratio of and the duration for which they invested the money is in the ratio of respectively. Who will get the maximum profit at the end of the year ? **(a)** F only **(b)** G only **(c)** Both F and G **(d)** Neither F nor G

56. **[CDS (I) Paper 16 Q35]** Incomes of Mahesh and Kamal are in the ratio I : 2 and their expenses are in the ratio 1 : **(a)** 11 31. If2p+3q= 12 and4p2+4pq—3q2 — 126, then what is the value of p + 2q ? 29. There are twelve friends A, B, C, D, E, 33. What is the unit digit of 7139 F, G, H, I, J, K and L who invested money in some business in the ratio of and the duration for which they invested the money is in the ratio of respectively. Who will get the maximum profit at the end of the year ? **(b)** F only **(c)** G only **(d)** Both F and G

57. **[CDS (I) Paper 16 Q39]** The speeds Of three cars are in the ratio 2 : 3 : **(a)** 12:45 hours **(b)** 5:00 p.m. **(c)** 11:45 a.m. **(d)** 11:50 a.m. 38. A train is travelling at 48 kmfhour completely crosses another train having half its length and travelling in opposite direction at 42 km/hour in 12 s. It also passes a railway platform in 45 s. What is the length of the platform ?

58. **[CDS (I) Paper 16 Q4]** What is the ratio between the times taken by these cars to travel the same distance ? **(a)** 12:45 hours **(b)** 5:00 p.m. **(c)** 11:45 a.m. **(d)** 11:50 a.m. 38. A train is travelling at 48 kmfhour completely crosses another train having half its length and travelling in opposite direction at 42 km/hour in 12 s. It also passes a railway platform in 45 s. What is the length of the platform ?

59. **[CDS (I) Paper 17 Q29]** 9 11 A sum Of 8,400 was taken as a loan. This is to be paid in two equal instalments. If the rate Of interest is 10% per annum, compounded annually, then the value Of each instalment **(a) (b) (c) (d)** 18 20 25 30 28.

60. **[CDS (I) Paper 17 Q36]** 37. 38. e of (IO-A) A 225 m long train is running at a speed Of 30 kmmour. How much time does it take to Cross a man running at 3 km/hour in the same direction ? **(a) (b) (c) (d)** 40 seconds 30 seconds 25 seconds 15 seconds Which one among the following is the largest ? **(a)** 9 11 **(b)** 14 10 The difference between the simple and the compound interest on a certain sum of money at 4% per annum in 2 years is

61. **[CDS (I) Paper 17 Q74]** 75. and (20 -A) The angles Of a triangle are in the ratio 2 : 4 **(a) (b) (c) (d)** •BLS-S-FDE If x = a cos B + b sin and y = a sin — b cos 6, 72. 68. then What is + Y2 equal to ?

62. **[CDS (I) Paper 17 Q75]** The areas of two circular fields are in the ratio 16 : **(a) (b) (c) (d)** •BLS-S-FDE If x = a cos B + b sin and y = a sin — b cos 6, 72. 68. then What is + Y2 equal to ?

63. **[CDS (I) Paper 18 Q7]** 8. **(a) (b) (c)** STS-M-TPT 2 2 3 What is the positive value of m for which the roots of the equation 12x2 + mx + 5 = 0 are in the ratio 3 : 2 ? **(b)** 12 5 **(c)** 12 12 **(d)** 5 Let fix) and g(x) be two polynomials (with real coemcients) having degrees 3 and 4 respectively. What is the degree of ? G.

64. **[CDS (I) Paper 18 Q45]** There are 350 boys in the first three standards. The ratio of the number of boys in first and second standards is 2 : 3, while that of boys in second and third standards is 4 : **(a) (b) (c) (d)** 4 km 8 km 16 km 20 km

65. **[CDS (I) Paper 18 Q43]** The difference between the compound interest (compounded annually) and simple interest on a sum of money deposited for 2 years at 5% per annum is **(a) (b) (c) (d)** 4 km 8 km 16 km 20 km

66. **[CDS (I) Paper 18 Q96]** The pie diagrams on the monthly expenditure of two families A and B are drawn with radii Of two circles taken in the ratio 16 : 9 to compare their expenditures. Which one of the following is the appropriate data used for the above mentioned pie diagrams ? **(a) (c) (d)** 16,000 and 9,000 8,000 and 4,500 e 25,600 and 8,100 e 4,000 and e 3,000 Consider the following statements : Statement I: The value of a random variable having the highest frequency is mode. Statement 11 : Mode is unique. Which one of the following is correct in respect of the above statements ? **(a) (b) (c) (d)** Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I Both Statement I and Statement 11 are true but Statement 11 is not the correct explanation of Statement I Statement I is true but Statement 11 is Statement I is false but Statement 11 is Which one of the following is not correct ? The proportion of various items in a pie diagram is proportional to the **(a)** Areas of slices **(b)** Angles of slices **(c)** Lengths of the curved arcs of the sl ices **(d)** Perimeters of the slices STS-M-TPT (30 ure 97. adii

67. **[CDS (I) Paper 18 Q94]** 95. 96. (30 —A) The pie diagrams on the monthly expenditure of two families A and B are drawn with radii Of two circles taken in the ratio 16 : 9 to compare their expenditures. Which one of the following is the appropriate data used for the above mentioned pie diagrams ? **(a)** 12 6 **(b) (c) (d)** 16,000 and 9,000 8,000 and 4,500 e 25,600 and 8,100 e 4,000 and e 3,000 The total number of live births in a specific locality during different months of a specific year was obtained from the omce Of the Birth Registrar. This set of data may be called Consider the following statements : Statement I: The value of a random variable having the highest frequency is mode. Statement 11 : Mode is unique. Which one of the following is correct in respect of the above statements ?

68. **[CDS (I) Paper 19 Q44]** 45. An alloy A contains two elements, copper and tin in the ratio Of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3 : **(a) (b) (c)** 8 years 9 years IO years 11 years 43. An alloy A contains two elements, copper and tin in the ratio Of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3 : 4. If 20 kg Of alloy A, 28 kg of alloy B and some more pure N **(d)** 3 kg

69. [CDS (I) Paper 19 Q43] An alloy A contains two elements, copper and tin in the ratio Of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3 : **(a) (b) (c)** 8 years 9 years IO years 11 years 43. An alloy A contains two elements, copper and tin in the ratio Of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3 : 4. If 20 kg Of alloy A, 28 kg of alloy B and some more pure N **(d)** 3 kg

70. [CDS (I) Paper 19 Q50] 51. A, B, C, D and E start a partnership firm. Capital contributed by A is three times that contributed by D. E contributes half Of A's Contribution, B contributes one-third of E's contribution and C contributes two-third of A's contribution. [f the difference between the combined shares Of A, D and E and the combined shares Of B and C in the total profit of the firm is 13,500, what is the combined share of B, C and E ? (The shares are supposed to be proportional to the contri bu tions) **(c)** 13,500 e 18,000 e 19,750 20,250 A Pie Chart is drawn for the following data : Sector Agriculture and Rural Develo t Irrigati on Energy Industry and Minerals Transport and Communication Social Services Percen tage 12-9 12-5 272 15-4 15-9 16•1 What is the angle (approximately) subtended by the Social Services Sector at the centre Of the circle ? **(c)** 450 460 580 980 SPSS-T-XHI (160 n **(a) (b) (c)** 15, 5 12, 8 16, 4 18, 2 **(d)** 13,500 e 18,000 e 19,750 20,250 53. 54. The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10•5 and the arithmetic mean of the last 6 observations is 11•5. What is the sixth observation ? A Pie Chart is drawn for the following data : Sector Agriculture and Rural Develo t Irrigati on Energy Industry and Minerals Transport and Communication Social Services Percen tage 12-9 12-5 272 15-4 15-9 16•1

71. [CDS (I) Paper 19 Q22] If a = xyp—l, b c aq ¯ ¯ P cP — q is equal to = zxr —1, then **(a)** ab 1 **(b)** abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to **(c)** abc **(d)** xyz

72. [CDS (I) Paper 19 Q25] 26. The sum of all products taken two at a time out of the numbers ± 1, ± 2, ± 3, ± 4 is **(a) (b) (c) (d)** - 30 30 55 The remainder when 3xa — 2x2y — 13xy2 + 10y3 is divided by (x — 2y) is equal to **(a) (c) (d)** Zero If ab + bc + ca = O, then the value of **(a) (b) (d)** 2 What is the principal amount which earns 210 as compound interest for the second year at 5% per annum ? **(a) (b) (c) (d)** GTHY-T-XMT 2000 3200 4000 4800 t

73. [CDS (I) Paper 19 Q38] X bought 4 bottles of lemon juice and Y bought one bottle of orange juice. Orange juice per bottle costs twice the cost of lemon juice per bottle. Z bought nothing but contributed 50 for his share of the drink which they mixed together and shared the cost equally. If Ts e 50 is covered from his share, then what is the cost of one bottle of orange juice ? **(c)** 75 50 46 e 30 Ten (10)' years before, the ages of a mother and her daughter were in the ratio 3 : **(a) (b)** 1075 1100 e 1125 1150 **(c)** 75 50 46 e 30 40. 41. Twelve **(d)**

74. [CDS (I) Paper 19 Q1] Tn another IO years from now, the ratio Of their ages will be 13 : **(a) (b)** 1075 1100 e 1125 1150 **(c)** 75 50 46 e 30 40. 41. Twelve **(d)**

75. [CDS (I) Paper 19 Q48] 15-A) There are n zeros appearing immediately after the decimal point in the value of It is given that the value of log102 = 0•30103. The value of n is **(b) (c)** 25 19 18 17 The ratio of the sum and difference of the ages of the father and the son is 11 : **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11

76. [CDS (I) Paper 19 Q2] The ratio of their ages is 8 : **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11

77. [CDS (I) Paper 19 Q5] The ratio of their ages after the son attains twice the present age will bell : **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11

78. [CDS (I) Paper 19 Q47] 48. The ratio of the sum and difference of the ages of the father and the son is 11 : **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11

79. [CDS (I) Paper 19 Q57] 58. Data on ratings of hotels in a city is measured **(a) (b) (c) (d)** Nominal scale Ordinal scale Interval scale Ratio scale The average marks of section A are 65 and that of section B are

80. [CDS (I) Paper 19 Q56] 57. 58. (19-A) **(a) (b) (c) (d)** Nominal scale Ordinal scale Interval scale Ratio scale The average marks of section A are 65 and that of section B are

81. [CDS (I) Paper 19 Q1] The ratio of area of T to that of S is 2 : **(a) (b) (c)** x +3 cos x +3 cos x + cos6x? 1 8 Walls **(d)** 12 16

82. [CDS (I) Paper 19 Q87] In a triangle ABC, the sides AB, AC are produced and the bisectors Of exterior angles of Z ABC and Z ACB intersect at D. Z BAC 500, then Z BDC is equal to If **(a) (b) (c)** 1150 650 550 400 cones have their heights in the ratio 1 If the radii of their bases are in the ratio 3 : , then the ratio of their volumes will be **(a) (b)** If two lines AB and CD intersect at O such that Z AOC at O are **(b) (c) (d)** 400 300 306 600 GTHY-T-XMT = 5 Z AOD, then the four angles , 400, 1400, 1400 , 300, 1500, 1500 , 450, 750, 2100 , 600, 1200, 1200 (27- If a point P moves such that the sum of the 88. squares Of its distances from two fixed points A and B is a constant, then the locus of the point p is If

83. [CDS (I) Paper 20 Q44] 45. What is the least number Of complete years in which a sum of money put Out at annual compound interest will be more than **(a) (b) (d)** 3 4 5 6 A person divided a sum Of 17,200 into three parts and invested at and 9% per annum simple interest. At the end of two years, he got the same interest on each part of money. What is the money invested at 9% ? **(a) (c) (d)** e 3,200 4,800 e 5,000 DFGT-T-HTM (150

84. **[CDS (I) Paper 21 Q45]** LA two liÜ p üld q be puallel. Corwider two pints j and C on the line p and two E On the line q. Ille line thr•agh B E line C D at A in betwem two p g. If then is the ratio of ara of tiangle ABC to thü Of ADE ? **(c)** 16:81 **(a)** 612 **(b)** 212 43. A of wire in of a Etor Of a circle Of 20 cm, suM01ding an mgle 1500 at If it is ban ih 6rm Of a circle. what will be its ? **(c)** None Of the P, Q R Üe mid-points Of si&s of a &iangle of 128 cm2. If a &imøe ABC is &awn by joining of side Of riangle POR, what is of timgle ABC? **(d)** 4 cm2 O, 8 cm2

85. **[CDS (I) Paper 21 Q46]** An dianÜ a squze constucted using *Ilic wire of length. WhM is ratio of ara of üimøe to Of ? **(c)** 46•.9 **(d)** 26 **(a)** 612 **(b)** 212 43. A of wire in of a Etor Of a circle Of 20 cm, suM01ding an mgle 1500 at If it is ban ih 6rm Of a circle. what will be its ? **(c)** None Of the P, Q R Üe mid-points Of si&s of a &iangle of 128 cm2. If a &imøe ABC is &awn by joining of side Of riangle POR, what is of timgle ABC? **(d)** 4 cm2 O, 8 cm2

86. **[CDS (I) Paper 21 Q49]** Two cyli&s of have in ratio 2 : **(a)** 32% **(b)** 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ? **(c)** cm2 **(d)** 50

87. **[CDS (I) Paper 22 Q28]** The ratio of speeds of Xand Y is 5 : **(a)** 49x2 -f- 3x + 2 **(b)** 49x2 _ 3x + 2 **(c)** 49++2 **(d)** 59++2 24. If x varies asyz, theny varies inversely

88. **[CDS (I) Paper 22 Q32]** A sum was put at simple interest at certain rate for 2 years. Had it been put at higher rate of interest, it would have fetched e24 more. What is the sum ? 11 **(c)** ooo **(d)** 0200 **(a)** 50 knvhr **(b)** 40 km/hr **(c)** 35 knvhr **(d)** 25 km/hr 30. If 6 men and 8 women can do a piece of work in IO days; and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman ?

89. **[CDS (I) Paper 22 Q33]** The population of two villages is 1525 and 2600 respectively. If the ratio of male to female population in the first village is 27 : 34 and the ratio of male to female population in the second village is 6 : 7, then what is the ratio of male to female population of these two villages taken together ? 33 41 85 82 71 90 5 6 **(a)** 50 knvhr **(b)** 40 km/hr **(c)** 35 knvhr **(d)** 25 km/hr 30. If 6 men and 8 women can do a piece of work in IO days; and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman ?

90. **[CDS (I) Paper 22 Q34]** In a class room the ratio of number of girls to that of boys is 3: **(a)** 50 knvhr **(b)** 40 km/hr **(c)** 35 knvhr **(d)** 25 km/hr 30. If 6 men and 8 women can do a piece of work in IO days; and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman ?

91. **[CDS (I) Paper 22 Q68]** ABCD is a plate in the shape of a parallelogram. EF is the line parallel to DA and passing through the point of intersection O of the diagonals AC and BD. Further, E lies on DC and F lies on AB. The triangular portion DOE is cut out from the plate ABCD. What is the ratio of area of remaining portion of the plate to the whole ? 5 8 5 3 4 7 **(a)** +1 **(b)** + 3 **(c)** l **(d)** 266+1

92. **[CDS (I) Paper 22 Q76]** The length and breadth of a rectangle are in the ratio 4 : **(a)** — •E **(b)** litre **(c)** litre , **(d)** — litre 10

93. **[CDS (I) Paper 23 Q64]** 65. (23-A) ABC is an triangle. Ihe side BC is trisected at D such that BC = 3 BD. What is the ratio Of AD2 AB2 ? **(c) (d)** Consider the following statements **(a) (b) (c)** 300 900 1200 , z B -90•and64. O, then what is 65. ABC is an triangle. Ihe side BC is trisected at D such that BC = 3 BD. What is the ratio Of AD2 AB2 ? **(d)**

94. **[CDS (I) Paper 23 Q2]** 'me diagonals Of a divide each other proportionally. Any line drawn parallel to the parallel sides of a trapezium divides the nan-parallel sides proportionally. Which of the above statements Ware correct ? **(a) (b) (c) (d)** 1 only 2 only Both I and 2 Neither I nor 2 If H, C and V are respectively the height, curved surface area and volume of a cone, then what is 37tVHa 9V2 equal to ? **(c)** c2H2 5 c2H1 7 c2H2 In a quadrilateral ABCD AB2 + BC2 + CD2 - AD2 **(a) (b) (d)** 300 900 1200 , z B -90•and64. O, then what is

95. **[CDS (I) Paper 23 Q65]** ABC is an triangle. Ihe side BC is trisected at D such that BC = 3 BD. What is the ratio Of AD2 AB2 ? **(c) (d)** Consider the following statements In a A ABC, AC 12 cm, AB 16 and AD 62. is the bisector of Z A. If BD 4 em, then what is DC equal to ? (23-A) **(a) (b) (c)** 300 900 1200 , z B -90•and64. O, then what is 65. ABC is an triangle. Ihe side BC is trisected at D such that BC = 3 BD. What is the ratio Of AD2 AB2 ? **(d)**

96. **[CDS (I) Paper 23 Q4]** What is the ratio of their volumes ? 1:16 1:12 1:10 The length, breadth and height of a brick are 20 em, 15 cm and IO em respectively. The number Of bricks required to construct a wall With dimensions 45 m length, 0•15 m breadth and 3 m height is **(a) (b) (c) (d)** 12450 11250 6750 None of the

97. **[CDS (I) Paper 23 Q5]** What is the ratio Of their heights ? **(b) (d)** 25: 64 16:25 9:16 DZOL-T-LKM (27- lar **(a) (b)** Hexagon Nonagon Decagon 79. A bicycle wheel makes 5000 revolutions in moving 11 km. What is the radius of the wheel ? **(c) (d)**

98. **[CDS (I) Paper 23 Q80]** The volumes Of two cones are in the ratio I and their diameters are in the ratio 4 What is the ratio Of their heights ? In a ABC, if 2 z A 3 6 Z C, then what is z A + Z C equal to ? 900 1350 1500 If the perimeter of a circle and a are equal, then what is the ratio of the area of the circle to that Of the square ? **(a) (b) (c)** The lengths Of the sides Of a right-angled triangle are consecutive even integers (in em). What is the product of these integers ? **(b) (d)** 25: 64 16:25 9:16 **(c)** 60 120 360 480 DZOL-T-LKM

99. [CDS (I) Paper 24 Q2] If x is directly proportional to z and y is directly proportional to z, then (x 2 — y2) is directly proportional to za. If x is inversely proportional to z and y is inversely proportional to z, then (xy) is inverseb' Which Of the above statements is/are correct? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither 1 nor 2

100. [CDS (I) Paper 24 Q28] The incomes Of A, B and C are in the ratio 7 : 9 : 12 and their expenditures are in the ratio 8: 9 : **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

101. [CDS (I) Paper 24 Q15] If A's saving is one- fourth Of his income, then the ratio of Of A, B and C is savings 56: 99 69 99: 56 69 69:56 99 99:69 56 **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

102. [CDS (I) Paper 24 Q32] 33 A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? . The e2,ooo 1,800 e 1,600 e 1,200 difference between the compound interest (compounded annually) and the simple interest on a certain sum Of money at 12% per annum for 2 years is **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

103. [CDS (I) Paper 24 Q38] In a mixture of 80 litres Of a liquid and water, 25% of the mixture is the liquid. How much water should added to the mixture so that the liquid becomes 20% of the mixture? **(c) (d)** 15 litres 20 litres 24 litres 25 litres **(a) (b)** 48 minutes 50 minutes 54 minutes 60 minutes **(c) (d)** 15 litres 20 litres 24 litres 25 litres 36. The cost price of 100 manoes is equal to the selling price Of 80 manoes. What is the profit percentage? 16% 20% 24% 25% 37. X sells his goods 25% cheaper than Y and 25% dearer than Z. How much percentage is Z's goods cheaper than Y? 100 200 XVWS-T-MTK/SOA 39. If 20 can clean 20 noors in 20 days, then in how many days can 16 persons clean 16 floors?

104. [CDS (I) Paper 24 Q63] The ratio of the radius of base to the height of a cylinder is 2 : **(a) (b)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 66. 63. The ratio of the radius of base to the height of a cylinder is 2 : 3. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? **(c)** 242 cm **(d)** 385 cm 2

105. [CDS (I) Paper 24 Q2] What is the ratio Of its radius to slant height? (Take 5 27 **(a)** 28 m **(b)** 44 m **(c)** 66 m 72. A sphere of diameter 6 cm is dropped into a cylindrical vessel partly filled With water. The radius Of the vessel is 6 cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised? 73. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55 cm, then What is the area of the sector? **(d)** 577•5 cm2

106. [CDS (I) Paper 24 Q77] AABC is similar to ADEF. The perim eters of AABC and ADEF are 40 cm and 30 cm respectively. What is the ratio of (BC* CA) to (EF+FD) equal to? XVWS-T—MTK/SOA **(a)** 3850 m **(b)** 7700 m 15400 m 76. In a trapezium ABCD, AB is parallel to DC. The diagonals AC and BD intersect at P. If AP : PC 4 : Bx — 4 **(c)** 11:23 **(d)** 23:25

107. [CDS (I) Paper 24 Q78] Two isosceles triangles have equal vertical angles and their areas are in the ratio 4-84 : 5•29. What is the ratio of their corresponding height'? (a/ 11:23 23:25 22:23 **(d)** 484 : 529 **(a)** 3850 m **(b)** 7700 m 15400 m 76. In a trapezium ABCD, AB is parallel to DC. The diagonals AC and BD intersect at P. If AP : PC 4 : Bx — 4 **(c)** 11:23 **(d)** 23:25

108. [CDS (I) Paper 24 Q81] The sides Of a right-angled trianøe are in the ratio x : (x — I) (x — What is the perimeter Of the trianøe? **(a) (b) (c)** 28 units 42 units 56 units 84 units 82. ABC is a right angled at R Let M and N be two points on AB such that AM - MN-NB Let P and points on AC such that PM to QN and QN is parallel Q two is parallel to If BC 12 cm, then equal to? what is 8S. The sides Of a triangle ABC are 4 cm, 6 cm and 8 cm. With the vertices of the triangle as centres, three circles are drawn each touching the Other two externally. What is the sum of the radii Of the three circles? **(d)**

109. [CDS (I) Paper 24 Q89] ABCD is a trapezium in which AB is parallel to DC and 2AB=3DC The diagonals AC and BD intersect at Q What is the ratio Of the area of MOB to that of ADOC? **(a)** 24 cma 27 cm2 30 Cm 36 cm 2 88. ABC is a triangle right angled at C Let p be the length Of the perpendicular drawn from C on AR If BC = 6 cm and CA 8 cm, then What is the value Of p ? 4-8 cm XVWS-T-MTK/SOA 89. ABCD is a trapezium in which AB is parallel to DC and 2AB=3DC The diagonals AC and BD intersect at Q What is the ratio Of the area of MOB to that of ADOC? 90. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? **(b)** 14 cm **(c)** 13 cm **(d)** 12 cm

110. [CDS (I) Paper 24 Q99] The ratio Of drop in total production in 2004 compared to 2001 to that in 2000 compared to 2001, is **(a)** Type 11 **(b)** Type 111 **(c)** Type **(d)** Type V XVWS-T-MTK,'SOA 98. In Which one Of the following pairs of years, the difference in total number Of tablets produced between them is m inimum?

111. [CDS (I) Paper 25 Q26] If three positive numbers are in the ratio 2 : 3 : 5 and the sum of their squares is 1368, then what is sum of numbers ? A - NYRG-T-MTK **(a)** 30 **(b)** 45 **(c)** 60 **(d)** 75 23. What IS value of N?

112. **[CDS (I) Paper 25 Q37]** A sum of money compounded annually doubles itself in 5 years. In how many years will it become four times of itself? **(a)** IO years **(b)** 12 years **(c)** 15 years **(d)** 20 years 11 11

113. **[CDS (I) Paper 26 Q66]** 67. The radius and height Of a right circular cone are in the ratio 3 **(a)** 100 The radius and height Of a right circular cone are in the ratio 3 7. If the volume Of the cone is 528 cm3 , then What is the height of the cone? **(b)** 10•5 cm **(c)** 140 cm A square sheet is forrned by joining n identical Square sheets of same size. If the length of the diagonal of the bigger square sheet formed is m, then What is the side length Of a smaller square sheet? **(d)** A cylindrical pipe has inner diameter Of 14 cm. Water flows through it at a rate of 154 litres per minute. What is the speed Of water in km/hr? wake 2?

114. **[CDS (I) Paper 26 Q92]** If the ratio Of taxpayers to Other persons in the same age group below 30 years is I : 2, then What is the number of taxpayers (in lakhs) in that age group? **(d)** 0-4 0-7 1>85 105 1 P.T.O. 86. **(a)** 21 90. In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? **(b)** 12 cm **(c)** 12•2 cm **(d)** 12•4 cm

115. **[CDS (I) Paper 27 Q6]** If (ag + b") is proportional to (a2 — b2) then **(a) (b) (c)** 4 5 **(d)**

116. **[CDS (I) Paper 27 Q7]** If , then what is the value of ? then What is x equal to ? **(a) (b) (d)** 4 5 **(a) (b) (c) (d)** 9 10 12 What is the HCF (x7 — ya + x5y2_x2y5)? **(b)** (x3 — ya _ x2y + **(c)** (xa — + x2y — xy2) **(d)** of and where a c, then which one of the following is correct ? **(a) (b) (c) (d)** a + b=c+d a+c=b+d a + b+c•d If the Sum of the squares of the rtxjts Of the equation x2 — 14x + k = O is 100, then what is the value Of k ? **(a) (b) (c) (d)** 42 48 52 56 If (ag + b") is proportional to (a2 _ b2), then If (x + k) is the HCF Of x2 px + q and x2 + qr • p, where p q, then What is the value of k ? **(a) (b) (c)** — ab + b2) is proportional to (a + ab + b) (aa-b3) **(c) (d)** 2 F DGT.T-EMT

117. **[CDS (I) Paper 27 Q40]** If C is the compound interest on 10,000 for one year at 4% annum when compounded quarterly, then which one of the following is correct ? **(b) (c)** 100 c»e400 **(a)** vd **(b) (c) (d)**

118. **[CDS (I) Paper 28 Q6]** However, the time invested by them is in the ratio 6 : 5 : 4 : 3 : **(a)** 34 **(b)** 90 **(c)** 127 **(d)** 129 What is the largest power Of IO that divides the product 29 x 28 x 27 x In a party Of 150 75 persons take tea, 60 persons take coffee and 50 take milk. 15 of them take both tea and coffee, but no one taking milk takes tea. If each in the party takes at least one drink, then What is the number of persons taking milk only ?

119. **[CDS (I) Paper 28 Q41]** 42. What is the difference between simple interest and compound interest on 10,000 for two years at 20% per annum compounded half-yearly ? **(a) (c)** e 842 e 756 e 641 e 542 Consider the following statements in respect of the polymomial a(b — c) (x — b) (x — c) + The coefficient of x2 is O. The coefficient ofx is (a — b) (b — c) (c — a). Which Of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 JNBY-D-MTK (15-0

120. **[CDS (I) Paper 28 Q9]** 72. CA = 6 cm. CA z 7 cm 75. -A) Question : What are the unique values Of a, b and c if 2 is a root of the equation ax2 + bx + c = O ? Statement : Statement 11 : Ratio of c to a is I. Ratio of b to a is 5/2). Question : Is m > n, where m, n are nori-zero numbers ? Statement I : Statement 11 : Question : Can a circle be drawn through the points A, B and C ? Statement : Statement 11 : AB=5cm, BC AB=3cm, BC 4 em, x and y are consecutive Odd integers. Question : Can the value Of (x y) determined uniquely ? be Staternent I : Statement 11 : (x + 256. (x < **(a) (b) (c) (d)** Choose this option if the Question can be answered by one of the Statements alone but not by the Other. Choose this option if the Question can be answered by either Statement alone. Choose this option if the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. Choose this option if the Question cannot be answered even by using both Statements together. Question : Is m > n, where m, n are nori-zero numbers ? Statement I : Statement 11 : Question : Can a circle be drawn through the points A, B and C ? Statement : Statement 11 : AB=5cm, BC AB=3cm, BC 4 em, A number 277XY5

121. **[CDS (I) Paper 29 Q5]** 6. is 7. in 3-A) A person borrowed 10,000 at 12% rate Of interest per annum compounded quarterly for a period Of 9 months. What is the interest paid by him to settle his account after 9 months ? **(a) (c)** 92727 947•47 e 967.67 e 987•87 For What relation between a and b is the equation gin B possible ? **(a) (b) (c) (d)** ab ag b Three A, B and C together can do a piece of work in 36 days. A and B together can do five times as much w•ork as C alone; B and C together can do as much work as A alone. If A and C together can do n times as much work as B alone, then what is the value **(b) (c) (d)** 16 2 3 If a2 — bc = u, b2 — ac = B, c2 — ab = y, then what is equal to ? **(a) (b) (c) (d)** If (x — IP is a factor Of x4 + + ßx2 + —

122. **[CDS (I) Paper 29 Q21]** I. If (a + b) is directly proportional to (a — b), then (a2• b2) is directly proportional to ab. **(a) (b) (c)** I only 2 only Both I and 2 Neither I nor 2 **(d)**

123. **[CDS (I) Paper 29 Q2]** If a is directly pr0JXjrtional to b, then (a2 — b2) is directly proportional to ab. Which of the statements given above Ware correct ? **(a) (b) (c)** I only 2 only Both I and 2 Neither I nor 2 **(d)**

124. **[CDS (I) Paper 29 Q25]** I. If (a + b) is directly proportional to (a — b), then (a2• b2) is directly proportional to ab. **(a) (b) (c)** I only 2 only Both I and 2 Neither I nor 2 **(d)**

125. **[CDS (I) Paper 29 Q43]** 44. (x — — mb) What is x mab What is x equal to ? **(a) (b) (c)** equal to ? of the sum of two parts, one part is proportional to number Of rmms and the other part is proportional to number of units consumed. 400 is the monthly electricity bill for a house with 8 and consuming 240 units and 320 is the monthly electricity bill for a house with 6 and consuming 200 units. 45. 46. What is the monthly electricity bill for a house with m rooms and consuming n units ? **(d)**

126. **[CDS (I) Paper 29 Q55]** A lamp shade is in the shape Of a part Of a cone and its top and bottom ends are circles whose circumferences are respectively 30 em and 40 cm. The perpendicular distance between the ends is 6 cm. If the cone were to be completed, then hmv far would its vertex be from the top end ? **(a) (b) (c) (d)** 20 cm 18 cm 12 em 9 cm The corners Of an equilateral triangular plate were cut in such a manner that it forms a regular hexagonal plate. What is the ratio of the area of the triangular plate to the area of the hexagonal plate ? **(a) (b) (c) (d)** Two equal arcs of different circles Cl and C2 subtend angles Of 600 and 750 respectively, at the centres. What is the ratio Of the radius Of Cl to the radius of C2 ? ABC is a triangle With sides AB 41 em, BC = 28 cm and CA = 15 cm. [f D, E and F are the mid-points Of AB, BC and CA then What is the area of the triangle DEF ? A sum Of money at 20% rate of compound interest per annum becomes more than 100 times in n years. What is the least value 3=0-477)

127. **[CDS (I) Paper 29 Q67]** 203: 117 203 : 157 172 : 91 165: 88 **(a) (b) (c) (d)** 0-42 cm 0-46 cm 0-50 cm Three glasses P, Q and R have capacities in the ratio I : 2 :

128. **[CDS (I) Paper 29 Q3]** All these glasses are completely filled With mixtures Of milk and water. The ratio of milk to water in P is I : 2, in Q it is 2 : 3 and in R it is 3 : I. If the content of all three glasses are put into a bigger container, what will be the ratio of milk to water in the container ? **(a) (b) (c)** JNUP-T-EMT (25 -g What is the LCM of + x2y2 y4, + y', **(a) (c) (d)** — ya) (x-4 + x2y2 + y-4)2 (x2 _ (x4 + 2x2y2 + y4) (x6 — y6) 69. t

129. **[CDS (I) Paper 29 Q69]** 70. What is —y —z —2yz — z2 + 2xy equal to ? **(a) (b)** 2y 2z — Y2 — z2 — 2yz — 2xz **(a) (b) (c) (d)** 0-42 cm 0-46 cm 0-50 cm Three glasses P, Q and R have capacities in the ratio I : 2 :

130. **[CDS (I) Paper 29 Q90]** 16 years; the average age of A and E is 11 years; the average age of B and C is also 11 years. B and C have equal weight A's weight is IO kg less than that of B; D is 4 kg heavier than E; E is 4 kg heavier than F; F is 4 kg heavier than G. Further, D has age-weight ratio of 9 : 20, where age is in years and weight in kg; A has age-weight ratio Of 2 **(a)** 15 **(b)** 16 **(c)** 18 JNUP-T-EMT In a triangle ABC, angle B = 900 and p is the 87. 84. length Of the perpendicular from B to AC. If BC IO cm and AC = 12 em, then what is the value Of p ? 5Jfi **(d)** 1 oJii

131. **[CDS Paper 30 Q4]** The speeds of four cars are 2u, 3u, 4u and Xu and the time taken by them to cover the same distance is xt, 4t, 3t and 2t respectively, where 6. x, u, t are real numbers. What is the value Of x? **(c)** 8 : 4, then what is (2m + "p) : (n + 3q) equal to ? **(b)** If the rate Of interest is 5%, then What would be the difference between compound interest and simple interest received on e 10,000 (each) after 3 years from now ? **(a) (b) (c) (d)** 6

132. **[CDS Paper 30 Q6]** 7. If the difference between the interior and exterior angles Of a regular polygon is 1440, then What is the number of sides of the polygon ? 4, then what is **(a) (b) (c) (d)** 12 16 18 20 (2m + "p) : (n + 3q) equal to ? **(b)** If the rate Of interest is 5%, then What would be the difference between compound interest and simple interest received on e 10,000 (each) after 3 years from now ? If the sum and product Of the roots Of a quadratic equation are 2 and — 100 respectively, then which one of the following is correct ? **(a) (c)** 175.25 e 152-25 e 76-25 24•25 **(a) (b) (c) (d)** There are infinitely many such equations having different roots. 'lhere is only one such equation which is + — 100 O. There is only one such equation which is 100 = o. There is no such equation. SRSU-T.EMT

133. **[CDS Paper 30 Q81]** 32. 33. —A) A bottle contains spirit and water in the ratio I : 4 and another identical bottle contains spirit and water in the ratio 4 : **(a) (b)** 12 18 24 28 **(c) (d)**

134. **[CDS Paper 30 Q1]** In what ratio should the mixtures in the two bottles be mixed to get a new mixture in which the ratio of spirit to water is I : 3 ? **(a) (b) (d)** 10:1 11:1 If 3 sin B + 5 cos = 5, then what is the value of 5 sin cos 0? **(a) (c)** Consider the following in of the polynomial x4k + x4k*2 + x4k•4 + x4k•6 . The remainder is zero when the polynomial is divided by x2 I. The remainder is zero when the polynomial is divided by x4 + I. Which of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

135. **[CDS Paper 30 Q46]** 47. A sum of money invest«l at simple interest triples itself in 8 years and becomes n times in 20 years. What is the value Of n ? **(c)** 5 6 75 9 If the work done by x men in (x + I) days is equal to the work done by (x * 5) men in (x — 2) days, then What is the value ? **(c)** 5 6 8 If (a + b) : c) a) z 5: 7 ; 6, then what is the value Of (a —b + c) : (a b — c) ? **(a) (b) (c)** Let x the compound interest at the end Of 3 years on a sum of 1000 at the rate Of annually and y be the simple interest at the end of 3 years on a sum Of 1000 at the annual rate of 11%. What is the difference between x and y ? **(a) (c) (d)** 16 e 15 In a right-angled triangle ABC, AB 15 em, 44•

136. **[CDS (I) Paper 31 Q11]** Consider the Question and the Statements and mark the correct option. The ratio of P's salary Question : to Q's salary is 6 : **(a)** , The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements 67. A question is given followed by two statements I and 11. Consider the Question and the Staternents and mark the correct option. The largest of five Question : different integers is 8 and least is 2. What is the average of these integers ? • The sum of all the Statement-I 5 integers is a multiple of 5 Statement-II : The number of odd integers is odd Which one of the following is correct in respect of the above Question and the Statements ?

137. **[CDS (I) Paper 31 Q5]** How much is PS expenditure ? • The ratio of P's saving Statement-I to Q's saving is 3 : 2 The ratio Of P's Statement-II : expenditure to Q's expenditure is 1 : 1 Which one of the following is correct in respect of the above Question and the Statements ? **(a)**, The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements 27

138. **[CDS (I) Paper 31 Q81]** Three amounts x, y, z are such that y is the compound interest on x; and z is the compound interest on y. The rate of interest per annum and the time period in years are same. Which one of the following is correct ? **(b)** Y2 = zx **(c)** z2 — xy **(a)** 81. Three amounts x, y, z are such that y is the compound interest on x; and z is the compound interest on y. The rate of interest per annum and the time period in years are same. Which one of the following is correct ? **(b)** Y2 = zx **(c)** z2 — xy 82. There are n concentric squares. The area of the innermost square is 1 unit and the distance between corres- ponding comers of any two consecu- tive squares is 1 unit. Consider the following statements : I. The diagonal of the nth square is 11. The area included between nth square and **(d)** I only

139. **[CDS (I) Paper 31 Q86]** There are two containers A and B. In container A. the ratio of milk and water is I : 3 and in container B, the ratio of milk and water is m : n. If the mixture in the containers A and B are mixed in the ratio 2 : 3 to get 20 litres of a mixture having milk and water in the ratio 3 : 7, then what is the value of 2 4 4 37 37 **(a)** 96 square cm **(b)** 84 square cm **(c)** 80 square cm **(d)** 72 square cm 86. There are two containers A and B. In container A. the ratio of milk and water is I : 3 and in container B, the ratio of milk and water is m : n. If the mixture in the containers A and B are mixed in the ratio 2 : 3 to get 20 litres of a mixture having milk and water in the ratio 3 : 7, then what is the value of 2 4 4 37 87. A cone, a hemisphere and a" cylinder stand on equal base of radius r and have the same height. If the sum of volumes of cone, the hemisphere and the cylinder is equal to volume of a R3 sphere of radius R, then what is equal to ?

140. **[CDS (I) Paper 32 Q24]** 25. 26. -A) The minimum power required to lift a mass of 50 kg up to a vertical distance Of 8 m in IO s is : (Take g z IO nv's2) **(b) (c)** 400 w 40 w 50 w 500 w Which one among the following statements is not correct ? **(a) (b) (c) (d)** All bodies in the universe attract each Other due to the gravitational force. Newton's law Of gravitation follows from Newton's laws Of motion. The value of g (acceleration due to gravity) at the centre of the Earth is zero. Mass and wæight are measured in different units. A body floats in a liquid if the buoyant force is : **(a) (b) (c) (d)** equal to its weight. less than its weight. greater than its weight. zero.

141. **[CDS (I) Paper 32 Q46]** Which one of the following statements about Rig Vedic goddesses is correct ? **(a) (b) (c)** Ushas represents freedom from sickness. Sinivali bestows children. Aditi is goddess of dawn. Raka is goddess Of speech. Which one of the following statements about the Buddhist Goddess Hariti is correct ? **(a) (b) (c) (d)** She was converted by the Buddha from a blood-thirsty yakshi to a benevolent maternal figure. She is the feminine personification Of She protects people from eight great She is a warrior goddess associated with the dawn. Consider the following statements about the literary language Apabhramsha : It appeared in recognizable forms by the sixth century. It developed out of mixtures of regional vernaculars With the Prakrits and Which of the statements given above Ware correct ? **(a) (b) (c)** I only 2 only Both I and 2 Neither I nor 2

142. **[CDS (I) Paper 32 Q58]** 59. 60. (19-A) Rutherford's a-particle scattering experiment on gold foil is responsible for the discovery of : 61. **(a) (b) (c) (d)** Neu tron Electron Proton Atomic nucleus Which one of the following is a metalloid ? **(a) (b) (c) (d)** Germani um Coke Iodine Sugar Which one of the following is not explained by the Dalton 's atomic theory ? **(a) (b) (c)** Law of conservation of mass Law of constant composition Presence Of different kind Of sub-atomic particles in an atom Law of multiple proportion Which one of the following contains all Lewis acids ? **(a)** BF3 **(b)** BCIa, NHa, OH- **(c)** Mg2+,H20, **(d)** BF3, NH3, H20 Number of electrons present in the species H;, He, H2 and 02 respectively are • **(a) (b) (d)** 1, 2, 2,15 1, 2, 1, 14 2, 2, 2, 16 0, 2, 1, 14 CEBN-S-GKL

143. **[CDS (I) Paper 32 Q2]** 3. Global Value Chains (GVC) World Integrated Trade Solutions Curre nt Account Deficit (CAD) International Product Sharing Database on GVC related Trade Foreign Direct Investment Select the answer using the code given below : **(a) (b) (c) (d)** I and 2 only 2 and 3 only 3 only 1, 2 and 3 How can currency depreciation stimulate an increase in net exports ? I. By reducing exmmt costs

144. **[CDS (I) Paper 32 Q68]** International Product Sharing Database on GVC related Trade Foreign Direct Investment **(a) (b)** Honeywell Citibank Texas Instruments SAP Labs India **(c) (d)**

145. **[CDS (I) Paper 33 Q58]** 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount Of each installment (approximately) ? **(a)** e 4,437 **(b)** e 4,237 **(c)** r 4,021 **(d)** r 3,811

146. **[CDS (I) Paper 33 Q92]** TO find the average ratio like price/unit, work done/ hour, kilometre/hour under certain conditions, the suitable measure Of central tendency applicable is **(a)** 500 **(b)** 300 **(c)** 60 92. **(d)**

147. **[CDS (I) Paper 34 Q12]** In what duration of time will *3300 become *3399 at 12% per annum interest compounded quarterly ? **(a)** 3 months **(b)** 6 months **(c)** 9 months **(d)** 1 year

148. **[CDS (I) Paper 34 Q20]** The ratio of the present age of X to that of Y is 4 : 5. k (less than 10) years ago, this ratio was I : **(a)** 10 minutes **(b)** 12 minutes **(c)** 1 S minutes **(d)** 20 minutes 18. A person goes downstream at x km/h and upstream at y km/h by the same boat. What is the ratio of speed of boat in still water to speed of •water ? Y 19. Let Q=x2 + bx + c. If the sum of the roots is equal to product Of the roots of the equation Q = 0, then which of the following statements is/are correct ? 20. 21. 11. Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below :

149. **[CDS (I) Paper 34 Q35]** Two numbers P and Q are such that . P : Q —x: y. If 1 is added to both the numbers, the ratio becomes 7 : IO. If 3 is added to both the numbers, the ratio becomes 3 : **(a)** 100 **(b)** 81 **(c)** 64 **(d)** 49 35. Two numbers P and Q are such that . P : Q —x: y. If 1 is added to both the numbers, the ratio becomes 7 : IO. If 3 is added to both the numbers, the ratio becomes 3 : 4. What is

150. **[CDS (I) Paper 34 Q89]** What is the initial rate of interest per annum ? **(a)** Rs. 1,200 **(b)** Rs. 1,440 **(c)** Rs. 1,500 **(d)** Rs. 1,800

151. **[CDS (I) Paper 34 Q90]** What is the ratio of the interest on the first loan to the interest on the second loan ? For the next two (02) items that follow : Let 12(tane + cot0) = 25, where 450 < 0 < **(a)** 1/2 **(b)** 1/√2 **(c)** √3/2 **(d)** 1

152. **[CDS (I) Paper 34 Q94]** What is p4q2 + p2q4 equal to ? A - KPRS-T-MTE For the next two (02) items that follow : A person borrows a sum Of e 16,000 on simple interest at the beginning of a year. Aner 4 months, *24,000 more is borrowed at a rate of interest double the previous one. At the end of the year, the total interest on both the loans is *4800. **(a)** 4 : 5 **(b)** 5 : 6 **(c)** 3 : 4 **(d)** 2 : 3

153. **[CDS (I) Paper 35 Q47]** What is the ratio Of the greatest value Of 51. sin2 x + 2 (where O S x S —) to its least value ? **(a) (b) (c)** 2 3 2 2 **(d)**

154. **[CDS (I) Paper 35 Q49]** 50. (17 —A) What is the ratio Of the greatest value Of sin2 x + 2 (where O S x S —) to its least value ? **(a) (b) (c)** 2 3 2 2 **(d)**

155. **[CDS (I) Paper 35 Q95]** 96. The average weekly wages of male employees in a company is 4,200 and that Of females is 3,200. If the average weekly wage of all employees is 4,000, then What is the ratio of male to female employees ? **(b) (c) (d)** Consider the observations x, x + 4, 30, 33, 74, 78, 49, 52, 98, **(a) (b)** 8-5 6-5 5-5 **(c)** The average weekly wages of male employees in a company is 4,200 and that Of females is 3,200. If the average weekly wage of all employees is 4,000, then What is the ratio of male to female employees ? **(d)**

156. **[CDS (I) Paper 35 Q100]** The following data gives the investment in different categories by a governmental organization Investment Item in crores) 2 3 4 Which Science, Technology and Environment Transport and C ommunication Rural Development Industry and Minerals Agricultural Services 1,500 3,800 2,500 of the following statements Ware correct ? l. **(a)** 1369 13-39 13-19 1309 and Environment Transport and C ommunication Rural Developme nt Industry and Minerals Agricultural Services 3,000 1,500 3,800 2,500 Consider the following incomplete frequency No. 2 4 Which of the following statements Ware distribution with a total frequency Of 100 : correct ? Class Frequency 10 20 l. 11. The percentage investment in the 'Science, Technol **(b) (c) (d)**

CHAPTER 05

TIME, SPEED & DISTANCE, TIME & WORK

Total Archive Questions: 224 Questions

Typical Exam Weightage: 10 - 15 Questions (10 - 15 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q44] What is the sum of all nine numbers ? **(a)** 450 **(b)** 501 **(c)** 504 **(d)** 540 What is the value Of **(b)** 1/3 Two persons P and Q Start at the same time from city A for city B, 60 km away. p travels 4 km."hr slower than Q. Q reaches city B and at once tums back meeting P, 12 km from city B. What is the speed of P? **(a)** 8 km/hr **(b)** 12 km/hr **(c)** 16 km/hr **(d)** 20

2. [CDS (I) Paper 1 Q30] 31. A boy walks from his house to school at 2-5 km/hr and arrives 12 minutes late. The next day he walks at 4 km,rhr and reaches the school 15 minutes earlier. What is the distance from his house to school ? **(a)** 2 km **(b)** km **(c)** 3 km **(d)** 3-5 km A and B can do a piece Of work in 8 days, B and C can do the same work in 12 days. If A, B and C can complete the same work in 6 days, in how many days can A and C complete the same work ? **(b)** 10 **(c)** 12 **(d)** 16 The compound interest On a sum for 2 years is Rs. 832 and the simple interest On the same sum at the same rate for the same period is Rs.

3. [CDS (I) Paper 1 Q39] Two taps can fill a tub in 5 minutes and 7 minutes respectively. A pipe can empty it in 3 minutes. If all the three are kept open simultaneously, when will the tub be full ? **(a)** 60 min **(b)** 85 min **(c)** 90 min **(d)** 105 min If (x/y) — (zJw), then what is (xy + zw)2 equal to ? **(a)** (x2 + zz) + w') **(b)** x2y2 + z2w2 **(c)** x2w2 + y2z2 **(d)** (x2 + w2) (ya + z2) 1009 2 1, then what is Y+2 2+1009 the value of x +1 y +2 z+1009 Suppose y is equal to the sum of two quantities of which one varies directly as x and the other inversely as x. If y = 6 when x = 4, and y 10/3 when x 3, then what is the relation between x and y ? **(a) (b) (c) (d)** y — —2x + (4/x) — 2x + (8/x) y y 2x - (8/x) 10

4. [CDS (I) Paper 1 Q42] 43. 10 A train Of length 150m takes IO s to cross another train 100 m long coming from the opposite direction. If the speed of first train is 30 kmph, what is the speed of second train ? **(a)** 72 kmph **(b)** 60 kmph **(c)** 54 kmph **(d)** 48 kmph There are some coins and rings of either gold or silver in a box. 60% of the objects are coins, 40% of the rings are of gold and 30% of the coins are of silver. What is the percentage Of gold articles ? **(a)** 16 **(b)** 27 **(c)** 58 **(d)** 70 What is the total number of three digit numbers with unit digit 7 and divisible by 11? What is the sum of positive integers less than 100 which leave a remainder 1 when divided by 3 and leave a remainder 2 when divided by 4 ? **(a) (b) (c) (d)** 416 620 1250 1314

5. [CDS (I) Paper 2 Q6] 7. 16% re to the 8• ided 9. „ 55 hich The mess charges for 35 students for 24 days is Rs. 6,300. In how many days Will the mess charges bc Rs. 3,375 for 25 students ? **(a)** 12 **(b)** 15 18 **(d)** 21 • A person P started a business with a capital of Rs. 2,525 and another person Q joined P after some months with a capital of Rs. 1,200. Out Of the total annual profit Of Rs. 1,644, P's share was Rs. 1,212. When did Q join as partner ? **(a)** After 2 months **(b)** After 3 m Onths **(c)** After 4 months **(d)** After 5 months The number is a rational number less than O•OI **(a) (b)** a rational number an irrational number **(d)** neither a rational number nor an irrational number If y z : t —2: 5 and = 3 : 4, then what is thc value Of k ? 1/2 **(a)** 1/3 A person walks a distance in 114 days, when he rests 9 hours a day. How long will he tako to walk twice thé distance if he walks twice as fast and rests twice as long cach day as before ? **(a) (b) (c) (d)** 57 days 228 days 285 days 324 days p and Q can do a job in 2 days; Q and R can do it in 4 -days; and P and R in 12/5 days. What is the number Of days required for P alone to do the job ? **(a) (c) (d)** 5/2 3 14/5 6

6. [CDS (I) Paper 2 Q14] In an army camp ration is available for 100 soldiers for IO days. After 2 days, 60 soldiers joined. Then for how many more days will the remaining ration last ? **(a) (b) (c)** 6 5 4 Two men p and Q start from a place walking at 5 km and km por hour respectively. What is the time they will take to bc 92 km apart if they walk in opposite directions ? **(a) (b) (c)** The 2 hr 6 hr number Of Workers in the employment guarantee scheme increased by 15 which resulted into an increase of 20%. What was the **(a) (b) (c) (d)** initial number of workers ? 60 75 80 90 P-DETBO-FN 14 — at 15. on ds ±tely 16. for Ong

7. [CDS (I) Paper 2 Q50] 51. A right triangle ABC With sides 5 cm, 12 cm and 13 cm is revolved about the side 12 cm. What is the volume Of the solid so obtained ? **(b) (c)** 501t cm3 100k cm3 1257t Cm 150% cm3 How many litres Of Water flow out of a pipe having an area Of cross-section of 5 crn2 in one minute, if the speed of water in the pipe is 30 cm/seé ? **(a) (b) (c) (d)** 90 litres 15 litres 9 litres litres A wire is in the form of a cinle Of radius 42 cm. If it is bent into a square, then what is the side Of the squåre ? **(a) (b) (c)** 66 em 42 cm 36 cm 33 Crn q Seven semi-circular areas are removed from the rectangle ABCD as shown in the figure above, in which AB 2 cm and AD 0-5 cm. The radius Of each semi-circle r, s, t, u, v is half of that df semi-circle p or q. What is the area of the remaining portion ? **(a) (b) (d)** _(128 - 137)/128 cm (125 - 1310/125 cm (128 = 1510/128 cm2 None of the above

8. [CDS (I) Paper 2 Q89] 90. 91. of 92. (24 A) The angles Of a triangle are in AP and the greatest angle is double the least. What is the ratio of •angles in thc radian measure ? **(c) (d)** For what value Of e, is (sin e + cosec O) = 2 5 where O < 90' ? **(a) (b) (c)** 300 60 • 900 If O < •O < < 90•, then which one Of the following is correct ? **(a)** (sine cos > 2 **(b)** (sin2 + cos2 2 **(c)** (sin2B + cos2 •) < 2 **(d)** (sin2 B + cos2 $) > 2 Which one Of the fallowing is not correct in respect of the sets A and B ? **(a) (c)** If A r; B, If A n B then B U A = B then A n (A -B) then B n A = O, then either A Or A boy travels a distance Of 8 km at the rate of 4 km per hour, 6 km at the rate of 3 km per hour' and 4 km at the rate Of 2 km per hour. What is the average speed for the entire journey ? **(a) (b) (c) (d)** 2 kmph 3 kmph 4 kmph 6 kmph

9. [CDS (I) Paper 3 Q26] The shadow of a pole 6 metre high is 15 metre long and at the same time shadow Of a tree is 25 metre long. What is the height of tree ? **(a)** 21 m **(b)** 10m **(c)** 35 m **(d)** None of the abo ve 27. speed of a boat in still water is 11 km/h. It can go 12 km upstream and return downstream to tiE engine point in 2 hours 45 minutes. What is the speed Of stream ? **(a)** 5 km/h **(b)** 4 km/h **(c)** 3 km/h **(d)** 2 km/h c In the figure given above, AB is a dia- meter of a circle and CD is perpendicular to AB, if AB = IO cm and AE=2cm, then what is the length Of ED ? T.DETA-K-FN - A 8 **(c)** cm

10. [CDS (I) Paper 3 Q27] The speed of a boat in still water is 11 km/h. It can go 12 km upstream and return downstream to the engine point in 2 hours 45 minutes. What is the speed of stream ? **(a)** 5 cm **(c)** cm **(d)** em **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

11. [CDS (I) Paper 3 Q32] If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the sarru in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work ? **(a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** 7 days

12. [CDS (I) Paper 3 Q59] A person travelled by train for 1 hour at a speed of 50 kmph. He then travelled by a taxi for 30 minutes at a speed of 32 kmph to complete his journey. What is the average speed at which he travelled during the journey ? d) 44 kmph **(b)** 42 kmph **(c)** 41 kmph **(d)** 33 kmph **(a)** 1911-21 **(b)** 1921-31 **(c)** 1951-61 **(d)** 1961-71 55. The middle points of ihe parallel sides AB and CD of a parallelogram ABCD are p and Q respectiveiy. If AQ and CP divide the diagonal BD into three parts BX, XY and YD, then which one of the following is correct ?

13. [CDS (I) Paper 3 Q86] The inner and outer radii of a 7 m long hollow iron right circular cylindrical pipe are 2 cm and 4 cm respectively. If 1000 cni3 of iron weighs 5 kg, what is the weight of the pipe ? **(a)** 264 kg **(b)** 132 kg **(c)** 396 kg **(d)** None of the above

14. [CDS (I) Paper 4 Q15] A father and his son start at a point A With speeds Of 12 kmph and 18 kmph respectively, and reach another point B If his son starts 60 minutes after his father at A and reaches B 60 minutes before his father, What is the distance between A and B? **(a)** 20 years **(b)** 22 years **(c)** 24 years **(d)** 25 years IO. A mixture contains niilk and water in the ratio 5 : On Adding 5 litres Of water, the ratio Of' milk and water becomes 5 : 2. Wh#t is the quantity Of milk in the original mixture?

15. [CDS (I) Paper 4 Q17] Two train* of lengths 100m and 150 m afe traveling in opposite directions I at speeds of 75 kmph and 50 kmph respectively. What is the time {aken by them to cross each 0th* **(b)** 72 s 68 s 18, If (a then what is (a2 +b2) equal to? 5: 13 13 : 5 **(a)** 72 s 68 s 18, If **(b) (c) (d)** Myz 2xyz 3xyz —3xyz 21. If the 14th term Of an arithmetic series is 6 and 6th term is 14, then What is the 95th term? @ -75

16. [CDS (I) Paper 4 Q33] A motorboat takes 2 hours to travel a distance of 9 km down the current and it takes 6 hours to travel the same distance against the current. What is the speed of the boat in still water in kmph? **(c)** 1-5 **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

17. [CDS (I) Paper 4 Q34] A can do a piece of work in 24 days. If B is 60% more efficient than A, then how many days does B require to do the same work? **(d)** 18 **(a)** 6 years and 3 months **(b)** 8 years and 3 months **(c)** 9 years and 6 months On selling an article for Rs 240, a trader loses 4%. In order to gain 10%, he must sell the article for 32. 14:11 28 : 25 25 : 28 34. **(d)** 18 fraction?

18. [CDS (I) Paper 4 Q45] TWO poles Of heights 6 m and 11 m stand vertically upright on a plane ground. If the distance between their feet is 12 m, What is the distance betweer; their tops? **(a)** 1/4 40. What is the value of 510gio 25 -210g10 3 + 10310 18 ? 43. If cos02+ in the first quadrant, then Which One Of the following is correct? 44. What is thc value of cos I • cos 20 cos 3' **(b)** . The 2 product Of a rational number cos 90' ? **(c)** 1/2 41 42. and an frrational number is **(d)** a natural number

19. [CDS (I) Paper 4 Q73] A wheel of a bicycle has inner diameter 50 cm and thickness IO What is the speed of the bicycle if it makes IO revolutions in 5 seconds? **(a) (b) (c)** None of the above 69. Half of a large cylindrical tank open at the top is fillpd with water and identical heavy *pherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius Of a ball, What is the rpaximum number of balls that can pe dropped? **(d)** 12

20. [CDS (I) Paper 5 Q3] 4. 4 9 21 then What is the where x O and y O, value of x + y ? **(a) (b) (c)** 2 4 8 trains each 200 m long move towards each other on parallel lines With velocities 20 kmph and 30 kmph respectively. What is the time that elapses when they first meet until they have cleared each Other ? **(a) (b) (c)** 20 s 24-8 s 28-8 s 30 s X' is twice 8s Old as Y 3 years ago, when was as old as CY' today. If the difference between their ages at present is 3 years, how old is X' at present ? **(a) (c) (d)** 18 years 12 years 9 years 8 years P-PQFA-L-RZ its

21. [CDS (I) Paper 5 Q32] 33. x2 — 6) 34. **(a) (d)** 4 2 What is the number of circles passing through all the vertices Of a given triangle ? **(a) (c) (d)** One Two Three Infinite Four taps can individually fill a cistern Of water in hour, 2 hours, 3 hours and 6 hours respectively. If all the four taps are opened simultaneously, the cistern can be filled in how many minutes ? **(a) (b) (c) (d)** 20 30 35 40 What is the HCF of 4x2 + 3) ? **(a) (b) (d)** P-PQFA-L-RZ (x4 and (10

22. [CDS (I) Paper 5 Q2] Let P be a point On a straight line L. Let Q, R, S be the points On the same plane containing the line L such that PQ, PR, PS are perpendicular to L. Then there exists no triangle With vertices Q, R, S. Let C be a circle passing through three distinct points D, E, F such that the tangent at D to the circle C is parallel to EF. Then DEF is an isosceles triangle. Which Of the statements given above is/are correct ? **(a) (c) (d)** I only 2 only Both and 2 Neither 1 nor 2 completes a job in 2 days and SY' completes it in 3 days and 'Z' takes 4 days to complete it. If they work together and get 3,900 for the job, then how much amount does Y get ? **(a) (b) (c) (d)** 1,800 1,200 900 800

23. [CDS (I) Paper 5 Q2] The height Of the rectangle is determined by the frequency Of the class it represents. Which Of the statements given above Ware correct ? **(a) (b) (c) (d)** I Only 2 only Both 1 and 2 Neither I nor 2 A cistern 6 m long and 4 m Wide contains water to a depth Of 125 m. What is the area of wetted surface ? **(a) (b) (c) (d)** 40 rn2 45 m 49 m 73 m2 The length Of a minute hand Of a wall clock is 9 cm. What is the area swept (in cm2) by the minute hand in 20 minutes ? (Take 3-14) **(a) (b) (c) (d)** 88-78 8478 67-74 5778 What is the number Of prime factors Of **(a) (b) (c)** Four Five None of the above P-PQFA.L-Rz

24. [CDS (I) Paper 5 Q96] A man can walk uphill at the rate of kmph If and downhill at the rate of 325 kmph. If the total time required to walk a certain distance up the hill and return to the starting position is 4 hr 36 min, What is the distance he walked n **(a)** 15 cm **(b)** 20 cm **(c)** 25 cm **(d)** 30 cm Let D, E be the points on sides AB and AC respectively Of a triangle ABC such that DE is parallel to BC. Let AD 2 cm, DB = I em, AE = 3 cm and area' Of triangle ADE 3 em What is EC equal to ?

25. [CDS (I) Paper 6 Q11] 12. A lawn 30 m long and 16 m wide is surrounded by a path 2 m wide. What is the area of the path ? **(a)** 200 rn2 **(b)** 280 rn2 **(c)** 300 rn2 **(d)** 320 tn2 The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance Of I OO m from its base is

26. [CDS (I) Paper 6 Q23] 0-3 **(b)** 23 **(c)** The earth takes 24 hours to rotate about 24. own axis. Through what angle will it turn in 4 hours and 12 minutes ? **(b)** 64 **(c)** 65 **(d)** 700 **(a)** 630

27. [CDS (I) Paper 6 Q24] If (x, y) = (4, 1) is the solution of the pair of linear equations mx + y = 2x + ny 5, then what is m + n equal to ? (2-3) -0.027 What is the value of **(a)** 0-3 **(b)** 2-3 The earth takes 24 hours to rotate about own axis. Through what angle will it turn in 4 hours and 12 minutes ? **(a) (a) (b) (c) (d)** cm cm cm 630

28. [CDS (I) Paper 6 Q30] 31. How many men will be required to plough 100 acres of land in 10 days if IO men require 8 days to plough 20 acres of land ? **(a)** 30 **(b)** 40 **(c)** 50 **(d)** 60 A train 280 m long is moving at a speed of 60 kmph. What is the time taken by the train to Cross a platform 220 m long ? **(a)** 45 s **(b)** 40s **(c)** 35 s **(d)** 30s = O, then what is xyz equal to ? **(c)** xy + + zx **(d)** None of the above What is the remainder when (xil + 1) is divided by (x + 1) ? **(a) (c) (d)** 11 12

29. [CDS (I) Paper 6 Q48] 50. 51 17' 43 IT 45 17' 47 1 7' 49 A, B, C can do a piece ofwork individually in 8, IO and 15 days respectively. A and B Start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed ? **(a)** 53/9 days **(b)** 34/7 days **(c)** 85/13 days **(d)** 53/10 days 49.. The radii of the circular ends Of a bucket Of height 40 em are of lengths 35 cm and 14 cm. What is the volume Of the bucket ? **(a)** 60060 cubic cm **(b)** 70040 cubic cm **(c)** 80080 cubic cm **(d)** 80160 cubic cm If S is the total surface area Of a cube and V is its volume, then which one of the following is correct ? **(a)** 216 S2 **(b)** 216 V2 **(d)** S2 - 36 '•,'3 If A, B, C, D are the successive angles Of a cyclic quadrilateral, then what is cos A + cos B + cos C + cos D equal to ?

30. [CDS (I) Paper 6 Q79] A car travels along the four sides of a square at speeds v, 2v, 3v, 4v respectively. If u is the average speed of the car in its travel around the square, then which one of the following is correct ? **(a)** u = 2•25v **(b)** u = 3v **(c)** 3v < u < 4v 80. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? **(d)** 256

31. [CDS (I) Paper 7 Q26] A wheel of radius m ofa vehicle makes 75 revolutions in one minute. What is the speed Of the vehicle ? **(a) (b) (c) (d)** 78 kmph 594 kmph 37 •4 kmph 35 •4 kmph

32. [CDS (I) Paper 7 Q33] 34. What is, the mean proportional between and (27 ? **(b)** 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? **(a)** 54 days **(b)** 60 days **(c)** 64 days **(d)** 72 days 9 2 3 = 2 and If x+Y x+y x —Y then what is ¯ equal to ? **(a)** 3/2 **(c)** 2/3 **(d)** 1/5 4

33. [CDS (I) Paper 7 Q43] 44. 45. x4 + xy3 + x3y + xz3 + ya + yz3 is divisible by . **(a)** (x + y) only **(b)** (x3+y3 + z') only **(c)** both (x + y) and (x' + + z3) **(d)** None of the abo ve 'A' can do a piece of work in 'x' days and 'B' can do the same work in 3x days. To finish the work together they take 12 days. What is equal to ? **(b)** 10 **(c)** 12 **(d)** 16 ABCD is a rectangle. Let E be a point on AB and F a point on CD such that DE is parallel to BF. If AE = 3 cm and if the area Of triangle BFC = 6 Square cm. Consider the following statements :

34. [CDS (I) Paper 7 Q86] 87. There are two taps A and B to fill up a water tank. The tank can be filled in 40 minutes if both taps are on. The same tank can be filled in 60 minutes iftap A alone is on. How much time will tap B alone take to fill up the same tank ? **(a)** 64 minutes **(b)** 80 minutes **(c)** 96 minutes **(d)** 120 minutes The pair of numbers which are relatively prime to each other is : **(a)** (68, 85) **(b)** (65, 91) **(c)** (92, 85) **(d)** (102, 153) How many rational numbers are there between I and 1000 ? **(a)** 998 **(b)** 999 **(c)** 1000 **(d)** Infinite ne sum of two numbers is 20 and their product is

35. [CDS (I) Paper 8 Q4] A cloth store is offering "Buy 3, get 1 free". What is the net percentage discount being offered by the store ? **(b)** 25% **(d)** 334% X can do a piece of work in 25 days. Y is 25% more efficient than X. Ihe number of days taken by Y is • **(a)** 15 days **(b)** 20 days **(c)** 21 days **(d)** 30 days A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of the train ? **(a)** 400 m **(b)** 300m **(c)** 200m **(d)** 100m A sum of money becomes 3 times in 5 years. In how many years will the same sum become 6 times at the same rate of simple interest ? **(a)** 15 years **(b)** 12.5 years **(c)** IO years **(d)** 7.5 years

36. [CDS (I) Paper 8 Q5] If there are 2400 students in the school, how many girls are there ? **(a)** 500 **(b)** 700 **(c)** 800 **(d)** 1000 45 people take 18 days to dig a pond. If the pond would have to be dug in 15 days, then the number of people to be employed will be **(a)** 50 **(b)** 54 **(c)** 60 **(d)** 72 A merchant eams a profit of 200/0 by selling a basket containing 80 apples which cost Rs.

37. [CDS (I) Paper 8 Q23] A car is traveling at a constant rate of 45 km per hour. The distance travelled by car from 10.40 am to 1.00 pm is **(a)** 165 km **(b)** 150 km **(c)** 120 km **(d)** 105 km A number consists of two digits whose sum is

38. [CDS (I) Paper 8 Q28] Three planets revolve round the sun once in 200, 250, days respectively in their own orbits. When do they all come relatively to the same position as at a certain point of time in their orbits ? **(a)** After 3000 days **(b)** After 2000 days **(c)** After 1500 days **(d)** After 1200 days 8

39. [CDS (I) Paper 8 Q2] Distance traveled by a circular wheel of diameter 2d Crn in one revolution is greater than Sd cm. Which of the above statements is/are correct ? **(a) (b) (c) (d)** I only 2 only Both 1 and 2 Neither I nor 2 B The angles xo, ao, c0 and (7t — c are indicated in the figure given above. Which one of the following is correct ? 20 O

40. [CDS (I) Paper 9 Q5] 6. sly 'X' comp)ete a days. If X' and Work t.ogclher, can complete job 6— day'. alone can complete the job in **(a)** days 15 18 d eye if a body B at the rate Of x and gnother distance at or y km/hr, thon the average 'Deed in krn'hr is XN-Y 2 A sails a distance 48 elong the of a river in 8 it takes 12 to return the distance, then the epeed of the Lhe river is O•ti Y. k'n/br A a in hours, Arter working for 6 hours, he look the help of boy finished goather 5 The time that the alone ecanplete Gbe both Mathematics and Physics. The of teachers i.eat:hing only emetics is 12 -A **(b) (c)** 30 45 60 hrs 5. 6. sly 'X' comp **(d)** days 15 18 d eye if a body B at the rate Of x and gnother distance at or y km/hr, thon the average 'Deed in krn'hr is XN-Y 2 A sails a distance 48 elong the of a river in 8 it takes 12 to return the distance, then the epeed of the Lhe river is O•ti Y. k'n/br A a in hours, Arter working for 6 hours, he look the help of boy finished goather 5 The time that the alone ecanplete Gbe

41. [CDS (I) Paper 9 Q6] 5 'X' comp)ete a days. If X' and Work t.ogclher, can complete job 6— day'. alone can complete the job in **(a)** days 15 18 d eye if a body B at the rate Of x and gnother distance at or y km/hr, thon the average 'Deed in krn'hr is XN-Y 2 A sails a distance 48 elong the of a river in 8 it takes 12 to return the distance, then the epeed of the Lhe river is O•ti Y. k'n/br A a in hours, Arter working for 6 hours, he look the help of boy finished goather 5 The time that the alone ecanplete Gbe both Mathematics and Physics. The of teachers i.eat:hing only emetics is 12 -A **(b) (c)** 30 45 60 hrs 5. 6. sly 'X' comp **(d)** days 15 18 d eye if a body B at the rate Of x and gnother distance at or y km/hr, thon the average 'Deed in krn'hr is XN-Y 2 A sails a distance 48 elong the of a river in 8 it takes 12 to return the distance, then the epeed of the Lhe river is O•ti Y. k'n/br A a in hours, Arter working for 6 hours, he look the help of boy finished goather 5 The time that the alone ecanplete Gbe

42. [CDS (I) Paper 9 Q33] If one Of the is 297, the other i. 216 **(b)** 264 d) 192 travels distarre at 3 krwhr reach. 15 minutes late. he travels knvhr, he reaches 15 earlier. The to travel 4-S km 72 O-DET 14. two digu which gum of and product Of the digits the and be Of quotients 18 one, is 14 32 The short and Jang hand' of em 6 Theo distance. by Of •bort hand in days long band in 3 days 725k 725 - X 371 1960 1096 the of solutions N 1 2 Of above **(a)** 1 **(b)** $\sqrt{3}$ **(c)** $1/\sqrt{3}$ **(d)** 2

43. [CDS (I) Paper 9 Q30] If 3 men 4 buys can a piece Of work in 8 days, then 6 men and B eon in 2 as•ys 4 days e days i 6 days X. Y and had taken dinner together. cost Of the meal Of Z 2W more than that Of Y and the east the meal Of X was 5/6 aB much as the cost Of the of Z. If paid then what W•øg the that the three Of them had paid ? 285 300 35b None of the above **(a)** 32 36 40 **(b)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only **(c)** 32 36 40 30. If 3 men 4 buys can a piece Of work in 8 days, then 6 men and B eon in 2 as•ys 4 days e days i 6 days X. Y and had taken dinner together. cost Of the meal Of Z 2W more than that Of Y and the east the meal Of X was 5/6 aB much as the cost Of the of Z. If paid then what W•øg the that the three Of them had paid ? 285 300 35b None of the above 26. Consider the i" Of A and B 2. Which Of the above are ? and 2 only Z and 3 only 3 •ancy iB the HCF of the poiynomiaiz **(d)** O.DETA-N. None ot the above Bet of natural numbers is closed under addition subtraction mui tipiieation division OF the above correct ? only gnd S 2 and and 4 only

44. **[CDS (I) Paper 9 Q79]** S m deep is filled by flowing pipe at the rate of 30, SOO cm time taken to fin the swimming pool is 71/2 hourg hours 250/3 hour. None of the *bove a regular 966 square em, then its 36 c. 48. em 54 em 64 em a.DETA.N.FMN 20 -A if arms of one angle are parallel to the arms Of another angh then ate **(a)** neither eq nor supplemen ta_ry **(b)** nol equa t supplemen tar-y **(c)** equal but n ot **(d)** either egual or supplementary eide BC Of a triangle ABC iB pmau&d to D; bisectors the angles ABC and ACD meet at P. if Z BPC and BAC which one Of the ? **(c)** + y• ISOC There are 45 male and IS female employees in an office. Jf the mean of the 60 employees ig 4.800 and the Salary of the male employees is .5,ONJ, then the mean salary of the female employees is **(b)** 4,500 **(d)** 6,000

45. **[CDS (I) Paper 10 Q6]** A can do a piece of work in 4 days and B Can complete the same work in days. What is the number of days required to do the same work together? **(a)** 1398 142% **(b)** 1484 4. What iS the HCF of a2b4 5 and **(c)** ab **(d)** a2b3 . What iS the LCM of x2 +2x-8, 2. 3. Consider the following statements . 2. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7. Every composite number less than 121 is divisible by a prime number less than 11. x

46. **[CDS (I) Paper 10 Q30]** 2 men and 1 woman can complete a piece of work in 14 days, while 4 women and 2 men can do the same work in 8 days. If a man gets 90 per day, what should be the wages per day Of a woman? **(d)** 48 72 e 135 **(a)** HCF **(b)** HCF **(c)** HCF **(d)** None Of the above If b is the largest square divisor Of c and a2 divides C, then which one of the following is correct?

47. **[CDS (I) Paper 10 Q33]** A train running at the speed of 72 kmph goes past a pole in 15 seconds. What is the length of the train? **(b)** 200 m **(c)** 300 m **(d)** 350 m **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

48. **[CDS (I) Paper 10 Q34]** 18 men can earn e 360 in 5 days. How much money will 15 men earn in 9 days? (Q) e 600 **(b)** 540 **(c)** e 480 **(d)** e 360 B-BCRB-M-NBU/48A 10 **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

49. **[CDS (I) Paper 10 Q37]** 10 **(c)** so 60 70 A can finish a work in IS days, B in 20 days and Cin 25 days. All these three worked together and earned e 4,700. The share of c is **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

50. **[CDS (I) Paper 10 Q39]** TWO odd integer even integer rational number None of the above cars A and B start simultaneously from a certain place at the speed of 30 km/hr and 45 km/hr respectively. The Car B reaches the destination 2 hours earlier than A. What is the distance between the starting point and des tination ? **(a) (b) (c) (d)** 39. TWO odd integer even integer rational number None of the above cars A and B start 41. 43. A man cycles with a speed of IO kmph and reaches his office at 1 p.m. However, when he cycles with a speed of 15 kmph, he reaches his omce at 11 a.m. At what speed should he cycle so that he reaches his office at 12 noon?

51. **[CDS (I) Paper 10 Q40]** 4 goats or 6 sheep can graze a field in 50 days. 2 goats and 9 sheep can graze the field in **(a) (b) (c) (d)** 100 days 75 days 50 days 25 days B-BCRB-M-NBU / 48A 12

52. **[CDS (I) Paper 10 Q41]** 43. 12 A man cycles with a speed of IO kmph and reaches his office at 1 p.m. However, when he cycles with a speed of 15 kmph, he reaches his omce at 11 a.m. At what speed should he cycle so that he reaches his office at 12 noon? **(a) (b) (c) (d)** 39. TWO odd integer even integer rational number None of the above cars A and B start 41. 43. A man cycles with a speed of IO kmph and reaches his office at 1 p.m. However, when he cycles with a speed of 15 kmph, he reaches his omce at 11 a.m. At what speed should he cycle so that he reaches his office at 12 noon?

53. **[CDS (I) Paper 11 Q7]** 8. 9. is iral A positive number, when increased by IO, equals 200 times its reciprocal What is that number ? **(a)** 100 **(b)** 10 **(c)** 20 **(d)** 200 + + + 6 is divisible by **(a)** (x I) only **(b)** (x • 2) only **(c)** (x + 3) only **(d)** All of the above The present age pf Ravi's father is-four times Ravi's present age. Five years back he was seven times as Old as Ravi was at that time. What is the present age Of Ravi's father ? **(a)** 84 years **(b)** 70 years **(c)** 40 years **(d)** 35 years The average age Of male employees in 8 firm i: 52 years and that Of female employees is 42 years. The mean age Of employees is 50 years. The percentage Of male and female employus are respectively **(a) (b) (c) (d)** and 20% and and 50% and 48% 15 men complete a work in 16 days. If 24 men are employed, then the time required to complete that work will be **(a) (b) (c) (d)** 7 days 8 days IO days 12 days What is the number Of integral solutions of the equations HCF (a, b) and R + b = 65 ? **(a)** None **(b)** Infinitely many **(c)** Less than 65 **(d)** Exactly one The difference of two consecutive cubes

54. **[CDS (I) Paper 11 Q9]** HCF equal to ? **(a)** (e) HCF (a, b) product Of four consecutive nßtural numbers plus one is **(a)** a non•square **(b)** always sum of two square numbers **(c)** a square **(d)** None of the above For any integers 'a' and with HCF(a, b) what is HCF (a b, a — b) equal to ? A positive number, when increased by IO, equals 200 times its reciprocal What is that number ? **(a)** 100 **(b)** 10 **(c)** 20 **(d)** 200 + + + 6 is divisible by **(a)** (x l) only **(b)** (x • 2) only **(c)** (x + 3) only **(d)** All of the above The present age pf Ravi's father is-four times Ravi's present age. Five years back he was seven times as Old as Ravi was at that time. What is the present age Of Ravi's father ? **(a)** 84 years **(b)** 70 years **(c)** 40 years **(d)** 35 years The average age Of male employees in 8 firm i: 52 years and that Of female employees is 42 years. The mean age Of employees is 50 years. percentage Of male and female employus are respectively (8) It is always 1 **(b)** It is always 2 **(c)** Either I or 2 **(d)** None of th e above The expression 2xa * x2 by **(a) (b) (c)** 2x-i **(a) (b) (c) (d)** and 20% and and and 48% — I is divisible 15 men complete a work in 16 days. If 24 men are employed, then the time required to complete that work will be **(a) (b) (c) (d)** 7 days 8 days IO days 12 days B-ABQA-N-Nau

55. **[CDS (I) Paper 11 Q14]** 15. A train takes 9 seconds to cross a pole. (f the speed of the train is 48 km/hr, the length of the train is **(a)** 150 m **(b)** 120m **(c)** 90 m **(d)** 80 m Ravi's brother is 3 years elder to him. His father was 28 years of age when his sister was born While his mother was 26 years Of age when he was born. If hig sister was 4 years of age when his brother was born, the ages of Ravi's father and mother respectively when his brother was born were **(a)** 32 years and 23 years **(b)** 32 years and 29 years **(c)** 35 years and 29 years (d) 35 years and 33 years In solving a problem, one student makes a mistake in the coefficient of the first degree term and Obtaink — 9 and — I for the roots. Another student makes a mistake in the constant term Of the equation and obtains 8 and 2 for the roots. The correct equation **(a)** *lox *9-0 **(c)** x2 —IOX + 9 O (d) None of the above If m and n are the roots Of the equation ax2 bx c O, then the equation whose roots are (mz 4 1)/m and (n2+ I)/n is **(a)** acx2 + (ab + + b2 (a **(c)** acx2 + (ab — bc) x + ba — (a **(d)** acx2 (ab • bc) x b2 — (a — B-A3QA-N-NBU (4 f the 16. th Of 17. was rs of es of

56. **[CDS (I) Paper 11 Q30]** A, B and C can do a piece of work individually in 8, 12 and 15 days respectively. A and B start working, but A quits after working for 2 days. After this, C joins B till the completion Of work. In how many days Will the work be completed ? **(a) (b) (c) (d)** x

57. **[CDS (I) Paper 11 Q41]** Out of 105 students taking an examination for English and Mathematics, 80 students pass in English, 75 students pass in Mathematics.and IO students fail in both the subjects. How many students pass in only One subject ? **(b) (c) (d)** 9 : 04 am 9 : 08 am 9: 14 am None of the above then **(a) (b) (c) (d)** 26 30 35 45 If A is a non-empty subset Of a set E, what is E — (A — e)) equal to ? **(b)** Complement of A then what is A : D equal to ? At What point of time after g o'clock, do the hour-hand and the minute-hand of a clcrk occur at right angles for the first time ? **(a) (b) (c) (d)** 9 0'clock 4 hours 37 — minutes 3 hours 30 — minutes 11 9 hours 32 — minutes If the average Of A and B is 30, the average of C and D is 20, then which of the following is/are ?

58. **[CDS (I) Paper 12 Q1]** 2. Pipe A can fill a tank in 3 hours. But there is a leakage also, due to which it takes 3-5 hours for the tank to be filled. HOW much time Will the leakage take in emptying the tank the tank is filled initially? **(a)** 21 hours **(b)** 20 hou **(c)** 18 hou rs **(d)** 10•5 hours A train takes IO seconds to cross a 2. pole and 20 seconds to cross a platform Of length 200 m. What is the length Of the Gain?

59. **[CDS (I) Paper 12 Q3]** A, B and C can do a piece of individually in 8, 12 and 15 daye respectively. A and B start workinE together but A quits after workinE for 2 days. After this, C joins anc works till completion Of the work In how many days Will the work bt completed? days 5 — days 5 days 6 days B-ABQB-N-LZS/ IA But 'hich to be nk if Lat is d ays rking rking work. rk be **(a)** 21 hours **(b)** 20 hou **(c)** 18 hou rs **(d)** 10•5 hours A train takes IO seconds to cross a 2. pole and 20 seconds to cross a platform Of length 200 m. What is the length Of the Gain?

60. **[CDS (I) Paper 12 Q7]** The distance between two points (A and B) is 110 km. X starts running from point A at a speed Of 60 km/hr and Y starts running from point B at a speed Of 40 km/hr at the same time. They • meet at a point C, somewhere on the line AB. is the ratio of AC to BC? A is thrice as efficient as B and hence completes a work in 40 days less than the number Of days taken by B. What Will be the number Of days taken by both Of them when working together? **(a)** 22-5 days **(b)** 15 day s **(c)** 20 days **(d)** 18 d ays If I O persons can dig 8 feet trench in 12 days, then how many days will 8 persons take to dig 6 feet trench? **(a)** I O days **(b)** 1005 days **(c)** I I days **(d)** I I •25 days The height Of a tree varies as the square root Of its age (between 5 to 17 years). When the age Of the tree is 9 years, its height is 4 feet. What will be the height of the tree at the age Of 16 years? **(a)** 5 feet 4 in ches **(b)** 5 feet 5 inches **(c)** 4 feet 4 inches **(d)** 4 feet 5 inches 2 Pipe A can fill a tank in 3 hours. But 1. there is a leakage also, due to which it takes 3-5 hours for the tank to be filled. HOW much time Will the leakage take in emptying the tank if the tank is filled initially? **(a)** 21 hours **(b)** 20 hou **(c)** 18 hou rs **(d)** 10•5 hours A train takes IO seconds to cross a 2. pole and 20 seconds to cross a platform Of length 200 m. What is the length Of the Gain? **(a)** 50 m **(b)** 100m **(c)** 150 m **(d)** 200 m

61. **[CDS (I) Paper 12 Q22]** If an 3 — 4m, then What + + an equal to? 1+2+3+4+...+n **(b) (d)** A train travels 2 -npn - 1) -npn + 1) at a speed is Of 40 km/hr and another train at a speed of 20 mis. What is the ratio of speed of the first train to that Of the **(b)** second train? 5 6 **(a)** 310 **(b)** 314 **(c)** 318 **(d)** 323 20. If every side Of an equilateral triangle is doubled, then the area Of new triangle becomes k times the area Of the Old One. What is k equal to? 17

62. [CDS (I) Paper 12 Q21] 22. If an 3 — 4m, then What + + an equal to? 1+2+3+4+...+n **(b) (d)** A train travels 2 -npn - 1) -npn + 1) at a speed is Of 40 km/hr and another train at a speed of 20 mis. What is the ratio of speed of the first train to that Of **(a) (b) (c) (d)** 001 cm 0•001 Cm the **(b)** second train? 5 B-ABQB-N-LZS/ IA 6

63. [CDS (I) Paper 12 Q33] A man rides one-third Of the distance from A to B at the rate Of x km/hr and the remainder at the rate Of 2y km/hr. If he had travelled at a uniform rate of 6z km/hr, he could have ridden from A to B and back again in the same time. Which one Of the following is correct? **(c) (d)** 2z x y X-ABQB-N-LZS/ IA 237. **(b)** 2ab l/hr e Of ould one **(a)** 37 **(b)** 2ab 34. 35. 10 If ax + by — 2 O and axby = I, Where a O, b O, then what is x + b 2 y **(c) (d)**

64. [CDS (I) Paper 13 Q20] A man rows downstream 32 km and 14 km upstream, and he takes 6 hours to cover each distance. What is the speed Of the current? **(a)** years — years years 2 — years **(b) (c) (d)** _g_f_ km _æ_ km Pc 20. A man rows downstream 32 km and 14 km upstream, and he takes 6 hours to cover each distance. What is the speed Of the current?

65. [CDS (I) Paper 13 Q21] A car travels the first one-third Of a certain distance with a speed of 10 km/hr, the next one-third distance with a speed of 20 km/hr and the last one-third distance with a speed of 60 km/hr. The average speed of the car for the whole journey is **(a)** years — years years 2 — years **(b) (c) (d)** _g_f_ km _æ_ km Pc 20. A man rows downstream 32 km and 14 km upstream, and he takes 6 hours to cover each distance. What is the speed Of the current?

66. [CDS (I) Paper 13 Q22] 24. Two persons A and B start simultaneously from two places c km apart, and walk in the same direction. If A travels at the rate of p km/hr and B travels at the rate of q I-a-n/hr, then A has travelled before he overtakes B a distance Of **(b) (c) (d)** _g_f_ km _æ_ km Pc **(a)** years — years years 2 — years **(b) (c) (d)** _g_f_ km _æ_ km Pc 20. A man rows downstream 32 km and 14 km upstream, and he takes 6 hours to cover each distance. What is the speed Of the current?

67. [CDS (I) Paper 13 Q26] In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. The duration of the flight is **(b) (c)** hour 2 hours 3 hours 4 hours **(a)** 200 m **(b)** 300 m **(c)** 270 m **(d)** 160 m 29. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of IO km/hr and I I km/hr respec- tively. What is the distance between them after 6 minutes?

68. [CDS (I) Paper 13 Q27] With a uniform speed, a car covers a distance in 8 hours. Had the speed been increased by 4 km/hr, the same distance could have been covered in 7 hours and 30 minutes. What is the distance covered? **(c)** 420 km 480 kni 520 km 640 km B-EFUA-O-PDW/ **(a)** 200 m **(b)** 300 m **(c)** 270 m **(d)** 160 m 29. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of IO km/hr and I I km/hr respec- tively. What is the distance between them after 6 minutes?

69. [CDS (I) Paper 13 Q29] A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of IO km/hr and I I km/hr respec- tively. What is the distance between them after 6 minutes? 12 — m 10 **(d)** 100 m 120 m 150 m 160 m **(a)** 200 m **(b)** 300 m **(c)** 270 m **(d)** 160 m 29. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of IO km/hr and I I km/hr respec- tively. What is the distance between them after 6 minutes?

70. [CDS (I) Paper 13 Q31] A light was seen regularly at an interval Of 13 seconds. It was seen for the first time at hour 54 minutes 50 seconds (a.m.) and the last time at 3 hours 17 minutes 49 seconds (a.m.). HOW many times was the light seen? **(b)** 378 **(d)** 384 **(a)** 378 **(b)** 384 32. 33. 34. 35. 36. Let A and B be finite non-empty sets With the number of elements in A m and number Of elements in B z n. Let m > n. If for some integer k l, the number Of nan-empty subsets of A 2k the number Of non•empty subsets Of B, then which one Of the following is correct? **(c)** 375 383 **(d)**

71. [CDS (I) Paper 14 Q8] If x then the value of 106 **(a) (b)** 206 **(c) (d)** 9. value of lx4 4 x 7 19 19 19 7X10 lox 19 10. trains are moving in the same direction at km/minute and 60 km/ hour respectively. A man in the faster train observes that it takes 27 seconds to cross the slower train. Ille length of the slower train is B-ABVB-O-GPU - A 4 **(a)** 225m **(b)** 230m **(c)** 240 m **(d)** 250m 4

72. [CDS (I) Paper 14 Q17] A and B are two taps which can fill a tank individually in 10 minutes and 20 minutes respectively. However, there is .a leakage at the bottom, which can empty a filled tank in 40 minutes. If the tank is empty initially, how much time •c will both the taps take to fill the tank with leakage ? **(a)** 2 minutes **(b)** mihutes **(c)** 5 minutes **(d)** 8 minutes

73. [CDS (I) Paper 14 Q26] A motor boat, whose speed is 15 km/ hour in still water goes 30 km down- stream and comes back in a total of 4 hour and 30 minutes. "Ihe speed of the stream is **(a)** 1000 10000 **(b)** 100 26. A motor boat, whose speed is 15 km/ hour in still water goes 30 km down- stream and comes back in a total of 4 hour and 30 minutes. "Ihe speed of the stream is **(c)** 4 km/hour a' **(d)** 6 km/hour

74. [CDS (I) Paper 14 Q27] If 4 men working 4 hours per day for 4 days complete 4 units of work, then how many units of work will completed by 2 men working for 2 hours per day in 2 days ? 2 B-ABVB-O-GPU - A 10 **(a)** 1000 10000 **(b)** 100 26. A motor boat, whose speed is 15 km/ hour in still water goes 30 km down- stream and comes back in a total of 4 hour and 30 minutes. "Ihe speed of the stream is **(c)** 4 km/hour a' **(d)** 6 km/hour

75. **[CDS (I) Paper 14 Q28]** By increasing the speed of his car by 15 km/hour, a person covers 300 km distance by taking an hour less than before. The original speed of the car **(a)** 45 km/hour **(b)** 50 km/hour **(c)** 60 kni/hour **(d)** 75 km/hour

76. **[CDS (I) Paper 14 Q29]** If m persons can paint a house in d days, how many days will it take for (m + 2) persons to paint the same house ? **(b)** md-2 **(d)** **(a)** 1000 10000 **(b)** 100 26. A motor boat, whose speed is 15 km/ hour in still water goes 30 km down- stream and comes back in a total of 4 hour and 30 minutes. "Ihe speed of the stream is **(c)** 4 km/hour a' **(d)** 6 km/hour

77. **[CDS (I) Paper 14 Q31]** Two trains, one is of 121 m in length at the speed of 40 km/hour and the other is of 99 m in length at the speed of 32 km/hour are running in opposite diregtions. In how much time will they be completely clear from each other from the moment they meet ? **(a)** 10 s **(b)** 16 s **(c)** 21 s 32. Three athletes run a 4 km race. Their srredsareintheratio 16:d5: 11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is **(d)** 1000m

78. **[CDS (I) Paper 14 Q11]** When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is **(a)** 1000m **(b)** 800m **(c)** 750m **(d)** 600 m

79. **[CDS (I) Paper 14 Q43]** An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from some point on the ground are 600 and 450 respectively. Then the vertical distance between the two planes is **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

80. **[CDS (I) Paper 14 Q48]** In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race ? **(b)** 204 **(c)** 208 **(d)** 3-2 16 **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

81. **[CDS (I) Paper 14 Q51]** A pipe with square cross-section is supplying water •to a cistern which was initially empty. The area of cross-section is 4 cma and the nozzle velocity of water is 40 mis. The dimensions of the cistern are 10m x 8m x 6m. Then the cistern will be fill in **(a)** 3600 **(b)** 6200 **(c)** 6028 **(d)** 6400 50. Consider all positive two digit numbers each of which when divided by 7 leavés a remainder 3. What is their sum ?

82. **[CDS (I) Paper 14 Q55]** A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is **(a)** 15-5 km/hour **(b)** 15•84 kn/hour **(c)** 16 km/hour **(d)** 16-36 km/hour

83. **[CDS (I) Paper 14 Q56]** 'lhere are 437 fruit plants in an orchard planted in rows. distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants.. niere is I m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of Rs. 100 per metre ? **(a)** Rs. 15600 **(b)** RS. 16,800 **(c)** Rs. 18,200 **(d)** More information is required

84. **[CDS (I) Paper 14 Q80]** A river 2-5m deep and 45m wide is flowing at the speed of km/hour. The amount of water that runs into the sea per minute is **(a)** 6650 **(b)** 6750 **(c)** 6850 **(d)** 6950 B-ABVB-O-GPU - A 28

85. **[CDS (I) Paper 15 Q22]** A man walking at 5 km/hour noticed that a 225 m long train coming in the opposite direction crossed him in 9 seconds. The speed Of the train is **(a)** 75 km/hour **(b)** 80 km / hour **(c)** 85 km / hour **(d)** 90 kin/ hour B-ABPA-P-JRw/ 54A

86. **[CDS (I) Paper 15 Q23]** 24. 25. sin 350 cos 550 2 COS 550 t 2 sin30a is equal sin 350 A cyclist moves non-stop from A to B, a distance of 14 km, at a certain average speed. If his average speed reduces by I km per hour, he takes 20 minutes more to cover the same distance. The original average speed Of the cyclist is **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 710 — 510 is divisible by 10

87. **[CDS (I) Paper 15 Q33]** A bike consumes 20 mL Of petrol per kilometre, if it is driven at a speed in the range of 25—50 km/hour and consumes 40 mL of petrol per kilometrc at any other speed. How much petrol is consumed by the bike in travelling a distance Of 50 km, if the bike is driven at a speed Of 40 km/hour for the first IO km, at a speed Of 60 km/hour for the next 30 km and at a speed of 30 km/hour for the last IO km? **(d)** IL 1-2 L 1-4 L 1-6 L B-ABPA-P-JRw / 54A **(a)** I + q + q2 = 3pq **(b)** p 3+ q + q2 3pq 2 3pq 34. 36. Consider the following statements : 2. 3. 4. Every natural number is a real number. Every real number is a rational number. Every integer is a real number. Every rational number is a real n umber. 32. What is the maximum value of m, the number divisible by 3m ? if **(c)** 8 7 6 5 33. A bike consumes 20 mL Of petrol per kilometre, if it is driven at a speed in the range of 25—50 km/hour and consumes 40 mL of petrol per kilometrc at any other speed. How much petrol is consumed by the bike in travelling a distance Of 50 km, if the bike is driven at a speed Of 40 km/hour for the first IO km, at a speed Of 60 km/hour for the next 30 km and at a speed of 30 km/hour for the last IO km? Which Of the above Statements are correc t? **(d)** I, 2 and 3

88. **[CDS (I) Paper 15 Q39]** A passenger train takes I hour less for a journey of 120 km, if its speed is increased by IO km/hour from its usual speed. What is its usual speed? 40 **(a)** 50 km/hour **(b)** 40 km/hour **(c)** 35 km/hour 30 km/hour . The value of k, for Which the system Of equations 3x — ku — 20 = O and 6x— 100+40=0 has no solution, is 10 5 **(d)** 3 B-ABPA-P.JRW / 54A

89. **[CDS (I) Paper 15 Q46]** Two pipes A and B can fill a tank in 60 minutes and 75 minutes respectively. There is also an outlet C. If A, B and C are opened together, thc tank is full in 50 minutes. How much time will be taken by C to empty the full **(a)** 9 < xs30 45. A cloth merchant buys cloth from a weaver and cheats him by using a scale Which is IO cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but While selling uses a scale which is IO Crn shorter than a normal metre scale. What is his gain? **(b)** 22—0/0 **(c)** 23—0/0 B-ABPA-PÄIRW / 54A 46. Two pipes A and B can fill a tank in 60 minutes and 75 minutes respectively. There is also an outlet C. If A, B and C are opened together, the tank is full in 50 minutes. How much time will be taken by C to empty the full **(d)**

90. **[CDS (I) Paper 15 Q47]** Ina race of 1000 m, A beats B by 100m or IO seconds. If they start a race of 1000 m simultaneously from the same point and if B gets injured after running 50 m less than hair the race length and due to Which his speed gets halved, then by how **(a)** 9 < xs30 45. A cloth merchant buys cloth from a weaver and cheats him by using a scale Which is IO cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but While selling uses a scale which is IO Crn shorter than a normal metre scale. What is his gain? **(b)** 22—0/0 **(c)** 23—0/0 B-ABPA-PÄIRW / 54A 46. Two pipes A and B can fill a tank in 60 minutes and 75 minutes respectively. There is also an outlet C. If A, B and C are opened together, the tank is full in 50 minutes. How much time will be taken by C to empty the full **(d)**

91. **[CDS (I) Paper 15 Q82]** A person travels 7 km eastwards and then turns right and travels 3 km and further turns right and travels 13 km. What is the shortest distance of the present position or the person from his starting point? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In a triangle PQR, point X is on PQ and point Y is on PR such that XP = I •5 units, 6 units, PY = 2 units and YR 8 units. Which Of the following are correct? 1. QR=5XY 2. OR is parallel to XY. 3. Triangle PYX is similar to triangle PRQ. Select the correct answer using the code given below.

92. **[CDS (I) Paper 16 Q11]** A boy saves Rs. 4•65 daily. What is the least number of days in which he will be able to save an exact number of rupees ? **(a)** 10 **(b)** 20 **(c)** 21 **(d)** 25

93. **[CDS (I) Paper 16 Q17]** A can do 50% more work than B in the same time. B alone can do a piece of work in 30 hours. B starts working and had already worked for 12 hours when A joins him. How many hours should B and A work together to complete the remaining work ? **(a)** 6 hours **(b)** 12 hours **(c)** hours **(d)** 7-2 hours

94. **[CDS (I) Paper 16 Q18]** When the speed 0fa train is increased by 20%, it takes 20 minutes less to cover the same distance. What is the time taken to cover the same distance with the original speed ? **(a)** 140 minutes **(b)** 120 minutes **(c)** 100 minutes **(d)** 80 minutes

95. **[CDS (I) Paper 16 Q26]** A person can row downstream 20 km in 2 hours and upstream 4 km in 2 hours. What is the speed of the current ? **(a)** 2 km/hour **(b)** 2-5 km,'hour **(c)** 3 km/hOur **(d)** 4 knvhour

96. **[CDS (I) Paper 16 Q37]** A tank can be filled by pipe X in 2 hours and pipe Y in 6 hours. At IO a.m. pipe X was opened. At what time will the tank be filled if pipe Y is opened at 11 a.m. ? **(a)** 12:45 hours **(b)** 5:00 p.m. **(c)** 11:45 a.m. **(d)** 11:50 a.m.

97. **[CDS (I) Paper 16 Q38]** A train is travelling at 48 kmfhour completely crosses another train having half its length and travelling in opposite direction at 42 km/hour in 12 s. It also passes a railway platform in 45 s. What is the length of the platform ? **(a)** 12:45 hours **(b)** 5:00 p.m. **(c)** 11:45 a.m. **(d)** 11:50 a.m. 38. A train is travelling at 48 kmfhour completely crosses another train having half its length and travelling in opposite direction at 42 km/hour in 12 s. It also passes a railway platform in 45 s. What is the length of the platform ?

98. **[CDS (I) Paper 16 Q67]** Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? **(a)** 4275 **(b)** 4350 **(c)** 4375 **(d)** 4450

99. **[CDS (I) Paper 16 Q99]** A motorist travels to a place 150 km away at an average speed of 50 km/hour and returns at 30 km/hour. What is the average speed for the whole journey ? **(a)** 35 km,fhour **(b)** 37 km,fhour **(c)** 37-5 **(d)** 40 km/hour

100. **[CDS (I) Paper 17 Q4]** If the loss in total value Of the diamond was 70,000, what was the price Of the original diamond ? **(b) (c)** In a 100 m race, A runs at a speed of — m/s. If A gives a start of 4 m to B and still beats him by 12 seconds, what is the speed of B ? **(a) (b) (d)** — mis 4

101. **[CDS (I) Paper 17 Q26]** If 15 men take 21 days Of 8 hours each to do a piece Of work, then what is the number of days of 6 hours each that 21 women would take, if 3 women would do as much work as 2 men ? **(a) (b) (c) (d)** 18 20 25 30 What number must be subtracted from both the numerator and the denominator of the 27 fraction — so that it becomes — ? **(a) (b) (c) (d)** 35 6 8 9 11 A sum Of 8,400 was taken as a loan. This is to be paid in two equal instalments. If the rate Of interest is 10% per annum, compounded annually, then the value Of each instalment **(a) (b) (c) (d)** BLS-S-FDE e 4,200 e 4,480 4,840 None of the above (8-Io a 27. if

102. **[CDS (I) Paper 17 Q28]** 29. Leela got married 6 years ago. Today her age is I — times her age at the time Of her marriage. Her son's age is —times her age. 10 What is the present age of her son ? **(a) (b) (c)** I year 2 years 3 years 4 years A and B working together can finish a piece of work in 12 days while B alone can finish it in 30 days. In how many days can A alone finish the work ? **(a) (b) (c) (d)** 18 days 20 days 24 days 25 days s is ded

103. **[CDS (I) Paper 17 Q30]** The values of x which satiso• the equation x 51 **(a) (b) (c) (d)** 18 20 25 30 28.

104. **[CDS (I) Paper 17 Q33]** 24. A passenger train departs from Delhi at 6 p.m. for Mumbai. At 9 p.m., an express train, whose average speed exceeds that Of the passenger train by 15 kmfhour leaves Mumbai for Delhi. Two trains meet each Other mid-route. At what time do they meet, given that the distance between the cities is 1080 km ? **(a) (b) (d)** 4 p.m. 12 midnight In a class Of 49 students, the ratio Of girls to boys is 4 : S. If 4 girls leave the class, the ratio of girls to boys would be **(a) (b) (c) (d)** 11:7 8:7 If a + b = 5 and ab = 6, value Of ag + bg ? then what is the **(a) (b) (c) (d)** 35 40 90 125 R.njermrn hnught A mobile with digrnunt: on the selling price. If the mobile cost him e 4,875, what is the original selling price of the mobile ? **(b) (c) (d)** BLS-S-FDE e 6,300 6,400 6,500 6,600 (10. ,.m. 35. the

105. **[CDS (I) Paper 17 Q35]** 31. for Mumbai. At 9 p.m., an express train, whose average speed exceeds that Of the passenger train by 15 kmfhour leaves Mumbai for Delhi. Two trains meet each Other mid-route. At what time do they meet, given that the distance between the cities is 1080 km ? **(a) (b) (c) (d)** 40 seconds 30 seconds 25 seconds 15 seconds

106. **[CDS (I) Paper 17 Q32]** 33. 24. (IO-A) A 225 m long train is running at a speed Of 30 kmmour. How much time does it take to Cross a man running at 3 km/hour in the same direction ? **(a) (b) (c) (d)** 40 seconds 30 seconds 25 seconds 15 seconds **(a) (b) (d)** 4 p.m. 12 midnight

107. **[CDS (I) Paper 17 Q45]** What is the value of 12 125 **(a) (b) (d)** 2 4 5 2 2— A thief is spotted by a policeman from a distance of 100 m. When the policeman starts the chase, the thief also starts running. If the speed of the thief is 8 km/hour and that of the policeman is 10 km/hour, then how far will the thiefhave to run before he is overtaken ? **(a) (b) (c) (d)** 200 m 300m 400 m 500m Aman and Alok attempted to solve a quadratic equation. Aman made a mistake in writing down the constant term and ended up in roots (4. 3). Alok made A down the coefficient of x to get roots (3, 2). The correct roots of the equation are **(a) (b) (c) (d)** 5 th part Of the population in a village are 9 males. If of the males are married, the percentage of unmarried females in the total population is

108. **[CDS (I) Paper 17 Q41]** 42. (12-A) A thief is spotted by a policeman from a distance of 100 m. When the policeman starts the chase, the thief also starts running. If the speed of the thief is 8 km/hour and that of the policeman is 10 km/hour, then how far will the thiefhave to run before he is overtaken ? **(b) (c) (d)** 25 26 27 28 **(a) (b) (c) (d)** 200 m 300m 400 m 500m What is the LCM of + 8, x2 + 5x + 6 x3 + 4x2 + 4x ? **(b)** x (x — (x — 3) (x2 + 2x + 4) **(c)** (x +2)2 (x + 3) (x2 — 2x + 4) and

109. **[CDS (I) Paper 18 Q14]** 15. -A) If a train crosses a km-stone in 12 seconds, how long will it, take to cross 91 km-stones completely if its Speed is 60 ? **(a) (b) (c) (d)** I hr 30 min I hr 30 min 12 sec 1 hr 51 min 1 hr 1 min 3 sec In a 100 m race, A runs at 6 km/hr. If A gives B a start of 8 m and still beats him by 9 seconds, what is thespeed ofB ? **(a) (b) (c) (d)** 4 •6 km/hr 4 •8 km./hr km/hr 5 •4 km/hr The quotient of 8xg 2xy + 4x2 + is — ya when divided by **(a) (b) (c) (d)** x + 2y 2x-y If (x + 2) is a common factor ofx2 + ax + b and x2 + bx + a, then the ratio a : b is equal to **(a) (b) (c) (d)** 1 2 3 4

110. **[CDS (I) Paper 18 Q11]** If the points p and Q represent the real 12. numbers 0-83 and on the number line, then the distance between P and Q is 21 **(a)** 90 19 90 21 100 56 **(d)** 90 Sudhir purchased a chair with three consecutive discounts of 20%, and 5%. The actual deduction will be If a train crosses a km-stone in 12 seconds, how long will it, take to cross 91 km-stones completely if its Speed is 60 km/hr ? **(a) (b) (c) (d)** I hr 30 min I hr 30 min 12 sec 1 hr 51 min 1 hr 1 min 3 sec

111. **[CDS (I) Paper 18 Q15]** In a 100 m race, A runs at 6 km/hr. If A gives B a start of 8 m and still beats him by 9 seconds, what is thespeed ofB ? **(a) (b) (c) (d)** 4 •6 km/hr 4 •8 km./hr km/hr 5 •4 km/hr **(a) (b) (c) (d)** The quotient of 8xg 2xy + 4x2 + is — ya when divided by 32% 35% **(a) (b) (c) (d)** x + 2y 2x-y A fruit seller has a certain number of mangoes of which 5% are rotten. He sells 75% of the remainder and he is left with 95 mangoes. HOW many mangoes did he have originally ? If (x + 2) is a common factor ofx2 + ax + b and x2 + bx + a, then the ratio a : b is equal to **(a) (b) (c) (d)** STS-M-TPT 500 450 400 350 **(a) (b) (c) (d)** 1 2 3 4

112. **[CDS (I) Paper 18 Q30]** To maintain 8 cows for 60 days, a milkman has to spend 6,400. To maintain 5 cows for n days, he has to spend e 4,800. What is the value of n ? **(a) (b) (c) (d)** STS-M-TPT 46 days 50 days 58 days 72 days

113. **[CDS (I) Paper 18 Q38]** 39. What is equal to ? + (y —z)3 + 3 3 3 If loglo 6 = and loglo 8 = 0•9031, then what is the value of log 8000 + loglo 600 ? **(a)** 4•6BL3 **(b)** 5-5813 **(c)** 1-5813 **(d)** 6-6813 30 men can complete a job in 40 days. However, after 24 days some men out of the assigned 30 left the job. The remaining people took another 40 days to complete the job. The number of men who left the job is **(a)** 24 **(b)** 18 12 4 goats or 6 sheep can graze a field in 50 days. 2 goats and 3 sheep will graze it in **(a)** 200 days **(b)** 150 days **(c)** 100 days **(d)** 50 days A tap can fill a tub in IO hours. Aft.er opening the tap for 5 hours it was found that a small outlet at the bottom of the tub was open and watcr wag leaking through it. It was then immediately closed. It took 7 hours to fill the tub after closing the outlet. What time will be taken by the outlet to empty the full tub of water ? **(c) (d) (a) (b) (c) (d)** 35 hours 25 hours 20 hours 17 hours

114. **[CDS (I) Paper 18 Q2]** 3. The speed of the man against the stream is km/hr. speed of the man in still water is 3•15 km.,hr. The speed ofthe stream is 1•35 km/hr. Which of the above statements are correct ? **(a) (b) (c) (d)** 1 and 2 only 2 and 3 only 1 and 3 only 1, 2 and 3 A boy went to his school at a speed of 43• 12 kmJhr and returned to his house at a speed of 8 km/hr. If he hae taken 50 minutes for the whole journey, what was the total distance walked ? The difference between the compound interest (compounded annually) and simple interest on a sum of money deposited for 2 years at 5% per annum is

115. **[CDS (I) Paper 18 Q93]** What is the ratio ofthe total number of boys to the total number Of girls ? A man travelled 12 km at a speed of 4 km/hr and further 10 km at a speed of 5 km/hr. What was his average speed ? **(a) (b) (c)** STS-M-TPT 67 : 83 17 : 26 27 : 19 189 : 179 **(a) (b) (c) (d)** km/hr 4•5 kmfhr 5 •O km/hr 2 •5 kmmr

116. **[CDS (I) Paper 19 Q26]** 27. and D : E 3 ; 4, then what is B : E equal to ? **(a) (c) (d)** A work when done by IO women is completed in 12 days. The same work can be completed in 8 days when done by 5 men. How many days Will it take to complete when 6 women and 3 men are employed to perform the same job ? **(a) (c) (d)** 12 10 A man undertakes to do a certain '"Ork in 150 days. He employs 200 men. He finds that only a quarter of the work is done in 50 days. HOW many additional men should he employ so that the whole work is finished in time ? **(b) (c)** 75 85 100 120 A train moving with a speed Of 60 km hour crosses an electric pole in 30 seconds. What is the length Of the train in metres ? **(b) (c)** 300 400 500 600 SPSS-T-XHI 3

117. **[CDS (I) Paper 19 Q30]** 120 is distributed among A, B and C so that A's share is 20 more than B's and 20 less than C's. What is B's share ? **(a) (c)** 10 15 20 25 A work when done by IO women is completed in 12 days. The same work can be completed in 8 days when done by 5 men. How many days Will it take to complete when 6 women and 3 men are employed to perform the same job ? In the following table of inverse variation, what are the values of A, B and C respectively ? **(a) (c) (d)** 12 10 **(a) (b) (c) (d)** 15 10,-30 1 -30, 10, -1 - 1, -30,10 c 60 A man undertakes to do a certain '"Ork in 150 days. He employs 200 men. He finds that only a quarter of the work is done in 50 days. HOW many additional men should he employ so that the whole work is finished in time ? **(b) (c)** 75 85 100 120 A person borrowed 5,000 at rate of interest annum and immediately lent it at 5-5%. After two years he collected the amount and settled his loan. What is the amount gained by him in this transaction ? A train moving with a speed Of 60 km hour crosses an electric pole in 30 seconds. What is the length Of the train in metres ? **(b) (b) (c)** 300 400 500 600 e 25 50 100 200 SPSS-T-XHI

118. **[CDS (I) Paper 19 Q32]** 33. 34. (IO—A) **(a) (b) (c) (d)** 44 years 45 years 50 years 52 years If 5 tractors can plough 5 hectares Of land in 5 days, then what is the number Of tractors required to plough 100 hectares in 50 days ? **(a) (b)** 100 20 10

119. **[CDS (I) Paper 19 Q41]** The age Of a woman is a twcvdigit integer. On reversing this integer, the new integer is the age of her husband Who is elder to her. The difference between their ages is one-eleventh of their sum. What is the difference between their ages ? **(a) (b) (d)** 8 years 9 years IO years 11 years A passenger train and a goods train are running in the same direction on parallel railway tracks. If the passenger train now takes three times as long to pass the goods train, as when they are running in opposite directions, then what is the ratio of the speed Of the passenger train to that Of the go-Ods (Assume that the trains run at uniform speeds) **(a) (b) (c)** All Odd prime numbers upto 110 are multiplied together. What is the unit digit in this product ? **(a) (b) (c) (d)** None of the above SPSS-T-XHI (12-A

120. **[CDS (I) Paper 19 Q46]** 48. water. If in a stream which is flowing at a speed of y knvhr it takes him z hours to row to a place and back, then what is the distance between the two places ? 2y z(x —y) **(c)** A water tank has been fitted with two taps P 47. and Q and a drain pipe R. Taps p and Q fill at the rate Of 12 litres per minute and IO litres per minute respectively. Consider the following statements SI, S2 and (14_A) A car has an average speed Of 60 km per hour while going from Delhi to Agra and has an average speed Of y km per hour while returning to Delhi from Agra (by travelling the same distance). If the average speed of the car for the whole journey is 48 km per hour, then what is the value Of y ? **(a) (b) (c) (d)** 30 km hour 35 km per hour 40 km per hour 45 km per hour SI : S2 ; Sa : Pipe R drains out at the rate of 6 litres per minute. If both the taps and the drain pipe are opened simultaneously, then the tank is filled in 5 hours 45 minutes. Pipe R drains out (fully) the filled tank in 15 hours 20 minutes.

121. **[CDS (I) Paper 19 Q10]** If x6 + —S = k x2 + 11. (5-A) — g and aß — , where 1 < p < q r, p then which one of the following is the **(b) (d)** Two workers 'A' and 'B' working together completed a job in 5 days. Had 'A' worked twice as efficiently as he actually did and 'B' worked one-third as efficiently as he actually did, the work would have completed in 3 days. In how many days could 'A' alone complete the job ? **(b) (d) (a) (b) (c)** 625 654 684 725 **(d)**

122. **[CDS (I) Paper 19 Q30]** 9-A) In an examination, 50% of the candidates failed in English, 40% failed in Hindi and 15% failed in both the subjects. The percentage of candidates who passed in both English and Hindi is **(b)** 25% A train 100 m long passes a platform 100 m long in 10 seconds. The speed of the train is **(a)** 36 kmph **(b)** 45 kmph **(c)** 54 kmph **(d)** kmph A cyclist covers his first 20 km at an average speed of 40 kmph, another 10 km at an average speed of IO kmph and the last RO km at an average speed of 40 kmph. Then the average speed of the entire journey is **(a) (b) (c) (d)** 20 kmph 26•67 kmph 28-24 kmph 30 kmph In a race of 1000 m, A beats B by 150 m, while in another race of 3000 m, C beats D by 400 m. Speed of B is equal to that of D. (Assume that A, B, C and D run with uniform speed in all the events). If A and C participate in a race of 6000 m, then which one of the following is correct ? **(a) (b) (c) (d)** A beats C by 250 m C beats A by 250 m A beats C by 115•38 m C beats A by 115-38 m

123. **[CDS (I) Paper 19 Q29]** If ab + bc + ca = O, then the value of In an examination, 50% of the candidates failed in English, 40% failed in Hindi and 15% failed in both the subjects. The percentage of candidates who passed in both English and Hindi is **(b)** 25% A train 100 m long passes a platform 100 m long in 10 seconds. The speed of the train is **(a)** 36 kmph **(b)** 45 kmph **(c)** 54 kmph **(d)** kmph A cyclist covers his first 20 km at an average speed of 40 kmph, another 10 km at an average speed of IO kmph and the last RO km at an average speed of 40 kmph. Then the average speed of the entire journey is **(a) (b) (d) (a) (b) (c) (d)** 20 kmph 26•67 kmph 28-24 kmph 30 kmph 2 What is the principal amount which earns 210 as compound interest for the second year at 5% per annum ? 30. (9-A) In a race of 1000 m, A beats B by 150 m, while in another race of 3000 m, C beats D by 400 m. Speed of B is equal to that of D. (Assume that A, B, C and D run with uniform speed in all the events). If A and C participate in a race of 6000 m, then which one of the following is correct ? **(a) (b) (c) (d)** GTHY-T-XMT 2000 3200 4000 4800 **(a) (b) (c) (d)** A beats C by 250 m C beats A by 250 m A beats C by 115•38 m C beats A by 115-38 m

124. **[CDS (I) Paper 19 Q44]** 45. The minute hand of a clock overtakes the hour hand after every 72 minutes of correct time. How much time does the clock lose or gain in a day of normal time ? **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11 47 A thief steals a car parked in a house and goes away with a speed Of 40 kmph. The theft was discovered after half an hour and immediately the owner sets off in another car with a speed Of 60 kmph. When Will the owner meet the thief ? **(a) (b) (c) (d)** 55 km from the owner's house and one hour after the theft 60 km from the owner's house and 1•5 hours aner the theft 60 km from the owner's house and 1 •5 hours after the discovery of the theft 55 km from the owner's house and 1 •5 hours after the theft X and Y together can finish a job in 6 days. X can alone do the same job in 12 days. How long will Y alone take to do the same job ? **(a)** 16 days **(b)** 12 days **(c)** IO days **(d)** 8 days Twelve (12) persons can paint 10 identical rooms in 16 days. In how many days can 8 persons paint 20 such rooms ? **(a) (b) (c) (d)** 12 24 36 48 GTHY-T-XMT •

125. **[CDS (I) Paper 19 Q43]** 44. 45. (15-A) There are n zeros appearing immediately after the decimal point in the value of It is given that the value of log102 = 0•30103. The value of n is **(a) (b) (c) (d)** Lose 121 — minutes 11 Lose 157 — minutes 11 Gain 121 — minutes 11 Gain 157 — minutes 11 **(b) (c)** 25 19 18 17 A thief steals a car parked in a house and goes away with a speed Of 40 kmph. The theft was discovered after half an hour and immediately the owner sets off in another car with a speed Of 60 kmph. When Will the owner meet the thief ?

126. **[CDS (I) Paper 20 Q5]** 3—A) Consider the multiplication 999 x abc def132 in decimal notation, where a, b, c, d, e and f are digits. What are the values of a, b, c, d, e and f respeetively ? **(b)** 8.6.8.6. 7.8 Three cars A. B and C started from a point at 5 p.m., 6 p.m. and 7 p.m. respectively and travelled at uniform speeds of 60 km/hr, 80 km/hr and x knvhr respectively in the same direction. If all the three met at another point at the game ingtnnt then what is the value Of x ? **(a) (d)** 120 110 105 100 Priya's age was cube of an integral number (different from l) four years ago and square Of an integral number after four years. HOW long should she wait so that her age becomes of a number in the previous year and cube of a number in the next year ? **(a) (b) (c) (d)** 7 years 12 years 14 years 21 years What is the remainder when (1729 + 1929) is divided by 18 ? **(c)** Consider the multiplication 999 x abc def132 in decimal notation, where a, b, c, d, e and f are digits. What are the values of a, b, c, d, e and f respeetively ? **(b)** 8.6.8.6. 7.8

127. **[CDS (I) Paper 20 Q84]** 11 —A) A has three parts. The speed and time required to complete the individual parts for a runner is displayed on the following chart • Part 1 Partn Part 111 Speed (kmph) 9 Time (minutes) 50 80 75 100 What is the average speed of thig runner ? **(a) (d)** 8• 17 kmph 7 •80 kmph 777 kmph , then which one Of the following Statements is correct ? **(a) (b) (d)** Each fraction is equal to I or —I. Each fraction is equal to — or I. — or —I. Each fraction is equal to — only The 3521 is divided by 8. is remainder ? **(a) (c)** 3 9 A prime number contains the digit X at unit's place. How many such digits Of X are possible ? **(a) (b) (d)** 3 4 5 6

128. **[CDS (I) Paper 20 Q28]** 29. 30. (11-A) If a : b c : d I : 6, then what is the value of A has three parts. The speed and time required to complete the individual parts for a runner is displayed on the following chart • Part 1 Partn Part 111 Speed (kmph) 9 Time (minutes) 50 80 75 100 What is the average speed of thig runner ? **(a)** 36 28. 29. 30. **(b) (c)** What is • equal t.o ? 8• 17 kmph 7 •80 kmph 777 kmph , then which one Of **(d)**

129. **[CDS (I) Paper 20 Q38]** If an article is sold at a gain of instead of a loss of 6%, the seller gets 6 more. What is the cost price of the article ? 18 36 42 e 50 A field can reaped by 12 men or 18 women in 14 days. In how many days can 8 men and 16 women reap it ? **(b) (c) (d)** 26 days 24 days 9 days 8 days If 4Y = 12', then z equal to **(a)** If (4a 7b) — 7d) (4a— 7b) (4c + 7d), then Which one Of the is ? 7b = DFGT-T-HTM (13- a

130. **[CDS (I) Paper 20 Q41]** 42. 13—A) Given that the polynomial (x2 • ax b) leaves the same remainder when divided by (x — I) Or (x 1). What are the values Of a and b **(a) (b)** 4 and O O and 3 3 and O O and any integer Tushar takes 6 hours to complete a piece of work, while Amar completes the same Work in IO hours. If both Of them work together, then what is the time required to complete the **(a) (c) (d)** a hours 15 minutes 3 hours 30 minutes 3 hours 45 minutes What is the value of 2 + 2 + ? 2 3 4 an candidates failed in English Mud failed in If 17% failed in both the subjects, then what percent passed in both the subjects ? **(b)** 77 58 48 23 If an article is sold at a gain of instead of a

131. **[CDS (I) Paper 20 Q36]** A field can reaped by 12 men or 18 women 40. in 14 days. In how many days can 8 men and 16 women reap it ? 38. (13 -A) Given that the polynomial (x2 • ax b) leaves the same remainder when divided by (x — I) Or (x 1). What are the values Of a and b **(a) (b)** 4 and O O and 3 3 and O O and any integer Tushar takes 6 hours to complete a piece of work, while Amar completes the same Work in IO hours. If both Of them work together, then what is the time required to complete the **(b) (c) (d)** 26 days 24 days 9 days 8 days **(a) (c) (d)** a hours 15 minutes 3 hours 30 minutes 3 hours 45 minutes If 4Y = 12', then z equal to **(a)** If (4a 7b) (4e— 7d) (4a— 7b) + 7d), then Which one Of the is ? 7b = DFGT-T-HTM

132. **[CDS (I) Paper 20 Q55]** 56. **(a)** 0 2 **(c)** sin 20 2 Suppose O < < 900, then for every O, 4 sin2 0 + I ig greater than or equal to There are two parallel streets each North to South. A in the first street travelling from South to North wishes to take the second street which is on his right side. At some place, he makes a 1500 turn to the right and he travels for 15 minutes at the speed Of 20 km/hr. After that he takes a lea turn Of 600 and travels for 20 minutes at the speed of JO km'hr in order to meet the second street. What is the distance between the two Streets ? **(a) (b) (d)** 4 sin 4 tan O **(b) (d)** 75 km 10•5 km 12,5 15 km DFGT-T-HTM

133. **[CDS (I) Paper 21 Q25]** A of gmiM is •uÖ 24 mn 48 days. How long will wne stock lut 160 ma ? **(a)** 17 12 32 19 21 24. N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ? **(b)** 2975 **(c)** 3094 **(d)** 3101

134. **[CDS (I) Paper 21 Q81]** A has avaage of 510 on md 240 days. What is the mnnb« Of visitors in a of 30 days b%inning Snu•y ? **(a)** 276 **(b)** 282 **(c)** 285 **(d)** 375 C - SDHY-T-MTR

135. **[CDS (I) Paper 21 Q91]** If (b — 6) is root Of the quüadc Nuaüon .x2 + b = O, whae b is intqer, is the rnAimun value of b2 ? O) 36 O) 49 81 93. is 15 4 15 of **(a)** 21 **(b)** 19 value of x2+5x+ O 94. mtio of (x 2) workas (x— 3) days to d01E by (x + 4) in (x — 2) is 3 : 4, valE of x ? **(b)** 10 **(c)** 12 **(d)** 15 95 WWCh of is **(c)** If pxq is p q ze always **(d)** is ü irrad&lal numba ,96. is of int. of n Br which + + 92 a per&ct squae ? **(d)** -19 31 C — SDHY-T-MTH 93. is D. IIE q O to p q whQ g O. Whn re the valuß Of p nd q ? 15 4 **(c)** -1.2 **(d)** -1,-2 4

136. **[CDS (I) Paper 22 Q6]** If Y allows X a Stan Of 70 m in a 1 •2 km race, then who will win the race and by what distance ? **(a) (b) (c) (d)** 9 X wins the race by 30 m Y wins the race by 90 m Y wins the race by 130m The race finishes in a dead heat A - PIXI-T-MTK

137. **[CDS (I) Paper 22 Q29]** A train takes two hours less for a journey of 300 km if its speed is increased by 5 knvhr from its usual speed. What is its usual speed ? **(a)** 50 knvhr **(b)** 40 km/hr **(c)** 35 knvhr **(d)** 25 km/hr

138. **[CDS (I) Paper 22 Q30]** If 6 men and 8 women can do a piece of work in IO days; and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman ? **(d)** 334 **(a)** 50 knvhr **(b)** 40 km/hr **(c)** 35 knvhr **(d)** 25 km/hr 30. If 6 men and 8 women can do a piece of work in IO days; and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman ?

139. **[CDS (I) Paper 22 Q70]** What is the difference between the maximum and minimum marks scored in the Internal Assessment ? **(a)** 15 **(b)** 17 **(c)** 20 **(d)** 24 Directions for the following three (03) items : Read the following information and answer the three items that follow : The following data presents count of released convicts who have served prison terms those Who have received some educational or technical training during their term (Y) and those who were offered Company placement (Z) respectively, from six different jails A, B, C, D, E and F, in the year 2010. A B c D E 86 1305 2019 1166 954 1198 45 940 869 544 464 z 25 461 474 416 254 174 13

140. **[CDS (I) Paper 22 Q39]** Jail with highest placement rate of trained convicts is PLKIÄ-MTK **(a)** 7-5 36. In a pi-diagram there are three sectors. If the ratio Of the angles Of the sectors is 1 : 2 : 3, then what is the angle of the largest sector ? **(b)** 2000 **(c)** 1800 **(d)** 1500

141. **[CDS (I) Paper 22 Q40]** Jails from which more than half of the trained convicts are offered jobs, are **(a)** A, B and C **(b)** A, B and D **(c)** A, E and F 41. The number of three digit numbers **(d)** Six

142. **[CDS (I) Paper 22 Q44]** In covering certain distance, the average speeds of X and Y are in the ratio 4 : **(a)** A, B and C **(b)** A, B and D **(c)** A, E and F 41. The number of three digit numbers **(d)** Six

143. **[CDS (I) Paper 22 Q87]** Out of 4 identical balls of radius r, 3 balls are placed on a plane such that each ball touches the other two balls. The 4th ball is placed on them such that this ball touches all the three balls. What is the distance Of centre Of 4th ball from the plane ? **(a)** 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ? **(b)** 4 square units **(c)** 245 square units **(d)** r square units

144. **[CDS (I) Paper 23 Q16]** The sum Of all possible products taken two at a time out of the numbers ± 1, ± 2, ± 3, ± 4, ± 5 **(a) (b) (c) (d)** - 30 - 55 55 A train of length 110 m is moving at a unifi)rm speed Of 132 kny'hr. The time requir«i to cross a bridge oflength 165 m is **(a) (b) (c) (d)** seconds 7 seconds 7 •5 seconds seconds The simple interest on a certain gum is one-murth of the sum. If the of years and the rate Of annual interest are numerically equal, then the number of **(a) (b) (c) (d)** 2-5 5 DZOL-T-LKM at

145. **[CDS (I) Paper 23 Q17]** 18. of A 60-page has n lines per page. If the of lines were reduced by 3 in each page, the number Of pages would have to be increased by IO to the same writing space. What is the value Of n ? **(a) (b) (c) (d)** 18 21 24 30 If x men working x hours day can do x units Of work in x days, then y men working y hours per day in y days would be able to do k units of work. What is the value of k ? **(a) (b) (c) (d)** x2y —S x3y —2 y2x-a y3x-2

146. **[CDS (I) Paper 23 Q18]** The simple interest on a certain gum is one-murth of the sum. If the of years and the rate Of annual interest are numerically equal, then the number of If x men working x hours day can do x units Of work in x days, then y men working y hours per day in y days would be able to do k units of work. What is the value of k ? **(a) (b) (c) (d)** 2-5 5 **(a) (b) (c) (d)** x2y —S x3y —2 y2x-a y3x-2 DZOL-T-LKM

147. **[CDS (I) Paper 23 Q24]** By inereasing the speed Of his ear by 15 km.'hr, a a distance of 300 km by taking an hour less than befire. What was the original speed of the car ? **(a) (b) (c) (d)** 45 km/hr 50 knv'hr 60 km/hr 75 kny'hr 'mree start a business with . The first capitals in the ratio person withdraw-s half his capital after 4 months. What is share Of profit if the business fitches an annual profit of 96,800 ? **(a) (b) (c)** e 32,000 e 34,500 36,000 e 36,800 DZOL-T-LKM /hr,

148. **[CDS (I) Paper 23 Q26]** Fith 27. (11-A) If x varies as y, then of the following Ware correct ? I. 2. x2 + varies as 12 — Y2 varies inversely ag y Fy varies as Select the correct answer using the given **(a) (b) (c) (d)** I and 2 only 2 and 3 only 3 only I, 2 and 3 Ena was born 4 years after her parents marriage. Her mother is 3 years younger than her father and 24 years older than Ena, who is 13 years old. At what age did Ena's father get married ? **(a) (b) (d)** 25 years 24 years 23 years 22 years By increasing the speed Of his ear by 15 km'hr,

149. **[CDS (I) Paper 23 Q40]** X, Y and Z travel from the same place with uniform 4 km.mr, 5 knvhr and 6 km.•'hr respectively. Y starts 2 hours after X. HOW long after Y must Z start in order that they overtake X at the same instant ? (e) 6 What is the square of 4x4 + 8x8 **(b) (c)** — hours — hours **(a) (b) (d)** 2x2 — 2x — I 2x2 x —I 2x2 + 2x + I 2x2 2x — I Hle Sum of the digits Of a two digit number is 13 and the difference between the number and that formed by reversing the digits is

150. **[CDS (I) Paper 23 Q41]** A car did a journey in t hours. Had the speed been x kmph greater, the journey would have taken y hours less. How long was the journey ? **(a) (b) (c)** square root Of an integer cube of an integer Square Of an HCF Of two integers **(d)**

151. **[CDS (I) Paper 23 Q42]** When a ball is allowed to the time it takes to any distance varies as the square root Of the distance and it takes 4 seconds to fall 7840 m. How long would it take to fall 122-50 m ? 43. **(a) (b) (d)** 5 seconds seconds 6•5 seconds If 63 41+5 8 (Given log102 0•301 and loglo 3 = 0477), then which one of the ft'llowing is correct ? **(a) (b) (c) (d)** DZOL-T-LKM the

152. **[CDS (I) Paper 23 Q2]** Its volume is rational. Select the correct answer using-the code given **(a) (b) (c) (d)** I only 2 only Both 1 and 2 Neither I nor 2 If — sin e = m and e — cos e then what is m 3 n 3 + man g equal to ? **(a)** m2n2 If cos O • sec O Sin2 e — ? **(c) (d)** k. then what is the value of A car did a journey in t hours. Had the 44• 41. speed been x kmph greater, the journey would have taken y hours less. How long was the journey ? x (t — y) ty x (t — y) ty¯

153. **[CDS (I) Paper 23 Q53]** A road Curve is to be laid out on circle. What radius should be used if the track is to change direction by 420 in distance of 44 m ? 22 (Assume **(a) (b)** Consider the following statements : 49. 50. **(c) (d)**

154. **[CDS (I) Paper 24 Q29]** A train 200 m long pages a 100 m long in IO seconds. What is the speed of the train? **(b)** 99 **(d)** 101 **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

155. **[CDS (I) Paper 24 Q30]** If 40 m/ s 30 mis 25 m/s 20 m/ s lx2 2x3 3x4 then What is the value Of n ? 98 100 SOA 100 **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

156. **[CDS (I) Paper 24 Q34]** A train travels 600 km in 5 hours and the next 900 km in IO hours. What is the average speed of the train? **(a)** 99 **(b)** 101 31. A trader gives successive discounts Of 20%, 10% and 5% respectively. What is the **(c)** overall discount? 316% 32. 33 34. A sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched e 500 more. What was the principal amount? **(d)**

157. **[CDS (I) Paper 24 Q39]** If 20 can clean 20 noors in 20 days, then in how many days can 16 persons clean 16 floors? **(b) (c) (d)** 25 days 24 days 20 days 16 days Let the work done by (x —l) men in (x + l) days be y. Ikt the work done by (x men in (x— l) days be z. If y: z = 9 : then What is the value of x? 9 10 12 IP.T.o. 3 S. Walking at Of his usual speed, a man is 12 minutes late for his office. What is the usual time taken by him to cover that distance ? **(a) (b)** 48 minutes 50 minutes 54 minutes 60 minutes **(c) (d)** 15 litres 20 litres 24 litres 25 litres 36. The cost price of 100 manoes is equal to the selling price Of 80 manoes. What is the profit percentage? 16% 20% 24% 25% 37. X sells his goods 25% cheaper than Y and 25% dearer than Z. How much percentage is Z's goods cheaper than Y? 100 200 XVWS-T-MTK/SOA 39. If 20 can clean 20 noors in 20 days, then in how many days can 16 persons clean 16 floors?

158. **[CDS (I) Paper 24 Q64]** The difference between the outside and the inside surface area Of a cylindrical pipe 14 cm long is 44 cm **(a) (b)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 66. 63. The ratio of the radius of base to the height of a cylinder is 2 : 3. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? **(c)** 242 cm **(d)** 385 cm 2

159. **[CDS (I) Paper 25 Q6]** 20 men are supposed to complete a work in 10 days. Aner working for 5 days, they realise that only one-fourth of the work is done. How many more men they need to employ to finish the work on time ? **(a)** 40 **(b)** 30 **(c)** 20 **(d)** 15

160. **[CDS (I) Paper 25 Q14]** The LCM of two prime numbers p and q is 2231, where p > q. What is the value of p — q ? **(a)** 67 **(b)** 70 **(c)** 74 **(d)** 81 IS. Three runners are running in a circular track, and they complete one round in 20, 30 and 35 minutes respectively. When will they next meet at the starting point ? **(a)** After 3 hour 30 minutes **(b)** After 4 hour 30 minutes **(c)** After 3 hours **(d)** Aner 7 hours

161. **[CDS (I) Paper 25 Q28]** Let work done by (3n — 1) men in (2n + 1) days be x and work done by (3n + 1) men in (4n —3) days be y. If x : y = 6: 11, then what is the value of **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 only when Ware correct ? 28. Let work done by

162. **[CDS (I) Paper 25 Q33]** A motor boat has 30 km/hr in still water. It goes 60 km down stream and comes back in hours. What is the **(a) (b) (c) (d)** 9 speed of the stream ? 5 km/hr 8 km/hr 10 km/hr 12 km/hr A - NYRG-T-MTK

163. **[CDS (I) Paper 25 Q39]** Sheela can stitch a suit in 2 days. while Meena can stitch a suit in (hys. How many days will both take in stitching 30 suits ? **(a)** 32 days **(b)** 33 days **(c)** 35 days **(d)** 40 days

164. **[CDS (I) Paper 25 Q42]** A agrees to work for 30 days, on a condition that for every day's work he should receive 5(H), and that for every day's absence from work he should forfeit **(a)** 20 **(b)** 21 **(c)** 24 **(d)** 25 43. A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ?

165. **[CDS (I) Paper 25 Q100]** At the end of the time he received 11,400. How many days did he work ? **(a)** 20 **(b)** 21 **(c)** 24 **(d)** 25

166. **[CDS (I) Paper 25 Q44]** A person rode one-third of a journey at 60 km/hr, one-third at 50 km/hr and the rest at 40 km/hr. Had the person ridden half of the journey at krnfhr and the rest at 40 km/hr, he would have taken 4 minutes longer to complete the journey. What distance did the person ride ? **(a)** 180 km **(b)** 210 km **(c)** 240 km **(d)** 300 km

167. **[CDS (I) Paper 25 Q62]** Two circles touch externally. The sum of their areas is 411t square cm. If the distance between their centres is 9 cm, then what is difference between their diameters ? **(b)** I •5 cm **(c)** 2 cm **(a)** sin 9 **(b)** cose **(c)** tan 0 **(d)** cot O 59. If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C

168. **[CDS (I) Paper 26 Q7]** A man walks at an average speed of 3 km/hr from his residence and reaches Office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence and office ? **(a)** 6 km **(c)** IO km **(b)** 8 km **(d)** 12 km

169. **[CDS (I) Paper 26 Q19]** A car travels from A to B at a speed Of 40 km/hr, travels back from B to A at a speed of 30 km/hr and again goes from A to B at a speed of 60 km/hr. What is the average speed of the car? **(a)** km/hr **(b)** 42 km/hr **(c)** 40 km/hr **(d)** km/hr

170. **[CDS (I) Paper 26 Q27]** 8 men or 12 women can do a piece Of work in 24 days. In how many days can the work be done by 8 men and 12 women? **(a) (b) (d)** 12 days 18 days 24 days Cannot be determined due to insufficient data A car takes p minutes to travel a distance of 350 km With an average speed Of u km/hr. Another car takes q minutes to travel the same distance with an speed of v km/hr. If u —v = 5 and q — p 140 then what is the value of u? **(a)** 35 **(c)** 25 **(b)** 30 **(d)** 20

171. **[CDS (I) Paper 26 Q29]** HOW many minutes are there in x weeks and x days? **(a)** 11520x **(c)** 480x **(b)** 5760x **(d)** 192x

172. **[CDS (I) Paper 26 Q40]** A train X takes 2 hours less than a train Y to cover a distance of 192 km between two cities. Their average speeds differ by 16 km/hr. HOW long does the faster train take to cover the journey? **(a)** 3 hours **(c)** 5 hours 9 **(b)** 4 hours **(d)** 6 hours [P.T.O.

173. **[CDS (I) Paper 26 Q53]** An following? aeroplane is Observed to be approaching the airport. It is at a distance Of IO km from the point of observation on the ground and makes an angle Of elevation e. If the aeroplane is at a height Of 8 km above the ground, then Which one of the following is corre ct? **(a)** 120' **(b)** 135' **(c)** 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is **(d)** following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

174. **[CDS (I) Paper 26 Q83]** ABC is a, triangle right angled at B. Let D be the midpoint On AC If BD = 6 •5 cm, then What is AB2 + BC2 equal to? **(a)** 144 square cm **(b)** 169 square cm **(c)** 196 square cm **(d)** 225 square cm Water is trickling out of a completely filled cylindrical tank of height I m and diameter 2 m. Every second a spherical droplet Of I Crn radius trickles down from the bottom Of the tank. The tank Will be emptied in approximately **(a)** 280 hours **(b)** 260 hours **(c)** 230 hours **(d)** 210 hours

175. **[CDS (I) Paper 27 Q15]** X and Y can do a piece Of work in 45 days and 40 days respectively. They begin to work together, but X leaves after n days and then Y completes the remaining work in 23 days. What is n equal to ? **(c)** 480 8 9 10 12 'Prsons working IO hours per day complete one-fourth of a work in IO days. How many additional are to employed in order to complete the remaining work in 20 days, working 8 houry per day ? **(a) (c) (d)** 400 420 480 500 12 women and 16 men can do a piece of work in 5 days. 13 women and 24 men can do it in 4 days. 'How long Will 25 women and 50 men take to do it ? **(a) (b) (c) (d)** I day 2 days 3 days 4 days FDGT-T-EMT

176. **[CDS (I) Paper 27 Q16]** 17. k Ther•e are two stations X and Y, 1320 km apart. A train starts from station X at 6 a.m. and moves at an average speed of 60 km/hr. At 2 p.m. another train starts from Y towards X and moves at an average speed of 80 knv'hr. When do they meet ? **(a) (b) (c) (d)** 6 p.m. 7 p.m. 9 p.m. TWO glasses Of equal volume are filled With a mixture Of alcohol and water in the ratio 3 : 2 and 4 : I, respectively. These glasses are emptied into a third glass. What is the ratio Of alcohol and water in the third glass ? **(b) (c) (d)**

177. **[CDS (I) Paper 27 Q17]** 12 women and 16 men can do a piece of work in 5 days. 13 women and 24 men can do it in 4 days. 'How long Will 25 women and 50 men take to do it ? **(a) (b) (c) (d)** I day 2 days 3 days 4 days TWO glasses Of equal volume are filled With a mixture Of alcohol and water in the ratio 3 : 2 and 4 : I, respectively. These glasses are emptied into a third glass. What is the ratio Of alcohol and water in the third glass ? **(b) (c) (d)** FDGT-T-EMT

178. **[CDS (I) Paper 27 Q36]** In a flight Of 2800 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 100 kmfhr and time Of flight increased by 30 minutes. What was the original average speed Of the aircraft ? **(a) (b) (c) (d)** 700 km/hr 750 km,'hr 800 km/hr 900 knv'hr If x b + c, y c + a, z = a + b, then what is (x + y + — 24xyz equal to ? **(a) (b) (c)** a3 + b3 None of the above A person X starts from place A and another person Y starts simultaneously from another place B which is d km away from A. They walk in the same direction. X walks at an average speed of u knv'hr and Y walks at an average speed Of v km/hr. How far will X have walked before he Overtakes Y ? ud **(a)** vd (ud — vd) (ud + vd) FDGT-T-EMT (15-

179. **[CDS (I) Paper 27 Q39]** 40. 5-A) If p is the difference between a number and its reciprocal and q is the difference between the square Of the same number and the square of its reciprocal, then what is p4 + 4p2 equal to ? **(a) (b) (c)** If x 4q 8q 4q2 = 2 + 212, then what is the value of **(d)** + 16x-4? 152 144 136 132 X takes 3 hours longer thali Y to walk 30 km. • If X doubles his he takes 2 hours less than Y. What is the speed of Y ? **(a) (b) (c)** 3 km/hr 4 knvhr 4 — km/hr 4 — km.mr If C is the compound interest on 10,000 for one year at 4% annum when compounded quarterly, then which one of the following is correct ? **(b) (c)** 100 c»e400 A person X starts from place A and another person Y starts simultaneously from another place B which is d km away from A. They walk in the same direction. X walks at an average speed of u knv'hr and Y walks at an average speed Of v km/hr. How far will X have walked before he Overtakes Y ? ud **(a)** vd (ud — vd) (ud + vd) FDGT-T-EMT

180. **[CDS (I) Paper 27 Q34]** In a flight Of 2800 km, an aircraft was slowed 37• down due to bad weather. Its average speed for the trip was reduced by 100 kmfhr and time Of flight increased by 30 minutes. What was the original average speed Of the aircraft ? **(a)** vd **(b) (c) (d)**

181. **[CDS (I) Paper 27 Q35]** 36. (15-A) If p is the difference between a number and its reciprocal and q is the difference between the square Of the same number and the square of its reciprocal, then what is p4 + 4p2 equal to ? **(a) (b) (c) (d)** 700 km/hr 750 km,'hr 800 km/hr 900 knv'hr 38. **(a) (b) (c)** If x 4q 8q 4q2 = 2 + 212, then what is the value of If x b + c, y c + a, z = a + b, then what is (x + y + — 24xyz equal to ? **(a) (b) (c)** a3 + b3

182. **[CDS (I) Paper 27 Q88]** Two circles touch externally. The sum of their 91. areas is 89m square em and the distance between their centres is 13 cm. What is the difference in their radii ? **(a) (b) (c)** IO em FDGT-T-EMT 88. Two circles touch externally. The sum of their 91. areas is 89m square em and the distance between their centres is 13 cm. What is the difference in their radii ? 89. 90. **(d)**

183. **[CDS (I) Paper 28 Q8]** 9. IO. A can do a certain work at twice the speed Of 11 B. Further, B can do the same work at I •5 times the speed Of C. All Of them together can finish the work in 12 days. In how many days can C alone finish the work ? **(a) (b) (c) (d)** 36 days 45 days days 66 days The sum of digits of a 2-digit number is

184. **[CDS (I) Paper 28 Q12]** When the digits are reversed, the number becomes greater by eighteen. What is the difference between the digits in the number ? **(a) (b) (c) (d)** 2 4 The time taken by a train to cross a man travelling in another train is IO seconds, when the Other train is travelling in the opposite direction. However, it takes 20 seconds, if both the trains are travelling in the same direction. The length of the first train is 200 m and that of the second train is 150 m. What is the speed Of the first train ? **(a) (b) (c)** 60 km/hr 56 km."hr 54 km/hr 52 km/hr JNBY-D-MTK if

185. **[CDS (I) Paper 28 Q11]** B. Further, B can do the same work at I •5 times the speed Of C. All Of them together can finish the work in 12 days. In how many days can C alone finish the work ? If a, b, c, d, e and f satisfy 2a = 3b = 6c = 9d = 12e = 18f, then What is the value of (a + + d + e + fi ? **(a) (b) (c) (d)** 36 days 45 days days 66 days **(b) (c) (d)** 4/7 2 5/2 9/2

186. **[CDS (I) Paper 28 Q86]** If 17 women and 24 men can do a piece of work in 5 days and 12 women and 23 men can do it in 6 days, then which one of the following is ? **(a) (b) (c) (d)** Effciency Of 13 women = Effciency Of 18 men = Efficiency Of Effciency Of 11 women men Emciency of Efficiency Of 13 women 17 men Effciency Of I I women = Efficiency Of 15 men Three A, B and C together can fill a tank in 6 hours. Tap C alone can fill the tank in 12 hours. To fill the tank, when it is empty , all the three taps are started together. After working t hours, tap C is closed and the tank is filled in 8 more hours. What is t equal to ? **(a) (b) (c) (d)** 4 6 A, B and C can complete a work in x, I •5x and 2x days respectively. If they complete the work together, in what ratio should they be paid ? **(a) (b) (d)** JNBY-D-MTK (13- in all

187. **[CDS (I) Paper 28 Q38]** What is the last digit Of the sum S **(a) (b) (c) (d)** Effciency Of 13 women = Effciency Of 18 men = Efficiency Of Effciency Of 11 women men Emciency of Efficiency Of 13 women 17 men Effciency Of I I women = Efficiency Of

188. **[CDS (I) Paper 28 Q40]** A, B and C can complete a work in x, I •5x and 2x days respectively. If they complete the work together, in what ratio should they be paid ? **(a) (b) (d)** JNBY-D-MTK The speed of a boat in still water is 15 knv'hr. If it can travel 42 km downstream and 28 km upstream in the same time, then what is the speed Of the stream ? **(a) (b) (c)** 2 •5 knu'hr 3 km/hr km."hr 6 km."hr

189. **[CDS (I) Paper 29 Q45]** What is the product Of the digits ? **(a) (b) (c) (d)** 5 2 If A and B can finish a work in IO days, B and C can finish the same work in 12 days, C and A can finish the same work in 15 days; then in how many days can A, B and C together finish half Of the work ? **(a)** 8 days **(b)** 5 days **(c)** 4 days **(d)** 3 days JNUP-T-EMT (30 en

190. **[CDS (I) Paper 29 Q6]** 7. For What relation between a and b is the equation gin B = possible ? **(a) (b) (c) (d)** ab ag b **(a) (b) (c) (d)** 5 2 If A and B can finish a work in IO days, B and C can finish the same work in 12 days, C and A can finish the same work in 15 days; then in how many days can A, B and C together finish half Of the work ? Three A, B and C together can do a piece of work in 36 days. A and B together can do five times as much work as C alone; B and C together can do as much work as A alone. If A and C together can do n times as much work as B alone, then what is the value **(a) (b)** 8 days 5 days 4 days 3 days **(b) (c) (d)** 16 2 3 JNUP-T-EMT

191. **[CDS (I) Paper 29 Q2]** If the sum of the two numbers is 45, then what is the product of the numbers ? **(a) (b) (c) (d)** 243 486 504 Cannot be determined insufficient data due to A man walks at an average speed of 3 km/hr from his home and reaches office 40 minutes early. If he walks at an average speed of 2 km/hr, he would reach office 40 minutes late. What is the distance between his home and office ? **(a) (b) (c) (d)** 8 km IO km 12 km 4b

192. **[CDS (I) Paper 29 Q78]** A train completely overtakes two persons, walking in the same direction with speeds 3 knu'hr and 4 km/hr in 9 seconds and 75 — seconds respectively. What is the length Of 8 the train ? **(a)** 20 A train completely overtakes two persons, walking in the same direction with speeds 3 knu'hr and 4 km/hr in 9 seconds and 75 — seconds respectively. What is the length Of 8 the train ? **(b) (c) (d)**

193. **[CDS Paper 30 Q31]** GM of these two numbers is lower than their AM by 20%, then What is the difference between the two numbers ? **(b) (c)** 12 18 24 28 50 men can complete a work in 40 days. They begin the work together but a batch of 5 men 32. left after each period of IO days. What is the time to complete the work ? **(a) (b) (c)** 45 days 50 days 55 days 60 days then which one Of the following is correct ? **(a) (b) (c) (d)** O < x < 0-5 SRSU-T-EMT (11 —A)

194. **[CDS Paper 30 Q45]** If the work done by x men in (x + I) days is equal to the work done by (x * 5) men in (x — 2) days, then What is the value ? Let the positive numbers at, a2, as, agn be in GP. IfPis the GMofa1, as, ...,anandQ is the GM of 1, an an +3, a3n, then What is the GM Of 3n numbers ? 46. p2Q PQ2 pV3 **(a)** 5 6 75 9 **(b) (c) (d)**

195. **[CDS Paper 30 Q55]** 57. Two trains A and B leave Delhi for Hyderabad 58. at 7:00 a.m. and 7:50 a.m. on the same day and travel at 80 kmph and 100 kmph respectively. After how many kilometers from Delhi will the two trains be together ? **(a) (b)** 3 100 km 3 1000 **(c) (d)**

196. **[CDS Paper 30 Q2]** PQ varies as R. 3. p2 — QZ varies as R. Select the correct answer using the code given **(a) (b) (c)** I and 2 only 2 and 3 only I and 3 only I, 2 and 3 p number of men can finish a piece Of work in q days. If there are more men, then the work will be finished 12 days earlier. What is the value of q ? **(d)** 48 40 36 Cannot be determined insufficient data due to a2 * 3a+I What is the minimum value Of b2+3b+1 **(a)** for a, b > O ? 16 25

197. **[CDS Paper 30 Q73]** How far from station Q will the two trains meet ? **(a) (b) (c)** — square cm 3 77 — square em square em 6 616 24 units. Question : Statement-I : Statement-II : 72. What are the values of r and r is an integer. R is an integer. square cm 9 If p is the perimeter of the smallest sectnr, then what is the value of 9p ? **(d)** 142 cm

198. **[CDS Paper 30 Q74]** 75. (25-A) **(a) (b) (c) (d)** 5:28 a.m. 5:44 a.m. 4:56 a.m. 6:24 a.m. If the lengths Of the two trains A and B are 400 m and 500 m respectively, then what is the time taken by them to cross each other ? **(a) (b) (c)** 21 •6 seconds 18•2 seconds 17 •4 seconds 15 •4 seconds Consider the following for the next three (03) items that follow : A triangle CEF is drawn inside a square ABCD as shown in the figure given belcm'. Given : CF = 8 em, EF = 6 cm and CE = IO cm. D SRSU-T-EMT

199. **[CDS (I) Paper 31 Q9]** Train X crosses a man standing on the platform in 24 seconds and train Y crosses a man standing on the platform in 18 seconds. They cross each other while running in opposite directions in 20 seconds. What is the ratio of speed of X to speed of Y? **(a)** 17 **(b)** 14 **(c)** 11 **(d)** 10 9. Train X crosses a man standing on the platform in 24 seconds and train Y crosses a man standing on the platform in 18 seconds. They cross each other while running in opposite directions in 20 seconds. What is the ratio of speed of X to speed of Y? 10. Let p, q be the roots of the equation x2 + mx — n 0 and m, n be the roots of the equation .r2 + px—q = O

200. **[CDS (I) Paper 31 Q11]** Consider the Question and the Statements and mark the correct option. Question Statement-I Statement-II In how many days can A, B and C together finish the work ? : A and B together can finish the work in 24 . B and C together can finish the work in 36 days Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements 23

201. **[CDS (I) Paper 31 Q77]** The average of the temperatures recorded at noontime from Monday to Sunday is 31 oc. Ifthe lowest tempera- ture recorded is 300C, then what is the maximum of temperature that is possible to record at noontime on any one of the days **(a)** 340C **(b)** ,350C **(c)** 360C **(d)** 370C 33

202. **[CDS (I) Paper 32 Q23]** If your image appears to be erect, no matter 24. how far you stand from a mirror, the mirror is likely to : **(a) (b) (c) (d)** Plane Concave Convex Either plane or convex The rule that determines the direction Of a 25. magnetic field produced around a straight conductor carrying current is : **(a) (b) (c) (d)** Right-hand thumb rule Flemings left-hand rule Flemin€s right-hand rule Hund's rule A motorcycle has initial velocity Of 5 m/s. After 3 seconds, the velocity is 7 The displacement of the motorcycle in 3 seconds **(a) (b) (c)** 21 m 18 m 36 m CEBN-S-GKL

203. **[CDS (I) Paper 32 Q31]** 2 x 10-4 is completely immersed in water. The buoyant force exerted by water on the sphere is : (Given : density of water 1000 kg/m3 , g = IO m/s2) **(c)** 0.5 N 1-5 N 2-5 N A light bulb rated as 60 W at 220 V has a potential difference of 110 V across its ends. The B)wer dissipated in this light bulb is : **(a) (c)** 30 w 45 w 15 w Which one among the following is used in a simple microscope ? **(a) (b) (c) (d)** Diverging lens Concave mirror Converging lens Plane mirror The force with which the Earth attracts an Object is called : **(a)** Weight of the obj ect **(b)** Mass of the object **(c)** Density of the object **(d)** Magnitude of the obj ect CEBN-S-GKL

204. **[CDS (I) Paper 32 Q69]** 70. Core inflation is : **(a) (b) (c) (d)** CPI headline inflation minus all non-food WPI inflation rninus all food and non -food items CPI headline inflation minus food and energy items WPI inflation minus food and energy items In alignment with the guidelines of NEP 2020, 'Poshan bhi Padhai bhi' (PBPB) was launched under which one among the following programmes ? **(a)** Integrat«l Teacher Training Programme **(b)** Vidya Pravesh (e) Early Childhood C are and Education **(d)** Samagra Shiksha Abhiyan Which one among the following defines flexi workers in India ? **(a) (b) (c)** Formal contract staff employed through organised contract Informal sector employees working in urban areas Agricultural labour migrating to urban Regular salaried workers working in registered companies Which one among the following is an integrated teacher training programme ? **(a) (c)** Vidya pravesh NISHTHA Vidyanjali DIKSHA Which one among the following established the 67• 64. first Global Capability Centre in India ?

205. **[CDS (I) Paper 32 Q70]** Which one among the following is an integrated teacher training programme ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 **(a) (c)** Vidya pravesh NISHTHA Vidyanjali DIKSHA CEBN-S-GKL

206. **[CDS (I) Paper 33 Q51]** TWO trains X and Y are travelling in the same direction at 100 km/hr and 60 km/hr respectively. Train X crosses a man in train Y in 9 seconds. What is the length of train X ? 80 m 100m 120m 150m **(a)** p q **(b)** 2p q **(c)** 2q p — l **(d)** 80 m 100m 120m 150 m 19 52. TWO persons X and Y leave place P for place Q at 7:00 a.m. and 7:10 a.m. respectively along the same path. X walks at a speed of 4 •8 km/hr and Y walks at a speed Of 6 km/hr. HOW many kilometres from place p Will X meet Y ?

207. **[CDS (I) Paper 33 Q52]** TWO persons X and Y leave place P for place Q at 7:00 a.m. and 7:10 a.m. respectively along the same path. X walks at a speed of 4 •8 km/hr and Y walks at a speed Of 6 km/hr. HOW many kilometres from place p will X meet Y ? **(a)** p q **(b)** 2p q **(c)** 2q p — l **(d)** 80 m 100m 120m 150 m 19 52. TWO persons X and Y leave place P for place Q at 7:00 a.m. and 7:10 a.m. respectively along the same path. X walks at a speed of 4 •8 km/hr and Y walks at a speed Of 6 km/hr. HOW many kilometres from place p Will X meet Y ?

208. **[CDS (I) Paper 33 Q68]** TWO poles are situated 24 m apart and their heights differ by I O m. What is the distance between their tips? **(a) (b)** 6-4 m 67. ABC is a triangle right anøed at B. Further , **(c) (d)** 75 units 100 square units 125 square units 150 units 68. TWO poles are situated 24 m apart and their heights differ by I O m. What is the distance between their tips?

209. **[CDS (I) Paper 33 Q100]** What is the mean Of the distribution? 21-625 22.225 **(c)** 23 225 24 625 P.T.O. For the following two (02) items : The following data represent the distance covered (in metres) by two groups Of athletic median value 24 children. It is known that the median distance in the first group is 20 • 8 metres while the mean distance in the second group is 17 • 3 metres. Some frequencies in the groups are **(a)** 20 **(b)** 25 For the following tvo **(c)** 3 98. What is the value of v? 5 8 ANFB-T-EMT/4SA 99. What is the value of k ? 22 30 100. What is the mean Of the distribution? 21-625 22.225 **(d)** 23 225 24 • 625 P.T.O. For the following tvo

210. **[CDS (I) Paper 34 Q9]** Eight men and 24 women can finish a piece of work in one day while 12 men and IS women can also finish it in one day. What is the time taken by 24 men and 72 women to finish the work ? 5 **(a)** x4 — 2a2x2 + a4 **(b)** x4 + 2a2x2 + a4 **(c)** xa — a2x2 4 a4 8. Which of the following expressions can divide both the polynomials x3+2x2—5x+2 and x3 +4x2+x—6 exactly ? 11. X+l 111. X+2 Select the correct answer using the code given below : **(d)** I and 11 only

211. **[CDS (I) Paper 34 Q14]** X can do a piece of work in 4 days and Y can do the same piece of work in 6 days. Which of the following statements is/ are correct ? I. If X and Y work alternately starting with X, then the piece of work will be finished on the fifth day. **(a)** 3 months **(b)** 6 months **(c)** 9 months **(d)** 1 year 13. If x varies directly as y, then which of the following is/ are correct ? I.

212. **[CDS (I) Paper 34 Q11]** If X and Y work alternately starting with Y, then the piece of work will be finished in less than 5 days. Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

213. **[CDS (I) Paper 34 Q15]** 12 men working 8 hours a day can completely build a wall of length 100 m, breadth 20 cm and height 5 m in 10 days. How many days will 16 men working 10 hours a day require to build a wall of length 200 m, breadth 60 cm and height 5 m ? **(a)** 18 **(b)** 24 **(c)** 27 **(d)** 36

214. **[CDS (I) Paper 34 Q16]** A • train P starts from station X for station Y and another train Q starts from station Y for station X at the same time. After passing each other, P takes 4 hours to reach Y and Q takes 1 hour to reach X. If the average speed of P is 60 km/h, then what is the average speed of Q ? **(b) (c) (d)** 72 km/h 80 km/h 96 km/h 120 km/h 7 A - KPRS-T-MTE **(a)** 3 months **(b)** 6 months **(c)** 9 months **(d)** 1 year 13. If x varies directly as y, then which of the following is/ are correct ? I.

215. **[CDS (I) Paper 34 Q17]** Without stoppages, a train travels at an average speed of 60 km/h and with stoppages it covers the same distance at an average speed Of 48 km/h. HOW many minutes per hour does the train stop ? **(a)** 10 minutes **(b)** 12 minutes **(c)** 1 S minutes **(d)** 20 minutes

216. **[CDS (I) Paper 34 Q18]** A person goes downstream at x km/h and upstream at y km/h by the same boat. What is the ratio of speed of boat in still water to speed of •water ? Y **(a)** 10 minutes **(b)** 12 minutes **(c)** 1 S minutes **(d)** 20 minutes 18. A person goes downstream at x km/h and upstream at y km/h by the same boat. What is the ratio of speed of boat in still water to speed of •water ? Y 19. Let Q=x2 + bx + c. If the sum of the roots is equal to product Of the roots of the equation Q = 0, then which of the following statements is/are correct ? 20. 21. 11. Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below :

217. **[CDS (I) Paper 34 Q26]** What is the greatest possible speed at which a person can walk 13-3 km and 209 km so that the time (in hours) required in each case is a whole number ? **(a)** 1-7 km/h **(b)** 1-9 km/h **(c)** km/h **(d)** 2•3 km/h

218. **[CDS (I) Paper 34 Q34]** A locomotive engine can go 40 km/hr. Its speed gets reduced by a quantity that varies directly as the square root of the number of wagons attached. It is known that its speed becomes 16 km/hr if 36 wagons are attached. What is the smallest number of wagons with which the engine is unable to move ? **(a)** 100 **(b)** 81 **(c)** 64 **(d)** 49

219. **[CDS (I) Paper 35 Q6]** Suppose p and q are positive integers where q is not a perfect square. If is a root of a quadratic equation having integer coemcients, then which Of the following statements Ware correct ? I. The sum of the roots is 2p times the product Of the roots. IL The sum Of the reciprocals Of the roots is equal to 2p. Select the answer using the code given below : **(a) (b) (c) (d)** If I only 11 only Both I and 11 Neither I nor 11 , then what is 2b + 2c 2c + 2a a2 b2 + c2 to ? **(a) (b) (c) (d)** 9c2 4c2 3c2 2c2 A train completely crossed two persons X and Y moving in the same direction as the train in 8 seconds and 8•1 seconds respectively. X is walking at a uniform speed of 3 km/hr and Y is walking at a uniform speed of 4 km/hr. What is the speed of the train ? **(a) (b) (c) (d)** 81 krn/hr 82 km/hr• 83 km/hr 84 km/hr TWO IXvsitive integers x and y are in the ratio 17 :

220. **[CDS (I) Paper 35 Q14]** Had I walked 0-5 knv'hr faster, I would have taken 8 minutes less to walk 2 km. What is my original speed ? **(a) (b) (c) (d)** 2 •5 knvhr 2 • 75 km.'•hr 3 knvhr 3 •25 km/hr

221. **[CDS (I) Paper 35 Q16]** 17. Had I walked 0-5 knv'hr faster, I would have taken 8 minutes less to walk 2 km. What is my original speed ? **(a) (b) (c) (d)** 2 •5 knvhr 2 • 75 km.'•hr 3 knvhr 3 •25 km/hr

222. **[CDS (I) Paper 35 Q24]** 25. 26. -A) A boatman can row to a place (Y) at a distance of 24 km from the starting point (X) and back in 14 hours. If he can row 4 km with the stream in the same time as 3 km against it, what is the speed of the stream ? **(a) (b) (c) (d)** 5 km/hr I km/hr 2 km/hr km/hr An alloy (X) of gold and silver is taken in the ratio I : 2 and another alloy (Y) Of gold and silver is taken in the ratio 2 :

223. **[CDS (I) Paper 35 Q26]** A boatman can row to a place (Y) at a distance of 24 km from the starting point (X) and back in 14 hours. If he can row 4 km with the stream in the same time as 3 km against it, what is the speed of the stream ? **(a) (b) (c) (d)** 960 gm 950 gm 925 gm 900 gm **(a) (b) (c) (d)** 5 km/hr I km/hr 2 km/hr km/hr If the roots of the equation px2 — 40K • 96 = O are even integers, then what is the value Of p? **(a) (d)** 2 3 4 A number N consists Of two digits. The digit in the tens place is 3 times the digit in the units place. The digits are reversed and the resulting number is denoted by R. If N x R = 3627, then what is the product of the digits of the N ? An alloy (X) of gold and silver is taken in the ratio I : 2 and another alloy (Y) Of gold and silver is taken in the ratio 2 :

224. **[CDS (I) Paper 35 Q85]** If the median Of the data is 67, then what is the value of x ? **(b) (c) (d)** 66 65 63 Cannot be determined insufficient data due to A cyclist pedals from her house to her Offce at a speed of 3 kmph and back from the Offce to her house at 6 kmph- What is the average speed ? **(a) (b) (c)** 5 kmph 4-5 kmph 4 kmph kmph BFVS.T-TME If the heights (in em) Of 9 students Of a class are 150, 165, 145, 149, 150, 147, 152, 144 and 148, then what is the algebraic sum of the heights measured from their arithmetic **(b)** 0 5 150 Consider the following grouped data : Class 0— 10 10-20 20-30 30-40 40-50

CHAPTER 06

Total Archive Questions: 559 Questions

ALGEBRA, POLYNOMIALS & LINEAR EQUATIONS

Typical Exam Weightage: 10 - 15 Questions (10 - 15 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q17] The product Of two alternate odd integeN exceeds three times the smaller by **(a)** I only **(b)** 2 only **(c)** O or I If x —I + •Æ, then what is the value of x' — 4x3 + ? What is the magnitude of difference Of the roots of x2 — ax + b = O ? a2 —4b What is the solution Of the equations x -y— and ll **(d)** x = and y

2. [CDS (I) Paper 1 Q18] If = b and — a' d then which one of the following is correct ? **(a)** I only **(b)** 2 only **(c)** O or I If x —I + •Æ, then what is the value of x' — 4x3 + ? What is the magnitude of difference Of the roots of x2 — ax + b = O ? a2 —4b What is the solution Of the equations x -y— and ll **(d)** x = and y

3. [CDS (I) Paper 1 Q19] A ball is dropped from a height 64 m above the ground and every time it hits the ground it rises to a height equal to half of the previous. What is the height attained after it hits the ground for the 16th time ? **(b) (c)** 2-10m 2-4 m **(d)** 4 **(a)** I only **(b)** 2 only **(c)** O or I If x —I + •Æ, then what is the value of x' — 4x3 + ? What is the magnitude of difference Of the roots of x2 — ax + b = O ? a2 —4b What is the solution Of the equations x -y— and ll **(d)** x = and y

4. [CDS (I) Paper 1 Q49] From a rectangular metal sheet Of sides 25 cm and 20 em, a circular sheet as large as possible is cut-off. What is the area of the remaining sheet ? **(a)** 186 **(b)** 144 **(c)** 93 cm2 **(d)** 72 cm2

5. [CDS (I) Paper 1 Q56] A cylindrical vessel of base radius 14 cm is filled with water to some height. If a rectangular solid of dimensions 22 cm x 7 cm x 5 Ctn is immersed in it, what is the rise in water level ? **(a)** 12 cml **(b)** 18 cm2 **(c)** 24 cm2 **(d)** 36 cml If A is the area of a triangle in cm2, whose sides are 9 cm, IO and 11 cm, then which one Of the following is correct ?

6. [CDS (I) Paper 1 Q76] 77. **(d)** x —y = 10 **(a)** x —y = 10 78. 79. 80. 81. 82. 83. 84. sin x cos x 1/2 t, sin x — cos x DE t **(b)** *rut A log IO **(c)** Tt/2 **(d)** Tt/3

7. [CDS (I) Paper 1 Q7] Assertion **(a)** Both A and R are individually true and R is the correct explanation of A **(b)** Both A and R are individually true but R is not the correct explanation of A **(c)** A is true but R is false **(d)** A is false but R is true 98. Assertion

8. [CDS (I) Paper 2 Q31] When x40 + 2 is divided by x' + I, the remainder ? What is **(a) (b) (c) (d)** 2 3 4 WT1ich one of the following statements is correct ? **(a) (c) (d)** Remainder theorem is a special case of Factor theorem. Factor theorem is a special case Of Remainder theorem. Factor theorem and Remainder theorem arc two independent results. None Of the above If x 2 + , then Bhat is (x2 + x equal Io ? 12 13 14 15 What is the ratio Of sum of squares Of roots to thc product Of the roots Of the equation 7X2 + 12K +18=0? When a polynomial is divided by a linear polynomial, then what is the remainder ? **(a) (b) (c)** Constant polynomial only Zero polynomial only . Either constant Or zero polynomial Linear polynomial P-DETB-J-FN is

9. [CDS (I) Paper 2 Q32] If ax by cz and abc = xy + ya + zx equal to ? **(a) (b) (c) (d)** 2 3 4 **(a) (c)** XYZ 0 WT1ich one of the following correct ? statements special case then what is then what is the **(a) (c) (d)** Remainder theorem is a Factor theorem. 33. of Of If -0037Y 10000, value of — — — ? Factor theorem is a special case Remainder theorem. Factor theorem and Remainder theorem arc two independent results. None Of the above **(a) (c) (d)** 1/2 1/4 If x 2 + , then Bhat is (x2 + x equal Io ? 12 13 14 15 What is the ratio Of sum of squares Of roots to thc product Of the roots Of the equation 7X2 + 12K +18=0? When a polynomial is divided by a linear polynomial, then what is the remainder ?

10. [CDS (I) Paper 2 Q37] 38. If a then what is equal to ? **(c)** If pq + qr + rp z O, p — qr — rp then what is the value Of r — pq The length Of a line segment AB is 2 units. It is divided into two parts at the point C such that AC2 An x CB. What is the length Of **(a) (b) (c)** 3 + units 3 — units 2 units units If x + y z **(b) (d)** 42 then what is the •value of P-DETB.J.FN

11. [CDS (I) Paper 2 Q6] P-DETBO.FN 112 — ' **(a) (b) (c)** 2a3 -2aa **(d)**

12. [CDS (I) Paper 2 Q73] 440 670 78•5' 600 500 c In the figure AB is parallel to CD and BE is parallel to PH. What is Z FHE equal to ? **(a) (b) (c)** 73. 440 670 78•5' 600 500 c **(d)**

13. [CDS (I) Paper 2 Q78] 79. 80. (23 - A) **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

14. [CDS (I) Paper 3 Q16] What is the value of [(1 — sin2 0) sec2 +',tan2 61 (cos2 9 + i) when 0 < 0 < 900 ? **(a)** 72 **(b)** 216 **(c)** 372 **(d)** 600 18. If r and s are any real, numbers such that 1, then what is the maximum value Of the product rs ? 19. r is a non•zero real Humber such that r75 > PO. This is possible only when 20. The median BD of Che triangle ABC meets AC at D. If then which one of the following is correct ?

15. [CDS (I) Paper 3 Q22] What is the value of **(a)** 5•9949 **(b)** 5-9894 **(c)** 59984 **(d)** 5-9994

16. [CDS (I) Paper 3 Q2] What was the original wage bill of the factory if the present bill is Rs. 5,000/- ? **(b)** Rs. 6000/- **(c)** Rs. 8,000,'- **(d)** None of the above **(a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** days 33. If of

17. [CDS (I) Paper 3 Q35] If a quantity y varies as the sum of three quantities of which the first varies as x, the second varies as —x+x2, the third varies as x3 —p then what is y equal to ? **(a)** where k is a constant **(b)** kx+lx2+nu3, where k, l, m are constants **(c)** kP where k is a constant **(d)** kx where k is a constant T-DETA-K.FN - A 10

18. [CDS (I) Paper 3 Q41] What is the „ + sin275ø ? + sin2250 + **(a)** tan2150 + + tan2250 tan2750 **(b)** cos2150 + cos220e + cos2250 . cos2750 **(c)** + cot225• — + cot2750 **(d)** sec2200 + sec2250

19. [CDS (I) Paper 3 Q47] What is the value of I •Sä+4.li ? 12 **(a)** tan2t50 + tan2200 + tan2250 . + tan275Ø **(b)** cos2150 + cos2200 + cos2250 cos2750 **(c)** cot2150 — cot220e + cot2250 — + cot2750 **(d)** sec2150 + sec2200 + sec2250 + sec2750 42. Consider the following data : 5 9 2 The arithmetic mean of the above distribution is 296. What is the missing frequency ? 43. If a is an acute angle and then what is tan equal to ?

20. [CDS (I) Paper 3 Q54] Consider the following data • '911-21 1921-3' 385 31 a 35-9 14-8 For which period is-the natural growth rate minimum ? **(a)** 1911-21 **(b)** 1921-31 **(c)** 1951-61 **(d)** 1961-71

21. [CDS (I) Paper 3 Q66] If + y + z) 9, + y z) 16 and 144, then what is x equal to ? 18 **(a)** One **(b)** Only two **(c)** More than two, but finite **(d)** Infinitely many; 62. Two circles touch each other externally at p. Two secants APB and CPD are drawn through P to meet the circle , at A, C and B, D respectively. Then, which one of the following is correct ?

22. [CDS (I) Paper 3 Q72] A rectangular tank whose length and breadth are 2-5m and 1-5m respec- tively is half full of water. If 750 litre more water is poured into the tank, what is the height through which water level further goes up? **(a)** Rs. 500 @ 10%, Rs. 800 @ 12% **(b)** Rs. 700 @ 10%, Rs.600 @ **(c)** Rs. 800 @ 10%, Rs. 400 @ **(d)** @ 10%, ROOO @ 68. A man is watching from the top of a tower a boat sl*eding away from the tower. The boat makes an angle Of depression of 450 with the man's eye when at a distance of 60m from the bottom of tower. After 5 seconds, the angle of depression becomes 300. What is the approximate speed of the boat assuming that it is running in still water ? 70. The marks of the students of a class who appeared for a test in English are represented in the following frequency table : 11-20 21-30 31-40 41-50 51.60 What is/are the modal class

23. [CDS (I) Paper 3 Q76] A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 **(a)** cm3 **(b)** 3006 cm3 **(c)** cm3 1078 **(d)** cm3 T.DETA-K-FN - A 22

24. [CDS (I) Paper 3 Q77] Smaller lead shots are to be prepared by using the material of a spherical lead shot of radius I cm. Some possibilities are listed in the statements given below: . l. The material is just sufficient to prepare 8 shots each of radius cm. **(a)** 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? **(b)** 9th **(c)** 15m **(d)** I •5m

25. [CDS (I) Paper 3 Q2] A shot of radius 0-75 cm and a second shot of radius 0-8 cm can be prepared from the available materiaL Which of the above statements is/ are correct ? **(a)** 1 Only **(b)** 2 Only **(c)** Both I and 2 **(d)** Neither nor 2

26. [CDS (I) Paper 3 Q85] The (ZADB—ZAEB-900) Consider the following in respect Of the above figure '1. ADAC-AEBC **(a)** 45 **(b)** 225 then what is the value Of tan 6 ? one **(c)** **(d)**

27. [CDS (I) Paper 3 Q91] Consider the following statements : l. HCFof x + y and is **(a)** 264 kg **(b)** 132 kg **(c)** 396 kg **(d)** None of the above 87. Consider the . following statements in respect of a quadrilateral - I. The line segments joining the midpoints of the two pairs of opposite sides bisect each other at the point of intersection. 2. The area of the quadrilateral formed by joining the midpoints of the four adjacent sides is half of the total area of the quadrilateral. Which of the statements given above is/are correct ?

28. [CDS (I) Paper 3 Q3] The HCFof x —y and is **(a)** 264 kg **(b)** 132 kg **(c)** 396 kg **(d)** None of the above 87. Consider the . following statements in respect of a quadrilateral - I. The line segments joining the midpoints of the two pairs of opposite sides bisect each other at the point of intersection. 2. The area of the quadrilateral formed by joining the midpoints of the four adjacent sides is half of the total area of the quadrilateral. Which of the statements given above is/are correct ?

29. [CDS (I) Paper 3 Q4] The HCFof x —y and xlO—y10 is Which of the statements given above Correct ? are **(a)** 1 and 2 **(b)** 2 and 3 **(c)** 1 and 4 **(d)** 2 and 4 26

30. [CDS (I) Paper 3 Q92] If 0 < — and p = see e, then which one of the following is correct ? **(a)** IO cm 5 cm **(b)** cm **(c)** 56 94. When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the 2 2 97. IA there three simultaneous linear equations in two unknowns. What can be the number of solutions **(d)** One or infinite

31. [CDS (I) Paper 3 Q98] What is the sum of the squarés of all the six numbers ? **(a)** 40 **(b)** 42 **(c)** 45 **(d)** 48

32. [CDS (I) Paper 4 Q4] What are the factors Of **(a)** (x —2y—4) and (x —2Y+2) **(b)** (x —y and (x —4 y + 4) **(c)** (x —y and (x —4 y —4) **(d)** (x (x +2Y+2) 26k Q-DETB-K-FN /

33. [CDS (I) Paper 4 Q14] If px and = p z = n, then which one Of the following is correct? XLV = YZ **(c)** **(a)** 20 years **(b)** 22 years **(c)** 24 years **(d)** 25 years IO. A mixture contains niilk and water in the ratio 5 : On Adding 5 litres Of water, the ratio Of' milk and water becomes 5 : 2. Wh#t is the quantity Of milk in the original mixture?

34. [CDS (I) Paper 4 Q16] If the eqyation (a2 + b2)x? + d 2) —O has equål roots, then which one of the following is correct? **(a)** 72 s 68 s 18, If **(b) (c) (d)** Myz 2xyz 3xyz —3xyz 21. If the 14th term Of an arithmetic series is 6 and 6th term is 14, then What is the 95th term? @ -75

35. [CDS (I) Paper 4 Q19] If x varies as. the mth power Of y, y varies as the nth power Of z and x varies as the: pth power Of z, then which one Of the following is correct ? **(b) (d)** None Of the above Q-DETB-K-FN / 26A 20 If x = (b —d (a — d), y z (a — b) (c — then what is + y3 equal to? of **(b)** 2xyz **(c)** 3xyz **(d)** —3xyz **(a)** 72 s 68 s 18, If **(b) (c) (d)** Myz 2xyz 3xyz —3xyz 21. If the 14th term Of an arithmetic series is 6 and 6th term is 14, then What is the 95th term? @ -75

36. [CDS (I) Paper 4 Q21] If the 14th term Of an arithmetic series is 6 and 6th term is 14, then What is the 95th term? 'h @ -75 (b' 75 80 - 80 **(a)** 72 s 68 s 18, If **(b) (c) (d)** Myz 2xyz 3xyz —3xyz 21. If the 14th term Of an arithmetic series is 6 and 6th term is 14, then What is the 95th term? @ -75

37. [CDS (I) Paper 4 Q23] 6 If x = (b —d (a — d), y z (a — b) (c — then what is + y3 equal to? **(a) (b) (c)** Myz 2xyz 3xyz —3xyz

38. [CDS (I) Paper 4 Q30] Which one Of the following is a non-termmating and repeating decimal ? **(a)** 13/8 **(b)** 3/16 **(c)** 3/11 **(d)** 137/25

39. [CDS (I) Paper 4 Q39] If one root Of the equation ax is —l, then What is the other root? **(a)** 1/4 40. What is the value of 510gio 25 -210g10 3 + 10310 18 ? 43. If cos02+ in the first quadrant, then Which One Of the following is correct? 44. What is thc value of cos I • cos 20 cos 3' **(b)** . The 2 product Of a rational number cos 90' ? **(c)** 1/2 41 42. and an frrational number is **(d)** a natural number

40. [CDS (I) Paper 4 Q43] If cos02+ in the first quadrant, then Which One Of the following is correct? **(a)** 1/4 40. What is the value of 510gio 25 -210g10 3 + 10310 18 ? 43. If cos02+ in the first quadrant, then Which One Of the following is correct? 44. What is thc value of cos I • cos 20 cos 3' **(b)** . The 2 product Of a rational number cos 90' ? **(c)** 1/2 41 42. and an frrational number is **(d)** a natural number

41. [CDS (I) Paper 4 Q51] ABC BC D ZACD- 1200, ZABC- -2? **(a)** 47. 48. 49. 14 sm I — sine **(b)** 9 — tan 9 **(c)** sece+ tan 8 **(d)** cosece + cot 9

42. [CDS (I) Paper 4 Q69] Half of a large cylindrical tank open at the top is fillpd with water and identical heavy *pherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius Of a ball, What is the rpaximum number of balls that can pe dropped? **(a) (b) (c)** None of the above 69. Half of a large cylindrical tank open at the top is fillpd with water and identical heavy *pherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius Of a ball, What is the rpaximum number of balls that can pe dropped? **(d)** 12

43. [CDS (I) Paper 4 Q75] In order to fix an electric pole along a roadside, a pit With dimensions 50 50 cm is dug With the help of a spade. The pit is prepared by removing earth by 250 strokes Of spade. If one stroke Of spade removes SOO cma of earth, then What is the depth Of the pit? **(a) (b) (c)** None of the above 69. Half of a large cylindrical tank open at the top is fillpd with water and identical heavy *pherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius Of a ball, What is the rpaximum number of balls that can pe dropped? **(d)** 12

44. [CDS (I) Paper 4 Q77] What is the average marks Of all the candidates? **(a)** 758 **(b)** 75-8 **(c)** 760. **(d)** 768'.

45. [CDS (I) Paper 4 Q2] If two tridngles have equal area, then they are siinilar. Which Of the statements given above is/are corect? **(a)** I only **(b)** 2 only I **(c)** Both I and **(d)** Neither I nér 2

46. [CDS (I) Paper 5 Q80] If the larger number exceeds four times the smaller by 5, what is the smaller number ? **(a) (b) (c)** Null set Singleton set Infinite set None Of the above 20. 21. 22. 23. **(d)**

47. [CDS (I) Paper 5 Q2] 203 Which Of the above numbers is/are prime ? **(a) (b) (c)** Null set Singleton set Infinite set None Of the above 20. 21. 22. 23. **(d)**

48. [CDS (I) Paper 5 Q23] 1 9 , then what is 9x2 + If x— — — equal to ? **(a) (b) (c)** Null set Singleton set Infinite set None Of the above 20. 21. 22. 23. **(d)**

49. [CDS (I) Paper 5 Q37] 2x — 3y + I x + 4y 8 If 2 3 -7Y+2 5 then what is (x + y) equal to ? **(a) (b)** 4 2 **(c) (d)** 3 -2 Consider the following statements : What is the number of circles passing through all the vertices Of a given triangle ?

50. [CDS (I) Paper 5 Q36] 37. 2x — 3y + I x + 4y 8 If 2 3 -7Y+2 5 then what is (x + y) equal to ? **(a) (b)** 4 2 **(c) (d)** 3 -2 Consider the following statements : What is the number of circles passing through all the vertices Of a given triangle ?

51. [CDS (I) Paper 5 Q86] 87. **(a)** 121-7 121-75 **(c)** 121-75 ABC t, X Y AX.AYur **(a)** 16 2 **(b)** 20 cm 2 **(c)** 25 cm2 **(d)** 30 cm2 fiqfäfä7 $1fäv afre, PQR PQ•crn, QR.4cm RP 5 cm I PQR : PQR fhF I 3#t5 **(a) (c)** 3tTR a;ZF-11 t, 31k a'i2FF11 t, 7521+11, 7527-1 I t, 3tx t I a-NXR-I I P-PQFA-L-RZ 25 -

52. [CDS (I) Paper 5 Q89] AD2 — 3CD2 AC2 = AD2 AD2 — 4CD2 - AD2 - 5CD2 90. **(a) (b) (c) (d)** ACBD ACBD t ACBD B qz ABC 3åq BC D t (BD > BC) **(a) (b) (c) (d)** AC2 AC2 _ AC2 — — 2CD2 LMNP 3tz LPiq•q NR t RNPäi RP 6 LR ? **(a) (b) (c) (d)** 15 cm 12 cm

53. [CDS (I) Paper 5 Q85] 86. AD2 — 3CD2 AC2 = AD2 **(a)** 121-7 121-75 **(b)** 121-75 ABC t, X Y AX.AYur **(c)** 16 2 **(d)** 20 cm 2

54. [CDS (I) Paper 5 Q87] AD2 — 4CD2 - AD2 - 5CD2 27 x x 5-5262 x **(a)** 121-7 121-75 **(c)** 121-75 ABC t, X Y AX.AYur **(a)** 16 2 **(b)** 20 cm 2 **(c)** 25 cm2 **(d)** 30 cm2 fiqfäfä7 $1fäv afre, PQR PQ•crn, QR.4cm RP 5 cm I PQR : PQR fhF I 3#t5 **(b)** 31k t, 7521+11, 7527-1 I **(a) (b) (c) (d)** ACBD ACBD t ACBD

55. [CDS (I) Paper 6 Q20] What is the sum of their reciprocals ? **(a)** 2 **(b)** 2 **(c)** 2 **(d)** 2-562 + b2

56. [CDS (I) Paper 6 Q95] In the figure given above, AB is parallel to CD. If Z DCE = x and Z ABE = y, then what is Z CEB equal to ? **(c)** x + y — (11/2) 980 1440 Iri the figure given above, AB is parallel to CD. If Z BAF 980 and AFC = 1440, then what is L ECD equal to ? **(a)** 620 **(b)** 640 **(c)** 820 **(d)** 840

57. [CDS (I) Paper 6 Q96] How many degrees are there in an angle which equals two-third of its complement ? **(a)** 360 450 **(b)** 48 **(c)** 600 97. 470 670 N In the figure given above, PQ is parallel to RS. What is Z NMS equal to ? **(d)** 270

58. [CDS (I) Paper 7 Q4] A number consists of two digits. The sum of the digits is IO. On reversing the digits of the number, the number decreases by **(a)** 21 **(b)** 24 **(c)** 36 **(d)** 42 Which one among the following is not correct ?

59. [CDS (I) Paper 7 Q2] Colour of the eye Of the person **(a)** 21 **(b)** 24 **(c)** 36 **(d)** 42 Which one among the following is not correct ?

60. [CDS (I) Paper 7 Q3] Number of times a }xrson correctly hits a target in ten attempts Which of the following in respect of the above is correct ? **(a)** I is a continuous variable, 2 is not a variable and 3 is a discrete variable **(b)** 1 is a continuous variable, 2 and 3 are discrete variables **(c)** I and 3 are discrete variables and 2 is not a variable **(d)** •I, 2 and 3 all are discrete variables Consider the followiög •statements :

61. [CDS (I) Paper 7 Q1] The data collected by investigator to be used by himself are called primary data **(a)** 21 **(b)** 24 **(c)** 36 **(d)** 42 Which one among the following is not correct ?

62. [CDS (I) Paper 7 Q49] 3a-2 **(a)** I unit **(b)** 2 umts 4 units **(c) (d)** 8 units If then what is — + — equal to ? -4a-2 dar lel If Of is **(d)** afb

63. [CDS (I) Paper 7 Q2] The area of circle is less than the area oftriangle. Which of the statements Ware correct ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither 1 nor 2 The expression sin2x + cos2x — O is satisfied by how many values of x ? 53 58 59 **(a) (b) (c) (d)** Only one value Of x TWO values Of x Infinite values of x NO value Of X 116

64. [CDS (I) Paper 7 Q61] 62. 1001x1001xi001+0999xo.999xo•999 1001xlOOl-1001x0999+O•999xo.999 If p(x) is a common multiple Of degree 6 Of the polynomials (x) + x2 — x — I and g(x) — x2 + X— I then which Of the following is correct: ? **(b)** 1) (x2+l)2 Ifthe roots of the equation x2 — 2ax •ea2 + a — 3 O are real and less than 3, then which One of the following is correct ? **(a)** p **(b)** 3 and —4 **(c)** —3 and—4 **(d)** —3 and 4

65. [CDS (I) Paper 7 Q66] Consider the following : l. 10+ I **(a)** p **(b)** 3 and —4 **(c)** —3 and—4 **(d)** —3 and 4

66. [CDS (I) Paper 7 Q94] 95. c In the figure given above ZABC = ZAED= **(a)** 93. 92. 600 **(b)** 750 **(c)** 950 **(d)** 17/8 x In the quadrilateral ABCD shown above, LDAB = ZDCX - 1200. If ZABC- 1050, what is ZADC equal to ? 450 c o 94. 95. c In the figure given above ZABC = ZAED= 900. Consider the following statements : I. ABC and ADE are similar triangles. 2. The four points B, C, E and D may lie on a circle. Which of the above Statements is/are correct ?

67. [CDS (I) Paper 7 Q100] If a set Acontains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does A u B contain ? **(a)** 130 **(b)** 100 **(c)** 80 **(d)** 70 28

68. [CDS (I) Paper 8 Q24] If two factors of — 2a3 — 9a2 + 2a + 8 are (a + l) and (a — l), then what are the Other two factors ? **(a)** (a — 2) and (a + 4) **(b)** (a + 2) and (a + 4) **(c)** (a + 2) and (a — 4) **(d)** (a — 2) and (a —4)

69. [CDS (I) Paper 8 Q26] If 0.764 y 1.236 x, then what is the Y¯x value Of **(a)** x- 10 **(b)** x = 11 **(c)** x = 11.5 If 0.764 y = 1.236 x, Y—x value Of **(d)** 0.764

70. [CDS (I) Paper 8 Q38] 39. 40. **(d)** None Of the above Consider the following table : Y 100 90 70 40 How are x and y related ? = 100 — 5x — 5x2 4 10 **(a) (b)** 3 6 If the expression x3 + 4x2 + 4x + k has **(c)** 16 **(d)** 32 - 32 35. If

71. [CDS (I) Paper 8 Q45] 46. If x + y 9 and xy = 8, then what is one Of the values of (x' — y') ? **(a)** 4025 **(b)** 4065 **(c)** 4095 **(d)** 4905 There are some benches in a class room having the number of rows 4 more than the number Of columns. If each bench is seated with 5 students, there are two seats vacant in a class of 158 students. The number of rows is . **(d)** 10

72. [CDS (I) Paper 8 Q73] In the figure given above, AB is parallel to CD. ZABC = 650, ZCDE = 150 and AB - AE. What is ZAEF equal to ? 300 **(a) (b)** 350 **(c)** 400 **(d)** 45

73. [CDS (I) Paper 8 Q75] 200 400 B In the figure given above EC is parallel to AB, LECD - 700 and ZBDO = **(a)** 300 **(b)** 500 950 A In the figure given above ABC is a triangle. BC is parallel to AE. If BC = AC, then what is ZCAE equal to ? **(c) (d)** 75. 200 400 76. 77. 78. B In the figure given above EC is parallel to AB, LECD - 700 and ZBDO = 200. What is LOBD equal to ?

74. [CDS (I) Paper 8 Q78] B In the figure given above EC is parallel to AB, LECD - 700 and ZBDO = **(a)** 300 **(b)** 500 950 A In the figure given above ABC is a triangle. BC is parallel to AE. If BC = AC, then what is ZCAE equal to ? **(c) (d)** 75. 200 400 76. 77. 78. B In the figure given above EC is parallel to AB, LECD - 700 and ZBDO = 200. What is LOBD equal to ?

75. [CDS (I) Paper 8 Q99] Which of the following is/are correctly matched ? **(a)** Mean Median **(b)** Mean Mode **(c)** Mean > Median **(d)** Mean > Mode What is the harmonic mean 25, 40 and 50 ?

76. [CDS (I) Paper 8 Q2] Educational qualification of the person Continuous variable Attribute 28 Select the correct answer using the code given below • **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither 1 nor 2

77. [CDS (I) Paper 9 Q25] The Of two number' is 6 10 12 15 Of N:uation• y O a unique x unique X infinite solutions -2+4 Za-l 2.3—28-1 2.8—'28 + 292 •a + — Qy) only O.OETA **(a)** k = 2 **(b)** k = 4 **(c)** k = -2 **(d)** k = ±4

78. [CDS (I) Paper 9 Q18] 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf — 5x2 — 81 ia ta) equgi to ? a unique x a unique solution x — infinite solutions 2.3-28-1 2.8-'28 291 + **(a)** x = 1, y = 2 **(b)** x = 2, y = 1 **(c)** x = 3, y = 1 **(d)** x = 1, y = 3

79. [CDS (I) Paper 9 Q32] Which One of by n, ? 45678940 54857266 87524398 93455120 Ten chairs e fonoWiog is ai tables together cost three chairs nd two together ,900. The and 5 cable* **(a)** a3r03 **(b)** b4/a4 following is 35. tables together cost nd two together 36. 4 chai-rs and 5 cable* proporvion, $7. differenee in Of equation 2x2- 4 3-5 and **(c) (d)** The

80. [CDS (I) Paper 9 Q1] Which the Only 2 only Neither Q-DETA.N.FMN Stet-em correct ? **(a)** x = 1, y = 2 **(b)** x = 2, y = 1 **(c)** x = 3, y = 1 **(d)** x = 1, y = 3

81. [CDS (I) Paper 9 Q7] If each is increased by 2, then the new **(b)** 12 20 9 8 **(a) (b)** equal to equal to r greater than 1 less than I •us 91. and tio 92. ABCD is AD and the following statements I. me La ngent Of a circle ig a line the t meets the circle in one and Only one g. The Of circle at the end p t of the diameter perrgdiculaT t. the Which Of the above ? 1 only **(c)** 2 only Booa and 2 Neither I nor 2 triangle Z C 900 ana CD is perpendicular to A.B. If ABxCD-CAxCE, then cna IS equal t. In the figure given above. EF is parallel to y equal 460 **(d)** 93. 1 The mean observation 1 if CA > CB CA2 of 7 observations is 7. If each is increased by 2, then the new In the figure given above, Z PQR and QL is a median, PQ 5 cm and QR 12 an. QL is equal to G em

82. [CDS (I) Paper 10 Q11] Which of the above statements is/are correct? **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither nor 2 Which one is one of the factors of **(a) (c)** B-BCRB-M-NBU / 48A

83. [CDS (I) Paper 10 Q12] In a class of 110 students, x students take both Mathematics and Statistics, 2x+ 20 students take Mathematics and 2x+ 30 students take Statistics. There • are no students who take neither Mathematics nor Statistics. What is x equal to? **(a)** 15 **(c)** 25 **(b)** 20 **(d)** 30 o, lal

84. [CDS (I) Paper 10 Q13] If then what•is one of the values of x equal to? **(a)** 15 **(b)** 25 **(c)** 20 **(d)** 30 9. 10. 11. What is equal to?

85. [CDS (I) Paper 10 Q14] The quantity which must be added to (1— x 2) to obtain x 3 is 2X3 + 3X2 + I 2x3+x + x —1 **(b)** Consider the following statements • **(a)** 15 **(b)** 25 **(c)** 20 **(d)** 30 9. 10. 11. What is equal to?

86. [CDS (I) Paper 10 Q20] If x Y = 4 and + _ = 1 then what is x + y equal to? **(a)** O only **(b)** 20 only **(c)** both 10 and 20 **(d)** neither IO nor 20 If

87. [CDS (I) Paper 10 Q28] 8 How many 200 mm lengths can be cut from IO m of ribbon? **(a)** 50 **(b)** 40 **(c)** 30 **(d)** 20

88. [CDS (I) Paper 10 Q29] What is 262 +972 equal to? @ 272 +932 **(b)** 342 +932 (C) 822 +412 **(d)** 792 + 622 **(a)** HCF **(b)** HCF **(c)** HCF **(d)** None Of the above If b is the largest square divisor Of c and a2 divides C, then which one of the following is correct?

89. [CDS (I) Paper 10 Q36] A bus starts with some passengers. At the first stop, one-fifth Of the passengers gets down and 40 passengers get in. At the second stop, half of the passengers gets down and 30 get in. The number of passengers now is **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

90. [CDS (I) Paper 10 Q31] 33. **(a)** 40 Whiéh is the smallest number among the following? **(b) (c) (d)** pair of rational numbers that between — and — is 4 4 262 752 1000' 1000 24 74 100 100 9 31 252 748 1000' 1000

91. [CDS (I) Paper 10 Q45] If a equation xa — x — 1 O, then what is equal to? **(a)** 56 **(b)** 60 cm **(c)** 64 cm **(d)** 1 only

92. [CDS (I) Paper 10 Q84] 26 If tang + sec8= m, then what sec9 equal to? is **(c)** 2m **(d) (a)** 2 only **(b)** Both 1 and 2 **(c)** Neither 1 nor 2 84. 26 If tang + sec8= m, then what sec9 equal to? is 80. If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 **(d)** 2m

93. [CDS (I) Paper 10 Q85] If 5sin9+ 12cose= 13, then what is 5cos8 — 12sin9 equal to? **(a)** 2 only **(b)** Both 1 and 2 **(c)** Neither 1 nor 2 84. 26 If tang + sec8= m, then what sec9 equal to? is 80. If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 **(d)** 2m

94. [CDS (I) Paper 10 Q2] The value of sinB+cose is always greater than **(a)** 150' If . O, then sin 4 B + cos 49 equal to? what is 91. What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? 4 2 88. What is cosec **(b)** 2sin0 If triangle ABC is right angled at C, then what is cos **(c)** 2 92. Consider the following statements • 1. tan B increases faster than sin O as B increases, 2. The value of sinB+cose is always greater than 1. Which of the above Statements is/ are correct? 1 only **(d)** 2 only

95. [CDS (I) Paper 10 Q1] Which of the above Statements is/ are correct? 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 **(a)** 150' If . O, then sin 4 B + cos 49 equal to? what is 91. What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? 4 2 88. What is cosec **(b)** 2sin0 If triangle ABC is right angled at C, then what is cos **(c)** 2 92. Consider the following statements • 1. tan B increases faster than sin O as B increases, 2. The value of sinB+cose is always greater than 1. Which of the above Statements is/ are correct? 1 only **(d)** 2 only

96. [CDS (I) Paper 10 Q95] 96. **(a)** 30 **(b)** 35 of the which **(d)** h=2r centre of the balloon, then one of the following is Methods of presentation of data are **(a)** tables only **(b)** graphs only **(c)** diagrams only **(d)** All of the abov e

97. [CDS (I) Paper 10 Q97] The average Of u, v, w, x, y, z is IO. What is thc average of u + 10, v + 20, x+40, Y+50, 2+60? 40 B-BCRB-M-NBU /48A **(a)** tan e **(b)** cote A spherical balloon of radius r subtends angle 60' at the eye of an observer. If the angle of elevation of its centre is 600 and h is the height 95. 96. **(c)** 30 **(d)** 35 of the which

98. [CDS (I) Paper 10 Q99] What is x equal to? 6 ZOO. What is the and y ? **(a)** 2x = 3y **(b)** 3x 2y **(c)** **(d)** 2x=y relation between

99. [CDS (I) Paper 11 Q33] 84. **(a)** **(b)** **(c)** **(d)** x) and 28 (x6*I)? 4(x4-x2+1) 2(x4-x2+ 1) None of the above TWO chairs and one table cost 700 and I chair and 2 tables cost

100. [CDS (I) Paper 11 Q25] Select the correct answer using the code given **(a)** **(b)** **(c)** 9 : 04 am 9 : 08 am 9: 14 am None of the above then **(d)**

101. [CDS (I) Paper 12 Q23] 24. **(d)** 11 25. **(d)** 184 tx + y) (x - y) 3: 5 and xyz positive imply that **(a)** 2X+1 B-ABQB-N-LzS/ IA 23. 24. **(b)** 11 25. **(c)** 184 tx + y **(d)**

102. [CDS (I) Paper 12 Q34] 35. If ax + by — 2 O and axby = I, Where a O, b O, then what is x + b 2 y) equal to? **(c)** Consider the following statements • 2. (a — b — c) is one Of the factors Of 3abc•b3 + c3 — a (b + c— I) is one Of the factors Of 3bc+ b3 + c3 —I. Which Of the above is / are correct? **(a)** 37 **(b)** 2ab 34. 35. 10 If ax + by — 2 O and axby = I, Where a O, b O, then what is x + b 2 y **(c)** **(d)**

103. [CDS (I) Paper 12 Q43] There are 48 cricket balls, 72 hockey of balls and 84 tennis balls, and they have to be arranged in several rows in such a way that every row ch contains the same number Of balls of one type. Whåt is the minimum number of rows required for this to happen? **(b)** 16 of **(b)** 14a **(a)** 16 42. **(b)** 14a If m and n are the roots Of the equation x + + b = O, and m 2 and are the roots Of the equation x — cx+d O, then which of the following is/ are correct? 43. There are 48 cricket balls, 72 hockey balls and 84 tennis balls, and they have to be arranged in several rows in such a way that every row contains the same number Of balls of one type. Whåt is the minimum number of rows required for this to happen? 2b — a2 c Select the correct answer using the code given below. **(c)** 1 only **(d)** 2 only Both 1 and 2

104. [CDS (I) Paper 12 Q64] If sinx + cosecx=a then what **(a)** sin x **(b)** cosx 62. What is **(c)** sin x + cos x **(d)** sin x — cos x 63. Consider the following statements • l. sin > sin I 2. COS < COS Which Of the above Stater-nents 66. What is the number of pairs of perpendicular planes in a cuboid?

105. [CDS (I) Paper 12 Q1] ADM DCM ABCD t, Afa AB-CDtl **(a)** cos x **(b)** sin X + cos x **(c)** sin x — cos x 63. • l. sin r > sin 1 2. cos l' cos I **(d)** 4q 66. 67.

106. [CDS (I) Paper 12 Q24] What is the value Of x? **(a)** 18 **(b)** 20 **(c)** 24 Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range 98. When we take class intervals On the x-axis and corresponding frequencies on the y-axis, and draw rectangles with the areas proportional to the frequencies Of the respective class intervals, the graph so Obtained is called 2. The cumulative frequency polygon gives an equivalent representation Of frequency distribution table. The cumulative frequency polygon is a closed polygon with one horimntal and one vertical side. The have nan-negative Which of the above is/ are correct? 1 only **(d)** 2 only Both 1 and 2

107. [CDS (I) Paper 13 Q5] The digit in the units place Of the product 81 x 82 x 83 x 84 is 4 6 **(a)** 3 Cannot be determined due to insufficient data 4. 5. 6. For what value Of k is **(b)** 2 3 B-EFUA-O-PDW/S2A 2 is 2 4. What is equal to? 5. The digit in the units place Of the product 81 x 82 x 83 x 84 is 4 6 6. The square root of **(c)** 2 3 I. What is the remainder when 496 is divided by 6? 2. 3. 4 3 2 37 13 where x, y, z are natural numbers, then what is z equal to? **(d)** 3 Cannot be determined due to insufficient data For what value Of k is

108. **[CDS (I) Paper 13 Q25]** He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was **(a)** 2 3 **(b)** 7 6 None of the above 8. Consider the following statements : 2. The equation 1990x — 173y= I I has no solution in integers for x and y. The equation 3x— 12y= 7 has no solution in integers for x and y. The difference between the squares Of two consecutive odd integers is always divisible by Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only Both I and 2 Neither 1 nor 2 A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was

109. **[CDS (I) Paper 13 Q23]** For which value of k does the pair of equations x — = O and = yield a unique positive solution Of x ? A person goes to a market between 4 p.m. and 5 p.m. When he comes back, he finds that the hour hand and minute hand have interchanged their positions. For how much time (approximately) was he out or his hou se? **(b) (c) (d)** 55-25 minutes 55 • 30 minutes 55, 34 minutes 55 • 38 minutes **(a)** years — years years 2 — years **(b) (c) (d)** _g_f_ km _æ_ km Pc 20. A man rows downstream 32 km and 14 km upstream, and he takes 6 hours to cover each distance. What is the speed Of the current?

110. **[CDS (I) Paper 13 Q37]** The HCF and LCM of two polynomials are (x + y) and (3x5• 3x2y3— 5xy4 —'2yS) respectively. If one of the polynomials is (x2 — then the other polynomial is —2y **(b) (d)** 3x4 — 8x3y+IOx2y 3x4 — 8x3y— IOx2y2 +7 xy + 2y 3x4 +7xy3 * 2y 3x4 8x3y— IOX2y +7xy3 + 2y4 **(a) (b)** 3x4 — 8x3y+IOx2y 3x4 — 8x3y— IOx2y2 +7 xy + 2y 3x4 +7xy3 * 2y 3x4 8x3y— IOX2y +7xy3 + 2y4 38. Let p be a prime number Other than 2 Or 5. One would like to express the vulgar fraction l/ p in the form of a recurring decimal. Then the decimal will be Which Of the above statements is/ are correct? **(c)** I only **(d)** 2 only

111. **[CDS (I) Paper 13 Q43]** There are five parties A, B, C, D and E in an election. Out of total I OCIOOO votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allocated for party E in the pie Ch art? **(c) (d) (a)** 100 If G and are the arithmetic, geometric and harmonic means between a and b respectively. then Which One of the following relations is correct? 43. There are five parties A, B, C, D and E in an election. Out of total I OCIOOO votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allocated for party E in the pie Ch art? **(b)** . The G is the geometric between A and H A is the arithmetic between G and H H is the harmonic between A and G None of the above geometric mean Of m ean mean mean three **(c) (d)** 44. Out in 4S. If 540 60S 72 or 532 savings accounts held a post office, 218 accounts have deposits over 10,000 each, Further, in 302 accounts, the first or sole depositors are men, of which the deposits exceed 10,000 in 102 accounts. In how many accounts the first or sole depositors are women and the deposits are up to e 10,000 only? positive numbers a, b, c is 3 and the geometric mean Of another three positive numbers d, e, f is 4. Also, at least three elements in the set {a, b, c, d, e, f} are distinct. Which One Of the following inequalities gives the best information about M, the arithmetic mean Of the six numbers?

112. **[CDS (I) Paper 13 Q44]** Out in 4S. If 540 60S 72 or 532 savings accounts held a post office, 218 accounts have deposits over 10,000 each, Further, in 302 accounts, the first or sole depositors are men, of which the deposits exceed 10,000 in 102 accounts. In how many accounts the first or sole depositors are women and the deposits are up to e 10,000 only? 4. **(c)** 100 ng **(b) (d)** 116 114 Cannot be determined from the given data then x + y + z 3 + 3xyz equal to **(a) (b) (d)** 6 xyz 12 xyz XYZ

113. **[CDS (I) Paper 13 Q56]** If a Star figure is formed by elongating the sides of a regular pentagon, then the measure Of each angle at the angular points of the star figure is **(a)** 360 **(b)** 350 **(c)** 320 **(d)** 300

114. **[CDS (I) Paper 13 Q92]** Let on, OC and OD are rays in the anticlockwise direction such that ZAOB= ZCOD= 1 ocp, ZBOC= 820 and ZAOD= **(a) (b)** 2 cm Cannot be determined due to insufficient data 92. Let on, OC and OD are rays in the anticlockwise direction such that ZAOB= ZCOD= 1 ocp, ZBOC= 820 and ZAOD= 780. Consider the following statements 89. From a circular piece of cardboard of radius 3 cm, two sectors of4 **(c)** I I square cm **(d)** 22 square cm

115. **[CDS (I) Paper 13 Q2]** AOC and BOD are lines _ ZBOC and ZAOD supplem entary. are Which of the above is / are correct? statements **(a) (b)** 2 cm Cannot be determined due to insufficient data 92. Let on, OC and OD are rays in the anticlockwise direction such that ZAOB= ZCOD= 1 ocp, ZBOC= 820 and ZAOD= 780. Consider the following statements 89. From a circular piece of cardboard of radius 3 cm, two sectors of4 **(c)** I I square cm **(d)** 22 square cm

116. **[CDS (I) Paper 14 Q6]** The average of 'm numbers is n4 and the average of n numbers is m4. The average of (m + n) numbers is **(c)** mn(m2 + n2) (4) mn(m2 + n2 — mn) 7. ne digit in the units place of the resulting number of the expression (234)100 + (234/01 is 2 **(a)** 51 **(b)** 53 **(c)** 55 **(d)** 57 2. What is the sum of digits Of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11 respectively as the remainders ? 3. A milkman claims to sell milk at its cost price only. Still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture ?

117. **[CDS (I) Paper 14 Q11]** A tin of oil was — full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was — fill]. Oil of. how many EX)ttles can the tin contain ? (All bottles are of equal volurne) **(a)** 35 **(b)** 40 **(c)** 45 **(d)** 50

118. **[CDS (I) Paper 14 Q13]** Consider the following in respect of the equation y = l. if x > 1 2. if x < 1 3. y exists for all values of x Which of the above statements is/are correct ? **(a)** I only **(b)** 2 only **(c)** I and 2 only l, 2 and Y is/are 14. If **(d)** 2 minutes

119. **[CDS (I) Paper 14 Q16]** If a, b and c satisfy the equation — 3x2+ 41 = 0 then what is the value of —+—+_ ? 2 2 **(a)** I only **(b)** 2 only **(c)** I and 2 only l, 2 and Y is/are 14. If **(d)** 2 minutes

120. **[CDS (I) Paper 14 Q20]** If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ? **(b)** 5<k$11 **(a)** 43-2% **(b)** 56-8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ? **(c)** 5<k$11 21. The number of pairs **(d)** Zero

121. **[CDS (I) Paper 14 Q24]** In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m. Which of the following is / are conect ? •l. The length Of the race is 300 m. **(a)** 43-2% **(b)** 56-8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ? **(c)** 5<k$11 21. The number of pairs **(d)** Zero

122. **[CDS (I) Paper 14 Q42]** Select the correct •answer using the code given below • **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 8

123. **[CDS (I) Paper 14 Q30]** If and b. c 2a : 3b : 7c is equal to **(a)** 42 : 105 : 320 **(b)** 15 : 21 : 35 **(c)** 6: 15:40 **(d)** 30; 21 : 350• 10 = 7: 8, then

124. **[CDS (I) Paper 14 Q36]** Consida the following statements : I + tan20 I + cot2e all 4 2. is for O — 450 only. Which of the statements is/are correct ? **(a)** 10 s **(b)** 16 s **(c)** 21 s 32. Three athletes run a 4 km race. Their srredsareintheratio 16:d5: 11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is **(d)** 1000m

125. **[CDS (I) Paper 14 Q40]** If O is an acute angle and sinecos0= 2cos30 — 1-5cosO, then what is sine equal to ? 6-1 4 1-6 **(b)** 4 4 4 **(a)** tan29 **(b)** sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine **(c)** coso tano + I tang — I **(d)** sin 6 39. If tan

126. **[CDS (I) Paper 14 Q41]** Consider the following statements : 1. sin660 is less than cos660. 2„ sin260 is less than cos260. Which of the above statements is/are corect ? **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2

127. **[CDS (I) Paper 14 Q42]** If a and b are positive, then the relation sino 3b **(b)** Byssible only if a b **(c)** possible if a > b **(d)** possible if a < b • 4 **(a)** tan29 **(b)** sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine **(c)** coso tano + I tang — I **(d)** sin 6 39. If tan

128. **[CDS (I) Paper 14 Q44]** A pole is standing erect on the ground which is horizontal tip of the_pole is tied tight with a roly of length VI 2 m to a point on the ground. If the rope is making 300 with the horizontal, then the height of the pole is **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

129. **[CDS (I) Paper 14 Q45]** If tano + = 2, then tang is equal to 3 4 5 4 2 5 **(d)** 2 B-ABVB-O-GPU - A 16 **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

130. **[CDS (I) Paper 14 Q7]** If the digits are reversed,' the number is increased by **(a)** 100066-1 **(b)** 10000m **(c)** 100+-6 **(d)** 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

131. **[CDS (I) Paper 14 Q52]** A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height IO cm each that can be packed in the drum ? **(a)** 60 **(b)** 70 **(c)** 80 **(d)** 90

132. **[CDS (I) Paper 14 Q1]** LOAC+ZOCB+ZOBA=900 **(a)** 15-5 km/hour **(b)** 15•84 kn/hour **(c)** 16 km/hour **(d)** 16-36 km/hour 56. 'lhere are 437 fruit plants in an orchard planted in rows. distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants.. niere is I m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of Rs. 100 per metre ?

133. **[CDS (I) Paper 14 Q3]** The petpendiculars drawn from any point on OM to AB and AC are always equal Which of the above statements are correct ? **(a)** - 1 and 2 only **(b)** 2 and 3 only **(c)** 1 and 3 only **(d)** l, 2 and 3

134. **[CDS (I) Paper 14 Q83]** D In the above figure, ABCD is a parallelo- gram. p is a point on BC such that PB : PC=I : **(a)** 20 square t **(b)** 30 square unit **(c)** 40 square unit **(d)** None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

135. **[CDS (I) Paper 14 Q87]** The value of (063+0-37) is 100 99 1000 **(d)** 999 30 In the above figure, ABCD is a parallelo- gram. p is a point on BC such that PB : PC=I : **(a)** 20 square t **(b)** 30 square unit **(c)** 40 square unit **(d)** None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

136. **[CDS (I) Paper 14 Q90]** The difference of maximum values of the expressions (6 + 5x — x2) and (y — 6 — y2) for any real values of x and **(a)** 16 **(b)** 17 **(c)** 18 **(d)** 19

137. **[CDS (I) Paper 14 Q91]** A clock is started at noon. By 10 minutes past 5, through what angle, the hour hand rnoves ? **(a)** 1600 **(b)** 1450 **(c)** 1500 **(d)** 1550 B-ABVB-O-GPU - A 32 92. minimum value of cos2x + cos2y —cos2z

138. **[CDS (I) Paper 14 Q93]** Ihe •alue of 32cot2 —8sec2 — +8cos3 — 3 4 is equal to **(b)** 20 **(d)** 36 91 , then the value of **(a)** Rs. 1000 **(b)** RK1025 **(c)** Rs. 1050 **(d)** RS. 1100 a mixture of milk and water of 89. In volume 30 litre, the ratio of milk and water is 7 : 3. The quantity of water to be added to the mixture to make the • ratio of milk and water 1 : 2 is

139. **[CDS (I) Paper 14 Q94]** If x— 216 (I—xyn 9 9 4 4 32 **(a)** Rs. 1000 **(b)** RK1025 **(c)** Rs. 1050 **(d)** RS. 1100 a mixture of milk and water of 89. In volume 30 litre, the ratio of milk and water is 7 : 3. The quantity of water to be added to the mixture to make the • ratio of milk and water 1 : 2 is

140. **[CDS (I) Paper 14 Q96]** The average weight of students in a class is 43 kg. Four new students are admitted to the class whose weights are 42 kg. 36•5 kg, 39 kg and 42-5 kg respectively. Now the average weight, of the students of the class is 426 kg. The number of students in the beginning was **(a)** 24 **(b)** a —25 **(c)** 28 **(d)** a— 1 25

141. **[CDS (I) Paper 14 Q97]** Four years ago, the average age of A and B was 18 years. Now the average age Of A, B and C is 24 years. After 8 years, the age of C will be **(a)** 32 years **(b)** 28 years **(c)** 36 years **(d)** 40 years B-ABVB-O-GPU - A **(a) (b) (c)** 34

142. **[CDS (I) Paper 15 Q3]** If 2 + g m and _ + — n, then what is — equal to? **(a)** 3 7. 8. 9. The sum and difference of two expressions are 5x2 — x — 4 and x 2 + 9x — 10 respectively. The HCF of the two expressions Will be **(b) (c)** -15 If then value of b2+C2 will be If the polynomial x + px + qx — x the 3. If 2 + g m and _ + n, then is — equal to? **(d)** np — mq 4. what and and is divisible by

143. **[CDS (I) Paper 15 Q4]** If O, O and by—c2 O, then the value Of will be **(d)** If the equations x2 — px q O and x 2 + qx — p O have a common root, then which one of the following is correc t? p +q—2=O p-q-l=o B-ABPA-P.JRW / s 4A **(a)** 3 7. 8. 9. The sum and difference of two expressions are 5x2 — x — 4 and x 2 + 9x — 10 respectively. The HCF of the two expressions Will be **(b) (c)** -15 If then value of b2+C2 will be If the polynomial x + px + qx — x the 3. If 2 + g m and _ + n, then is — equal to? **(d)** np — mq 4. what and and is divisible by

144. **[CDS (I) Paper 15 Q6]** If x •23+2¯Å, then the 2x3 —6x—5 is equal to value of **(a)** 3 7. 8. 9. The sum and difference of two expressions are 5x2 — x — 4 and x 2 + 9x — 10 respectively. The HCF of the two expressions Will be **(b) (c)** -15 If then value of b2+C2 will be If the polynomial x + px + qx — x the 3. If 2 + g m and _ + n, then is — equal to? **(d)** np — mq 4. what and and is divisible by

145. **[CDS (I) Paper 15 Q13]** If thc linear factors — (a2 + I)x+a are p and q, then p q is equal to **(a)** R is a non-zero even integer **(b)** R is odd, if m is Odd **(c)** R s 2 for some integer s, if n is **(d)** R = t 3 for some integer t, if 3 divides n If4X2Y 128 and 33*32y— 9XY O, then the value Of x + y can be equal to 13. If thc linear factors —

146. **[CDS (I) Paper 15 Q2]** For (p + q) to be greater than (p — q), both p and q must be positive. Which of the above Statements is/ are correct? **(b)** 2 only **(c)** Both I and 2 Neither I nor 2 **(a)** R is a non-zero even integer **(b)** R is odd, if m is Odd **(c)** R s 2 for some integer s, if n is **(d)** R = t 3 for some integer t, if 3 divides n If4X2Y 128 and 33*32y— 9XY O, then the value Of x + y can be equal to 13. If thc linear factors —

147. **[CDS (I) Paper 15 Q3]** If each Of them saves 2,000 per year, What is the difference in their annual incomes? **(d)** e 4,000 4,500 e 5,000 5,500 B-ABPA-P.JRW / 54A 8 **(a) (b)** 6 None of the above B'E'2+CF'2 Which one Of the following is correct in respect of the above statements? **(c)** Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I **(d)** Both Statement I and Statement 11 are true but Statement 11 is not thc correct explanation of Statement I

148. **[CDS (I) Paper 15 Q29]** A clock strikes once at 1 0'clock, twice at 2 0'clock and thrice at 3 0'clock, and so on. If it takes 8 seconds to strike at 5 0'clock, the time taken by it to strike at IO o'clock is **(a) (b) (c) (d)** 14 seconds 16 seconds 18 seconds None of the above

149. **[CDS (I) Paper 15 Q37]** The solution Of the inequation is (given that x O) -I-JS -ltvs **(c) (a)** 38. Which of the points P **(b)** Q and R o nly **(c)** P and R o nly **(d)** P and Q o Illy

150. **[CDS (I) Paper 15 Q38]** Which of the points P(5, — I), QI 3, —2) and R(I, I) lie in the solution of the system Of inequations x + y 4 and x-y>2? **(a)** Q and R o nly **(b)** P and R o nly **(c)** P and Q o Illy **(d)** P, Q and R

151. **[CDS (I) Paper 15 Q42]** There are three brothers. The sums of ages of two Of them at a time are 4 years, 6 years and 8 years. The age difference between the eldest and the youngest is **(a)** 3 years **(b)** 4 years **(c)** 5 years **(d)** 6 years

152. **[CDS (I) Paper 15 Q43]** A person goes to a market between 4 p.m. and 5 p.m. When he comes back, he finds that the hour hand and the minute hand Of the clock have interchanged their positions. For how much time (approximately) was he out of his house? **(a) (b) (c) (d)** 55•38 minutes 55•48 minutes 55•57 minutes 5567 minutes

153. **[CDS (I) Paper 15 Q44]** In a gathering Of 100 people, 70 of them can speak Hindi, 60 can speak English and 30 can speak French. Further, 30 or thcm can speak both Hindi and English, 20 can speak both Hindi and French. If x is the number of people who can speak both English and French, then which one Of the following is correct? (Assume that everyone can speak at least one Of the three Ian) **(a)** 9 < xs30 45. A cloth merchant buys cloth from a weaver and cheats him by using a scale Which is IO cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but While selling uses a scale which is IO Crn shorter than a normal metre scale. What is his gain? **(b)** 22—0/0 **(c)** 23—0/0 B-ABPA-PÄIRW / 54A 46. Two pipes A and B can fill a tank in 60 minutes and 75 minutes respectively. There is also an outlet C. If A, B and C are opened together, the tank is full in 50 minutes. How much time will be taken by C to empty the full **(d)**

154. **[CDS (I) Paper 15 Q48]** The salary of a person is increased by Of his original salary. But hc received the samc amount even after increment. What is the pcrcentage of his salary he did not receive? (100 (90 / **(a)** 9 < xs30 45. A cloth merchant buys cloth from a weaver and cheats him by using a scale Which is IO cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but While selling uses a scale which is IO Crn shorter than a normal metre scale. What is his gain? **(b)** 22—0/0 **(c)** 23—0/0 B-ABPA-PÄIRW / 54A 46. Two pipes A and B can fill a tank in 60 minutes and 75 minutes respectively. There is also an outlet C. If A, B and C are opened together, the tank is full in 50 minutes. How much time will be taken by C to empty the full **(d)**

155. **[CDS (I) Paper 15 Q63]** Consider the following : I — cose cosec9 — Cote I + cose I + COS = cosec9 + cote I cose Which Of the above is/are identity/ iden tities? **(a)** 9600 square units **(b)** 10000 square units **(c)** 12000 square units **(d)** None of the above 58. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to

156. **[CDS (I) Paper 15 Q64]** If p = cotB+ tane and q = sece — then (p2q)5 —(q2p)S is equal to **(a)** 2 -Be , thenC+2B **(b)** 0-1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e **(c)** only **(d)** 2 only

157. **[CDS (I) Paper 15 Q66]** If **(c)** 0-1 0+1 cos26 — 3cos8+2 sin2 where O < e < then which of the following statements is/ are correct? I. There are two values of e satisfy ing the above equation. **(a)** 2 -Be , thenC+2B **(b)** 0-1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e **(c)** only **(d)** 2 only

158. **[CDS (I) Paper 15 Q2]** 9 260' is satisfied by the above Select the correct answer using the code given below. **(b) (d)** only 2 only Both I and 2 Neither I nor 2 / S4A **(a)** 2 -Be , thenC+2B **(b)** 0-1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e **(c)** only **(d)** 2 only

159. **[CDS (I) Paper 15 Q69]** There are five lines in a no two of which are parallel. The maximum number of points in which they can intersect is **(c)** 4 10 None Of the above **(a)** 2 -Be , thenC+2B **(b)** 0-1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e **(c)** only **(d)** 2 only

160. **[CDS (I) Paper 15 Q98]** The value of the expressi n (243 64712 243x243+647x 7 is equal to **(a)** AM < Median **(b)** GM Median **(c)** AM = Me dian Suppose for Which one of the corre ct? **(d)** GM>1/6

161. **[CDS (I) Paper 15 Q2]** Ifa < b, then — Select th e correct answer sing the code given below. **(b)** 2 only **(c)** 3 only **(d)** 2 and 3 **(a)** AM < Median **(b)** GM Median **(c)** AM = Me dian Suppose for Which one of the corre ct? **(d)** GM>1/6

162. **[CDS (I) Paper 16 Q4]** If e —y + z, = z + x and then what is the value of SDC-S-TRD - A **(a)** x = 100 **(b)** x = IO **(c)** 330 **(d)** 340

163. **[CDS (I) Paper 16 Q1]** What is the solution of loglo l— **(a)** x = 100 **(b)** x = IO **(a)** 330 **(b)** 340 **(c)** 355 **(d)** 366 -1 2

164. **[CDS (I) Paper 16 Q5]** What would be the maximum value of Q in the equation 5P9+ 3R7 + 2Q8= 1114? **(a)** x = 100 **(b)** x = IO **(c)** 330 **(d)** 340

165. **[CDS (I) Paper 16 Q7]** In an examination, a student was asked to divide a certain number by **(a)** x = 100 **(b)** x = IO **(c)** 330 **(d)** 340

166. **[CDS (I) Paper 16 Q8]** By mistake he multiplied it by 8 and got the answer 2016 more than the correct answer. What was, the number ? **(a)** 252 **(b)** 256 **(c)** 258 **(d)** 260

167. **[CDS (I) Paper 16 Q3]** What is to ? 62 +72 +8 7 + 46 — 2 + 92 + 4 + 26 = z + x and then what is the value of SDC-S-TRD - A **(a)** x = 100 **(b)** x = IO **(c)** 330 **(d)** 340

168. **[CDS (I) Paper 16 Q10]** Let A = {7, 8, 9, 10, 11, 12} and B = {7, IO, 14, 15}. What is the number of elements in (A —B) and (B —A) respectively ? **(a)** 2 and 4 **(b)** 4 and 2 **(c)** 2 and 2 **(d)** 4 and 4

169. **[CDS (I) Paper 16 Q14]** If a and are the two zeros of the polynomial 25x2 — 15x + 2, then what is a quadratic polynomial whose zeros are (2a)-l and (2ß)-l ? **(a)** x2+30x+2 **(b)** 8x2-30X+25 **(c)** 8x2-30x **(d)** x2+30x

170. **[CDS (I) Paper 16 Q1]** Which of the following statements are correct ? l. Both p and q may be prime. **(a)** 1 and 2 only **(b)** 2 and 3 Only **(c)** 1 and 3 only **(d)** l, 2 and 3 25. What is 23. A shopkeeper increases the cost price Of an item by 20% and offers a discount Of 10% on this marked price. What is his percentage gain ?

171. **[CDS (I) Paper 16 Q3]** Both the numbers may be com- posite. Select the correct answer using the code given below : **(a)** 1 and 2 only **(b)** 2 and 3 Only **(c)** 1 and 3 only **(d)** l, 2 and 3

172. **[CDS (I) Paper 16 Q25]** What is 32 **(b)** 2007 22 20072 2008 - 22 2 equal to ? 2008 **(a) (c) (d)** 2008 2007 2007 2008 2008 2009 SDC-s-TRD - A 8

173. **[CDS (I) Paper 16 Q4]** Which of the above statements is/are correct ? **(a)** 1 Only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither 1 nor 2

174. **[CDS (I) Paper 16 Q31]** If2p+3q= 12 and4p2+4pq—3q2 = 126, then what is the value of p + 2q ? 25 4 99 16 **(a)** 11 31. If2p+3q= 12 and4p2+4pq—3q2 — 126, then what is the value of p + 2q ? 29. There are twelve friends A, B, C, D, E, 33. What is the unit digit of 7139 F, G, H, I, J, K and L who invested money in some business in the ratio of and the duration for which they invested the money is in the ratio of respectively. Who will get the maximum profit at the end of the year ? **(b)** F only **(c)** G only **(d)** Both F and G

175. **[CDS (I) Paper 16 Q32]** The number Of digits in 330 is n and it is given that log103 -04771. What is the value Of n ? **(a)** 13 **(b)** 14 **(c)** 15 **(d)** 16 SDC-S-TRD - A 10

176. **[CDS (I) Paper 16 Q34]** If4x+3a 4 32 then what is the value of x — ax+a **(a)** 11 31. If2p+3q= 12 and4p2+4pq—3q2 — 126, then what is the value of p + 2q ? 29. There are twelve friends A, B, C, D, E, 33. What is the unit digit of 7139 F, G, H, I, J, K and L who invested money in some business in the ratio of and the duration for which they invested the money is in the ratio of respectively. Who will get the maximum profit at the end of the year ? **(b)** F only **(c)** G only **(d)** Both F and G

177. **[CDS (I) Paper 16 Q45]** Into a conical tent Of radius 8•4m and vertical height m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is **(a)** 264 **(b)** 201 **(c)** 132 **(d)** 105

178. **[CDS (I) Paper 16 Q49]** What is the number of rounds that a wheel of diameter -N m will make in traversing 7 km ? **(a)** 3300 **(b)** 3500 **(c)** 4400 **(d)** 4900

179. **[CDS (I) Paper 16 Q57]** Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? **(a)** Rs. 8 **(b)** Rs. 9 **(c)** Rs. 13 **(d)** Rs. 14 53. If a quadrilateral has an Inscribed then the sum of a pair Of opposite sides equals

180. **[CDS (I) Paper 16 Q71]** If sin6+cos9= 2 sing— coso equal to ? **(a)** 4275 **(b)** 4350 **(c)** 4375 **(d)** 4450 68. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ?

181. **[CDS (I) Paper 16 Q75]** What is the value of cosec2680 + sec2560 — cot2340 — **(a)** 1006m **(b)** 75-6 m 3 **(c)** 606 m 75. What is the value of cosw2680 + sec2560 — cot2340 — tan2220 ? 76. If 2y Coso x Sino and 2r sece — y cosec9 = 3, then what is x2 + 4y2 equal to ? 77. IfsinO+ coso= where O < 9 < — 2' 2 then what is tan O + coto equal to ? 4 SDC-S-TRD - A **(d)** 78. 4 If A sin20 + cos40 where < — 2' then which one of the following is correct ?

182. **[CDS (I) Paper 16 Q76]** If 2y coso = x Sino and 2x seco — y cosecO 3, tan222ö ? then what is x2 + 4y2 equal to ? **(a)** 1006m **(b)** 75-6 m 3 **(c)** 606 m 75. What is the value of cosw2680 + sec2560 — cot2340 — tan2220 ? 76. If 2y Coso x Sino and 2r sece — y cosec9 = 3, then what is x2 + 4y2 equal to ? 77. IfsinO+ coso= where O < 9 < — 2' 2 then what is tan O + coto equal to ? 4 SDC-S-TRD - A **(d)** 78. 4 If A sin20 + cos40 where < — 2' then which one of the following is correct ?

183. **[CDS (I) Paper 16 Q77]** IfsinO+cosO= where 0 < < — 2' 2 then what is tano + coto equal to ? 4 SDC-S-TRD - A 22 4 **(a)** 1006m **(b)** 75-6 m 3 **(c)** 606 m 75. What is the value of cosw2680 + sec2560 — cot2340 — tan2220 ? 76. If 2y Coso x Sino and 2r sece — y cosec9 = 3, then what is x2 + 4y2 equal to ? 77. IfsinO+ coso= where O < 9 < — 2' 2 then what is tan O + coto equal to ? 4 SDC-S-TRD - A **(d)** 78. 4 If A sin20 + cos40 where < — 2' then which one of the following is correct ?

184. **[CDS (I) Paper 16 Q74]** 300 600 **(a)** 100-Jjm **(b)** 750S m 10046 3 **(c)** 60-6 m 75. cosec2680 + sec2560 — cot2340 — tan2220 76. 2y cos9= x sing 3åq 2x secØ—y cosecO= 3 t, x2 + 4y2 ? 77. sin9+ 2 2 t, tano + cote ? 4 **(d)** 73 4 78. = sin2Ö+cos4et, t, 4 13 —s AS2 16 13 16 cot A +cosec A — 1 79. cot A — cosec A + 1 1 + cos A sin A I — cos A

185. **[CDS (I) Paper 16 Q89]** A cricketer has a certain average Of IO innings. In the eleventh inning he scored 108 runs, thereby increasing his average by 6 runs. What is his new average ? **(a)** 42 **(b)** 47 **(c)** 48 **(d)** 60

186. **[CDS (I) Paper 16 Q91]** Number of credit cards held by an Indlvldual can be treated as **(a)** Qualitative data **(b)** Discrete data **(c)** Categorical data **(d)** None of the above

187. **[CDS (I) Paper 16 Q92]** In a class of 100 students, there are 70 boys whose average marks in a subject are **(a)** x-1-25 **(b)** x-o.5 **(c)** 1-25 88. The median of set Of 9 distinct observations is If each Of the largest 4 observations of the set is increased by 2, then the median Of the new set **(d)** is increased by 2

188. **[CDS (I) Paper 16 Q1]** The classes of type 15-19, 20-24, 25—29 etc. are exclusive classes. **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 95. Suppose the class interval 10—15 has frequency 30, then what is the frequency density of this class interval ? 98. 99. 100. 28 W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ?

189. **[CDS (I) Paper 16 Q2]** The classes of type 15-20, 20-25, 25—30 etc. are inclusive classes. Which Of the above statements is/are correct ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2

190. **[CDS (I) Paper 16 Q100]** If equal to ? 7 11 10 17 10 = 2, then what is x **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 95. Suppose the class interval 10—15 has frequency 30, then what is the frequency density of this class interval ? 98. 99. 100. 28 W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ?

191. **[CDS (I) Paper 17 Q9]** 11. If x- then what is bx2 — 2ax + b equal to (b O) ? **(a) (b) (c) (d)** 2ab What is the value Of (443 + 5402 + (443-5402 **(a) (b) (c) (d)** 443* 443+ 547x547 0 2 3 If x = tt¯l and y = what is the relation between x and y ? **(a) (b) (c)** BLS-S-FDE x1Jy xl'y = yl}x (4_

192. **[CDS (I) Paper 17 Q14]** 15. If A. • B = 3 : 4, then what is the value of the 3A2 +4B .3A - 4B2 43 **(a)** 55 43 55 47 **(c)** 55 **(d)** Cannot be determined If A = (x : x is a multiple of 71, B = : x is a multiple Of 51 and C (x x is a multiple Of 351, then which one Of the following is a null set ? **(a) (b) (c) (d)** (A-D-C If x = 2 + + 21/3 , then what is the value Of xa — 6x2 + 6x ? **(a) (b) (c) (d)** 3 2 1 + — and x + y 26, is the value Of XY ? then what **(a) (b) (c) (d)** 15 25 30

193. **[CDS (I) Paper 17 Q18]** 19. What is the solution Of the equ ation + logz03 - **(a) (b) (c) (d)** 10 1 If and are the roots Of the equation x2 + px + q O, then What is a2 + ß2 equal t.o ? **(a) (b) (c) (d)** p2-2q q2 2p p2 + 2q q2_q If a3 335 + b3 and a = 5 + b, then what is the value Of a + b (given that a > O and b > O) ? **(d)** 7 9 16 49 If 9X3Y-2187 and 22y-4xy_ what can be the value Of (x + y) ? **(a) (b) (c)** BLS-S-FDE 3 7 — O, then (6- it is

194. **[CDS (I) Paper 17 Q40]** Sunil wants to spend 200 on two types Of sweets, costing 7 and IO respectively. What is the maximum number Of sweets he can get so that no money is left over ? **(a) (b) (c) (d)** 2 20 9 27 9 7 27 9 2 29— 9

195. **[CDS (I) Paper 17 Q48]** 49. 20 — 40 40-60 60-80 80 — 100 **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

196. **[CDS (I) Paper 17 Q59]** S Of rD = 90R 901) 60. (16-A) 8 10 4 8 18 29 45 What are the values offl and f2 respectively ? **(a) (b) (d)** 10 and 17 17 and 10 11 and 16 16 and 11 If D is the number Of degrees and R is the number Of radians in an angle O, then which one Of the following is correct ? **(b) (c) (d)** = IBOR 1801) What is the minimum value Of 9 tan2 + 4 cot2 9 ? **(a) (c) (d)** 6 9 12 13

197. **[CDS (I) Paper 17 Q60]** If D is the number Of degrees and R is the number Of radians in an angle O, then which one Of the following is correct ? **(b) (c) (d)** = IBOR 1801) c 100 1 •50 200 50 20 What is the minimum value Of 9 tan2 + 4 cot2 9 ? It is known that the average price per pencil is 1•25. M'llat is the value Of x ? **(a)** 10 **(b)** e 30 **(c)** e 40 **(d)** 60 BLS-S-FDE **(a) (c) (d)** 6 9 12 13

198. **[CDS (I) Paper 17 Q72]** 68. then What is + Y2 equal to ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

199. **[CDS (I) Paper 17 Q73]** From the top of a building 90 m high, the 69. angles of depression of the top and the bottom Of a tree are 300 and 450 respectively. What is the height Of the tree ? **(a) (b) (c) (d)** •BLS-S-FDE If x = a cos B + b sin and y = a sin — b cos 6, 72. 68. then What is + Y2 equal to ?

200. **[CDS (I) Paper 18 Q21]** 22. The product nf the polynomials (x 9), (x — 2), (x3 — 2x2 + 4x — 8) and (xa + 2x2 + + 8) is **(a) (b) (c) (d)** STS-M-TpT —256 (X4 — 16)2 (xd + 16)2 (x2 —

201. **[CDS (I) Paper 18 Q31]** Which one of the following is a correct 35. statement ? - 5} = {01 — 51 = (IO-A) **(a) (b) (c) (d)** If ab + bc + ca 2 — bc b2 a **(a)** 3 **(b) (c)** 1 5)

202. **[CDS (I) Paper 18 Q5]** What is the total number of boys in first and third standards ? **(a) (b) (c) (d)** 3TS-M-TpT 302 280 242 230

203. **[CDS (I) Paper 18 Q15]** What is the sum of money deposited ? **(a) (b) (c) (d)** 4 km 8 km 16 km 20 km

204. **[CDS (I) Paper 18 Q52]** 53. (16-A Consider the following statements : **(a) (b) (c) (d)** STS-M-TPT 52. 53.

205. **[CDS (I) Paper 18 Q80]** 81. -A) Consider the following statements in respect of three straight lines A, B and C on a plane : **(a) (b) (c)** STS-M-TpT 76. TWO rectangular sheets Of sizes 27t x 47t and 79. x 57t are available. A hollow right circular cylinder can be formed by joining a pair of parallel sides of any sheet. What is the maximum possible volume of the circular cylinder that can be formed this way ? 77. 78. PQ = 2XY PX=QY PQ- 3XY **(d)**

206. **[CDS (I) Paper 18 Q86]** 26 -A) Consider the following statements : **(a) (b) (c) (d)** 4 cm There are 8 lines in a plane, no two Of which 83. are parallel. What is the maximum number of points at which they can intersect ? 84.

207. **[CDS (I) Paper 18 Q85]** 82. off by a line DE drawn parallel to BC such that D is on AB and E is on AC. If BC 10 cm, then what is DE equal to ? **(a) (b) (c) (d)** 4 cm There are 8 lines in a plane, no two Of which 83. are parallel. What is the maximum number of points at which they can intersect ? 84. (26 -A) Consider the following statements :

208. **[CDS (I) Paper 18 Q92]** Mathematics and Physics Mathematics and Statistics A small company pays each of its 5 category workers 20,000, each of its 3 category 'B' workers 25,000 and a category 'A' worker 65,000. The number Of workers earning less What is the difference between the number of boys studying Mathematics and the number Of girls studying Physics ? **(a) (b) (c) (d) (b) (c) (d)** 8 5 4 3 20 30 60 80

209. **[CDS (I) Paper 19 Q4]** If a + b = 2c, then what is the value Of **(a) (b) (c)** If x y = zlå and z = Where x I, y l, z I, then what is the value of abc ? 7. **(a) (b) (c) (d)** 3 If 2b a • c and = xz, xb—e b equal to ? **(a) (c) (d)** then what is SPSS-T-XHI

210. [CDS (I) Paper 19 Q13] 14. 15. -A) Which one of the following is a zero of the polynomial 3xg • 4x2 — 7 ? **(a) (b) (c) (d)** 2 There are two numbers which are greater than 21 and their LCM and HCF are and 21 respectively. What is the sum of these numbers ? **(a) (c) (d)** 504 508 514 528 If u and are the roots of the equation ax2 + bx + c = O, then what is the value of the expression (0 + l) (p + l) ? **(a) (a) (b) (c) (d)** 13. ¯ = 5, then What is the value of 125 -125 140 - 140 Which one of the following is a zero of the polynomial 3xg • 4x2 — 7 ? **(a) (b) (c) (d)** 2 IO.

211. [CDS (I) Paper 19 Q17] 18. 19. (6-A k? **(b) (d)** -14 14 If fix) and g(x) are polynomials of degree p and q respectively, then the degree of If(x) ± (if it is non•zero) is **(a) (b) (c)** Greater than min(p, q) Greater than max(p, q) Less than or equal to max(p, q) Equal to min(p, q) What is the value Of **(c)** What is the value of I+xa¯b + xc—b where x O ? **(c) (d)** 3 is

212. [CDS (I) Paper 19 Q31] At present the average of the ages Of a father 35. and a son is 25 years. After seven years the son will be 17 years old. What will be the age of the father after IO years ? **(a) (b) (c) (d)** 44 years 45 years 50 years 52 years 36.

213. [CDS (I) Paper 19 Q40] 41. (12-A) **(a) (b) (c)** 8 years 9 years IO years 11 years 43. An alloy A contains two elements, copper and tin in the ratio Of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3 : 4. If 20 kg Of alloy A, 28 kg of alloy B and some more pure N **(d)** 3 kg

214. [CDS (I) Paper 19 Q90] SPSS-T-XHI (26-A S **(a)** 4:31 SPSS-T-XHI 83. What is the largest power of IO that divides 87. the product I x 2 x 3 x 4 ..x23x24x25? 84. 85. 86. **(b) (c)** 2 4 5 None of the above Walls **(d)**

215. [CDS (I) Paper 19 Q7] If x2 — 6x — 27 > O, then which one of the following is correct ? **(a) (b) (c) (d)** x < 9 or x > —3 x < — 3 only If x 2 + 22/3 + 21/3, then the value of the expression x3 — 6x2 + 6x will be The number Of divisors Of the number 38808, exclusive of the divisors 1 and itself, is **(b) (c) (d)** 2 —2 8. **(b) (c) (d)** 74 72 70 68 How many five-digit numbers of the form XXYXX Ware divisible by 33 ? HCF and LCM of two polynomials are (x + 3) and (xa — 9x2 — x + 105) respectively. If one of the two polynomials x2 - — 21, then other is **(b) (c) (d)** 3 5 Infinite **(a) (b) (c) (d)** X2 + — 21 x2 + 2x + 15 x2-2x-15 x2—x— 15 GTHY-T-XMT

216. [CDS (I) Paper 19 Q3] One of them is a pentagon and the other is a rectangle. One of them is a decagon and the other is an octagon. The sum of their exterior angles is 7200 Which of the above statements Ware correct ? **(a) (b) (c) (d)** The I only 2 only 1 and 3 2 and 3 minimum value of the expression 2x2 + + 5 is **(b) (c) (d)** 15/8 - 15/8 If H is the harmonic mean of P and Q, then H 11 the value Of — + — is **(a) (b)** 2

217. [CDS (I) Paper 19 Q35] TIs of of (11 -A) A person bought two articles X and Y from a departmental store. The sum of prices before sales tax was **(a) (b) (c)** 44 45 years 46 years 48 years **(d)**

218. [CDS (I) Paper 19 Q84] 35. -A) A person bought two articles X and Y from a departmental store. The sum of prices before sales tax was **(a) (b) (c)** 44 45 years 46 years 48 years **(d)**

219. [CDS (I) Paper 19 Q7] What are their present ngeg ? **(a) (b) (c) (d)** 39 years, 21 years 55 years, 25 years 75 years, 25 years 49 years, 31 years In a class of 60 boys, there are 45 boys who play chess and 30 boys who play carrom. If every boy of the class plays at least one of the two games, then how many boys play Carrom only ? **(a) (b) (d)** 30 20 15 10 GTHY-T-XMT (13

220. [CDS (I) Paper 19 Q39] 36. one bottle of orange juice. Orange juice per bottle costs twice the cost of lemon juice per bottle. Z bought nothing but contributed 50 for his share of the drink which they mixed together and shared the cost equally. If Ts e 50 is covered from his share, then what is the cost of one bottle of orange juice ? **(a) (b)** 1075 1100 e 1125 1150 **(c)** 75 50 46 e 30 40. 41. Twelve **(d)**

221. [CDS (I) Paper 19 Q8] Which of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both 1 and 2 Neither 1 nor 2 The solution of linear inequalities x + y 2 5 and x — y s 3 lies **(a) (b) (c) (d)** Only in the first quadrant In the first and second quadrants In the second and third quadrants In the third and fourth quadrants The minute hand Of a clock overtakes the hour

222. [CDS (I) Paper 19 Q4] Which of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both 1 and 2 Neither 1 nor 2 Consider the following statements :

223. [CDS (I) Paper 19 Q77] 78. 79. (25-A) A rectangular block Of length 20 em, breadth 15 and height IO is cut up into exact number of equal cubes. The least possible number of cubes will be **(a) (b) (c) (d)** 3497t m2 2897t m2 2097t m2 1497t m2 **(a) (b) (c)** 12 16 20 24

224. [CDS (I) Paper 20 Q100] If one Of them exceeds another by 36, then one Of the numbers is **(c) (d)** 17 29 43 Nono of the above **(a)** 19 20 21 22 The sum of three prime numbers is 100. If one Of them exceeds another by 36, then one Of the numbers is Proportion of WSpgp« 11 Both I and 11 Both 11 and 111 Both 1 and 111 All the three 1% **(b) (c)** 17 16. 29 43 Nono of the above 17. **(d)**

225. [CDS (I) Paper 20 Q16] 17. 18. **(a) (c) (d)** 20,000 25,000 30,000 35,000 What is the number of people who read at least two ? **(b) (d)** 12,000 13,000 14,000 15,000 What is the number of people who do not read any of these three newspapers ? **(c) (d)** 62,000 64,000 66,000 68,000

226. [CDS (I) Paper 20 Q18] What is the number of people who do not read any of these three newspapers ? 2 2-33 **(c) (d)** 62,000 64,000 66,000 68,000 DFGT-T-HTM **(a)** 19 20 21 22 The sum of three prime numbers is 100. If one Of them exceeds another by 36, then one Of the numbers is Proportion of WSpgp« 11 Both I and 11 Both 11 and 111 Both 1 and 111 All the three 1% **(b) (c)** 17 16. 29 43 Nono of the above 17. **(d)**

227. [CDS (I) Paper 20 Q50] 52. 53. (16-A) 52n _ .23n **(a) (b) (c) (d)** 2 tan 890 ? sin 46' cog 440 + cos 460 sin 44' ?

228. [CDS (I) Paper 20 Q57] 0—10 10-20 20-30 **(a) (b)** 61. , then What Consider the distribution : following grouped 10 frequency What is the value Of sin2 250 + sin2 65' ? **(c) (d)** 2 4 What is the value Of sing e 0 3 sin2 0 cos2 0 — I ? If the mean of the above data is 25•2, then what is the value Of p ?

229. [CDS (I) Paper 20 Q67] 21 -A) The monthly expenditure Of a is e The distribution of expenditure on various items is as follows : **(a) (b) (c) (d)** 44 55 66 If the above data is represented by a percentage bar diagram of height 15 em, then what are the lengths of the two segments of the bar diagram corresponding to education and miscellaneous respectively ? Diagrammatic representation Of data ineludes which of the following ? I.

230. [CDS (I) Paper 20 Q69] Which one of the following pairs is correctly matched ? **(a) (b)** The Median Mean Geometric mean Mode Graphical location Graphical location Ogive 73. **(c)** NO change following pairs relate to frequency The volume of a spherical balloon is increased by 700'h What is the percentage increase in its surface area ? distribution Of a discrete variable and its frequency polygon. Which one Of the following pairs is not correctly matched ? **(d)** 300% 400% 450% 500%

231. [CDS (I) Paper 20 Q3] AC-AB<BC BC-AC<AB AB-BC<AC Which Of the above aie correct ? 91. **(a) (b)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 **(c) (d)** 21: 16 14:11

232. [CDS (I) Paper 21 Q3] Numba of o' Y by E is wlnt p.« cat of the mamba of wld by ±owroorn C ? **(a)** 52 54 **(c)** 55 **(d)** 56 4. of ni—Øld by B is what cat mom than of of X Old showromn A ? **(a)** 78} **(b)** 83} 86} **(c) (d)** 88} C — SDBY-T•yrH 02) : frlbwidg "Gution fir NO of Bwa **(a)** 336 **(b)** 35-6 **(c)** 37•6 the that fillow CIUS interwi 10-20 40-50 Serja-I, 20 15 Series-a 8 4 I. is åe of freg—y Of Saia-I ? 2. is Of frqny Of Sa•ie-ll ? **(a)** 26 **(b)** 36 **(c)** 46 **(d)** 56 (00 ; 61bWing itans • La Of Of of X and Y mld 5 (A, B, C, D nd in a be by Sl and distri$u€on oftnmlba of "0Mers of only X by five in tie be nmber Ocooters 12 64m 19 21 15 24 20 8 30

233. [CDS (I) Paper 21 Q10] What is numba of daily mad nide1Ü in 2017 ? **(a)** 10 **(b)** 12 19 **(d)** 21 C - SDHY-T-MTH S. What is Of woot«s Of Y by showrooms A, C E ? **(a)** 461} 0) 441} **(c)** 426} **(d)** 416} What is bawn manba of Of boff —ia by A toul Of Of X by **(a)** 416 **(b)** 426 **(c)** 432 **(d)** 436 Red Ellowing md ne data ±ows **(a)** 10 **(b)** 12 19 **(d)** 21

234. [CDS (I) Paper 21 Q8] Adding I Of two it WEch Of **(a)** 1,2and3 **(b)** I nd 2 Mily **(c)** 2 3 mxly **(d)** I 3 Mly IS. The rüe of two is it is But in Of the half is in dre nuany. Equl amouMs ue in If Of returns after 2 years is 482, what is in •æh **(c)** enooo

235. [CDS (I) Paper 21 Q16] For value of k x3 + — + 6 be C SDHY-T-MA **(a)** be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? **(b)** 80 **(c)** 60 36 • **(d)** 24 14. WEch Of

236. [CDS (I) Paper 21 Q14] WEch Of **(a)** 1,2and3 **(b)** I 2 Mily **(c)** 2 3 mxly **(d)** I 3 Mly IS. TIE rüe of two is it is But in Of the half is in dre

237. [CDS (I) Paper 21 Q18] WbatistheLOAof- —7 3' 6 9 27 18 27 10 27 20 **(a)** 26 18 **(b)** 42 mimnes 36 **(c)** 45 minutæ **(d)** 46 mimit—12 sæonds 2 4 18. WbatistheLOAof,-, —, — ? 9 27 18 27 10 27 20 19. If qudons whA is vahE Of k ? 10 3 10 5 3 10 9. 20. A Int to B at tlr time me to C sne 7% sinple intaut. Aner 4 yeas A 01200 intæt B md C. HOW mueh A to C ?

238. [CDS (I) Paper 21 Q24] N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ? **(a)** 2975 **(b)** 3094 **(c)** 3101 **(d)** 3269

239. [CDS (I) Paper 21 Q29] Which orz of åe mllowing is ca•ræt in Of a light riangle ? **(a)** Its lia ixuide O **(b)** Its lies on the triülgle **(c)** It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? **(d)** AB<RX

240. [CDS (I) Paper 21 Q30] Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? **(a)** Its lia ixuide O **(b)** Its lies on the triülgle **(c)** It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? **(d)** AB<RX

241. [CDS (I) Paper 21 Q32] If = I, then What is value of + 3iniOx + + ? •13 **(a)** Its lia ixuide O **(b)** Its lies on the triülgle **(c)** It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? **(d)** AB<RX

242. [CDS (I) Paper 21 Q33] If sine x —cota What is value of + 3x + + 81++243+ ? **(a)** 941 **(b)** tooo **(c)** 1220 **(d)** 1365 M. The of Of the tops Of two Of h and 2h from a pint p the line Pining the Eet of åe two tlr distarm of of üle pitlan from P x y Evely, thal which of Nloüg is ? **(a)** 2h2 **(d)** 2h2

243. [CDS (I) Paper 21 Q36] The Of a frimgle is 22 em. Thmugh orthe a lilE puallei to the side is What is the of timgle by line ? **(a)** 33 cm **(b)** 44 **(c)** 66 cm 37. The sides AD, BC of a ü•apdum ABCD prallel md the di.N1s AC BD at O. If the of Gangle AOB is 3 cm2 the Of tiangle BDC is 8 then what is åe of aianøe AOD ? **(d)** 8 cm2

244. [CDS (I) Paper 21 Q39] The perimeaz oftwo 'imilü ringle ABC PQR Ue 7$ cm nd 50 r«pætively. If the Of one side of the &ianøe PQR is 20cm, thal what is the Ingh of side of the U-iuwle 4BC ? **(a)** 33 cm **(b)** 44 **(c)** 66 cm 37. The sides AD, BC of a ü•apdum ABCD prallel md the di.N1s AC BD at O. If the of Gangle AOB is 3 cm2 the Of tiangle BDC is 8 then what is åe of aianøe AOD ? **(d)** 8 cm2

245. [CDS (I) Paper 21 Q40] IA PQRS be whose diagonals PR at O. If QRS is tiangle having a side of length 10 em, then wut is the l•.of **(b)** **(c)** 156 cm **(d)** 20-6 cm **(a)** 33 cm **(b)** 44 **(c)** 66 cm 37. The sides AD, BC of a ü•apdum ABCD prallel md the di.N1s AC BD at O. If the of Gangle AOB is 3 cm2 the Of tiangle BDC is 8 then what is åe of aianøe AOD ? **(d)** 8 cm2

246. [CDS (I) Paper 21 Q2] If Of fre lalgths of is 12 what is arn of ? **(a)** 9 an-2 **(b)** 12 cm2 **(c)** 16 **(d)** 25 cma C - SDHY-T-MTB

247. [CDS (I) Paper 21 Q63] If one Of a aingle (with all sides intega•s) is 15 an, thm maximum Of uiangle ? **(a)** 240 cm **(b)** 225 **(c)** 113 cm **(d)** 112 cm

248. [CDS (I) Paper 21 Q1] If p is relatively to of g r, p is relatiüely prime to poduct qr. **(a)** 1 only **(b)** 96" cm2 **(c)** 2 only **(d)** Boå 1 2 69T cm2

249. [CDS (I) Paper 21 Q73] Rmi Five yen back, the of Radha Was tima that of Rni, but buk age of WE pf Rmi. is age them' ? **(c)** 10 **(d)** 11 **(a)** 1 only **(b)** 96" cm2 **(c)** 2 only **(d)** Boå 1 2 69T cm2

250. [CDS (I) Paper 21 Q74] A e 500 wmts to md outof it. of anle 5 of is e 7, what is numba of ways in wh#h a pæ buy total amount ? **(a)** 10 0) 14 (C) 15 **(d)** 17 Given y is to , x 36 whal is åe value of x 54 ? **(a)** 54 27 16 C - SDHY-T-M'N A circular cme has If radius of base is 6 what is its ? **(a)** 1 only **(a)** 96" cm2 **(b)** 2 only **(c)** Boå 1 2 69T cm2 **(c)** an2

251. [CDS (I) Paper 21 Q76] Whatis åe qure root of 16+66 ? 0) 4-47 **(c)** 3+4i **(d)** 3-N5 **(a)** 3+4i **(b)** 3-N5 77. What is the numb' in 125, 823 md 920 ? [Givn hg102 = 0-301, log. —&477, 10867 - 0845] **(c)** 276 **(d)** 282

252. [CDS (I) Paper 21 Q90] How pairs Of ze such that åe of thü Nuar— 35 ? **(a)** -1.2 **(b)** -1,-2 4 90. How pairs Of 15 ze such that åe of thü Nuar— 35 ? 91. If **(c)** 21 **(d)** 19 value of x2+5x+ O 94. mtio of

253. [CDS (I) Paper 22 Q7] Which one of the following is the largest number among 22222, 22222, 22222, 22222 ? **(a)** a prime number **(b)** not a prime, but power of a prime **(c)** not a power of a prime, but a composite even number **(d)** not a of a prime, but a composite odd number 2. Two unequal pairs of numbers satisfy the following conditions :

254. [CDS (I) Paper 22 Q13] What are the values of p and q respectively, if (x — 1) and (x +2) divide the polynomial x3+4x2+px+q ? **(a)** 13 **(b)** 39 **(c)** 117 5 11. A two-digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two-digit number is 8, then what is the number ? **(d)** 81

255. [CDS (I) Paper 22 Q15] If — then what is the value of 102x? **(b)** 1000 **(c)** 100000 **(d)** 1000000 **(a)** 1000 **(b)** 100000 **(c)** 1000000 16. A number divides 44333 so as to remainder in each 19. What is **(d)** 2am-n equal to ? a 12288, 28200 and leave the same case. What is that 20. 22. If x = •a, y=VÄ and z=fi, then which one of the following is correct ? number ? 272

256. [CDS (I) Paper 22 Q17] If x2+9y2 equal to ? then what is y : x **(a)** 1000 **(b)** 100000 **(c)** 1000000 16. A number divides 44333 so as to remainder in each 19. What is **(d)** 2am-n equal to ? a 12288, 28200 and leave the same case. What is that 20. 22. If x = •a, y=VÄ and z=fi, then which one of the following is correct ? number ? 272

257. [CDS (I) Paper 22 Q19] What is **(d)** 2am-n **(a)** 1000 **(b)** 100000 **(c)** 1000000 16. A number divides 44333 so as to remainder in each 19. What is **(d)** 2am-n equal to ? a 12288, 28200 and leave the same case. What is that 20. 22. If x = •a, y=VÄ and z=fi, then which one of the following is correct ? number ? 272

258. [CDS (I) Paper 22 Q20] If x=u, y = VÄ and z=fi, then which one of the following is correct ? a-ß equal to ? a **(a)** 1000 **(b)** 100000 **(c)** 1000000 16. A number divides 44333 so as to remainder in each 19. What is **(d)** 2am-n equal to ? a 12288, 28200 and leave the same case. What is that 20. 22. If x = •a, y=VÄ and z=fi, then which one of the following is correct ? number ? 272

259. [CDS (I) Paper 22 Q21] If 10500 and y what is logy equal to ? **(a)** 4-2500 **(b)** 2-5625 **(c)** 1-5625 **(d)** 1-2500

260. [CDS (I) Paper 22 Q24] If x varies asyz, theny varies inversely (x:) **(a)** 49x2 -f- 3x + 2 **(b)** 49x2 _ 3x + 2 **(c)** 49++2 **(d)** 59++2 24. If x varies asyz, theny varies inversely

261. [CDS (I) Paper 22 Q26] What is the point on the xyplane satisfying 5x +2y 7xy and **(b)** **(c)** I Ox + 3y ? '6 6' 6' **(a)** 49x2 -f- 3x + 2 **(b)** 49x2 _ 3x + 2 **(c)** 49++2 **(d)** 59++2 24. If x varies asyz, theny varies inversely

262. [CDS (I) Paper 22 Q31] Students of a class are made to sit in rows of equal number of chairs. If number of students is increased by 2 in each row, then the number of rows decreases by **(a)** 50 knvhr **(b)** 40 km/hr **(c)** 35 knvhr **(d)** 25 km/hr 30. If 6 men and 8 women can do a piece of work in IO days; and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman ?

263. [CDS (I) Paper 22 Q3] If number of students is increased by 4 in each row, then the number of rows decreases by **(a)** 50 knvhr **(b)** 40 km/hr **(c)** 35 knvhr **(d)** 25 km/hr 30. If 6 men and 8 women can do a piece of work in IO days; and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman ?

264. [CDS (I) Paper 22 Q4] The average height of students in the class is 4-6 feet. Ifthe average height of the boys in the class is feet, then what is the average height of the girls in the class ? **(a)** Less than 4-2 feet **(b)** More than 4•2 feet but less than 4-3 feet **(c)** More than 4•3 feet but less than 4-4 feet **(d)** More than 44 feet but less than feet 11 A - PIXI-T-MTK

265. [CDS (I) Paper 22 Q42] HOW many integral values Of x and y satisfy the equation 5x + 9y 7, where -500 < x < 500 and -500 500 ? **(a)** 110 **(b)** 111 **(c)** 112 **(d)**

266. [CDS (I) Paper 22 Q43] Let None of the above XYZ be a 3-digit number. Let S-XYZ+YZX+ZW. Which of the following statements Ware correct ? **(a)** A, B and C **(b)** A, B and D **(c)** A, E and F 41. The number of three digit numbers **(d)** Six

267. [CDS (I) Paper 22 Q5] If X takes 45 minutes more than Y to reach the destination, then what is the time taken by Y to reach the destination ? **(a)** 135 minutes **(b)** 150 minutes **(c)** 180 minutes **(d)** 225 minutes

268. [CDS (I) Paper 22 Q48] In a water tank there are two outlets. It takes 20 minutes to empty the tank if both the outlets are opened. If the first outlet is 0FEned, the tank is emptied in 30 minutes. What is the time taken to empty the tank by second outlet ? **(a)** 30 minutes **(b)** 40 minutes **(c)** 50 minutes **(d)** minutes 49, If (P —1) is a fictor of ax4 +bx3+cx2 +dx+ e, then which one of the f)llowing i' correct ? **(a) (c) (d)** 17 x18) SO. If x' + - — — 47 , what is the value of **(a)** 36 **(b)** 27 **(c)** 18

269. [CDS (I) Paper 22 Q51] A wheel makes 360 revolutions in one minute. What is the number of radians it tums in one second ? **(a)** 10 **(b)** 16i **(c)** 36 **(d)** 27

270. [CDS (I) Paper 22 Q53] Let the **(d)** 17 A - PLKI-T-MTK **(a)** 10 **(b)** 16i **(c)** 36 **(d)** 27

271. [CDS (I) Paper 22 Q55] A tree of height 15m is broken by wind in such a way that its top touches the ground and makes an angle 300 with the ground. What is the height from the ground to the point where tree is broken ? **(a)** 10m 56. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75 feet, then what is the height of the taller tower **(b)** 85 feet **(c)** 110 feet **(d)** 125 feet

272. [CDS (I) Paper 22 Q64] What is the value of 1 —2 sin2 8cos2 6 +4 equal to ? sin4 8 + cos4 9 **(a)** 2sine **(b)** 2cos0 61. What is **(c)** cosec x + cot x 63. If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 **(d)**

273. [CDS (I) Paper 22 Q80] HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

274. [CDS (I) Paper 22 Q85] A ladder 5 m long is placed in a room so as to reach a point 4•8m high on a wall and on turning the ladder over to the opposite side of the wall without moving the base, it reaches a point 1 •4 m high. What is the breadth of the room ? **(d)** 7-5 m **(a)** 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ? **(b)** 4 square units **(c)** 245 square units **(d)** r square units

275. [CDS (I) Paper 22 Q93] In the given figure, if 6 and = 5, then what is the value of x ? **(a)** 450 **(b)** 300 **(c)** 150 **(d)** 100

276. [CDS (I) Paper 22 Q97] In the given figure AB is parallel to CD and AC is parallel to BD. If 550, ZHAB-xO, then what is the value of x ? **(a)** 85 **(b)** 80 **(c)** 75 **(d)** 65 A - PIXI-T-MTK Angles are shown in the given figure. What is value of ZI + Z2+Z3+Z4 + **(a)** 2400 **(b)** 3600 **(c)** 5400 **(d)** 7200 33 In the given figure PQ is parallel to RS, ZAEF=950, ZBHS- 1106, and LABC = -ro. Then what is the value of x? **(a)** 15 **(b)** 25 **(c)** 30 **(d)** 35

277. [CDS (I) Paper 23 Q2] Au + 2 is alw ays an number. **(a)** 25% 40% **(b) (c) (d)** 1 and 3 only 22. I and 2 only 2 and 3 only I, 2 and 3 A river 3 m deep and 40 m wide is flowing at the rate Of 2 km/hr and falls into the sea. What is the amount of water in litres that will fall into the æa from this river in a minute ? If An Pn + l, where pn is the product of the first n prime numbers, then consider the following statements • I. An is alw ays a composite number. 2. Au + 2 is alw ays an number. 3. An I is always an even number. Which of the above statements Ware correct ?

278. [CDS (I) Paper 23 Q29] Mahesh is 60 years old. Ram is 5 years younger to Mahesh and 4 years elder to Babu is a younger brother Of Raju and he is 6 years younger. What is the age difference between Mahesh and Babu ? **(a) (b) (c)** 18 years 15 years 13 years The number Of items in a t»oklet is N. In the first year there is an increase of in this number and in the subsequent year there is a decrease of At the end of the two years, what will be the number of items in the booklet ? **(b) (c) (d)** Less than N Equal to N More than N It depends on the value of N ab xy — Xb O and bc + yz — cy O, then equal ? what is — — **(a) (c)** DZOL-T-LKM (13 rs

279. [CDS (I) Paper 23 Q85] 36. If — 37. **(a) (b)** — hours — hours **(c) (d)**

280. [CDS (I) Paper 23 Q27] What is the prt:xluct Of the digits of the number ? **(d)** 35 40 45 54 then one Of the follmving is correct ? **(d)** x + 2y + 3z O DZOL-T-LKM (15 **(a) (b)** — hours — hours **(c) (d)**

281. [CDS (I) Paper 23 Q36] If — 37. (15-A) What is the value of **(a) (b)** — hours — hours **(c) (d)**

282. [CDS (I) Paper 23 Q51] 52. 53. (19-A) The difference between two angles is 150 and the sum Of the angles in radian is — bigger angle is k times the smaller angle. What k to ? **(a) (b)** Consider the following statements : 49. 50. **(c) (d)**

283. [CDS (I) Paper 23 Q76] 77. (27 —A) **(a) (b)** Hexagon Nonagon Decagon 79. A bicycle wheel makes 5000 revolutions in moving 11 km. What is the radius of the wheel ? **(c) (d)**

284. [CDS (I) Paper 23 Q79] A bicycle wheel makes 5000 revolutions in moving 11 km. What is the radius of the wheel ? (Assume It **(a) (b) (c) (d)** 17-5 cm 35 em 70 cm 140 cm

285. [CDS (I) Paper 23 Q93] Fifteen candidates appeared in an examination. The marks Of the candidates who passed in the examination are 9, 6, 7, 8, 8, 9, 6, 5, 4 and **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

286. [CDS (I) Paper 23 Q91] Fifteen candidates appeared in an 94. examination. The marks Of the candidates who passed in the examination are 9, 6, 7, 8, 8, 9, 6, 5, 4 and **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

287. [CDS (I) Paper 24 Q3] How many pairs of (x. y) can chosen from the set 8, 9) such that 2, Where x Y ? **(a)** 3 6 7 **(b)** How 2. What is the remainder when 21000000 divided by 7? 2 3 4 5 many terms are there in the **(c) (d)** 2 How many pairs of

288. [CDS (I) Paper 24 Q6] How many such pairs are there ? **(d)** S. How 2 3 4 5 many terms are there in the following product? (al + a2 + b2 + b3 + b4) 15 30 45 60 **(a)** 3 6 7 **(b)** How 2. What is the remainder when 21000000 divided by 7? 2 3 4 5 many terms are there in the **(c) (d)** 2 How many pairs of

289. [CDS (I) Paper 24 Q7] If a b + c O, then which of the following are correct? a 3 + b 3 c 3 3abC —2(ab + bc + ca) = —3ab(a + b) Select the correct answer using the code given below. **(c)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 3q-2 then what is the value of p2 o XVWS—T-MTK/SOA - 3pq+2? 5 **(a)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 3q-2 then what is the value of p2 o XVWS—T-MTK/SOA - 3pq+2? 5 9. What is the unit digit in the expansion of 6732? **(b)** 3 IO. What is the value of x, if 2bx z a? b 2 — 2bx **(c)** P.T.O. 9. What is the unit digit in the expansion of 6732? 3 IO. What is the value of x, if 2bx z a? b 2 — 2bx P.T.O. 7. If a b + c O, then which of the following are correct? a 3 + b 3 c 3 3abC —2 **(d)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 3q-2 then what is the value of p2 o XVWS—T-MTK/SOA - 3pq+2? 5

290. [CDS (I) Paper 24 Q17] What is 8x equal to? 2 4 XVWS-T-MTK/50A 9 **(a)** Which Of the above statements is/are correct? **(b)** I only **(c)** 2 only **(d)** Both I and 2

291. [CDS (I) Paper 24 Q18] For what integral value of x is **(b) (a)** Which Of the above statements is/are correct? **(b)** I only **(c)** 2 only **(d)** Both I and 2

292. [CDS (I) Paper 24 Q19] 5 3 2 — + I = k 2 , then which One of the is a possible expression k P — 3X+I **(a)** Which Of the above statements is/are correct? **(b)** I only **(c)** 2 only **(d)** Both I and 2

293. [CDS (I) Paper 24 Q20] What is bc(a — b) (a — equal to ? **(a)** Which Of the above statements is/are correct? **(b)** I only **(c)** 2 only **(d)** Both I and 2

294. [CDS (I) Paper 24 Q45] If 5*-3 then What is x equal to? I — loglo 2 3 1 loglo 2 **(d)** What I-log102 5 1 — loglo 2 is the least value Of 3sin2e *4cos2 e ? 3 2 **(a)** 16 44. I — loglo 2 3 log102 I-loglo 2 5 **(b)** 1-log102 What is the 3sin2e +4cos2 e ? 3 2 least value Of . The sum Of the reciprocals of two What alternate natural numtrrs is is the sum Of the numbers? **(c)** 12 13 14 If n is any natural then 52n — 1 is always divisible by how many natural numbers? 47. If sinecose k, where os 9 s then which one of the following is correct? **(d)**

295. [CDS (I) Paper 24 Q47] If sinecose k, where os 9 s then which one of the following is correct? **(b) (c) (d)** 17 Osksl O S k 5 only O •5 k I only P.T.O. **(a)** 16 44. I — loglo 2 3 log102 I-loglo 2 5 **(b)** 1-log102 What is the 3sin2e +4cos2 e ? 3 2 least value Of . The sum Of the reciprocals of two What alternate natural numtrrs is is the sum Of the numbers? **(c)** 12 13 14 If n is any natural then 52n — 1 is always divisible by how many natural numbers? 47. If sinecose k, where os 9 s then which one of the following is correct? **(d)**

296. [CDS (I) Paper 24 Q58] 6 + secB 8 — 6 tane **(a)** 11 **(b)** 22 **(c)** 77 **(d)** 99 k sece

297. [CDS (I) Paper 24 Q57] What is (I + cote equal to? 3 2 XVWS-T-MTK/50A SA If 6+ 8tane sece and 8 — 6tane then What iB the value of k 27 11 22 77 **(d)** 99 **(a)** 99 56. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß 2 and sin u + sinß where O a , O sßs9 **(b)** Cannot be determined due insumcient data to 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre If + where O S S W, then what is the value of sin 2 ? 2 57. What is **(c)** 99 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre 60. A vertical tower standing at the comer Of a rectangular field subtends angles of 60' and 45' at the two nearer comers. If e is the angle that the tower subtends at the farthest comer, then What is cote equal to? 2 1 P.T.O. 54. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß—2 and Bina + O, where O a 9 **(d)** 0 2 Cannot be determined due to insumcient data If + where O S S W, then what is the value of sin 2 ? 2 57. What is

298. [CDS (I) Paper 24 Q59] A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre **(a)** 99 56. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß 2 and sin u + sinß where O a , O sßs9 **(b)** Cannot be determined due insumcient data to 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre If + where O S S W, then what is the value of sin 2 ? 2 57. What is **(c)** 99 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre 60. A vertical tower standing at the comer Of a rectangular field subtends angles of 60' and 45' at the two nearer comers. If e is the angle that the tower subtends at the farthest comer, then What is cote equal to? 2 1 P.T.O. 54. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß—2 and Bina + O, where O a 9 **(d)** 0 2 Cannot be determined due to insumcient data If + where O S S W, then what is the value of sin 2 ? 2 57. What is

299. [CDS (I) Paper 24 Q66] A conical vessel whose internal radius is 5 cm and height 24 Crn is full Of water. The water is emptied into a cylindrical vessel with internal radius IO cm. What is the height to which the water rises? I cm **(c) (a) (b)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 66. 63. The ratio of the radius of base to the height of a cylinder is 2 : 3. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? **(c)** 242 cm **(d)** 385 cm 2

300. [CDS (I) Paper 24 Q82] ABC is a right angler* at R Let M and N be two points on AB such that AM MN NB. 1k t P and Q two points on AC such that PM is parallel to QN and QN is parallel to CH If BC 12 cm, then what is (PM + QN) equal to? **(a) (b) (c)** 28 units 42 units 56 units 84 units 82. ABC is a right angled at R Let M and N be two points on AB such that AM - MN-NB Let P and points on AC such that PM to QN and QN is parallel Q two is parallel to If BC 12 cm, then equal to? what is 8S. The sides Of a triangle ABC are 4 cm, 6 cm and 8 cm. With the vertices of the triangle as centres, three circles are drawn each touching the Other two externally. What is the sum of the radii Of the three circles? **(d)**

301. [CDS (I) Paper 24 Q91] Consider the -following data With regard to production Of cars (in lakhs) : Year 2015 Year 2016 Coun Coun Coun Coun 38 88 **(a)** In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? **(b)** and E **(c)** A, C and D only **(d)** C D and E

302. [CDS (I) Paper 24 Q95] The marks obtained by 5 students are 21, 27, 19, 26, **(a)** In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? **(b)** and E **(c)** A, C and D only **(d)** C D and E

303. [CDS (I) Paper 24 Q97] Which product is produced least over the years 2000-2005? **(a)** Type 11 **(b)** Type 111 **(c)** Type **(d)** Type V XVWS-T-MTK,'SOA

304. [CDS (I) Paper 24 Q98] In Which one Of the following pairs of years, the difference in total number Of tablets produced between them is m inimum? **(a)** (2003, 2005) **(b)** (2001, 2005) **(c)** g003, 2004) **(d)** (2000, 2002)

305. [CDS (I) Paper 24 Q100] In Which year, the production I is more than the sum of the production Of Type and Type TV? **(a)** Type 11 **(b)** Type 111 **(c)** Type **(d)** Type V XVWS-T-MTK,'SOA 98. In Which one Of the following pairs of years, the difference in total number Of tablets produced between them is m inimum?

306. [CDS (I) Paper 25 Q8] What is the sum of the linear factors (in x and y) of the expression 2x-2 + — 3y2 ? **(a)** 2t-3y **(b)** 3x-2y **(c)** 3x + 2y **(d)** 2r+3y

307. [CDS (I) Paper 25 Q11] For which values of k, does the equation x2 — kx + 2 0 have real and distinct solutions ? **(b)** only **(c)** k>2v5 only **(d)** k<-24i or k>2Vi **(a)** only **(b)** k>2v5 only **(c)** k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is **(d)** I—aßy

308. [CDS (I) Paper 25 Q12] If a + D + Y aß+ "y + ya, then what is (1 equal to? **(a)** only **(b)** k>2v5 only **(c)** k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is **(d)** I—aßy

309. [CDS (I) Paper 25 Q16] 5 pencils, 6 notebooks and 7 erasers cost *250; whereas 6 pencils, 4 note- books and 2 erasers cost **(a)** only **(b)** k>2v5 only **(c)** k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is **(d)** I—aßy

310. [CDS (I) Paper 25 Q17] How many zeros are there in the product 150 x 249 x 348* x 501 ? **(a)** 262 **(b)** 261 **(c)** 246 **(d)** 235

311. [CDS (I) Paper 25 Q20] If 5-t¯l then what is the value of x ? **(b)** logr02 **(c)** log105 **(d)** 210g105 **(a)** 30 **(b)** 45 **(c)** 60 **(d)** 75 23. What IS value of N?

312. [CDS (I) Paper 25 Q21] If 96-64a3 3 = what is alt + 4a3 equal to ? then **(a)** 30 **(b)** 45 **(c)** 60 **(d)** 75 23. What IS value of N?

313. [CDS (I) Paper 25 Q22] If A+B= equal to ? x2 —4x+4 2x2 -7X+3 2x-1 2x2+7x-3 2x-1 then what is B 7 the 7 **(a)** 30 **(b)** 45 **(c)** 60 **(d)** 75 23. What IS value of N?

314. [CDS (I) Paper 25 Q24] If then what is (a + b) : (a — b) equal to ? **(a)** 3 only **(b)** 5 only **(c)** —3 or 3 **(d)** —5 or 5

315. [CDS (I) Paper 25 Q25] If b+c—a b—c—a a+b—c then what is equal to? **(a)** 30 **(b)** 45 **(c)** 60 **(d)** 75 23. What IS value of N?

316. **[CDS (I) Paper 25 Q27]** Consider the following inequalities : a2—b2 a—b > — where a > b > O a2+b2 a a3+b3 a2+b2 only when 2. a2+b2 a+b Which of the above is/are correct ? **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2

317. **[CDS (I) Paper 25 Q29]** If ay — bx _ cx — az _ bz ¯ then which of the following is/are correct ? x+y+z_z 2. a+b+c¯c Select the correct answer using the code given below • **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither 1 nor 2 9

318. **[CDS (I) Paper 25 Q34]** Ihe present age of a father is equal to sum of the ages of his 4 children. Aner ten years the sum of the ages of the children will be 1-6 times the age of their father. What is the present age of father ? **(a)** 36 years **(b)** 40 years **(c)** 42 years **(d)** 45 years

319. **[CDS (I) Paper 25 Q36]** If the system of equations 7x + ky 27 and kx + 7y = 19 have unique solution, then which one of the following is correct ? **(b)** 13 **(d)** k-13 **(a)** 36 years **(b)** 40 years **(c)** 42 years **(d)** 45 years 35. The sum of numerator and denominator of a fraction is 10. If the numerator is increased by 3 and denominator is decreased by 1 , the fraction becomes I. What is the difference between numera- tor and denominator of the fraction ? 36. If the system of equations 7x + ky 27 and kx + 7y = 19 have unique solution, then which one of the following is correct ?

320. **[CDS (I) Paper 25 Q38]** Between 3 and 4 0'clock, both hour hand and minute hand will coincide past 3 0'clock between **(a)** 15-16 minutes **(b)** 16-17 minutes **(c)** 17-18 minutes **(d)** 18-19 minutes

321. **[CDS (I) Paper 25 Q40]** If 2r-3y- 7-0, then what is the value of -27+340? **(a)** 36 years **(b)** 40 years **(c)** 42 years **(d)** 45 years 35. The sum of numerator and denominator of a fraction is 10. If the numerator is increased by 3 and denominator is decreased by 1 , the fraction becomes I. What is the difference between numera- tor and denominator of the fraction ? 36. If the system of equations 7x + ky 27 and kx + 7y = 19 have unique solution, then which one of the following is correct ?

322. **[CDS (I) Paper 25 Q45]** A person saves more than he did the previous year. If he saves *2000 in the first year, in how many years will he save 170000 ? 13 **(a)** 16 years **(b)** 17 years **(c)** 18 years **(d)** 19 years

323. **[CDS (I) Paper 25 Q46]** What is the minimum value of cos30+ sec39 where 00$9<900 ? **(d)** None of the above **(a)** 20 **(b)** 21 **(c)** 24 **(d)** 25 43. A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ?

324. **[CDS (I) Paper 25 Q47]** If 14 sin29+ 10 cos2e= 11 where 00 < O < 900 , then what is the value of tan0+ cote ? 4 2 **(d)** 20 sin30 + cos30 **(a)** 20 **(b)** 21 **(c)** 24 **(d)** 25 43. A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ?

325. **[CDS (I) Paper 25 Q52]** If sin2 where 00 < e < 900 then what is sin20 + coso equal to ? 4 3 2 7 4 **(a)** 9<300 **(b)** 450 < 0<600 **(c)** 600 < 9<900 cos20 — 3cos6 +2 52. If sin2 where 00 < e < 900 then what sin20 + coso equal to ? 4 3 2 7 4 53. Consider the following : IS 2. sin4Ø+cos49=l+2sin20cos20 15 Which Of the above are identities ? and 2 only and 3 only and 3 only **(d)** 1, 2 and 3 is the value of 54. What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B 55. If x and z = p cos A, then what is the value of x2 + y2+z2 ?

326. **[CDS (I) Paper 25 Q65]** The minute hand of a clock is 21 cm long. What is the area on the face of the clock described by the minute hand between 10-10 a.m and 1030 a.m ? (take z **(a)** 231 cm2 **(b)** 331 cm2 **(c)** 462 cm2 **(d)** 492 cm2 The length and breadth of a room are 21 m and 16 m respectively. If the length of the longest rod that can be placed in the room is 29 m, then what is height of the room ? **(a)** 10m **(b)** 11m **(c)** 12m **(d)** 13m

327. **[CDS (I) Paper 25 Q67]** A hemispherical bowl of intemal radius 18 cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3 cm and internal height 4 cm. What is the number of bottles used to empty the bowl ? **(a)** 54 **(b)** 81 **(c)** 108 **(d)** 135

328. **[CDS (I) Paper 25 Q68]** A hollow spherical shell is made of a metal of density 7 g/cm3. If its internal and extemal radii are 3 cm and 6 cm respectively, then what is the mass of the shell ? (take **(a)** 2772 g **(b)** 3322 g **(c)** 4433 g **(d)** 5544 g 19 19

329. **[CDS (I) Paper 25 Q84]** Which one of the following is correct ? **(a)** AB2-AD2=ADXBD **(b)** AC2-AD2=BDxCD **(c)** AB2-AD2 =2ADxBD **(d)** AC2-AD2-2BDXCD

330. **[CDS (I) Paper 25 Q86]** What is AB2 + 4BQ2 equal to ? **(a)** 41•25cm2 **(b)** 42•25 cm2 **(c)** 43-75 cm2 **(d)** 44•25 cm2

331. **[CDS (I) Paper 25 Q87]** What is AQ2 + CP2 equal to ? **(b)** **(c)** **(d)** 1-2 AC2 1-25 AC2 1-5 AC2 A - NYRG-T-MTK **(a)** 10•05 cm2 **(b)** 10•15cm2 **(c)** 11-05 cm2 **(d)** 11•15cm2 Consider the following for the next three

332. **[CDS (I) Paper 25 Q91]** There are two categories of employees in 2019 whose total strength remained the same in 2020. What are these two categories ? **(a)** C and B **(b)** E and A **(c)** E and F **(d)** C and D

333. **[CDS (I) Paper 26 Q3]** If x 7 + 40, then what is the value of + — ? **(a)** 2 7. A man walks at an average speed of 3 km/hr from his residence and reaches Office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence 3 2. 4143 4343 is divisible by and office ? **(b)** 6 km **(c)** IO km **(d)** 8 km

334. **[CDS (I) Paper 26 Q6]** What is the value of the following? (5.4)3 -0.064 (5-4) +2' to **(b)** 44 **(d)** 54 5 GHSA-T-MTK/ 4SA 3 **(a)** 2 7. A man walks at an average speed of 3 km/hr from his residence and reaches Office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence 3 2. 4143 4343 is divisible by and office ? **(b)** 6 km **(c)** IO km **(d)** 8 km

335. **[CDS (I) Paper 26 Q8]** What is the value of the following? 746+6.9 **(b)** 2 6 **(a)** 2 7. A man walks at an average speed of 3 km/hr from his residence and reaches Office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence 3 2. 4143 4343 is divisible by and office ? **(b)** 6 km **(c)** IO km **(d)** 8 km

336. **[CDS (I) Paper 26 Q9]** If x 9999, then What is the value of the following? 4x3 — x 3333 **(b)** 2222 **(d)** 6666 3 **(a)** 2 7. A man walks at an average speed of 3 km/hr from his residence and reaches Office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence 3 2. 4143 4343 is divisible by and office ? **(b)** 6 km **(c)** IO km **(d)** 8 km

337. **[CDS (I) Paper 26 Q13]** If43Xx47Y then what is the value of the following? **(c)** 25 4xy+ x + y 2xy-x-y **(b)** 15 **(d)** 45 **(a)** 6 units **(b)** 8 units **(c)** IO units **(d)** 12 units 13. If43Xx47Y then what is the value of the following? 4xy+ x + y 2xy-x-y 16. 17. 18. 20. 5 What is the least value of n if 194480* nzm4, where m and n are natural numbers? If + yv'Q ¯ then What is the value Of ?

338. **[CDS (I) Paper 26 Q24]** What is equal to? **(d)** éöii-l **(a)** 12 **(b)** 10 **(c)** 21 26. 28. If -2 then what is the value Of as + b3? 2n.2 + m and q •2 — m Let p 2 **(d)**

339. **[CDS (I) Paper 26 Q25]** If then what is the Value Of the following? 7x2 -3x+7 3 7 3 **(c)** GHSA-T-MTK/45A 5 12 29 **(a)** 12 **(b)** 10 **(c)** 21 26. 28. If -2 then what is the value Of as + b3? 2n.2 + m and q •2 — m Let p 2 **(d)**

340. **[CDS (I) Paper 26 Q26]** 28. If -2 then what is the value Of a **(a)** 12 **(b)** 10 **(c)** 21 26. 28. If -2 then what is the value Of as + b3? 2n.2 + m and q •2 — m Let p 2 **(d)**

341. **[CDS (I) Paper 26 Q35]** What is the number Of divisors Of 1000 (excluding I and 1000)? **(a)** 12 **(c)** 14 GHSA-T-MTK/45A **(b)** 13 **(d)** 16 9

342. **[CDS (I) Paper 26 Q41]** Consider a question and two statements : Question • Is 3x +2y positive? Statement—I : —29 • 8 Statement—II : y3 3x Which one Of the following is correct in respect of the question and the statements? **(a) (b) (c)** Statement—I alone is sufficient to answer the question Statement—II alone is sufficient to answer the question Both Statement—I and Statement—II are together sufficient to answer the question Both Statement—I and Statement—II are not sufficient to answer the question **(d)**

343. **[CDS (I) Paper 26 Q44]** 11 Consider a question and tqro statements : Question : Is always greater than Statement—I : x > y Statement—II : x 2 + > 2xy Which one Of the following is correct in respect Of the question and the State merits? **(a)** Statement—I alone is required to answer the question **(b)** Statement—II alone is required to answer the question **(c)** Both Statemen and Statement—II are required to answer the question **(d)** Neither Statement—I nor State- ment—ll is required to answer the qu estion P.T.O.

344. **[CDS (I) Paper 26 Q48]** What is the minimum value sm4B+cos4 —2sin ecos2 9? 0 Minimum value does not exist **(d)** Of **(a)** Zero **(b)** One **(c)** Two **(d)** Infinitely many 46. If I — cos 9 + sine I + sine sine + cose — 1 then what is 48. 49. 50. What is the minimum value sm4B+cos4 —2sin ecos2 9? value does not exist Of

345. **[CDS (I) Paper 26 Q56]** If tane + sece 3, then What is the value of 3 tan8+ 9sece? **(b)** 17 **(a)** 120' **(b)** 135' **(c)** 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is **(d)** following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

346. **[CDS (I) Paper 26 Q59]** If O, where then what is 'the value Of 12cot2 B + 3cosece? **(a)** (0+112 **(b)** (0+2)2 **(c)** p" +1)2 **(d)** (30+1)2

347. **[CDS (I) Paper 26 Q70]** A tall cylindrical container With circular base Of radius 18 cm contains a good quantity of water. Metal balls each Of radius 0-9 cm are immersed in it. HOW many balls are required to raise the water level by 3 cm? **(a)** 100 The radius and height Of a right circular cone are in the ratio 3 7. If the volume Of the cone is 528 cm3 , then What is the height of the cone? **(b)** 10•5 cm **(c)** 140 cm A square sheet is forrned by joining n identical Square sheets of same size. If the length of the diagonal of the bigger square sheet formed is m, then What is the side length Of a smaller square sheet? **(d)** A cylindrical pipe has inner diameter Of 14 cm. Water flows through it at a rate of 154 litres per minute. What is the speed Of water in km/hr? wake 2?

348. **[CDS (I) Paper 26 Q71]** An equilateral triangular sheet is forrned by joining 9 equilateral triangular sheets each of area 90 cm2. What is the height of the biger triangular shee t? **(a)** 12 **(b)** 943 cm **(c)** 18 cm **(d)** 27 cm 72. A farmland is in the shape Of a rhombus. The perimeter of the land is 100 m and the length of one of the diagonals is 40 m. The land is divided into four equal parts. What is the area of each 75. 76. 77. 21 If the perimeter of an isosceles right triangle is 4

349. **[CDS (I) Paper 26 Q91]** What is the total population of the city (in lakhs)? 18 15 12 **(a)** 21 90. In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? **(b)** 12 cm **(c)** 12•2 cm **(d)** 12•4 cm

350. **[CDS (I) Paper 26 Q93]** HOW many times the increase in expenditure in a year exceeded by more than as compared to previous **(a)** 306 under : Year 2011 2012 2013 2014 2015 2016 2017 138 15-4 10-4 13-1 15•8 172 19-4 **(b) (c)** Direction e 36048 lakhs e 300-36 lakhs 04032 lakhs e 18040 lakhs : Consider the following for the 93. HOW many times the increase in expenditure in a year exceeded by more than as compared to previous 94. In Which year, the percentage increase in expenditure is maximum as compared to its previous year? **(d)** 2012

351. **[CDS (I) Paper 26 Q95]** How much less amount is allocated to A and C together as compared to D and E together? **(a)** 0004 lakhs **(b)** 0005 lakhs **(c)** 6008 lakhs **(d)** 0010 lakhs GHSA-T-MTK/ 45K

352. **[CDS (I) Paper 26 Q96]** HOW inuch amount will be increased on A if the total budget is increased by three times? **(b) (d)** Direction e 36048 lakhs e 300-36 lakhs 04032 lakhs e 18040 lakhs : Consider the following for the (04) items that follow : 500 candidates appeared in an examination comprising tests in English, Hindi and Mathematics. 30 candidates failed in English only; 75 failed in Hindi only; 50 failed in Mathematics only; 15 failed in both English and Hindi; 17 failed in both Hindi and Mathematics; 17 failed in both Mathematics and English; 5 failed in all three tests. **(a)** 306 under : Year 2011 2012 2013 2014 2015 2016 2017 138 15-4 10-4 13-1 15•8 172 19-4 **(b) (c)** Direction e 36048 lakhs e 300-36 lakhs 04032 lakhs e 18040 lakhs : Consider the following for the 93. HOW many times the increase in expenditure in a year exceeded by more than as compared to previous 94. In Which year, the percentage increase in expenditure is maximum as compared to its previous year? **(d)** 2012

353. **[CDS (I) Paper 27 Q1]** 2. 3. then What is x equal to ? **(a) (b) (d)** 4 5 5. where a c, then which one of the following is correct ? **(a) (b) (c) (d)** a + b=c+d a+c=b+d a + b+c•d

354. **[CDS (I) Paper 27 Q11]** 12. Consider the following statements : l. x' (y —z) is positive if x > y > z. g. x' (y — z) (z— (x — y) is negative if x < y < z. Which Of the at»ve statements Ware correct ? **(a) (b) (c) (d)** What to? **(a) (c)** Only I Only 2 Both I and 2 Neither I nor 2 x 6.— 24x4 + 144x2 is (x2 + - x2 x2_2 x2-2v'ä What is x (x — yHx — z) equal to ? **(d)** FDGT-T-EMT

355. **[CDS (I) Paper 27 Q12]** Y (y — zKy — x) **(a) (b) (c) (d)** What to? **(a) (c)** Only I Only 2 Both I and 2 Neither I nor 2 x 6.— 24x4 + 144x2 is (x2 + - x2 x2_2 x2-2v'ä

356. **[CDS (I) Paper 27 Q22]** 23. Consider the number N = 126 x 38 x **(a)** y = 180* y 270* y = 360x the one 21. LCM of Of the Consider the number N = 126 x 38 x 53. Which Of the following statements Ware correct ? I. The number Of Odd factors of N is 60. 2. The number Of even factors of N is 720. Select the correct answer using the code' given below : **(b) (c) (d)**

357. **[CDS (I) Paper 27 Q33]** Ifa z 3b, 4b equal to ? 27 43 43 27 43 43 27 = 5c, 6c 7d, then what is What are the values of k for which the polynomial (k 3) x2 — kx — I has no real linear factors ? **(a) (b) (d)** If bc + cd 2bdand a + c 2b, then which one Of the following is correct ? **(a) (b) (c) (d)** ab ad -cd=o —bdz0 -bc-o ad + bc

358. **[CDS (I) Paper 27 Q32]** 33. What are the values of k for which the polynomial (k 3) x2 — kx — I has no real linear factors ? **(a) (b) (d)** If bc + cd 2bdand a + c 2b, then which one Of the following is correct ? **(a) (b) (c) (d)** Statement-I alone is sufficient to answer the question Statement-2 alone is sufficient to answer the question Both Statement-I and Statement-2 are sufficient to answer the question Both Statement-I and Statement-2 are not sufficient to answer the question **(a) (b) (c) (d)** ab ad -cd=o —bdz0 -bc-o ad + bc F DGT-T-EMT (13-A)

359. **[CDS (I) Paper 27 Q41]** 42. What is one Of the J»ssible values of x — **(a) (b)** What is the radius Of the circle ? What is one Of the J»ssible values of x — — ? **(c) (d)** What is one of the possible values Of + X 45.

360. **[CDS (I) Paper 27 Q46]** 47. '-A) **(a) (b) (c)** 300 450 Carinot be determined insufficient data due to What is PC equal to ? **(a) (b) (c) (d)** H metres -vfiH metres -vfäH metres 2H metres Consider the equation 6 6x2 — 25x +

361. **[CDS (I) Paper 27 Q54]** 55. 56. (21 —A) If x equal to ? , then What is tane — I **(a) (b)** sin 0 sin x, then what is equal to, **(c) (d)**

362. **[CDS (I) Paper 27 Q76]** 2 (AB - AC) BC2 4 (AB + (AB - AC) = 3 BC2 **(a) (b) (c)** 12 12 **(d)** 2 2 78. 79. 80. Triangle ABC is right-angled at C AC BC. What is Z ABC equal to ? and

363. **[CDS (I) Paper 27 Q2]** CD-DE Select the correct answer using the code given **(a)** — 't **(b) (c) (d)** Only I and 2 Only 2 and 3 Only I and 3 l, 2 and 3

364. **[CDS (I) Paper 28 Q70]** What are the average marks (approximately) of both the sections if it was detected later that an entry of 47 marks was wrongly made as 74 ? **(b) (c) (d)** 67-28 6768 68-11 68-63 **(a) (b) (c) (d)** 5 divides

365. **[CDS (I) Paper 28 Q21]** 23. If a and are the r€Kits Of the equation — 7x + I = O, then What is the value of 04+34? **(a) (b) (c)** 2207 2247 2317 25. 2337 **(d)** Of x ?

366. **[CDS (I) Paper 28 Q23]** If a and are the r€Kits Of the equation 24. — 7x + I = O, then What is the value of 04+34? WhatistheHCFof ? 39-1 **(a) (b) (c)** 2207 2247 2317 25. 2337 **(d)** Of x ?

367. [CDS (I) Paper 28 Q5] 3. n5 — 5n3 4n is divisi ble by **(a) (b) (c) (d)** Effciency Of 13 women = Effciency Of 18 men = Efficiency Of Effciency Of 11 women men Emciency of Efficiency Of 13 women 17 men Effciency Of I I women = Efficiency Of

368. [CDS (I) Paper 28 Q44] The coefficient of x2 is O. The coefficient ofx is (a — b) (b — c) (c — a). Which Of the statements given above Ware correct ? **(a) (b)** Consider the following statements in respect of the polynomial I — x — xn + xn +1, Where n is a natural number : I. It is divisible by I — 2x + 2. It is divisible by I — xn. Which of the statements given above Ware correct ? e 842 e 756 e 641 e 542 **(c) (d)**

369. [CDS (I) Paper 28 Q50] 121 **(a)** What is the minimum value of 6 — 4 sm e, osesE? **(b) (c) (d)**

370. [CDS (I) Paper 28 Q71] For the next ten (IO) items that follow : Each item contains a Question followed by two Statements. Answer each item using the following instructions : (25 -A) Choose this option if the Question can be answered by one of the Statements alone but not by the Other. Choose this option if the Question can be answered by either Statement alone. Choose this option if the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. Choose this option if the Question cannot be answered even by using both Statements together. **(a) (b) (c) (d)** Choose this option if the Question can be answered by one of the Statements alone but not by the Other. Choose this option if the Question can be answered by either Statement alone. Choose this option if the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. Choose this option if the Question cannot be answered even by using both Statements together. Question : Is m > n, where m, n are nori-zero numbers ? Statement I : Statement 11 : Question : Can a circle be drawn through the points A, B and C ? Statement : Statement 11 : AB=5cm, BC AB=3cm, BC 4 em, A number 277XY5

371. [CDS (I) Paper 28 Q16] Each item contains a Question followed by two Statements. Answer each item using the following instructions : **(a) (b) (c) (d)** Choose this option if the Question can be answered by one of the Statements alone but not by the Other. Choose this option if the Question can be answered by either Statement alone. Choose this option if the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. Choose this option if the Question cannot be answered even by using both Statements together. Question : Is m > n, where m, n are nori-zero numbers ? Statement I : Statement 11 : Question : Can a circle be drawn through the points A, B and C ? Statement : Statement 11 : AB=5cm, BC AB=3cm, BC 4 em, A number 277XY5

372. [CDS (I) Paper 29 Q55] Further, the difference Of the given number and the number obtained by reversing the order Of the digits Of the number is **(a) (b) (c) (d)** If

373. [CDS (I) Paper 29 Q3] 4. (3-A) **(a) (b) (c) (d)** If

374. [CDS (I) Paper 29 Q15] 16. 17. + 38-x **(a) (b) (b) (c) (d)** t? **(a) (b) (c) (d)** 4 3 2 - 2b3 — 2ab3 2bg 2ab3 — 18xy + *lix • v'5y — 3x2 + 2y2 — v"iÄx — 3x2 + 2y2 + + JNUP-T-EMT — •'Gxy + 6 — OGxy + 6 (6-

375. [CDS (I) Paper 29 Q45] 46. What is the monthly electricity bill for a house with m rooms and consuming n units ? **(a) (b) (c)** equal to ? of the sum of two parts, one part is proportional to number Of rmms and the other part is proportional to number of units consumed. 400 is the monthly electricity bill for a house with 8 and consuming 240 units and 320 is the monthly electricity bill for a house with 6 and consuming 200 units. 45. 46. What is the monthly electricity bill for a house with m rooms and consuming n units ? **(d)**

376. [CDS (I) Paper 29 Q52] 53. (21 -A) **(a) (b) (c) (d)** 10-5 cm 14 cm 21 cm 28 cm

377. [CDS (I) Paper 29 Q3] What is the value of x when y ? **(a) (b) (c) (d)** 18 21 24 27

378. [CDS (I) Paper 29 Q89] 90. -A) What is the average age (in years) Of B, C, D, E and G? **(a)** 12 **(b)** 13 **(d)** 15 What is the difference between the weights (in kg) of G and C ? **(a) (c)** What is the average weight (in kg) of A, B, C, **(a) (c)** 201 197 30 32 Consider the following statements : I.

379. [CDS (I) Paper 29 Q2] 3. The age Of F cannot be determined due to insufficient data. The average weight of D and F is equal to weight Of E. The weight difference is maximum for D Which Of the statements given above are correct ? **(a) (b) (c) (d)** I and 2 only 2 and 3 only I and 3 only 1.2 and 3 Consider the following for the next (05) items that follou' : A, B, D, E, F and G are cousins. D is thrice as Old as A. Further, C is as many years younger to B, as G to E and E to D. The average age of D and G is

380. [CDS (I) Paper 29 Q1] JNUP-T-EMT (33 —A I) **(a)** if the question can be answered by using one Of the statements alone, but cannot be answered using the other statement alone. Mark option **(b)** if the question can answered by using either statement alone. Mark option **(c)** if the question can be answered by using both the statements together, but cannot be answered using either statement alone. Mark option **(d)** if the question Cannot be 94. answered even by using both the statements

381. [CDS Paper 30 Q8] 9. IO. If 2 is a zero of the polynomial p(x) = x3 + 3x2 — 6x — a, then what is the sum of the squares Of the other zeros Of the polynomial ? **(b)** 17 **(c)** 21 **(d)** 37 If t = cos 790, then What is cosec 790 (I — cos 790) equal to ? 1+t **(a) (d)** Suppose p(x) = x' + a2X2 + alX + ao and q(x) = xa + b3X3 + b2X2 + blX + bo are the polynomials. If N, Y, 6 are zeros Of and a, ß, •r, are zeros of q(x), then what is p(x) — q(x) equal to ? **(c)** SRSU-T-EMT

382. [CDS Paper 30 Q13] 14. -A) If the equation x cos O p has a real solution for every O where O S O E, then which one of the following is correct ? **(a) (b) (c) (d)** p 1/8 p 21/8 What is' the difference between the greatest value and the least value Of cos2 0 3 sin2 + 2 ? **(a) (c) (d)** 4 2 ABC is a right-angled triangle, right-angled at B such that AB 6 cm and BC 8 cm. What is the perimeter of the inscribed in the triangle ABC with maximum area ? **(a) (b) (c) (d)** 24/7 em 96/7 cm 24 cm 32 cm What is the greatest value Ofk for which 2x2 — 4x + k = O has real roots ? **(a) (b)** 2 **(c) (d)** 4 q(x) = xa + b3X3 + b2X2 + blX + bo are the polynomials. If N, Y, 6 are zeros Of and a, ß, •r, are zeros of q(x), then what is p(x) — q(x) equal to ? **(c)** SRSU-T-EMT If 2 is a zero of the polynomial

383. [CDS Paper 30 Q36] If 93 is replaced by 94, then consider the following statements : **(a) (b)** 4 6 3. old median is I. The difference between new Old mean is less than 0-1. The difference between new old mode is zero. mean mode above Which of the statements given correct ? and and are 19. When the number **(c) (d)**

384. [CDS Paper 30 Q48] 49. ABCD å, AB 6 BC = 18 em, CD z 6 em DA = IO em BD z x, ? **(a)** 8 < X < 12 **(c)** 16 < 18 18 x 20 **(a) (b) (c) (d)** (v'fi +1): 1

385. [CDS Paper 30 Q50] ABCDÅ, AB = BC CD = DA; BDz12cmt 30 arf em **(c)** SRSU-T-EMT em, **(a)** SRSU-T-EMT 48. ABCD AB = 6 BC = 18 em, 51. 49. **(b) (c)** z 6 em DA = IO em BD z x, ? 8 < X < 12 16 < 18 18 x 20 52. tan **(d)** 250

386. [CDS Paper 30 Q51] 48. BC = 18 em, CD Ifthe diagonal BD x, then which one of the following is correct ? **(a) (b) (c)** 8 x < 12 12 < x < 16 16 < x < 18 18 x < 20 If tan **(d)** 250

387. [CDS Paper 30 Q5] What is the share of P? **(c) (d)** e 2307 e 2376 e 2508 2896 SRSU-T-EMT **(a) (b)** 3 100 km 3 1000 **(c) (d)**

388. [CDS Paper 30 Q60] 19-A) If p2 varies as R and Q2 varies as R, (P Q), then which Of the foll(Wing are correct ? l. • p2 varies as R. **(a) (b)** 3 100 km 3 1000 **(c) (d)**

389. [CDS Paper 30 Q65] If the Question can be answered by either Statement alone. If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. If the Question cannot be answered even by using both Statements together. **(a) (b) (c) (d)** If the Question can be answered by one Of the Statements alone, but not by the 65. If the Question can be answered by either Statement alone. If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. If the Question cannot be answered even by using both Statements together. 66. Statement-II : y z Let a, b and c be the sides of a triangle ABC. Question : Statement •I : Statement-II Is the triangle equilateral ? a2 + b2 + c2

390. [CDS Paper 30 Q84] If the people coming from a particular State belonging to S are and in number, then What is the total number of migrating people belonging to the age group B ? **(b) (d)** 1-2 lac 125 lac 1-30 lac 1-50 lae What is the maximum Of differences between the number Of people coming from different groups P, Q, R and S ? **(a) (b) (c) (d)** I •6 lac Iac 2•4 Iac 2•6 lac SRSU-T.EMT

391. [CDS (I) Paper 31 Q18] Which one of the following is correct ? **(a) (b) (c)** 100 -0-125 **(d)** 0125 2. The difference of the square of two natural numbers m and n

392. [CDS (I) Paper 31 Q11] What is the maximum value of 8sin9 — 4sin20 ? **(d)** 12 **(a)** 17 **(b)** 14 **(c)** 11 **(d)** 10 9. Train X crosses a man standing on the platform in 24 seconds and train Y crosses a man standing on the platform in 18 seconds. They cross each other while running in opposite directions in 20 seconds. What is the ratio of speed of X to speed of Y? 10. Let p, q be the roots of the equation x2 + mx — n 0 and m, n be the roots of the equation .r2 + px—q = O

393. [CDS (I) Paper 31 Q12] What is (1 + tana tanß)2 + (tana — tanß)2 equal to ? **(a)** tanza tan2ß **(b)** sec2a sec2ß **(c)** tan2a cot2ß **(d)** sec2a tan2ß

394. [CDS (I) Paper 31 Q13] Consider the following statements : I. tan500 — cot500 is positive 11. cot250 — tan250 is negative Which of the statements is/are correct ? **(a)** Ioniy **(b)** 11 only Both I and 11 Neither I nor 11 A - RAKU-T-EMT equal to ? **(a)** 17 **(b)** 14 **(c)** 11 **(d)** 10

395. [CDS (I) Paper 31 Q15] What is the value of sin2e cos29 (s«2e + cosæ2e) equal to ? sinle **(a)** 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e **(b)** 2h **(c)** Htana•tanß **(d)** h cota•cOtß

396. [CDS (I) Paper 31 Q16] If 64 64 - 16 where O O — , 2 then what is the value of + coto ? **(a)** 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e **(b)** 2h **(c)** Htana•tanß **(d)** h cota•cOtß

397. [CDS (I) Paper 31 Q17] If cosece — Cote — m and sece — tane n, then what is cosece + sece qual to ? **(d) (a)** 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e **(b)** 2h **(c)** Htana•tanß **(d)** h cota•cOtß

398. [CDS (I) Paper 31 Q18] From a point X on a bridge across a river, the angles of dqression of two points P and Q on the banks on Opposite side of the river are a and respectively. If the point X is at a height h above the surface of the river, what is the width of thc river if a are complementary ? **(a)** 2h(tana + Cota) **(b)** H tana•tanß **(c)** h cota•cOtß **(d)** h seca• coseca

399. **[CDS (I) Paper 31 Q23]** A person purchases one kg of tea powder from each of the four places A, B, C, D at the rate per 1 kg, 2 kg, 4 kg, 5 kg. If on an average he purchased x kg of tea powder per 1000, then what is the approximate value of x ? 9 9 **(a)** 195 **(b)** 2-00 **(c)** 205 **(d)** 2•10

400. **[CDS (I) Paper 31 Q11]** If A is odd, then B is odd Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 1 730 1 65

401. **[CDS (I) Paper 31 Q29]** If x3+_j___ and y3+ 27 ' then which one of the following is a value of xy ? 11 **(a)** 16 **(b) (c) (d)**

402. **[CDS (I) Paper 31 Q31]** A 4-digit number N has exactly 15 distinct divisors. What is the total number of distinct divisors of ? **(a)** 16 **(b)** 30 **(c)** 45 **(d)** 225

403. **[CDS (I) Paper 31 Q33]** What is the minimum a4b4 where a > 0, b > 0? value of 11 A - RAKU-T-EMT **(a)** 16 **(b) (c) (d)**

404. **[CDS (I) Paper 31 Q10]** If the LCMisx (a 2-digit number), then how many values of x are possible ? **(a)** Only one **(b)** Only two **(c)** Only three **(d)** More than three 13

405. **[CDS (I) Paper 31 Q37]** WhatisHCFof a4+2a3 + 3a2 + 2a + 1 arxi 4—2a3+l ? **(a)** 20 **(b)** 25 **(c)** 30 **(d)** 35 35. 36. Let ABC be a triangle with area 36 square cm. If AB=9 cm, BC' 12 cm and LABC = O, then what is cos9 equal to ? 4 3 2 Let n be a natural number. The HCF of n, n +10 is 10. If the LCM is x

406. **[CDS (I) Paper 31 Q40]** What is equal to ? 2 cos30 — cose **(a)** sun29 **(b)** cos2e **(c)** cot9 **(d)** tano A - RAKU-T-EMT

407. **[CDS (I) Paper 31 Q44]** A trolley with two wheels one metre apan is moved clockwise on the circular track around a ground with radius 50 m (described by right wheel). If the size of each wheel is of 1 foot radius and the right wheel turns 1000 times, how many times will the other wheel tum ? **(a)** 1010 **(b)** 1015 **(c)** 1020 **(d)** 1025

408. **[CDS (I) Paper 31 Q45]** What is the rernainder when 70 x 71 x 72 x 73 x 74 x 75 x 76 x 77 x 78 x 79 is divided by 1000? **(a)** IO mimin **(b)** 5 mimin **(c)** 10 cm/sec **(d)** 5 cm/sec 43. A spherical wooden ball of radius r is to be divided into eight identical parts by cutting by planes passing through the same diameter. What is the surface area of each final piece ? 3 2 2m•2 3 4%r2 A trolley with two wheels one metre 44. apan is moved clockwise on the circular track around a ground with radius 50 m

409. **[CDS (I) Paper 31 Q46]** A vertical pole of length 80m is situated on the horizontal plane. The base of the pole is at P. There are two points A and B such that p, A, B are on the same straight line. Let the angles of elevation of top of the pole from A and B be a and (a>ß) respectively. If PA 64 m and AB 36 m, then what is (a +ß) equal to ? **(a)** 600 **(b)** 900 **(c)** 1200 **(d)** 1350 15 A - RAKU-T-EMT

410. **[CDS (I) Paper 31 Q52]** If LACP= to ? 2 2 17 y, then what is tany equal A - RAKU-T-EMT **(a)** xu•2 48. A square is drawn inside a square of side 14 cm in such a way that the corners of the inner square coincide with the mid points of the sides of the outer square. What is the area lying between the two squares ? **(b)** 98 square cm **(c)** 56 square cm **(d)** 49 square cm

411. **[CDS (I) Paper 31 Q56]** If LBAC— then what is sine + cos9 equal to ? 71 65 73 65 79 **(d)** 65 **(a)** I and 11 only **(b)** 11 and Ill only **(c)** I and Ill only **(d)** 1, 11 and 111 Consider the following for the next two

412. **[CDS (I) Paper 31 Q57]** If BD is the perpendicular on the side AC, then what is the length of BD ? 1008 65 **(b)** cm 65 A - RAKU-T-EMT **(a)** I and 11 only **(b)** 11 and Ill only **(c)** I and Ill only **(d)** 1, 11 and 111 Consider the following for the next two

413. **[CDS (I) Paper 31 Q61]** A question is given followed by two statements I and **(a)** 5 cm **(b)** 6 cm 59. What is PM equal to ? **(c)** IO cm **(d)** 12 cm

414. **[CDS (I) Paper 31 Q62]** A question is given followed by two statements I and **(a)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements the

415. **[CDS (I) Paper 31 Q63]** A question is given followed by two statements I and **(a)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements the

416. **[CDS (I) Paper 31 Q64]** A question is given followed by two statements I and **(a)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using any of the 65. A question is given followed by two statements I and 11 Consider the Question and the Statements and mark the correct option. : If the price of petrol Question goes up by 20%, by what percentage should the consumption be reduced so that the expenditure remains the same ? price of petrol per litre Statement-I was Rs. 90 Consumption was 24 Statement-II . litre before price hike Which one of the following is correct in respect of the above Question and the Statements ?

417. **[CDS (I) Paper 31 Q66]** A question is given followed by two statements I and **(a)** , The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements 67. A question is given followed by two statements I and 11. Consider the Question and the Staternents and mark the correct option. The largest of five Question : different integers is 8 and least is 2. What is the average of these integers ? • The sum of all the Statement-I 5 integers is a multiple of 5 Statement-II : The number of odd integers is odd Which one of the following is correct in respect of the above Question and the Statements ?

418. **[CDS (I) Paper 31 Q67]** A question is given followed by two statements I and **(a)** , The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements 67. A question is given followed by two statements I and 11. Consider the Question and the Staternents and mark the correct option. The largest of five Question : different integers is 8 and least is 2. What is the average of these integers ? • The sum of all the Statement-I 5 integers is a multiple of 5 Statement-II : The number of odd integers is odd Which one of the following is correct in respect of the above Question and the Statements ?

419. **[CDS (I) Paper 31 Q68]** A question is given followed by two statements I and **(a) (b) (c) (d)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question cannot be answered even by using both the Statements together 29

420. **[CDS (I) Paper 31 Q69]** A question is given followed by two statements I and **(a) (b) (c) (d)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question cannot be answered even by using both the Statements together 29

421. **[CDS (I) Paper 31 Q70]** A question is given followed by two statements I and **(a) (b) (c)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question cannot be answered even by using both the Statements together **(d)** 800

422. **[CDS (I) Paper 31 Q11]** Consider the Question and the Statements and mark the correct option. Is p2 +pq+q2 odd Question : where p, q are inte- gers ? Statement-I : p + qis even Statement-II : pq is odd Which one of the following is correct in respect of the above Question and the Statements ? **(a) (b) (c)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question cannot be answered even by using both the Statements together **(d)** 800

423. **[CDS (I) Paper 31 Q78]** If 33 xy and x + z 2y, then what is equal to ? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 77. The average of the temperatures recorded at noontime from Monday to Sunday is 31 oc. Ifthe lowest tempera- ture recorded is 300C, then what is the maximum of temperature that is possible to record at noontime on any one of the days

424. **[CDS (I) Paper 31 Q89]** XIZ is a 3-digit number, where X, Y, Z are distinct non-zero digits. The dif- ference between the two 3-digit numbers XYZ and YXZ is **(a)** 121 **(b)** 120 **(c)** 119 **(d)** None of the above 91. What is the HCF 0f236-1 and 245-1 ?

425. **[CDS (I) Paper 31 Q90]** How many possible values exist for the sum **(a)** 121 **(b)** 120 **(c)** 119 **(d)** None of the above 91. What is the HCF 0f236-1 and 245-1 ?

426. **[CDS (I) Paper 31 Q97]** How many numbers of the form 2n — 1 and less than 2000 are prime ? 41 **(a)** 1100 **(b)** 1200 **(c)** 1300 **(d)** 1400 96. The number 199 can be written as m2 — n2, where m, n are natural numbers

427. **[CDS (I) Paper 31 Q98]** In a class of 160 students, each of them opt at least one language from among English, Hindi and Sanskrit. It is found that 130 students opt English, 120 students Hindi and 110 Sanskrit. If the students opt either only one language or all three languages, then what is the number of students who study all three languages ? **(a)** 40 **(b)** 60 **(c)** 80 **(d)** 100

428. **[CDS (I) Paper 32 Q11]** It is Often found in basaltic rocks. Ill. Magnesium, iron and silica are its maj or elements. Select the answer using the code given below : **(a)** Amphibolite **(b)** Feldspar **(c)** Pyroxene **(d)** Olivine Which Of the following statements with reference to Salt Weathering Ware correct ? Salt crystalli•atinn is the most effective of all the salt weathering processes. Salt crystal growth is more effective over sedimentary rocks as compared to igneous rocks. Select the answer using the code given below : **(a)** I only **(b)** 2 only **(c)** Both I and 2 (d) Neither I nor 2

429. **[CDS (I) Paper 32 Q11]** 12. 18. -A) Which Of the following statements With reference to Easter Island, a UNESCO World Heritage Site, Ware correct ? I. It is part Of Polynesia group Of is lands. **(a) (b) (c)** Which of the following statements With reference to Humus Ware correct ? I. It is a dark coloured amorphous 12. substance. 2. Being colloidal in nature , it senes as a reservoir of nutrients. Select the answer using the code given below : Which Of the following statements With reference to Easter Island, a UNESCO World Heritage Site, Ware correct ? I. It is part Of Polynesia group Of is lands. 2. It is a territory of the South American nation Chile. Select the answer using the code given below : **(d)**

430. **[CDS (I) Paper 32 Q1]** 2. It is a I andlocked country. Select the answer using the code given below : **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

431. **[CDS (I) Paper 32 Q16]** Which one among the following placeg is 17. situated close to the Pearl River Estuary in Asia ? **(a) (b) (c) (d)** Bangkok Hong Kong Macau Singapore Consider the following statements with regard to Australia : I. The Coral Sea separates the Great Barrier Reef from the South Pacific.

432. **[CDS (I) Paper 32 Q2]** The Great Dividing Range extends from Melbourne to Perth in Australia. Which Of the statements given above Ware correct ? **(a) (b) (c) (d)** Bangkok Hong Kong Macau Singapore Which of the following statements With reference to saline soil in India Ware correct ? I. They are rich in nitrogen. 2. They

433. **[CDS (I) Paper 32 Q4]** Karakoram Kunlun Ladakh Zaskar Select the answer using the code given below : 1,2, **(c) (d)** CEBN-S-GKL **(a) (b) (c) (d)** Bangkok Hong Kong Macau Singapore Which of the following statements With reference to saline soil in India Ware correct ? I. They are rich in nitrogen. 2. They

434. **[CDS (I) Paper 32 Q19]** 20. -A) Which of the following statements With reference to saline soil in India Ware correct ? I. They are rich in nitrogen. **(a) (b) (c) (d)** Bangkok Hong Kong Macau Singapore Which of the following statements With reference to saline soil in India Ware correct ? I. They are rich in nitrogen. 2. They

435. **[CDS (I) Paper 32 Q26]** A body floats in a liquid if the buoyant force is : **(a) (b) (c)** 21 m 18 m 36 m **(a) (b) (c) (d)** equal to its weight. less than its weight. greater than its weight. zero. CEBN-S-GKL

436. **[CDS (I) Paper 32 Q33]** 84. Which of the following is one of the 'three gems' (triratna) of Jainism ? **(a) (b) (c) (d)** Brahmacharya Samyag-darshana Sunrita Asteya Which one among the following motifs does not appear on Harappan seals ? **(a) (b) (c) (d)** Humped Bull Elephant Lion Crocodile Which one among the following is the earliest text ? **(a) (b) (c) (d)** Natyashastra Of Bharata Sushruta Samhita Ashtadhyayi Manu-smriti Which one of the following statements about Ashoka' s inscriptions is correct ? **(a) (b) (c) (d)** The Major Rock Edict found in Kerala is in Sanskrit. Minor Rock Edicts in Bihar are found only in Pataliputra. The Dhauli Major Rock Edict makes a mention of the Kalinga War. Kalsi in Uttarakhand is a site of Major Rock Edicts.

437. **[CDS (I) Paper 32 Q29]** 30. (11 -A) Which of the following is one of the 'three gems' (triratna) of Jainism ? **(a) (b) (c) (d)** Brahmacharya Samyag-darshana Sunrita Asteya **(c)** 0.5 N 1-5 N 2-5 N

438. **[CDS (I) Paper 32 Q84]** Converging lens Plane mirror Which one of the following statements about Ashoka' s inscriptions is correct ? The force with which the Earth attracts an Object is called : **(a)** Weight of the obj ect **(b)** Mass of the object **(c)** Density of the object **(d)** Magnitude of the obj ect CEBN-S-GKL **(a) (b) (c) (d)** The Major Rock Edict found in Kerala is in Sanskrit. Minor Rock Edicts in Bihar are found only in Pataliputra. The Dhauli Major Rock Edict makes a mention of the Kalinga War. Kalsi in Uttarakhand is a site of Major Rock Edicts.

439. **[CDS (I) Paper 32 Q3]** The Konarak horse sculpture (c. 1250) shows a ring-stirrup. A broad, probably wooden, stirrup Khajuraho sculptures (probably 10th century). The iron-stirrup was used in the Delhi Sultans' armies from the thirteenth century onw ards. How many of the abo.•e statements Ware correct ? **(a) (c)** 2 3 Nil In the Rig Veda Samhita, use Of the terms ishukrit and isukara referred to the maker Of ; **(a) (b) (c) (d)** Arrows Swords Vajra (Thunderbolt) Maces CEBN-S.GKL

440. **[CDS (I) Paper 32 Q43]** Which one of the following is listed in the Ain-i **(a) (b) (c) (d)** Ayurveda Vyakarana Ganita Jyotisha

441. **[CDS (I) Paper 32 Q44]** Akbari as one of the 'Indian sciences' prescribed by Akbar for scholars to learn at school ? **(a) (b) (c) (d)** Ayurveda Vyakarana Ganita Jyotisha Consider the following statements about the arms used by Europeans in warfare in India

442. **[CDS (I) Paper 32 Q2]** The flint-lock muskets could work even in wet weather. The addition of bayonet made musket a more effective and dangerous weapon in close combat. The wrought-iron cannon barrels were easy to move and mancæuvre. How many of the above statements Ware correct ? **(c)** 3 Nil Which one of the following statements about the grammarian Katyayana is correct ? **(a) (b) (c) (d)** Ayurveda Vyakarana Ganita Jyotisha

443. **[CDS (I) Paper 32 Q48]** 49. 50. (17 -A **(a) (b) (c) (d)** Bhattavatara Jonaraja Shrivara

444. **[CDS (I) Paper 32 Q52]** Vachaspati In which one Of the following works has Amir Khusrau praised the climate Of India, its languages, notably Sanskrit, its arts and music and even its animals ? **(a) (b) (c) (d)** Ijaz •i Khusrawi Dewal Rani Khizr Khani Khazain ul Futuh Nuh Sipihr

445. **[CDS (I) Paper 32 Q53]** Which one of the following works was written in early eighteenth century by the Bengali poet Rameshw ar Bhattacharya ? **(a) (b) (c) (d)** Dharmamangal Shi usankirttan Mansimha Vidyasundar Which one of the following present day territories was outside the limits of Ashoka's **(a) (b) (c) (d)** Brahmaputra valley , Assam Baluchistan North Bengal Kashmir valley CEBN-S-GKL

446. **[CDS (I) Paper 32 Q55]** Isobars Of elements have same : **(a) (b) (c) (d)** Atomic numbers. Number of electrons. Mass numbers. Number Of neutrons. Which one Of the following could not be explained by Rutherford's nuclear model ? **(a) (b) (c)** Stability of the atom Presence of nucleus in the centre Of an Comparison Of sizes of atom and nucleus Nucleus is positively charged If atomic number of an element is 11, the correct name, symbol and valency are : **(a) (b) (c) (d)** Lithium, Li, I Magnesium, Mg, 2 Boron, B, 3 Sodium, Na, I Which one of the following gas is used in making soft drinks ? **(a) (c)** NHa C02 Which one of the following is considered as a compound ? **(a) (c) (d)** Silicon Tin Sugar solution Calcium carbonate Who composed Zainavilas, a history Of the

447. **[CDS (I) Paper 32 Q54]** Dharmamangal Shi usankirttan Mansimha Vidyasundar Which one of the following gas is used in making soft drinks ? **(a) (b) (c) (d)** Bhattavatara Jonaraja Shrivara 52. Vachaspati

448. **[CDS (I) Paper 32 Q61]** 62. 63. -A) Consider the follmving statements with respect to performance of the merchandise export Of India in FY - 2024 : I. Share of capital goods such as m achinery, equipment and Other durable goods decreased substantially. **(a) (b) (c) (d)** Neu tron Electron Proton Atomic nucleus Which one of the following is a metalloid ? Consider the follmving statements with respect to performance of the merchandise export Of India in FY - 2024 : I. Share of capital goods such as machinery, equipment and Other durable goods decreased substantially. 2. Share of consumer goods and intermedi ate goods increased. Which of the statements given above Ware

449. **[CDS (I) Paper 32 Q2]** Share of consumer goods and intermedi ate goods increased. Which of the statements given above Ware **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 Which one among the following statements about Project Nexus is correct ? **(a) (b) (d)** It is an international agreement to enable labour movement between countries. It is an international initiative to enable instant cross-border retail payments. It is an international initiative to transfer technology between countries. It is an international agreement to create database on financial inclusion. While calculating the consumer price index, 'housing' as an item is classified as : **(a) (b) (c) (d)** Core consumer durable Core consumer non-durable Core capital goods Core service

450. **[CDS (I) Paper 32 Q56]** Rutherford's a-particle scattering experiment 61. on gold foil is responsible for the discovery of : **(a) (b) (c) (d)** Neu tron Electron Proton Atomic nucleus Which one of the following is a metalloid ? Consider the follmving statements with respect to performance of the merchandise export Of India in FY - 2024 : I. Share of capital goods such as machinery, equipment and Other durable goods decreased substantially. 2. Share of consumer goods and intermedi ate goods increased. Which of the statements given above Ware

451. **[CDS (I) Paper 32 Q59]** 60. (19-A) **(a) (b) (c) (d)** Neu tron Electron Proton Atomic nucleus Which one of the following is a metalloid ? Consider the follmving statements with respect to performance of the merchandise export Of India in FY - 2024 : I. Share of capital goods such as machinery, equipment and Other durable goods decreased substantially.

452. **[CDS (I) Paper 32 Q62]** 63. Which one among the following statements about Project Nexus is correct ? Which one of the following contains all Lewis acids ? **(a)** BF3 **(b)** BCIa, NHa, OH- **(c)** Mg2+,H20, **(d)** BF3, NH3, H20 Number of electrons present in the H;, He, H2 and 02 respectively are • **(a) (b) (d)** It is an international agreement to enable labour movement between countries. It is an international initiative to enable instant cross-border retail payments. It is an international initiative to transfer technology between countries. It is an international agreement to create database on financial inclusion. species While calculating the consumer price index, 'housing' as an item is classified as : **(a) (b) (d)** 1, 2, 2,15 1, 2, 1, 14 2, 2, 2, 16 0, 2, 1, 14 **(a) (b) (c) (d)** Core consumer durable Core consumer non-durable Core capital goods Core service CEBN-S-GKL

453. **[CDS (I) Paper 32 Q64]** 65. 66. (Global Capability Centre) ? **(a)** (Honeywell) **(b)** (Citibank) **(c)** (Texas Instruments) **(d)** (SAP Labs India) (GVC) (CAD) 1, Gvc **(c)** (Currency depreciation) (net exports) t? 1. **(d)** qi 1, 4t12 CEBN-s-GKL (20-

454. **[CDS (I) Paper 32 Q69]** 70. (20-A) (Core inflation) : **(a) (b) (c) (d)** CPI WPI CPI WPI NEP 2020 (PBPB), **(a) (b)** (flexi workers) t ? **(c)** (NISHTHA) (DIKSHA) t? 1. **(d)** qi 1, 4t12 CEBN-s-GKL

455. **[CDS (I) Paper 32 Q65]** 66. (20-A) (Core inflation) : **(a) (b) (c) (d)** CPI WPI CPI WPI (GVC) (CAD)

456. **[CDS (I) Paper 32 Q65]** 66. (21 -A **(a) (b)** Honeywell Citibank Texas Instruments SAP Labs India **(c) (d)**

457. **[CDS (I) Paper 32 Q73]** 74. 75. (blood pressure) (manually) **(a)** (Sphygmomanometer) **(b)** (Centrifuge) **(c)** (Electropho resis) **(d)** (Thermal cycler) q-rfiqT (Myopia) ? **(c)** (Cataract) **(d)** (Presbyopia) (Endoplasmic Reticulum) (rough)' **(a)** (Gibberellins) **(b)** (Abscisic acid) **(d)** (Thyroxine) **(a)** Hg 80 mm **(b)** Hg 100 mm **(d)** Hg 140 mm CEBN-S-GKL (22-

458. **[CDS (I) Paper 32 Q78]** 79. (Sponges) **(a)** (Porifera) **(b)** (Platyhelminthes) **(c)** (Cnidaria) **(d)** (Annelida) (Lactose) t ? **(b)** (Isotopes) **(c)** (I somers) **(d)** (Coenzymes) 80. (photoreceptor cells) I **(a) (b) (c)** 22-A) (Fovea) fq (Blind spot) (Dark spot) (Orbit) **(a)** (Gibberellins) **(b)** (Abscisic acid) **(d)** (Thyroxine) **(a)** Hg 80 mm **(b)** Hg 100 mm **(d)** Hg 140 mm CEBN-S-GKL 71. (blood pressure) 76. (manually) **(a)** (Sphygmomanometer) **(b)** (Centrifuge) **(c)** (Electropho resis) **(d)** (Thermal cycler) 77. q-rfiqT (Myopia) ? 72. **(c)** (Cataract) **(d)** (Presbyopia)

459. **[CDS (I) Paper 32 Q79]** 80. (Sponges) **(a)** (Porifera) **(b)** (Platyhelminthes) **(c)** (Cnidaria) **(d)** (Annelida) (Lactose) t ? **(a)** (Isoenzymes) **(b)** (Isotopes) **(c)** (I somers) **(d)** (Coenzymes) (photoreceptor cells) I **(a) (b) (c)** (Fovea) (Blind spot) (Dark spot) (Orbit)

460. **[CDS (I) Paper 32 Q71]** Which instrument is used to measure the 76. pressure manually ? **(a) (b) (c) (d)** Sphygmomanometer Centrifuge Electrophoresis Thermal cycler

461. **[CDS (I) Paper 32 Q74]** 75. (23 -A: **(a) (b) (c) (d)** Sphygmomanometer Centrifuge Electrophoresis Thermal cycler Myopia is also known as 77. **(a) (b) (c)** Fapsightedness Near-sightedness Cataract Presbyopia Why does the Rough Endoplasmic Reticulum look 'rough' under a micrcvscope ? **(a) (b) (c) (d)** Due to the presenc* Of ribosomes on its Due to the absence Of ribosomes on its Due to the presence Of lysosomes on its Due to the presence of mitochondria on its surface Which one among the following is not a plant horrnone ? **(a) (b) (c) (d)** Gibbere Ilins Abscisic acid Auxins Thyroxine The normal systolic pressure in human beings 80. **(a) (b) (c) (d)** 80 mm Of Hg 100 mm of Hg 120 mm of Hg 140 mm of Hg CEBN-S-GKL

462. **[CDS (I) Paper 32 Q78]** 79. 80. -A) Sponges belong to which one among the following phylum ? **(a) (b) (c) (d)** Porifera Platyh elminthes Cnidaria Annelida Lactose is made up of : **(a) (b) (c) (d)** Glucose only Glucose and Fructose Glucose and Sucrose Glucose and Galactose The enzymes which catalyze the same reaction but have different amino acid compositions are commonly referred as : **(a) (b) (c) (d)** Isoenzymes Isomers Coenzymes How many Carbon atoms are present in a Pyru-v•ate molecule ? **(a) (b) (d)** The 2 3 4 5 human eye has a point that lacks photoreceptor cells. This is identified as : **(a) (b)** (e) Fovea Blind spot Dark spot Orbit

463. **[CDS (I) Paper 32 Q73]** 74. 75. (23-A) Which instrument is used to measure the pressure manually ? Sponges belong to which one among the following phylum ? **(a) (b) (c) (d)** Sphygmomanometer Centrifuge Electrophoresis Thermal cycler **(a) (b) (c) (d)** Porifera Platyh elminthes Cnidaria Annelida Myopia is also known as Lactose is made up of : **(a) (b) (c)** Fapsightedness Near-sightedness Cataract Presbyopia **(a) (b) (c) (d)** Glucose only Glucose and Fructose Glucose and Sucrose Glucose and Galactose Why does the Rough Endoplasmic Reticulum look 'rough' under a micrcvscope ? **(a) (b) (c) (d)** Due to the presenc* Of ribosomes on its Due to the absence Of ribosomes on its Due to the presence Of lysosomes on its Due to the presence of mitochondria on its surface

464. **[CDS (I) Paper 32 Q80]** The enzymes which catalyze the same reaction but have different amino acid compositions are commonly referred as : **(a) (b) (c) (d)** Isoenzymes Isomers Coenzymes Which one among the following is not a plant horrnone ? How many Carbon atoms are present in a Pyru-v•ate molecule ? **(a) (b) (c) (d)** Gibbere Ilins Abscisic acid Auxins Thyroxine The normal systolic pressure in human beings **(a) (b) (d)** The 2 3 4 5 human eye has a point that lacks photoreceptor cells. This is identified as : **(a) (b) (c) (d)** 80 mm Of Hg 100 mm of Hg 120 mm of Hg 140 mm of Hg **(a) (b)** (e) Fovea Blind spot Dark spot Orbit CEBN-S-GKL

465. **[CDS (I) Paper 32 Q3]** Central Vigilance Commissioner is appointed by the President of India. The Committee to recommend the appointment of Central Vigilance Commissioner has the Minister of Home Affairs as one Of the members. In certain cir•cumstanees the Leader Of the single largest group in opposition in Lok Sabha can be on the Committee to recommend the appointment of Central Vigilance Commissioner. How many of the above statements Ware correct ? **(c) (d)** Nil 2 3 CEBN-S-GKL **(a) (b) (c) (d)** Constitution

466. **[CDS (I) Paper 32 Q84]** International Day Of Democracy, as decided vide a UN General Assembly resolution, is observed on : **(a) (b) (c) (d)** Constitution

467. **[CDS (I) Paper 32 Q2]** Prime Minister of India is the Chairperson of its Governing Council. Lieutenant Governor Of Ladakh is a member of its Governing Council. Select the answer using the code given below ; **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 Which of the following statements about National Commission for Women are correct ?

468. **[CDS (I) Paper 32 Q85]** Constitution (One Hundred and Sixth Amendment) Act **(a) (b) (c) (d)** Constitution

469. **[CDS (I) Paper 32 Q86]** Which of the following statements about National Commission for Women are correct ? 3. appointed by the President of India. The Committee to recommend the appointment of Central Vigilance Commissioner has the Minister of Home Affairs as one Of the members. In certain cir•cumstanees the Leader Of the single largest group in opposition in Lok Sabha can be on the Committee to recommend the appointment of Central Vigilance Commissioner. **(a) (b) (c) (d)** Constitution

470. [CDS (I) Paper 32 Q88] 89. (27 -A **(a) (b) (c) (d)** Maithili N epali Konkani Manipuri Which one among the following states has the highest number of seats in the Council Of States ? **(a) (b) (c) (d)** Andhra Pradesh Bihar Madhya Pradesh Tamil Nadu Which Of the following statements about the Indus Waters Treaty Ware correct ? The term 'Eastern Rivers' means the Sutlej, the Beas and the Ravi taken together. The term 'Western Rivers' means the Indus, the Jhelum and the Chenab taken Select the answer using the code given below : **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 CEBN-S-GKL

471. [CDS (I) Paper 32 Q90] 91. -A) Which of the following statements Ware correct as the Constitution of India ? The Constitution (Eighty -sixth Amendment) Act has added the Right to Education to the chapter of Directive Principles of State policy. Right to Work and Right to Education are both, Fundamental Rights and Dir•ective Principles of State Policy. Select the answer using the code given below : **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 As defined in Article 12 of the Constitution Of India, 'State' includes :

472. [CDS (I) Paper 32 Q4] Parliament Of India State Governme nts State Legislatures Local authorities in India Select the answer using the code given below : **(a) (b) (c) (d)** I and 3 only I, 2 and 3 only I, 2, 3 and 4 2 and 4 only

473. [CDS (I) Paper 32 Q91] Which Of the following statements about the Indus Waters Treaty Ware correct ? The term 'Eastern Rivers' means the Sutlej, the Beas and the Ravi taken together. The term 'Western Rivers' means the Indus, the Jhelum and the Chenab taken Select the answer using the code given below : As defined in Article 12 of the Constitution Of India, 'State' includes : **(a) (b) (c) (d)** Maithili N epali Konkani Manipuri Which of the following statements Ware correct as the Constitution of India ? The Constitution

474. [CDS (I) Paper 32 Q92] Which Of the following descriptions With regard 95. to Parliamentary Privilege for Members of the Parliament of India Ware correct ? **(a) (b) (c)** Cyril Radcliffe Mortimer Durand Henry McM ahon Arthur Belfour Select the answer using the eode given below : **(d)**

475. [CDS (I) Paper 32 Q93] 94. (29-A) I. Speaker is empowered to refer Suo moto any question Of privilege for investigation A question of privilege is accorded priority over other items in the List Of Business The privilege extends to freedom from preventive arrest or detention Select the answer using the eode given below : **(a) (b) (c) (d)** I only 2 and 3 only I and 2 only 1, 2 and 3 Which one among the following statements about an adjournment motion in the Parliament Of India is not correct ? **(a) (b) (c) (d)** Its objective is to draw attention of the House to any recent matter of urgent public importance. Adjournment motion is an extraordinary This motion can be invoked on any matter Of urgent importance that may even be within the jurisdiction of a State Gov ernment. The Chair has absolute discretion to refuse consent.

476. [CDS (I) Paper 32 Q97] The Kyoto Protocol specifically relates to which one among the following issues ? **(a) (b) (c)** Cyril Radcliffe Mortimer Durand Henry McM ahon Arthur Belfour Select the answer using the eode given below : **(d)**

477. [CDS (I) Paper 32 Q96] 97. Which one among the following statements about Dr. Muthulakshmi Reddy, a freedom fighter , is not Correct ? Which one among the following statements about an adjournment motion in the Parliament Of India is not correct ? **(a) (b) (c) (d)** She was the first girl student to be admitted into a men's college. She was the first woman legislator in British India. She was a medical professional. She was honoured With Bharat Ratna in 1956. **(a) (b) (c) (d)** Its objective is to draw attention of the House to any recent matter of urgent public importance. Adjournment motion is an extraordinary This motion can be invoked on any matter Of urgent importance that may even be within the jurisdiction of a State Gov ernment. The Chair has absolute discretion to refuse consent. "Since wars begin in the minds of men, it is in the minds of men that the defences of peace must be constructed." This observation can be found in the Constitution of which one Of the following Organs Of the United Nations ? The Kyoto Protocol specifically relates to which one among the following issues ? **(a) (b) (d)** Emission reduction Ozone layer depletion Chlorofluorocarbons Biodiversity management **(a) (b) (c) (d)** World Health Organisation United Nations Educational, Scientific and Cultural Organisation International Atomic Energy Agency United Nations Economic and Social Co uncil CEBN-S-GKL

478. [CDS (I) Paper 32 Q2] 1860 2023 I **(b) (d)** 7å1,7tt2 **(b)** CEBN-S-GKL (30 101. (firewall) **(a)** $ZüZ 102. ? **(a)** (Spreadsheet) **(c)** (MS PowerPoint) (d) (Linker) 103. *-qT, (Malicious software) ? **(b)** (Password Cracking) **(c)** (Trojan Horse) **(d)** (Firewall)

479. [CDS (I) Paper 32 Q2] 3. Moidam-The Mound-Burial system of the Ahom dynasty was inscribed in the UNESCO World Heritage lisu The Government Of India conferred the status of classical language to Pali, Assamese, Bengali and Maithili. Project PARI is an initiative Of Ministry of Education Select the answer using the code given below : **(a)** 2 only **(b)** I only **(c)** 2 and 3 only **(d)** 1,2and3

480. [CDS (I) Paper 33 Q11] Which of the above are correct? **(a)** I and 11 only **(b)** 11 and Ill only **(c)** I and Ill only **(d)** 1, 11 and 111

481. [CDS (I) Paper 33 Q3] Ixt p and q be natural such that q > p. What is the largest value Of p such that q2 — 5p —4 is negative? ANFB-T-EMT/45A **(a)** 3-2* Consider the following in Of a 2. positive real number x 11. 4. 0. 7. Let x each How and y natural numbers, less than 20, such that x , y, and x — y are prime numbers. many such combinations Of x + y, x — y **(b)** None If **(c)** I and 11 only **(d)** 11 and Ill only

482. [CDS (I) Paper 33 Q14] If P+q then Which one Of the following is correct? **(a) (b)** Either p O or p None Of the above 15. If n is natural numirr less than 7, then What is the number Of values Of n for which **(c) (d)** ANFB-T-EMT/4SA 7 17. Let x = n

483. **[CDS (I) Paper 33 Q15]** If n is natural numirr less than 7, then What is the number Of values Of n for which (12m + '2) and (8n + I) are relatively prime ? **(a)** **(b)** Either p O or p None Of the above 15. If n is natural numirr less than 7, then What is the number Of values Of n for which **(c)** **(d)** ANFB-T-EMT/4SA 7 17. Let x = n

484. **[CDS (I) Paper 33 Q21]** What is c2 + bc—a2 —ab equal to? **(a)** 12 21. What is c2 + bc—a2 —ab equal to? **(b)** -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ? **(c)** 2p **(d)** then what is then what is x 22. If ab ba , then what is equal to?

485. **[CDS (I) Paper 33 Q22]** If ab ba , then what is equal to? **(d)** ab ANFB-T-EMT/45A 9 n. If x-2+21/2 +23/2 x2 —4x— IO equal to? **(a)** 12 21. What is c2 + bc—a2 —ab equal to? **(b)** -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ? **(c)** 2p **(d)** then what is then what is x 22. If ab ba , then what is equal to?

486. **[CDS (I) Paper 33 Q24]** If equal to ? 2p 2p then what is then what is x a—b 2 a+b x + b O, then what **(a)** 12 21. What is c2 + bc—a2 —ab equal to? **(b)** -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ? **(c)** 2p **(d)** then what is then what is x 22. If ab ba , then what is equal to?

487. **[CDS (I) Paper 33 Q26]** Ifx-—- t is x 2, x > O; then **(a)** 40 If **(b)** 18 **(c)** 322 **(d)** 72VS ANFB-T-EMT A number N is such that when divided by 4, 6, 7 or 9, it leaves 3 as remainder. What is the smallest 4-digit number that satisfies this property?

488. **[CDS (I) Paper 33 Q27]** 28. equal to? **(b)** 40 If (a— (b — (c— a) 2 = 6 and a2 + b2 + c2 29, then What is c) equal to? , then What and q = If p ,rs-2 • s equal to? **(a)** 18 **(c)** 322 **(d)** 72VS ANFB-T-EMT

489. **[CDS (I) Paper 33 Q29]** What is the digit at hundreds place Of the number ? A number N is such that when divided by **(a)** 40 If **(b)** 18 **(c)** 322 **(d)** 72VS ANFB-T-EMT A number N is such that when divided by 4, 6, 7 or 9, it leaves 3 as remainder. What is the smallest 4-digit number that satisfies this property?

490. **[CDS (I) Paper 33 Q33]** If 2 tane=sec29 —2, where R 12, then what is cote equal to? **(c)** v'ä-l **(d)** 0+2 **(a)** 3/2 5/3 33. If 2 tane=sec29 —2, where R 12, then what is cote equal to? **(b)** v'ä-l **(c)** 0+2 I — sin 34. What is **(d)** mane

491. **[CDS (I) Paper 33 Q38]** If x4 + y4 14x2 , then consider the I. loglO(x 2 +) loglo x *loglo y t 210g10 2 **(a)** 11 only **(b)** Both I and 11 **(c)** Neither I nor 11 39. Which Of the is/ are the Of **(d)** Both I and 11

492. **[CDS (I) Paper 33 Q11]** loglO(x2 — ya) —loglo x + loglo y Which of the above is/are correct? **(a)** I only **(b)** 11 only **(c)** Both 1 and 11 **(d)** Neither I nor 11

493. **[CDS (I) Paper 33 Q1]** 13y) n. (x+19y) Select the correct answer using the code given below. **(a)** I only **(b)** 11 only **(c)** Both 1 and 11 **(d)** Neither I nor 11 ANFB-T-EMT/4SA

494. **[CDS (I) Paper 33 Q40]** What is +3 equal to? **(b)** 1/2 For the tvo items : Let psin2 +qcos2 a qsin2 pcos2 ß=n; p m, n and q m. n. **(a)** 11 only **(b)** Both I and 11 **(c)** Neither I nor 11 39. Which Of the is/ are the Of **(d)** Both I and 11

495. **[CDS (I) Paper 33 Q42]** If a and are complementary angles, then which one Of the following is **(a)** mn — I O For the following two **(b)** 2/3 For the following tvo **(c)** 1 P.T.O. For the following two **(d)** 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

496. **[CDS (I) Paper 33 Q43]** What is (p sine + qcos9)equal to? **(a)** mn — I O For the following two **(b)** 2/3 For the following tvo **(c)** 1 P.T.O. For the following two **(d)** 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

497. **[CDS (I) Paper 33 Q49]** What is the relation between p and q? **(a)** p q **(b)** 2p q **(c)** 2q p — I **(d)** 80 m 100m 120m 150 m 19 52. TWO persons X and Y leave place P for place Q at 7:00 a.m. and 7:10 a.m. respectively along the same path. X walks at a speed of 4 •8 km/hr and Y walks at a speed Of 6 km/hr. HOW many kilometres from place p Will X meet Y ?

498. **[CDS (I) Paper 33 Q57]** If one root Of the equation 2x2 — 5pxt2p2 O exceeds the Other by 4, then What is the value of p ? **(a)** py/x SS. X and Y are two alloys Of copper **(b)** 8/3 **(c)** 2/3 **(d)** 1/3 An amount Of 10,000 is borrowed at 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount Of each installment

499. **[CDS (I) Paper 33 Q59]** What is the solution Of the inequalities 5x + 3 8x— 9 and 2x +20 > 5x 2? 1 P.T.O. **(a)** py/x SS. X and Y are two alloys Of copper **(b)** 8/3 **(c)** 2/3 **(d)** 1/3 An amount Of 10,000 is borrowed at 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount Of each installment

500. **[CDS (I) Paper 33 Q66]** TWO poles Of I O m and 15 m are 25 m apart. What is the height of the point of intersection Of the lines joining the tip of each pole to the foot of the Other pole? **(c)** **(d)** 6-4 m **(a)** **(b)** 6-4 m 67. ABC is a triangle right anøed at B. Further , **(c)** **(d)** 75 units 100 square units 125 square units 150 units 68. TWO poles are situated 24 m apart and their heights differ by I O m. What is the distance between their tips?

501. **[CDS (I) Paper 33 Q74]** An angle is exactly one-fourth of its complementary angle. What is the value of angle B ? **(b)** 15 200 76. **(a)** 15 200 73. **(b)** 10 **(c)** 11 **(d)** 12 71. In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ?

502. **[CDS (I) Paper 33 Q81]** A Question is given followed by two Statements I and **(a) (b) (c)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any Of the Statements ANFB-T-EMT/45A **(d)**

503. **[CDS (I) Paper 33 Q82]** A Question is given followed by two Statements I and **(a) (b) (c)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any Of the Statements ANFB-T-EMT/45A **(d)**

504. **[CDS (I) Paper 33 Q83]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using One of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be an swered by using tx'th the Statements together, but cannot be answered using either Statement alone **(d)** The Question Can be answered even without using any of the ANFB-T-EMT/45A 83. ZB- ZC 84. A Question is given followed by Statements I and 11. Consider Question and the Statements. In a triangle ABC, ZA Is angle A acute? two the Statement—I ABC is Statement—II : not an Obtu se-angled Angle C is acute. Which One of the following is correct in respect of the above Question and the Statements?

505. **[CDS (I) Paper 33 Q3]** Which one Of the following is correct in respect Of the above Question and the Statements? **(a)** The Question can be answered by using One of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be an swered by using tx'th the Statements together, but cannot be answered using either Statement alone **(d)** The Question Can be answered even without using any of the ANFB-T-EMT/45A

506. **[CDS (I) Paper 33 Q84]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using One of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be an swered by using tx'th the Statements together, but cannot be answered using either Statement alone **(d)** The Question Can be answered even without using any of the ANFB-T-EMT/45A 83. ZB- ZC 84. A Question is given followed by Statements I and 11. Consider Question and the Statements. In a triangle ABC, ZA Is angle A acute? two the Statement—I ABC is Statement—II : not an Obtu se-angled Angle C is acute. Which One of the following is correct in respect of the above Question and the Statements?

507. **[CDS (I) Paper 33 Q85]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(b) (c) (d)**

508. **[CDS (I) Paper 33 Q86]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(b) (c) (d)**

509. **[CDS (I) Paper 33 Q87]** A is øven by two Statements I and **(a)** The Question can be answered by using One of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question Can be answered by using either Statement alone **(c)** The Question can be an swered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can answered even without using any of the Statements ANPB-T-EMT/4SA 88. A Question is given followed by two Statements I and 11. Consider the Question and the Statements. In a quadrilateral ABCD, AB -6 units, BC- 18 units, CD 6 units, DA 9 units. What is the length Of diagonal BD ? Statement—I : The length Of BD is an integer greater than 13. Statement—II : The length Of BD is an even integer. Which one Of the following is correct in respect Of the above Question and the Statements ?

510. **[CDS (I) Paper 33 Q88]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using One of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question Can be answered by using either Statement alone **(c)** The Question can be an swered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can answered even without using any of the Statements ANPB-T-EMT/4SA 88. A Question is given followed by two Statements I and 11. Consider the Question and the Statements. In a quadrilateral ABCD, AB -6 units, BC- 18 units, CD 6 units, DA 9 units. What is the length Of diagonal BD ? Statement—I : The length Of BD is an integer greater than 13. Statement—II : The length Of BD is an even integer. Which one Of the following is correct in respect Of the above Question and the Statements ?

511. **[CDS (I) Paper 33 Q89]** A Question is given followed by two Statements I and **(a) (b)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(c)** ANFB-T-EMT/45A The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question Can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even Without using any of the State ments I P.T.O. 90. A Question is given followed by two Statements I and 11. Consider the Question and the Statements. Question The diagonals of a rhombus ABCD are in the ratio 5:12. Is one Of the diagonals equal to side of the rhombu s? Statement—I : The sum Of the diagonals = 34 cm. Statement—ll . The length of a side 13 cm Which one of the following is correct in respect Of the above Question and the Sta tement s ? @ The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone **(d)** The Question can be answered by using either Statement alone

512. **[CDS (I) Paper 33 Q90]** A Question is given followed by two Statements I and **(a) (b)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(c)** ANFB-T-EMT/45A The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question Can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even Without using any of the State ments I P.T.O. 90. A Question is given followed by two Statements I and 11. Consider the Question and the Statements. Question The diagonals of a rhombus ABCD are in the ratio 5:12. Is one Of the diagonals equal to side of the rhombu s? Statement—I : The sum Of the diagonals = 34 cm. Statement—ll . The length of a side 13 cm Which one of the following is correct in respect Of the above Question and the Sta tement s ? @ The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone **(d)** The Question can be answered by using either Statement alone

513. **[CDS (I) Paper 33 Q12]** Is one Of the diagonals equal to side of the rhombu s? Statement—I : The sum Of the diagonals — 34 cm. Statement—ll . The length of a side 13 cm Which one of the following is correct in respect Of the above Question and the Sta tement s ? **(b) (c)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(d)** ANFB-T-EMT/45A The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question Can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even Without using any of the State ments I P.T.O. **(a) (b)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(c)** ANFB-T-EMT/45A The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question Can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even Without using any of the State ments I P.T.O. 90. A Question is given followed by two Statements I and 11. Consider the Question and the Statements. Question The diagonals of a rhombus ABCD are in the ratio 5:12. Is one Of the diagonals equal to side of the rhombu s? Statement—I : The sum Of the diagonals = 34 cm. Statement—ll . The length of a side 13 cm Which one of the following is correct in respect Of the above Question and the Sta tement s ? @ The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone **(d)** The Question can be answered by using either Statement alone

514. **[CDS (I) Paper 34 Q1]** Ifp+q+r=0, then what is equal to? **(c)** 22 **(d)** 23 **(a)** -23 **(b)** -17 **(c)** 17 **(d)** 23 which —5k + 4

515. **[CDS (I) Paper 34 Q2]** What is the value of k for which (k2— 5k + 4)x2 + (k2 —3k —4)x+ (k2— 4k) 0 is an identity ? 152 **(a)** -23 **(b)** -17 **(c)** 17 **(d)** 23 which —5k + 4

516. **[CDS (I) Paper 34 Q3]** If then what is the value of a4 + b4 + ca equal to ? **(a)** a2b2 + + c2a2 **(b)** 2(a2b2 c2a2) **(c)** 3(a2b2+ Fc2 + c2a2) **(d)** 4(a2b2 b2c2 + c2a2) 3

517. **[CDS (I) Paper 34 Q4]** If—-L — + , then what is pq x + p x equal to ? Fi¯i ¯ —q x—p x —q **(a)** -23 **(b)** -17 **(c)** 17 **(d)** 23 which —5k + 4

518. **[CDS (I) Paper 34 Q5]** Let P 55 x 1515 x 2525 x 3535 and 1010 x 4040. What is the number of consecutive zeros at the end of the sum (P + Q) ? **(a)** 100 **(b)** 65 **(c)** 50 **(d)** Zero

519. **[CDS (I) Paper 34 Q8]** Which of the following expressions can divide both the polynomials x3+2x2—5x+2 and x3 +4x2+x—6 exactly ? **(a)** x4 — 2a2x2 + a4 **(b)** x4 + 2a2x2 + a4 **(c)** xa — a2x2 4 a4 8. Which of the following expressions can divide both the polynomials x3+2x2—5x+2 and x3 +4x2+x—6 exactly ? 11. X+l 111. X+2 Select the correct answer using the code given below : **(d)** I and 11 only

520. **[CDS (I) Paper 34 Q10]** For what integral value of k, the system of equations kx— 5y + 6 = O and 12K +3=0 has no solution ? **(d)** No such value exists **(a)** x4 — 2a2x2 + a4 **(b)** x4 + 2a2x2 + a4 **(c)** xa — a2x2 4 a4 8. Which of the following expressions can divide both the polynomials x3+2x2—5x+2 and x3 +4x2+x—6 exactly ? 11. X+l 111. X+2 Select the correct answer using the code given below : **(d)** I and 11 only

521. **[CDS (I) Paper 34 Q13]** If x varies directly as y, then which of the following is/ are correct ? I. (x2 + y2) varies directly as ya. varies directly as y. Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11

522. **[CDS (I) Paper 34 Q2]** How many values of k are possible if the ages are expressed in complete years ? **(d)** More than 3 **(a)** 10 minutes **(b)** 12 minutes **(c)** 1 S minutes **(d)** 20 minutes 18. A person goes downstream at x km/h and upstream at y km/h by the same boat. What is the ratio of speed of boat in still water to speed of •water ? Y 19. Let Q=x2 + bx + c. If the sum of the roots is equal to product Of the roots of the equation Q = 0, then which of the following statements is/are correct ? 20. 21. 11. Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below :

523. **[CDS (I) Paper 34 Q27]** What is the sum of all the divisors of 256 ? **(a)** 511 **(b)** 510 **(c)** 256 **(d)** 255 A — KPRS-T-MTE

524. **[CDS (I) Paper 34 Q11]** If m and n are relatively prime, then m x n is always even. Which of the statements given above is/are correct ? **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11

525. **[CDS (I) Paper 34 Q29]** If and then what is log10(384) equal to ? **(b)** p—q+r **(d)** p+q—r **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11 29. If and then what is log10

526. **[CDS (I) Paper 34 Q30]** What is the digit in the unit place of the number 11 x 33 x 55 x 77 x 99 ? 13 **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11 29. If and then what is log10

527. **[CDS (I) Paper 34 Q4]** What is (x + y) : (x —y) equal to ? **(d)** Cannot be determined due to insufficient data **(a)** 100 **(b)** 81 **(c)** 64 **(d)** 49 35. Two numbers P and Q are such that . P : Q —x: y. If 1 is added to both the numbers, the ratio becomes 7 : IO. If 3 is added to both the numbers, the ratio becomes 3 : 4. What is

528. **[CDS (I) Paper 34 Q39]** The sum of the deviations of a set of n numbers Xl, x2, x3 , x meas- ured from 15 is —90 and the sum of the deviations of the same numbers measured from —3 is **(a)** 3-25 **(b)** 3-50 **(c)** 3-75 **(d)** 4-25 40. The numbers 59, 53, 51, 43, 36,

529. **[CDS (I) Paper 34 Q25]** What is/are the values ofx? a) —12 only **(b)** 4 only **(c)** —12, 4 **(d)** 12, -4

530. **[CDS (I) Paper 34 Q43]** If 3sine + 4cos9 —5, then what is a value of 4tan0 + 3cot0 ? **(a)** 3-25 **(b)** 3-50 **(c)** 3-75 **(d)** 4-25 40. The numbers 59, 53, 51, 43, 36,

531. **[CDS (I) Paper 34 Q11]** Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

532. **[CDS (I) Paper 34 Q11]** ZPAD+ ZPCB Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both landll **(d)** Neither I nor 11 A - KPRS-T-MTE

533. **[CDS (I) Paper 34 Q71]** A Question is given followed by two Statements I and **(a)** 60 square units **(b)** 62 square units **(c)** 64 square units **(d)** Cannot be determined due to insufficient data 71. A Question is given followed by two Statements I and 11. Consider the Question and the Statements and mark the correct option. 72.

534. **[CDS (I) Paper 34 Q72]** A Question is given followed by two Statements I and **(a)** 60 square units **(b)** 62 square units **(c)** 64 square units **(d)** Cannot be determined due to insufficient data 71. A Question is given followed by two Statements I and 11. Consider the Question and the Statements and mark the correct option. 72.

535. **[CDS (I) Paper 34 Q73]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Staternent alone **(d)** The Question can be answered even without using both the Statements Which one of the following is correct in respect of the above Question and the Statements ?

536. **[CDS (I) Paper 34 Q74]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Staternent alone **(d)** The Question can be answered even without using both the Statements Which one of the following is correct in respect of the above Question and the Statements ?

537. **[CDS (I) Paper 34 Q75]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements together Which one of the following is correct in respect of the above Question and the Statements ?

538. **[CDS (I) Paper 34 Q76]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements together Which one of the following is correct in respect of the above Question and the Statements ?

539. **[CDS (I) Paper 34 Q77]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements

540. **[CDS (I) Paper 34 Q78]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements

541. **[CDS (I) Paper 34 Q79]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using both the 80. A Question is given followed by two Statements I and 11. Consider the Question and the Statements and mark the correct option. Question : Let XYZ be a 3-digit number and the dif- ference between XYZ and ZYX is equal to PQR. Is

542. **[CDS (I) Paper 34 Q80]** A Question is given followed by two Statements I and **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using both the 80. A Question is given followed by two Statements I and 11. Consider the Question and the Statements and mark the correct option. Question : Let XYZ be a 3-digit number and the dif- ference between XYZ and ZYX is equal to PQR. Is

543. **[CDS (I) Paper 34 Q81]** What is the value of p ? **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can be answered even without using both the 80. A Question is given followed by two Statements I and 11. Consider the Question and the Statements and mark the correct option. Question : Let XYZ be a 3-digit number and the dif- ference between XYZ and ZYX is equal to PQR. Is

544. **[CDS (I) Paper 34 Q82]** What is the value of q ? next two (02) items that For the follow : 16 + **(a)** 487 **(b)** 483 **(c)** 481 **(d)** 480 85. What

545. **[CDS (I) Paper 34 Q84]** What is 65 8 61 8 59 For the follow : -257-0 equal to ? equal to ? next two (02) items that and y = Let x 39 **(a)** 487 **(b)** 483 **(c)** 481 **(d)** 480 85. What

546. **[CDS (I) Paper 34 Q92]** What is a value of (cosecO + secO) ? 35 12 25 12 For the next two (02) items that follow : cosece — sino = p' and seco — cose = q3 **(a)** 1/2 **(b)** 1/√2 **(c)** √3/2 **(d)** 1

547. **[CDS (I) Paper 34 Q96]** What is cot2ß equal to ? p For the next two (02) items that follow : p + q coto = 3cosec0 and q — p cote 2cosec9 **(a)** 96. What is cot2ß equal to ? p For the next two **(b)** 10 **(c)** 13 For the next two **(d)** 2pq — —

548. **[CDS (I) Paper 34 Q97]** What is p2 + q2 equal to ? **(d)** 13 **(c)** 10 43 **(a)** 96. What is cot2ß equal to ? p For the next two **(b)** 10 **(c)** 13 For the next two **(d)** 2pq — —

549. **[CDS (I) Paper 34 Q99]** Which one Of the following is correct ? **(a)** 96. What is cot2ß equal to ? p For the next two **(b)** 10 **(c)** 13 For the next two **(d)** 2pq — —

550. **[CDS (I) Paper 34 Q100]** What is 2 17 4 9 2 21 4 equal to ? A - KPRS-T-MTE **(b)** **(a)** 96. What is cot2ß equal to ? p For the next two **(b)** 10 **(c)** 13 For the next two **(d)** 2pq — —

551. **[CDS (I) Paper 35 Q4]** 5. 6. **(a) (b)** p q 2. 3. 4. 5. 6. Suppose p and q are positive integers where q is not a perfect square. If is a root of a quadratic equation having integer coemcients, then which Of the following statements Ware correct ? I. The sum of the roots is 2p times the product Of the roots. IL The sum Of the reciprocals Of the roots is equal to 2p. Select the answer using the code given below : data q = k, then what is **(c) (d)**

552. **[CDS (I) Paper 35 Q3]** How many parts Of X and Y must be taken to obtain a new alloy Z consisting of gold and silver in the rati04:7? **(a) (b) (c) (d)** 960 gm 950 gm 925 gm 900 gm

553. **[CDS (I) Paper 35 Q33]** I-A) Consider the follmving statements : I. The set Of all birds living on the Earth is an infinite set. **(a) (b) (c)** 6% 8% 31. 32. 33. Consider the follmving statements : I. The set Of all birds living on the Earth is an infinite set. 11. The set Of all re al numbers between O and IO is a finite set. Which Of the statements given above Ware correct ? **(d)**

554. **[CDS (I) Paper 35 Q61]** A Question is given followed by two Statements I and **(a) (b) (c)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement The Question can be answered by using either Statement alone. The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone. The Question can be even without using both the Statements. **(d)**

555. **[CDS (I) Paper 35 Q62]** A Question is given followed by two Statements I and **(a) (b) (c)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement The Question can be answered by using either Statement alone. The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone. The Question can be even without using both the Statements. **(d)**

556. **[CDS (I) Paper 35 Q4]** Statement 11 BC = 15 cm. Which one Of the following is correct in respect Of the above Question and Statements ? (23-A **(a) (b) (c) (d)** The Question can be answered by using one of the Statements alone, but cannot be answered using the Other Statement The Question can be answered by using either Statement alone. The Question can be answered by using both the Statements together, but cannot answered using either Statement alone. The Question can be answered even without using both the Statements. BFVS-T-TME

557. **[CDS (I) Paper 35 Q64]** A Question is given followed by two Statements I and **(a) (b) (c) (d)** The Question can be answered by using one of the Statements alone, but cannot be answered using the Other Statement The Question can be answered by using either Statement alone. The Question can be answered by using both the Statements together, but cannot answered using either Statement alone. The Question can be answered even without using both the Statements.

558. **[CDS (I) Paper 35 Q65]** 66. (25-A) A Question is given followed by two Statements I and **(a)** AB:BC **(b)** AB2 Consider the following statements : 11. The areas Of two similar triangles are in the ratio Of the squares of the corresponding medians. The areas of two similar triangles are in the ratio of the squares Of the corresponding heights. **(c) (d)**

559. **[CDS (I) Paper 35 Q91]** 92. 0— 10 10-20 20-30 30-40 **(a) (b)** 8-5 6-5 5-5 **(c)** The average weekly wages of male employees in a company is 4,200 and that Of females is 3,200. If the average weekly wage of all employees is 4,000, then What is the ratio of male to female employees ? **(d)**

CHAPTER 07

Total Archive Questions: 51 Questions

QUADRATIC EQUATIONS, ROOTS & PROGRESSIONS

Typical Exam Weightage: 5 - 8 Questions (5 - 8 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q13] 14. 15. = 12, then what is the value **(a)** I only **(b)** 2 only **(d)** O or I If x —I + •Æ, then what is the value of x' — 4x3 + ? What is the magnitude of difference Of the roots of x2 — ax + b = O ? a2 —4b What is the solution Of the equations x -y— and ll(x + **(a)** x = and y **(b)** x I and y **(c)** x = 2 and y = I •I **(d)** x = I •2 and y = Pooja started her job With certain monthly salary and gets a fixed increment every year. If her salary was Rs. 4200 after 3 years and Rs. 6800 aner 8 years of service, then what are her initial salary and the annual increment respectively ? **(a)** **(b)** **(c)** **(d)** 2640, 320 2460, 320 2460, 520 2640, 520 4

2. [CDS (I) Paper 2 Q95] 96. What is the value of **(a)** **(b)** **(c)** **(d)** 1/2 2 loglo 13 There are four numbers forming a GP which the third term is greater than the first by 9 and the second term is greater than fourth by

3. [CDS (I) Paper 3 Q90] If one of the roots of the equation ax2 is —1-5, then what is the value Of a ? **(a)** 264 kg **(b)** 132 kg **(c)** 396 kg **(d)** None of the above 87. Consider the . following statements in respect of a quadrilateral - I. The line segments joining the midpoints of the two pairs of opposite sides bisect each other at the point of intersection. 2. The area of the quadrilateral formed by joining the midpoints of the four adjacent sides is half of the total area of the quadrilateral. Which of the statements given above is/are correct ?

4. [CDS (I) Paper 3 Q94] When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the following is correct ? **(a)** IO cm 5 cm **(b)** cm **(c)** 56 94. When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the 2 2 97. IA there three simultaneous linear equations in two unknowns. What can be the number of solutions **(d)** One or infinite

5. [CDS (I) Paper 3 Q95] What are the roots of the equation 45) -2? 11, —5 **(d)** -11, 5 **(a)** IO cm 5 cm **(b)** cm **(c)** 56 94. When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the 2 2 97. IA there three simultaneous linear equations in two unknowns. What can be the number of solutions **(d)** One or infinite

6. [CDS (I) Paper 3 Q96] The sum of the roots of the equation = — is zero'. What is the product of the roots of the equation ? 2 **(b)** T-DETA-K-FN - A 28 2 2 **(a)** IO cm 5 cm **(b)** cm **(c)** 56 94. When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the 2 2 97. IA there three simultaneous linear equations in two unknowns. What can be the number of solutions **(d)** One or infinite

7. [CDS (I) Paper 4 Q26] What are the roots of the equation 4X -3.2X±2 **(a)** divisible by 24 **(b)** d ivisible 8 but may no t be divisible 24 **(c)** divisible by 4 but may not be divisible by 8 **(d)** divisible by 2 but may not be divisible by 4 25. If salary of X is more than salary Y, then by how much percentöge is salary Of Y less than X?

8. [CDS (I) Paper 4 Q27] If and are the roots of the equation —X—I = O, then What is the value Of (a 4 7 2 **(d)** None of the above Q-DETB-K-FN 8 Under what condition do the 28. equations k.x — y 2 and 6x —2y— 3 have a unique solution? **(a)** divisible by 24 **(b)** d ivisible 8 but may no t be divisible 24 **(c)** divisible by 4 but may not be divisible by 8 **(d)** divisible by 2 but may not be divisible by 4 25. If salary of X is more than salary Y, then by how much percentöge is salary Of Y less than X?

9. [CDS (I) Paper 4 Q29] If sum as well as product of roots of a quadratic equation is 9, then what is the equation? **(b)** X2 -18X+9=o **(c)** x2+9X+9=o **(d)** x2-9X+9-o **(a)** divisible by 24 **(b)** d ivisible 8 but may no t be divisible 24 **(c)** divisible by 4 but may not be divisible by 8 **(d)** divisible by 2 but may not be divisible by 4 25. If salary of X is more than salary Y, then by how much percentöge is salary Of Y less than X?

10. [CDS (I) Paper 7 Q6] B starts with a wrong value Of q and gets the roots as 2 —9. What the correct roots of the equation ? **(a)** 3 and — 4 **(b)** —3 and 4 **(c)** —3 and 4 **(d)** 3 and 4 64. of

11. [CDS (I) Paper 7 Q62] 63. What is the value Of 1001xlOOlxiOOI+0999x0999xo•999 1001x1001-1001x0999+0999xo-999 If p(x) is a common multiple of degree 6 Of the (x) = x' + x2 — x — I and g(x) — — x2 + X— I then which Of the following is correct? **(a)** p **(b)** 3 and —4 **(c)** —3 and—4 **(d)** —3 and 4

12. [CDS (I) Paper 8 Q39] 40. a b —y, then what and —+ is x equal to ? a+b **(b)** b **(d)** None of the above The quadratic equation whose roots are 3 and —I is • **(c)** x2+2x-3-o Consider the following statements • **(a)** **(b)** 3 6 If the expression x3 + 4x2 + 4x + k has **(c)** 16 **(d)** 32 - 32 35. If

13. [CDS (I) Paper 10 Q44] Consider the following statements in respect of the quadratic equation ax2 + bx + b = O, where a O I. The product of the roots is equal to the sum of the roots. **(a)** 56 **(b)** 60 cm **(c)** 64 cm **(d)** 1 only

14. [CDS (I) Paper 10 Q2] The roots of the equation are always unequal and real, Which Of the above statements is/ are correct? **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 and ß are the roots or the

15. [CDS (I) Paper 12 Q45] If m 4 NO set Of m and n is possible satisfying the given conditions and > n) are the roots Of the equation 71Y+2a)2 +3a2 = 5a(7x+23a) where a > O, then What is 3m — n equal to? 12a 15a 14 41. **(b)** 16 42. **(b)** 14a If m and n are the roots Of the equation x + + b = O, and m 2 and are the roots Of the equation x — cx+d O, then which of the following is/ are correct? **(a)** 16 42. **(b)** 14a If m and n are the roots Of the equation x + + b = O, and m 2 and are the roots Of the equation x — cx+d O, then which of the following is/ are correct? 43. There are 48 cricket balls, 72 hockey balls and 84 tennis balls, and they have to be arranged in several rows in such a way that every row contains the same number Of balls of one type. Whåt is the minimum number of rows required for this to happen? 2b — a2 c Select the correct answer using the code given below. **(c)** 1 only **(d)** 2 only Both 1 and 2

16. [CDS (I) Paper 12 Q93] If one Of the roots Of the equation px + qx+r=O is three times the Other, then Which one of the following relations is correct? **(b) (d)** 3q2 = 16pr q2 •24pr p+q+r=I **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither nor 2 93. If one Of the roots Of the equation px + qx+r=O is three times the Other, then Which one of the following relations is correct? **(d)** 63 m 2 96 m 2 126 m2 252 m 2

17. [CDS (I) Paper 14 Q25] If the roots of the quadratic equation x2—4x— are all real, then the minimum value of N is 100 **(b)** 1000 10000 **(d)** 100 **(a)** 1000 10000 **(b)** 100 26. A motor boat, whose speed is 15 km/ hour in still water goes 30 km down- stream and comes back in a total of 4 hour and 30 minutes. "Ihe speed of the stream is **(c)** 4 km/hour a' **(d)** 6 km/hour

18. [CDS (I) Paper 15 Q16] If the sum of the roots of ax + bx+c—O is equal to the sum of the squares of their reciprocals, then which one Of the following relations is correct? **(a)** bc 2 = 2a2c **(b)** ac2+ 2b 2 a **(c)** ab2+ bc 2' a 2c **(d)** a2+b2+c2=I

19. [CDS (I) Paper 15 Q30] In a circle of radius 2 units, a diameter AB intersects a chord of length 2 units perpendicularly at P. If AP > BP, then AP is equal to **(a) (b) (c) (d)** (2 + units (2 + O) units (2 + units 3 units 8

20. [CDS (I) Paper 15 Q31] Under what condition on p and one Of the roots Of the equation x px+q= O is the square of the Other? **(a)** I + q + q2 = 3pq **(b)** p 3+ q + q2 3pq 2 3pq 34. 36. Consider the following statements : 2. 3. 4. Every natural number is a real number. Every real number is a rational number. Every integer is a real number. Every rational number is a real n umber. 32. What is the maximum value of m, the number divisible by 3m ? if **(c)** 8 7 6 5 33. A bike consumes 20 mL Of petrol per kilometre, if it is driven at a speed in the range of 25—50 km/hour and consumes 40 mL of petrol per kilometrc at any other speed. How much petrol is consumed by the bike in travelling a distance Of 50 km, if the bike is driven at a speed Of 40 km/hour for the first IO km, at a speed Of 60 km/hour for the next 30 km and at a speed of 30 km/hour for the last IO km? Which Of the above Statements are correc t? **(d)** I, 2 and 3

21. [CDS (I) Paper 15 Q2] If CP m cm and PD n cm, then m and n are the roots of the quadratic equation x2— 1 O. **(a)** I and 2 0 nly **(b)** 2 and 3 only **(c)** and 3 0 **(d)** I, 2 and 3 Let ABC be a triangle in which AB AC Let L be the locus of points X inside or on the triangle such that BX CX. Which of the following statements are corre ct? L is a straight line passing through A and in-centre of triangle ABC is L is a straight line passing through A and orthocentre of triangle ABC is a point on L L is a straight line passing through A and centroid of triangle ABC is a point on Select the correct answer using the code given below. 72. 73. Consider a circle With centre at C Let OP, OQ denote respectively the tangents to the circle drawn from a point O outside the circle. Let R be a point on OP and S be a point on OQ such that OR x SQ 0SxRP. Which of the following statements is/ are correct? If X is the circle with centre at O and radius OR, and Y is the circle With centre at O and radius OS then X -Y. 2. zpoct zoco- 9cr Select the correct answer using the code given below.

22. [CDS (I) Paper 16 Q20] If p and q are the roots ofx2+px+q=0, then which one of the following is correct ? **(b)** p -loniy = —2 or O = —2 Ohly **(a)** p -loniy 20. 6 If p and q are the roots of x2+px+q = 0, then which one of the following is correct ? remaining work ? **(b)** 6 hours **(c)** 12 hours **(d)** hours

23. [CDS (I) Paper 19 Q18] If and are the roots of the equation 19, ax2 + bx + c = O, then the value of 1 au + b aß + b bc **(b)** 20 ab 1 **(d)** abc Consider the following statements in respect of three a-digit numbers xyz, yzx and ZXY : **(a)** ab 1 **(b)** abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to **(c)** abc **(d)** xyz

24. [CDS (I) Paper 20 Q48] 49. IS-A) What is to ? **(a) (b) (c) (d)** 0 3 3 If ax by and b2 **(a) (b)** ac, then what is If p and q are the roots of the equation x2 — 15x + r O and p — q I, then what is the value Of r ? **(a) (b)** 55 56 60 64 For the inequation — 7x 12 > O, which one Of the following is ? **(a) (b) (c)** — x 3 only 4 x < Only A man who recently died left a sum of 46. e to be divided among his wife, five and four daughters. He directed that each son should receive 3 times as much as each daughter receives and that each daughter should receive twice as much as their mother receives. What was the wife's share ? What is **(a)** to ? **(a) (b) (c) (d)** 0 3 3 14,000 47. e 10,000 If ax by and b2 **(a) (b)** ac, then what is

25. [CDS (I) Paper 22 Q12] If a and are the roots ofthe quadratic equation x2 + kx — 15 = 0 such that a —p 8, then What is the positive value Of k ? **(a)** 13 **(b)** 39 **(c)** 117 5 11. A two-digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two-digit number is 8, then what is the number ? **(d)** 81

26. [CDS (I) Paper 23 Q2] 3. The number of (a, b, c), where a, b, c are positive such that abc 30, is **(a) (b) (d)** 30 27 9 8 If the roots Of the quadratic equation — — loglo N O are real, then what is the minimum value of N ? 10 **(d)** 10000 DZOL-T-LKM (3-

27. [CDS (I) Paper 24 Q76] In a trapezium ABCD, AB is parallel to DC. The diagonals AC and BD intersect at P. If AP : PC 4 : Bx — 4) and then what is the value of x ? 2 **(a)** 3850 m **(b)** 7700 m 15400 m 76. In a trapezium ABCD, AB is parallel to DC. The diagonals AC and BD intersect at P. If AP : PC 4 : Bx — 4 **(c)** 11:23 **(d)** 23:25

28. [CDS (I) Paper 25 Q9] Which one of the following equations does not have real roots ? **(b)** 2r2+10x-1-o **(d)** •4x2 + A - NYRG-T-MTK **(a)** 29, 17, 37 **(b)** 31, 17, 47 **(c)** 19, 37, 13 **(d)** 23, 31, 37 2. Which one of the following is a factor of the

29. [CDS (I) Paper 25 Q18] If p and q (p > q) are the roots of the equation xa — 60x + 899 = O, then which one of the following is correct ? -24+27=0 **(c)** 2p—q 30 0 **(d)** 3p—2q 43 0 A - NYRG-T-MTK **(a)** only **(b)** k>2v5 only **(c)** k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is **(d)** l—aßy

30. [CDS (I) Paper 25 Q19] If the roots of the equation x2—4x— O are real, then what is the minimum value of N? **(a)** 0-1 **(b)** 0-01 **(c)** 0-001 **(d)** 0-0001

31. [CDS (I) Paper 25 Q31] If one Of the roots of the equation is 4, then what is the sum of the squares of the roots ? 15 2 2 19 2 2 **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 only when Ware correct ? 28. Let work done by

32. [CDS (I) Paper 26 Q31] If a and ß are the roots Of the quadratic equation x 2 t + O, where (I then what is the value Of a — ß ? **(a)** 15 **(b)** 17 **(c)** 16 **(d)** 18 32. 34. A shopkeeper marks the price Of an article at ? 200. After allowing a discount Of 10%, he still gains 20% on the cost price. What is the cost price 37. 38. 39 40. 9 A piece of cleth costs 10,000. If a 2 m longer piece Of the same cloth is purchased for the same amount, it would cost 250 less per metre. What is the original length of the piece Of the article?

33. [CDS (I) Paper 26 Q36] If the sum Of the roots of the equation x 2 —k2X+30kx— Ois zero, then What is the difference of the rrx'ts? **(a)** 15 **(c)** 17 **(b)** 16 **(d)** 18

34. [CDS (I) Paper 26 Q42] Consider a question and two statements Question : Does the equation ax2 bx c O have real roots of opposite sign? Statement—I : The discriminant D > O Statement—II : c/ a > O Which one Of the following is correct in respect Of the question and the statements? **(b) (c) (d)** Statement—I alone is sufficient to answer the question Statement—II alone is sufficient to answer the question Both Statement—I and Statement—II are together sufficient to answer the question Both Statement—I and Statement—ll are not sufficient to answer the question GHSA-T-MTK/ 45A 11 **(a) (b) (c)** Statement—I alone is sufficient to answer the question Statement—II alone is sufficient to answer the question Both Statement—I and Statement—ll are together sufficient to answer the question Both Statement—I and Statement—II are not sufficient to answer the question **(d)**

35. [CDS (I) Paper 26 Q50] If y — cos2 x + sec x, where O S x < — 2' then Which one Of the following is correct? **(a) (c) (d)** 0-5SY<1 I Sy < 2 P.T.O. How many quadratic equations have the 45. sum of their roots equal to the product of their roots? **(a)** Zero **(b)** One **(c)** Two **(d)** Infinitely many

36. [CDS (I) Paper 28 Q7] If the roots of the equation x2 differ by 5, then which One of the following is JNBY-D-MTK **(a)** 34 **(b)** 90 **(c)** 127 **(d)** 129 What is the largest power Of IO that divides the product 29 x 28 x 27 x In a party Of 150 75 persons take tea, 60 persons take coffee and 50 take milk. 15 of them take both tea and coffee, but no one taking milk takes tea. If each in the party takes at least one drink, then What is the number of persons taking milk only ?

37. [CDS (I) Paper 29 Q47] 48. If average weekly wages earned by a worker is 3,520, then what is the value of k ? **(a)** 1, ' **(b) (c)** 10 50. 12 15 20 Consider the following statements : 2. One of the roots Of the equation is always less than I if a, b and c are all positive. One Of the roots of the equation is always negative if a, b and c are all negative. What is the median **(d)**

38. [CDS (I) Paper 29 Q49] 50. What are the roots of the equation ? **(a)** 1, ' **(b) (c)** 10 50. 12 15 20 Consider the following statements : 2. One of the roots Of the equation is always less than I if a, b and c are all positive. One Of the roots of the equation is always negative if a, b and c are all negative. What is the median **(d)**

39. [CDS (I) Paper 29 Q2] One of the roots Of the equation is always less than I if a, b and c are all positive. One Of the roots of the equation is always negative if a, b and c are all negative. Which of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 Consider the following for the next two (02) Consider the following for the next two (02) 2050 - 2550 2550 - 3050 3050 - 3550 3550 - 4050 4050 - 4550 4550 - 5050 (IS-A) items that follow : A grouped frequency distribution is given below : Weekly w ages in Numbers Of Rupees (e) items that follow : A quadratic equation is given by (a + b c) x2 — (2a + 2b) x + (a b —c) O; where a, b and c are real and distinct. workers k 10

40. [CDS (I) Paper 29 Q74] 75. 76. , then What is (p q equal to ? **(a) (b) (d)** 9 4 Which one of the following is a factor of a2 —b2—c2 2bC+ a + b —c ? **(a) (b) (c)** a +b+c—l **(d)** Let and be the roots Of the equation x+a+b Which one of the following is a quadratic equation whose rcujts are a2 and 152 ? **(a) (b)** (e) **(d)** + (a2 + b2) x + a2b2 = O x2 — (a2 b2) x + a2b2 0 x2 (a2 b2) x -a2b2=o -a2b2-o JNUP-T-EMT

41. [CDS (I) Paper 29 Q91] 92. Mark option **(a)** if the question can be answered by using one Of the statements alone, but cannot be answered using the other statement alone. Mark option **(b)** if the question can answered by using either statement alone. Mark option **(c)** if the question can be answered by using both the statements together, but cannot be answered using either statement alone. Mark option **(d)** if the question Cannot be answered even by using both the statements Question : Is m > n if m, n are real numbers ? Statement•I : m = (I — p) (p2 + p + I) and Statement-II : m pn Question What is the other root Of the quadratic equation with real if one of the is Staternent-I : The product Of the roots is Statement-II : The sum of roots Of quadratic equation is —

42. [CDS (I) Paper 29 Q92] Consider the following for the next ten (IO) 93. items that follow Mark option **(a)** if the question can be answered by using one Of the statements alone, but cannot be answered using the other statement alone. Mark option **(b)** if the question can answered by using either statement alone. Mark option **(c)** if the question can be answered by using both the statements together, but cannot be answered using either statement alone. Mark option **(d)** if the question Cannot be 94. answered even by using both the statements (33—A) Question : Is m > n if m, n are real numbers ? Statement•I : m = (I — p) (p2 + p + I) and Statement-II : m pn Question What is the other root Of the quadratic equation with real if one of the is Staternent-I : The product Of the roots is Statement-II : The sum of roots Of quadratic equation is —

43. [CDS (I) Paper 31 Q10] Let p, q be the roots of the equation x2 + mx — n 0 and m, n be the roots of the equation .r2 + px—q = O (m, n, p, g are non-zero numbers). Which of the following statements is/ are correct ? L m(m + n) = IL p+q-l **(c) (d)** 5 Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11

44. [CDS (I) Paper 31 Q20] If p and q are the roots of the equation x2 — sin29x — cos2B 0, then what is the minimum value of p2 + q2 ? 3 2 **(a)** nM — k **(b)** n **(c)** 81 **(d)** 105

45. [CDS (I) Paper 31 Q38] If the roots of the equation — (k —2)x + (k + 1) —0 are equal, then what are the values of k ? **(a)** 20 **(b)** 25 **(c)** 30 **(d)** 35 35. 36. Let ABC be a triangle with area 36 square cm. If AB=9 cm, BC' 12 cm and LABC = O, then what is cos9 equal to ? 4 3 2 Let n be a natural number. The HCF of n, n +10 is 10. If the LCM is x

46. [CDS (I) Paper 33 Q25] If are the roots Of this equation? 2' 2b [P.T.O. **(a)** 12 21. What is c2 + bc—a2 —ab equal to? **(b)** -1/2 23. 24. 25. If x -2+21/2 +23/2 x2 —4x— IO equal to? If equal to ? **(c)** 2p **(d)** then what is then what is x 22. If ab ba , then what is equal to?

47. [CDS (I) Paper 33 Q37] If a and are the roots Of the equation then what is equal to? **(a)** 16 **(b)** 25 **(c)** 36 **(d)** 49 1 P.T.O.

48. [CDS (I) Paper 33 Q11] Consider the Question and the Statements. Question • AB and CD are chords Of a circle intersecting at P. If AP • 48 square units. then What is cpx PD equal to? AP 8 units CP IO units Which orE Of the following is correct in resF*Ct Of the above Question and th Statements? **(a)** The Question can be answered by using One of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question Can be answered by using either Statement alone **(c)** The Question can be an swered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question can answered even without using any of the Statements ANPB-T-EMT/4SA 88. A Question is given followed by two Statements I and 11. Consider the Question and the Statements. In a quadrilateral ABCD, AB -6 units, BC- 18 units, CD 6 units, DA 9 units. What is the length Of diagonal BD ? Statement—I : The length Of BD is an integer greater than 13. Statement—II : The length Of BD is an even integer. Which one Of the following is correct in respect Of the above Question and the Statements ?

49. [CDS (I) Paper 34 Q19] Let Q=x2 + bx + c. If the sum of the roots is equal to product Of the roots of the equation Q = 0, then which of the following statements is/are correct ? **(a)** 10 minutes **(b)** 12 minutes **(c)** 1 S minutes **(d)** 20 minutes 18. A person goes downstream at x km/h and upstream at y km/h by the same boat. What is the ratio of speed of boat in still water to speed of •water ? Y 19. Let Q=x2 + bx + c. If the sum of the roots is equal to product Of the roots of the equation Q = 0, then which of the following statements is/are correct ? 20. 21. 11. Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below :

50. [CDS (I) Paper 34 Q36] ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? 15 **(a)** 443 cm **(b)** 9 cm **(c)** I Ocm **(d)** cm

51. [CDS (I) Paper 34 Q67] Two triangles ABC right-angled at A and DBC right-angled at D are drawn such that AC and DB intersect at P. If AP=x, pc=y and BP=z, then what is (AC + BD) equal to ? (xy + yz + zx) (xy + yz+ zx + ZI) **(b)** (xy + yz+ zx + **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 66. A metallic sheet of rectangular shape is 48 cm x 36 cm. From each corner a square of side x cm is cut off and an open box is made of the remaining sheet. If the box holds 5•12 litre of water, then what is the value of x ?

CHAPTER 08

TRIGONOMETRY & HEIGHTS AND DISTANCES

Total Archive Questions: 264 Questions

Typical Exam Weightage: 12 - 16 Questions (12 - 16 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q78] 79. Let AB and AC be two rays intersecting at A. Let D, E be the points lying on , respectively and p be the point such that P divides the line DE such that PD : PE AD : AE. What is the locus of the point P ? **(a)** The angle bisector of angle A **(b)** The angle trisector Of angle A **(c)** bisector of angle A **(d)** None of the above What is log (tan 10) + log (tan 20) + log (tan 30) + . + log (tan 890) equal Consider the following equations : coseca x + see x = cosec2 x see x 2. sec2 x + tan2 x = sec2 x tana x 3. coseca x 4 tana x — c0t2 x + Sec2 x Which of the above statements are ? **(a)** I and 2 only **(b)** 2 and 3 only **(c)** . I and 3 only **(d)** 1, 2 and 3 If cos x + cosi x — l, then what is the value of sina x + sin' x ? **(a) (b) (c)** 2

2. [CDS (I) Paper 1 Q80] If sin x cos x = 1/2, then what is the value of sin x — cos x ? nt lat of Iftan2 y cosec2 x — 1 tana y, then which 81. one of the following is correct ? **(b)** x = 2y **(c)** y — 2x **(a)** The angle bisector of angle A **(b)** The angle trisector Of angle A **(c)** bisector of angle A **(d)** None of the above What is log

3. [CDS (I) Paper 1 Q83] 84. 22 COS x cos x 2 , which one I + cosec x cosecx — of the following is one Of the values **(a)** The angle bisector of angle A **(b)** The angle trisector Of angle A **(c)** bisector of angle A **(d)** None of the above What is log

4. [CDS (I) Paper 1 Q78] 79. DE t PD : PE AD : AE I p **(a)** *rut A log IO) + log (tan 20) + log (tan 39 + . + log (tan 89» ? I. x + sec2x = coseca x see x 2. sec2 x + tan 2 x = seca x tana x 3. coseca x + tani x = cota x + sec2 x **(a) (b) (c) (d) (a)** 1, 230X3 cos x + x I t, sina x + sin' x

5. [CDS (I) Paper 1 Q84] 23 sin x cos x 1/2 t, sin x — cos x 80. tan2 y coseca x — 1 = tan2 y t, 81. **(d)** x —y = 10 **(a)** x —y = 10 78. 79. 80. 81. 82. 83. 84. sin x cos x 1/2 t, sin x — cos x DE t **(b)** *rut A log IO **(c)** Tt/2 **(d)** Tt/3

6. [CDS (I) Paper 1 Q82] 1 + cosec x cosec x — I **(a)** Tt/2 **(b)** Tt/3 **(d)** It/6 83. x + y 900 sin x : sin y **(a) (b) (c) (d)** (m2 — **(a) (b) (c)** cos x Sin X sin y cos Y n') sin' y ?

7. [CDS (I) Paper 1 Q87] 5 sin 750 sin 770 + 2 cos 130 Cos 150 cos 150 sin 770 7 sin 810 cos 90 88. radio transmitter antenna Of height 100 m stands at the top of a tall building. At a point on the ground, the angle of elevation Of bottom of the antenna is 450 and that of top of antenna is **(a)** Sini x < 1/2 and cos2 x > 1/2 **(b)** sin2 x > 1/2 and cos' x < 1/2 **(c)** At least one of sin2 x, cos2 x is less than I If p = tani x + coti x, then which one Of the following is correct ? What is the value Of 5 sin 750 sin 770 + 2 cos 130 Cos 150 cos 150 sin 770 7 sin 810 cos 90 89. 90. 91. 24 The angle Of elevation of the top of unfinished pillar at. a point 150 m from its base is 300. If the angle of elevation at the same point is to be 450, then the pillar has to be raised to a height Of how many meters ? **(d)** 59•4 m

8. [CDS (I) Paper 1 Q89] The angle Of elevation of the top of unfinished pillar at. a point 150 m from its base is **(a)** Sini x < 1/2 and cos2 x > 1/2 **(b)** sin2 x > 1/2 and cos' x < 1/2 **(c)** At least one of sin2 x, cos2 x is less than I If p = tani x + coti x, then which one Of the following is correct ? What is the value Of 5 sin 750 sin 770 + 2 cos 130 Cos 150 cos 150 sin 770 7 sin 810 cos 90 89. 90. 91. 24 The angle Of elevation of the top of unfinished pillar at. a point 150 m from its base is 300. If the angle of elevation at the same point is to be 450, then the pillar has to be raised to a height Of how many meters ? **(d)** 59•4 m

9. [CDS (I) Paper 1 Q90] 91. 24 has to be raised to a height Of how many meters ? **(a)** 59•4 m **(b)** 61•4 m **(c)** 62•4 m **(d)** 63•4 m Which one of the following represents statistical data ? **(a)** The names of all owners of shops located in a shopping complex **(b)** A list giving the names of all states of India **(c)** A list Of all European countries and their respective capital cities **(d)** The volume Of a rainfall in certain geographical area, recorded every month for 24 consecutive months If sin x + Sin y a and cos x + cos y — b, what is sin x • sin y + cos x • cos y equal to ? **(b)** a + b + ab a2+b2—2 **(d)** 2

10. [CDS (I) Paper 1 Q91] 24 The angle Of elevation of the top of unfinished pillar at. a point 150 m from its base is **(a)** Sini x < 1/2 and cos2 x > 1/2 **(b)** sin2 x > 1/2 and cos' x < 1/2 **(c)** At least one of sin2 x, cos2 x is less than I If p = tani x + coti x, then which one Of the following is correct ? What is the value Of 5 sin 750 sin 770 + 2 cos 130 Cos 150 cos 150 sin 770 7 sin 810 cos 90 89. 90. 91. 24 The angle Of elevation of the top of unfinished pillar at. a point 150 m from its base is 300. If the angle of elevation at the same point is to be 450, then the pillar has to be raised to a height Of how many meters ? **(d)** 59•4 m

11. [CDS (I) Paper 1 Q87] 88. **(a)** sin2 x < 1/2 cos2 x > 1/2 **(b)** sin2 x > 1/2 cosi x < 1/2 **(c)** sin2 x < 1/2 cosi x < 1/2 **(d)** sina x, cos2 x p x + cot2 x 5sin 750 sin 770+2 cos 130cos150 cos 150 sin 770 7 sin 810 Cos 90 450 fif 600 t I ? 100 m **(a) (b)** 50 m 500 1) m **(c)** 5066-1) m **(d)** 25

12. **[CDS (I) Paper 1 Q89]** 150m 300 I 450 **(a)** 59•4 m **(b)** 61 •4m **(c)** 62•4 m **(d)** 63•4 m 90. w, **(c)** 2vif 24 91. sin x + sin y a cos x + cos y = b t, sinx siny+cosx **(a) (b) (c)** 25 a + b — ab

13. **[CDS (I) Paper 1 Q91]** 150m gr 300 I 450 **(a)** 59•4 m **(b)** 61 •4m **(c)** 62•4 m **(d)** 63•4 m **(a) (b) (c) (d)** 24 sin x + sin y — a cos x + Cosy = b t, sinx siny+cosx **(a) (b) (c) (d)** a + b — ab 2

14. **[CDS (I) Paper 1 Q97]** Data on percentage distribution Of area of land in acres by households in two districts of a particular State are as follows • Land holding District—A District-B 0-01—0-99 7-5—12-49 12 •5—19-99 20-0—29-99 50 18-35 47-12 19-34 7-21 2•36 13-53 21•84 39-32 12-15 7-43 5•73 What is the appropriate diagram to represent the above data ? **(a)** pie diagram **(b)** Histogram **(c)** Bar Chart **(d)** None Of the above If is the angle Of first quadrant such that cosec' a = 17 + cot' a, then what is the value Of sin a ? **(d)** 1/16 If x + (l/x) 2 cos then what is the value Of + (I/XZ) ? **(a)** 4 cos* **(b)** 4 cos' — I **(c)** 2 cos2 a — 2 Sini a **(d)** cos2 a — Sini

15. **[CDS (I) Paper 2 Q79]** If cot B I — cos I + cos O 8/15, then -What is the Of where e is a positive acute angle ? **(a) (b) (c) (d)** 1/5 2/5 3/5 4/5 What is the angle subtended at the centre a circle Of radius 8 metres after traversing metres along its circumference ? **(a) (b) (c) (d)** 27t/3 311/4 Consider the following . cos e — sin O cos + sin 2B cos20 (I tan B) (I I + sin O = (tan B + sec

16. **[CDS (I) Paper 2 Q2]** I — sin B — tan e) Which of the statements given above is/are correct ? **(a) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 cos O I — sin O cos O 2 is satisfied by which I + sin O one of the followini values Of O ? **(a) (b) (c) (d)** 7t/6 P.DETB-J.FN 22 of 81. re of rsmg

17. **[CDS (I) Paper 2 Q82]** 89. slare 84. vhi Ch 22 -A) The difference Of the two angles in degree measure is I and their sum in circular measure is also I. What are the angles in circular measure ? 360 90 **(b) (d)** 2 2 2 360 ' 2 1 2 (1 None of these If and 450 < 90', then which one Of the following is correct ? - tan2 ? **(a) (b) (c) (d)** sin x sin y sin x sin y What is the value of sin 600 cot 30' -2 sec2 45' 3 cos 600 tan 45' 35/8 **(a) (b)** -35/8 -11/8 **(c)** 11/8 **(d)** then What is If tan O equal to ? p sec B + q cosecO **(b) (c) (d)** 1

18. **[CDS (I) Paper 2 Q81]** 1 — cos 9 cot 8/15, **(a)** 1/5 **(b)** 2/5 **(c)** 4/5 8 a-rt 4't 81. 83. 84. 2 2 90 180 2 2 2 3i7Z 1 t 360 90 180 o < x < 45' 450 < y < 90', **(d)**

19. **[CDS (I) Paper 2 Q77]** I + cos O **(a)** 1/5 **(b)** 2/5 **(c)** 4/5 8 a-rt 4't 81. 83. 84. 2 2 90 180 2 2 2 3i7Z 1 t 360 90 180 o < x < 45' 450 < y < 90', **(d)**

20. **[CDS (I) Paper 2 Q2]** 2n/3 3r/4 cog2B — sin20 cosZ9 + sin29 cos2e(I + tan O) (I — tan O) I sin (tan + sec 1 — sin e COs B cos O I — sin I + sin e **(a) (b) (c) (d)** rt/3 P-DETB-J-FN 2 2 90 180 2 2 2 3i7Z 1 t 360 90 180 o < x < 45' 450 < y < 90', **(a)** s.n x Sin y **(b)** sm x < sm y **(c)** sin x S sin y sins 600 cot 300 — 2 sec2 45' + 3 cos 600 tan 45' — tan2 60 **(a) (c) (d)** 35/8 -35/8 -11/8 11/8 tan z p 1 p see B — q cogee9 psec9 + q cosecB

21. **[CDS (I) Paper 2 Q79]** 80. 1 — cos 9 cot 8/15, I + cos O **(a)** 1/5 **(b)** 2/5 **(c)** 4/5 8 a-rt 4't 81. 83. 84. 2 2 90 180 2 2 2 3i7Z 1 t 360 90 180 o < x < 45' 450 < y < 90', **(d)**

22. **[CDS (I) Paper 2 Q84]** 2 2 90 180 2 2 2 3i7Z 1 t 360 90 180 o < x < 45' 450 < y < 90', **(a) (b) (c) (d)** 2n/3 3r/4 cog2B — sin20 cosZ9 + sin29 **(a)** s.n x Sin y **(b)** sm x < sm y **(c)** sin x S sin y sins 600 cot 300 — 2 sec2 45' + 3 cos 600 tan 45' — tan2 60 cos2e(I + tan O) (I — tan O) I sin (tan + sec

23. **[CDS (I) Paper 2 Q87]** The value of cosec20 — always 2 + sin e is **(a) (b) (c) (d)** If to ? less than zero n on-negative zero then- what is cos 9 equal The length Of the shadow of a person s cm tall when the angle Of elevation Of the sun is-a is p em. It is q cm •when the angle of elevation Of the sun is Which one the following is correct when 3a ? **(a) (b) (c) (d)** P-DETm-FN tan — tan aa tan tan a tan aa — tana g tan tan a tan — tan tan aa tan tan 2a tan tan a (24 88. qual

24. **[CDS (I) Paper 2 Q85]** 86. 87. (24 The value of cosec20 — always 2 + sin is **(a) (b) (c) (d)** If to ? less than zero n on-negative zero then- what is cos 9 equal

25. **[CDS (I) Paper 2 Q92]** The angles Of a triangle are in AP and the greatest angle is double the least. What is the ratio of •angles in thc radian measure ? **(c) (d)** For what value Of e, is (sin e + cosec O) = 2 5 where O < 90' ? **(a) (b) (c)** 300 60 • 900 If O < •O < < 90•, then which one Of the following is correct ? **(a)** (sine cos > 2 **(b)** (sin2 + cos2 2 **(c)** (sin2B + cos2 •) < 2 **(d)** (sin2 B + cos2 $) > 2 Which one Of the fallowing is not correct in respect of the sets A and B ? The length of the shadow of a person s cm tall when the angle Of elevation Of the sun is-a is p em. It is q cm •when the angle of elevation Of the sun is Which one of the following is correct when 3a ? **(a) (c)** If A r; B, If A n B then B U A = B then A n (A -B) then B n A = O, then either A Or **(a) (b) (c) (d)** P-DETm-FN tan — tan aa tan tan a tan aa — tana g tan tan a tan — tan tan aa tan tan 2a tan tan a A boy travels a distance Of 8 km at the rate of 4 km per hour, 6 km at the rate of 3 km per hour' and 4 km at the rate Of 2 km per hour. What is the average speed for the entire journey ? **(a) (b) (c) (d)** 2 kmph 3 kmph 4 kmph 6 kmph

26. **[CDS (I) Paper 2 Q86]** 87. 25 -A) **(a) (b) (c) (d)** cot cos Vâi s cm p cm t, 89. 90. 91. 92.

27. [CDS (I) Paper 2 Q91] Våi s cm p cm t, **(b) (c) (d)** p tan — tan 3rx tan tan tan 3a — tan 3 tan tan tan — tana tan tan tan 2a tan 311 tan P-DETB-J-FN **(c)** (sin e) — 2-5 300 **(c)** 600 **(d)** 900 at o 900, **(a)** • (sin O + cos > 2 (sin 2 e + 0) S 2 (sin2 + < 2 **(d)** (sin2e + **(b)** Ag B, An (A —B) = $ (Ch AG B, * B n A—A 8 km 4 km 6 km 3 km 4 km 2 km tia 1 **(a) (b) (c)** 2 kmph 3 kmph 4 kmph 6 kmph cosec29—2+ sin20

28. [CDS (I) Paper 3 Q3] If cos p and cos 890 q, then which one of the following is correct ? **(a)** Last digit is 4 **(b)** Last digit is 8 **(c)** Last digit is 2 the expansion is a positive S. Consider the following assumption and two statements : A number •ABCDE' is Assumption divisible by 11. E-D+C-B+A is Statement — I divisible by 11. • E-D+C-B+A=O. Statement — , Which one Of the following is corect ? **(d)** Last two digits are zero 2. What is the locus Of centres of circles which touch a given line at a given point ?

29. [CDS (I) Paper 3 Q11] Which one of the following is correct ? **(a)** There is only one g with 00 < < 900 such that sin = a, where a is a real number **(b)** There is more than one O with 00 < e 900 such that sin = a, where a is a real number T.DETA-K-FN - A 4 **(c)** There is no 9 with 00 < 900 such that sin a. where a is a real **(d)** There are exactly two Os with < 900 such that sin = a, where a is a real number

30. [CDS (I) Paper 3 Q31] If and cos 0+6 sin 6 —2, then what is the value of ? **(d)** 2 8 25. of a work is completed in half a day by 5 persons. Then, of the work can be completed by 6 in how many days **(d) (a)** 26. The shadow of a pole 6 metre high is 15 rnetre long and at the same time the shadow of a tree is 25 metre long. What is the height of the tree ? **(b)** 21 m **(c)** 10m **(d)** 35 m

31. [CDS (I) Paper 3 Q43] If •a is an Elute angle and , then what is tan a equal to ? **(a)** tan2t50 + tan2200 + tan2250 . + tan275Ø **(b)** cos2150 + cos2200 + cos2250 cos2750 **(c)** cot2150 — cot220e + cot2250 — + cot2750 **(d)** sec2150 + sec2200 + sec2250 + sec2750 42. Consider the following data : 5 9 2 The arithmetic mean of the above distribution is 296. What is the missing frequency ? 43. If a is an acute angle and then what is tan equal to ?

32. [CDS (I) Paper 3 Q44] Suppose the angle of elevation of the top of a tree at a point E due east of the tree is 600 and that at a point F due west of the tree is **(a)** tan2t50 + tan2200 + tan2250 . + tan275Ø **(b)** cos2150 + cos2200 + cos2250 cos2750 **(c)** cot2150 — cot220e + cot2250 — + cot2750 **(d)** sec2150 + sec2200 + sec2250 + sec2750 42. Consider the following data : 5 9 2 The arithmetic mean of the above distribution is 296. What is the missing frequency ? 43. If a is an acute angle and then what is tan equal to ?

33. [CDS (I) Paper 3 Q68] A man is watching from the top of a tower a boat sl*eding away from the tower. The boat makes an angle Of depression of 450 with the man's eye when at a distance of 60m from the bottom of tower. After 5 seconds, the angle of depression becomes **(a)** Rs. 500 @ 10%, Rs. 800 @ 12% **(b)** Rs. 700 @ 10%, Rs.600 @ **(c)** Rs. 800 @ 10%, Rs. 400 @ **(d)** @ 10%, ROOO @ 68. A man is watching from the top of a tower a boat sl*eding away from the tower. The boat makes an angle Of depression of 450 with the man's eye when at a distance of 60m from the bottom of tower. After 5 seconds, the angle of depression becomes 300. What is the approximate speed of the boat assuming that it is running in still water ? 70. The marks of the students of a class who appeared for a test in English are represented in the following frequency table : 11-20 21-30 31-40 41-50 51.60 What is/are the modal class

34. [CDS (I) Paper 3 Q81] If 7 cos2 9+3 sin2 2 then what is the value Of tan 6 ? **(a)** 45 **(b)** 225 then what is the value Of tan 6 ? one **(c) (d)**

35. [CDS (I) Paper 4 Q40] What is the value of 510gio 25 -210g10 3 + 10310 18 ? **(c)** . The 2 product Of a rational number 41 42. and an frrational number is **(a)** a natural number **(b)** an irrationa I number **(c)** a cognpos ite number **(d)** a rational n umber The angle Of elevation Of the top Of a tower from the bottom Of a building is twice that from its top. What is the height Of the building if the height Of the tower is 75 m and the angle Of elevation Of the top Of the tower from the bottom of the building is 60'? **(b) (c) (d)** 25 m 37.5 50 m 60 m Q-DETB-K-FN / 26A

36. [CDS (I) Paper 4 Q44] What is thc value of cos I • cos 20 cos 3' . cos 90' ? **(a)** 1/4 40. What is the value of 510gio 25 -210g10 3 + 10310 18 ? 43. If cos02+ in the first quadrant, then Which One Of the following is correct? 44. What is thc value of cos I • cos 20 cos 3' **(b)** . The 2 product Of a rational number cos 90' ? **(c)** 1/2 41 42. and an frrational number is **(d)** a natural number

37. [CDS (I) Paper 4 Q46] If sinO+cosO=i, then what is the value of sine dose? 2 **(d)** 1/2 I + sin a **(a)** 1/2 47. **(b)** 60 **(c)** 45 **(d)** 30

38. [CDS (I) Paper 4 Q47] What is equal to? I—sin0 **(b)** secG+ tane **(c)** cosece + cot é **(d)** cosec — c Ot **(a)** 1/2 47. **(b)** 60 **(c)** 45 **(d)** 30

39. [CDS (I) Paper 4 Q48] The shadow Of a tower is 15 m when the Sun's altituqe is **(a)** 1/2 47. **(b)** 60 **(c)** 45 **(d)** 30

40. [CDS (I) Paper 4 Q49] 14 sm t? I — sine **(a)** 9 — tan 9 **(b)** sece+ tan 8 **(c)** cosece + cot 9 **(d)** cosecO — cot 9 30' t, 15 m 60• t, **(d)** 100 **(b) (c)** 8905 83 192 33 88-55 8755 89-55 Q-DETB-K-FN/ 26A IS **(c)**

41. [CDS (I) Paper 4 Q52] PAB pc **(a)** 47. 48. 49. 14 sm I — sine **(b)** 9 — tan 9 **(c)** sece+ tan 8 **(d)** cosece + cot 9

42. [CDS (I) Paper 5 Q9] 11. A bicycle is running straight towards north. What is the locus Of the centre of the front wheel Of the bicycle whose diameter is d ? **(a) (b) (c)** A line parallel to the path Of the wheel Of the bicycle at a height d cm A line parallel to the path Of the wheel Of the bicycle at a height d/2 cm A circle Of radius d/2 A circle of radius d cm TO pass an examination, a candidate needs 40% marks. All questions carry equal marks. A candidate just passed by getting IO answers correct by attempting 15 Of the total questions. HOW many questiong are there in the examination ? **(a) (b) (c)** 25 ao 40 45 Assume the Earth to be a sphere of radius R. What is the radius Of the circle of latitude 400 S ? **(a) (b) (c) (d)** R cos 400 R sin 800 R sin 400 R tan 400 P-PQFA.L.Rz h

43. [CDS (I) Paper 5 Q26] AB is a straight line. C is a point whose perpendicular distance from AB is 3 cm. What are the number Of points which are at a perpendicular distance of I cm from AB and at a distance 4 cm from C ? **(a) (b)** Straight line A circle concentric With C A circle touching C A circle intersecting C at two points **(c) (d)** If 2 — cos B tan A then sin B what is 28. 29. 30. What is the value of 2-6 — lö ?

44. [CDS (I) Paper 5 Q30] What is the value of 2-6 — lö ? **(a) (b) (c) (d)** 0-6 03 0-7 0-7 2 tan A equal to ? I — tan A **(c)** (tan B)/2 2 tan B tan B 4 tan B A garrison Of 'n' men had enough food to Jast for 30 days. After IO days, 50 more men joined them. If the food now lasted for 16 days, what is the value of n ? **(b) (c) (d)** 200 240 280 320 What is the value of **(c)** 2 What is the ratio between times taken by a train 240 m long to cross an electric pole and a bridge Of 80 m length ? **(a) (c) (d)** P-PQFA-L-RZ

45. [CDS (I) Paper 5 Q80] The angle of elevation of the top Of a tower ata Fl point on level ground is **(a)** If **(b) (c)** Out 10 **(d)**

46. [CDS (I) Paper 5 Q97] 98. 60 —cos 9 99. up the hill ? **(a)** km **(b)** km **(c)** 56 km **(d)** 6•5 km What is cot 150 cot 200 cot 700 cot 750 equal If sin 30 cos (O — 20) where 39 and (e are acute angles, what is the value Of B ? - 20) **(a) (b) (c) (d)** 220 230 240 250 What is 2 equal to ? e — cos e cos4 6 **(a) (d)** sind e I — sin2 0 cos2 9 I + sin2 9 eos2 B I — 3 sin2 e cos2 e

47. [CDS (I) Paper 5 Q100] Consider the following : I. tan2 B — sin 2 9 tan 2 B sin 2 B 2. (cosec 9 — sin e) (sec O cos 9) (tan B + cot O) Which Of the above Ware correct ? **(a)** 15 cm **(b)** 20 cm **(c)** 25 cm **(d)** 30 cm Let D, E be the points on sides AB and AC respectively Of a triangle ABC such that DE is parallel to BC. Let AD 2 cm, DB = I em, AE = 3 cm and area' Of triangle ADE 3 em What is EC equal to ?

48. [CDS (I) Paper 5 Q94] 1000 5 mm **(a)** 22 m **(b)** 20 m **(c)** 18 m 95. qt«rq (cm2 t ? 16Jä -8T 16Jä - **(a) (b) (c) (d)** 16 NG P-PQFA-L-RZ 27 - ' g 96. **(a)** 3-5 km **(b)** km **(c)** 5•5 km **(d)** 6-5 km cot 19 cot 200 cot 700 cot 75' '

49. [CDS (I) Paper 5 Q97] 98. sin cos (9-20) 6 0 — cos e — cos I — sin2 B cos2 e 2-5 kmph 3aqå 3-25 kmph t I 4 36 **(a)** 16 cm **(b)** 20 em **(c)** 25 cm **(d)** 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

50. [CDS (I) Paper 5 Q99] 100. **(a) (b) (c) (d)** sin sin **(a) (b) (c) (d)** 220 240 250 sin 4 e — cos e + sin2 0 cos2 0 I — 3 sin2 0 cos2 e I. tan2 e — sin 2 0 tan2 e sin2 e 2. e — sin 9) (sec O — cos O) (tan'9 + cot•9) I **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

51. [CDS (I) Paper 5 Q93] 6 0 — cos e — cos **(a)** 16 cm **(b)** 20 em **(c)** 25 cm **(d)** 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

52. [CDS (I) Paper 5 Q98] 99. 2-5 kmph 3aqå 3-25 kmph t I 4 36 **(a)** 3-5 km **(b)** km **(c)** 5•5 km **(d)** 6-5 km cot 19 cot 200 cot 700 cot 75' ? 1000 5 mm **(a)** 22 m **(b)** 20 m **(c)** 18 m 100. qt«rq (cm2 t ? **(a) (b) (c) (d)** sin sin **(a) (b) (c) (d)** sin cos (9-20) 220 240 250 sin 4 e — cos e + sin2 0 cos2 0 I — 3 sin2 0 cos2 e **(a) (b) (c) (d)** 16 NG I. tan2 e — sin 2 0 tan2 e sin2 e 2. e — sin 9) (sec O — cos O) (tan'9 + cot•9) I **(a) (b) (c) (d)** P-PQFA-L-RZ

53. [CDS (I) Paper 6 Q39] The point of concurrence of the altitudes of' a triangle is called **(a)** Circum-centre **(b)** Ortho-centre **(c)** In-centre **(d)** Centroid sin O cos O If cos O sin O - 2 with o < e < 900, then what is 0 equal to ? **(a)** 300 **(b)** 450 **(c)** 600 **(d)** 750 Ifthe population figures are given for each State Of India, then the data can be classified as **(a)** Qualitative **(b)** Quantitative **(c)** Chronological **(d)** Geographical A student moves x km east from his residence and then moves x km north. He then goes x km north-east and finally he takes a turn of 900 towards right and moves a distance x km and reaches his school. What is the shortest distance Of the school from his residence ? (20 + km **(a) (b)** 3x km **(c)**

54. [CDS (I) Paper 6 Q38] 39. **(a) (b) (c) (d)** square cm 26 square Crn 36 square cm 46 square cm The point of concurrence of the altitudes of' a triangle is called **(a)** Circum-centre **(b)** Ortho-centre **(c)** In-centre **(d)** Centroid sin O cos O - 2 with o < e < 900, If cos O sin O then what is 0 equal to ? **(a)** 300 **(b)** 450 **(c)** 600 **(d)** 750 Ifthe population figures are given for each State Of India, then the data can be classified as **(a)** Qualitative **(b)** Quantitative **(c)** Chronological **(d)** Geographical A student moves x km east from his residence and then moves x km north. He then goes x km north-east and finally he takes a turn of 900 towards right and moves a distance x km and reaches his school. What is the shortest distance Of the school from his residence ? (20 + km **(a) (b)** 3x km **(c)**

55. [CDS (I) Paper 6 Q46] 47. From a solid cylinder of height 4 cm and radius 3 cm, a conical cavity of height 4 cm and of base radius 3 cm is hollowed out. What is the total surface area of the remaining solid **(a)** 1 square cm **(b)** 22m square cm **(c)** square cm **(d)** 487t square cm If A = 't/6 and B = It/3, then which of the following is/are correct ? l. sin A + sin B = cos A + cos B 2. tan A + tan B cot A + cot B Select the correct answer using the code given below • **(a)** I Only **(b)** 2 only **(c)** Both 1 and 2 **(d)** . Neither nor 2 If a + b + c = 6, + + = 26, then what is ab + bc + ca equal to ? Which among the following is the largest four digit number that is divisible by 88 ? **(a)** 9988 **(b)** 9966 **(c)** 9944 **(d)** 8888

56. [CDS (I) Paper 6 Q74] The length of shadow of a tree is 16 m when the angle of elevation of the sun is **(a)** 286 **(b)** 296 73. The areas of two circles are in the ratio I : 2. If the two circles are bent in the form Of squares, what is the ratio of their areas ? The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? **(c)** 3 units **(d)** 5 units

57. [CDS (I) Paper 6 Q87] What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ? **(c)** r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca • cosec ß sin cosß equal to ? **(d)** None of the abo ve Consider the following statements : **(a)** 280 **(b)** 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ? **(c)** r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca • cosec ß sin cosß equal to ? **(d)** None of the abo ve Consider the following statements : 88. 89. 90. 91. Which one of the following is a prime number ?

58. [CDS (I) Paper 6 Q85] 86. 87. **(b)** 280 **(d)** 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ? **(c)** r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca • cosec ß sin cosß equal to ? **(d)** None of the abo ve Consider the following statements : **(a)** 280 **(b)** 52 What is the length of the perpendicular drawn from the centre Of circle of radius r On the chord of length r ? **(c)** r/2 If a and ß are complementary angles, then sin a Cosa - i what is coseca • cosec ß sin cosß equal to ? **(d)** None of the abo ve Consider the following statements : 88. 89. 90. 91. Which one of the following is a prime number ?

59. [CDS (I) Paper 7 Q20] 21. Ihe angle of elevation of the tip of a tower from a point On the ground is **(a)** 16 In the triangle ABC, the base BC is trisected at D and E. The line through D, parallel to AB, meets AC at F and the line through E parallel to AC meets AB at G Let EG and DF intersect at H. What is the ratio of the sum Of the area of parallelogramÅGEF and the area of the triangle DHE to the area Of the triangle ABC ? **(b)** 1/6 Ifx — 2 is a factor + kx2 what is the value of k ? — 2K —24 then AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number ofpoints which are at a distance of I cm from AB and 5 cm from C ? 18. 19. 20. 21. Ihe angle of elevation of the tip of a tower from a point On the ground is 450. Moving 21 m directly towards the base ofthe tower, the angle ofelevation changes to 600. What is the height Of the tower, to the nearest meter ? **(c)** 48 m **(d)** . 49 m

60. [CDS (I) Paper 7 Q52] 53. From the top of a cliff 200 m high, the angles of depression of the top and bottom Of a tower are observed to be 300 and 450 respectively. What is the height Of the tower ? **(a)** 400m **(b)** 4006 m **(c)** 400/6 m **(d)** None of the above If x3,'+c 5x2 + 10k leaves remainder —2x when divided by x2 + 2, then what is the value Of k ? If one of the roots of quadratic equation 7x2 — 50x + k = 0 is 7, then what is the value Of k ? **(c)** **(d)** 50/7 7/50 If a' b, by c ahd xyz= l, then what is cz equal to ? **(a)** **(c)** 14

61. [CDS (I) Paper 7 Q65] 66. of 18 Consider the following : 1. sin2 I 2. sec2330 — coe 570 — cosec2370 — 530 Which of-the above statements 'is/are correct ? **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 If p = g sin x + b cos x and q a cos x — b sin x, then what is p2 + q2 equal to ? **(b)** ab **(c)** a2 + b2 Statement-I . Statement-ll : Let the side DE ofa triangle DEF be divided at S so that DS 1 . Ifa line through S parallel to EF meets DF at T, then the area of triapgle DEF is twice the area of the triangle DST. The areas of the similar triangles are proportional to the Squares on the corresponding sides. Which one of the following is correct in respect Of the above statem?Ats ? **(a)** Both statement-I and statement-II are true and statement-II is the correct explanation of statement-I **(b)** Both statement-I and statement-II are true but statement-II iS not the correct explanation of statement-I **(c)** Statement-I is true, but statement-II is false **(d)** Statement•ll is true, but statement-I is false

62. [CDS (I) Paper 7 Q5] What is the difference between the two angles ? **(a)** 600 **(b)** 900 **(c)** 1200 **(d)** can not be determined with the given data What is the logarithm with respect to base IO? Consider the following statements : I. There is Only one value of x in the first quadrant that satisfies sin x + cos x =

63. [CDS (I) Paper 7 Q2] 2. There is only one Value Of x in the first quadrant that satisfies sin x — cos x = **(a)** 64 minutes **(b)** 80 minutes **(c)** 96 minutes **(d)** 120 minutes The pair of numbers which are relatively prime to each other is :

64. [CDS (I) Paper 7 Q98] Consider the following : cot 300 + 1 = 2 (cos 300+1) cot 300-1 **(a)** I only **(b)** 2 only **(c)** Both land2 **(d)** Neither nor 2 If 3 sin x + 5 cos x = 5, then.what is the value of

65. [CDS (I) Paper 7 Q2] 2 sin 450 cos 450 — tan 450 cot 450 O Which of the above identities is/are correct ? **(a)** I only **(b)** 2 only **(c)** Both land2 **(d)** Neither nor 2 If 3 sin x + 5 cos x = 5, then.what is the value of (3 cos x — 5 sin x) ? How many between —l I and I I are multiples of 2 or 3 ? **(a)** 11 **(b)** 14 **(c)** 15 **(d)** None Of the above 28

66. [CDS (I) Paper 8 Q47] If tan O 3/4 and O is acute, then what is sin 0 equal to ? **(a)** ac I If f **(b)** -20 **(c)** 20 **(d)** 12

67. [CDS (I) Paper 8 Q48] What is vane of — O) . sin 9 450 ? **(a)** ac I If f **(b)** -20 **(c)** 20 **(d)** 12

68. **[CDS (I) Paper 8 Q54]** 55. 14 A wheel makes 12 revolutions per minute. The angle in radian described by a spoke Of the wheel in I second is : **(a)** 57t/2 **(b)** 27t/5 (C) 37t/5 **(d)** 47t/5 A ladder 20 m long is placed against a wall so that the foot of the ladder is IO m from the wall. The angle of inclination of the ladder to the horizontal will be • **(a)** 300 **(b)** 450 **(c)** 600 **(d)** 750 If 0 < O < 900, then all the trigonometric ratios can be obtained when **(a)** only sin is given **(b)** only cos 9 is given **(c)** only tan O is given **(d)** any one of the six ratios is given 'What is sin A cos A tan A + cos A sin A cot A equal to ? **(a)** sin2A + cosA **(b)** sin2A + tan2A **(c)** sin2A + cot2A **(d)** cosec2A — cot2A On walking 120 m towards a chimney in 49. a horizontal line through its base the angle of elevation of tip of the chimney changes from 300 to

69. **[CDS (I) Paper 8 Q56]** 57. 58. sin 0 What is 1 + cos O **(a)** 2 cosec 0 **(b)** 2 sec O **(c)** sec 0 **(d)** cosec O 1 + cos equal to ? sin O The angles Of elevation of the top of a tower from two points which are at distances of 10 m and 5 m from the base of the tower and in the same straight line with it are complementary. The height of the tower is : **(b)** 15m The angles of elevation of the top of an inaccessible tower from two points on the 59 60 61 same straight line from the base Of the tower are 300 and 600 respectively. If the points are separated at a distance of 62 100 m, then the height Of the tower is close to **(a)** 86.6m **(b)** 84.6 m **(c)** 82.6 m **(d)** 80.6 m 16

70. **[CDS (I) Paper 8 Q62]** 100 m, then the height Of the tower is close to **(a)** 86.6m **(b)** 84.6 m **(c)** 82.6 m **(d)** 80.6 m 16 In a triangle ABC, LBCA = 600 and AB2 = BC2 + CA2 + X. What is X equal **(a) (b) (c) (d)** Zero

71. **[CDS (I) Paper 9 Q40]** 41. if 2m, then m equals **(a) (d)** consider statements io r ue Of cos i ayg than l. 2. v cot e is gteacer than **(a)** 0 **(b)** 1 **(c)** -1 **(d)** 2

72. **[CDS (I) Paper 9 Q44]** 45. A— sin A What is gin3 A €9) — cos A see A sec A cos A None Of the ain A and is an angle. A equal if Sih following correct ? cos B COS * cos COS 2 sin O cos O 2 sin O' sec B tan b one Of A — sin A 42 **(a)** 1 **(b)** √3 **(c)** 1/√3 **(d)** 2

73. **[CDS (I) Paper 9 Q50]** Ing 51. **(d)** 15 15" 30 + 1,5./ä m None Of the above and the ang2e of Of the foot tower as seen the top the budding. then What is tan O to ? .3V3 above 46, If A + then 2 tan A): + equal to ? 8jn2 A A see A None cf the ab ions : Consider the following tic next iota," 50. if , what is • tan to ? **(a) (b)** None of above If and h m, tien what is the digtauce between the of the bunding and the base of the tower ? **(c)** 15 15" 30 + 1,5./ä m None Of the above **(d)**

74. **[CDS (I) Paper 10 Q80]** If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 2 **(d)** 9T+A If sine + cos9=Vä, then what tan B + equal to? What is the angle of elevation of the sun when the shadow of a pole of height x m is m? 450 750 B-BCRB-M-NBU/ 48A 26 **(a)** 2 only **(b)** Both 1 and 2 **(c)** Neither 1 nor 2 84. 26 If tang + sec8= m, then what sec9 equal to? is 80. If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 **(d)** 2m

75. **[CDS (I) Paper 10 Q83]** What is cos2(456 + 9) + (450—9) tan(6cr +0) tan(3CP -8) equal to? **(a)** 2 only **(b)** Both 1 and 2 **(c)** Neither 1 nor 2 84. 26 If tang + sec8= m, then what sec9 equal to? is 80. If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 **(d)** 2m

76. **[CDS (I) Paper 10 Q86]** If 4 tan 9 = 3, then what is 4 sine — coso 4sine + 9cos9 equal to? 2 4 3 6 **(a)** 2 only **(b)** Both 1 and 2 **(c)** Neither 1 nor 2 84. 26 If tang + sec8= m, then what sec9 equal to? is 80. If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 **(d)** 2m

77. **[CDS (I) Paper 10 Q88]** What is cosec(750 + e) — equal to? **(c)** 2sin0 If triangle ABC is right angled at C, then what is cos(A+ B) + sin(A+ B) equal to? 2 **(a)** 150' If . O, then sin 4 B + cos 49 equal to? what is 91. What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? 4 2 88. What is cosec **(b)** 2sin0 If triangle ABC is right angled at C, then what is cos **(c)** 2 92. Consider the following statements • 1. tan B increases faster than sin O as B increases, 2. The value of sinB+cose is always greater than 1. Which of the above Statements is/ are correct? 1 only **(d)** 2 only

78. **[CDS (I) Paper 10 Q90]** Ifa, ß, are agute angles such that — — , tan Y I 2 then what is a + •Y equal to? **(b)** 105' 120' 135' **(a)** 150' If . O, then sin 4 B + cos 49 equal to? what is 91. What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? 4 2 88. What is cosec **(b)** 2sin0 If triangle ABC is right angled at C, then what is cos **(c)** 2 92. Consider the following statements • 1. tan B increases faster than sin O as B increases, 2. The value of sinB+cose is always greater than 1. Which of the above Statements is/ are correct? 1 only **(d)** 2 only

79. **[CDS (I) Paper 10 Q91]** What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? **(c)** 2 **(a)** 150' If . O, then sin 4 B + cos 49 equal to? what is 91. What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? 4 2 88. What is cosec **(b)** 2sin0 If triangle ABC is right angled at C, then what is cos **(c)** 2 92. Consider the following statements • 1. tan B increases faster than sin O as B increases, 2. The value of sinB+cose is always greater than 1. Which of the above Statements is/ are correct? 1 only **(d)** 2 only

80. **[CDS (I) Paper 10 Q92]** 28 Consider the following statements • 1. tan B increases faster than sin O as B increases, **(a)** 150' If . O, then sin 4 B + cos 49 equal to? what is 91. What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? 4 2 88. What is cosec **(b)** 2sin0 If triangle ABC is right angled at C, then what is cos **(c)** 2 92. Consider the following statements • 1. tan B increases faster than sin O as B increases, 2. The value of sinB+cose is always greater than 1. Which of the above Statements is/ are correct? 1 only **(d)** 2 only

81. [CDS (I) Paper 10 Q93] What is (sine +cose) (tan 9 + cot9) sece + cosec9 equal to? **(c) (d)** sin cos B 87 , 89. — — , tan Y I **(d)** 150' If . O, then sin 4 B + cos 49 equal to? what is **(a)** 150' If . O, then sin 4 B + cos 49 equal to? what is 91. What is sin 6 9 + cos 6 9 + 3sin2 ecos2 0 equal to? 4 2 88. What is cosec **(b)** 2sin0 If triangle ABC is right angled at C, then what is cos **(c)** 2 92. Consider the following statements • 1. tan B increases faster than sin O as B increases, 2. The value of sinB+cose is always greater than 1. Which of the above Statements is/ are correct? 1 only **(d)** 2 only

82. [CDS (I) Paper 10 Q87] Q sine-cose=O t, sin4 + 0 t? 2 3 4 4 88. **(a) (c)** 2 sin **(d)** 2cos9 89. fiF ABC, c t, + B) + sin(A+ B) t? 90. sin a , cosß=—, tany 2 **(a) (b)** 1050 1200 1350 150• B-BCRB-M-NBU 148A 29 91. sin6e 3sin2 Bcos29 **(b)** 4 92. . 1. e tan e 2. sinO+cose 1 (sin9+cos9) (tanB+cot9) 93. sece + cosec9 **(b)** 2 **(c)** sin **(d)** cos 9 29

83. [CDS (I) Paper 10 Q92] 93. 29 1. e tan e 2. sinO+cose 1 (sin9+cos9) (tanB+cot9) sece + cosec9 **(b)** 2 **(c)** sin **(d)** cos 9 B-BCRB-M-NBU 148A **(a)** 4 2 88. cosec **(b) (c)** 2 sin **(d)** 2cos9 — sec

84. [CDS (I) Paper 10 Q94] What is (1 + sece — tan 9) coso (l + secB+ tanB) (1 — sinB) equal to? **(c)** tan e **(d)** cote A spherical balloon of radius r subtends angle 60' at the eye of an observer. If the angle of elevation of its centre is 600 and h is the height **(a)** tan e **(b)** cote A spherical balloon of radius r subtends angle 60' at the eye of an observer. If the angle of elevation of its centre is 600 and h is the height 95. 96. **(c)** 30 **(d)** 35 of the which

85. [CDS (I) Paper 11 Q52] In a trapezium, the two non-parallel sides are equal in length, .each being of 5 units. The parallel sides are at a distance of 3 units apart. If the smaller side of the parallel sides is Of length 2 units, then the sum of the diagonals of the trapezium is **(a) (b) (c) (d)** I OJS units 60S units 5-v'€ units units The shadow of a tower standing On a level plane is found to be 50 m longer when the Sun's elevation is 300 than when it is

86. [CDS (I) Paper 11 Q54] 55. of 56. (14 _ A) If Cos x sec x 2, then What is cosn x + see x equal to, where n is a positive integer ? **(a) (c)** 2 20-2 2n-1 What is sin 250 sin 350 sec 650 see 550 to ? The value of cos 250 — sin 250 is equal **(a) (b) (d)** positive but less than I positive but greater than I negative a right-angled triangle ABC, right-angled 4 at B, if cos — , then what is sin C equal to ? **(a) (c)**

87. [CDS (I) Paper 11 Q55] 56. If Cos x sec x 2, then What is cosn x + see x equal to, where n is a positive integer ? **(a) (c)** 2 20-2 2n-1 **(a) (b) (c) (d)** I OJS units 60S units 5-v'€ units units What is sin 250 sin 350 sec 650 see 550 to ? The value of cos 250 — sin 250 is equal The shadow of a tower standing On a level plane is found to be 50 m longer when the Sun's elevation is 300 than when it is

88. [CDS (I) Paper 11 Q59] 60. If u and are complementary angles, then what is equal to ? **(a) (b) (c) (d)** If 2 cot e — equal to ? **(c)** If sin O cos = equal to ? sec COS a — tan ß 2 — sin 6 — 3, then What is 2 g 4 sin O — , then what is sin + cos B **(b) (c)** 2 3 If sec tan B sec ? (e) B-ABQA-N-NBu what is the value of (16- [en

89. [CDS (I) Paper 11 Q61] 62. 63. of 64. (16-A) What is (750 e) equal to ? **(d)** — (550 + e) + cot — O) If sin where O — , then what is 2 sin — cos 6 equal to ? **(a) (c)** If tan 80 = eot 29, where O < 89 — what is the value of tan 50 ? **(b) (d)** , then A square is inscribed in a circle of diameter 2a and another square is circumscribing the circle. The difference between the areas Of outer and inner squares is **(a) (b) (c) (d)** 2a2 4a2

90. [CDS (I) Paper 11 Q63] 64. What is (750 e) equal to ? **(d)** — (550 + e) + cot — O) **(a) (b) (c) (d)** sec COS a — tan ß If 2 cot e — equal to ? **(c)** If sin O cos = equal to ? 2 — sin 6 — 3, then What is 2 g 4 sin O — , then what is sin + cos B If sin where O — , then what is 2 sin — cos 6 equal to ? **(a) (c)** If tan 80 = eot 29, where O < 89 — what is the value of tan 50 ? **(b) (d)** , then **(b) (c)** 2 3 If sec tan B sec ? (e) B-ABQA-N-NBu what is the value of A square is inscribed in a circle of diameter 2a and another square is circumscribing the circle. The difference between the areas Of outer and inner squares is **(a) (b) (c) (d)** 2a2 4a2

91. [CDS (I) Paper 11 Q59] 60. — sin ? **(a) (b) (d)** sec COS a Sin a — tan 2 cot0=3 **(a) (c)** 4 sin e cos — 2 3 see 9 + tan G 2 O — sin 9 2 cos + sin O t, sin6 e cos O sec O **(a) (c) (d)** 5 B-ABOA.N-N3U

92. [CDS (I) Paper 11 Q62] 64. 17 -A) cosec (750 + e) — sec (150 — -tan sin6.2coso=1.78t 2sine-cos0 2 2 **(d)** tan 80 = 0't2e e, E, tan 50 ? 2a **(b)** (e) **(d)** 2a2 3a2 4aZ **(a) (b) (c)** sec COS a Sin a — tan 2 cot0=3 **(d)**

93. [CDS (I) Paper 11 Q57] 58. 59. — sin ? **(a) (b) (d)** sec COS a Sin a — tan 2 cot0=3 **(a) (c)** 4 sin e cos — 2 O — sin 9 2 cos + sin O sin6 e cos sec O

94. [CDS (I) Paper 11 Q63] 64. (17-A) cosec (750 + e) — sec (150 — A) — tan (550 + e) cot (350 — e) **(a)** (e) **(d)** sin 642 cos B=IÄ 2 sine-cose tan 86 = cot 20 0 89 tan 50 ? **(a) (b) (d) (a) (c) (d)** 2 3 see 9 + tan G 5 **(a) (d)** 2a2 3a2 B-ABOA.N-N3U

95. [CDS (I) Paper 11 Q76] If sin I, where O < B < 450, what is cos (A —B) equal to ? **(a) (b) (c) (d)** 8 cm and IO cm 9 and 11 cm IO cm and 12 em ll.cm and 13 cm sin 2B sin B cos 2B cos B For a plot Of land Of 100 m x 80 m, the level is to be raised by spreading the earth from a stack Of a rectangular base IO m x 8 m with vertical section being a trapezium Of height 2 m. The top of the stack is 8m X 5 m. By how many centimeters can the level be raised ? A cylinder is surmounted by a cone at One end, a hemisphere at the Other end. The radius is cm, the height of the cylinder Lis 6-5 cm and the total height or the structure is 12•8 cm. The volume V Of the structure lies between

96. [CDS (I) Paper 11 Q79] 20-A) then **(a)** sin 2B sin B cos 2B cos B For a plot Of land Of 100 m x 80 m, the level is to be raised by spreading the earth from a stack Of a rectangular base IO m x 8 m With vertical section being a trapezium Of height 2 m. The top of the stack is 8m X 5 m. By how many centimeters can the level be raised ? **(a) (b) (c) (d)** 3 cm AB is a straight line; C and D are points on the same side Of AB such that AC is perpendicular to AB and DB perpendicular to AB. Let AD and BC meet at E. AE BE What is _ + equal to ? **(a) (b)** AD BC 1-5 N one Of the above Consider an equilateral triangle Of a Of unit length. A new equilateral triangle is formed by joining the mid-points Of the first one, then a third equilateral triangle is formed by joining the mid-points of the second. The process is continued. fie sum perimeter of all triangles thus formed is **(a) (c) (d)** 2 units 3 units 6 units Infinity 73. what is cos (A —B) equal to ?

97. [CDS (I) Paper 12 Q53] A railroad curve is to be laid on a circle. What radius (approximate) Should be used if the track is to direction by 250 in a distance of 120 m? **(a)** 300 m **(b)** 280 m **(c)** 275 m **(d)** 264 m B-ABQB-N-LZS/ IA If then what equal to ? **(b) (c)** cose — sin O sine — cose — sin8) COs sin e SS. If tan9+cot9— 2, then sine + cose equal to? 2 What is

98. [CDS (I) Paper 12 Q56] What is equal to? cotx tan x **(a)** sin x **(b)** cos x **(c)** tan x **(d)** cot x 18

99. [CDS (I) Paper 12 Q57] From a certain point on a straight road, a person observes a tower in the west direction at a distance of 200 m. He walks some distance •along the road and finds that the same tower is 3CO m south of him. What is the shortest distance of the tower from the road? 300 soo **(c)** 900 **(d)** 13 sin x —cos x + I What is equal to? sin X + cos x — I sin x — I COS X sin x + I **(b)** sin x — cos x + I sin X +1 **(d)** COS x + 1 B-ABQB-N-LZS/ IA 59. in of 60. (sin x cos y- cos xsin y) What is (sin x —cos equal to? sm x cos **(a) (c) (d)** sin 4 x — COS sin x — cos Cos x — sin sm x — cos What is (sin x cos y + cos x sin y) equal to? **(a)** cos x — cos — Sin COS X sm x — Y 57. (sin x cos y- cos xsin y) X — Sin From a certain point on a straight road, a person observes a tower in the west direction at a distance of 200 m. He walks some distance •along the road and finds that the same tower is 3CO m south of him. What is the shortest distance of the tower from the road? 300 soo **(c)** 900 **(d)** 13 sin x —cos x + I What is equal to? sin X + cos x — I sin x — I COS X sin x + I **(b)** sin x — cos x + I sin X +1 **(d)** COS x + 1

100. [CDS (I) Paper 12 Q59] 60. What is (sin x —cos equal to? sm x cos **(a)** 900 **(b)** 13 sin x —cos x + I What is equal to? sin X + cos x — I sin x — I COS X sin x + I **(c)** sin x — cos x + I sin X +1 **(d)** COS x + 1 59. 60. What is

101. [CDS (I) Paper 12 Q57] 200 m 300m 300 Jfä 500 600 4Tä 900 **(d)** sin x —cos x + I sin X 4-cosx — I sin X — I **(a)** sin x —cos x + I sin X 4-cosx — I cos x —cos y sin X — I **(b)** COS X sin x + I cos x y COS X sin x —I sin x —cos y **(c)** COS X + I sin x + I **(d)** sin 2 x — sin y cosx+ i B-ABQB-N-LZS/ IA 59. 60. 21

102. [CDS (I) Paper 12 Q59] 60. cos x —cos y cos x y sin x —cos y sin 2 x — sin y 21 **(b) (d)** sin x — COS sin 6 x — COS cos x X —COS (sin x cos y + cos x sin y) (sin xcos y —cos x sin y) t? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

103. [CDS (I) Paper 12 Q61] What (1 4 cot x — + tan x + secx) equal to? **(c)** sin x **(d)** cosx **(a)** sin x **(b)** cosx 62. What is **(c)** sin x + cos x **(d)** sin x — cos x 63. Consider the following statements • l. sin > sin I 2. COS < COS Which Of the above Stater-nents 66. What is the number of pairs of perpendicular planes in a cuboid?

104. [CDS (I) Paper 12 Q62] What is (cosecx — sin x) (secx —cos X) (tan x + cot x) equal to? **(a)** sin x **(b)** cosx 62. What is **(c)** sin x + cos x **(d)** sin x — cos x 63. Consider the following statements • l. sin > sin I 2. COS < COS Which Of the above Stater-nents 66. What is the number of pairs of perpendicular planes in a cuboid?

105. [CDS (I) Paper 12 Q63] Consider the following statements • l. sin > sin I **(a)** sin x **(b)** cosx 62. What is **(c)** sin x + cos x **(d)** sin x — cos x 63. Consider the following statements • l. sin > sin I 2. COS < COS Which Of the above Stater-nents 66. What is the number of pairs of perpendicular planes in a cuboid?

106. [CDS (I) Paper 12 Q2] COS < COS Which Of the above Stater-nents is / are correct? **(a)** I Only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2

107. [CDS (I) Paper 12 Q65] Sin **(a) (c)** If San p **(a)** x + 9 x equal to? 2 512 **(b)** 18 **(d)** 1024 sin x + cos x P x + cos 3 x q, then what — 3 p equal to? **(d)** 4q B-ABQB-N-LZS/ IA is 22

108. [CDS (I) Paper 12 Q2] DCM + ABM ABCD e I 61. (l t? (e) sin x **(d)** cos x (cosecx — sin x) (sec x — cos x) (tan x + cot*) **(a)** cos x **(b)** sin X + cos x **(c)** sin x — cos x 63. • l. sin r > sin 1 2. cos l' cos I **(d)** 4q 66. 67.

109. [CDS (I) Paper 12 Q66] 67. **(c)** 12 **(c) (d)** 64. sin **(a)** 65. sin t, sin x + cosecx 2 2 512 **(b)** 18 **(d)** 1024 sin x + COs x p X+cos3 x z q t, ABCD t, AB, 68. •

110. [CDS (I) Paper 13 Q62] If COSA — tan B, cos B tan C and cos C = tan A, then sin A is equal to JB-I **(a) (b)** , where — , where K < — or K > 3 **(c) (d)** I only 2 only Both I and 2 Neither I nor 2 62. If COSA — tan B, cos B tan C and cos C = tan A, then sin A is equal to JB-I

111. **[CDS (I) Paper 13 Q65]** The angle Of elevation Of a cloud from a point 200 m above a lake is 300 and the angle of depression Of its reflection in the lake is **(a)** 200 m 300 m 400 m 600 m tan A + cotA 4 68. Consider the following cos A sin A I — tan A I —cot A z sin A + cos A 2. **(b)** 2 only Both I and 2 Neither I nor 2 66. 67. If -cot2 330 - tan2 23' then tan 4 A + cot 4 A is equal to 110 191 194 — sin x I — sin x I + sin x cos x I * sin x then which Of the following is/ are correct? 1. p-q=r 2 - p2 qr Select the correct answer using the code given below. 69. 70. **(c) (d)** ABC is a triangle right angled at B and What is sinA•sinB+sinC equal to? 5 The value Of

112. **[CDS (I) Paper 13 Q66]** 67. If then tan 4 A + cot 4 A is equal to 110 191 194 — sin x I — sin x I + sin x cos x I * sin x then which Of the following is/ are correct? 1. p-q=r 2 - p2 qr Select the correct answer using the code given below. **(a)** 200 m 300 m 400 m 600 m tan A + cotA 4 68. Consider the following cos A sin A I — tan A I —cot A z sin A + cos A 2. **(b)** 2 only Both I and 2 Neither I nor 2 66. 67. If -cot2 330 - tan2 23' then tan 4 A + cot 4 A is equal to 110 191 194 — sin x I — sin x I + sin x cos x I * sin x then which Of the following is/ are correct? 1. p-q=r 2 - p2 qr Select the correct answer using the code given below. 69. 70. **(c) (d)** ABC is a triangle right angled at B and What is sinA•sinB+sinC equal to? 5 The value Of

113. **[CDS (I) Paper 13 Q68]** Consider the following cos A sin A I — tan A I —cot A z sin A + cos A 2. (I —sinA — cos A J 2 — +cosA) Which of the above is / are identity/ identities? I only **(b)** 2 only Both I and 2 **(c) (d)** Neither I nor 2 **(a)** 200 m 300 m 400 m 600 m tan A + cotA 4 68. Consider the following cos A sin A I — tan A I —cot A z sin A + cos A 2. **(b)** 2 only Both I and 2 Neither I nor 2 66. 67. If -cot2 330 - tan2 23' then tan 4 A + cot 4 A is equal to 110 191 194 — sin x I — sin x I + sin x cos x I * sin x then which Of the following is/ are correct? 1. p-q=r 2 - p2 qr Select the correct answer using the code given below. 69. 70. **(c) (d)** ABC is a triangle right angled at B and What is sinA•sinB+sinC equal to? 5 The value Of

114. **[CDS (I) Paper 13 Q70]** The value Of cosec 6T + sec2 5T -cot2 330 - tan2 23' **(b) (c) (a)** 200 m 300 m 400 m 600 m tan A + cotA 4 68. Consider the following cos A sin A I — tan A I —cot A z sin A + cos A 2. **(b)** 2 only Both I and 2 Neither I nor 2 66. 67. If -cot2 330 - tan2 23' then tan 4 A + cot 4 A is equal to 110 191 194 — sin x I — sin x I + sin x cos x I * sin x then which Of the following is/ are correct? 1. p-q=r 2 - p2 qr Select the correct answer using the code given below. 69. 70. **(c) (d)** ABC is a triangle right angled at B and What is sinA•sinB+sinC equal to? 5 The value Of

115. **[CDS (I) Paper 13 Q1]** There exists at least one value Of x between O and — which satisfies the equation sin x —2sin x 2 _ sin 5 is greater th an cos I - S. Which Of the above statements is/are correct? **(a)** only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 From the top Of a tower, the angles Of depression of two objects p and Q (situated on the ground on the same side of the tower) separated at a distance of 10013 — N'S) m are 450 and 6CP respectively. The height of the tower is 200 m 250 m 300 m None Of the above sin x + cos x c then sin x + cos6 x is equal to 1+6c2 -3c4 **(a)** 16 1+6c2 — 3c4 **(b)** 1+6c2 + 3c4 **(c)** 16 1 6c2 4 4 B-EFUA-O-PDW / 52A

116. **[CDS (I) Paper 13 Q74]** 75. 26 Consider the following statements There exists no value Of x such that I — sin x 2. sin x = 3 sin 2 x does not hold good for any real x. Which Of the above statements is / are correct? **(a) (b) (c) (d)** 1 only 2 only Both 1 and 2 Neither I nor 2 A triangle ABC is inscribed in a circle. If sum Of the squares Of sides Of the triangle is equal to twice the square of the diameter, then sin A + sin B + sin C is equal to **(d)** 2 3 4 None of the above Consider the following statements There exists no value Of x such that I — sin x 72. **(b) (c)**

117. **[CDS (I) Paper 14 Q37]** If x=acosO and y=bcot9, then + by-I) is equal to **(c)** tan29 **(d)** sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine **(b)** coso tano + I tang — I **(d)** sin 6 **(a)** tan29 **(b)** sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine **(c)** coso tano + I tang — I **(d)** sin 6 39. If tan

118. **[CDS (I) Paper 14 Q39]** If where B — 2 tan(x + + 20) — x) Otan(SO — = I, then the value of x is equal to **(a)** tan29 **(b)** sin2e COS 9 38. is equal to I — sine tang— I tang +1 1 + sine **(c)** coso tano + I tang — I **(d)** sin 6 39. If tan

119. **[CDS (I) Paper 15 Q61]** 62 A square is inscribed in a right-angled triangle with legs p and q, and has a Common right angle with the triangle. The diagonal of the square is given by **(a)** opq **(c) (d)** x tan 9 + Y = I, then tan e and — the value of _ is **(a)** 2sec2 e **(b)** sec e **(c)** cos 2 2cos2 B

120. **[CDS (I) Paper 15 Q65]** Two observers are stationed due north Of a tower (of height x metre) at a distance y metre from each other. The angles Of elevation of the tower observed by them are 30' and 45' respectively. Then x y is equal to **(a)** 2 -Be , thenC+2B **(b)** 0-1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e **(c)** only **(d)** 2 only

121. **[CDS (I) Paper 15 Q96]** The value of -loglo 3125-410g10 2 32 5 is If tane + cote , where O < < then 2 sin + cos 9 is equal to 0-1 **(c)** 2 B-ABPA-P-JRW / 54A **(a)** AM < Median **(b)** GM Median **(c)** AM = Me dian Suppose for Which one of the corre ct? **(d)** GM>1/6

122. **[CDS (I) Paper 16 Q72]** If sin x + sin2x=l, then what is the value of cossx + 2cos6x + cos x ? **(a)** 4275 **(b)** 4350 **(c)** 4375 **(d)** 4450 68. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ?

123. **[CDS (I) Paper 16 Q73]** A man from the top of a 100m high tower sees a car moving towards the tower at an angle of depression **(a)** 4275 **(b)** 4350 **(c)** 4375 **(d)** 4450 68. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ?

124. **[CDS (I) Paper 16 Q74]** Two men on either side of a tower 75 m high observe the angle of elevation Of the top Of the tower to be 300 and **(a)** 1006m **(b)** 75-6 m 3 **(c)** 606 m 75. What is the value of cosw2680 + sec2560 — cot2340 — tan2220 ? 76. If 2y Coso x Sino and 2r sece — y cosec9 = 3, then what is x2 + 4y2 equal to ? 77. IfsinO+ coso= where O < 9 < — 2' 2 then what is tan O + coto equal to ? 4 SDC-S-TRD - A **(d)** 78. 4 If A sin20 + cos40 where < — 2' then which one of the following is correct ?

125. **[CDS (I) Paper 16 Q78]** If Æ=sin29 + cos40 where 2 then which one of the following is correct ? **(a)** 1006m **(b)** 75-6 m 3 **(c)** 606 m 75. What is the value of cosw2680 + sec2560 — cot2340 — tan2220 ? 76. If 2y Coso x Sino and 2r sece — y cosec9 = 3, then what is x2 + 4y2 equal to ? 77. IfsinO+ coso= where O < 9 < — 2' 2 then what is tan O + coto equal to ? 4 SDC-S-TRD - A **(d)** 78. 4 If A sin20 + cos40 where < — 2' then which one of the following is correct ?

126. **[CDS (I) Paper 16 Q79]** What is cot A —cosec A + 1 1 + cos A sin A 1 —cos A **(b)** sin A 1+ sin A cos A 1 — sin A cos A **(a)** 1006m **(b)** 75-6 m 3 **(c)** 606 m 75. What is the value of cosw2680 + sec2560 — cot2340 — tan2220 ? 76. If 2y Coso x Sino and 2r sece — y cosec9 = 3, then what is x2 + 4y2 equal to ? 77. IfsinO+ coso= where O < 9 < — 2' 2 then what is tan O + coto equal to ? 4 SDC-S-TRD - A **(d)** 78. 4 If A sin20 + cos40 where < — 2' then which one of the following is correct ?

127. **[CDS (I) Paper 16 Q80]** Consider the following : 1. sin I n > sin 1 c **(a)** 1006m **(b)** 75-6 m 3 **(c)** 606 m 75. What is the value of cosw2680 + sec2560 — cot2340 — tan2220 ? 76. If 2y Coso x Sino and 2r sece — y cosec9 = 3, then what is x2 + 4y2 equal to ? 77. IfsinO+ coso= where O < 9 < — 2' 2 then what is tan O + coto equal to ? 4 SDC-S-TRD - A **(d)** 78. 4 If A sin20 + cos40 where < — 2' then which one of the following is correct ?

128. **[CDS (I) Paper 16 Q2]** COS IO < COS I C 3. tan 10 > tan IC Which Of the above are not correct ? **(a) (b) (c) (d)** 22 I and 2 only 2 and 3 only I and 3 only 1, 2 And 3

129. **[CDS (I) Paper 16 Q76]** 2y cos9= x sing 3åq 2x secØ—y cosecO= 3 t, x2 + 4y2 ? 77. sin9+ 2 2 t, tano + cote ? 4 **(b)** 73 23 4 78. = sin2Ö+cos4et, t, 4 13 —s AS2 16 13 16 cot A +cosec A — I 79. cot A — cosec A + 1 I + cos A sin A I — cos A **(b)** sin A 1 + sin A cos A — sin A **(d)** Cos A 80. : I. sin 10 > sin I c 2. cos 10 < cos I c 3. tan 10 > tan IC **(a) (c) (d)** 23 2 3fr(3 1, 2 3å<3 A - SDC-s-TRD

130. **[CDS (I) Paper 16 Q81]** If tan2x+ = 2 and 00 < x < 900, tan x then what is the value of x ? **(a)** 150 **(b)** 300 **(c)** 450 **(d)** 600

131. **[CDS (I) Paper 16 Q82]** Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ? **(a)** 150 **(b)** 300 **(c)** 450 **(d)** 600

132. **[CDS (I) Paper 16 Q83]** Consider the following : COS 750 sin 150 cos 350 2. sin 550 sin 800 3. cos sin 120 cos 180 cos 780 sin 720 sin 1 10 cos 790 cos 280 cosec 620 = 1 —sin 590 sec 310=0 Which Of the above are correct ? **(a)** 1 and 2 only **(b)** 2 and 3 only **(c)** I and 3 only **(d)** 1, 2 and 3 SDC-S-TRD - A 24

133. **[CDS (I) Paper 16 Q86]** 83C I I z 85 I 3iqs 79 C, 81 g I X, **(b)** 805 **(a) (c) (d)** 25 81-5 79-0 78-0 A - SDC-S-TRD 84. (ZÜFT) ABC Qi 81. tan x **(a)** 150 **(b)** 300 **(c)** 450 **(d)** 600 82. **(b)** 805 **(c)** 85. tan10 tan40. **(a)** 150 **(b)** 300 **(c)** 450 **(d)** 600 83. : .. tan890 cos 750 sin 150 cos 350 2. sin 550 sin 800 3. cos 100 sin 120 cos 180 COs 780 sin 720 sin 1 IO cos 790 cos 280 cosec 620 = -sin 590 sec310=O

134. **[CDS (I) Paper 17 Q63]** 64. 2z tan If x sin y = , then what is I — tan2 e 4z2 (x2 + y2) equal to ? **(a) (b) (c)** (x2 + y.2)3 (x2 — y2)2 (x2 —y2)3 If cos 91 + cos 92 + cos 3, then what is sin GI + sin + sin 03 equal to ? **(a) (b) (c)** 2 3 What is the value Of G which satisfies the cos + **(a) (b) (c) (d)** 450 600 What is the value Of **(a) (c) (d)** BLS-S-FDE sin x tan x 1 (18-

135. **[CDS (I) Paper 17 Q67]** A — sin 4 A What is equal to ? **(a) (b) (c) (d)** ms A — sin2A cos2 A — sin2 A cos A — sin A 2 If 7 sin2 x + 3 cos2 x = 4, O < x < 900, then what is the value Of tan x ? **(b) (d)** 2 An aeroplane flying at a height Of 300 m above the ground passes vertically above another plane at an instant when the angles Of elevation Of the two planes from the same point On the ground are 600 and 450 respectively. What is the height of the lower plane from the ground ? **(a) (b) (c) (d)** 100 Mfm 100 50•.16 m 150 (vfi+l)m

136. **[CDS (I) Paper 17 Q70]** 71. If x a cos B + b sin O and y = a sin — b cos 6, then What is + Y2 equal to ? **(a) (b) (c) (d)** From the top of a building 90 m high, the angles of depression of the top and the bottom Of a tree are 300 and 450 respectively. What is the height Of the tree ? **(a) (b) (c) (d)** 30 Jäm 90-30 Jim 90 + m GO * m Which one Of the following triples does not represent the sides Of a triangle ? **(b) (c) (d)** (4, 7, 10) If the perimeter Of a rectangle is IO cm and the area is 4 cm2, then its length is **(a) (b) (c) (d)** •BLS-S-FDE 6 cm (20 - so,

137. **[CDS (I) Paper 17 Q100]** From an aeroplane vertically over a straight horizontal road, the angles of depression of two consecutive kilometre-stones on the opposite sides Of the aeroplane are observed to be and B. The height of the aeroplane above the road is tan a + tan **(a) (b) (c)** 2, 3 and 4 only I, 2 and 4 only I, 3 and 4 only 1 and 2 only **(d)**

138. **[CDS (I) Paper 18 Q46]** If a triangle has sides 5, 13 and 12 units and 0 49 is the acute angle of the triangle, then what is the value of (sin 0 + cos 0) ? 5 13 7 **(b)** 13 12 13 17 13 **(a)** STS-WTPT 46. If a triangle has sides 5, 13 and 12 units and 0 49. is the acute angle of the triangle, then what is the value of **(b)** 13 12 13 17 13 If O < x < — , then **(c) (d)**

139. **[CDS (I) Paper 18 Q47]** If O < x < — , then (sin x + cosecx) **(a) (b) (c) (d)** 48. (14 -A

140. [CDS (I) Paper 18 Q51] If sin O and O < < — , 2 then) **(a)** STS-WTPT 46. If a triangle has sides 5, 13 and 12 units and 0 49. is the acute angle of the triangle, then what is the value of **(b)** 13 12 13 17 13 If O < x < — , then **(c) (d)**

141. [CDS (I) Paper 18 Q5] 5 Esos— 5 5 7Tt 15 15 If sin O what is the value of 2mn **(c)** STS-WTPT **(a)** STS-WTPT 46. If a triangle has sides 5, 13 and 12 units and 0 49. is the acute angle of the triangle, then what is the value of **(b)** 13 12 13 17 13 If O < x < — , then **(c) (d)**

142. [CDS (I) Paper 18 Q50] 51. and 450 600 sec 450 — tan 450 , then which one of the u)sec 450 + cot 450 following is correct ? and cos 0 ? then 2' **(a) (b) (c) (d)** If 0 measured in radians is the angle between the hour hand and the minute hand of a clock when the time is 4 : 36 pm, then which one of the following is correct ? **(a) (b) (d)** 47t 5 5 37t

143. [CDS (I) Paper 18 Q1] 1 + cos G = x2 for If O' < < 450, then I — cos 9 some real number x > **(a) (b) (c) (d)** STS-M-TPT 52. 53.

144. [CDS (I) Paper 18 Q56] A man, standing at a point X on the bank XY Of a river that cannot be crossed, observes a tower to be N ao E on the opposite parallel bank. He then walks 200 m along the bank to the point Y towards East, and finds the tower to be N W. From these observations, the breadth of the river will be (Given that tan = 2 and tan 0-5) **(a) (b) (c) (d)** 60 m 70 m 80 m 90 m sin 1 e The value of where IC represents sin IC 1 radian is **(a) (b) (c) (d)** Equal to 1 Less than 1 Greater than 1 but less than 2 Greater than 2 The diameters of two given circles are in the ratio 12 : 5 and the sum Of their areas is equal to the area of a circle of diameter 65 cm. What are their radii ? **(a) (b) (c) (d)** 12 and 5 24 cm and 10 cm 60 cm and 25 cm 30 cm and 12•5 cm **(a) (b) (c) (d)** STS-M-TPT

145. [CDS (I) Paper 19 Q57] What is cot 10 equal to ? **(b) (c)** cot 230 cot 450 cot 670 cot 890 **(a) (b)** What angle does the hour hand of a clock 56. describe in IO minutes of time ? 59. 57. **(c) (d)** 50 100 Each side of a square subtends an angle of 600 at the tip of a tower of height h metres standing at the centre Of the square. If I is the length of each side of the square, then what' is h2 equal to ?

146. [CDS (I) Paper 19 Q58] What angle does the hour hand Of a clock describe in IO minutes of time ? **(a) (c)** 50 100 Consider the following statements : I. (sec2B — I) (I — I 2. sin e (I + cos + (I + cos e) (sin = Which Of the above Ware correct ? **(b) (c) (d)** only 2 only Both I and 2 Neither I nor 2 SPSS-T-XHI 2 cosec (18-A 9

147. [CDS (I) Paper 19 Q60] Each side of a square subtends an angle of 600 at the tip of a tower of height h metres standing at the centre Of the square. If I is the length of each side of the square, then what' is h2 equal to ? **(a)** 212 12 **(b)** 312 2 212 **(d)** 3 From a height Of h units, a man observes the angle of elevation as and angle of depression as Of the top and the bottom respectively Of a tower of height H 4h). To what further height should he climb so that the values of angle Of elevation and angle Of depression get interchanged for the top and bottom of the tower ? **(b) (d)** H — h units H — 2h units H — 3h units H — 4h units If sec x cosec x 2, then what is tann x + cotn x equal to ? **(a) (b) (c) (d)** 2n—1 What is cot 10 cot 230 cot 450 cot 670 cot 890

148. [CDS (I) Paper 19 Q59] 57. (18-A) **(a) (c)** 50 100 Each side of a square subtends an angle of 600 at the tip of a tower of height h metres standing at the centre Of the square. If I is the length of each side of the square, then what' is h2 equal to ? **(a)** 212 12 **(b)** 312 2 212 **(d)** 3 From a height Of h units, a man observes the angle of elevation as and angle of depression as Of the top and the bottom respectively Of a tower of height H 4h). To what further height should he climb so that the values of angle Of elevation and angle Of depression get interchanged for the top and bottom of the tower ? Consider the following statements : I. (sec2B — I) (I — I 2. sin e (I + cos + (I + cos e) (sin = 2 cosec Which Of the above Ware correct ? **(b) (d)** H — h units H — 2h units H — 3h units H — 4h units

149. [CDS (I) Paper 19 Q63] 64. If cos x + cos2 x = I, then what is sin x sin4 x equal to ? **(a) (c)** 1-5 3 If sin A + cos A = p and sin3 A + cos3 A then which one of the following is correct ? **(a) (b) (c)** qS-3q+2p = o p3 — 3p 2q O p3 • 3p + 2q = O sec2 0 — tan O If x , then which one of the sec2 B + tane following is correct ? **(a) (b) (c) (d)** — sxsa ABC is a right angled triangle with base BC and height AB. The hypotenuse AC is four times the length Of the perpendicular drawn to it from the opposite vertex. What is tan C equal to ? **(b) (c)** SPSS-T-XHI 2-1ä Jä-l Jä+l (20-A

150. [CDS (I) Paper 19 Q63] 64. where O < 6 < — If cos 0 2 2 tan 0 is equal to 1— tan2 0 **(a)** 4/3 **(b)** -4/3 **(c)** 1/3 , then If 0 < 0 < 900, O < 900 and cos 9 < cos 'i, then which one of the following is correct ? **(c)** 900 **(d)** No conclusion can be drawn On the top of a hemispherical dome of radius r, there stands a flag of height h. From a point on the ground, the elevation of the top of the flag is

151. [CDS (I) Paper 19 Q1] 2. cos 0 cos 0 = 4 , where If 1 — sin 0 1 + sin 6 O < O < 900, then O = **(a)** 4/3 **(b)** -4/3 **(c)** 1/3 , then 2 65. 66. 67. 1. 2. cos 0 cos 0 = 4 , where If 1 — sin 0 1 + sin 6 O < O < 900, then O = 600. If 3 tan O + cot O = 5 cosec O, where O < 9 < 900, then 0 = 600. If 0 < 0 < 900, O < 900 and cos 9 < cos 'i, then which one of the following is correct ? **(d)** 900

152. [CDS (I) Paper 19 Q72] If sin2 x + sin x = 1, then what is the value of cos **(b) (c) (d)** x +3 cos x +3 cos x + cos6x? 1 8 If 3 sin 0 + 5 cos = 4, then what is the value of (3 cos O — 5 sin ? **(b)** 12 16 **(d)** 18 If cot (1 + sin 0) = 4m and cot 0 (1 — sin B) 4n, then which one of the following is correct ? **(a)** (m2 + n2)2 mn **(b)** (m2 — n2)2 = mn **(c)** (m2 — n2)2 = m 2n2 **(d)** (m2 + n2)2 = m2n2 If base and hypotenuse of a right triangle are (u-2 — v2) and (u2 + v2) respectively and the area of the triangle is 2016 square units, then the perimeter of the triangle may **(a)** 224 units **(b)** 288 units **(c)** 448 units **(d)** 576 units A circle is inscribed in an equilateral triangle of side of length l. The area of any square inscribed in the circle is **(a)** 12 4 4 **(d)** 6 GTHY-T-XMT (23- of

153. **[CDS (I) Paper 19 Q73]** 68. If 3 sin 0 + 5 cos = 4, then what is the value **(a) (b) (c)** x +3 cos x +3 cos x + cos6x? 1 8 Walls **(d)** 12 16

154. **[CDS (I) Paper 19 Q71]** 72. (23-A) cos **(b) (c) (d)** x +3 cos x +3 cos x + cos6x? 1 8 Walls (excluding roofs and floors) of 5 identical rooms having length, breadth and height 6 m, 4 m and m respectively are to be painted. Out of five rooms, two rooms have one square window each having a side of 2-5 m. Paints are available only in cans of 1 litre; and 1 litre of paint can be used for painting 20 square metres. The number of cans required for painting is of (3 cos O — 5 sin ? **(b)** 12 16 **(d)** 18 If cot (1 + sin 0) = 4m and cot 0 (1 — sin B) 4n, then which one of the following is correct ? **(a)** (m2 + n2)2 mn **(b)** (m2 — n2)2 = mn **(c)** (m2 — n2)2 = m 2n2 **(d)** (m2 + n2)2 = m2n2 If base and hypotenuse of a right triangle are (u-2 — v2) and (u2 + v2) respectively and the area of the triangle is 2016 square units, then the perimeter of the triangle may **(a)** 224 units **(b)** 288 units **(c)** 448 units **(d)** 576 units A circle is inscribed in an equilateral triangle of side of length l. The area of any square inscribed in the circle is **(a)** 12 4 4 **(d)** 6 GTHY-T-XMT **(d)** 10 12 13 14

155. **[CDS (I) Paper 20 Q53]** 52n _ .23n 3 7 17 tan x l, O < x 2 Sin x cos x **(d) (a)** (e) sin 46' cog 440 + cos 460 sin 44' ? **(a) (b) (c) (b)** sin 20 2 0 < 0 900 t, 2 4 sin e COS 4 tan DFGT-T-HTM (16-$

156. **[CDS (I) Paper 20 Q56]** 16—A) ABCDEFV(I B 300 t, c 450 1 B C 1:2Jä tan IO tan 20 tan 30 **(a) (b) (c) (d)** 2 tan 890 ? sin 46' cog 440 + cos 460 sin 44' ?

157. **[CDS (I) Paper 20 Q54]** ABCDEFV(I B 300 t, c 450 1 B C **(d) (a) (b) (c) (d)** 2 tan 890 ? sin 46' cog 440 + cos 460 sin 44' ?

158. **[CDS (I) Paper 20 Q55]** 56. 1:2Jä tan IO tan 20 tan 30 **(a) (b) (c) (d)** 2 tan 890 ? sin 46' cog 440 + cos 460 sin 44' ?

159. **[CDS (I) Paper 20 Q52]** 53. The expression 52n — 23n has a factor **(a) (d)** 3 7 17 None Of the above If tan x I, O < x < 900, then what is the value Of 2 sin x Cos x ? (e) What is thc value of sin 46' cos 440 + cos 460 sin 44' ? **(c)** sin 20 2 Suppose O < < 900, then for every O, 4 sin2 0 + I ig greater than or equal to **(a) (b) (d)** 4 sin 4 tan O DFGT-T-HTM (17-

160. **[CDS (I) Paper 20 Q56]** Consider a regular hexagon ABCDEF. TWO towers are situated at B and C. The angle of elevation from A to the top of the tower at B is 30', and the angle Of elevation to the top Ofthe tower at C is **(a) (b)** 3 7 17 None Of the above If tan x I, O < x < 900 value Of 2 sin x Cos x ? **(c) (d)**

161. **[CDS (I) Paper 20 Q59]** If 3 tan O cot O where O is the value Ofo ? **(a) (c)** , then What What is the value Of sin2 250 + sin2 65' ? **(a) (b)** 2 4 What is the value Of sing e 0 3 sin2 0 cos2 0 — I ? **(a) (d)** 2 4 Consider the following for real numbers a, B,

162. **[CDS (I) Paper 20 Q2]** 4. sec a 1/4 tan 20 cos z 2 HOW many Of the above statements not possible ? **(a) (c) (d)** One TWO Three Four DFGT-T-HTM are (19 Consider the distribution : 0—10 10-20 20-30 following grouped frequency If the mean of the above data is 25•2, then what is the value Of p ? **(b) (c) (d)** 9 10 11 12

163. **[CDS (I) Paper 20 Q58]** 59. (19 —A) If 3 tan O cot O where O is the value Ofo ? **(a) (c)** 61. , then What Consider the distribution : following grouped 10 frequency What is the value Of sin2 250 + sin2 65' ? **(a) (b)** 2 4 What is the value Of sing e 0 3 sin2 0 cos2 0 — I ? If the mean of the above data is 25•2, then what is the value Of p ? **(b) (c) (d)** 9 10 11 12 62. **(a) (d)** 2 4 Consider the following for real numbers a, B,

164. **[CDS (I) Paper 21 Q35]** What isthevalueof2-!-E cos730 eos71P Sin 170 C - SDHY-T-MTH **(a)** Its lia ixuide O **(b)** Its lies on the triülgle **(c)** It 30. Let the of BAC of a aiangle ABC BC in X. Which one Of Blbwing is ? **(d)** AB<RX

165. **[CDS (I) Paper 22 Q52]** What is the least value of (25 cosec2x + sec2x) ? **(a)** 40 **(b)** 36 **(c)** 26 **(d)** 24 and 1009-900. If a cot n9, then which one of following is correct ?

166. **[CDS (I) Paper 22 Q54]** If tan 60* cot 20, where then what is the value of sec 40 ? 2 4 3 **(a)** 10m 56. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75 feet, then what is the height of the taller tower **(b)** 85 feet **(c)** 110 feet **(d)** 125 feet

167. **[CDS (I) Paper 22 Q56]** On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75 feet, then what is the height of the taller tower (approximately) ? **(a)** 85 feet **(b)** 110 feet **(c)** 125 feet **(d)** 140 feet 19 **(a) (c) (d)** 19

168. **[CDS (I) Paper 22 Q57]** The angles Of elevation Of the top Of a tower from two points at distances p and q from the base and on the same straight line are 270 and 630 respec- tively. What is the height of the tower ? **(a)** 10m 56. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75 feet, then what is the height of the taller tower **(b)** 85 feet **(c)** 110 feet **(d)** 125 feet

169. **[CDS (I) Paper 22 Q58]** What is the value of sin260 + sin2120 + sin2180 + + sin2840 + sin2900 ? cos 9 + — equal to ? **(a)** 10m 56. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75 feet, then what is the height of the taller tower **(b)** 85 feet **(c)** 110 feet **(d)** 125 feet

170. **[CDS (I) Paper 22 Q59]** What is I + sin O cote cosec0 **(b)** seco sec9 + cosecO cosece — cot O A - PIXI-T-MTK **(a)** 10m 56. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75 feet, then what is the height of the taller tower **(b)** 85 feet **(c)** 110 feet **(d)** 125 feet

171. **[CDS (I) Paper 22 Q65]** 48 cm 14cm I 775 168 725 168 375 84 325 84 66. (x+y+z)2 **(b)** xy+ yz+zx 20 sin 9 — coso + I sin O + I 60. sin 9 + cose — I cose **(c)** 2sin0 **(d)** 2cos9 61. (tan x + cot x cot y) + (cot x + cot — tan x tan y), sec x — tan x 62. secx + tan x sm x + cosx **(b)** tan x + cot x secx + tan x **(d)** cosec x + cot x 63. e, cote=— t, (sin9+cose) 16 69 65 79 65 I—2sin2 9cos2 0 sin' 9 + cos' 9 **(a)** 2sin0 **(b)** 2cos9 61. **(c)** tan x + cot x secx + tan x **(d)** cosec x + cot x 63. e, cote=— t,

172. **[CDS (I) Paper 22 Q60]** What is sine *coso — 1 cos O equal to ? **(c)** 2sine **(d)** 2cos0 **(a)** 2sine **(b)** 2cos0 61. What is **(c)** cosec x + cot x 63. If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 **(d)**

173. **[CDS (I) Paper 22 Q61]** What is (tan x + cot y) + (cot x + cot x tan y) equal to ? secx— tan x **(a)** 2sine **(b)** 2cos0 61. What is **(c)** cosec x + cot x 63. If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 **(d)**

174. **[CDS (I) Paper 22 Q62]** What is equal to ? Sec x + tan x sin x + cosx tan x + cot x secx + tan x **(d)** cosec x + cot x **(a)** 2sine **(b)** 2cos0 61. What is **(c)** cosec x + cot x 63. If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 **(d)**

175. **[CDS (I) Paper 22 Q63]** If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 (Sino + cos0) ? 69 65 79 65 21 • **(a)** 2sine **(b)** 2cos0 61. What is **(c)** cosec x + cot x 63. If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 **(d)**

176. **[CDS (I) Paper 22 Q66]** What is the area of the triangle having side lengths — + _ , **(a)** 2sine **(b)** 2cos0 61. What is **(c)** cosec x + cot x 63. If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 **(d)**

177. **[CDS (I) Paper 23 Q50]** ABC is a trianglé inscribed in a semicircle of diameter AB. unlat is cos (A + B) + sin (A B) equal to ? **(a) (d)** Consider the following statements : sin e x + — is B»ssible for some real value of x. cos O x — is possible for some real value of x. Which of the above statements Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 What is the magnitude (in radian) Of the interior angle of a regular pentagon ? DZOL-T-LKM (19- , of

178. **[CDS (I) Paper 23 Q2]** The equation 2 sin2 e — cos B + 4 = O is possible for all e. tan e + cot e cannot less than 2, where Which of the atN)ve statements Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 A road Curve is to be laid out on circle. What radius should be used if the track is to change direction by 420 in distance of 44 m ? 22 (Assume **(a) (c) (d)** 60 m 66 m 75m 80 m

179. **[CDS (I) Paper 23 Q48]** ABC is a trianglé inscribed in a semicircle of 51. diameter AB. unlat is cos (A + B) + sin (A B) equal to ? **(a) (b)** Consider the following statements : 49. 50. **(c) (d)**

180. **[CDS (I) Paper 23 Q49]** 50. (19-A) The difference between two angles is 150 and the sum Of the angles in radian is — bigger angle is k times the smaller angle. What k to ? **(a) (b)** Consider the following statements : 49. 50. **(c) (d)**

181. **[CDS (I) Paper 23 Q56]** 57. What is the maximum value Of 3 sin e **(b) (c) (d)** If sin e + cos e qi, then what sin6 9 + eos6 + 6 sin2 e cos2 0 equal to ? **(a) (b) (c)** qmat is the least value Of 9 sin2 + 16 cos2 e ? **(a) (c) (d)** 0 9 16 25 If cos 470 + sin 470 k, then what is the value of cos2 470 — sin2 470 ? -k 2-k2 **(c)** OZOL-T-LKM

182. **[CDS (I) Paper 23 Q58]** 59. e? (21 —A) If cosec e — sin O ps and sec B — then what is the value of tan O ? cos e **(c) (d)** p2q2 If O a, 900 such that cos (a — B) = I, then what is sin a — sin cog a — cos equal to ? **(a) (b) (c) (d)** 0 2 Consider the tdlowing statemene • The value of cos 610 sin 290 cannot exceed I. The value Of tan 230 — cot 670 is less than O. Which of the above statements Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

183. **[CDS (I) Paper 24 Q49]** What is the ratio of the greatest to the smallest value Of x, 3 SO. If the equation x2 + Y2 — 2xysin2 e = O contains real solution for x and y, then XVWS-T-MTK/SOA Consider the following inequalities : 2. cos6C > sin 5 T Which Of the above is/are correct? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 . If p • sece—tan9 and q = cowce+cote, then what is p + — l) equal to? 0 52 19 Sa. If cot B equal to? What is P.T.O. If p sin 2 + cos4 e for O s e s then consider the following statements I. p can less than 2. p can be more than I. Which Of the above statements is/ are correct? **(d)** I only 2 only Both and 2 Neither I nor 2 49. x, Consider the following inequalities : 2. cos6C > sin 5 T Which Of the above is/are correct? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 . If p • sece—tan9 and q = cowce+cote, then what is p + — l) equal to? 0 What is the ratio of the greatest the smallest value Of 2 —2 sinx — sin 3 52 19 Sa. If cot B equal to? What is SO. If the equation x2 + Y2 — 2xysin2 e = O contains real solution for x and y, then XVWS-T-MTK/SOA P.T.O.

184. **[CDS (I) Paper 24 Q54]** 55. cosa + cosß 2 3m sina sinß- O t, Osa s 9C, O S ß S 909 cosm -cos2ß t? 56. sece+cosB-—, os0S900 t, sin 2 e t? 2 3 57. (1+cot8 t sece) 4 3 2 XVWS-T-MTK/SOA **(a)** 11 **(b)** 22 **(c)** 77 **(d)** 99 k sece 59. 60' 60. 60• 45'

185. **[CDS (I) Paper 24 Q54]** ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß—2 and Bina + O, where O a 9(Y, O What is the value of cosm — ? 56. **(d)** 0 2 Cannot be determined due to insumcient data If + where O S S W, then what is the value of sin 2 ? 2 **(a)** 99 56. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß 2 and sin u + sinß where O a , O sßs9 **(b)** Cannot be determined due insumcient data to 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre If + where O S S W, then what is the value of sin 2 ? 2 57. What is **(c)** 99 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre 60. A vertical tower standing at the comer Of a rectangular field subtends angles of 60' and 45' at the two nearer comers. If e is the angle that the tower subtends at the farthest comer, then What is cote equal to? 2 1 P.T.O. 54. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß—2 and Bina + O, where O a 9 **(d)** 0 2 Cannot be determined due to insumcient data If + where O S S W, then what is the value of sin 2 ? 2 57. What is

186. **[CDS (I) Paper 24 Q60]** A vertical tower standing at the comer Of a rectangular field subtends angles of 60' and 45' at the two nearer comers. If e is the angle that the tower subtends at the farthest comer, then What is cote equal to? 2 1 P.T.O. 54. **(d)** 99 **(a)** 99 56. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß 2 and sin u + sinß where O a , O sßs9 **(b)** Cannot be determined due insumcient data to 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre If + where O S S W, then what is the value of sin 2 ? 2 57. What is **(c)** 99 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre 60. A vertical tower standing at the comer Of a rectangular field subtends angles of 60' and 45' at the two nearer comers. If e is the angle that the tower subtends at the farthest comer, then What is cote equal to? 2 1 P.T.O. 54. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß—2 and Bina + O, where O a 9 **(d)** 0 2 Cannot be determined due to insumcient data If + where O S S W, then what is the value of sin 2 ? 2 57. What is

187. **[CDS (I) Paper 24 Q56]** ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß 2 and sin u + sinß where O a , O sßs9(Y. What Is the value of cosm 0 2 SA If 6+ 8tane sece and 8 — 6tane then What iB the value of k 27 11 22 77 **(d)** Cannot be determined due insumcient data to **(a)** 99 56. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß 2 and sin u + sinß where O a , O sßs9 **(b)** Cannot be determined due insumcient data to 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre If + where O S S W, then what is the value of sin 2 ? 2 57. What is **(c)** 99 59. A pole on the ground leans at 60' With the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtend the same angle. If the pole and the point are in the same vertical plane, then what is the length Of the pole? x metre 2x metre 60. A vertical tower standing at the comer Of a rectangular field subtends angles of 60' and 45' at the two nearer comers. If e is the angle that the tower subtends at the farthest comer, then What is cote equal to? 2 1 P.T.O. 54. ABC be a triangle right angled at C, then what is tan A + tan B equal to? SS. Let Cosa + cosß—2 and Bina + O, where O a 9 **(d)** 0 2 Cannot be determined due to insumcient data If + where O S S W, then what is the value of sin 2 ? 2 57. What is

188. **[CDS (I) Paper 25 Q48]** What is sine + cose equal to ? sin 3 e—cos39 sin 0 — coso 13 **(a)** 20 **(b)** 21 **(c)** 24 **(d)** 25 43. A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ?

189. **[CDS (I) Paper 25 Q49]** A ladder 10m long reaches a point 10m below the top of a vertical flagstaff. From foot of the ladder, the elevation of top of the flagstaff is **(a)** 20 **(b)** 21 **(c)** 24 **(d)** 25 43. A EErson bought a chair and a table for *750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% on the whole. What is original cost of table ?

190. **[CDS (I) Paper 25 Q57]** 00 900 9 tan0=l t, sin29—2sin9cos9 14 50. t, 00<9 < 900 51. frå 1200 m 300 0 I 3000m t, **(a)** 9<300 cos20 — + 2 52. sin20 00 < 900 sin20+ 4 2 7 4 53. : 2. sin40+cos40=l+2sin29cos20 3. tan46+tan20=sec O—sec20 NYRG-T-MTK - A 54. sin240sin660 + tan 240 tan 660 — cot 240 cot 660 x=psinAcosB, y = psinA sinB z=pcosAt, **(b)** 2p2 56. qt x—msecA+ntanA y = m tan" + n secAt, **(c)** m2 — n2 + mn 57. 00 900 9 tan0=l t, sin29—2sin9cos9 14 54. sin240sin660 + tan 240 tan 660 — cot 240 cot 660 x=psinAcosB, y = psinA sinB z=pcosAt, **(d)** 2p2 56. qt x—msecA+ntanA y = m tan" + n secAt,

191. **[CDS (I) Paper 25 Q50]** What is the maximum Value Of 1 + 2 sin2 cos2 — sin 40 —cos4ø where (f < 900 ? **(a)** 9<300 **(b)** 450 < 0<600 **(c)** 600 < 9<900 cos20 — 3cos6 +2 52. If sin2 where 00 < e < 900 then what sin20 + coso equal to ? 4 3 2 7 4 53. Consider the following : IS 2. sin4Ø+cos49=l+2sin20cos20 15 Which Of the above are identities ? and 2 only and 3 only and 3 only **(d)** 1, 2 and 3 is the value of 54. What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B 55. If x and z = p cos A, then what is the value of x2 + y2+z2 ?

192. **[CDS (I) Paper 25 Q53]** Consider the following : 2. sin4Ø+cos49=l+2sin20cos20 15 15 Which Of the above are identities ? **(a)** I and 2 only **(b)** 2 and 3 only **(c)** 1 and 3 only **(d)** 1, 2 and 3 M. What is the value of sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ?

193. **[CDS (I) Paper 25 Q55]** If x —psinAcosB, y = psinA sinB and z = p cos A, then what is the value of x2 + y2+z2 ? **(c)** p2 **(d)** 2p2 **(a)** 9<300 **(b)** 450 < 0<600 **(c)** 600 < 9<900 cos20 — 3cos6 +2 52. If sin2 where 00 < e < 900 then what sin20 + coso equal to ? 4 3 2 7 4 53. Consider the following : IS 2. sin4Ø+cos49=l+2sin20cos20 15 Which Of the above are identities ? and 2 only and 3 only and 3 only **(d)** 1, 2 and 3 is the value of 54. What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B 55. If x and z = p cos A, then what is the value of x2 + y2+z2 ?

194. **[CDS (I) Paper 25 Q56]** If x m sec.4 + n tan" and y = m tanA + n secA, then what is x2 _ equal to ? **(d)** m2 + mn **(a)** 9<300 **(b)** 450 < 0<600 **(c)** 600 < 9<900 cos20 — 3cos6 +2 52. If sin2 where 00 < e < 900 then what sin20 + coso equal to ? 4 3 2 7 4 53. Consider the following : IS 2. sin4Ø+cos49=l+2sin20cos20 15 Which Of the above are identities ? and 2 only and 3 only and 3 only **(d)** 1, 2 and 3 is the value of 54. What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B 55. If x and z = p cos A, then what is the value of x2 + y2+z2 ?

195. **[CDS (I) Paper 25 Q57]** If for some 9 lying between 00 and 900, tan I, then what is the value of sin20— 2sinecos0? A - NYRG-T-MTK **(a)** 9<300 **(b)** 450 < 0<600 **(c)** 600 < 9<900 cos20 — 3cos6 +2 52. If sin2 where 00 < e < 900 then what sin20 + coso equal to ? 4 3 2 7 4 53. Consider the following : IS 2. sin4Ø+cos49=l+2sin20cos20 15 Which Of the above are identities ? and 2 only and 3 only and 3 only **(d)** 1, 2 and 3 is the value of 54. What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B 55. If x and z = p cos A, then what is the value of x2 + y2+z2 ?

196. **[CDS (I) Paper 25 Q54]** What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B **(a)** 9<300 **(b)** 450 < 0<600 **(c)** 600 < 9<900 cos20 — 3cos6 +2 52. If sin2 where 00 < e < 900 then what sin20 + coso equal to ? 4 3 2 7 4 53. Consider the following : IS 2. sin4Ø+cos49=l+2sin20cos20 15 Which Of the above are identities ? and 2 only and 3 only and 3 only **(d)** 1, 2 and 3 is the value of 54. What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B 55. If x and z = p cos A, then what is the value of x2 + y2+z2 ?

197. **[CDS (I) Paper 25 Q58]** What is cos9—2cos3e' equal to ? **(a)** sin 9 **(b)** cose **(c)** tan 0 **(d)** cot O

198. **[CDS (I) Paper 26 Q11]** What is the value of the following? 2 sin 680 2cot 150 cos 220 5 tan 750 3 tan20' tan4cr tan 450 tan 5 The perpendicular dropped from a vertex 12. of a right-angled triangle the hypotenuse divides it into two segments Of lengths 9 units and 16 units respectively. What is the length Of the perpe n dicular? **(a)** 6 units **(b)** 8 units **(c)** IO units **(d)** 12 units

199. **[CDS (I) Paper 26 Q46]** If I — cos 9 + sine I + sine sine + cose — 1 then what is cos e **(d)** **(a)** Zero **(b)** One **(c)** Two **(d)** Infinitely many 46. If I — cos 9 + sine I + sine sine + cose — 1 then what is 48. 49. 50. What is the minimum value sm4B+cos4 —2sin ecos2 9? value does not exist Of

200. **[CDS (I) Paper 26 Q47]** If cos(x + y) O and sin (x equal to? ¯ y) Where x, y E Q — , then what is the value Of cot Px — y) ? 2 GHSA-T-MTK/45A **(a)** Zero **(b)** One **(c)** Two **(d)** Infinitely many 46. If I — cos 9 + sine I + sine sine + cose — 1 then what is 48. 49. 50. What is the minimum value sm4B+cos4 —2sin ecos2 9? value does not exist Of

201. **[CDS (I) Paper 26 Q49]** If cos9+sece—2 O, where — 2' then what is the value Of cos 4 9 +sec4 —2? **(a)** Zero **(b)** One **(c)** Two **(d)** Infinitely many 46. If I — cos 9 + sine I + sine sine + cose — 1 then what is 48. 49. 50. What is the minimum value sm4B+cos4 —2sin ecos2 9? value does not exist Of

202. **[CDS (I) Paper 26 Q54]** The angle of elevation Of the top Of a tower of height x metre from a point on the ground is found to be 60•. By going y metre away from that point, it becomes **(a)** 120' **(b)** 135' **(c)** 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is **(d)** following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

203. **[CDS (I) Paper 26 Q57]** Consider the following • l. tan e + cote, where O < < where O < e < 9CP Which of the atx»ve is/are an identity/ identities? 1 only **(a)** **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 S I. If A, C are acute angles and sin(B+C- A) -cos(C+A- B) -tan(A+B-C)- then what is (A + B + C) equal to? **(b)** 120' **(c)** 135' **(d)** 150•

204. **[CDS (I) Paper 26 Q58]** What is the length Of the chord Of a unit circle Which subtends an angle 20 at the centre, Where e < 450 ? sin 2B **(b)** cos29 **(c)** 2 sin B **(d)** 2cOsB **(a)** cos29 **(b)** 2 sin B **(c)** 2cOsB 59. If O, where then what is 'the value Of 12cot2 B + 3cosece? 61. Considér the following statements : 2. If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. **(d)**

205. **[CDS (I) Paper 26 Q60]** If cos B -2 coecB + I coæcB — I where O < e 9(P, then What is the value of sin 4 cos4 9? **(b)** 2 4 GHS'ET-MTK,'45A 17 **(a)** cos29 **(b)** 2 sin B **(c)** 2cOsB 59. If O, where then what is 'the value Of 12cot2 B + 3cosece? 61. Considér the following statements : 2. If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. **(d)**

206. **[CDS (I) Paper 26 Q74]** ABCD is a parallelogram with AB 15 cm and AD 8 cm. If e is the acute angle between AB and AD, then what is the area of the parallelogram in square cm? **(a)** 12 **(b)** 943 cm **(c)** 18 cm **(d)** 27 cm 72. A farmland is in the shape Of a rhombus. The perimeter of the land is 100 m and the length of one of the diagonals is 40 m. The land is divided into four equal parts. What is the area of each 75. 76. 77. 21 If the perimeter of an isosceles right triangle is 4

207. **[CDS (I) Paper 27 Q44]** 45. What is sin2 A + sin 2 B + sin2 C equal to ? 2 **(b) (c)** What is the radius Of the circle ? **(a) (b)** What is the radius Of the circle ? What is one Of the J»ssible values of x — — ? **(c) (d)** What is one of the possible values Of + X 45.

208. **[CDS (I) Paper 27 Q51]** 52. 53. q If tan e = ; O O < 900, then what is see 9 + cos + 2 equal to ? pq **(c)** pq If 11 sin e + 60 cos O 61; O 900, then what is the value of (tan g + cot e) ? **(a) (c)** 61 610 122 What is the diameter Of a circle inscribed in a regular polygon Of 15 sides With side length unity ? **(b) (d)** 120 cot 120 0-5 tan 120 tan 120

209. **[CDS (I) Paper 27 Q52]** 53. **(a) (c)** 61 610 122 **(b) (c) (d)** 32 34 36 38 What is the diameter Of a circle inscribed in a regular polygon Of 15 sides With side length unity ? 120 What is the medign Of the distribution ? **(a) (c) (b) (d)** cot 120 0-5 tan 120 tan 120 FDGT-T-EMT

210. **[CDS (I) Paper 27 Q56]** I + sino tane + seco — I cos O. tang — + I **(a) (b) (c)** 2x 2 tan O sin e Bec O + I tan e — sin e seco — I ajS e —sin cos2 0 sin 2 0 **(d)** 1,Rf12 sin G cose sin O — cose 2 sine 3 cose 2tan B t, tan 29+1 **(b)** 3 sine — 2 coso 2 12 F DGT-T-EMT 4 (20-

211. **[CDS (I) Paper 27 Q59]** 4 20-A) cosec O — O sin O cosec O cot O sine **(a) (c)** 2 sin6 + 2 **(a) (b) (c)** 2 — 3 sin4 e - 3 cos' o 3600 sec2 0 + 121 cosec2 e **(a) (a) (b) (c) (d)** 1320 2401 3721 5041 1 sine cos O tan O cot O 2 tan e 2 cot e cos O I + sin O

212. **[CDS (I) Paper 27 Q56]** I + sino tane + — I If x , then What is tane — I equal to ? **(a) (b)** sin 0 sin x, then what is equal to, **(c) (d)**

213. **[CDS (I) Paper 27 Q59]** 60. If cosec e — cot cosec 9 + cot e **(b) (c)** sin 0 sin x, then what is equal to, What is 2 sin6 e 2 cos6 0 — 3 sin4 e — 3 cos' B equal to ? **(a) (b)** sin 0 sin x, then what is equal to, **(c) (d)**

214. **[CDS (I) Paper 28 Q2]** 3. cos — sin O = cosec O + cot O = cos2e-sin2ez 2 tan B I — tan2 I — tan 1 + tan Which Of the above equations are identities ? **(a) (b) (c) (d)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 then what is the value Of If sin 13' (tan e + sec (cosec e — cot O ? 2

215. **[CDS (I) Paper 28 Q48]** 49. 50. -A) If tans O cots e m, then What is the value of tan O+cote? **(c)** What is the minimum value of 6 — 4 sm e, osesE? **(a)** What is the minimum value of 6 — 4 sm e, osesE? **(b) (c) (d)**

216. **[CDS (I) Paper 28 Q3]** 2 tan B cos — sin O = I — tan2 cosec O + cot O = I — tan cos2e-sin2ez 1 + tan Which Of the above equations are identities ? If tans O cots e m, then What is the value of tan O+cote? **(c)** What is the minimum value of 6 — 4 sm e, osesE? **(a) (b) (c) (d)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 the value

217. **[CDS (I) Paper 28 Q49]** Of 50. **(a) (c)** 2 4 6 If sin then what is 13' (tan e + sec (cosec e — cot O — ? 121 **(a)** 4 169 **(b)** 196 225 What is the value of x that satisfies 300+2xsin300-cot2300- **(a) (c) (d)** 6 tan 150 tan 750 = O ? 3 6 JNBY-D-MTK

218. **[CDS (I) Paper 28 Q53]** 54. cos 320 + cos 2 580 What is the value Of sec2 500 40' 4 tan 130 tan 370 tan 530 tan 770 ? **(a) (b) (c)** What is the value Of (I cot2 e) (I + e) (I — cos O) — (I + tan2 e) (I + sin B) (I **(c) (d)** If 2 cos O + sin O — 2 What is the value of O ? **(a)** 4 **(d)** A person on the top Of observes a car moving at coming directly unvards 6 minutes for the angle — sin O) ? , then a vertical tower 5' a uniform speed it. If it takes Of depression to change from 300 to 450, and further t minutes to reach the tower, which one of the following is correct ? **(b) (c) (d)** 8-3 8-3 8-6 8-6 < t < JNBY-D-MTK 55. r

219. **[CDS (I) Paper 28 Q57]** 19-A) A woman is standing on the deck Of a ship, which is h (in metres) above water level. She observes the angle of elevation of the top Of a tower as 600 and the angle Of depression Of the base Of the tower as **(a) (b) (c)** What is the value Of **(d)**

220. **[CDS (I) Paper 28 Q52]** 53. 54. (19-A) cos 320 + cos 2 580 What is the value Of sec2 500 40' 4 tan 130 tan 370 tan 530 tan 770 ? A woman is standing on the deck Of a ship, which is h (in metres) above water level. She observes the angle of elevation of the top Of a tower as 600 and the angle Of depression Of the base Of the tower as **(a) (b) (c)** What is the value Of **(d)**

221. **[CDS (I) Paper 28 Q91]** 92. -A) What is the length Of QC ? **(a) (b) (c) (d)** 3•6 cm 3•2 cm If Z ABD = e, then What is sin equal to ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

222. **[CDS (I) Paper 29 Q22]** 28. 24. (94 **(a) (b) (c)** I only 2 only Both I and 2 Neither I nor 2 (3a — 6b + c — 2d) (3a + 6b —c — 2d), then which one of the following is correct ? **(a) (b) (c) (d)** ab = cd ac bd ad bc ad + bc O If 3 sin O + 5 5, then what is the value of 5 sin O — 3 cos O equal to ? **(a)** -2 The combined age Of a man and his Wife is 6 times the combined age of their children. Two years ago their combined age was IO times the tx.mbined age of their children; and six years later their combined age will be 3 times the combined age Of their children. How many children do they have if each child is at least 2 years old ? **(a) (c)** JNUP-T-EMT

223. **[CDS (I) Paper 29 Q27]** What is 3 (sin x — cos + 6 (sin x + cos 4 (sin + 4 (cos equal to ? **(b) (c)** 9 11 13 15 What is the value of sin O • cos O, if O satisfies the equation cot2 9 — (qi + l) cot O O; **(b)** 4 2 vfä+l is Which one Of the following is a value Of O, if O satisfies the equation tan 29 tan 40 — I O; **(a) (b) (c)** I only 2 only Both I and 2 Neither I nor 2 **(d)**

224. **[CDS (I) Paper 29 Q26]** 27. What is the value of sin O • cos O, if O satisfies the equation cot2 9 — (qi + l) cot O O; **(a) (b) (c) (d)** ab = cd ac bd ad bc ad + bc O **(b)** 4 2 vfä+l If 3 sin O + 5 5, then what is the value of 5 sin O — 3 cos O equal to ? **(a)** -2 The combined age Of a man and his Wife is 6 times the combined age of their children. Two years ago their combined age was IO times the tx.mbined age of their children; and six years later their combined age will be 3 times the combined age Of their children. How many children do they have if each child is at least 2 years old ? **(a) (c)** Which one Of the following is a value Of O, if O satisfies the equation tan 29 tan 40 — I O; **(a)** 12 **(b)** 15 **(c)** 6 JNUP-T-EMT

225. **[CDS (I) Paper 29 Q29]** 30. (sin2 0 — 4 sin 0 + 3) (4 — cos2 + 4 sin e) = O, **(c)** x O + y O = sin O cos 0 x sine-ycoso ? **(a) (b)** sin B + cos O tan x sin O — cos O sin x ? sin O + cos O sin e — cos e sin B + ms O sine — e 2 29. 30. **(c)** x O + y O = sin O cos 0 x sine-ycoso ? 31. 32. **(d)**

226. **[CDS (I) Paper 29 Q31]** 32. IO-A) **(a) (b) (c) (d) (a) (b) (c) (d)** (02) H tan y — h tan h tan •t — H tan H tan — h tan a h tan — H tan a H (cot — P) H (tan — tan B) 2 d = H (cot a — cot B) d = H (tan a — tan P) **(a) (b)** sin B + cos O tan x sin O — cos O sin x ? sin O + cos O sin e — cos e sin B + ms O sine — e 2

227. **[CDS (I) Paper 29 Q28]** 29. 30. sin e + cos B If tan x sin O — cos 0 4 what is sin x equal to ? , then **(a) (b) (d)** sin O + cos O sin B — cos e sin e + os B sin O — O 2 How many values Of e Will satisfy the equation (sin2 e — 4 sin e + 3) (4 — eos2 0 + 4 sin O) O, where O O — ? **(a) (b) (c) (d)** 31 None Only one Onle two Only three If x y sin cos B and x sin B —y cos e 20, foreveryee O, — , then what is x2 + Y2 equal to ? **(a) (b) (c)** JNUP-T-EMT Consider the following for the next two (02) items that follow : A flagstaff stands on the top of a vertical tower. The angle of elevation of the top Of the flagstaff from a certain place on the same horizontal level with the base of the tower is found to be a. Advancing a distance d towards the tower in the same horizontal plane, the angle of elevation Of the top of the flagstaff is observed to be 13 and that Of the top of the tower is observed to be y. Let H be the height of the top of the flagstaff from the base of the tower and h be the heizht Of the tower.

228. **[CDS (I) Paper 29 Q31]** 32. Which one of the following is correct ? **(a) (b) (c) (d)** H tan •r — h tan h tan y — H tan H tan •r — h tan a h tan y — H tan a Which one Of the following is correct ? **(a) (b) (c)** H (tan — tan P) d = H (cot a — cot p) d = H (tan — tan ß)

229. **[CDS (I) Paper 29 Q37]** 38. a-Ai What is px2 + U2 + rz2 equal to ? **(a)** 4 12 12 88. 84. **(b) (c) (d)**

230. **[CDS (I) Paper 29 Q70]** 25 -A) What is —y —z —2yz — z2 + 2xy equal to ? **(a) (b)** 2y 2z — Y2 — z2 — 2yz — 2xz If tan A eot A = 2, where O < A < 900 , then what is the value of tan2 A + tana A + tan4 A + + tann A ? **(a) (b) (c) (d)** 65. spheres ? **(c)**

231. **[CDS (I) Paper 29 Q73]** Which of the following Ware identity/identities ? sin 3 + cos O sin O Os g **(a) (b) (c)** 9 4 **(d)**

232. **[CDS (I) Paper 29 Q71]** 72. 73. (27-A) Which of the following Ware identity/identities ? sin 3 + cos O sin O Os g **(a) (b) (c)** 9 4 **(d)**

233. **[CDS Paper 30 Q9]** IO. (5-A) If the equation x cos O has a real solution for every O where O S O E, then which one of the following is correct ? **(a) (c) (d)** p z 1/8 p 21/8 **(b) (c) (d)** If t 17 21 37 = cos 790 , then What is -cos 790) equal to ?

234. **[CDS Paper 30 Q38]** Sin **(a) (b) (c)** A + 5sinA +1 sin A 9 for O < A s — 22 36. 37. **(d)**

235. **[CDS Paper 30 Q51]** 52. 53. tan (3A) (A - 220), SA W **(a)** 250 **(b)** 270 **(c)** 280 **(d)** 300 sin e — Cos + I z p sec 9 + q tan O, sin cos 54. :

236. **[CDS Paper 30 Q1]** ABC#, sin A + sin B + sin C = **(a)** SRSU-T-EMT 48. ABCD AB = 6 BC = 18 em, 51. 49. **(b) (c)** z 6 em DA = IO em BD z x, ? 8 < X < 12 16 < 18 18 x 20 52. tan **(d)** 250

237. **[CDS Paper 30 Q2]** ABC#, cos A + cos B + cos C = 16 —A) 39 , **(a)** SRSU-T-EMT 48. ABCD AB = 6 BC = 18 em, 51. 49. **(b) (c)** z 6 em DA = IO em BD z x, ? 8 < X < 12 16 < 18 18 x 20 52. tan **(d)** 250

238. **[CDS Paper 30 Q53]** 54. -A) If tan (3A) = cot (A — 220), where 3A is an acute angle, then what is the value of A ? **(a)** 250 **(b)** 270 **(c)** 280 **(d)** 300 sin e — cos O I If p sec O q tan O, sin where O < e < — , then what is- p * q equal to ? **(a) (d)** 2 4 The angles of elevation of the top Of a tower from two points A and B at a distance Of x m and (x + 5) m from the base of the tower of height 6 m and in the same straight line with it are complementary. What is the value of x ? **(a) (b) (c) (d)** Consider the following statements :

239. [CDS Paper 30 Q2] In a triangle ABC, if sin A + sin B + sin C = triangle can be equilateral. In a triangle ABC, if A + cos B + cos C triangle can be equilateral. , then the 3 — , then the 2 Which Of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 In a quadrilateral ABCD, AB 6 em,

240. [CDS Paper 30 Q52] 53. 54. to ? **(a) (d)** In a quarter circle of radius R, a circle of radius r is inscribed. What is the ratio Of R **(a) (c) (d)** In a quadrilateral ABCD, AB = BC and CD = DA; AC and BD are diagonals such that AC = 6 em and BD 12 cm. What is the area of the quadrilateral ? 2 4 The angles of elevation of the top Of a tower from two points A and B at a distance Of x m and (x + 5) m from the base of the tower of height 6 m and in the same straight line with it are complementary. What is the value of x ? **(a) (b) (c) (d)** Consider the following statements :

241. [CDS Paper 30 Q78] What is tan + tan equal to ? **(a) (b) (c) (d)** 5:28 a.m. 5:44 a.m. 4:56 a.m. 6:24 a.m. If the lengths Of the two trains A and B are 400 m and 500 m respectively, then what is the time taken by them to cross each other ?

242. [CDS (I) Paper 31 Q14] If o 2' + P) -.13 and tan(a— ± , then what is tana•cot2ß equal to ? **(a)** 15. What is the value of 18. From a point X on a bridge across a sin29 cos2e **(b)** 2h **(c)** Htana•tanß **(d)** h cota•cOtß

243. [CDS (I) Paper 31 Q39] What is cose — sin 9 + 1 (cot 9 — cosece) coso + sin O — 1 equal to ? sino — 2 sin30 **(a)** 20 **(b)** 25 **(c)** 30 **(d)** 35 35. 36. Let ABC be a triangle with area 36 square cm. If AB=9 cm, BC' 12 cm and LABC = O, then what is cos9 equal to ? 4 3 2 Let n be a natural number. The HCF of n, n +10 is 10. If the LCM is x

244. [CDS (I) Paper 31 Q42] Two towers A and B of height 23 m and 11 m respectively, stand 9 m apart. A straight rod is joined to the two tops of the towers. A monkey sitting on the top of A, climbs the rod to reach the top of B. If the monkey takes 5 minutes to reach the other end, what is the average speed of the monkey ? **(a)** IO mimin **(b)** 5 mimin **(c)** 10 cm/sec **(d)** 5 cm/sec

245. [CDS (I) Paper 32 Q95] 96. 97. -A) Who among the following was responsible fir demarcating the borders between India and Pakistan during the partition in 1947 ? **(a) (b) (c)** Cyril Radcliffe Mortimer Durand Henry McM ahon Arthur Belfour Which one among the following statements about Dr. Muthulakshmi Reddy, a freedom fighter , is not Correct ? **(a) (b) (c) (d)** She was the first girl student to be admitted into a men's college. She was the first woman legislator in British India. She was a medical professional. She was honoured With Bharat Ratna in 1956. "Since wars begin in the minds of men, it is in the minds of men that the defences of peace must be constructed." This observation can be found in the Constitution of which one Of the following Organs Of the United Nations ? **(a) (b) (c) (d)** World Health Organisation United Nations Educational, Scientific and Cultural Organisation International Atomic Energy Agency United Nations Economic and Social Co uncil

246. [CDS (I) Paper 33 Q34] What is (sece-tan9) to? **(b)** mane **(d)** sine + cos B ANFB-T-EMT/45A I — sin I + sin 9 6, then What is as. If cote coec2e — sec2 e coxc29+sec e equal to? **(a)** 1/2 **(b)** 1/3 **(c)** 2/3 **(d)** 3/4 difference between the two acute

247. [CDS (I) Paper 33 Q44] What is p2q2(p2 + q2 + 3) equal to? ANFB-T-EMT For the following two (02) items : **(d)** 2/3 sina 4 u sin # cosa cosß 9 **(a)** mn — I O For the following two **(b)** 2/3 For the following tvo **(c)** 1 P.T.O. For the following two **(d)** 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

248. [CDS (I) Paper 33 Q45] What is tan 2 equal to? **(a)** mn — I O For the following two **(b)** 2/3 For the following tvo **(c)** 1 P.T.O. For the following two **(d)** 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

249. [CDS (I) Paper 33 Q46] What is tan 2 ß equal to? 1/3 For the following tvo (02) items : I + sin e COS 9 **(a)** mn — I O For the following two **(b)** 2/3 For the following tvo **(c)** 1 P.T.O. For the following two **(d)** 2/3 sina 4 u sin # cosa cosß 9 45. What is tan 2 equal to? 46. What is tan 2 ß equal to? 1/3 For the following tvo

250. [CDS (I) Paper 33 Q48] What is tan e equal to? For the following two (02) items • Let sin B + cos O p and + q , where p I. **(a)** p q **(b)** 2p q **(c)** 2q p — I **(d)** 80 m 100m 120m 150 m 19 52. TWO persons X and Y leave place P for place Q at 7:00 a.m. and 7:10 a.m. respectively along the same path. X walks at a speed of 4 •8 km/hr and Y walks at a speed Of 6 km/hr. HOW many kilometres from place p Will X meet Y ?

251. [CDS (I) Paper 34 Q41] If cos I —sina cos O = 4, then which I + Sin one of the following is a value of (tan29 + cot29) ? 10 17 **(b) (d)** 17 **(a)** 3-25 **(b)** 3-50 **(c)** 3-75 **(d)** 4-25 40. The numbers 59, 53, 51, 43, 36,

252. [CDS (I) Paper 34 Q42] For O < —, consider the following : I. (tan40 + + c0160) = sec29 cosec29 tan 0 + sin 0 = cor29(sec9 + 11. tan — sin O Which of the above is/are identities ? **(a)** I only **(b)** 11 only **(c)** Both land 11 **(d)** Neither I nor 11

253. [CDS (I) Paper 34 Q44] At a point on level ground, the tangent of the angle of elevation of the top of a tower is found to be S. On walking 70 m towards the tower, the tangent of the angle of elevation of the top of the tower is found to be Y. What is the height of the tower ? **(a)** 3-25 **(b)** 3-50 **(c)** 3-75 **(d)** 4-25 40. The numbers 59, 53, 51, 43, 36,

254. [CDS (I) Paper 34 Q45] Two persons are on diametrically opposite sides of a tower. They measure the angles of elevation of the top of the tower as 300 and 600 respectively. If the height of the tower is 100m, what is the approximate distance between the two persons ? **(a)** 220 m **(b)** 225 m **(c)** 230m **(d)** 235 m

255. **[CDS (I) Paper 34 Q98]** What is tane equal to ? (29+3p) **(a)** 96. What is cot2ß equal to ? p For the next two **(b)** 10 **(c)** 13 For the next two **(d)** 2pq — —

256. **[CDS (I) Paper 35 Q36]** 37. What should be added to 2x3+ax2-1 x2-1 **(a)** If x + y + z = O, then what is to get x(y + + + + + equal to ? **(a) (b) (c) (d)** xyz 2xyz 3xyz 6xyz If tan 60 . tan 30 = I, where 0 < e < 300, then what is O equal to ? **(c)** 50 15' No such value exists I — tan x I + tan x then what is 2 tan x 1 tan2 x (sin x + cos x) equal to ? **(a) (c)** 3 BFVS-T.TUE

257. **[CDS (I) Paper 35 Q40]** 13-A) If cos2 x + cos4 x = I, where O < x < — , then what is (sin2 x + sing x) + (sing x sin4 x) equal to ? 21 126 **(a)** If x + y + z = O, then what is 38. to get 39. x **(b)** 16 **(c)** 9 8 If sind x + eos4 x = I, where O x , then what is sin x cos x equal to ? **(d)**

258. **[CDS (I) Paper 35 Q46]** 14 -A) t? **(b) (c) (d)** 306 600 1 2vfi km Q@qTVT ß t, t? **(a) (b) (c) (d)** h tan a tan h cot a tan h tan a cot h cot a cot tan + cot e ? 65 169 313 156 371 sin cosec — cot O p 3 13 12 13 BFVS-T-TME 41. cot 9 + cos z m, cot e — cos 0 n,

259. **[CDS (I) Paper 35 Q44]** 42. 43. (14_A) **(a) (b) (c) (d)** 2 4 COæc 6 cot e sin O t? **(b) (c) (d)** 300 600 1 2vfi km

260. **[CDS (I) Paper 35 Q45]** 46. Q@qTVT ß t, t? **(a) (b) (c) (d)** h tan a tan h cot a tan h tan a cot h cot a cot , sin O 13, - , å tan + cot e ? 65 169 313 156 371

261. **[CDS (I) Paper 35 Q43]** If cot O + cos O m, cot O — cos = n, where O < O < — , then what is **(a) (b) (c) (d)** If 2 4 cosec e + cot e equal to ? = p, then What is sin O equal to ? sin e cosec O — cot e **(c)** If 8 sin e — cos O z 4, where what is sin e equal to ? 3 **(a)** 13 5 12 BFVS-T-TME , then (

262. **[CDS (I) Paper 35 Q45]** 46. 15 —A) A walking along a straight road obsenes that at two consecutive kilometer-stones the angles of elevation of a hill in front of him are 300 and 600 respectively. What is the height Of the hill ? 3Jä km km **(b) (c)** A tower subtends an angle a at a point P on the same level as the foot of the tower. Q is a point vertically above P and PQ h. If the angle of depression Of the foot of the tower measured from Q is B, then what is the height of the tower ? **(a) (b) (c) (d)** h tan a tan h cot a tan h tan cot h cot a cot B If 5 sin + 12 cos O 13, where O — , then what is tan B + cot equal to ? 194 **(a) (b)** 313 371

263. **[CDS (I) Paper 35 Q46]** A walking along a straight road obsenes that at two consecutive kilometer-stones the angles of elevation of a hill in front of him are 300 and 600 respectively. What is the height Of the hill ? **(a) (b) (c) (d)** If 2 4 cosec e + cot e 3Jä km km **(b) (c)** sin O equal to ? sin e **(c)** cosec O — cot e A tower subtends an angle a at a point P on the same level as the foot of the tower. Q is a point vertically above P and PQ h. If the angle of depression Of the foot of the tower rneasured from Q is B, then what is the height of the tower ? **(a) (b) (c) (d)** h tan tan h eot a tan h tan cot h cot a cot B If 8 sin e — cos O z 4, where what is sin e equal to ? 3 **(a)** 13 5 12 , then (15 If 5 sin + 12 cos O 13, where — , then what is tan B + cot equal to ? 194 **(a) (b)** 313 371 BFVS-T-TME

264. **[CDS (I) Paper 35 Q48]** 49. 50. (17 —A **(a) (c) (d)** 2 3 2 2 If A + B + C (A, B, C > O) and the angle C is obtuse, then which of the following ware correct ? l. sin A sin B < I 11. tan A. tan B > I Select the answer using the code given below . **(b) (c) (d)** I only [I only Both 1 and 11 Neither I nor 11 If (cos 350 + cos 550) = P, sin 350 . cos 350 equal to ? (I—p2) **(a)** If p sec + tan and q then what 54. = cosec O — cot O then what is (p — q — pq) equal to ? **(a) (c) (d)** 2 BFVS-T.TME

CHAPTER 09

Total Archive Questions: 287 Questions

GEOMETRY: LINES, ANGLES, TRIANGLES & POLYGONS

Typical Exam Weightage: 10 - 15 Questions (10 - 15 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q48] Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length **(a)** 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ? **(b)** 47 **(c)** 43 **(d)** 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

2. [CDS (I) Paper 1 Q62] 63. APQR is right angled at Q, PR — 5 cm and QR = 4 cm. Ifthe lengths of sides of another triangle ABC are 3 cm, 4 em and 5 em, then which one of the following is correct ? **(a)** Area of APQR is double that of AABC **(b)** AreaofAABC is double that ofAPQR 2 **(d)** Both triangles are congruent Which one Of the following figures has only one line of symmetry ? **(a)** Rhombus **(b)** Rectangle **(c)** Isosceles trapezium **(d)** Parallelogram A ladder 25 m long is leaning against a wall which is perpendicular to the level ground. The bottom of the ladder is 7 m from the base of the wall. If the top Of the ladder slips down 4 m, how much will the bottom Of the ladder slip ? **(d)** 15m If C, and C, and r, and are respectively the centroids and radii of incircles of two congruent triangles, then which one Of the following is correct ? **(a)** C, and C2 are the same point and **(b)** C, and C2 are not necessarily the same point and rt = rz **(c)** Cl and Ca are the same point and r, is not necessarily equal to rz **(d)** C, and Cz are not newssarily the point and r 1 is not necessarily equal to r, 16

3. [CDS (I) Paper 1 Q66] 16 In the figure given above, P is a point on AB and PQ is parallel to AC. What is the number of pairs of distinct similar triangles in the figure ? If the medians Of two equilateral triangles are in the ratio 3 : 2, then what is the ratio Of their sides ? The centroid and the orthocenter are coincident for which one of the following triangles ? **(a) (b) (c) (d)** Scalene triangle Isosceles triangle Equilateral triangle Right angled triangle

4. [CDS (I) Paper 1 Q75] In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? **(a)** 500 **(b)** 600 **(c)** 700 **(d)** 800

5. [CDS (I) Paper 1 Q98] Assertion **(a)** Both A and R are individually true and R is the correct explanation of A **(b)** Both A and R are individually true but R is not the correct explanation of A **(c)** A is true but R is false **(d)** A is false but R is true 98. Assertion

6. [CDS (I) Paper 1 Q99] ABC is a triangle. Let D, E denote the mid points Of BC, CA respectively. Let AD and BE intersect at G Let O be a point on AD such that AO : OD 2 : **(a)** Both A and R are individually true and R is the correct explanation of A **(b)** Both A and R are individually true but R is not the correct explanation of A **(c)** A is true but R is false **(d)** A is false but R is true 98. Assertion

7. [CDS (I) Paper 1 Q100] ABC is a triangle. AD, BE, CF are altitudes of AABC. Assertion (A) : Reason (R) 28 (AB2 + BC2 + CAZ) > (ADI BE2 + CF2) (CD2 - CE2) O Directions : Each Of the next THREE (03) items consists of two statements, one labelled as Assertion (A)' and the other as 'Reason (R)'. You are to examine these two statements carefrlly and select the answers to these items using the codes given below . Codes : **(a)** Both A and R are individually true and R is the correct explanation of A **(b)** Both A and R are individually true but R is not the correct explanation of A **(c)** A is true but R is false **(d)** A is false but R is true

8. [CDS (I) Paper 2 Q64] 65. A point p .moves such that its distances from two given points A and B are equal. Then what is the locus Of the point P ? . In the triangle ABC, AB 2 cm, BC = 3 cm and AC 4 cm. D is the middle point of AC. If a square is constructed on the side BD, what is thc area Of the square ? **(a) (b) (c) (d) (c) (d)** 4-5 Cm cm2 625 etn2 Nonc Of the above A straight line t."ich is the right bisector Of AB A circle With céntre at A A circle with centre -at B A straight line passing through either A or B A solid cone Of height 8 Crn and base radius 6 cm is melted and recast into identical cones, each Of height' 2 cm and radius I cm. What is thc number Of cones formed ? ABC is a triangle. X is a point outside the triangle ABC such that CD = CX, where D is the point of intersection Of BC and AX; and Z BAX Z XAC. Which one Of the following is correct ? **(a) (c) (d)** 36 72 144 180 **(a) (b) (c) (d)** Triangles ABD, ACX are similar Z ABD < Z ACD AC CX > z DXC P-DETB-J-FN t 16

9. [CDS (I) Paper 2 Q3] Congruent triangles are similar. Similar triangles are congruent. If the hypotenuse and a side Of One right triangle are equal to the hypotenuse and a side of another right triangle respectively, then the two right triangles are congruent. Which Of the Statements given above Ware correct ? **(a) (c)** I only 2 only 2 and 3 and 3 ABC is a triangle and the perpendicular drawn from A meets BC in D. If AD2 = BD . DC, then which one of the following is correct ? **(c) (d)** ABC must be an obtuse angled triangle ABC must be an acute angled triangle Either z B 2450 or Z C 2 450 AC2 = AB2 + BC2 ABC is a triangle in which AB AC. Let BC be produced to D. From a point E on the line AC let EF be a straight hne such that EF is parallel to AB. Consider the quadnlateral ECDF thus formed. If Z ABC 650 and Z EFD 80', then what is Z FDC equal to ? **(a) (b) (c) (d)** P-DETE.J-FN 18 - 69. ght

10. **[CDS (I) Paper 2 Q70]** 71. 18 -A) The bisectorS of the angles ABC and BCA Of a triangle ABC meet in a point O. What is the angle at O facing the side BC ? **(a) (b) (c) (d)** 900 + (A/2) 900 - A What is the number of points in the plane of a triangle ABC which are at equal distance from the vertices Of the triangle ? **(a) (c) (d)** 2 o In the figure given above, O is the centre Of the circle. The line UTV is a tangent to thc circle at T, VTR 52- and triangle PTR is an isosceles triangle such that TP = TR. VVhat is Z x + Z y + zz equal to ? **(a) (b) (c) (d)** 1750 208' 2180 2500

11. **[CDS (I) Paper 2 Q69]** The bisectorS of the angles ABC and BCA Of a triangle ABC meet in a point O. What is the angle at O facing the side BC ? **(a) (b) (c) (d)** 900 +

12. **[CDS (I) Paper 2 Q74]** 75. x In the figure given above, ABCD is a Square in which AO AX. What is Z XOB ? **(a) (b) (c)** The 22-50 300 45 quadrilateral formed by joining the mid-points Of the sides AB, BC, CD, DA Of a quadrilateral ABCD is **(a) (b) (c) (d)** a trapezium but not a parallelogram a quadrilateral but not a trapezium a parallelogram only a rhombus 106' c In the figure given above, O is the centre of the circle and Z AOD = 106'. What is Z BCD **(b) (d)** 530 435 370 20 o In the figure given above, Z AOB 46'; AC and 0B intersect each other at right angles. What is the measure of Z OBC ? (O is the centre Of the circle)

13. **[CDS (I) Paper 3 Q6]** Staternent—I : I.Rt LMNbea triangle. Let P, Q be the mid- points of the sides LM, LN respectively. If PQ2&MP2+NQ2, then LMN is a fight angled triangle right angled at L. Statement—II : If in a triangle ABC, .4B2 > BC + CA2, then LACB is obtuse. Which of the following is correct in the light of the above statements ? **(a)** Both statement — I and statement 11 are correct and statement — 11 is the reason for statement — I **(b)** Both statement — I and statement — 11 are corect and statemeht — 11 is not the reason for statement — I **(c)** Statement—I is true, but state: ment — 11 is false **(d)** Statement — I is false, but state— ment — 11 is true 2

14. **[CDS (I) Paper 3 Q8]** What is tlE median of the ages Of minor children ? **(b)** 5 years **(c)** 7 years **(d)** Cannot determined due to in- sufficient data **(a)** 3 years **(b)** 4 years **(c)** 5 years **(d)** 6 years 8. What is the median of the ages of minor children ?

15. **[CDS (I) Paper 3 Q20]** The median BD of ghe triangle ABC meets at D. If then which one Of the following is correct ? **(a)** ZACB = I right angle **(b)** LBAC = I right angle **(c)** LABC I right angle **(d)** None of ttw abové T-DETA-K-FN - A 6

16. **[CDS (I) Paper 3 Q24]** The three sides of a triangle are IO, 100 and x. Which one of the following is correct ? **(c)** 90 6 **(a)** 72 **(b)** 216 **(c)** 372 **(d)** 600 18. If r and s are any real, numbers such that 1, then what is the maximum value Of the product rs ? 19. r is a non•zero real Humber such that r75 > PO. This is possible only when 20. The median BD of Che triangle ABC meets AC at D. If then which one of the following is correct ?

17. **[CDS (I) Paper 3 Q39]** ABC is a right triangle with right angle at A. If the value of tan B = any.real k the length of the hypotenuse is of the form **(b)** 2k **(d)** 9k **(c)** 5k 10 **(a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** days 33. If of

18. **[CDS (I) Paper 3 Q53]** The following sets of conditions relate to two triangles ABC and DEF. Which set of conditions does not guarantee the congruence of ABC and DEF ? 14 **(a)** 12 49. ABCD is a square. p. Q, R, S are points on the sidesAB, BC, CD, DA respectively such that CR=DS. What is ZSPQ equal ? **(b)** 300 **(c)** 450 **(d)** 600

19. **[CDS (I) Paper 3 Q55]** The middle points of ihe parallel sides AB and CD of a parallelogram ABCD are p and Q respectiveiy. If AQ and CP divide the diagonal BD into three parts BX, XY and YD, then which one of the following is correct ? **(d)** XY-2BX **(a)** 1911-21 **(b)** 1921-31 **(c)** 1951-61 **(d)** 1961-71 55. The middle points of ihe parallel sides AB and CD of a parallelogram ABCD are p and Q respectiveiy. If AQ and CP divide the diagonal BD into three parts BX, XY and YD, then which one of the following is correct ?

20. **[CDS (I) Paper 3 Q56]** Consider the following statements in respect of any triangle : **(a)** 1911-21 **(b)** 1921-31 **(c)** 1951-61 **(d)** 1961-71 55. The middle points of ihe parallel sides AB and CD of a parallelogram ABCD are p and Q respectiveiy. If AQ and CP divide the diagonal BD into three parts BX, XY and YD, then which one of the following is correct ?

21. **[CDS (I) Paper 3 Q1]** The three medians of a triangle divide it into six triangles of equal **(a)** 1911-21 **(b)** 1921-31 **(c)** 1951-61 **(d)** 1961-71 55. The middle points of ihe parallel sides AB and CD of a parallelogram ABCD are p and Q respectiveiy. If AQ and CP divide the diagonal BD into three parts BX, XY and YD, then which one of the following is correct ?

22. **[CDS (I) Paper 3 Q2]** The perimeter of a triangle is greater than the sum of the lengths of its three medians. Which of the statemepts given above is/are correct ? **(a)** only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 T-DETA-K-FN - A 16

23. **[CDS (I) Paper 3 Q87]** Consider the . following statements in respect of a quadrilateral - I. The line segments joining the midpoints of the two pairs of opposite sides bisect each other at the point of intersection. **(a)** 264 kg **(b)** 132 kg **(c)** 396 kg **(d)** None of the above 87. Consider the . following statements in respect of a quadrilateral - I. The line segments joining the midpoints of the two pairs of opposite sides bisect each other at the point of intersection. 2. The area of the quadrilateral formed by joining the midpoints of the four adjacent sides is half of the total area of the quadrilateral. Which of the statements given above is/are correct ?

24. **[CDS (I) Paper 3 Q2]** The area of the quadrilateral formed by joining the midpoints of the four adjacent sides is half of the total area of the quadrilateral. Which of the statements given above is/are correct ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither nor 2

25. [CDS (I) Paper 3 Q93] In a triangle ABC, ZABC=900, ZACB-300, AB=5cm What is the length of AC? **(a)** IO cm 5 cm **(b)** cm **(c)** 56 94. When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the 2 2 97. IA there three simultaneous linear equations in two unknowns. What can be the number of solutions **(d)** One or infinite

26. [CDS (I) Paper 4 Q60] In the figure given above, ABCD is a parallelogram. p is a point on DC such that DP produced meets AB produced at Q If the area Of the triangle BPQ is 20 square units, what is the area of the triangle DCP? **(a) (b) (c) (d)** 20 square units 30 square units 40 square units None Of the above 18 c In the figurel given above, L is any point on the bisector Of the acute angle ABC Knd the fine ML is • parallel to Which onc Of the following is cprrect? **(a)** The triangle BML is equilateral **(b)** The triangle BML is isosceles but right-angled **(c)** The triangle BML is but not right-angled **(d)** The triangle BtvfL isosceles isosceles is not In • the figure given above, PQ is a diameter Of the circle whose centre is at O. If EROS then what is the value of LRTS ?

27. [CDS (I) Paper 4 Q61] In the figure given above, AB is parallel to CD. What is zxOY? **(b)** 90 **(c)** 95 62. **(d)** 100' What is the radius of the circle inscribed in la triangle having side lengths 35 ctn, 44 cm and 75 cm? **(d)** 6 cm **(a)** 90 **(b)** 95 62. **(c)** 100' What is the radius of the circle inscribed in la triangle having side lengths 35 ctn, 44 cm and 75 cm? **(d)** 6 cm 63. What is the maximum length of a rod that Can be placed inside a box having the hape Of a cuboid Of length 30 cm, breadth 24 cm and height 18 em?

28. [CDS (I) Paper 4 Q88] Consider the follOW'ing statements • If two mangles are equi- angular, then they are similar. **(a)** The area Of the circle is e qual to that of square **(b)** The area o' the circle is larger than that Of square **(c)** The area of the circle is less than that bf square **(d)** No conclusiion can be drawn 87• If

29. [CDS (I) Paper 4 Q90] ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? **(a)** The area Of the circle is e qual to that of square **(b)** The area o' the circle is larger than that Of square **(c)** The area of the circle is less than that bf square **(d)** No conclusiion can be drawn 87• If

30. [CDS (I) Paper 4 Q98] 99. What is BE equal to? **(a)** 30 cm **(b)** 3045 cm **(c)** 300 cm **(d)** None Of the above What is the area Of the triangle ADC equal to? **(a)** 300 cm **(b)** 450 cm 2 **(c)** 600 cm 2 **(d)** N one of the above

31. [CDS (I) Paper 5 Q8] If e 2,600 is divided among three persons A, B and C in the ratio — , how much does A get ? **(a) (b) (c)** 600 800 1,000 1,200 For a positive integer n, define d(n) the number Of positive divisors Of n. What is the value of ? **(a) (c) (d)** None Of the above In a triangle ABC, side AB is extended beyond B, side BC beyond C and side CA beyond A. What is the sum of the three exterior angles ? **(a) (c) (d)** 2700 3050 360' 5400 PQR is a triangle right-angled at Q. If X and Y are the mid-points of the sides PQ and QR respectively, then which one Of the following is not correct ? **(a) (b)** RX2 py2 = 5 xy2 RX2 + py2 = xy2 PR2 4 + py2) = 5 PR2 + py2 - 3 (PQ2 + QR2) The diagonal of a cube is 4 cm. What is its 5. volume ? If e 2,600 is divided among three persons A, B and C in the ratio — , how much does A get ? **(b) (c)** 16 cm 32 cma 64 em 3 192 crn3 **(a) (b) (c)** 21 600 800 1,000 1,200 2. and — + What is the

32. [CDS (I) Paper 5 Q7] 8. In a triangle ABC, side AB is extended beyond B, side BC beyond C and side CA beyond A. What is the sum of the three exterior angles ? **(a) (b) (c)** 20 s 24-8 s 28-8 s 30 s **(a) (c) (d)** 2700 3050 360' 5400 X' is twice 8s Old as Y 3 years ago, when was as old as CY' today. If the difference between their ages at present is 3 years, how old is X' at present ? PQR is a triangle right-angled at Q. If X and Y are the mid-points of the sides PQ and QR respectively, then which one Of the following is not correct ? **(a) (c) (d)** 18 years 12 years 9 years 8 years **(a) (b)** RX2 py2 = 5 xy2 RX2 + py2 = xy2 PR2 4 + py2) = 5 PR2 + py2 - 3 (PQ2 + QR2) P-PQFA-L-RZ -A)

33. [CDS (I) Paper 5 Q15] The centroid Of a triangle ABC is 8 Cm from the vertex A. What is the length Of the median of the triangle through A ? **(a) (b) (d)** 20 cm 16 cm 12 cm IO Cm The equation whose roots are twice the roots of the equation x2 — 2x + 4 O is **(a) (c)** x - 2x +4 20 -2X+16.o -4X+8=o 4X+ 16-0 The outer and inner diameters a circular pipe are 6 cm and 4 cm respectively. If its length is IO em, then what is the total surface area in square centimeters ? **(a) (c) (d)** 55 150 None of the above A cylindrical rod Of iron whose radius is one-fourth of its height is melted and cast into spherical balls Of the same radius as that Of the cylinder. What is the number of spherical balls ? **(a) (b) (c) (d)** 2 3 4

34. [CDS (I) Paper 5 Q81] 82. 83. **(b) (d)** 1200 1050 900 850 What is the difference in the sectorial angles of the largest and smallest sectors ? **(a) (c) (d)** 850 82-50 77.50 In a triangle ABC, a line PQ is drawn parallel to BC, points p, Q being on AB and AC respectively. If AB 3AP, then what is the ratio of the area Of triangle APQ to **(a) (b)** the area Of triangle ABC ?

35. [CDS (I) Paper 5 Q86] 87. VVhat is one of the square roots of What is 27 x 1-2 x 5-5262 x equal to ? **(a)** 121-57 **(b)** 121-75 **(c)** 121-7S **(d)** None of the above ABC is an equilateral triangle inscribed in a circle with AB 5 cm. Let the bisector Of the angle A meet BC in X and the circle in 'Y. What is the value Of AX . AY ? **(a)** 16 cm **(b)** 20 cm2 **(c)** 25 cm2 **(d)** 30 cm2 Consider the following statements _ Statement-I.- Let PQR be a triangle in which P Q 3 em, QR 4 cm and RP 5 cm. If D is a point in the plane of the triangle such that D is either outside it or inside it, then DP + DQ + 6 Statement-II : PQR is a right-angled triangle. Which one of the following is correct in respect of the above two statements ? **(b) (c)** Both statement-I and statement211 are individually true and statement•li is the correct explanation of statement-I. Both statement-I and statement-II are individually true and statement-ll is not the correct explanation of statement•l_ Statement-I is true and statement-II is false. Statement-I is false and statement-n is 24 - Y. nd

36. **[CDS (I) Paper 5 Q88]** TWO unequal circles are touching each Other externally at P. APB and CPD are two secants cutting the circles at A, B, C and D. Which one of the following is correct ? 89. = AD2 — 4CD2 **(a) (c) (d)** ACBD is a parallelogram ACBD is a trapezium ACBD is a rhombus None Of the above ABC is a triangle right-angled at B and D is a point On BC produced (BD > BC), such that BD - 2 DC. is correct ? **(a) (b) (c)** AC2 = AC2 AC2 AC2 Which One of the following AD2 - 3CD2 ADZ — 2CD2 AD2 — 5CD2

37. **[CDS (I) Paper 5 Q90]** Let LMNP be a parallelogram and NR be perpendicular to LP. If the area Of the parallelogram is six times the area of triangle RNP and RP 6 cm, what is LR equal to ? **(a)** 121-57 **(b)** 121-75 **(c)** 121-7S **(d)** None of the above 88. TWO unequal circles are touching each Other externally at P. APB and CPD are two secants cutting the circles at A, B, C and D. Which one of the following is correct ?

38. **[CDS (I) Paper 5 Q92]** Let D, E be the points on sides AB and AC respectively Of a triangle ABC such that DE is parallel to BC. Let AD 2 cm, DB = I em, 9, AE = 3 cm and area' Of triangle ADE 3 em What is EC equal to ? **(a)** 15 cm **(b)** 20 cm **(c)** 25 cm **(d)** 30 cm Let D, E be the points on sides AB and AC respectively Of a triangle ABC such that DE is parallel to BC. Let AD 2 cm, DB = I em, AE = 3 cm and area' Of triangle ADE 3 em What is EC equal to ?

39. **[CDS (I) Paper 5 Q95]** Three congruent circles each or radius 4 cm touch one another. What is the area (in crn2) Of the portion included between them ? 87t 16Jä -47t 16Jä -21 **(c)** P.PQFA-L-RZ 87t 26 to **(a)** 15 cm **(b)** 20 cm **(c)** 25 cm **(d)** 30 cm Let D, E be the points on sides AB and AC respectively Of a triangle ABC such that DE is parallel to BC. Let AD 2 cm, DB = I em, AE = 3 cm and area' Of triangle ADE 3 em What is EC equal to ?

40. **[CDS (I) Paper 6 Q42]** 43. The lengths of two sides of a right angled triangle which contain the right angle are Sa' and 'b' respectively. Three squares are drawn On the three sides of the triangle on the outer side. What is the total area Of the triangle and the three squares ? **(a)** 2(a2 + b2) + ab **(b)** 2(a2 + b2) + 2•5ab **(c)** 2(a2 + b2) + 0•5ab **(d)** 2-562 + b2) What is — 14576 equal to ? **(a)** 2•3100191 **(b)** 20101091 **(c)** '2-3110091 **(d)** 2•3110901 Ihe sum of two numbers is IO and their product is

41. **[CDS (I) Paper 6 Q24]** What is their LCM ? **(b)** 286 **(d)** 296 77. **(a) (c)** The 280 288 area of a triangle ABC is equal to area Of square of side length 6 cm. What is the length of the altitude to AB where **(b) (c) (d)** 9 cm ? 14 cm 12 cm 8 cm

42. **[CDS (I) Paper 7 Q36]** What is the product of the two digits ? **(a)** 21 **(b)** 24 **(c)** 36 **(d)** 42 Which one among the following is not correct ? **(a)** Two congruent Ctriangles are necessarily similar **(b)** All equiangular triangles are similar **(c)** Two isosceles right triangles are similar **(d)** All isosceles triangles are similar ABCD is a rhombus with diagonals AC and BD. Then which one among the following is correct ? **(a)** AC and BD bisect each other but not necessarily perpendicular to each other **(b)** AC and BD are perpendicular to each other but,not necessarily bisect each **(c)** AC and BD bisect each Other fid p&rpendicular to each Other **(d)** AC and BD neither bisect each other nor perpendicular to each other The Sum Of a number and its reciprocal is 10/3 then the numbers are: **(a) (b)** 3 1/3 gits 36. not are lilar

43. **[CDS (I) Paper 7 Q8]** What is the value of vitiate X ? **(d)** 16 In the triangle ABC, the base BC is trisected at D and E. The line through D, parallel to AB, meets AC at F and the line through E parallel to AC meets AB at G Let EG and DF intersect at H. What is the ratio of the sum Of the area of parallelogramÅGEF and the area of the triangle DHE to the area Of the triangle ABC ? **(d)** 1/6 Ifx — 2 is a factor + kx2 what is the value of k ? -2x-24 then AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number ofpoints which are at a distance of I cm from AB and 5 cm from C ? **(a)** 16 In the triangle ABC, the base BC is trisected at D and E. The line through D, parallel to AB, meets AC at F and the line through E parallel to AC meets AB at G Let EG and DF intersect at H. What is the ratio of the sum Of the area of parallelogramÅGEF and the area of the triangle DHE to the area Of the triangle ABC ? **(b)** 1/6 Ifx — 2 is a factor + kx2 what is the value of k ? — 2K —24 then AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number ofpoints which are at a distance of I cm from AB and 5 cm from C ? 18. 19. 20. 21. Ihe angle of elevation of the tip of a tower from a point On the ground is 450. Moving 21 m directly towards the base ofthe tower, the angle ofelevation changes to 600. What is the height Of the tower, to the nearest meter ? **(c)** 48 m **(d)** . 49 m

44. **[CDS (I) Paper 7 Q29]** 30. 8 If a positive integer leaves remainder 28 when divided by 143, then what is the remainder obtained on dividing the same number by 13 ? **(d)** 10 What is the compound interest on Rs. 1600 at 25% per annum for 2 years compourided annually ? **(a)** Rs. 700 **(b)** Rs. 750 **(c)** Rs. 800 **(d)** Rs. 900 The graphs of ax + by c, dx + ey — f will be l. parallel if the system has no solution 2. coincident if the system has finite number of solutions 3. intersecting if the system has only one sol ution Which Of the above statements are correct ? **(a)** I and 2 Only **(b)** 2 and 3 only **(c)** 1 and 3 Only **(d)** t,2and3 The medians Of a triangle ABC intersect at G Which one of the following is correct ? **(a)** Five times the area of triangle ÅGB is equal to four times the area of triangle ABC. **(b)** Four times the area Of triangle AGB is equal to three times the area of triangle ABC. **(c)** -rhree times the area of triangle AGB is equal to the area Of triangle ABC **(d)** None Of the above

45. **[CDS (I) Paper 7 Q47]** Two transversals S and T cut a set Of distinct parallel lines. S cuts the parallel lines in points A, B, C and D and T cuts the parallel lines in points E, F, If AB -4, CD 12, then what is length of GH ? **(a)** afb A cardboard sheet in the form of a circular sector of radius 30 cm and central angle 1440 is folded to make a cone. What is the radius of the cone ? **(b)** 12 cm **(c)** 18 cm **(d)** 21 cm

46. **[CDS (I) Paper 7 Q56]** Let ABCD be a•parållelogram. Let m, n be positive integers such that n < m < 2n. Let AC = 2mn and BD = m2 — n'. Let AB + n2)/2. Statement-I : AC > BD Statement-II : ABCD is a rhombus. Which one of the following is correct in res}rct of the above statements ? **(a)** Both statement-I and statement-ll are true and statement-II is the correct explanation of statement-I **(b)** Both statement-I and statement-II are true but statement-II is not the correct e*iplanation of statement-I **(c)** Statement-I is true, but statement-II is **(d)** Statement-ll is true, but statement-I is false A Square, a circle and an equilateral triangle have same perimeter. Consider the following statements : I. The area of square is greater than the area of the triangle.

47. **[CDS (I) Paper 7 Q4]** What is the product of the two numbers ? **(a)** 360 **(b)** 480 **(c)** 486 **(d)** 512 Consider the following statements in respect Of an equilateral triangle ABC.

48. **[CDS (I) Paper 7 Q2]** 330 — 570 370 — tana 530 Which Of-the above statements is/are correct ? **(a)** I only **(b)** 2 only **(c)** Both land 2 **(d)** Neither I nor 2 Ifp=asin x + b cos x and q z a cos x —b sin x, then whit is p2 + q2 equal to ? **(d)** a2-b2 Statement-I • Statement-II' : Let the side DE ofa triangle DEF divided at S so that DS 1 Ifa line through i5E Vi S parallel to EF meets DF at T, then the area of triapgle DEF is twice the area of thc triangle DS T. The areas Of the similar triangles are proportional to the squares on the corresponding sides. Which Of the following is correct in respect Of the above statemfqts ? **(a) (b) (c) (d)** Both statement-I and statement-II are true and statement-ll is the correct explanation of statement-I Both statement-I and statement-ll are true but statement-II iS not the correct explanation of statement-I Statement-I is tme, but statement-II is Statement-ll is true, but statement-I is

49. **[CDS (I) Paper 7 Q93]** 92. x In the quadrilateral ABCD shown above, LDAB = ZDCX - 1200. If ZABC- 1050, what is ZADC equal to ? 600 **(c)** 750 **(d)** 950 450 c o In the figure given above AO = CD Where O is the centre of the circle. What is ZAPB equal to ? **(a)** 93. 92. 600 **(b)** 750 **(c)** 950 **(d)** 17/8 x In the quadrilateral ABCD shown above, LDAB = ZDCX - 1200. If ZABC- 1050, what is ZADC equal to ? 450 c o 94. 95. c In the figure given above ZABC = ZAED= 900. Consider the following statements : I. ABC and ADE are similar triangles. 2. The four points B, C, E and D may lie on a circle. Which of the above Statements is/are correct ?

50. **[CDS (I) Paper 8 Q61]** 62. If sine cose = 0/4, then the value of sin4 9 + cos4 0 is **(a)** 7/8 **(c)** 3/8 **(d)** 1/8 The locus of a point in rhombus ABCD which is equidistant from A and C is . **(a)** a fixed point on diagonal BD **(b)** diagonal BD **(c)** diagonal AC **(d)** None of the above In a triangle ABC, 2 ZC equal to ? **(a)** 350 **(b)** 400 **(c)** 600 **(d)** 700 — 800 , then what is In a triangle ABC, LBCA = 600 and AB2 = BC2 + CA2 + X. What is X equal **(d)** zero

51. **[CDS (I) Paper 8 Q2]** The locus of points which are equidistant from two parallel lines is a line parallel to both of them and drawn midway between them. The perpendicular distances Of any point on this locus line from two original parallel lines are equal. Further, no point outside this locus line has this property. Which Of the above statements is/are correct ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 If the diagonals of a quadrilateral are equal and bisect each other at right angles, then the quadrilateral is a **(a)** rectangle **(b)** square **(c)** rhombus **(d)** trapezium If two parallel lines are cut by two distinct transversals, then the quadrilateral formed by the four lines is always a **(a)** square **(b)** parallelogram **(c)** rhombus **(d)** trapezium 18

52. **[CDS (I) Paper 8 Q67]** 68. ABCD is a parallelogram. Ifthe bisectors of the angle A and angle C meet the diagonal BD at points P and Q respectively, then which one of the following is correct ? **(a)** PCQA is a straight line **(b)** Triangle APQ is similar to triangle PCQ Let X be any point within a square ABCD. On AX a square AXYZ is described such that D is within it. Which one of the following is Correct ? **(b)** ZADZ = ZBAX **(c)** AD **(d)** BX = DZ In a triangle ABC, XY is drawn parallel to BC, cutting sides at X and Y where AB = 4.8 em, BC = 7.2 cm, BX- 2 cm. What is the length of XY ? **(b)** 4.1 cm **(c)** 4.2 cm **(d)** 4.3 cm

53. **[CDS (I) Paper 8 Q68]** 18 ABCD is a parallelogram. Ifthe bisectors of the angle A and angle C meet the diagonal BD at points P and Q respectively, then which one of the following is correct ? **(a)** PCQA is a straight line **(b)** Triangle APQ is similar to triangle PCQ Let X be any point within a square ABCD. On AX a square AXYZ is described such that D is within it. Which one of the following is Correct ? **(b)** ZADZ = ZBAX **(c)** AD **(d)** BX = DZ In a triangle ABC, XY is drawn parallel to BC, cutting sides at X and Y where AB = 4.8 em, BC = 7.2 cm, BX- 2 cm. What is the length of XY ? **(b)** 4.1 cm **(c)** 4.2 cm **(d)** 4.3 cm

54. **[CDS (I) Paper 8 Q71]** Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? **(a)** AD is always greater than 4 cm but less than 5 cm **(b)** AD is always greater than 5 cm **(c)** AD is always less than 4 cm **(d)** None of the above Consider the following statements

55. **[CDS (I) Paper 8 Q72]** In the figure given above YZ is parallel to MN, xy is parallel to LM and XZ is parallel to LN. Then MY is • **(a)** the median of triangle LMN **(b)** the angular bisector of ZLMN **(c)** perpendicular to LN **(d)** perpendicular bisector Of LN

56. **[CDS (I) Paper 8 Q69]** 70. 71. **(b)** 350 **(c)** 400 **(d)** 45 Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? **(a)** AD is always greater than 4 cm but less than 5 cm **(b)** AD is always greater than 5 cm **(c)** AD is always less than 4 cm **(d)** None of the above Consider the following statements

57. **[CDS (I) Paper 8 Q74]** 650 **(b)** 300 **(d)** 500 950 A In the figure given above ABC is a triangle. BC is parallel to AE. If BC = AC, then what is ZCAE equal to ? **(a) (c)**

58. **[CDS (I) Paper 9 Q54]** 55. 56. {b) **(c) (d)** O None Of the above What is c•Nee (C — D B) equal to ? Consider the following statements ; I. ABCD is a cycbc quadrilateral. Z. sin (B -A) - cog (D - Ci W hich Of the atN3ve Atatements Ware **(a)** I only **(b)** 2 only **(c)** Both 2 **(d)** Neither I nor 2 W hat is see D — tan2 D equal to ? **(c)** 1/2 2/3 Nonc of the above An isosceles right-angled triangle ha area I square What is its ? **(b)** 2" + I units **(c)** + units + 1) units a-DETA-N-FMN 16 bur

59. **[CDS (I) Paper 9 Q84]** 85. of Of ig 22 Let side **(a)** ABC be an equilateral triangle. If the BC is produced to the point D So Olat 2CT **(b)** 40 em 45 em 50 em 80 em 83. ABC ia a right-angled triande such that a point On BC such that ÉD AB. he ratio of BD : DC for any value Of a and b given by ABC is a tnangle where BC = 2AB, C 30' and A Thc magnitude of the side AC 3 3BC BC of triangle ABC incel in J. What is BJC equal to ? 900 4 soo 2 ABC, - 900 CD is perpendicular AB. if An 4 cm ana 9 em, ehe of DC O.DETA.N+MN 22 84. 85. of Of ig 22 Let side **(c)** ABC be an equilateral triangle. If the BC is produced to the point D So Olat 2CT **(d)** 40 em 45 em 50 em 80 em 83. ABC ia a right-angled triande such that a point On BC such that ÉD AB. ratio of BD : DC for any value Of a and b given by O.DETA.N+MN 22

60. **[CDS (I) Paper 9 Q90]** Consider a circle with O r. Points A and B lie on its circum a point M hes outside of it Bueh that M, and O lie on the same straight line. then the. tio or MA to MB is (e) **(b)** (e) **(d)** equal to equal to r greater than 1 less than I In the figure given above. ABCD is EF is parallel to AD and y equal 460 **(d)** In the figure given above, Z PQR and QL is a median, PQ 5 cm and QR 12 an. QL is equal to G em Q.DETA.N-FMN 24 ius 91. fio **(a) (b)** equal to equal to r greater than 1 less than I •us 91. and tio 92. ABCD is AD and the following statements I. me La ngent Of a circle ig a line the t meets the circle in one and Only one g. The Of circle at the end p t of the diameter perrgdiculaT t. the Which Of the above ? 1 only **(c)** 2 only Booa and 2 Neither I nor 2 triangle Z C 900 ana CD is perpendicular to A.B. If ABxCD-CAxCE, then cna IS equal t. In the figure given above. EF is parallel to y equal 460 **(d)** 93. 1 The mean observation 1 if CA > CB CA2 of 7 observations is 7. If each is increased by 2, then the new In the figure given above, Z PQR and QL is a median, PQ 5 cm and QR 12 an. QL is equal to G em

61. **[CDS (I) Paper 10 Q49]** One side Of a parallelogram is 8•06 cm and its perpendicular distance from opposite side is 2•08 cm. What is the approximate area of the parallelogram? **(a)** -432x 51. **(b) (c)** 12•56 cm 14•56 cm2 16•76 22-56 cma SO. What is the volume of a cone having a base of radius IO cm and height 21 cm? 53. 54. 16 **(d)** y4

62. **[CDS (I) Paper 10 Q51]** What is the area of a circle whose area is equal to that of a triangle with sides 7 cm, 24 cm and 25 cm? **(a)** -432x 51. **(b) (c)** 12•56 cm 14•56 cm2 16•76 22-56 cma SO. What is the volume of a cone having a base of radius IO cm and height 21 cm? 53. 54. 16 **(d)** y4

63. **[CDS (I) Paper 10 Q56]** 57. If x is the curved surface area and y is the volume of a right circular cylinder, then which one of the following is correct? **(a) (b) (c) (d)** The ratio of height to radius of the cylinder is independent of x only The ratio of height to radius of the cylinder is independent of y only Either **(a)** Or **(b)** Neither **(a)** nor **(b)** A triangle DEP is formed by joining the midpoints of the sides of triangle ABC. Similarly a triangle PQR is formed by joining the mid- points of the sides of the triangle DEF. if the sides of the triangle PQRare Of lengths l, 2 and 3 units, what is the perimeter Of the triangle ABC ? **(a) (b) (c) (d)** 18 units 24 units 48 units Cannot be determined due to insufficient data B-BCRB-M-NBU/48A 18

64. **[CDS (I) Paper 10 Q65]** If the diagonals of a rhombus are cm and 1 •4 cm, then what is the perimeter of the rhombus? **(a)** 18 cm **(b)** 16 cm **(c)** 14 cm **(d)** 12 cm If the total surface area Of a cube is 6 square unit, then what is the volume of the cube? 65. If the diagonals of a rhombus are cm and 1 •4 cm, then what is the perimeter of the rhombus?

65. **[CDS (I) Paper 10 Q66]** Consider the following statements in respect of two chords XY and ZT of a circle intersecting at P • 1. px.PY-æ.PT 2. pxz and triangles. Which of the is/ are correct? Pry are similar above statements **(a)** 18 cm **(b)** 16 cm **(c)** 14 cm **(d)** 12 cm If the total surface area Of a cube is 6 square unit, then what is the volume of the cube? 65. If the diagonals of a rhombus are cm and 1 •4 cm, then what is the perimeter of the rhombus?

66. **[CDS (I) Paper 10 Q67]** ABCD is a quadrilateral such that BC = BA and CD > AD. Which one of the following is corTcct? **(d)** LEAD ZBCD ZBAD < EBCD ZBAD > ZBCD 2ZBAD ZBCD **(a)** 18 cm **(b)** 16 cm **(c)** 14 cm **(d)** 12 cm If the total surface area Of a cube is 6 square unit, then what is the volume of the cube? 65. If the diagonals of a rhombus are cm and 1 •4 cm, then what is the perimeter of the rhombus?

67. **[CDS (I) Paper 10 Q68]** ABC and XYZ are two similar triangles with ZZ, whose areas are respectively 32 Crn2 and 60-5 crn2. If XY = cm, then what is AB equal to? **(a)** C4.CD **(b)** AD DB B-BCRB-M-NBU / 48A 68. 69. 70. **(c)** 35 **(d)** 110 ABC and XYZ are two similar triangles with ZZ, whose areas are respectively 32 Crn2 and 60-5 crn2. If XY = cm, then what is AB equal to?

68. **[CDS (I) Paper 10 Q69]** 70. 22 **(a) (b) (c)** 56 cm 5-8 cm 6-0 cm A quadrilateral ABCD is inscribed in a circle. If AB is parallel to CD and AC BD, then the quadrilateral must be **(c) (d)** paral lelogram rhom bus trapezium None of the above

69. **[CDS (I) Paper 10 Q71]** In a triangle ABC, LB = 9th and then what is AB2 equal to? 2BC2 3BC2 **(b)** 4 BCI **(c) (d)** 5BC2 **(a)** C4.CD **(b)** AD DB B-BCRB-M-NBU / 48A 68. 69. 70. **(c)** 35 **(d)** 110 ABC and XYZ are two similar triangles with ZZ, whose areas are respectively 32 Crn2 and 60-5 crn2. If XY = cm, then what is AB equal to?

70. **[CDS (I) Paper 10 Q72]** The heights of two trees are x and y, where x > y. The tops of the trees are at a distance z apart. If S is the shortest distance between the trees, then what is s2 equal to? 73 **(a)** 35 **(d)** 110 22 **(b) (c) (d)** . The — z2 —2xy + — 2xy — y +2xy side AC of a triangle ABC is produced to D such that BC = CD. If ZACB is 700, then what is ZADB equal to? 4 50 ABC is a triangle right angled at A and a perpendicular AD is drawn on the hypotenuse BC What is BC. AD equal to? AB.AC **(b)** C4.CD **(d)** AD DB B-BCRB-M-NBU / 48A

71. **[CDS (I) Paper 10 Q2]** If the diagonals of a parallelogram ABCD are perpendicular, then ABCD may be a rhombus. If thc diagonals of a quadrilateral ABCD are equal and perpendicular, then ABCD is a square. Which of the statements given above is/are correct? **(a) (b) (c)** 1 only 2 only Both 1 and 2 Neither 1 nor 2 75. Consider the following statements • 1. The perpendicular bisector of a chord of a circle does not pass through the centre of the 2. The angle in a semicircle is a right angle. Which of the statements given above is/ are correct? **(d)** I only

72. **[CDS (I) Paper 10 Q76]** E is the midpoint of the median AD of a triangle ABC If BE produced meets the side AC at F, then CF is equal to **(a)** AC/3 **(b)** 2AC/3 **(c)** AC/2 **(d)** None of the above

73. **[CDS (I) Paper 10 Q77]** PQR is an equilateral triangle. O is the point of intersection of altitudes PL, QM and RN. If OP-8 cm, then what is the perimeter of the triangle mR ? (a' cm **(c)** 16-·/ä cm **(d)** 24Jä cm **(a) (b) (c)** 1 only 2 only Both 1 and 2 Neither 1 nor 2 75. Consider the following statements • 1. The perpendicular bisector of a chord of a circle does not pass through the centre of the 2. The angle in a semicircle is a right angle. Which of the statements given above is/ are correct? **(d)** I only

74. **[CDS (I) Paper 10 Q2]** If G is the centroid Of triangle ABC, then GA GC If H is the orthocentre of triangle then Which of the statements given above is/ are correct? only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 **(a)** 2 only **(b)** Both 1 and 2 **(c)** Neither 1 nor 2 84. 26 If tang + sec8= m, then what sec9 equal to? is 80. If the bisectors BI and CI of the angles B and C of a triangle ABC meet at the point I, then what is ZBIC equal to? 2 **(d)** 2m

75. **[CDS (I) Paper 11 Q44]** 45. AB and CD are two diameters Of a circle Of radius r and they are mutually perpendicular. What is the ratio Of the area of the circle to the area of the triangle ACD.? **(a) (d)** A circle of radius IO cm has an equilateral triangle inscribed in it. The length Of the perpendicular drawn from the centre to any side of the triangle is **(a)** None Of the above The dimensions Of a field are 15 m by 12 m. A pit 8 m long, 2-5 m wide and 2 m deep is dug in one corner of the field and the earth removed is evenly spread over the remaining area of the field. The level Of the field is raised by **(a) (b)** 15 Cm 20 cm 25 Cm 200 The sides of a right-angled triangle are equal to three consecutive numbers expressed in centimeters. What can be the area of such a triangle ? **(a) (b) (c)** 12 em 3-ABOA-N-NBU (12-, 46. of 47. ral

76. **[CDS (I) Paper 11 Q50]** 51. 52. (14 _ A) In a trapezium, the two non-parallel sides are equal in length, .each being of 5 units. The parallel sides are at a distance of 3 units apart. If the smaller side of the parallel sides is Of length 2 units, then the sum of the diagonals of the trapezium is **(a) (b)** 2 20-2 2n-1 **(c) (d)**

77. **[CDS (I) Paper 11 Q78]** 79. If sin I, where O < B < 450, then **(a) (a) (c) (d)** 8 cm and IO cm 9 and 11 cm IO cm and 12 em ll.cm and 13 cm sin 2B sin B cos 2B cos B For a plot Of land Of 100 m x 80 m, the level is to be raised by spreading the earth from a stack Of a rectangular base IO m x 8 m with vertical section being a trapezium Of height 2 m. The top of the stack is 8m X 5 m. By how many centimeters can the level be raised ? A cylinder is surmounted by a cone at One end, a hemisphere at the Other end. The radius is cm, the height of the cylinder Lis 6-5 cm and the total height or the structure is 12•8 cm. The volume V Of the structure lies between **(a) (b) (c) (d)** 3 cm AB is a straight line; C and D are points on the same side Of AB such that AC is perpendicular to AB and DB perpendicular to AB. Let AD and BC meet at E. AE BE What is _ + equal to ? **(b) (c) (d)** 370 cm3 and 380 em 380 cm3 and 390 cm 390 cm and 400 em None Of the above **(a) (b)** AD BC 1-5 N one Of the above In a triangle ABC, AB BC = CA. The ratio of the radius Of the Circumcircle to Of the **(a) (b) (c)** None of the above B-A3QA-N-NB'J Consider an equilateral triangle Of a Side Of unit length. A new equilateral triangle is by joining the mid-points Of the first one, then a third equilateral triangle is formed by joining the mid-points of the second. The process is continued. fie sum perimeter of all triangles thus formed is **(a) (c) (d)** 2 units 3 units 6 units Infinity

78. **[CDS (I) Paper 11 Q82]** Let ABCD be a parallelogram. Let P, Q R, S be the mid-points Of sides AB, BC, CD, DA respectively. Consider the following statements : I. Area Of trian gle APS < Area Of tri angle DSR if BD < AC. **(a) (b)** IO em 7 Crn **(c)** 1 only **(d)** 2 only

79. **[CDS (I) Paper 11 Q85]** 86. **(a) (b)** IO em 7 Crn **(c)** 1 only **(d)** 2 only

80. **[CDS (I) Paper 11 Q84]** ABC is triangle right•angled at B. If AB = 6 cm and BC 8 cm, then what is the length of the circum-radius Of the triangle ABC ? **(a) (b)** IO em 7 Crn **(a)** 1 only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 AB and CD are two chords Of a circle meeting externally at P. Then Which of the following Ware correct ?

81. **[CDS (I) Paper 11 Q89]** 90. ABCD is a parallelogram. E is a point on BC such that BE EC m : n. If and DB intersect in F, then what is the ratio of the area of triangle FEB to the area Of triangle 9 **(a) (b) (c)** (m/n)2 (n/m)2 [m/(m + n)12 If triangles ABC and DEF are similar such that ZAB DE and BC = 8 em, then What is EF equal to ? **(a) (c) (d)** 16 cm 12 cm A rectangle Of maximum area is drawn inside a circle of diameter 5 cm. What is the maximum area of such a rectangle ? **(a) (b)** 25 Cm 12-5 cm 12 Cm None of the above A drainage tile is a cylindrical shell 21 cm long. The inside and outside diameters are em and cm respectively, is the volume of the clay required for the tile ? **(c) (d)** 6-96m cubic cm None of the above B-ABQA-N-NBU (24 _ , C

82. **[CDS (I) Paper 12 Q20]** If every side Of an equilateral triangle is doubled, then the area Of new triangle becomes k times the area Of the Old One. What is k equal to? 17 **(d)** . The 7 4 3 sides Of a triangle are in the ratio — : If its perimeter is 52 cm, then What is the length Of the smallest side? **(a)** 9 cm **(b)** I O Crn **(c)** 11 cm **(d)** 12 cm

83. **[CDS (I) Paper 12 Q67]** How many equilateral triangles can be formed by joining any three vertices Of a cube? **(c) (d)** 8 None of the above For the next two (2) items that follow ABCD is a trapezium in which AB is parallel to CD. Let M be the midpoint of BC **(a)** sin x **(b)** cosx 62. What is **(c)** sin x + cos x **(d)** sin x — cos x 63. Consider the following statements • I. sin > sin I 2. COS < COS Which Of the above Stater-nents 66. What is the number of pairs of perpendicular planes in a cuboid?

84. **[CDS (I) Paper 12 Q2]** 'Area Of triangle ADM — Area of triangle ABM' is always equal to area Of triangle DCM , if AB- CD. Half Of area Of triangle ABM is equal to one-eighth Of area Of trapezium ABCD, if AB = CD. Which of the above statements is / are correct? **(a) (b) (c) (d)** I and 2 only 2 and 3 only 1 and 3 only 1, 2 and 3 Which of the above statements is / are correct?

85. **[CDS (I) Paper 12 Q70]** ABCD is a parallelogram. p and R are the midpoints of DC and BC respectively. The line PR intersects the diagonal AC at Q The distance CQ Will be equal to **(a)** AC/4 **(b)** BD/3 **(c)** BD/4 **(d)** AC/3 Consider the following Statements in respect Of an equilateral triangle • The altitudes are congruent.

86. **[CDS (I) Paper 12 Q72]** Consider the following ABC and DEF are triangles in a plane such that AB is parallel to DE, BC is parallel to EF and CA is parallel to FD Statement—I : If angle ABC is a right angle, then angle DEF is also a right angle. Statement—ll : Triangles Of the type ABC and DEF are always congruent. Which One of the following is correct in respect of the above statements? **(a) (b) (c) (d)** I and 2 only 2 and 3 only 1 and 3 only 1, 2 and 3 Which of the above statements is / are correct?

87. **[CDS (I) Paper 12 Q73]** 74. Let the incircle to a triangle ABC touch BC, AC and AB respectively at the points X, Y and Z. Stateme n If AB>BC then AB+AZ < BC+XC Statement—ll : AZ AY Which One of the following is correct in respect Of the above statements? **(d)** Statement—I and Statement—ll are correct and Statement—n is the correct explanation of Statement—I Statement—I and Statement—ll are correct and Statement—ll is not the correct explanation Of Statement—I Statement—I is correct and Statement—ll is incorrect Statement—I is incorrect and Statement—II is correct Let ABC be a triangle in which and Let the circle With centre at C and radius x meet BC at D. Let CF be the perpendicular drawn from C meeting AD at F. Statement—I : Triangle ACD is isosceles but not equilateral. Statement—ll : x 12 B-ABQB-N-LZS/IA Which one of the following is correct in respect Of the above statements? **(a) (b) (c) (d)** Statement—I and Statement—II are correct and Statement—II is the correct explanation Of Statement—I Statement—I and Statement—ll are correct and Statement—ll is not the Correct explanation Of Statemen Statement—I is correct and Statement—II is incorrect Statement—I is incorrect and Statement—ll is correct

88. **[CDS (I) Paper 12 Q75]** Bisectors Of two adjacent angles A and B Of a quadrilateral ABCD intersect each Other at a point P. Which one Of the following is correct? **(a) (b) (c) (d)** Statement—I and Statement—II are correct and Statement—II is the correct explanation Of Statement—I Statement—I and Statement—ll are correct and Statement—ll is not the Correct explanation Of Statemen Statement—I is correct and Statement—II is incorrect Statement—I is incorrect and Statement—ll is correct 75.

89. **[CDS (I) Paper 12 Q78]** In a triangle ABC, AD is the median through A and E is the midpoint Of AD, and BE produced meets AC at F. Then AF is equal to **(a)** AC/5 **(c)** AC/3 **(b)** AC/4 **(d)** AC/2 Three straight lines are drawn through the three vertices Of a triangle ABC, the line through each vertex being parallel to the opposite side. The triangle DEF is bounded by these parallel lines. Consider the following statements in respect Of the triangle DEF I. Each side Of triangle DEF is double the side of triangle ABC to Which it is parallel.

90. **[CDS (I) Paper 12 Q2]** Area Of triangle DEF is four times the area Of triangle ABC. Which of the above statements is/ are correct? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 Let ABCD be a parallelogram. Ikt X and Y be the midpoints Of the sides BC and AD respectively. Let M and N be the midpoints Of the sides AB and CD respectively. Consider the following statements : I. The straight I ine MX cannot be parallel to YN

91. **[CDS (I) Paper 12 Q86]** What is the volume Of the toy? **(a)** Crn 3 **(b)** 240rt cm3 **(c)** cm3 **(d)** 3207t cm 3 B-ABQB-N-LZS/ For the next three (3) items that follow A right triangle having hypotenuse 25 cm and legs in the ratio 3 : 4 is made to revolve about its hypotenuse. (It = 314)

92. **[CDS (I) Paper 12 Q90]** What is the parallelogram ? 91. area Of the **(c)** 63 m 2 96 m 2 126 m2 252 m 2 What is the shorter height Of the paral ogram? **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither nor 2 93. If one Of the roots Of the equation px + qx+r=O is three times the Other, then Which one of the following relations is correct? **(d)** 63 m 2 96 m 2 126 m2 252 m 2

93. **[CDS (I) Paper 12 Q2]** The difference between the diagonals Of the parallelogram is more than 20 m. The difference between the heights of the parallelogram is more than IO m. B-ABQB-N-LZS/ IA Which of the is/ are correct? above statements **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither nor 2 93. If one Of the roots Of the equation px + qx+r=O is three times the Other, then Which one of the following relations is correct? **(d)** 63 m 2 96 m 2 126 m2 252 m 2

94. **[CDS (I) Paper 13 Q47]** If each interior angle Of a regular polygon is 1350, then the number of diagonals of the polygon is equal to 54 **(b)** 48 **(c)** 20 **(d)** 18 **(a)** 12 Cm **(b)** 24 cm **(c)** 30 cm **(d)** 36 Crn 47. If each interior angle Of a regular polygon is 1350, then the number of diagonals of the polygon is equal to 54

95. **[CDS (I) Paper 13 Q53]** Let a, b, c be the sides of a right triangle, where c is the hypotenuse. The radius Of the circle which touches the sides Of the triangle is **(b) (c)** (a +2b +2c) /'2 **(a)** 12 Cm **(b)** 24 cm **(c)** 30 cm **(d)** 36 Crn 47. If each interior angle Of a regular polygon is 1350, then the number of diagonals of the polygon is equal to 54

96. **[CDS (I) Paper 13 Q2]** Let D be a point on the side BC of a triangle ABC If area Of triangle ABD area Of triangle ACD, then for all points O on AD, area Of triangle ABO area of triangle ACO If G is the point Of concurrence Of the medians Of a triangle ABC, then area Of triangle ABG=area of triangle BCG = area Of triangle ACG Which of the above statements is/ are correct? **(a) (b)** , where — , where K < — or K > 3 **(c) (d)** I only 2 only Both I and 2 Neither I nor 2 62. If COSA — tan B, cos B tan C and cos C = tan A, then sin A is equal to JB-I

97. **[CDS (I) Paper 13 Q69]** ABC is a triangle right angled at B and What is sinA•sinB+sinC equal to? 5 **(a)** 200 m 300 m 400 m 600 m tan A + cotA 4 68. Consider the following cos A sin A I — tan A I —cot A z sin A + cos A 2. **(b)** 2 only Both I and 2 Neither I nor 2 66. 67. If -cot2 330 - tan2 23' then tan 4 A + cot 4 A is equal to 110 191 194 — sin x I — sin x I + sin x cos x I * sin x then which Of the following is/ are correct? 1. p-q=r 2 - p2 qr Select the correct answer using the code given below. 69. 70. **(c) (d)** ABC is a triangle right angled at B and What is sinA•sinB+sinC equal to? 5 The value Of

98. **[CDS (I) Paper 13 Q87]** The sides Of a triangle are 25 cm, 39 cm and 56 cm. The perpen- dicular from the opposite vertex on the side Of 56 cm is **(a) (b) (c) (d)** 12 cm 15 cm 16 cm

99. **[CDS (I) Paper 13 Q88]** The diagonals of a trapezium are at right angles, and the slant sides, if produced, form an equilateral triangle With the greater of the two parallel sides. If thc area Of the trapezium is 16 square cm, then the distance between the parallel sides is **(a) (b)** 2 cm Cannot be determined due to insufficient data 92. Let on, OC and OD are rays in the anticlockwise direction such that ZAOB= ZCOD= 1 ocp, ZBOC= 820 and ZAOD= 780. Consider the following statements 89. From a circular piece of cardboard of radius 3 cm, two sectors of4 **(c)** I I square cm **(d)** 22 square cm

100. **[CDS (I) Paper 13 Q98]** What is the median class? 38 33 40 • 66 **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither I nor 2 Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 Cm is tangent to the inner circle. The radius Of the inner circle is **(d)** B-EFUA-O-PDW/52A 94. The complement angle of 80' is

101. **[CDS (I) Paper 14 Q57]** ABCD is a parallelogram with AB and AD as adjacent sides. If ZA = 600 and AB 2AD, then the diagonal BD will be equal to **(b)** OAD **(c)** 2AD **(d)** 3AD B-ABVB-O-GPU - A 20 **(a)** 15-5 km/hour **(b)** 15•84 kn/hour **(c)** 16 km/hour **(d)** 16-36 km/hour 56. 'lhere are 437 fruit plants in an orchard planted in rows. distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants.. niere is I m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of Rs. 100 per metre ?

102. **[CDS (I) Paper 14 Q58]** 'lhe EX)int O is equidistant fiorn the three sides of a triangle ABC. Consider the following statements : **(a)** 15-5 km/hour **(b)** 15•84 kn/hour **(c)** 16 km/hour **(d)** 16-36 km/hour 56. 'lhere are 437 fruit plants in an orchard planted in rows. distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants.. niere is I m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of Rs. 100 per metre ?

103. **[CDS (I) Paper 14 Q65]** An equilateral triangle BOC is drawn inside . a square ABCD. If angle AOD= 20, What is tane equal to ? **(a)** 2-0 **(b)** 2+6 66. The diameter Of a wheel that makes 452 revolutions to move 2 km and 26 decametre is equal to 67. If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow **(c)** I only **(d)** 2 only

104. **[CDS (I) Paper 14 Q68]** The two adjacent sides of a cyclic quadrilateral are 2 cm and 5 cm and the angle between them is **(a)** 2-0 **(b)** 2+6 66. The diameter Of a wheel that makes 452 revolutions to move 2 km and 26 decametre is equal to 67. If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow **(c)** I only **(d)** 2 only

105. **[CDS (I) Paper 14 Q69]** Let A denote the Set of quadrilaterals having two diagonals equal and bisecting each other. Let B denote the set of quadrilaterals having diagonals bisecting each other at **(a)** 2-0 **(b)** 2+6 66. The diameter Of a wheel that makes 452 revolutions to move 2 km and 26 decametre is equal to 67. If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow **(c)** I only **(d)** 2 only

106. **[CDS (I) Paper 14 Q85]** A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is **(b)** 128 **(b) (d)** 2 None of the above **(a)** 20 square t **(b)** 30 square unit **(c)** 40 square unit **(d)** None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

107. **[CDS (I) Paper 14 Q95]** How many right angled triangles can formed by joining the vertices of a cuboid ? **(a)** 24 **(b)** 28 **(c)** 32 **(d)** None of the above

108. **[CDS (I) Paper 14 Q98]** If a variable takes discrete values a + 4, a-3•5, a-2•5, a-3, a-2, a+O•5, a +5 and a —08 where a > O, then the median of the data set is **(a)** 24 **(b)** a —25 **(c)** 28 **(d)** a— 1 25

109. **[CDS (I) Paper 15 Q26]** Let ABC and A' B' C' be two triangles in which AB>A'B', BC>B'c• and CA > CA'. Let D, E and F be the mid- points of the sides BC, CA and AB respectively. Let D', E' and F' be thc midpoints of the sides BC', CA' and A' B' respectively. Consider the following statements . Statement I : AD>A'D', BE>B'E' are always true. Statement 11 : AB2 + CA2 AD2+BE2+CF2 and CF>CF' C'A'2 B'E'2+CF'2 Which one Of the following is correct in respect of the above statements? **(a)** Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I **(b)** Both Statement I and Statement 11 are true but Statement 11 is not thc correct explanation of Statement I **(c)** Statement I is true bu t Statement 11 is false **(d)** Statement I is false but Statement 11

110. **[CDS (I) Paper 15 Q4]** A' PW and CQ'D' are similar triangles, but need not bc Congruent A 'PW and C Q' D' are congruent triangle s. A 'PW and C' Q' D' are triangles of A'P'B' and C' Q' D' are triangles of same perimeter. Select the correct answer using the code given below. **(a) (b) (c) (d)** 2 and 3 only I and 3 only 1, 2 and 4 only 1, 2, 3 and 4

111. **[CDS (I) Paper 15 Q54]** If each interior angle Of a regular polygon is 140', then the number of vertices of **(a)** — unit **(b)** umt **(c)** —L unit **(d)** unit 53. A rhombus is formed by joining midpoints of thc sides of a rectangle in the suitable order. If the area of the rhombus is 2 square units, then the area Of the rectangle is

112. **[CDS (I) Paper 15 Q52]** Suppose ABC is a triangle with ABof unit length. D and E are the points lying on AB and AC respectively such that BC and DE are parallel. If the area Of triangle ABC is twice the area of triangle ADE, then the length Of AD is **(b)** umt **(c)** —L unit **(d)** unit **(a)** 10 **(a)** — unit the **(c)** polygon is equal to 8 B-ABPA-P-JRW / 54A

113. **[CDS (I) Paper 15 Q55]** Let the triangles ABC and DEF be such that LABC- ZDEF, LACB- ZDFE and ZBAC— ZEDF. Let L be the midpoint Of BC and M be the midpoint of EF. Consider the following statements . Statement I : Triangles ABL and DEM are similar. Statement 11 : Triangle ALC is congruent to triangle DMF even if DP. Which one of the following is correct in of the above statements? 56. **(a) (b) (c) (d)** The Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I Both Statement I and Statement 11 are true but Statement 11 is not the correct explanation of Statement I Statement I is true but Statement 11 is false Statement I is false but Statement 11 number Of rounds that a wheel of diameter 7/ I I metre will make in traversing 4 km Will be **(a) (b) (d)** 500 1000 1700 2000

114. **[CDS (I) Paper 15 Q59]** ABC and DEF are similar triangles. If the ratio Of side AB to side DE is a, then the ratio of area of triangle ABC to that Of triangle DEF is **(a)** 9600 square units **(b)** 10000 square units **(c)** 12000 square units **(d)** None of the above 58. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to

115. **[CDS (I) Paper 15 Q67]** In a triangle ABC if A is equal to **(c)** 2 -B- , thenC+2B **(a)** 2 -Be , thenC+2B **(b)** 0-1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e **(c)** only **(d)** 2 only

116. **[CDS (I) Paper 15 Q3]** The ratio of the area of triangle OPR to the arca of triangle OMP is : **(a)** I and 2 0 nly **(b)** 2 and 3 only **(c)** and 3 0 **(d)** I, 2 and 3 Let ABC be a triangle in which AB AC Let L be the locus of points X inside or on the triangle such that BX CX. Which of the following statements are corre ct? L is a straight line passing through A and in-centre of triangle ABC is L is a straight line passing through A and orthocentre of triangle ABC is a point on L L is a straight line passing through A and centroid of triangle ABC is a point on Select the correct answer using the code given below. 72. 73. Consider a circle With centre at C Let OP, OQ denote respectively the tangents to the circle drawn from a point O outside the circle. Let R be a point on OP and S be a point on OQ such that OR x SQ 0SxRP. Which of the following statements is/ are correct? If X is the circle with centre at O and radius OR, and Y is the circle With centre at O and radius OS then X -Y. 2. zpoct zoco- 9cr Select the correct answer using the code given below.

117. **[CDS (I) Paper 15 Q20]** Select the correct answer using the code given below. **(a)** I and 2 0 nly **(b)** 2 and 3 only **(c)** and 3 0 **(d)** I, 2 and 3 Let ABC be a triangle in which AB AC Let L be the locus of points X inside or on the triangle such that BX CX. Which of the following statements are corre ct? L is a straight line passing through A and in-centre of triangle ABC is L is a straight line passing through A and orthocentre of triangle ABC is a point on L L is a straight line passing through A and centroid of triangle ABC is a point on Select the correct answer using the code given below. **(a) (b) (c)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 B-ABPA-P-JRW / 54A

118. **[CDS (I) Paper 15 Q2]** If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n **(a)** 165% 21 B-ABPA-P-JRW/ 54A 74. Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(b)** 12% 76. If 61 where x, y and z are natural numbers, then What is z equal to? **(c)** 75. The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 77. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times thc arca Of the smaller ring, then n is equal to **(d)** 10 78. Consider the following statements : 2. If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n 3. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is

119. **[CDS (I) Paper 15 Q3]** Then the sum Of the interior angles of a regular polygon Of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is (m Which Of the above statements is/ are correct? **(a)** 165% 21 B-ABPA-P-JRW/ 54A 74. Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(b)** 12% 76. If 61 where x, y and z are natural numbers, then What is z equal to? **(c)** 75. The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 77. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times thc arca Of the smaller ring, then n is equal to **(d)** 10 78. Consider the following statements : 2. If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n 3. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is

120. **[CDS (I) Paper 15 Q2]** There exists a regular polygon whose exterior angle is **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In a triangle PQR, point X is on PQ and point Y is on PR such that XP = I •5 units, 6 units, PY = 2 units and YR 8 units. Which Of the following are correct? 1. QR=5XY 2. OR is parallel to XY. 3. Triangle PYX is similar to triangle PRQ. Select the correct answer using the code given below.

121. **[CDS (I) Paper 15 Q5]** Then the exterior angle Of any regular polygon Of n sides is acute. **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In a triangle PQR, point X is on PQ and point Y is on PR such that XP = I •5 units, 6 units, PY = 2 units and YR 8 units. Which Of the following are correct? 1. QR=5XY 2. OR is parallel to XY. 3. Triangle PYX is similar to triangle PRQ. Select the correct answer using the code given below.

122. **[CDS (I) Paper 15 Q80]** 81. Which Of the above statements is/are corre ct? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In a triangle PQR, point X is on PQ and point Y is on PR such that XP = I •5 units, 6 units, PY = 2 units and YR 8 units. Which Of the following are correct?

123. **[CDS (I) Paper 15 Q3]** Triangle PYX is similar to triangle PRQ. Select the correct answer using the code given below. **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In a triangle PQR, point X is on PQ and point Y is on PR such that XP = I •5 units, 6 units, PY = 2 units and YR 8 units. Which Of the following are correct? 1. QR=5XY 2. OR is parallel to XY. 3. Triangle PYX is similar to triangle PRQ. Select the correct answer using the code given below.

124. **[CDS (I) Paper 15 Q1]** The area of triangle ABC is equal to four times the area Of triangle BED. **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In a triangle PQR, point X is on PQ and point Y is on PR such that XP = I •5 units, 6 units, PY = 2 units and YR 8 units. Which Of the following are correct? 1. QR=5XY 2. OR is parallel to XY. 3. Triangle PYX is similar to triangle PRQ. Select the correct answer using the code given below.

125. **[CDS (I) Paper 15 Q88]** ABC is an equilateral triangle and X, Y and Z are the points on BC, CA and AB respectively such that BX CY = AZ. Which Of the following is/ are correct? I. XYZ is an equilateral triangle. **(a)** 24 16 = then the In the figure given above, AD CD What is the value of Z CDB? **(b)** 320 **(c)** 640 BC ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct? **(d)** AQ2+AB2 2 pQ2 BP2

126. **[CDS (I) Paper 15 Q2]** Triangle XYZ is similar to triangle Select the correct answer using thc code given below. only **(a)** 24 16 = then the In the figure given above, AD CD What is the value of Z CDB? **(b)** 320 **(c)** 640 BC ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct? **(d)** AQ2+AB2 2 pQ2 BP2

127. **[CDS (I) Paper 15 Q85]** If — and x — y value of xy is equal to 36 **(b)** 24 16 = then the In the figure given above, AD CD What is the value of Z CDB? **(a)** 320 **(b)** 640 BC ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct? **(a)** AQ2+AB2 2 pQ2 BP2 **(b)** AQ2+PQ2 **(c)** AQ2 2 _ + **(d)** AQ2+AP2 BK2+KQ2 B-ABPA-P-JRW / S4A **(d)** Cannot be determined due to insufficient data

128. **[CDS (I) Paper 15 Q89]** In the figure given above, p and q are parallel lines. What are the values of the angles x, y and z? **(a) (b) (c) (d)** 400, — 502, y- 40', y 5x-2y= 10 z- 1050 z-II(P The linear inequations, for which the shaded area in the figure given above is the solution set, are **(d)** 2x+6y•S21, 2x+6YS21, 2x.6Y221. 2X+6y>21, B-ABPA-P-ORW / 54A 5x-2ys 10 10 5x-2ys 10 5x-2y2 10 32

129. **[CDS (I) Paper 15 Q97]** Suppose = for O i IO, where X > Which one Of the following is correct? **(a)** AM < Median **(c)** GM Median **(d)** AM = Me dian Suppose — for i = I, 2, 3, . , Which one of the following is not corre ct? **(b)** GM>1/6 (C) HM>1/6 **(d)** Median HM

130. **[CDS (I) Paper 16 Q46]** Let ABC be a right angled triangle with BC = 5 and 12 cm. Let D be a point On the hypotenuse •AB such that ZBCD=300. What is length of CD? **(a)** 264 **(b)** 201 **(c)** 132 **(d)** 105 46. Let ABC be a right angled triangle with BC = 5 and 12 cm. Let D be a point On the hypotenuse •AB such that ZBCD=300. What is length of CD?

131. **[CDS (I) Paper 16 Q53]** If a quadrilateral has an Inscribed then the sum of a pair Of opposite sides equals **(a)** Half the sum of the diagonals **(b)** Sum of the other pair of opposite sides **(c)** Sum Of two adjacent sides **(d)** None of the above

132. **[CDS (I) Paper 16 Q56]** In an equilateral triangle another equilateral triangle is drawn inside joining the mid-points Of the sides of given equilateral triangle and the process is continued up to 7 times. What is the ratio of area of fourth triangle to that Of seventh triangle ? **(a)** 256 : I **(b)** 128 : 1 **(c)** 64: 1 **(d)** 16:1

133. **[CDS (I) Paper 16 Q61]** The sides of a triangle are given by c + a2 and (b + c) where a +122, a, b, c are positive. What is the area of the triangle equal to ? 2 a2b2 +b2c2 +c2a2 **(b)** 2 2 3(a2b2 +b2c2 +c2a2) **(d)** 2 **(a)** 367t cm3 **(b)** 187t cm3 **(c)** cm3 61. The sides of a triangle are given by c + a2 and **(d)** 2 2 3

134. **[CDS (I) Paper 16 Q68]** What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ? **(a)** 30 crn2 **(b)** 35 cm2 **(c)** 40 cm2 **(d)** 42

135. **[CDS (I) Paper 16 Q84]** A pole stands vertically inside a triangular park ABC. If the angle of elevation Of the top Of the pole from each corner of the park is same, then in the triangle ABC, the foot of the pole is at the **(a)** Centroid **(b)** Circumference **(c)** Incentre **(d)** Orthocentre

136. **[CDS (I) Paper 16 Q87]** For x > O, if a variable takes discrete values x + 4, x — 3•5, x —25, x— 3, x— 2, x x —0•5, x + 5, then what is the median ? **(a)** x-1-25 **(b)** x-o.5 **(c)** 1-25 88. The median of set Of 9 distinct observations is If each Of the largest 4 observations of the set is increased by 2, then the median Of the new set **(d)** is increased by 2

137. **[CDS (I) Paper 17 Q98]** 94. In the figure given below, PQRS is parallelogram. PA bisects angle p and SA bisects angle S. What is angle PAS equal to ? **(b) (c) (d)** S 600 750 1000 In the figure given below, Z A 800 Z ABC **(a) (b) (c)** S 600 750 1000 **(d)**

138. **[CDS (I) Paper 17 Q95]** 96. (28-A) In the figure given below, PQR is a non-isoseeles right-angled triangle, right angled at Q. If LM and QT are parallel and z RLMequa1to? **(c) (d)** Q Z PQT Z LRM z RML Qvr L R In the figure given below, PQ is parallel to RS and PR is parallel to QS. If Z LPR 350 and Z UST = 700, then what is Z MPQ equal to ? **(a) (b) (c)** S 600 750 1000 **(d)**

139. **[CDS (I) Paper 17 Q97]** 98. In the figure given below, ABC is a triangle with AB = BC and D is an interior point of the triangle ABC such that Z DAC = Z DCA. c Consider the following statements : **(a) (b) (c)** 2, 3 and 4 only I, 2 and 4 only I, 3 and 4 only 1 and 2 only **(d)**

140. **[CDS (I) Paper 17 Q1]** Triangle ADC is an isosceles triangle. **(a) (b) (c)** 2, 3 and 4 only I, 2 and 4 only I, 3 and 4 only 1 and 2 only **(d)**

141. **[CDS (I) Paper 17 Q2]** D is the centroid Of the triangle ABC. **(a) (b) (c)** 2, 3 and 4 only I, 2 and 4 only I, 3 and 4 only 1 and 2 only **(d)**

142. **[CDS (I) Paper 17 Q3]** Triangle ABD is congruent to the triangle CBD. Which Of the above statements are correct ? **(a) (c) (d)** I and 2 only 1 and 3 only 1, 2 and 3 the figure given below, M is the mid-point Of AB and Z DAB CBA and Z AMC Z BMD Then the triangle ADM is congruent to the triangle BCM by **(a) (b) (c)** BLS-S-FDE SAS rule SSS rule ASA rule AAA rule ogle 99. (30 A) ABCD is a square. X is the mid-point Of AB and Y is the mid-point of BC. Consider the following statements :

143. **[CDS (I) Paper 17 Q4]** Triangles ADX and BAY are congruent. Z DXA Z AYB. DX is inclined at an angle 600 With AY. DX is not perpendicular to AY. Which of the above statements are correct ? **(a) (b) (c)** 2, 3 and 4 only I, 2 and 4 only I, 3 and 4 only 1 and 2 only **(d)**

144. **[CDS (I) Paper 18 Q69]** If PL, QM and RN are the altitudes of triangle PQR whose orthocentre is O, then Q is the orthocentre of the triangle **(a) (b) (d)** OPQ OQR PLR OPR In triangle ABC, = 900 and CD is the perpendicular from C to AB. If = + then which one of the following is correct ? **(a) (b) (c) (d)** BC . CD AB . CA AB.BC=CD . CA CA2 + CB2 2 (AD2 + CD2) AB . CD = BC . CA If a point O in the interior of a rectangle ABCD is joined with each of the vertices A, B, C and D, then OB2 + OD2 will be equal to **(a) (b) (d)** 20C2 + OA2 OC2 -OA2 OC2 + OA2 OC2 + 20A2 **(a) (b) (c) (d)** STS-M-TPT The diagonals Of a cyclic quadrilateral ABCD

145. **[CDS (I) Paper 18 Q67]** 64. jntersect at P and the area ofthe triangle APB is 24 square cm. If AB 8 and CD = 5 cm, then what is the area of the triangle CPD ? **(a) (b) (c) (d)** STS-M-TPT The diagonals Of a cyclic quadrilateral ABCD 67. 64. jntersect at P and the area ofthe triangle APB is 24 square cm. If AB 8 and CD = 5 cm, then what is the area of the triangle CPD ? 65. 66.

146. **[CDS (I) Paper 18 Q65]** 66. (20-A) If PL, QM and RN are the altitudes of triangle PQR whose orthocentre is O, then Q is the orthocentre of the triangle **(a) (b) (c) (d)** 24 Square cm 15 square cm 12•5 square cm 9•375 square cm 68. **(a) (b) (d)** OPQ OQR PLR OPR In an equilateral triangle ABC, BD is drawn perpendicular to AC. What is BD2 equal to ? In triangle ABC, = 900 and CD is the perpendicular from C to AB. If = + then which one of the following is correct ? **(a) (b) (d)** The AD2 2AD2 3AD2 4AD2 distance between the centres of two **(a) (b) (c)** BC . CD AB . CA AB.BC=CD . CA CA2 + CB2 2 (AD2 + CD2) AB . CD = BC . CA circles having radii 9 cm and 4 cm is 13 cm. What is the length of the direct common tangent of these circles ?

147. **[CDS (I) Paper 18 Q77]** 78. PQ = 2XY PX=QY PQ- 3XY **(a) (b) (d)** 4 72 1•25 6•25 In a triangle ABC, the medians AD and BE intersect at G. A line DF is drawn parallel to BE such that F is on AC. If AC 9 em, then what is CF equal to ? **(a) (b) (c)** 2•25 cm 3 cm cm In a triangle PQR, X is a point on PR and Y is a point on QR such that PR = 10 cm, nx - 4 cm, YR 2 can, QR = 5 cm. Which one Of the following is correct ? XY is parallel to PQ (24 nd 79. of

148. **[CDS (I) Paper 18 Q1]** 2. 3. **(a) (b) (d)** 4 72 1•25 6•25 If A and C are parallel and B and C are parallel; then A and B are parallel. If A is perpendicular to C and B is perpendicular to C; then A and B are If the acute angle between A and C is equal to the acute angle between B and C; then A and B are parallel. Which of the above statements are correct ? In a triangle ABC, the medians AD and BE intersect at G. A line DF is drawn parallel to BE such that F is on AC. If AC 9 em, then what is CF equal to ? **(d)** 1, 2 and 3 I and 2 only I and 3 only 2 and 3 only **(a) (b) (c)** 2•25 cm 3 cm cm

149. **[CDS (I) Paper 18 Q84]** One-fifth of the area of a triangle ABC is cut off by a line DE drawn parallel to BC such that D is on AB and E is on AC. If BC 10 cm, then what is DE equal to ? **(a) (b) (c) (d)** 4 cm There are 8 lines in a plane, no two Of which are parallel. What is the maximum number of points at which they can intersect ? **(b) (c) (d)** 15 21 28 None Of the abov e A closed polygon has six sides and one of its angles is 300 greater than each of the other five equal angles. What is the value of one of the equal angles ? **(a)** 550 **(c) (d)** STS-M-TPT 1150 1500 1750 (26 - , ut

150. **[CDS (I) Paper 18 Q1]** 2. The point of intersection of the perpendicular bisectors of the sides of a triangle may lie outside the triangle. The point of intersection of the perpendiculars drawn from the vertices to the opposite sides of a triangle may lie Which Of the above statements is/are correct ? **(a) (b) (c) (d)** 1 only 2 only Both I and 2 Neither 1 nor 2 For the next five (05) items that follow : In a University there are 1200 students studying four different subjects, Mathematics, Statistics, Physics and Chemistry. 20% of the total number of students are studying Mathematics, one-fourth of the total number of students are studying Physics, 320 students are studying Statistics and remaining students are studying Chemistry. Three-fifth of the total number of students studying Chemistry are girls. 150 boys are studying Mathematics. 60% of students studying Physics are boys. 250 girls are studying Statistics. What is the total number of boys studying Statistics and Physics ? **(a) (b) (c) (d)** 180 240 250 310 One-fifth of the area of a triangle ABC is cut

151. **[CDS (I) Paper 19 Q3]** 4. The orthocentre of a triangle always lies inside the triangle. Centroid Of a triangle always lies inside the triangle. The Orthocentre of a right angled triangle lies on the triangle. The centroid of a right angled triangle lies on the triangle. Which Of the above statements are correct ? **(a) (b) (c)** I and 2 I and 4 2 and 3 2 and 4 The locus of a point equidistant from two intersecting lines is **(a)** A straight line **(b)** A circle **(c)** A pair of straight line s **(d)** None of the above SPSS-T-XHI (24 -A Of

152. **[CDS (I) Paper 19 Q82]** Consider the following statements ; Two triangles are said to be congruent, if **(a)** A pair of straight line s **(b)** None of the above SPSS-T-XHI The locus of the mid-points of the radii Of 80. 76. length 16 cm of a circle is 77. 78. 79. **(c) (d)**

153. **[CDS (I) Paper 19 Q81]** Consider the following statements . Which Of the above Statements are correct ? 1, 2 and 3 **(b)** l, 3 and 4 **(c)** l, 2 and 4 **(d)** 2, 3 and 4 Given that the angles Of a polygon are all equal and each angle is a right angle. **(a)** A pair of straight line s **(b)** None of the above SPSS-T-XHI The locus of the mid-points of the radii Of 80. 76. length 16 cm of a circle is 77. 78. 79. **(c) (d)**

154. **[CDS (I) Paper 19 Q97]** 98. In the figure given below, p, q, r are parallel lines; I and m are two transversals. p q c Consider the following : **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 100. In the equilateral triangle ABC given 98. AD DB and AE EC. If I is the length of a side Of the triangle, then What is the area of the shaded region ?

155. **[CDS (I) Paper 19 Q2]** ABxEF=BCxDE Which Of the above is/are correct ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 In the equilateral triangle ABC given AD DB and AE EC. If I is the length of a side Of the triangle, then What is the area of the shaded region ? **(a)** 16 312 16 32 **(d)** SPSS-LXHI (30- Ilel

156. **[CDS (I) Paper 19 Q100]** In the equilateral triangle ABC given **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 100. In the equilateral triangle ABC given 98. AD DB and AE EC. If I is the length of a side Of the triangle, then What is the area of the shaded region ?

157. **[CDS (I) Paper 19 Q4]** The difference between their interior angles is **(a)** ab 1 **(b)** abc equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to **(c)** abc **(d)** xyz

158. **[CDS (I) Paper 19 Q75]** 23-A) Walls (excluding roofs and floors) of 5 identical rooms having length, breadth and height 6 m, 4 m and m respectively are to be painted. Out of five rooms, two rooms have one square window each having a side of 2-5 m. Paints are available only in cans of 1 litre; and 1 litre of paint can be used for painting 20 square metres. The number of cans required for painting is **(d)** 10 12 13 14 Let S be the parallelogram obtained by joining the mid-points of the parallelogram T. Consider the following statements : **(a) (b) (c)** x +3 cos x +3 cos x + cos6x? 1 8 Walls **(d)** 12 16

159. **[CDS (I) Paper 19 Q74]** 75. Let S be the parallelogram obtained by joining the mid-points of the parallelogram T. Consider the following statements : **(a) (b) (c)** x +3 cos x +3 cos x + cos6x? 1 8 Walls **(d)** 12 16

160. **[CDS (I) Paper 19 Q83]** 25 -A) A rectangular block Of length 20 em, breadth 15 and height IO is cut up into exact number of equal cubes. The least possible number of cubes will be **(a) (b) (c)** 12 16 20 24 If the diagonal of a cube is of length l, then the total surface area of the cube is **(a) (c)** 312 2 12 An equilateral triangle, a square and a circle have equal perimeter. If T , S and C denote the area Of the triangle, area Of the square and area Of the circle respectively, then which one of the following is correct ? **(a) (b) (c) (d)** ser«c The areas Of two similar triangles are (7 — 4) cm2 and (7 4) cm2 respectively. The ratio Of their corresponding sides is **(a) (c) (d)** 7-3v'ä 5-v/ä There is a path of width 5 m around a circular

161. **[CDS (I) Paper 19 Q85]** 86. 87. (4 — 't)a2 - 2)a2 (27 -A) In a triangle ABC, the sides AB, AC are produced and the bisectors Of exterior angles of Z ABC and Z ACB intersect at D. Z BAC 500, then Z BDC is equal to **(a) (b) (c) (d)** 5 2 1/2 1/5

162. **[CDS (I) Paper 19 Q97]** 98 In the figure given below, ABC is a triangle With AB perpendicular to BC. Further BD is perpendicular to AC. If AD = 9 cm and DC = 4 cm, then what is the length of BD ? c **(a) (b) (c) (d)** 13/36 cm 36/13 13/2 In the figure given INIow, the diameter of bigger semicircle is 108 cm. What is the ar•ea of the shaded region ? **(a) (b) (c) (d)** 54 54 2017t cm2 186•3't cm2 405T cm2 7695m cm2 GTHY-T-XMT (31-

163. **[CDS (I) Paper 19 Q99]** In the figure given below, ABC is an equilateral triangle with each side Of length 30 cm. XY is parallel to BC, XP is parallel to AC and YQ is parallel to AB. If XY + xp + YQ is 40 cm, then the value of PQ is A **(a) (b) (c) (d)** 13/36 cm 36/13 13/2

164. **[CDS (I) Paper 20 Q4]** If the perimeter Of the triangle is k times its largest side, then What is the value ofk? **(d)** 2 2 ¯viå 2 **(a) (b)** Only one Only two Only three Infirutely many Two circles are drawn with the same centre. The circumference of the smaller circle is 44 cm and that of the bigger circle is double the smaller one. What is the area between these two circles ? 154 Square Square Cm 462 Square Cm Square Cm A rectangular red carpet of size 6 ft x 12 ft has a dark red border 6 inches wide. What is the area of the dark red border ? 82. The of a right-angled triangle is k times the shortest side. If the ratio of the other side to hym»tenuse is 4 : 5, then What is the value of k ? **(c) (d)** A 12 m long wire is cut into two pieces. one Of ' which is bent into a circle and the other into a square enclosing the circle. What is the radius Of the circle ? 12

165. **[CDS (I) Paper 20 Q88]** The hypotenuse Of a right-angled triangle IO cm and its area is 24 cm2. If the shorter side is halved and the longer side is doubled, the new hypotenuse becomes **(a) (b) (c)** cm 87. **(d)**

166. **[CDS (I) Paper 20 Q2]** Any cyclic parallelogram is a rectangle. Which of the above statements Ware correct ? **(b) (d)** 2 only Both I and 2 Neither I nor 2 **(a) (b) (c)** cm 87. **(d)**

167. **[CDS (I) Paper 20 Q89]** Consider the following inequalities in respect Of any triangle ABC : 90. (29-A) **(a) (b)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 **(c) (d)** 21: 16 14:11

168. **[CDS (I) Paper 20 Q96]** What is the area Of triangle ADC ? **(b) (c) (d)** 126 cm2 124 cm2 122 cm2 120 cm2 What is the area Of quadrilateral ABCD ? **(a) (b) (c) (d)** 300 cm2 306 cm2 312 cm2 316 em2 What is the difference between perimeter Of triangle ABC and perimeter Of triangle ADC ? **(b)** (e) **(d)** Consider the following for the next two (02) items : An equilateral triangle ABC is inscribed in a circle of radius 20 cm.

169. **[CDS (I) Paper 20 Q97]** What is the length of the side of the triangle ? **(a)** What is the area Of triangle ADC ? 15 Cm 20 em cm **(b) (c) (d)** 126 cm2 124 cm2 122 cm2 120 cm2 What is the area Of quadrilateral ABCD ? Consider the following for the next two

170. **[CDS (I) Paper 20 Q98]** 31 —A) The centroid of the triangle ABC is at a distance d from the vertex A. What is d equal **(b)** 20 em cm Consider the following for the next two (02) The sum of length, breadth and height Of a cuboid is 22 cm and the length Of its diagonal is What is the surface are a of the cuboid ? Of **(b) (c) (d)** 216 cm2 144 cm2 Cannot be determined insufficient data due to

171. **[CDS (I) Paper 21 Q48]** ABC is a tingle riÖt at B. If ABS 5 cm BC = What is of the p*iølar drawn from B the hypotenuse ? 243 (e) **(a)** 32% **(b)** 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ? **(c)** cm2 **(d)** 50

172. **[CDS (I) Paper 22 Q35]** What is the median of the data 3, 5, 9, 4, 6, 11, 18? **(d)** 7-5 **(a)** 7-5 36. In a pi-diagram there are three sectors. If the ratio Of the angles Of the sectors is 1 : 2 : 3, then what is the angle of the largest sector ? **(b)** 2000 **(c)** 1800 **(d)** 1500

173. **[CDS (I) Paper 22 Q67]** If the angles of a triangle are 300 and 450 and the included side is + 1) cm, then what is the area of the triangle ? **(a)** +1) cm2 **(b)** + 3) cm2 **(c)** l) cm2 **(d)** 266+1) cm2

174. **[CDS (I) Paper 22 Q3]** Then what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side ? **(b)** 4.3 **(c)** 16:9 **(a)** — •E **(b)** litre **(c)** litre , **(d)** — litre 10

175. **[CDS (I) Paper 22 Q78]** ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

176. **[CDS (I) Paper 22 Q79]** A line through the vertex A of a parallelogram ABCD meets DC in p and BC produced in Q. If P is the mid-point of DC, then which of the following Ware correct ? **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

177. **[CDS (I) Paper 22 Q81]** If the diagonals Of a rhombus are X and y, then what is its area ? 2 **(b)** 4 **(c)** xy **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

178. **[CDS (I) Paper 22 Q82]** The lengths of sides of a triangle are where If one of the angles is 900, then what are the minimum integral values of x, y, z respectively ? **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

179. **[CDS (I) Paper 22 Q83]** What the maximum number of circum-circles that a triangle can have ? **(d)** Infinite **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

180. **[CDS (I) Paper 22 Q89]** ABCD is a quadrilateral such that AD = DC CA = 20 units, BC 12 units and ZABC= **(a)** 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ? **(b)** 4 square units **(c)** 245 square units **(d)** r square units

181. **[CDS (I) Paper 22 Q94]** ABCD is a trapezium, where AB is parallel to DC. If AB—4 cm, BC—3 cm, CD — 7 cm and DA = 2 em, then what is the area of the trapezium ? **(a)** — cm2 **(b)** — cm2 **(c)** — E cm2 **(d)** cm2 92. 94. ABCD is a trapezium, where AB is parallel to DC. If AB—4 cm, BC—3 cm, CD — 7 cm and DA = 2 em, then what is the area of the trapezium ? 66 units

182. **[CDS (I) Paper 23 Q70]** How many solid lead balls each of diameter 2 mm can made from a solid lead ball Of radius S cm ? **(a) (b) (c) (d)** 512 1024 256000 512000 The two sides of a triangle are 40 cm and 41 cm. If the perimeter of the triangle is 90 cm, what is its area ? **(b) (c) (d)** 90 135 cm2 150 cm2 180 cm2 The diagonals of a rhombus dimr by 2 units and its perimeter exceeds the sum of the diagonals by 6 units. What is the area Of the rhombus ? **(a) (b)** (e) **(d)** 48 units 36 square units 24 square units 12 square units What is the area of a right-angled triangle, if the radius of the circumcircle is 5 em and altitude draw•n to the hypotenuse is 4 cm ? **(a) (b) (c) (d)** 20 cm2 18 cm2 16 em2 IO cm2 DZOL-T-LKM (25-A 'ter 71. is 72. re

183. **[CDS (I) Paper 23 Q67]** 69. 70. (25-A) How many solid lead balls each Of diameter 2 mm made from a solid lead ball of radius 8 Cm ? **(a) (b) (c)** 512 1024 2560m 512000 The two sideg Of a triangle 41 em. If the Of cm, What is its area ? are the 40 cm and triangle is 71. 72. 78. 74. In a triangle, value Of all the angles are integers **(d)**

184. **[CDS (I) Paper 23 Q77]** If the gum Of all interior angles of a regular polygon ig twice the sum of all its exterior angles, then the polygon is **(a) (d)** Hexagon Nonagon Decagon A bicycle wheel makes 5000 revolutions in moving 11 km. What is the radius of the wheel ? (Assume It **(a) (b) (c) (d)** 17-5 cm 35 em 70 cm 140 cm The volumes Of two cones are in the ratio I ; 4 and their diameters are in the ratio 4 :

185. **[CDS (I) Paper 23 Q82]** 83. **(a) (b) (c)** 50 units 55 units 60 units 65 units A triangle and a parallelogram have areas and equal bases. If the altitude of the triangle is k times the altitude of the parallelogram, then What is the value of k ? 4 **(c) (d)** DZOL-T-LKM (29-, It

186. **[CDS (I) Paper 23 Q85]** 86. 29-A) Areas of two squares are in the ratio m2 : n4. What is the ratio Of their perimeters ? **(a) (c)** AD is the median of the triangle ABC. If P is any point on AD, then which one of the following is correct ? **(a) (b)** (e) Area of triangle PAB is greater than the area of triangle PAC Area Of is to Of triangle PAC Area of triangle PAB is one-fourth of the area Of triangle PAC Area of triangle PAB is half of the area of triangle PAC What is the area Of a segment of circle of radius r subtending an angle at the centre ? **(a) (c) (d)** 2 sin— cos— ! r2 B—sm COS 0 — r2 sin cos — A circle is inscribed in a triangle ABC. It 84. touches the sides AB and AC at M and N respectively. If O is the centre of the circle and z A 700, then what is MON equal t.o ? 900 1000 1100

187. **[CDS (I) Paper 23 Q87]** ABC is a triangle at C. Let P be on Q point on BC. Which Of Correct ? AQ2+BP2-AB2+PQ2 **(a) (b) (c)** Ilx8 22x3 44xa 55xa **(d)**

188. **[CDS (I) Paper 23 Q99]** 75, 35-A) What is the median of 2, 4, 6, 100? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

189. **[CDS (I) Paper 24 Q74]** A wire is in the form of a circle of radius 70 cm. If it is bent in the form of a rhombus, then what is its side length? 22 (Take 55 cm **(b)** 75 cm **(c)** 95 cm **(d)** P.T.O. **(a)** 28 m **(b)** 44 m **(c)** 66 m 72. A sphere of diameter 6 cm is dropped into a cylindrical vessel partly filled With water. The radius Of the vessel is 6 cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised? 73. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55 cm, then What is the area of the sector? **(d)** 577•5 cm2

190. **[CDS (I) Paper 24 Q87]** If the perimeter of a right-angled triangle is 30 cm and thc hypotenuse is 13 cm. then what is the area of the triangle? **(c)** 24 cma 27 cm2 30 Cm 36 cm 2 **(a)** 24 cma 27 cm2 30 Cm 36 cm 2 88. ABC is a triangle right angled at C Let p be the length Of the perpendicular drawn from C on AR If BC = 6 cm and CA 8 cm, then What is the value Of p ? 4-8 cm XVWS-T-MTK/SOA 89. ABCD is a trapezium in which AB is parallel to DC and 2AB=3DC The diagonals AC and BD intersect at Q What is the ratio Of the area of MOB to that of ADOC? 90. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? **(b)** 14 cm **(c)** 13 cm **(d)** 12 cm

191. **[CDS (I) Paper 24 Q88]** ABC is a triangle right angled at C Let p be the length Of the perpendicular drawn from C on AR If BC = 6 cm and CA 8 cm, then What is the value Of p ? 4-8 cm XVWS-T-MTK/SOA **(a)** 24 cma 27 cm2 30 Cm 36 cm 2 88. ABC is a triangle right angled at C Let p be the length Of the perpendicular drawn from C on AR If BC = 6 cm and CA 8 cm, then What is the value Of p ? 4-8 cm XVWS-T-MTK/SOA 89. ABCD is a trapezium in which AB is parallel to DC and 2AB=3DC The diagonals AC and BD intersect at Q What is the ratio Of the area of MOB to that of ADOC? 90. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? **(b)** 14 cm **(c)** 13 cm **(d)** 12 cm

192. **[CDS (I) Paper 24 Q93]** What is the median Of the following data ? **(a)** In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? **(b)** and E **(c)** A, C and D only **(d)** C D and E

193. **[CDS (I) Paper 25 Q51]** From an aeroplane flying above a river at an altitude of 1200 m, it is observed that the angles of depression of opposite points on the two banks of a river are 300 and e. If the width of the river is 3000 m, then which one of the following is correct ? **(a)** 9<300 **(b)** 450 < 0<600 **(c)** 600 < 9<900 cos20 — 3cos6 +2 52. If sin2 where 00 < e < 900 then what sin20 + coso equal to ? 4 3 2 7 4 53. Consider the following : IS 2. sin4Ø+cos49=l+2sin20cos20 15 Which Of the above are identities ? and 2 only and 3 only and 3 only **(d)** 1, 2 and 3 is the value of 54. What sin 240 sin 660 — cos240 cos 66 + tan 240 tan 660 —cot 240 cot 660 ? psin A cosB, y = psinA sin B 55. If x and z = p cos A, then what is the value of x2 + y2+z2 ?

194. **[CDS (I) Paper 25 Q59]** If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C **(c)** sin A+B+C **(d)** tan 4 In a triangle ABC, right-angled at B, AB + BC + cm and length of the hypotenuse is 20 cm. What is the value of tanA + tan C ? 4 2 **(d)** 20 17 the 17 **(a)** sin 9 **(b)** cose **(c)** tan 0 **(d)** cot O 59. If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C

195. **[CDS (I) Paper 25 Q61]** The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? **(a)** cm2 **(b)** 80 cm2 **(c)** 100 cm2 **(d)** 120 cm2

196. **[CDS (I) Paper 25 Q74]** ABC is a triangle right-angled at B with AB 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surmce area of the cone so formed ? (take **(a)** 452 cm2 **(b)** 440 cm2 **(c)** 432 cm2 **(d)** 420 cm2

197. **[CDS (I) Paper 25 Q82]** What is the length of the altitude of the triangle drawn from vertex A on BC ? **(a)** 10•05 cm2 **(b)** 10•15cm2 **(c)** 11-05 cm2 **(d)** 11•15cm2 Consider the following for the next three

198. **[CDS (I) Paper 25 Q83]** If p, Q and R are the areas of sectors at A, B and C within the triangle respec- tively, then which of the following is/are correct ? l. P = Æcm2 **(a)** 10•05 cm2 **(b)** 10•15cm2 **(c)** 11-05 cm2 **(d)** 11•15cm2 Consider the following for the next three

199. **[CDS (I) Paper 25 Q2]** 9Q+4R— 36rcm2 Select the correct answer using the code given below : **(a)** 1 only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither 1 nor 2 23 Consider the following for the next two (02) items that follow : ABC is triangle in which AB = AC and D is any point on BC.

200. **[CDS (I) Paper 25 Q85]** If AD-5cm, BD-4cm and 6 cm, then what is AB equal to ? **(a)** 10•05 cm2 **(b)** 10•15cm2 **(c)** 11-05 cm2 **(d)** 11•15cm2 Consider the following for the next three

201. **[CDS (I) Paper 26 Q10]** If where x and y are real numbers, then what is the value of (x + u2? P.T.O. I. What is the ratio of interior angle to exterior angle of a regular polygon Of n sides? **(c)** 2 **(a)** 2 7. A man walks at an average speed of 3 km/hr from his residence and reaches Office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence 3 2. 4143 4343 is divisible by and office ? **(b)** 6 km **(c)** IO km **(d)** 8 km

202. **[CDS (I) Paper 26 Q37]** 38. 39 A piece of cleth costs 10,000. If a 2 m longer piece Of the same cloth is purchased for the same amount, it would cost 250 less per metre. What is the original length of the piece of cloth? **(c)** 12 m **(b)** IO m **(d)** 16 m What is the condition that the roots Of the equation ax bx c O are in the ratio c ; I ? **(b) (d)** . TWO a2 112 ab2 = sides of a triangle forming a right angle are 6x2 and (2x2 — l). If the area Of the triangle is 84 square units, then What is the perimeter of the triangle? **(a)** 51 units **(b)** 53 units **(c)** 56 units **(d)** 59 units

203. **[CDS (I) Paper 26 Q52]** If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of the **(b) (a)** 120' **(b)** 135' **(c)** 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is **(d)** following? aeroplane is Observed to be 56. If tane + sece 3, then What is the value of 3 tan8+ 9sece?

204. **[CDS (I) Paper 26 Q72]** A farmland is in the shape Of a rhombus. The perimeter of the land is 100 m and the length of one of the diagonals is 40 m. The land is divided into four equal parts. What is the area of each part? **(a)** 150 **(b)** 225 **(c)** 300 **(d)** 450

205. **[CDS (I) Paper 26 Q2]** The square and the triangle have the same perimeter. Four times the area of the square is equal to 30 Ernes the area of the triangle. Select the correct answer using the code given below. **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 Consider the following statements : l. The sum Of any two sides Of a triangle is less than twice the median drawn to the third side.

206. **[CDS (I) Paper 26 Q78]** Let D, E and F be the midpoints Of the sides BC, CA and AB respectively Of a triangle ABC Triangle DEF is congruent to which of the following **(a)** 2x-90' **(b)** 2x-18 **(c)** 18m-2x ABC is a triangle right angled at B with AC-2BC If ZA-x, then what is ZC equal to? 78. 80. Let D, E and F be the midpoints Of the sides BC, CA and AB respectively Of a triangle ABC Triangle DEF is congruent to which of the following 1. AEF 3. EDC Select the correct answer using the code given below. 81. 84. In a triangle ABC, AB— 16 cm , 12 cm and AD is the bisector Of ZA If BD —4 cm, then what is CD equal to? **(d)** 2-5 cm 82. An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct?

207. **[CDS (I) Paper 26 Q79]** In a 80. **(d)** I only 2 and 3 only 3 only 1, 2 and 3 triangle ABG AB AC and BC is **(a)** 2x-90' **(b)** 2x-18 **(c)** 18m-2x ABC is a triangle right angled at B with AC-2BC If ZA-x, then what is ZC equal to? 78. 80. Let D, E and F be the midpoints Of the sides BC, CA and AB respectively Of a triangle ABC Triangle DEF is congruent to which of the following 1. AEF 3. EDC Select the correct answer using the code given below. 81. 84. In a triangle ABC, AB— 16 cm , 12 cm and AD is the bisector Of ZA If BD —4 cm, then what is CD equal to? **(d)** 2-5 cm 82. An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct?

208. **[CDS (I) Paper 26 Q81]** 84. In a triangle ABC, AB— 16 cm , 12 cm and AD is the bisector Of ZA If BD —4 cm, then what is CD equal to? **(b)** 2-5 cm **(d)** 35 cm **(a)** 2x-90' **(b)** 2x-18 **(c)** 18m-2x ABC is a triangle right angled at B with AC-2BC If ZA-x, then what is ZC equal to? 78. 80. Let D, E and F be the midpoints Of the sides BC, CA and AB respectively Of a triangle ABC Triangle DEF is congruent to which of the following 1. AEF 3. EDC Select the correct answer using the code given below. 81. 84. In a triangle ABC, AB— 16 cm , 12 cm and AD is the bisector Of ZA If BD —4 cm, then what is CD equal to? **(d)** 2-5 cm 82. An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct?

209. **[CDS (I) Paper 26 Q90]** In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? **(a)** 12 cm **(b)** 12•2 cm **(c)** 12•4 cm **(d)** 126 cm

210. **[CDS (I) Paper 26 Q87]** In a triangle ABG AD is the bisector of ZBAC If 12 em, IO cmand DC — 5 cm. then what is the perimeter of the triangle? **(a)** 30 cm **(b)** 31 cm **(c)** 33 cm **(d)** 35 cm

211. **[CDS (I) Paper 27 Q62]** 63. 64. (23 —A **(a) (b) (c) (d)** XY + XZ — yz = O V + xz—2yz = O xy + xz + yz O xy + xz — 4yz = O What is the median of all possible factors Of 120 ? **(b) (c)** 11 12 135 The sum of deviations Of a set Of n values measured from 50 is — IO and the sum of deviations Of the values measured from 46 is

212. **[CDS (I) Paper 27 Q71]** A circle is inscribed in a triangle ABC. It 69. touches the sides BC, CA, AB at D, E, F respectively. What is Z EDF equal to ? 900 -A 900 — 900 -2A 21 The length Of a room is — times its breadth **(a) (b) (c)** 5 em 900 — 72. Hmv much water will remain in the vessel after the overflow ? **(d)**

213. **[CDS (I) Paper 27 Q74]** 75. 76. **(a) (c) (d)** 12 12 2 2 What is the area of the major segment ? **(a)** 12 Triangle ABC is right-angled at A and AD is perpendicular to BC. [f BD = 7-5 em and DC IO cm, then what is AD equal to ? **(a) (b) (c)** 5 cm IO em FDGT-T-EMT

214. **[CDS (I) Paper 27 Q79]** 80. -A) The perpendicular AD on the base BC Of a triangle ABC intersects BC at D that DB 3 CD. Which one Of the following is correct ? (e) **(d)** 2 (AB - AC) BC2 3 (AB + AC) (AB - AC) = 2 BC2 4 (AB + (AB - AC) 3 BC2 5 (AB + AC) (AB - AC) = 4 BC2 Triangle ABC is right-angled at C AC BC. What is Z ABC equal to ? and **(b)** 300 450 600 750 In a triangle ABC, A AB2 + AC2 - BC2 equal to ? = **(a) (b) (c)** 12 12 **(d)** 2 2 78. 79. 80. Triangle ABC is right-angled at C AC BC. What is Z ABC equal to ? and

215. **[CDS (I) Paper 27 Q80]** Triangle ABC is right-angled at C AC BC. What is Z ABC equal to ? and **(b)** 300 450 600 750 What is the area of the major segment ? **(a) (b) (c)** 12 12 **(d)** 2 2 78. 79. 80. Triangle ABC is right-angled at C AC BC. What is Z ABC equal to ? and

216. **[CDS (I) Paper 27 Q96]** In an equilateral triangle Of side em, a 97 circle is inscribed touching the sides. What is the area of the remaining portion of the triangle ? **(b)** (4v'ä — 't) square cm — r) square cm — 27t) square Cm A triangle has side lengths x em, x + 13 cm and x + 26 cm. If its area is 126 square cm. then what is the value Of x ? **(c)** 18 17 16 15 ABCD is a parallelogram. A circle through A, B and C intersects CD (produced) at E. Which of the following ware correct ? **(a)** — 't **(b) (c) (d)** Only I and 2 Only 2 and 3 Only I and 3 l, 2 and 3

217. **[CDS (I) Paper 27 Q3]** In an equilateral triangle, the centroid and centre of circumcircle coincide. Angle bisectors Of a cyclic quadrilateral form another cyclic quadrilateral. Every cyclic parallelogram is a Which of the statements given abm•e are **(a)** — 't **(b) (c) (d)** Only I and 2 Only 2 and 3 Only I and 3 l, 2 and 3

218. **[CDS (I) Paper 27 Q99]** ABCD is a cyclic quadrilateral. AB and DC when produced, meet in E. Which of the following statements Ware correct ? A EBC is similar to A EAD. 2. z CBE+ 1800. Select the correct answer using the code given below : (35—A) **(a) (b) (c) (d)** Only I Only 2 Both I and 2 Neither I nor 2 FDGT-T-EMT

219. **[CDS (I) Paper 27 Q100]** In a triangle ABC , DE is a line segment which intersects AB at D and AC at E such that DE is parallel to BC. The line segment divides the triangle in two parts Of equal area. What is — equal to ? -A) **(a) (b) (c) (d)** Only I Only 2 Both I and 2 Neither I nor 2 FDGT-T-EMT

220. **[CDS (I) Paper 29 Q56]** 57. The corners Of an equilateral triangular plate were cut in such a manner that it forms a regular hexagonal plate. What is the ratio of the area of the triangular plate to the area of the hexagonal plate ? **(a) (b) (c) (d)** Two equal arcs of different circles Cl and C2 subtend angles Of 600 and 750 respectively, at the centres. What is the ratio Of the radius Of Cl to the radius of C2 ? ABC is a triangle With sides AB 41 em, BC = 28 cm and CA = 15 cm. [f D, E and F are the mid-points Of AB, BC and CA then What is the area of the triangle DEF ? **(a) (b) (c) (d)** 63 square cm 45 square cm 31 •5 Square 22 •5 square em A triangle has sides 13 cm, 14 cm and 15 cm long. What is the length Of the smallest altitude of the triangle ? **(a) (b) (c) (d)** 11 cm 11•2 em 12 em 12,2 em

221. **[CDS (I) Paper 29 Q2]** I — sin6 e z cos2 0 (cos4 9 + 3 sin2 0) Select the correct answer using the code given **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 If 7 sin4 + 9 Cosa e + 42 sin2 9 — then What is tan O equal to ? **(a) (c)** An isosceles triangle has its base length 2.a and its height is h. On each side of the triangle a square is drawn external to the triangle. What is the area of the figure thus formed ? **(a) (b) (c) (d)** 6a2 + 2h2 • 2ah 6a2 + 2h2 + ah 4a2 2h2 + ah 6a2 + h2 + ah JNUP-T-EMT (27 —A

222. **[CDS (I) Paper 29 Q76]** If equal to ? (b—a)(c—a)' , then What is (p q **(a) (b) (d)** 9 4 **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 If 7 sin4 + 9 Cosa e + 42 sin2 9 — then What is tan O equal to ? **(a) (c)** An isosceles triangle has its base length 2.a and its height is h. On each side of the triangle a square is drawn external to the triangle. What is the area of the figure thus formed ? Which one of the following is a factor of a2 —b2—c2 2bC+ a + b —c ? **(a) (b) (c)** a +b+c—l **(d)** Let and be the roots Of the equation x+a+b Which one of the following is a quadratic equation whose rcujts are a2 and 152 ? **(a) (b) (c) (d)** 6a2 + 2h2 • 2ah 6a2 + 2h2 + ah 4a2 2h2 + ah 6a2 + h2 + ah **(a) (b)** (e) **(d)** + (a2 + b2) x + a2b2 = O x2 — (a2 b2) x + a2b2 0 x2 (a2 b2) x -a2b2=o -a2b2-o

223. **[CDS (I) Paper 29 Q86]** What is D's age (in years) ? **(a)** 15 **(b)** 16 **(c)** 18 JNUP-T-EMT In a triangle ABC, angle B = 900 and p is the 87. 84. length Of the perpendicular from B to AC. If BC IO cm and AC = 12 em, then what is the value Of p ? 5Jfi **(d)** 1 oJii

224. **[CDS (I) Paper 29 Q100]** Question : What is the area Of the triangle inscrii*2d in a semi-circle With the diameter as the base ? Statement-I The diameter Of semi-circle is 20 cm. Statement-II : Two shorter sides of the triangle are 12 em and 16 cm. difference between the second largest and the second smallest Of the numbers. Question : Is (a — b + c) > (a + b — c), where a, b and c are real numbers ? Statement-I : b is negative. c is negative. JNUP-T.EMT **(a)** 30° **(b)** 45° **(c)** 60° **(d)** 90°

225. **[CDS Paper 30 Q14]** What is' the difference between the greatest value and the least value Of cos2 0 3 sin2 + 2 ? cosec 790 (I 1+t **(a) (d)** Suppose p(x) = **(a) (c) (d)** 4 2 ABC is a right-angled triangle, right-angled at B such that AB 6 cm and BC = 8 cm. What is the perimeter of the inscribed in the triangle ABC with maximum area ? x' + a2X2 + alX + ao and **(a) (b) (c) (d)** 24/7 em 96/7 cm 24 cm 32 cm What is the greatest value Ofk for which 2x2 — 4x + k = O has real roots ? **(a) (b)** 2 **(c) (d)** 4

226. **[CDS Paper 30 Q41]** BC 20 em and AC z 25 cm. Further, BP is the perpendicular on AC. What is the difference in the area of triangles PAB and PCB? **(a)** 5 6 75 9 **(b) (c) (d)**

227. **[CDS Paper 30 Q2]** If p is a positive integer, then what is the remainder when pn is divided by p + I ? n IS even. p even. SRSU-T.EMT Question : Is xy positive ? 1: -0„19683 Statement-II : y z Let a, b and c be the sides of a triangle ABC. Question : Statement •I : Statement-II Is the triangle equilateral ? a2 + b2 + c2 (ab + bc + ca) 3a2 + + 4c2 = 2ab+4bc + 4ca Area Of a rectangle With length x and breadth y is p and area of a parallelogram (which is strictly not a rectangle) With adjacent sides of length x and y is Q. Question : Statement-I : Statement-II ; 1sP>Q? The angle between the two adjacent sides of the parallelogram is **(a) (b) (c) (d)** If the Question can be answered by one Of the Statements alone, but not by the 65. If the Question can be answered by either Statement alone. If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. If the Question cannot be answered even by using both Statements together. 66. Statement-II : y z Let a, b and c be the sides of a triangle ABC. Question : Statement •I : Statement-II Is the triangle equilateral ? a2 + b2 + c2

228. **[CDS Paper 30 Q66]** Statement-II : y z Let a, b and c be the sides of a triangle ABC. Question : Statement •I : Statement-II Is the triangle equilateral ? a2 + b2 + c2 (ab + bc + ca) 3a2 + + 4c2 = 2ab+4bc + 4ca **(a) (b) (c) (d)** If the Question can be answered by one Of the Statements alone, but not by the 65. If the Question can be answered by either Statement alone. If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone. If the Question cannot be answered even by using both Statements together. 66. Statement-II : y z Let a, b and c be the sides of a triangle ABC. Question : Statement •I : Statement-II Is the triangle equilateral ? a2 + b2 + c2

229. **[CDS Paper 30 Q68]** 69. 70. (23-A Question : Statement•I Staternent-II : What is the ratio Of the lengths Of diagonals Of a rhombus ? One diagonal Of the rhombus is equal to its side. The longer diagonal of the 71. rhombus is equal to times its side. The chord Of a cir-cle of radius R touches at a point on the circumference of a concentric circle Of radius r. The length of the chord is 24 units. Question : Statement-I : Statement-II : **(a) (b) (c)** — square cm 3 77 — square em square em 6 616 24 units. Question : Statement-I : Statement-II : 72. What are the values of r and r is an integer. R is an integer. square cm 9 If p is the perimeter of the smallest sectnr, then what is the value of 9p ? **(d)** 142 cm

230. **[CDS Paper 30 Q77]** 18•2 seconds 17 •4 seconds 15 •4 seconds Consider the following for the next three (03) items that follow : A triangle CEF is drawn inside a square ABCD as shown in the figure given belcm'. Given : CF = 8 em, EF = 6 cm and CE = IO cm. D SRSU-T-EMT **(a) (b) (c) (d)** 5:28 a.m. 5:44 a.m. 4:56 a.m. 6:24 a.m. If the lengths Of the two trains A and B are 400 m and 500 m respectively, then what is the time taken by them to cross each other ?

231. **[CDS Paper 30 Q81]** 82. **(a) (b) (c)** 38•5 41are cm 48 square em 52' 5 square cm 666 square cm 81. 82. **(d)**

232. **[CDS Paper 30 Q94]** 95. What is the length of the side of the triangle ABC ? **(a) (b) (c)** 2 unit 4 unit 6 unit 8 unit **(d)**

233. **[CDS (I) Paper 31 Q19]** In a triangle ABC, Z_.4BC = 600 and AD is the altitude. If AB — 6 cm and BC = 8 cm. then what is uea of the triangle ? **(a)** 12 Ctn 12-6 square cm **(b) (c)** 24 Crn 24-46 cm **(d)** A - RAKU-T-EMT

234. **[CDS (I) Paper 31 Q27]** In a triangle ABC, D is a point on BC. If AB.DC = AC.BD, ZBAD a and ECAD — 13 then which one of the following is correct ? **(a)** 16 **(b) (c) (d)**

235. **[CDS (I) Paper 31 Q32]** If p, q and r are the lengths (in cm) of the sides of a right-angled triangle, then (p — q — — r — — p — q) **(d)** 16 is always **(a)** Positive only **(b)** Negative only **(c)** Non-positive only **(d)** Non-negative only

236. **[CDS (I) Paper 31 Q35]** Let ABC be a triangle with area 36 square cm. If AB=9 cm, BC' 12 cm and LABC = O, then what is cos9 equal to ? 4 3 2 **(a)** 20 **(b)** 25 **(c)** 30 **(d)** 35 35. 36. Let ABC be a triangle with area 36 square cm. If AB=9 cm, BC' 12 cm and LABC = O, then what is cos9 equal to ? 4 3 2 Let n be a natural number. The HCF of n, n +10 is 10. If the LCM is x

237. **[CDS (I) Paper 31 Q49]** The base of a right-angled triangle is times the height of triangle. If the area of the triangle is 54 square cm, then what is the perimeter of the triangle ? **(a)** 30 cm **(b)** 32 cm **(c)** 36 cm **(d)** 40 cm

238. **[CDS (I) Paper 31 Q11]** The perimeter of the triangle APC is more than 46 cm Ill. The area of the triangle APC is 50 square cm Which of the statements given above are correct ? **(a)** I and 11 only **(b)** 11 and Ill only **(c)** I and Ill only **(d)** 1, 11 and 111 Consider the following for the next two (02) items that follow : A frequency distribution is as follows : Marks 18-26 27-35 3644 45-53 54-62 63-71 72-80

239. **[CDS (I) Paper 31 Q54]** What is the median of the distribu- tion ? **(a)** 44-9 **(b)** 45-5 **(c)** 45-9 **(d)** 46-3 19

240. **[CDS (I) Paper 31 Q55]** What is the mode of the distribution ? **(a)** I and 11 only **(b)** 11 and Ill only **(c)** I and Ill only **(d)** 1, 11 and 111 Consider the following for the next two

241. **[CDS (I) Paper 31 Q95]** The ratio of sum of interior angles to sum of exterior angles of a regular polygon of n sides is **(a)** 1100 **(b)** 1200 **(c)** 1300 **(d)** 1400 96. The number 199 can be written as m2 — n2, where m, n are natural numbers

242. **[CDS (I) Paper 31 Q4]** What is the measure of an interior angle of polygon ? **(a)** 1100 **(b)** 1200 **(c)** 1300 **(d)** 1400

243. **[CDS (I) Paper 31 Q100]** In a right triangle ABC, 900 and AD is perpendicular to BC. If ECAD 600 and BC = 6 cm, then what is AB equal to ? **(a)** 1100 **(b)** 1200 **(c)** 1300 **(d)** 1400 96. The number 199 can be written as m2 — n2, where m, n are natural numbers

244. **[CDS (I) Paper 33 Q36]** The angles in a right-angled triangle is — radian. One of the acute angles Of the triangle is **(a)** 600 **(b)** 57-50 **(c)** 52-50 **(d)** 47-50

245. **[CDS (I) Paper 33 Q61]** Let AD be the altitude of a triangle ABC If and (BD — CD) r, then what is BC equal to? **(b) (d)** qr / p pr / q pq/r **(a)** 20% **(b)** 40% **(c)** None of the above 61. Let AD be the altitude of a triangle ABC If and **(d)** I only

246. **[CDS (I) Paper 33 Q63]** ABC is a triangle right angled at B. P is the midpoint Of AB and Q is the mid- point of BC. Consider the following : I. AQ units 11. units Which of the atxwe is/ are required to determine the area Of the triangle? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** More information is needed

247. **[CDS (I) Paper 33 Q67]** ABC is a triangle right anøed at B. Further, (AB* BC) exceeAs AC by IO units. If the perimeter of the triangle is 60 units, then what is the area Of the triangle? **(b) (c)** 75 units 100 square units 125 square units 150 units **(a) (b)** 6-4 m 67. ABC is a triangle right anøed at B. Further , **(c) (d)** 75 units 100 square units 125 square units 150 units 68. TWO poles are situated 24 m apart and their heights differ by I O m. What is the distance between their tips?

248. **[CDS (I) Paper 33 Q69]** Let X, Y and Z be the midpoints Of the sides BC, CA and AB Of a triangle ABC respectively. Consider the following **(a) (b)** 6-4 m 67. ABC is a triangle right anøed at B. Further , **(c) (d)** 75 units 100 square units 125 square units 150 units 68. TWO poles are situated 24 m apart and their heights differ by I O m. What is the distance between their tips?

249. **[CDS (I) Paper 33 Q11]** The quadrilateral AAY is a The area Of the quadrilateral AZXY is half Of the area Of the triangle Which Of the statements given above is/ are correct? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

250. **[CDS (I) Paper 33 Q71]** In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ? **(a)** 21 cm **(b)** 210 cm2 **(c)** 42 cm **(d)** 420 cm2

251. **[CDS (I) Paper 33 Q72]** ABC is a triangle right angled at B. D is a point on AC such that BD is perpendicular to AC If AB p and BC v'äp, then what is BD equal to? **(b)** p 12 **(d)** Op/4 **(a)** 15 200 73. **(b)** 10 **(c)** 11 **(d)** 12 71. In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ?

252. **[CDS (I) Paper 33 Q73]** The difference between an interior angle and an exterior angle of a regular polygon is 1200. What is the number of sides of the polygon? 9 **(b)** 10 **(c)** 11 **(d)** 12 ANFB-T-EMT ,'45A **(a)** 15 200 73. **(b)** 10 **(c)** 11 **(d)** 12 71. In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ?

253. **[CDS (I) Paper 33 Q75]** The sides of a triangle are 11 cm, 60 cm and 61 cm. What is the area of the triangle forrned by joining the mid- points of the sides of the triangle? **(b) (d)** 165 cm 2 110 cm2 82 5 cm 72-5 cma A pendulum swings through an angle Of 300 and its end describes an arc Of length 55 cm. What is the length of the pendulum? (Take 22 / 7) **(b) (d)** 90 cm 100 cm 105 cm P.T.O. An angle is exactly one-fourth of its complementary angle. What is the value of angle B ? **(b)** 15 200 73. **(b)** 10 **(c)** 11 **(d)** 12 **(a)** 15 200 73. **(b)** 10 **(c)** 11 **(d)** 12 71. In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ?

254. **[CDS (I) Paper 33 Q79]** A wire is in the form of an equilateral triangle with an area Of 360 cm **(a) (b) (c)** — cm None of the above 80. A right circular cone and a hemisphere have equal base and equal volume. What is the ratio Of the height of the cone to the radius Of the hemisphere? ANFB-T-EMT/4SA Let the area of the largest possible square inscribed in a circle of unit radius be x. Let the area of the largest possible circle inscribed in a square of unit side length be y What is the relation between x and y? **(d)** NX 2y

255. **[CDS (I) Paper 33 Q11]** Consider the Question and the Statements. Question : In a triangle ABC right angled at B, AC 20 cm. What is the circum- radius Of the triangle? Statement—I : AB 12 Cm Statement—II : BC 16 cm Which one of the following is correct in respect Of the above Question and the Stateme nts? **(b)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts ANFB-T-EMT/45A 35 **(a)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(b) (c) (d)**

256. **[CDS (I) Paper 33 Q11]** Consider the Question and the Statements. Question : ABC is an isosceles triangle with AB AC IO units. If the area of the triangle is 48 square units, then What is the length of the base BC? Statement—I : The length of BC is an even integer. Statement—II : The height of the triangle is greater than the length of half Of the base. Which one of the following is correct in respect Of the above Question and the State ments? **(b) (c)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts ANFB-T-EMT/45A **(a) (b)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even without using any of the Statemen ts **(c)** ANFB-T-EMT/45A The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone The Question can be answered by using either Statement alone The Question Can be answered by using both the Statements together, but cannot be answered using either Statement alone The Question can be answered even Without using any of the State ments I P.T.O. 90. A Question is given followed by two Statements I and 11. Consider the Question and the Statements. Question The diagonals of a rhombus ABCD are in the ratio 5:12. Is one Of the diagonals equal to side of the rhombu s? Statement—I : The sum Of the diagonals = 34 cm. Statement—II . The length of a side 13 cm Which one of the following is correct in respect Of the above Question and the Sta tement s ? @ The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement alone **(d)** The Question can be answered by using either Statement alone

257. **[CDS (I) Paper 33 Q98]** What is the value of v? 5 8 ANFB-T-EMT/4SA For the following tvo (02) items Consider the following distribution' having median value 24 **(a)** 20 **(b)** 25 For the following tvo **(c)** 3 98. What is the value of v? 5 8 ANFB-T-EMT/4SA 99. What is the value of k ? 22 30 100. What is the mean Of the distribution? 21-625 22.225 **(d)** 23 225 24 • 625 P.T.O. For the following tvo

258. **[CDS (I) Paper 34 Q40]** The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is **(a)** 3-25 **(b)** 3-50 **(c)** 3-75 **(d)** 4-25 40. The numbers 59, 53, 51, 43, 36,

259. **[CDS (I) Paper 34 Q46]** In a cyclic quadrilateral ABCD, the diagonal AC bisects the angle C. Which of the following staternents is/are correct ? **(a)** 220 m **(b)** 225 m **(c)** 230m **(d)** 235 m 46. In a cyclic quadrilateral ABCD, the diagonal AC bisects the angle C. Which of the following staternents is/are correct ? 11. The diagonal BD is parallel to the tangent of the circle at A Select the answer using the code given below :

260. **[CDS (I) Paper 34 Q48]** ABC is a triangle in which AB = AC and D is any point on BC. Which of the following statements is/ are correct ? L AB2-AD2=CDXBD IL BC2+BD2-DC2-2BDXBC Select the answer using the code given **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

261. **[CDS (I) Paper 34 Q49]** ABC is a triangle right-angled at B with sides BC = a, CA b and AB = c. Let p be the length of the perpendicular from B to AC. Which of the following statements is/ are correct ? 11. pb Select the answer using the code given below : I only **(b)** 11 only Both 1 and 11 Neither I nor 11 **(d)** 19 A - KPRS-T-MTE **(a)** 220 m **(b)** 225 m **(c)** 230m **(d)** 235 m 46. In a cyclic quadrilateral ABCD, the diagonal AC bisects the angle C. Which of the following staternents is/are correct ? 11. The diagonal BD is parallel to the tangent of the circle at A Select the answer using the code given below :

262. **[CDS (I) Paper 34 Q50]** ABC is an equilateral triangle. Let PQRS be a square inscribed in it such that P is on AB and Q is on AC. Which of the following is/are correct ? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 51. The perimeter of a right-angled isosceles triangle is 20 units. If a is the area of the triangle, then

263. **[CDS (I) Paper 34 Q51]** The perimeter of a right-angled isosceles triangle is 20 units. If a is the area of the triangle, then **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 51. The perimeter of a right-angled isosceles triangle is 20 units. If a is the area of the triangle, then

264. **[CDS (I) Paper 34 Q53]** In a quadrilateral ABCD, the diagonals intersect at O. Let the area of the triangle ABD be p. If AO : OC = m : n, then what is the area of the triangle BCD ? n2p **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 51. The perimeter of a right-angled isosceles triangle is 20 units. If a is the area of the triangle, then

265. **[CDS (I) Paper 34 Q54]** Let ABCD be a parallelogram and E be the midpoint Of BC. The diagonal BD and line segment AE intersects at F. If BF = 2-4 cm, then what is BD equal to ? **(b)** 64 cm **(d)** 8-4 cm A - KPRS-T-MTE **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 51. The perimeter of a right-angled isosceles triangle is 20 units. If a is the area of the triangle, then

266. **[CDS (I) Paper 34 Q55]** ABC is a triangle right-angled at C. Let p be the midpoint of BC. If AP cm and AB 20 cm, then what is the perimeter of the triangle ABC? **(a)** 40 cm **(b)** 48 cm **(c)** 50 cm **(d)** 60 cm

267. **[CDS (I) Paper 34 Q56]** ABC is a triangle right-angled at B. Points M and N trisect BC. If AM = 6 cm and AN = 9 cm, then what is AC equal to ? **(a)** 4xfiö cm **(b)** 265 cm **(c)** 24 cm **(d)** 20 cm

268. **[CDS (I) Paper 34 Q57]** ABC is a triangle right-angled at B. and BC = which of the following statements is/are correct ? I. The value of AB is equal to the geometric mean of p and q. **(a)** 40 cm **(b)** 48 cm **(c)** 50 cm **(d)** 60 cm 56. ABC is a triangle right-angled at B. Points M and N trisect BC. If AM = 6 cm and AN = 9 cm, then what is AC equal to ?

269. **[CDS (I) Paper 34 Q11]** The perimeter of the triangle is Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 23

270. **[CDS (I) Paper 34 Q59]** ABCD is a quadrilateral with sides AB=9cm, BC=40cm, CD=28cm and DA 15 cm and one of the diagonals AC = 41 cm. Which of the following statements is/ are correct ? I. The vertices A, B, C and D of the quadrilateral lie on a circle. **(a)** 40 cm **(b)** 48 cm **(c)** 50 cm **(d)** 60 cm 56. ABC is a triangle right-angled at B. Points M and N trisect BC. If AM = 6 cm and AN = 9 cm, then what is AC equal to ?

271. **[CDS (I) Paper 34 Q64]** ABCD is a cyclic quadrilateral. AB and DC are produced to meet at P. Which of the following statements is/ are correct ? **(a)** 120 **(b)** 180 **(c)** 240 **(d)** 300 61. The diameter of a copper solid sphere is 6 cm. The sphere is melted and recast into a wire. If the diameter of the wire is cm, then what is the length of the wire ?

272. **[CDS (I) Paper 34 Q1]** APAD is similar to APBC **(a)** 120 **(b)** 180 **(c)** 240 **(d)** 300 61. The diameter of a copper solid sphere is 6 cm. The sphere is melted and recast into a wire. If the diameter of the wire is cm, then what is the length of the wire ?

273. **[CDS (I) Paper 34 Q11]** Area of triangle OAC is equal to 102 cm2. Select the answer using the code given **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

274. **[CDS (I) Paper 34 Q68]** In a triangle ABC, AB 18 cm, BC- 22 cm and AC = 15 cm. The bisector of ZBAC intersects BC at D. What is (BD DC) equal to ? **(a)** 121 cm2 **(b)** 120 cm2 **(c)** 117 cm2 **(d)** 96 cm2

275. **[CDS (I) Paper 34 Q11]** Consider the Question and the Statements and mark the correct option. Question : Statement-I Statement-II . ABCD is a quadrilateral such that AC bisects ZA and ZC. Is ABIDC equal to AD,'BC ? AB is parallel to DC AD is parallel to BC Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Staternent alone **(d)** The Question can be answered even without using both the Statements

276. **[CDS (I) Paper 35 Q58]** 54. In a cyclic quadrilateral ABCD, C - 3>0, Z D (5x - 10)'. What is L A + Z B equal to ? **(b) (c)** 2000 1900 1800 1700 The area of a rhombus is 720 cm2 and the sum of its diagonals is 98 cm. What is the perimeter of the rhombus ? **(a) (b) (c)** 160 em 164 cm 168 cm 172 cm A chord of a circle makes an angle of 900 at the centre of the circle. If the area Of the major segment is k times the area of the minor segment, then what is the value Of k ? (Take **(a) (b)** 11 ABC is a triangle in which (AB + BC) exceeds CA by IO em, (BC + CA) exceeds AB by 8 cm and (CA + AB) exceeds BC by 72 cm. What is the area Of the triangle ? **(a) (b) (c) (d)** 196 cm2 180 cm2 172 cm 160 cm2

277. **[CDS (I) Paper 35 Q11]** Consider the Question and the Statements and mark the correct Question : ABCD is a rhombus of side IO cm. What is the sum Of the squares Of the diagonals of the rhombus ? Statementi : Statement 11 : AC = 6 cm. Which one Of the following is correct in of the above Question and Statements ? **(a) (b) (c)** The Question can be answered by using one Of the Statements alone, but cannot be answered using the other Statement The Question can be answered by using either Statement alone. The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone. The Question can be even without using both the Statements. **(d)**

278. **[CDS (I) Paper 35 Q63]** Statements I and [l. Consider the Question and the Statements and mark the correct Question : ABC is a triangle right-angled at A and AD is perpendicular to BC. If AD 72 cm, then what is BD x CD equal to ? Statement I : AB : AC 3 : **(a) (b) (c) (d)** The Question can be answered by using one of the Statements alone, but cannot be answered using the Other Statement The Question can be answered by using either Statement alone. The Question can be answered by using both the Statements together, but cannot answered using either Statement alone. The Question can be answered even without using both the Statements.

279. **[CDS (I) Paper 35 Q70]** Which of the statements given above Ware correct ? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 If two similar scalene triangles ABC and PQR have the same area, then which Of the following statements Ware correct ? I. Both the tri angles are congruent.

280. **[CDS (I) Paper 35 Q75]** 76. **(a) (b) (c) (d)** 24 units 28 units 32 units 36 units A cubical wooden block of length IO cm is sliced to produce a cross-section of maximum area. What is the area of the cross-section ? **(a) (b) (c)** I (X) cm2 100 cm2 cm2 200 cm2 In a triangle ABC, AD is the bisector Of angle A that meets BC at D. If AB 21 cm, DC = 20 cm and AB : AC 3 : 4, then what is AC2 — BD2 equal to ? **(a)** 441 **(b)** 559 **(c)** 588 784 BFVS-T-TME

281. **[CDS (I) Paper 35 Q78]** 79. 80. (29-A) ABCD is a parallelogram with ABC z 1500 and the sides have integer values. If the area Of the parallelogram is 17•5 cm2, then consider the folloxving statements : l. It is possible to have a perimeter equal to 24 em. **(a) (b) (c) (d)** I only 11 only Both I and 11 Neither I nor 11

282. **[CDS (I) Paper 35 Q15]** If p, q, r, s and t are integer-s, then What is the area of the quadrilateral ? 83. **(a) (b) (c) (d)** 24 square units 36 units 48 units Cannot be determined insufficient data due to BFVS-T-TME For the next three (03) items that follow : The centre O of a circle inside a triangle ABC is at a distance of 13 cm from each of the vertices Of the triangle. The diameter of the circle is IO em and the circle touches only two sides Of the triangle, AB and

283. **[CDS (I) Paper 35 Q81]** 82. 83. -A) If x is the perimeter, in em, Of the triangle, then which one of the following is correct ? **(a) (b) (c) (d)** 69 cm < x < 65 em 65 cm x 67 cm 67 < x < 69 cm 69 cm < x < 71 cm If y is the area in cm2 of the triangle, then which one Of the following is correct ? **(a) (b) (c) (d)** 190 cm2 < y < 200 cm2 200 cm2 < y < 210 cm2 210 cm2 < y < 220 cm2 220 cm2 < y < 230 cm2 Consider the following statements : I. Z ABC lie s between 600 and

284. **[CDS (I) Paper 35 Q86]** What is AD DB to ? **(b) (c)** If x is the area of the circle and y is the area of the triangle such that z = — , one Of the following is correct ? **(b) (d)** What is the ratio of the area of the triangle ADO to the area of the quadrilateral BDOM ? **(a) (b) (c)**

285. **[CDS (I) Paper 35 Q90]** BFVS-T-TME For the next two (02) items that follow : ABC is an equilateral triangle. BP is perpendicular to AC. PR is perpendicular to AB, and PQ is perpendicular to BC. **(a) (b) (c)** If x is the area of the circle and y is the area of the triangle such that z = — , one Of the following is correct ? **(d)**

286. **[CDS (I) Paper 35 Q87]** 88. What is the ratio Of AB2 : BP2 : PR2 ? **(b) (d)** What is the ratio Of the area Of A PRB to the area of A PQC ? **(b) (c)** For the next two (02) items that follow : PQR is a triangle such that QP = QR = 15 cm and PR z 18 cm. PN, QM and RT are the altitudes of the triangle which intersect at O. **(a) (b) (c)** If x is the area of the circle and y is the area of the triangle such that z = — , one Of the following is correct ? **(d)**

287. **[CDS (I) Paper 35 Q89]** 90. What is QT : QO equal to ? **(a) (b) (c)** If x is the area of the circle and y is the area of the triangle such that z = — , one Of the following is correct ? **(d)**

CHAPTER 10

Total Archive Questions: 129 Questions

GEOMETRY: CIRCLES, CHORDS & TANGENTS

Typical Exam Weightage: 8 - 12 Questions (8 - 12 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q67] 0 In the figure given above, a circle is inscribed in a quadrilateral ABCD. Given that BC 38 em, QB -27 cm, DC 25 cm and AD is perpendicular to DC. What. is the radius of the circle ? **(b)** 14 cm **(c)** 15 cm **(d)** 16 cm 68. c **(b)** 450 In the figure given above, ZCBA ? what is **(a)** .**(c)** 300 500

2. [CDS (I) Paper 1 Q69] A, B, Ci D are four distinct points on a circle whose centre .is at O. If ZOBD - ZCDB = ZCBD - ZODB, then what is ZA equal to ? **(b)** 600 **(a)** 600 In the figure given above, a circle is inscribed in a quadrilateral ABCD. Given that BC 38 em, QB -27 cm, DC 25 cm and AD is perpendicular to DC. What. is 69. 70. 71. A, B, Ci D are four distinct points on a circle whose centre .is at O. If ZOBD - ZCDB = ZCBD - ZODB, then what is ZA equal to ? **(b) (c) (d)** PQ 450 1200 1350 is a common chord of two circles. the radius of the circle ?

3. [CDS (I) Paper 1 Q70] 71. 18 **(a) (c) (d)** PQ 450 1200 1350 is a common chord of two circles. APB is a secant line joining points A and B on the two circles. Two tangents AC and BC are drawn. If LACB = 450, then what is ZAQB equal to ? **(a)** 750 **(b)** 900 **(c)** 1200 **(d)** 1350 ABCD is a concyclic quadrilateral. •The tangents at A and C intersect each other at P. If LABC 1000, then what is ZAPC equal to ? **(a)** 100 **(b)** 200 **(c)** 300 400

4. [CDS (I) Paper 1 Q71] A, B, Ci D are four distinct points on a circle whose centre .is at O. If ZOBD - ZCDB = ZCBD - ZODB, then what is ZA equal to ? **(a) (c) (d)** PQ 450 1200 1350 is a common chord of two circles. the radius of the circle ? **(b)** 14 cm **(c)** 15 cm **(d)** 16 cm 68. c In the figure given above, ZCBA ? **(a)** 300 **(b)** 450 .**(c)** 500 **(d)** 600 what is APB is a secant line joining points A and B on the two circles. Two tangents AC and BC are drawn. If LACB = 450, then what is ZAQB equal to ? **(a)** 750 **(b)** 900 **(c)** 1200 **(d)** 1350 ABCD is a concyclic quadrilateral. •The tangents at A and C intersect each other at P. If LABC 1000, then what is ZAPC equal to ? **(a)** 100 **(b)** 200 **(c)** 300 400 18

5. [CDS (I) Paper 1 Q73] In the figure given above. M is the mid point of the side CD of the parallelogram ABCD What is ON : 0B ? o x In the figure given above, YAX is a tangent to the circle with centre O. If ZBAX 700 and ZBAQ 400, then what is LABQ equal to ? **(a) (b) (c)** 73. o x In the figure given above, YAX is a tangent to the circle with centre O. If ZBAX 700 and ZBAQ 400, then what is LABQ equal to ? **(d)** 200

6. [CDS (I) Paper 1 Q72] M 74. c In the figure given above. M is the mid point of the side CD of the parallelogram ABCD What is ON : 0B ? **(a) (c) (d)** 73. o x In the figure given above, YAX is a tangent to the circle with centre O. If ZBAX 700 and ZBAQ 400, then what is LABQ equal to ? **(a)** 200 **(b)** 300 **(c)** 350 **(d)** 400 20 c In the figure given above, AP — 3 cm. PB 5 cm, AQ — 2 cm and QC — What is the value Of x ? **(b) (c)**

7. [CDS (I) Paper 2 Q65] 16 -A) In thc figure given above, PT is a tangent to a circle Of radius 6 cm. If P is at a distance Of IO cm from the centre O and PB = 5 cm, then what is the length of the chord BC ? **(a) (b) (c)** 8-0 Crn 8-4 cm 90 cm A point p .moves such that its distances from two given points A and B are equal. Then what is the locus Of the point P ? **(a) (b) (c) (d)** A straight line t."ich is the right bisector Of AB A circle With céntre at A A circle with centre -at B A straight line passing through either A or B ABC is a triangle. X is a point outside the triangle ABC such that CD = CX, where D is the point of intersection Of BC and AX; and Z BAX Z XAC. Which one Of the following is correct ? **(a) (b) (c) (d)** Triangles ABD, ACX are similar Z ABD < Z ACD AC CX > z DXC

8. [CDS (I) Paper 3 Q2] What is the of centres of circles which touch a given line at a given point ? **(a)** A line pen*ndicular to thc given line, passing through the given **(b)** A line parallel to the given line **(c)** A circle tangent to the given line at the given point **(d)** A closed curve other -than a circle

9. [CDS (I) Paper 3 Q28] 5 km/h 4 km/h 3 km/h 2 km/h c In the figure given above, AB is a dia- meter of a circle and CD is perpendicular to AB, if AB = IO cm and AEz2cm, then what is the length of ED ? T-DETA-K-FN - A **(a)** 26. The shadow of a pole 6 metre high is 15 rnetre long and at the same time the shadow of a tree is 25 metre long. What is the height of the tree ? **(b)** 21 m **(c)** 10m **(d)** 35 m

10. [CDS (I) Paper 3 Q37] If ABC is a triangle right angled at C and having u units, v units, w units as the lengths of its sides opposite to the veltices A, B, C respectively, then what is the tangent of the angle at A + tangent of the angle at B equal to ? **(d) (a)** 4 days **(b)** 5 days **(c)** 6 days **(d)** days 33. If of

11. [CDS (I) Paper 3 Q61] What is the number of circles passing through a given pair of points ? **(a)** One **(b)** Only two **(c)** More than two, but finite **(d)** Infinitely many;

12. [CDS (I) Paper 3 Q62] Two circles touch each other externally at p. Two secants APB and CPD are drawn through P to meet the circle , at A, C and B, D respectively. Then, which one of the following is correct ? **(a)** AC is perpendicular to BD **(b)** AC intersects BD **(c)** AC is parallel to BD **(d)** None of the above 00 600 In the above figure, what is x equal to ? **(a)** 3 2 T-DETA-K-FN - A 18 In the figure given above, AB is parallel to 1M. What is angle a equal to ? **(a)** lt+b+c **(b)** 2z—b+c **(d)** 2R+b—c 6S. If the expressions pr3 + 3x2 —3 and — 5x + p when divided by x — 4 leave the same remainder, then what is the value of p ?

13. [CDS (I) Paper 3 Q69] 31-5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? **(a)** 7 crn2 **(b)** 28 cm2 **(c)** 14 cm2 **(d)** 21 Crn2 T-DETA-K.FN - A 20

14. [CDS (I) Paper 4 Q70] The diameters of It-wo circles are i 8 cm and 8 Cml The distance between their centres is 13 cm. What is the numper of common tangents? **(d)** None Of the above **(a) (b) (c)** None of the above 69. Half of a large cylindrical tank open at the top is fillpd with water and identical heavy *pherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius Of a ball, What is the rpaximum number of balls that can pe dropped? **(d)** 12

15. [CDS (I) Paper 4 Q74] A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? **(a)** 9 Crn **(b)** 8 Crn **(c)** 7 Crn **(d)** 6 Ctn

16. [CDS (I) Paper 4 Q76] In the figure given above, what is the sum of the angles formed around A, C except the angles Of the triangle 'ABC ? **(a)** 3600 **(b)** 720' 900' **(d)** looo• In the figure •given above, A and B are end points Of diameter Of a circle With centre at P, and C is a point on the circumference Of the circle such titat ZABC 350, then what is EPCA ? **(a) (b)** 300 350 550 Q-DETB-K-FN/26A 24 78. [n the figure given above, ABC is an equilateral triangle of side length 30 cm. XY is parallel to BC, xp is parallel to AC and YQ is parallel to AB. is 40 cm, then what is PQ equal to? **(b) (c) (d)** 12 cm 15 Cm None Of the above In the figure given above, AB is a line Of length 2a, with M as mid- point. Semicircles are drawn On one side With AM, MB and AB as diameters. A circle with centre O and radius r is drawn such that this circle touches all the three semicircles. What is the value Of r? **(a)** 2a/3

17. [CDS (I) Paper 4 Q86] A circle and a souare have the same perimeter. Which one of the following is correct? **(a)** The area Of the circle is e qual to that of square **(b)** The area o' the circle is larger than that Of square **(c)** The area of the circle is less than that bf square **(d)** No conclusiion can be drawn 87• If (x 4 + x — 32? What is One of the values of (x **(a)** 18 **(b)** 16

18. [CDS (I) Paper 5 Q29] 30. Let C be a circle. A point P moves such that the tangents from P to C include an angle of **(a) (b)** Straight line A circle concentric With C A circle touching C A circle intersecting C at two points **(c) (d)** If 2 — cos B tan A then sin B what is 28. 29. 30. What is the value of 2-6 — lö ?

19. [CDS (I) Paper 6 Q2] The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? **(a)** Statement- I statement-2 **(b)** Statement-2 statement-I **(c)** Statement-I statement-2 **(d)** Neither statement-I statement-2 nor statement-2 statement-I **(a) (c) (d)**

20. [CDS (I) Paper 6 Q97] 470 670 N In the figure given above, PQ is parallel to RS. What is Z NMS equal to ? **(a)** 200 **(b)** 230 **(c)** 2T **(d)** 470 B 98. c In the figure given above, O is the centre of the circle, OA 3 cm, AC = 3 cm and OM is perpendicular to AC. What is Z ABC equal to ? 450 **(c)** 300 **(a) (d)** None Of the above

21. [CDS (I) Paper 6 Q99] AC is the diameter Of the circum-circle of the cyclic quadrilateral ABCD. If Z BDC = 420, then what is Z ACB equal **(a)** 420 **(b)** 450 (C) 480 **(d)** 580

22. [CDS (I) Paper 7 Q35] 36. 37 38 The locus of the mid points of åll equal chords in a circle is : **(a)** the circumference of the circle concentric with the given circle and having radius equal to the length of the chords **(b)** the circumference of the circle concentric: with the given circle and having radius equal to the distange of the chords from the centre **(c)** the circümference of the circle concentric with the given circle and having radius equal to half of the radius of the given circle **(d)** the circumference Of the circle concentric with the given circle and having radius equal to half Of the distance of the chords from the centre 10

23. [CDS (I) Paper 7 Q48] Ifthe area 0fa circle is equal to the area Of a square with side 2vG units, what is the diameter of the circle ? **(a)** afb A cardboard sheet in the form of a circular sector of radius 30 cm and central angle 1440 is folded to make a cone. What is the radius of the cone ? **(b)** 12 cm **(c)** 18 cm **(d)** 21 cm

24. [CDS (I) Paper 7 Q2] The four points B, C, E and D may lie on a circle. Which of the above Statements is/are 95. **(b)** 17/8 correct ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither 1 nor 2 and 81 equal to ? **(a) (c) (d)** 17/16 17/4 15/4 3, then what is x **(a)**

25. [CDS (I) Paper 8 Q1] Area Of a segment of a circle is less than area of its corresponding sector. **(a)** 350 **(b)** 400 **(c)** 45 Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? **(d)** AD is always greater than 4 cm but less than 5 cm

26. [CDS (I) Paper 8 Q3] What is the area of the outer circle ? **(a)** 911 cm **(b)** 1611 em 2 **(c)** 2511 cm2 **(d)** 36r cm2 83 84 85 86 24

27. **[CDS (I) Paper 9 Q92]** 93. the following statements I. me La ngent Of a circle ig a line the t meets the circle in one and Only one g. The Of circle at the end p t of the diameter perrgdiculaT t. the Which Of the above ? 1 only **(b)** 2 only Booa and 2 Neither I nor 2 triangle Z C 900 ana CD is perpendicular to A.B. If ABxCD-CAxCE, then cna IS equal t. 1 The mean observation 1 if CA > CB CA2 of 7 observations is **(a) (b)** equal to equal to r greater than 1 less than I •us 91. and tio 92. ABCD is AD and the following statements I. me La ngent Of a circle ig a line the t meets the circle in one and Only one g. The Of circle at the end p t of the diameter perrgdiculaT t. the Which Of the above ? 1 only **(c)** 2 only Booa and 2 Neither I nor 2 triangle Z C 900 ana CD is perpendicular to A.B. If ABxCD-CAxCE, then cna IS equal t. In the figure given above. EF is parallel to y equal 460 **(d)** 93. 1 The mean observation 1 if CA > CB CA2 of 7 observations is 7. If each is increased by 2, then the new In the figure given above, Z PQR and QL is a median, PQ 5 cm and QR 12 an. QL is equal to G em

28. **[CDS (I) Paper 10 Q54]** 16 **(d)** 8/93 m 13/93 m 16/93 m 23/93 m The diameter ot a circle with centre at C is 50 cm. CP is a radial segtnent of the circle. AB is a chord perpendicular to CP and passes through p. CP produced intersects the circle at D. If DP = 18 cm, then what is the length of AD? **(a) (b) (c) (d)** 24 cm 32 cm 40 cm 48 Cm

29. **[CDS (I) Paper 10 Q60]** A conical cap has the base diameter 24 cm and height 16 cm. What is the cost of painting the surface of the cap at the ratc of 70 paisa per square cm? **(a) (b) (d)** e 520 e 524 e 528 e 532 55: A regular hexagon is inscribed in a circle of radius 5 cm. If x is the area inside the circle but outside the regular hexagon, then which one of the following is correct? **(a) (c)** 13 cm2 15 cm 15 Crn < x < 17 17 cm2 < x < 19 cm 19 cm < x < 2 lcm

30. **[CDS (I) Paper 10 Q1]** The perpendicular bisector of a chord of a circle does not pass through the centre of the **(a) (b) (c)** 1 only 2 only Both 1 and 2 Neither 1 nor 2 75. Consider the following statements • 1. The perpendicular bisector of a chord of a circle does not pass through the centre of the 2. The angle in a semicircle is a right angle. Which of the statements given above is/ are correct? **(d)** I only

31. **[CDS (I) Paper 10 Q78]** ABC is an equilateral triangle inscribed in a circle. D is any point on the arc BC What is ZADB equal to? **(c) (d)** 450 None of the above **(a) (b) (c)** 1 only 2 only Both 1 and 2 Neither 1 nor 2 75. Consider the following statements • 1. The perpendicular bisector of a chord of a circle does not pass through the centre of the 2. The angle in a semicircle is a right angle. Which of the statements given above is/ are correct? **(d)** I only

32. **[CDS (I) Paper 11 Q2]** Area of triangle ABC 4 (Area Of tri angle BPQ). Select the correct answer using the code given **(a)** 1 only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 AB and CD are two chords Of a circle meeting externally at P. Then Which of the following Ware correct ?

33. **[CDS (I) Paper 12 Q79]** The chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Which one Of the following is correct? y(2z— y) xa y(2Z— y) = 4X2 **(b)** 2ygz-y1 x2 **(c)** 4yPz— x2 **(d)** . In a triangle ABC, ZB-2ZC=2ZA. 80 **(a)** 2 only above statements **(b)** AC/5 **(c)** AC/3 **(d)** AC/4

34. **[CDS (I) Paper 13 Q22]** If the path is 5 m Wide, the diameter of the inner circle is **(a)** 12 Cm **(b)** 24 cm **(c)** 30 cm **(d)** 36 Crn 47. If each interior angle Of a regular polygon is 1350, then the number of diagonals of the polygon is equal to 54

35. **[CDS (I) Paper 13 Q54]** The area Of the largest triangle that can be inscribed in a semicircle of radius r is **(b)** 2r2 3r2 **(d)** 4r2 SS. AD is the diameter of a circle and AB is a chord. If AD 34 cm. AB— 30 cm, the distance Of AB from the centre Of the circle is **(a) (b) (c) (d)** 17 cm 15 cm 13 cm

36. **[CDS (I) Paper 13 Q77]** Two circles, each of radius r, with centres p and Q, are such that each circle passes through the centre Of the other circle. Then the area common to the circles is less than one-third of the sum Of the areas Of the two circles by 3 A spher and a cube have same surface area. The ratio of square of their volumes is **(a) (b) (c)** . Two area Of a rhombus With side cm and one diagonal IO cm 140 square cm 130 square cm 120 square cm I I O square cm circles, each of radius r, with 79. 80. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume Of the cone. The ratio Of the height Of the cone to the radius Of its base is **(d)** Water flows through a cylindrical pipe of internal diameter 7 cm at the rate Of 5 m/ s. The time, in minutes, the pipe would take to fill an empty rectangular tank 4m x 3m 31m is centres p and Q, are such that each circle passes through the centre Of the other circle. Then the area common to the circles is less than one-third of the sum Of the areas Of the two circles by 3 A spher and a cube have same surface area. The ratio of square of their volumes is

37. **[CDS (I) Paper 13 Q90]** Three equal circles each of diameter d are drawn on a plane in such a way that each circle touches the Other two circles, A big circle is drawn in such a manner that it touches each Of the small circles internally. The area Of the big circle nd2 + **(c)** nd 2 +2)2 12 B-EFUA-O-PDW / 52A **(a) (b)** 2 cm Cannot be determined due to insufficient data 92. Let on, OC and OD are rays in the anticlockwise direction such that ZAOB= ZCOD= 1 ocp, ZBOC= 820 and ZAOD= 780. Consider the following statements 89. From a circular piece of cardboard of radius 3 cm, two sectors of4 **(c)** I I square cm **(d)** 22 square cm

38. **[CDS (I) Paper 13 Q91]** If the angle between the radii of a circle is 13ff, then the angle between the tangents at the ends of the radii is **(a) (b)** 2 cm Cannot be determined due to insufficient data 92. Let on, OC and OD are rays in the anticlockwise direction such that ZAOB= ZCOD= 1 ocp, ZBOC= 820 and ZAOD= 780. Consider the following statements 89. From a circular piece of cardboard of radius 3 cm, two sectors of4 **(c)** I I square cm **(d)** 22 square cm

39. **[CDS (I) Paper 13 Q2]** If non-parallel sides Of a trapezium are equal, then it is cyclic. If the chord of a circle is equal to its radius, then the angle subtended by this chord at a point in major segment is 3CF. For the next four (4) items that follow : Consider the following frequency distribution Clas s 0-10 10-20 30—40 40_50 50-60 60-70 Freque 5 7 10 12 4 is **(a)** 37-2 **(b)** 38-1 **(c)** 39-2 **(d)** 40-1

40. **[CDS (I) Paper 13 Q99]** 100. **(a)** 20-30 **(b)** 30—40 **(c)** 40—50 **(d)** 50-60 What is the distribution? **(a)** 37 **(b)** 38 **(c)** 39 **(d)** 40 median Of the What is the mode Of the distribution? Which of the above statements is/ are correct? **(a)** I only **(b)** 2 only Both I and 2 **(d)** Neither I nor 2 Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 Cm is tangent to the inner circle. The radius Of the inner circle is **(b)** B-EFUA-O-PDW/52A

41. **[CDS (I) Paper 14 Q61]** Ihe circumference of a circle is 100cm. 'Ille side of the square inscribed in the Circle is **(a) (b)** 480 **(c)** 324 m 256 my **(d)** 63. Water is filled in a container in such a manner that its volume doubles every 5 minutes. If it takes 30 minutes for the container to be full, in how much time will it be one-fourth full ?

42. **[CDS (I) Paper 14 Q81]** O AB is a line segment of length 2a, with M as mid-point. Semicircles are drawn on one side with AM, MB and AB as diameter as shown in the above figure. A circle. with centre O -and radius r is drawn such that this circle touches all the three semicircles. The value of r is 3 4 **(a)** 4445 cm **(b)** 44-16 cm **(c)** 4745 cm **(d)** 476 cm 78. The area of a trapezium is 336cm2. If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is

43. **[CDS (I) Paper 14 Q82]** From an external point p tangents PA and PB are drawn to the circle as shown in the above figure. CD is the tangent to the circle at E. If AP 16 cm, then the perimeter of the triangle PCD is equal to **(a)** 24 cm **(b)** 28 cm **(c)** 30 cm **(d)** 32 cm 28

44. **[CDS (I) Paper 14 Q84]** Chord CD intersects the diameter AB of a circle at riW1t angle at a m)int p in the ratio I : **(a)** 20 square t **(b)** 30 square unit **(c)** 40 square unit **(d)** None of the above 84. Chord CD intersects the diameter AB of a circle at riW1t angle at a m

45. **[CDS (I) Paper 15 Q49]** A truck moves along a circular path and describes 100 m when it has traced out 360 at the centre. The radius of the circle is equal to 100 500 SO. Two poles are placed at p and Q on either side Of a road such that the line joining P and Q is perpendicular to the length Of the road. A person moves x metre away from P parallel to the road and places another pole at R. Then the moves further x metre in the same direction and turns and moves a distance y metre away from the road perpendicularly, where he finds himself, Q and R on the same line. The distance between P and Q (i.e., the Width of the road) in metre is B-ABPA-P-JRW/S4A Suppose chords AB and CD of a circle intersect at a point P inside the circle. TWO right-angled triangles A 'p 'B' and C' Q' D' are formed as shown in thc figures below such that A'p' AP, B'P'=BP, CQ'=cp, D'Q'-DP and ZA'P'B' = 90' - ZCQ'D' : Which of the following statements are not correct? **(a) (b) (c) (d)** 2 and 3 only I and 3 only 1, 2 and 4 only 1, 2, 3 and 4 B-ABPA-P-JRW/S4A Suppose chords AB and CD of a circle intersect at a point P inside the circle. TWO right-angled triangles A 'p 'B' and C' Q' D' are formed as shown in thc figures below such that A'p' AP, B'P'=BP, CQ'=cp, D'Q'-DP and ZA'P'B' = 90' - ZCQ'D' : Which of the following statements are not correct? 4. A' PW and CQ'D' are similar triangles, but need not bc Congruent A 'PW and C Q' D' are congruent triangle s. A 'PW and C' Q' D' are triangles of A'P'B' and C' Q' D' are triangles of same perimeter. Select the correct answer using the code given below.

46. **[CDS (I) Paper 15 Q60]** A tangent is drawn from an external point O to a circle of radius 3 units at P such that OP—4 units. If C is the centre Of the circle, then the sine Of the angle COP is **(c)** 3/5 B-ABPA-P-JRW / 54A **(a)** 9600 square units **(b)** 10000 square units **(c)** 12000 square units **(d)** None of the above 58. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to

47. **[CDS (I) Paper 15 Q70]** 71. Consider a circle with centre at O and radius 7 cm. Let QR be a chord Of length 2 cm and let P be the midpoint Of QR. Let CD be another chord or this circle passing through p such that ZCPQ is acute. If M is the midpoint Of CD and MP cm, then which Of the following statements are correct? 1. zQPD= 1350 **(a)** I and 2 0 nly **(b)** 2 and 3 only **(c)** and 3 0 **(d)** I, 2 and 3 Let ABC be a triangle in which AB AC Let L be the locus of points X inside or on the triangle such that BX CX. Which of the following statements are corre ct? L is a straight line passing through A and in-centre of triangle ABC is L is a straight line passing through A and orthocentre of triangle ABC is a point on L L is a straight line passing through A and centroid of triangle ABC is a point on Select the correct answer using the code given below. 72. 73. Consider a circle With centre at C Let OP, OQ denote respectively the tangents to the circle drawn from a point O outside the circle. Let R be a point on OP and S be a point on OQ such that OR x SQ 0SxRP. Which of the following statements is/ are correct? If X is the circle with centre at O and radius OR, and Y is the circle With centre at O and radius OS then X -Y. 2. zpoct zoco- 9cr Select the correct answer using the code given below.

48. **[CDS (I) Paper 15 Q72]** 73. Consider a circle With centre at C Let OP, OQ denote respectively the tangents to the circle drawn from a point O outside the circle. Let R be a point on OP and S be a point on OQ such that OR x SQ 0SxRP. Which of the following statements is/ are correct? If X is the circle with centre at O and radius OR, and Y is the circle With centre at O and radius OS then X -Y. 2. zpoct zoco- 9cr Select the correct answer using the code given below. **(a)** I only **(b)** 2 only Both I and 2 **(d)** Neither I nor 2 If m and n are distinct natural numbers, then which of the following is/are integer / integer s ? Select the correct answer using the code given below. **(a) (c) (d)** I and 2 2 only 2 and 3 3 only

49. **[CDS (I) Paper 15 Q100]** AD i s the diameter Of a rcle with area 707m2 and BC- as shown in the figure above. All curves inside the circle are semicirc s with their diameters on AD. What is the cost Of levelling the shaded regio at the rate of 63 per square metre? **(d)** e 29,700 e 22,400 14,847 None Of the above **(a)** AM < Median **(b)** GM Median **(c)** AM = Me dian Suppose for Which one of the corre ct? **(d)** GM>1/6

50. **[CDS (I) Paper 16 Q42]** If a square of side x and an equilateral triangle of side y are inscribed in a circle, then what is the ratio of x to y ? 2 3 **(b)** 2 3 3 **(a)** 12:45 hours **(b)** 5:00 p.m. **(c)** 11:45 a.m. **(d)** 11:50 a.m. 38. A train is travelling at 48 kmfhour completely crosses another train having half its length and travelling in opposite direction at 42 km/hour in 12 s. It also passes a railway platform in 45 s. What is the length of the platform ?

51. **[CDS (I) Paper 16 Q47]** Two circles touch externally and sum of their areas is 130m crn2 and the distance between their centres is 14 cm. What is the difference in the radii Of the circles ? **(a)** 264 **(b)** 201 **(c)** 132 **(d)** 105 46. Let ABC be a right angled triangle with BC = 5 and 12 cm. Let D be a point On the hypotenuse •AB such that ZBCD=300. What is length of CD?

52. **[CDS (I) Paper 16 Q48]** In a circle of radius 3 units, a diameter AB intersects a chord of length 2 units perpendicularly at P. If BP, then what is the ratio of AP to BP ? **(a)** 264 **(b)** 201 **(c)** 132 **(d)** 105 46. Let ABC be a right angled triangle with BC = 5 and 12 cm. Let D be a point On the hypotenuse •AB such that ZBCD=300. What is length of CD?

53. **[CDS (I) Paper 16 Q54]** A circle and a square have the same perimeter. Which one of the following is correct ? **(a)** Their areas are equal **(b)** The area Of the circle is larger **(c)** The area of the square is — times area of circle **(d)** The area of the square is It times area of circle

54. **[CDS (I) Paper 16 Q59]** If two tangents inclined at an angle 600 are drawn to a circle of radius 3 cm, then what is the length of each tangent ? **(a) (b) (c) (d)** 3--JS cm cm 20 cm

55. **[CDS (I) Paper 16 Q70]** ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r SDC-S-TRD - A 20 2-vfir 3 2 , then what is **(a)** 4275 **(b)** 4350 **(c)** 4375 **(d)** 4450 68. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ?

56. **[CDS (I) Paper 17 Q81]** 22-A) The Of a regular hexagon Of side 'a' is equal to **(a) (c)** — a square units 3 a square units 2 — a square units — a square units 2 ABCDEF is a regular polygon. rmv0 poles at C and D are standing vertically and subtend angles of elevation 300 and 600 at A respectively. What is the ratio of the height of the pole at C to that of the at D ? **(a)** 1:2Jä 2Jä Two parallel chords of a circle whose diameter is 13 cm are respectively 5 cm and 12 cm in length. If both the chords are on the same side of the diameter, then the distance between these chords is **(a) (b) (d)** 55 cm 3 em Let P, Q, R be the mid-points Of sides AB, BC,

57. **[CDS (I) Paper 17 Q87]** 83. square encloses an area Of 121 cm2. If the same Wire is bent in the form Of a circle, it encloses an area equal to **(a)** 2p2 100 **(b)** p 2+_p_ 100 A copper Wire when bent in the form Of a 87. 83. square encloses an area Of 121 cm2. If the same Wire is bent in the form Of a circle, it encloses an area equal to 84. 85. **(c) (d)**

58. **[CDS (I) Paper 18 Q61]** 62. 63. -A) Segment QR of length r is a tangent at Q to a circle of radius r with centre at P. What is the area of the part of the triangle PQR, which is outside the circular region ? 16 **(b)** 2 r2 2 4 8 rcr2 16 8 In a triangle ABC, AD is perpendicular on BC. If Z = 900, AB = c, BC = a, CA b and AD = p, then which one of the following is **(a) (b) (c) (d)** P -abc p2=bc bc ab AB and CD are parallel chords of a circle 3 cm apart. If AB = 4 em, CD = 10 cm, then what is the radius Of the circle ? **(a) (b) (c) (d)** 7 cm 14 em A hollow cube is formed by joining six

59. **[CDS (I) Paper 18 Q58]** Two straight lines AB and AC include an angle. A circle is drawn in this angle which touches both these lines. One more circle is drawn which touches both these lines as well as the previous circle. If the area Of the bigger circle is 9 times the area of the smaller circle, **(a)** 12 square cm **(b)** 24 square cm **(c)** 36 square cm **(d)** 48 square cm 58. Two straight lines AB and AC include an angle. A circle is drawn in this angle which touches both these lines. One more circle is drawn which touches both these lines as well as the previous circle. If the area Of the bigger circle is 9 times the area of the smaller circle, 59. 60.

60. **[CDS (I) Paper 18 Q3]** If A and C are parallel and B and C are parallel; then A and B are parallel. If A is perpendicular to C and B is perpendicular to C; then A and B are If the acute angle between A and C is equal to the acute angle between B and C; then A and B are parallel. Which of the above statements are correct ? **(d)** 1, 2 and 3 I and 2 only I and 3 only 2 and 3 only The diagonals of a rhombug arc of length 20 cm and 48 cm. What is the length of a side of the rhombus ? **(a) (b) (c) (d)** 13 26 cm 36 cm 39 cm An arc of a circle subtends an angle It at the centre. If the length of the arc is 22 em, then what is the radius of the circle ? (Take It = 22 7 **(a) (b) (c) (d)** 5 cm 7 cm 9 cm 11 cm **(a) (c) (d)** STS-M-TpT

61. **[CDS (I) Paper 19 Q61]** Then their curved surface areas are in the ratio **(a) (b) (d)** 16:9 64 : 61 81 : 64 In a room whose floor is a square Of side IO m, an equilateral triangular table Of side 2 m is placed. Four book-shelves of size 4 m x I m x 9 m are also placed in the room. If half Of the rest Of the area in the room is to be carpeted at the rate Of 100 per square metre, What is the cost of carpeting (approximately) ? **(b) (c) (d)** e 7,600 5,635 e 4,113 3/200 A region Of area A bounded by a circle C is divided into n regions, each Of area Mn, by drawing circl es of radii r r r that < r < r < ...r 1 concentric with the circle C. If p where m I, 2, 3, 75 (n — 2), then which one Of the following is correct ? **(a) (b) (c)** p increases as m increases p decreases as m increases p remains constant as m increases p increases for some values of m as m increases and then decreases thereafter

62. **[CDS (I) Paper 19 Q78]** 79. The locus of the mid-points of the radii Of length 16 cm of a circle is **(a) (b) (c) (d)** A concentric circle Of radius 8 cm A concentric circle Of radius 16 cm The diameter of the circle A straight line passing through the centre of the circle The curved surface area of a right circular is 1-76 m2 and its base diameter is 140 cm. What is the height of the cone ? **(a) (b) (d)** IO v'icrn IO q'äicm Consider the following statements .

63. **[CDS (I) Paper 19 Q80]** 76. length 16 cm of a circle is **(a)** A pair of straight line s **(b)** None of the above SPSS-T-XHI The locus of the mid-points of the radii Of 80. 76. length 16 cm of a circle is 77. 78. 79. **(c) (d)**

64. **[CDS (I) Paper 19 Q77]** 78. 79. (24-A) Consider the following statements ; Two triangles are said to be congruent, if **(a) (b) (c) (d)** A concentric circle Of radius 8 cm A concentric circle Of radius 16 cm The diameter of the circle A straight line passing through the centre of the circle The curved surface area of a right circular is 1-76 m2 and its base diameter is 140 cm. What is the height of the cone ?

65. **[CDS (I) Paper 19 Q99]** 100. of (30-A) In the figure given below, SVI' is a tangent to the circle at P and O is the centre of the circle. If Z QPT a, then what is Z POQ equal to ? o **(b) (c) (d)** s 20 900 —u 1800 -2a In the figure given below , two equal chords cut at point P. If AB = CD = IO em, OC = 13 cm (O is the centre Of the circle) and PB 3 em, then what is the length of OP ? o **(b)** In the figure given below, p, q, r are parallel

66. **[CDS (I) Paper 19 Q98]** AD DB and AE EC. If I is the length of a side Of the triangle, then What is the area of the shaded region ? **(a)** 16 312 16 32 **(d)** (30-A) SPSS-LXHI In the figure given below, SVI' is a tangent to the circle at P and O is the centre of the circle. If Z QPT a, then what is Z POQ equal to ? o **(b) (c) (d)** s 20 900 —u 1800 -2a In the figure given below , two equal chords cut at point P. If AB = CD = IO em, OC • 13 cm (O is the centre Of the circle) and PB 3 em, then what is the length of OP ? o **(b)**

67. **[CDS (I) Paper 19 Q89]** 90. (4 — It)a2 (a — 2)a2 27 -A) **(a) (b) (c) (d)** A straight line A circle Perpendicular bisector of AB An arbitrary curve If ABC is a right-angled triangle With AC as its hypotenuse, then which one of the following is correct ? **(c) (d)** AC3 < AB3 + BC AC3 > + BC AC3 AB3 + AC3 AB3 + BC3 The area Of the region bounded externally by a square of side 2a cm and internally by the circle touching the four sides of the square is **(b) (d)** (8 — It)a2/2 (z - 2)a2/2 The chord Of a circle is times its radius.

68. **[CDS (I) Paper 19 Q88]** The angle subtended by this chord at the minor arc is k times the angle subtended at the major arc. What is the value of k ? If a point P moves such that the sum of the squares Of its distances from two fixed points A and B is a constant, then the locus of the point p is **(a) (b) (c) (d)** 5 2 1/2 1/5 **(a) (b) (c) (d)** A straight line A circle Perpendicular bisector of AB An arbitrary curve

69. **[CDS (I) Paper 19 Q93]** In the figure given below, ABC is a 9 right-angled triangle where Z A 900, AB = p cm and AC = q On the three sides as diameters semicircles are drawn as shown in the figure. The ama of the shaded portion, m square cm, is c **(a) (b) (c) (d)** pq rt(p2 + q2)/2 It(p2 + q2) pq/2 In the figure given below, the radius Of the circle is 6 cm and AT = 4 cm. The length of tangent PT is A **(a) (b) (c) (d)** o 6 cm 8 cm 10 cm In the figure given below, ABCD is the diameter Of a circle Of radius 9 The lengths AB, BC and CD are equal. Semicircles are drawn on AB and BD as diameters as shown in the figure. What is the area Of the shaded region ? **(a) (b) (c) (d)** 97t 27m 3611 GTHY-T-XMT (29-, a

70. **[CDS (I) Paper 19 Q96]** 29-A) In the figure given below, what is Z BCD equal to ? c **(b) (c)** 750 800 900 In the figure given below, AB is the diameter of the circle whose centre is at O. Given that ECD Z EDC = 320, Z COF respectively are then Z CEF and c **(a)** (bj **(c) (d)** o 320, 640 640, 640 320, 320 640, 320 In the figure given below, A ABR A PQR. If PQ 3 em, AB = 6 em, BR = 82 cm and PR = em, then QR and AR are respectively **(a) (b) (c) (d)** cm, 10-4 cm cm, 6 cm em, 5-2 10•4

71. **[CDS (I) Paper 19 Q100]** 7 — cm 9— cm 31 -A) **(a) (b) (c) (d)** x 5 cm 12 cm 15 cm 10 cm Q c of In the figure given below, ABCD is a square of side 4 cm. Quadrants of a circle of diameter 2 cm are removed from the four corners and a circle of diameter 2 cm is also removed. What is the area of the shaded region ? c **(a) (b) (c) (d)** cm cm

72. **[CDS (I) Paper 20 Q82]** 25 -A) The of a right-angled triangle is k times the shortest side. If the ratio of the other side to hyptenuse is 4 : 5, then What is the value of k ? **(a) (b)** Only one Only two Only three Infirutely many Two circles are drawn with the same centre. The circumference of the smaller circle is 44 cm and that of the bigger circle is double the smaller one. What is the area between these two circles ? 154 Square Square Cm 462 Square Cm Square Cm A rectangular red carpet of size 6 ft x 12 ft has a dark red border 6 inches wide. What is the area of the dark red border ? 82. The of a right-angled triangle is k times the shortest side. If the ratio of the other side to hym»tenuse is 4 : 5, then What is the value of k ? **(c) (d)** A 12 m long wire is cut into two pieces. one Of ' which is bent into a circle and the other into a square enclosing the circle. What is the radius Of the circle ? 12

73. **[CDS (I) Paper 20 Q84]** 85. (27 —A) **(a) (b) (c)** cm 87. **(d)**

74. **[CDS (I) Paper 20 Q86]** 87. 88. —A) A ladder is resting against a vertical wall and its bottom is 2-5 m away from the wall. If it slips m down the wall, then its bottom will move away from the wall by m. What is the length of the ladder ? **(a) (b) (c)** cm 87. **(d)**

75. **[CDS (I) Paper 21 Q38]** A ggnalt AB is the diuneta of a circle with at O having mdius cm. Point p is in the of one circle Olat AP and jp —y. which one of fillowing cas— the pint P not lie On the circle ? **(b)** x — 12 cm y 5 **(d)** x cm y 13 **(a)** 33 cm **(b)** 44 **(c)** 66 cm 37. The sides AD, BC of a ü•apdum ABCD prallel md the di.N1s AC BD at O. If the of Gangle AOB is 3 cm2 the Of tiangle BDC is 8 then what is åe of aianøe AOD ? **(d)** 8 cm2

76. **[CDS (I) Paper 21 Q43]** A of wire in of a Etor Of a circle Of 20 cm, suM01ding an mgle 1500 at If it is ban ih 6rm Of a circle. what will be its ? **(d)** None Of the P, Q R Üe mid-points Of si&s of a &iangle of 128 cm2. If a &imøe ABC is &awn by joining of side Of riangle POR, what is of timgle ABC? **(a)** 612 **(b)** 212 43. A of wire in of a Etor Of a circle Of 20 cm, suM01ding an mgle 1500 at If it is ban ih 6rm Of a circle. what will be its ? **(c)** None Of the P, Q R Üe mid-points Of si&s of a &iangle of 128 cm2. If a &imøe ABC is &awn by joining of side Of riangle POR, what is of timgle ABC? **(d)** 4 cm2 O, 8 cm2

77. **[CDS (I) Paper 21 Q54]** A piu of wim of 33 is into an rc of a ofradius 14 is the angle by am at ca&e of circle ? **(c)** 1350 **(d)** 1500 C - SDHY T-MTÄ **(a)** 32% **(b)** 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ? **(c)** cm2 **(d)** 50

78. **[CDS (I) Paper 21 Q61]** A Mlow *ere Of extaul intanal diamaas 6 cm 4 an is into a Of base 8 m. What is åe of üw ? **(a)** 37, 11, 28, 16 **(b)** 37, 21, 11, 9 58. If — 3•3000, what is value of **(c) (d)** Q. IA XYZ be an Äiangle in 61. 21 which XY 7 cm. If A der»ta üe of the tinge, what is value of logwA4?

79. **[CDS (I) Paper 21 Q64]** A åin rod of 24 is Cut into Of qual size and joinal as to a cube. What is of of of **(a)** 25 squre Eet **(b)** 24 eet **(d)** 4 squae 23 **(a)** 23 Consid« a ABCD, in which AB is paallel to a) and AD perpendicular to AB. If treum which touches AB at E -and CD at F, EB 25 cm FC 16 cm, what is åe diameter of the circle ? **(a)** 16 **(b)** 25 cm **(c)** 36 cm 40 Three of radii 3 em, 4 cm and 5 an to a large spkk-e. Wlnt ? **(a)** 12 **(b)** 10 cm

80. **[CDS (I) Paper 22 Q69]** Two circles of radii 20 cm and 16 cm intersect and the length of common chord is 24 cm. If d is the distance between their centres, then which one Of the following is correct ? 23 **(a)** d < 26 cm **(b)** 26 cm < d < 27 cm **(c)** 27 **(d)** d > 28 cm

81. **[CDS (I) Paper 22 Q70]** In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? **(a)** +1 **(b)** + 3 **(c)** I **(d)** 266+1

82. **[CDS (I) Paper 22 Q11]** Two circles touch internally. The sum of their areas is 1367t cm2 and distance between their centres is 4 What are the **(a)** +1 **(b)** + 3 **(c)** I **(d)** 266+1

83. **[CDS (I) Paper 22 Q73]** A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? **(a)** — •E **(b)** litre **(c)** litre , **(d)** — litre 10

84. **[CDS (I) Paper 22 Q84]** If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

85. **[CDS (I) Paper 22 Q91]** What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? **(a)** — cm2 **(b)** — cm2 **(c)** — E cm2 **(d)** cm2

86. **[CDS (I) Paper 23 Q62]** In a A ABC, AC 12 cm, AB 16 and AD is the bisector of Z A. If BD 4 em, then what is DC equal to ? **(a) (b) (c) (d)** 3 em ABCD ig a cyclic quadrilateral. bisectors of the angles A, B, C and D cut the circle at p, Q, R and S respectively. What is Z RSP equal ? **(d)** 900 1800 2700 DZOL-T-LKM (23- nd

87. **[CDS (I) Paper 24 Q83]** AB and CD are the diameters of a circle which intersect at P. Join AG CB, BD and DA. If ZPAD 604, then what is LBPD equal to? 30• **(d)** 120 XvWS-T-MTK/50A **(a) (b) (c)** 28 units 42 units 56 units 84 units 82. ABC is a right angled at R Let M and N be two points on AB such that AM - MN-NB Let P and points on AC such that PM to QN and QN is parallel Q two is parallel to If BC 12 cm, then equal to? what is 8S. The sides Of a triangle ABC are 4 cm, 6 cm and 8 cm. With the vertices of the triangle as centres, three circles are drawn each touching the Other two externally. What is the sum of the radii Of the three circles? **(d)**

88. **[CDS (I) Paper 24 Q84]** 86 An equilateral triangle ABC and a scalene triangle DBC are inscribed in a circle on same side of the arc. What is ZBDC equal to? 300 8S. The sides Of a triangle ABC are 4 cm, 6 cm and 8 cm. With the vertices of the triangle as centres, three circles are drawn each touching the Other two externally. What is the sum of the radii Of the three circles? **(d)** . Let **(b)** 7 cm IO cm PAB be a secant to a circle intersecting the circle at A and H Lat be the tangent segrnent. If PA 9 cm and PT' 12 cm, then What is AB equal to ? **(c) (d)** 5 cm 6 cm 9 cm P.T.O. **(a) (b) (c)** 28 units 42 units 56 units 84 units 82. ABC is a right angled at R Let M and N be two points on AB such that AM - MN-NB Let P and points on AC such that PM to QN and QN is parallel Q two is parallel to If BC 12 cm, then equal to? what is 8S. The sides Of a triangle ABC are 4 cm, 6 cm and 8 cm. With the vertices of the triangle as centres, three circles are drawn each touching the Other two externally. What is the sum of the radii Of the three circles? **(d)**

89. **[CDS (I) Paper 25 Q79]** What is the approximate area of minor segment of the circle ? **(a)** 2•71 cm2 **(b)** 2•42 cm2 **(c)** 1•91 cm2 **(d)** 1•71cm2 A - NYRG-T-MTK

90. **[CDS (I) Paper 25 Q80]** What is the approximate area of major segment of the circle ? **(a)** 10•05 cm2 **(b)** 10•15cm2 **(c)** 11-05 cm2 **(d)** 11•15cm2 Consider the following for the next three (03) items that follow : ABC is a triangle with sides AB = 6 cm, BC — IO cm and CA 8 cm. With verti- ces A, B and C as centres, three circles are drawn each touching the other two externally.

91. **[CDS (I) Paper 25 Q81]** What is the sum of the radii of the circles ? **(a)** 104 cm **(b)** 11-2 cm **(c)** 12 cm **(d)** 13 cm

92. **[CDS (I) Paper 25 Q88]** What is 4(CP2-AQ2) equal to? **(a)** 101 •39 cm2 **(b)** 111-39cm2 **(c)** 121 •39cm2 **(d)** 131-39cm2 Consider the following for the next two (02) items that follow : AB is a diameter of a circle with centre O. Radius OP is perpendicular to AB. Let Q be any point on arc PB.

93. **[CDS (I) Paper 26 Q2]** If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. Which Of the above statements is/are correct? **(a)** I Only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2

94. **[CDS (I) Paper 26 Q62]** A circle Of radius 25 cm has a chord of length 48 cm. What is the length Of the perpendicular drawn from the centre of the circle to the chord? **(a)** cos29 **(b)** 2 sin B **(c)** 2cOsB 59. If O, where then what is 'the value Of 12cot2 B + 3cosece? 61. Considér the following statements : 2. If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. **(d)**

95. **[CDS (I) Paper 26 Q82]** An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct? **(c)** y = 2x **(a)** 2x-90' **(b)** 2x-18 **(c)** 18m-2x ABC is a triangle right angled at B with AC-2BC If ZA-x, then what is ZC equal to? 78. 80. Let D, E and F be the midpoints Of the sides BC, CA and AB respectively Of a triangle ABC Triangle DEF is congruent to which of the following 1. AEF 3. EDC Select the correct answer using the code given below. 81. 84. In a triangle ABC, AB— 16 cm , 12 cm and AD is the bisector Of ZA If BD —4 cm, then what is CD equal to? **(d)** 2-5 cm 82. An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct?

96. **[CDS (I) Paper 27 Q43]** A triangle ABC with sides AB z 15 cm , BC = 9 em, CA z 12 cm,is inscribed in a circle. What is cos2 A + cos2 B + cos2 C equal to ? **(a) (b)** What is the radius Of the circle ? What is one Of the J»ssible values of x — — ? **(c) (d)** What is one of the possible values Of + X 45.

97. **[CDS (I) Paper 28 Q79]** Statement I : Statement [I : 2p + q is Odd. q — 2p IS odd Question : What is the area of the circle C ? Statement I Statement 11 : An arc Of length 7 cm subtends an angle 300 at the centre of C. A chord of length IO cm subtends an angle 900 at the centre Of C. Question : Is triangle A right angled ? Statement I : Statement 11 : JNBY-D-MTK The length of the line segment joining the mid-points Of two sides of is half Of the third side of A. The angles Of A are in the ratio I : 2:3. (27 -A) **(a)** 1 **(b)** √3 **(c)** 1/√3 **(d)** 2

98. **[CDS (I) Paper 28 Q77]** 78. (27 -A) Statement I : Statement : 2p + q is Odd. q — 2p is odd. The lengths of two longer sides of the triangle are 25 cm and 24 cm. Question: What is the length of the shortest side ? Question : What is the area of the circle C ? Statement I : Statement 11 : Statement I Statement 11 : The angles of A are in the The length Of the perpendicular drawn on the longest side Of A from its opposite vertex is 6•72 cm. An arc Of length 7 cm subtends an angle 300 at the centre of C. A chord of length IO cm subtends an angle 900 at the centre Of C. **(a)** 15° **(b)** 30° **(c)** 45° **(d)** 60°

99. **[CDS (I) Paper 28 Q80]** Question : Is triangle A right angled ? A chord PQ of the circle C divides it into two segments such that 3 times the area Of the major segment is 4 times the area of the minor segment. Question : What is the radius of C ? The length of the line segment joining the mid-points Of two sides of is half Of the third side of A. The angles Of A are in the ratio I : 2:3. Statement I : Statement 11 : Area Of the minor segment is 66 square cm. Area Of the major segment is 88 square Cm. **(a)** 30° **(b)** 45° **(c)** 60° **(d)** 90°

100. **[CDS (I) Paper 28 Q93]** What is the radius Of the circle ? (33-A) **(a) (b) (c) (d)** 4•8 Cm that

101. **[CDS (I) Paper 28 Q94]** S What is the area Of the circle ? **(a) (b) (c) (d)** square cm 5011 cm JNBY-D-MTK

102. **[CDS (I) Paper 28 Q96]** Consider the following for the next three (08) items that follow : In the figure given below, a circle is inscribed in a PQRS. A rectangle at the corner P that measures 4 cm x 2 cm and a square at the corner R are drawn. 93. (33-A) What is the radius Of the circle ? **(a) (b) (c) (d)** 4•8 Cm

103. **[CDS (I) Paper 29 Q88]** 84. **(a) (b) (c) (d)** Cannot be concluded due to insumcient data Which one of the following is correct ? **(a) (b) (c) (d)** h2=y2-x2 following for the next two (02) Consider the items that : Two parallel chords AB and CD of a cir•cle are Of lengths 60 cm and 80 em respectively. They are on the same side of the centre O and IO cm apart.

104. **[CDS (I) Paper 29 Q35]** What is the diameter of the circle ? **(a) (b) (c) (d)** 120 cm 110 cm 100 cm 90 cm JNUP-T-EMT

105. **[CDS (I) Paper 29 Q36]** If the chord AB subtends an angle a and chord CD subtends an angle at the centre O, then what is the value of tan — **(a)** 4 12 12 88. 84. **(b) (c) (d)**

106. **[CDS (I) Paper 29 Q89]** 40. What is the height of the cone ? **(a)** 2R2 2 **(b)** What is the radius of the base of the cone ? 2r2 tan e 41. 42. **(c)** 2R2 tan e 2 **(d)**

107. **[CDS (I) Paper 29 Q61]** 58. diameter by 16•8 cm. What is the diameter of the circle ? (Take **(a) (b) (c)** 6•24 cm 6•42 cm 7•84 cm **(d)**

108. **[CDS (I) Paper 29 Q59]** 60. (23-A) **(a) (b) (d)** 6•24 cm 6•42 cm 7•84 cm The hypotenuse AC Of a right-angled ABC is 3v'6 em. If AB is doubled and BC is tripled such that ABC remains a right-angled triangle, the hypotenuse becomes 15 em. What is BC equal to ? **(a) (b) (c) (d)** IO cm 9 cm 2N'6 cm What is the area of the region between two concentric circles if the chord Of the outer circle of length 14 cm is a tangent Of the inner circle ? (Take = **(a) (b) (c) (d)** 125 square cm 132 square em 144 square cm 154 square cm JNUP-T-EMT

109. **[CDS (I) Paper 29 Q63]** 64. -A) A pendulum swings through an angle of 90 and its end descrit:æs an arc of length 14 •3 cm. What is the length of the pendulum ? (Take = **(a) (b)** 88 em 91 em 95 em 98 em The arch Of a bridge is in the form Of an arc Of a circle. If the span Of the bridge is 40 m and height in the middle is 8 m, then what is the radius Of cunature Of the bridge ? **(a) (b) (c) (d)** 25 m 27 m 29 m 31 m If a, b and c are the sides Of a right-an" triangle, where a > b > c, then What is the value of the expression **(a)** 4b2c2 **(b) (c) (d)** — 4b2c2 — 2a2b2 — 4a2b2 The cube root of x varies inversely as the square root Of y. x = 8 when y

110. **[CDS (I) Paper 29 Q58]** 59. 60. (23-A) The circumference of a cir-cle exceeds the diameter by 16•8 cm. What is the diameter of the circle ? (Take **(a) (b) (c)** 6•24 cm 6•42 cm 7•84 cm **(d)**

111. **[CDS (I) Paper 29 Q64]** The arch Of a bridge is in the form Of an arc Of a circle. If the span Of the bridge is 40 m and height in the middle is 8 m, then what is the radius Of cunature Of the bridge ? The hypotenuse AC Of a right-angled ABC is 3v'6 em. If AB is doubled and BC is tripled such that ABC remains a right-angled triangle, the hypotenuse becomes 15 em. What is BC equal to ? **(a) (b) (c) (d)** 25 m 27 m 29 m 31 m **(a) (b) (c) (d)** IO cm 9 cm 2N'6 cm If a, b and c are the sides Of a right-an" triangle, where a > b > c, then What is the value of the expression **(a)** 4b2c2 What is the area of the region between two concentric circles if the chord Of the outer circle of length 14 cm is a tangent Of the inner circle ? (Take = **(b) (c) (d)** — 4b2c2 — 2a2b2 — 4a2b2 The cube root of x varies inversely as the square root Of y. x = 8 when y

112. **[CDS (I) Paper 29 Q83]** A sphere of radius 5 cm is dropped in a right circular cylindrical vessel partly filled with water. The radius of the cylindrical vessel is IO cm. If the sphere is completely submerged in water, by how much will the level of water rise in the cylindrical vessel ? **(a) (b) (c) (d)** I em Consider the following statements : The angle in a sector greater than a semi-circle is less than a right angle. If two sides of a pair of opposite sides Of a cyclic quadrilateral are equal, then its diagonals are also equal. Which Of the statements given above Ware correct ? **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2 If a, b, c, x, y, z are real numbers such that (a • b + e)' — 3 (ab + bc + ca) + 3 (x2 + z2) = O, then which one of the following is correct ? **(a) (b) (c) (d)** -A) azbzc, x zyzz*O a x zyzz=O 6

113. **[CDS Paper 30 Q40]** From a circular metal plate Of radius 7 em and thickness 0•16 mm, a is cut Off containing an angle 1500. The remaining piece is moulded into a spherical bead Of radius r. What is the value of r in Cm ? **(a)** 0-35 105 14 The chord AB Of a circle With centre at O is 2&/ä times the height of the minor segment. If p is the area Of the sector OAB and Q is the area of the minor segment of the circle, then what is the approximate value of — ? (Take Jä 3.14) 14 26 What is the area Of the region between two concentric circles, if the length of a chord Of the outer circle touching the inner circle at a particular point Of its circumference is 14 cm ? (Take = **(a) (b) (c)** 7 154 square cm 144 square em 132 square cm Cannot be determined insufficient data due to What is the minimum value Sin **(a) (c) (d)** A + 5sinA +1 sin A 9 for O < A s — 22

114. **[CDS Paper 30 Q91]** 89. What is the radius of the circle with centre at 01 ? 92. **(d)** 3-6 **(a)** IO cm **(b)** 12 cm **(c)** 13 cm **(d)** 14 cm

115. [CDS (I) Paper 31 Q11] Consider the Question and the Staternents and mark the correct option. What is the integral Question value of k for which the expression 4x2 — kx + I is positive ? Statement-I Statement-II : k > —4 Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements together A - RAKU-T-EMT Consider the following for the next three (03) items that follow : Let MN be a chord of length 16 cm of a circle with centre at O and radius 10 cm. The tangents at M and N intersect at a point P. Further, OP intersects MN perpendi- cularly at Q.

116. [CDS (I) Paper 31 Q84] The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? **(a)** cm **(b)** I cm **(c)** cm **(d)** cm A - RAKU-T-EMT

117. [CDS (I) Paper 31 Q94] A circle is inscribed in a triangle ABC right-angled at B. If AB 5 cm and BC = 12 cm, then what is the radius of the circle ? **(a)** 121 **(b)** 120 **(c)** 119 **(d)** None of the above 91. What is the HCF 0f236-1 and 245-1 ?

118. [CDS (I) Paper 33 Q65] In a circle of radius 14 cm, APB is a shorter arc and P is the midpoint of the arc. Ikt C be the midpoint Of the chord AB and PC 7 cm. What is the length Of chord AP ? the **(d)** 14 cm 1 P.T.O. **(a)** 20% **(b)** 40% **(c)** None of the above 61. Let AD be the altitude of a triangle ABC If and **(d)** I only

119. [CDS (I) Paper 33 Q2] If it is changed into a semicircle, then what is its radius? **(a) (b) (c)** — cm None of the above 80. A right circular cone and a hemisphere have equal base and equal volume. What is the ratio Of the height of the cone to the radius Of the hemisphere? ANFB-T-EMT/4SA Let the area of the largest possible square inscribed in a circle of unit radius be x. Let the area of the largest possible circle inscribed in a square of unit side length be y What is the relation between x and y? **(d)** NX 2y

120. [CDS (I) Paper 33 Q80] Let the area of the largest possible square inscribed in a circle of unit radius be x. Let the area of the largest possible circle inscribed in a square of unit side length be y What is the relation between x and y? **(a) (b) (c)** — cm None of the above 80. A right circular cone and a hemisphere have equal base and equal volume. What is the ratio Of the height of the cone to the radius Of the hemisphere? ANFB-T-EMT/4SA Let the area of the largest possible square inscribed in a circle of unit radius be x. Let the area of the largest possible circle inscribed in a square of unit side length be y What is the relation between x and y? **(d)** NX 2y

121. [CDS (I) Paper 34 Q11] The diagonal BD is parallel to the tangent of the circle at A Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both I 11 **(d)** Neither I nor 11

122. [CDS (I) Paper 34 Q52] Let O be the centre of a circle. Let A, B and C lie on the circle such that LABC= 1530. What is the value of LAOC ? 21 **(a)** 540 **(b)** 520 **(c)** 390 **(d)** 270

123. [CDS (I) Paper 34 Q63] A triangle PQR is inscribed in a circle with its centre at O. A tangent PT is drawn at P such that ZQPT= **(a)** 120 **(b)** 180 **(c)** 240 **(d)** 300 61. The diameter of a copper solid sphere is 6 cm. The sphere is melted and recast into a wire. If the diameter of the wire is cm, then what is the length of the wire ?

124. [CDS (I) Paper 34 Q65] Let O be the centre of a circle. Let the chords AB = IO cm and CD = 24 cm be two parallel chords of the circle. The two chords are on opposite side of the centre and the distance between them is 17 cm. Let p be the midpoint of AB and Q be the midpoint of CD. Further, the points O and A, the points O and C, the points A and C are joined. Which of the statements given below is/are correct ? L Area of triangle OAP is equal to area of triangle OCQ. **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 66. A metallic sheet of rectangular shape is 48 cm x 36 cm. From each corner a square of side x cm is cut off and an open box is made of the remaining sheet. If the box holds 5•12 litre of water, then what is the value of x ?

125. [CDS (I) Paper 34 Q11] Consider the Question and the Statements and mark the correct option. Question : Statement-I Statement-II A circle is inscribed in the equilateral triangle A. What is the radius of the circle ? Area of A is equal to 16-6 cm2. Perimeter of A is 24 cm. Which one of the folloWing is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone 29 **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements

126. [CDS (I) Paper 34 Q11] Consider the Question and the Statements and mark the correct option. Question : Statement-I Statement-II Two circles with radii p, q (p > q) in cm touch externally, where p, q are natural numbers and each greater than l. What is the value of (p —q) ? The sum of their areas is 1307t cm2 The distance between the centres of the two circles is 14 cm Which one of the following is correct in respect of the above Question and the Statements ? **(a)** The Question can be answered by using one of the statements alone, but cannot be answered using the other statement alone **(b)** The Question can be answered by using either Statement alone **(c)** The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone **(d)** The Question cannot be answered even by using both the Statements together

127. **[CDS (I) Paper 35 Q56]** 57. (19-A **(b) (c) (d)** 3-5 m A wrell is dug With a diameter of m and 16 m depth. The earth soil so excavated is spread in the form of a right circular cone of radius 7 m. What is the height of the cone ? **(a) (b) (c) (d)** 1•5 m The chord of a circle is 20 cm and the height of the minor segment is 5 cm. What is the diameter of the circle ? **(a) (b) (c)** 24 cm 25 cm 26 em 30 Cm BFVS-T-TME

128. **[CDS (I) Paper 35 Q11]** Consider the Question and the Statements and mark the correct Question : ABCD is a quadrilateral with AB = 6 em, BC = 8 em, CD z 46 cm, DA 2v'S em and inscribed in a circle. What is the diameter of the circle ? State me nt I : The area of the quadrilateral is 44 cm . Statement 11 : Z ABC **(a) (b) (c) (d)** The Question can be answered by using one of the Statements alone, but cannot be answered using the Other Statement The Question can be answered by using either Statement alone. The Question can be answered by using both the Statements together, but cannot answered using either Statement alone. The Question can be answered even without using both the Statements.

129. **[CDS (I) Paper 35 Q11]** Consider the Question and the Statements and mark the correct option. Question : ABC is a triangle inscribed in a semi-circle such that AC coincides with the diameter of the semi-circle. Let P be any point on the arc Of the semi-circle. What is Ap2 + Cp2 + AC2 equal to ? Statement I : The radius Of the circle is Statement 11 : AB 6 em and BC = 8 em. Which one Of the following is correct in Of the above Question and Statements ? **(a)** AB:BC **(b)** AB2 Consider the following statements : 11. The areas Of two similar triangles are in the ratio Of the squares of the corresponding medians. The areas of two similar triangles are in the ratio of the squares Of the corresponding heights. **(c) (d)**

CHAPTER 11

Total Archive Questions: 148 Questions

MENSURATION 2D: AREAS & PERIMETERS

Typical Exam Weightage: 6 - 10 Questions (6 - 10 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q46] 47. What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ? **(a)** 47 **(b)** 43 (C) 37 **(d)** 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ? **(a)** 1000 m2 **(b)** 1250 m2 **(c)** 2500 m' **(d)** 5000 The difference between the area 0fa square and that of an equilateral triangle on the same base is 1/4 cm2. What is the length of side Of triangle ? **(a)** (4 — cm **(c)** (4 — cm **(d)** (4 + cm A horse is tied to a pole fixed at one corner ofa 50 m x 50 m square field Of grass by means of a 20 m long rope. What is the area Of that part of the field which the horse can graze ? **(a) (b) (c) (d)** 1256 942 m' 628 mz 314 m2 12

2. [CDS (I) Paper 3 Q45] Let VVXYZbe a square. Let P, Q, R be the midpoints of WX, XY and ZW tively and K, L be the midpoints of PQ and PR respectively. What is the value of area of triangle PKL area of square WXYZ **(a)** tan2t50 + tan2200 + tan2250 . + tan275Ø **(b)** cos2150 + cos2200 + cos2250 cos2750 **(c)** cot2150 — cot220e + cot2250 — + cot2750 **(d)** sec2150 + sec2200 + sec2250 + sec2750 42. Consider the following data : 5 9 2 The arithmetic mean of the above distribution is 296. What is the missing frequency ? 43. If a is an acute angle and then what is tan equal to ?

3. [CDS (I) Paper 3 Q49] ABCD is a square. p. Q, R, S are points on the sidesAB, BC, CD, DA respectively such that CR=DS. What is ZSPQ equal ? **(a)** 300 **(b)** 450 **(c)** 600 **(d)** 900

4. [CDS (I) Paper 3 Q52] ABC is a triangle. The internal bisector of LABC and the external bisector of ZACB meet at D Which one of the following is correct ? **(a)** ZBDC= ZBAC **(b)** ZBDC **(c)** ZBDC ZDBC **(d)** ZBDC-IZABC

5. [CDS (I) Paper 3 Q58] A parallelogram and a rectangle stand on the same base and on the same side of the base with the same height. If 11, be the perimeters of the parallelogram and the rectangle respectively, then which one of the following is correct ? **(a)** 11 12 **(b)** 11 = 12 **(c)** 12 but 11 *212 **(d)** 11 -212

6. [CDS (I) Paper 3 Q2] CA/CB-CD/CE Which of the above are correct ? **(a)** 45 **(b)** 225 l, 2 only 1, 3 only 2, 3 only table below gives the number of members of a club classified by sex and nativity ativity Incals Migrants Total Sex Male Female Total 85 35 120 45 35 80 130 70 200 The above data are represented by a pie diagram. What is the sectorial angle of the area representing male-migrant category ? **(c) (d)** 24 810 67-50

7. [CDS (I) Paper 3 Q88] Which one of the following numbers is not a square of any natural ? **(a)** 5041 **(b)** 9852 **(c)** 1936 **(d)** 6241 T-DETA.K.FN - A 26

8. [CDS (I) Paper 3 Q89] What is the maximum area of a rectangle, the IVimeter of which is 18cm? **(a)** 2025 cm2 **(b)** 20-00 cm2 **(c)** 19-75 cm2 **(d)** 19-60 cm2

9. [CDS (I) Paper 4 Q52] Let PAB be a secant to a circle intersecting at points A and and PC is a tangent. Which one of the following is correct? 33 33 **(c) (d)** The area Of rectangle With PA, PBas sides is equal to the area of square with PC as side The area Of rectangle With PA, PC as sides is equal to the area Of square With PB as side The area Of rectangle with PG PB as sides is equal to the area Of square with PA as side The perimeter of rectangle with PA, PB as sides is equal to the perimeter of square with PC as side **(a)** 1/2 47. **(b)** 60 **(c)** 45 **(d)** 30

10. [CDS (I) Paper 4 Q53] The line segments AB and CD intersect at O. .OF is the internal bisector Of obtuse angle BOC and OE is the internhl bisector Of acute angle AOC If zåoc- 130', what is the measure Of KFOE ? **(b) (c) (d)** In 900 110' 120' the figure given above, if ZBAD= 60', ZADC* 1050, then what is ZDPC equål to? @ 40 **(c)** Q-DETB-K-FN / 26A **(a) (b) (c)** In 900 110' 120' the figure given above, if In the figure given above, ABCD is a quadrilateral with AB parallel to DC and AD parallel to BC ADC is a right angle. If the perimeter of the triangle ABE is 6 units, what is the area of the quadrilateral? **(d)** 2&fi square units

11. [CDS (I) Paper 4 Q55] 16 In the figure given above, ABCD is a quadrilateral with AB parallel to DC and AD parallel to BC ADC is a right angle. If the perimeter of the triangle ABE is 6 units, what is the area of the quadrilateral? **(a)** 2&fi square units **(b)** 4 square units **(c)** 3 square units **(d)** 443 square units s In the figure given above, PQ is parallel to RS What is the angle between the lines PQ and LM ? **(a) (c) (d)** 1750 1770 1795 180'

12. [CDS (I) Paper 5 Q47] 48. 49. (14 -A **(a) (b) (c)** A certain amount of money has to be divided between two persons P and Q in the ratio 3 : 5. But it Was divided in the ratio Of 2 : 3 and thereby Q loses IO. What was the amount ? None Of the above Consider the following statements . If p is a prime such that p + 2 is algo a prime, **(d)**

13. [CDS (I) Paper 6 Q19] 20. **(b)** 64 **(c)** 65 **(d)** 700 What is the smallest number that must be added to 1780 to make it a perfect square ? **(a)** 39 **(b)** 49 **(c)** 59 **(d)** 69 What are the roots of the quadratic equation a2b2x2 — (a2 + b2) x + 1 = O ? A cylindrical vessel Of height IO cm has base radius 60 cm. Ifd is the diameter of a spherical vessel of equal volume, then what is d equal to ? **(a)** 30 cm **(b)** 60 cm **(c)** 90 cm **(d)** 120 cm The surface area of a sphere is 616 square Cm. If its radius is changed so that the area gets reduced by 75%, then the radius becomes From a lighthouse the angles Of depression of two ships on opposite sides of the light- house are observed to be 300 and

14. [CDS (I) Paper 6 Q34] 35. If (x — 3) is a factor of (x2 + 4px then what is the value of p ? - Ilp), What is the LCM Of a3b and ab(a + b) ? **(a)** a2b2(a2 — b2) **(b)** ab(a2 — b2) **(c)** a2b2 + ab3 **(d)** a3b3(a2 b2) ab3, a3b2+ a2b3 A circle circumscribes a rectangle with sides 16 cm and 12 cm. What is the area of the circle ? **(a)** 48 7t square cm **(b)** 50 It square cm **(c)** 100 7t square cm **(d)** 200 square cm What is the area of an equilateral triangle having altitude equal to 26 cm ? **(a) (b) (c) (d)** square cm 26 square Crn 36 square cm 46 square cm

15. [CDS (I) Paper 6 Q71] The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? **(a)** 3 units **(b)** 5 units **(c)** 7 units **(d)** IO units

16. [CDS (I) Paper 6 Q81] IO cylindrical pillars of a building have to be painted. The diameter Of each pillar is 70 cm and the height is 4 m. What is the cost of painting at the rate Of Rs. 5 RS. 500 per **(a) (b) (c) (d)** square metre ? 400 440 480

17. [CDS (I) Paper 6 Q100] The diameter Of a circle circumscribing a Square is 1545 cm. What is the length of the side of the square ? **(a)** 360 450 **(b)** 48 **(c)** 600 97. 470 670 N In the figure given above, PQ is parallel to RS. What is Z NMS equal to ? **(d)** 270

18. [CDS (I) Paper 7 Q2] Area of rectangle ABCD can be of the form pq2 square cm where p, q are distinct primes. Area ofthe figure EBFD of the form r2 square cm where r is rational but not an integer. Which of the above statements is/are correct ? **(a) (b) (c) (d)** 12 I only 2 only Both 1 and 2 Neither 1 nor 2

19. [CDS (I) Paper 7 Q59] The area of a rectangle whose length is 5 more than twice its width is 75 square units. What is the perimeter of the rectangle ? **(a)** 40 units **(b)** 30 units **(c)** 24 units **(d)** 20 units Two number* in the ratio 2

20. [CDS (I) Paper 7 Q2] Which of the statements is/are correct ? I only **(a) (b)** 2 only Both I and 2 **(c)** Neither 1 nor 2 **(d)** If2x + 3y.S 6, x 2 0, y 0, then one of the 75. solutions is : **(a)** x = —2 and y 3 **(b)** x = I and y = 2 **(c)** x — I and y = 1 **(d)** x = —1 and y ——1 The curved surface area of a -right circular 76. cone of radius 14 cm is 440 square cm. What is the slant height ofthe cone ? **(a)** I Ocm **(b)** 11 cm **(c)** 12 cm **(d)** 13 cm If the •volume of a cube is 729 cubic cm, 77. what is the length of its diagonal ? **(b)** 96 Crn **(c)** 18 cm **(d)** 186 cm The altitude Of an equilateral triangle is 78. cm. What is its perimeter ? **(b)** 36 cm **(c)** 6 cm **(d)** 6-6 cm The area of a rectangle whose one side is 79. is 2a2. What is the area of a square having one of the diagonals of the rectangle as side ? **(b)** 3a2 **(d)** 5a2

21. [CDS (I) Paper 8 Q81] 82. What is the height of a solid cylinder of radius 5 cm and total surface area is 660 cm2 ? **(a) (b) (c) (d)** IO cm 12 cm 15 cm 16 cm The diameter of base of a right circular cone is 7 cm and slant height is IO em, then what is its lateral surface area ? **(a)** 110 square cm **(b)** IOO square Crn **(c)** 70 square cm **(d)** 49 square cm The perimeter of a rectangle is 82 m and its area is 400 m2. What is the breadth Of the rectangle ? **(a)** 18m **(b)** 16m **(c)** 14m **(d)** 12 m The area enclosed between the circumferences of two concentric circles is 167t cm2 and their radii are in the ratio 5 :

22. [CDS (I) Paper 9 Q21] The Of y if one Of the Of the — bx + c O the square at the then Which one the following is ? **(a)** 120 cm^3 **(b)** 150 cm^3 **(c)** 180 cm^3 **(d)** 216 cm^3

23. [CDS (I) Paper 9 Q59] 80. Jf the heights and the area- Of the base a right circular cone and a pyramid with square base are thc same, then they have **(a) (b) (c)** same volume and same surface area same surracc area bul different volumes same volume but different surface areas different volumes and different surface A circular •water fountain m in diameter is surrounded by a path Of Width 1 •5 m. area of this path in square metres is **(c) (d)** 13-62% 19-151 None of the above The area Of a rectangular field ig 4500 square metres. If its length and brcadth are in the ratio 9 ; 5, then its perimeter is 150 m 360 m The area of a square inscribed in a eircle of radius 8 em is **(b)** 32 square cm 64 square em 128 square em 256 square cm The angleq A. B, C. .D Of a ABCD are in the ratio : 4 :

24. [CDS (I) Paper 10 Q46] The perimeter of a rectangle having area equal to 144 cm and sides in the ratio 4 : 9 is 52 cm **(b)** 56 **(c)** 60 cm **(d)** 64 cm **(a)** 56 **(b)** 60 cm **(c)** 64 cm **(d)** 1 only

25. [CDS (I) Paper 10 Q48] What is the area between a square Of side IO cm and two inverted semicircular cross-sections each of radius 5 cm inscribed in the square? **(a)** 56 **(b)** 60 cm **(c)** 64 cm **(d)** 1 only

26. [CDS (I) Paper 10 Q52] If the area of an equilateral triangle is x and its perimeter is then which one of the following is correct? @ y4 = 432x2 **(b)** y4 **(c)** -432x **(d)** None of the above **(a)** -432x 51. **(b) (c)** 12•56 cm 14•56 cm2 16•76 22-56 cma SO. What is the volume of a cone having a base of radius IO cm and height 21 cm? 53. 54. 16 **(d)** y4

27. [CDS (I) Paper 10 Q53] A rectangular field is 22 m long and IO m widé. Two hemispherical pitholes Of radius 2 m are dug from two places and the mud is spread over the remaining part of the field. The rise in the level Of the field is **(a)** -432x 51. **(b) (c)** 12•56 cm 14•56 cm2 16•76 22-56 cma SO. What is the volume of a cone having a base of radius IO cm and height 21 cm? 53. 54. 16 **(d)** y4

28. [CDS (I) Paper 10 Q59] 18 What is the whole surface area of a cone of base radius 7 cm and height 24 cm? **(a) (b) (c) (d)** 654 square cm 704 square Cm 724 square cm 964 square cm

29. [CDS (I) Paper 11 Q74] 75. (20-A) The area of a rectangle lies E*twæn 40 cm2 and 45 cm . If one of the sides is 5 em, then its diagonal lies between **(a) (b) (c) (d)** 8 cm and IO cm 9 and 11 cm IO cm and 12 em ll.cm and 13 cm sin 2B sin B cos 2B cos B For a plot Of land Of 100 m x 80 m, the level is to be raised by spreading the earth from a stack Of a rectangular base IO m x 8 m with vertical section being a trapezium Of height 2 m. The top of the stack is 8m X 5 m. By how many centimeters can the level be raised ? A cylinder is surmounted by a cone at One end, a hemisphere at the Other end. The radius is cm, the height of the cylinder Lis 6-5 cm and the total height or the structure is 12•8 cm. The volume V Of the structure lies between

30. [CDS (I) Paper 11 Q93] 94. 24 -A) The diameter Of the base Of a cone is 6 cm. altitude. is 4 cm. What is the approximate curved surface area of the cone ? **(a)** 45 square cm **(b)** 47 Square c m **(c)** 49 square cm **(d)** 51 Square cm What is the area of the larger ægment of a circle formed by a chord of length 5 cm subtending an angle of 900 at the centre ? 25 **(b)** 25 **(c)** 4 — + I cm **(d)** None of the abov e Consider the following in of the variate which takes values 2, 2, 2, 7, 7, 2, 7, 7 : I. The medi an ig equal to mean.

31. [CDS (I) Paper 11 Q91] 92. The diameter Of the base Of a cone is 6 cm. altitude. is 4 cm. What is the approximate curved surface area of the cone ? **(a)** 45 square cm **(b)** 47 Square c m **(c)** 49 square cm **(d)** 51 Square cm What is the area of the larger ægment of a circle formed by a chord of length 5 Cm subtending an angle of 900 at the centre ? If triangles ABC and DEF are that ZAB DE and BC = 8 em, EF equal to ? similar such then What is 25 **(b)** 25 **(c)** 4 **(a) (c) (d)** 16 cm 12 cm — + I cm A rectangle Of maximum area is drawn inside a circle of diameter 5 cm. What is the maximum area of such a rectangle ? **(a) (b)** 25 Cm 12-5 cm 12 Cm None of the above

32. [CDS (I) Paper 13 Q48] The sides of a triangular field are 41 m, 40 m and 9 m. The number Of rose beds that can be prepared in the field if each rose bed, on an average, needs goo square cm space, is 49 **(b)** . The 2000 1800 900 800 ratio Of thc outer and inner perimeters of a circular path is 23 : **(a)** 12 Cm **(b)** 24 cm **(c)** 30 cm **(d)** 36 Crn 47. If each interior angle Of a regular polygon is 1350, then the number of diagonals of the polygon is equal to 54

33. [CDS (I) Paper 13 Q57] Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is **(a)** I Ctn **(b)** 22 cm **(c)** 33 cm **(d)** 44 B-EFUA-O-PD W/ 52A 20 58. is equal to — I, then **(a)** 0

34. [CDS (I) Paper 13 Q76] 78. The area Of a rhombus With side 13 cm and one diagonal 10 cm will be **(a) (b) (c)** . Two area Of a rhombus With side cm and one diagonal IO cm 140 square cm 130 square cm 120 square cm I I O square cm circles, each of radius r, with 79. 80. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume Of the cone. The ratio Of the height Of the cone to the radius Of its base is **(d)** Water flows through a cylindrical pipe of internal diameter 7 cm at the rate Of 5 m/ s. The time, in minutes, the pipe would take to fill an empty rectangular tank 4m x 3m 31m is centres p and Q, are such that each circle passes through the centre Of the other circle. Then the area common to the circles is less than one-third of the sum Of the areas Of the two circles by 3 A spher and a cube have same surface area. The ratio of square of their volumes is

35. [CDS (I) Paper 13 Q81] The total outer surface area Of a right circular cone of height 24 cm with a hemisphere of radius 7 cm upon its base is **(d)** 327r square cm 307m square cm 293R square cm 273r square Cm **(a) (b) (c)** . Two area Of a rhombus With side cm and one diagonal IO cm 140 square cm 130 square cm 120 square cm I I O square cm circles, each of radius r, with 79. 80. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume Of the cone. The ratio Of the height Of the cone to the radius Of its base is **(d)** Water flows through a cylindrical pipe of internal diameter 7 cm at the rate Of 5 m/ s. The time, in minutes, the pipe would take to fill an empty rectangular tank 4m x 3m 31m is centres p and Q, are such that each circle passes through the centre Of the other circle. Then the area common to the circles is less than one-third of the sum Of the areas Of the two circles by 3 A spher and a cube have same surface area. The ratio of square of their volumes is

36. [CDS (I) Paper 13 Q78] The 13 will be **(a) (b) (c)** . Two area Of a rhombus With side cm and one diagonal IO cm 140 square cm 130 square cm 120 square cm I I O square cm circles, each of radius r, with 79. 80. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume Of the cone. The ratio Of the height Of the cone to the radius Of its base is **(d)** Water flows through a cylindrical pipe of internal diameter 7 cm at the rate Of 5 m/ s. The time, in minutes, the pipe would take to fill an empty rectangular tank 4m x 3m 31m is centres p and Q, are such that each circle passes through the centre Of the other circle. Then the area common to the circles is less than one-third of the sum Of the areas Of the two circles by 3 A spher and a cube have same surface area. The ratio of square of their volumes is

37. [CDS (I) Paper 13 Q83] 84. A rectangular block Of wood having dimensions 3 m x 2m x •75m has to be painted On all its faces. The layer of paint must be O I mm thick. Paint comes in cubical boxes having their edges equal to IO cm. The minimum number Of boxes of paint to be purchased is **(c) (d)** 3 2 The diagonals of three faces Of a cuboid are 13, and 20 linear units. Then the total surface area of the cuboid is **(a) (b) (c) (d)** 650 square units 658 square units 664 square units 672 square units B-EFUA-O-PDW/ S2A Of

38. [CDS (I) Paper 13 Q89] From a circular piece of cardboard of radius 3 cm, two sectors of4(r each have been cut Off. The area Of the remaining portion is **(a)** I I square cm **(b)** 22 square cm **(c)** 33 square cm **(d)** 44 square cm

39. [CDS (I) Paper 14 Q54] A square and an equilateral triangle have equal perimeter. If the diagonal of .the square is 12-45 cm, then the area of the triangle is **(a)** cm2 **(b)** 240 CITI 2 **(c)** 480 cm2 **(d)** 6443 cm2 18

40. [CDS (I) Paper 14 Q67] If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow **(a)** I only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 B-ABVB-O-GPU - A 24

41. [CDS (I) Paper 14 Q70] A square is inscribed in a right triangle with legs x and y and has commOn right angle with the triangle. The perimeter of the square is giveri by 2 **(b)** x2 + 4xy x2 + Y 24 **(a)** 2-0 **(b)** 2+6 66. The diameter Of a wheel that makes 452 revolutions to move 2 km and 26 decametre is equal to 67. If X is any point within a square ABCD and on AX a square AXYZ is described, which Of the following is/are corect ? 2 ZABX=ZADZor ZADX= ZABZ Select the correct answer using the code given txlow **(c)** I only **(d)** 2 only

42. [CDS (I) Paper 14 Q76] Three rectangles RI, Ra and Rs have the same area. Their lengths x, , xz 'and x, respectively are such that x, < x, < xy If VI , V2 and V3 are the volumes of the cylinders formed from the rectangles RI, R2 and Rs respectively by joining the parallel sides along the breadth, then which one of the following is correct ? **(a)** will be 75. Let **(b)** 52% **(c)** 540/0 **(d)** 98 cm 108 cm2

43. [CDS (I) Paper 14 Q2] DP and when produced meet at Q. If area of triangle BPQ is 20 square unit, the area of triangle DC? is **(a)** 20 square t **(b)** 30 square unit **(c)** 40 square unit **(d)** None of the above

44. [CDS (I) Paper 15 Q53] A rhombus is formed by joining midpoints of thc sides of a rectangle in the suitable order. If the area of the rhombus is 2 square units, then the area Of the rectangle is **(a)** 242 square units **(b)** 4 square units **(c)** 40 square units **(d)** 8 square units

45. [CDS (I) Paper 15 Q57] The base Of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area Of the triangle is **(a)** 9600 square units **(b)** 10000 square units **(c)** 12000 square units **(d)** None of the above

46. [CDS (I) Paper 15 Q58] Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to **(a)** 7/6 cm **(b)** 5/6 cm **(c)** 112 cm **(d)** 5/11 cm

47. [CDS (I) Paper 15 Q77] A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times the arca Of the smaller ring, then n is equal to **(d)** 10 **(a)** 165% 21 B-ABPA-P-JRW/ 54A 74. Outside a meeting room, Madhukar was told by a person that each meeting takes place after 13/4 hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 p.m. At What time did Madhukar receive this information? **(b)** 12% 76. If 61 where x, y and z are natural numbers, then What is z equal to? **(c)** 75. The 10:20 a.m. 11:30 a.m. 11:40 a.m. 11:50 a.m. expenditure Of a household for a 77. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times thc arca Of the smaller ring, then n is equal to **(d)** 10 78. Consider the following statements : 2. If n 3 and m 2 3 are distinct positive integers, then the sum Of the exterior angles Of a regular polygon Of m sides is different from the sum Of the exterior angles of a regular polygon Of n sides. Let m, n be integers such that m > n 3. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is

48. [CDS (I) Paper 16 Q52] The cost of painting a spherical vessel of diameter 14 cm is Rs. 8008. What is the cost Of painting per square centimetre ? **(a)** Rs. 8 **(b)** Rs. 9 **(c)** Rs. 13 **(d)** Rs. 14

49. [CDS (I) Paper 16 Q62] What is area of. largest triangle inscribed in a semi-circle Of radius r units ? **(a)** square units **(b)** 2r2 square units **(c)** 3? square units **(d)** 4r2 square units

50. [CDS (I) Paper 16 Q66] How many square metres Of canvas are used ? **(a)** 14450 **(b)** 14480 **(c)** 14580 **(d)** 14850 18

51. [CDS (I) Paper 16 Q69] If the perimeter of circle is equal to that Of a square, then what is the ratio of area of circle to that of square ? **(a)** 22: 7 **(b)** 14:11 **(c)** 7:22 **(d)** 11:14

52. **[CDS (I) Paper 17 Q3]** The smallest angle of the triangle is **(a)** 200 **(b)** 400 **(c)** 500 **(d)** 600 A ball Of radius I cm is put into a cylindrical pipe so that it fits inside the pipe. If the length of the pipe is 14 m, what is the surface area of the pipe ? **(a) (b) (c) (d)** 2200 square cm 4400 square cm 8800 square em 17600 square cm The areas of two circular fields are in the ratio 16 :

53. **[CDS (I) Paper 17 Q49]** If the radius of the bigger field is 14 m, then what is the radius Of the smaller field ? **(a) (b)** Let ABCD be a rectangle. Let P, Q, R, S be the mid-points ot' sides AB, BC, CD, DA respectively. Then the quadrilateral PQRS is **(a) (b) (c) (d)** Square Rectangle, but need not a square Rhombus, but need not be a square Parallelogram, but need not be rhombus **(a) (b) (c) (d)** •BLS-S-FDE If x = a cos B + b sin and y = a sin — b cos 6,

54. **[CDS (I) Paper 17 Q76]** 77. Let P, Q, R be the mid-points Of sides AB, BC, CA respectively of a triangle ABC. If the area Of the triangle ABC is 5 square units, then the area of the triangle PQR is **(a) (b) (d)** — square units square units 2 — square units I squat-e unit If each Of the dimensions Of a rectangle is increased by 200%, the area is increased by **(a) (b) (d)** 300% 400% 600% 800% Three circles each Of radius cm touch one 8 another. The' area subtended between them **(a) (b) (c) (d)** BLS-S-FDE 6 (— 2) square units 6 (2n —) square units 49 — (2 — It) square units 8 — (— n) square units (22

55. **[CDS (I) Paper 17 Q79]** 76. CA respectively of a triangle ABC. If the area Of the triangle ABC is 5 square units, then the area of the triangle PQR is 77. (22-A) The Of a regular hexagon Of side 'a' is equal to **(a) (b) (d)** — square units square units 2 — square units I squat-e unit 80. **(a) (c)** — a square units 3 a square units 2 — a square units — a square units 2 If each Of the dimensions Of a rectangle is increased by 200%, the area is increased by **(a) (b) (d)** 300% 400% 600% 800% Three circles each Of radius cm touch one 81. another. The' area subtended between them ABCDEF is a regular polygon. rmv0 poles at C and D are standing vertically and subtend angles of elevation 300 and 600 at A respectively. What is the ratio of the height of the pole at C to that of the at D ? **(a)** 1:2Jä 2Jä Two parallel chords of a circle whose diameter is 13 cm are respectively 5 cm and 12 cm in length. If both the chords are on the same side of the diameter, then the distance between these chords is **(a) (b) (c) (d)** BLS-S-FDE 6 (— 2) square units 6 (2n —) square units 49 — (2 — It) square units 8 — (— n) square units **(a) (b) (d)** 55 cm 3 em

56. **[CDS (I) Paper 17 Q89]** 90. ABCD is a rectangle. The diagonals AC and BD intersect at O. If AB = 32 cm and AD = 24 em, then what is OD equal to ? **(a)** 22 cm **(b)** 20 cm **(c)** 18 cm **(d)** 16 cm A field is divided into four regions as shown in the given figure. What is the area Of the field in square metres ? **(b) (d)** BLS-S-FDE (26- nd 91. eld 92. (26 -A) In the figure given below, D is the diameter of each circle. What is the diameter of the shaded circle ? D (Jä-l) **(b) (c)** In the figure given below, AC is parallel to ED and AB = DE = 5 cm and BC 7 cm. What is the area ABDE : area BDE : area BCD equal to ? **(a) (b) (d)** c

57. **[CDS (I) Paper 17 Q91]** BD intersect at O. If AB = 32 cm and AD = 24 em, then what is OD equal to ? **(a)** 22 cm **(b)** 20 cm **(c)** 18 cm **(d)** 16 cm A field is divided into four regions as shown in the given figure. What is the area Of the field in square metres ? 92. **(b) (d)** In the figure given below, D is the diameter of each circle. What is the diameter of the shaded circle ? D (Jä-l) **(b) (c)** In the figure given below, AC is parallel to ED and AB = DE = 5 cm and BC 7 cm. What is the area ABDE : area BDE : area BCD equal to ? **(a) (b) (d)** c BLS-S-FDE (26 -A)

58. **[CDS (I) Paper 18 Q4]** 2222 1 IGG4 343343 220347 Which of the above are not perfect squares ? **(a) (b) (c) (d)** 1, 2 and 3 1, 2 and 4 2, 3 and 4 1, 3 and 4 19,

59. **[CDS (I) Paper 18 Q59]** 60. A hollow cube is formed by joining six 61. identical squares. A rectangular cello tape of length 4 cm and breadth em is used for joining each pair Of edges. What is the total area Of cello tape used ? **(a)** 12 square cm **(b)** 24 square cm **(c)** 36 square cm **(d)** 48 square cm Two straight lines AB and AC include an angle. A circle is drawn in this angle which touches both these lines. One more circle is drawn which touches both these lines as well as the previous circle. If the area Of the bigger circle is 9 times the area of the smaller circle, then what must be the angle A ? **(a) (b) (d)** 450 600 750 900 An isosceles triangle is drawn outside on one of the sides of a square as base in such a way that the perimeter of the complete figure is times the perimeter of the original square. 6 What is the ratio of area of the triangle to the area Of the original square ? What is the area of the triangle whose sides are 51 cm, 37 cm and 20 ? **(a)** 300 square **(b)** 305 square cm (C) 306 square Cm **(d)** 307 square STS-M-TPT (18-A

60. **[CDS (I) Paper 18 Q62]** 63. In a triangle ABC, AD is perpendicular on BC. If Z = 900, AB = c, BC = a, CA b and AD = p, then which one of the following is An isosceles triangle is drawn outside on one of the sides of a square as base in such a way that the perimeter of the complete figure is times the perimeter of the original square. 6 What is the ratio of area of the triangle to the **(a) (b) (c) (d)** P -abc p2=bc bc ab area Of the original square ? What is the area of the triangle whose are 51 cm, 37 cm and 20 ? **(a)** 300 square **(b)** 305 square cm (C) 306 square Cm **(d)** 307 square STS-M-TPT AB and CD are parallel chords of a circle 3 cm apart. If AB = 4 em, CD = 10 cm, then what is the radius Of the circle ? sides **(a) (b) (c) (d)** 7 cm 14 em

61. **[CDS (I) Paper 18 Q66]** The diagonals of a cyclic quadrilateral ABCD 61 jntersect at P and the area ofthe triangle APB is 24 square cm. If AB 8 and CD = 5 cm, then what is the area of the triangle CPD ? **(a) (b) (c) (d)** 24 Square cm 15 square cm 12•5 square cm 9•375 square cm In an equilateral triangle ABC, BD is drawn perpendicular to AC. What is BD2 equal to ? **(a) (b) (d)** AD2 2AD2 3AD2 4AD2 The distance between the centres of two circles having radii 9 cm and 4 cm is 13 cm. What is the length of the direct common tangent of these circles ? **(a) (b) (c) (d)** STS-M-TPT 12 Crn 11 cm 10 cm 9-5 cm (204 D

62. **[CDS (I) Paper 19 Q22]** 23. -A) The sum of a number and its square -is **(a) (b)** -14 14 21. **(c) (d)**

63. **[CDS (I) Paper 19 Q85]** 86. What is the largest power of IO that divides the product I x 2 x 3 x 4 ...x23x24x25? **(a) (d)** 2 4 5 None of the above Walls (excluding their roofs and floors) Of 5 identical rooms having length, breadth and height 6 m, 4 m and m respectively are to be painted. Paints are available only in cans Of I L and one litre of paint can be used for painting 20 square metres. What is the number Of cans required for painting ? **(a) (b) (c)** 10 12 13 14 A rectangular pathway having width m and length IO m will have to be tiled using square tiles Of side 50 cm. Each packet of such tiles contains 20 pieces and costs

64. **[CDS (I) Paper 19 Q87]** 88. If the ratio of the circumference of the base Of a right circular cone Of radius r to its height is 3 : l, then What is the area of the curved surface Of the cone ? 2nr2 4712 + 9 **(b)** A wire is in the form of a circle Of radius 98 cm. A square is formed out Of the wire. What is the length of a side Of the square ? (Use 22/7) **(a) (b) (c) (d)** 146 cm 152 em 154 156 cm Consider the following for the next tuo (02)

65. **[CDS (I) Paper 19 Q84]** 85. 86. (26-A) **(a) (d)** 2 4 5 None of the above Walls (excluding their roofs and floors) Of 5 identical rooms having length, breadth and height 6 m, 4 m and m respectively are to be painted. Paints are available only in cans Of I L and one litre of paint can be used for painting 20 square metres. What is the number Of cans required for painting ? 88. **(a) (b) (c)** 10 12 13 14 If the ratio of the circumference of the base Of a right circular cone Of radius r to its height is 3 : l, then What is the area of the curved surface Of the cone ? 2nr2 4712 + 9 **(b)** A wire is in the form of a circle Of radius 98 cm. A square is formed out Of the wire. What is the length of a side Of the square ? (Use 22/7) A rectangular pathway having width m and length IO m will have to be tiled using square tiles Of side 50 cm. Each packet of such tiles contains 20 pieces and costs

66. **[CDS (I) Paper 19 Q93]** What is the area Of the largest circular disc cut from a squ are of side units ? **(a) (b) (c) (d)** square units I square unit square units 2 square units The product Of the lengths of the diagonals of a square is 50 square units. What is the length of a side ofthe square ? **(a) (b) (c) (d)** 5 units 5 units IO units 2 units The surface area Of a closed cylindrical box is 352 square cm. If its height is IO cm, then what is its diameter ? (Use It = **(a) (b) (c) (d)** SPSS-T-XHI 9•12cm 19 •26 cm ISC

67. **[CDS (I) Paper 19 Q96]** 28—A) A square and an equilateral triangle have the same perimeter. If the diagonal Of the square iS 6 v'5 cm, then what is the area Of the triangle ? **(a) (b)** 12 v'5 cm2 12 cm2 What is the area Of the region bounded internally by a square of side of length 'a' and externally by a circle passing through the four corners Of the ? **(a) (b) (c) (d)** (r — l) a2 square units square units (It — 2) a2 square units square units In the figure given below, XA and XB are two tangents to a circle. If Z AXB z 500 and AC is parallel to XB, then What is Z ACB equal to ? x **(b) (d)** 700 650 550 **(a) (b) (c) (d)** SPSS-T-XHI What is the area Of the largest circular disc

68. **[CDS (I) Paper 19 Q92]** 93. (28—A) A square and an equilateral triangle have the same perimeter. If the diagonal Of the square iS 6 v'5 cm, then what is the area Of the triangle ? **(a) (b) (c) (d)** square units I square unit square units 2 square units 95. **(a) (b)** 12 v'5 cm2 12 cm2 What is the area Of the region bounded internally by a square of side of length 'a' and externally by a circle passing through the four corners Of the ? The product Of the lengths of the diagonals of a square is 50 square units. What is the length of a side ofthe square ? **(a) (b) (c) (d)** 5 units 5 units IO units 2 units **(a) (b) (c) (d)** (r — l) a2 square units square units (It — 2) a2 square units square units

69. **[CDS (I) Paper 19 Q80]** 76. plot of land whose area is 144m m2. The total area of the circular plot including the path surrounding it is **(a) (b) (c) (d)** 3497t m2 2897t m2 2097t m2 1497t m2

70. **[CDS (I) Paper 20 Q75]** In a rectangle, length is three times its breadth. If the length and the breadth of the rectangle are increased by and If the lengths of two parallel chords in a circle of radius IO cm are 12 cm and 16 em, then what is the distance between these two Chords 2 em or 14 em 3 or 21 Cm 4 cm or 28 em Considering two opposite vertices of a square of side 'a' as centres, two circular arcs are drawn Within Square joining the Other two vertices, thus forming two sectors. What is the common area in these two sectors ? , its by DFGT-T-HTM **(a) (b)** The Median Mean Geometric mean Mode Graphical location Graphical location Ogive 73. **(c)** NO change following pairs relate to frequency The volume of a spherical balloon is increased by 700'h What is the percentage increase in its surface area ? distribution Of a discrete variable and its frequency polygon. Which one Of the following pairs is not correctly matched ? **(d)** 300% 400% 450% 500%

71. **[CDS (I) Paper 20 Q78]** 79. The corners of a square of side •a' are cut away so as to form a regular octagon. What is the side Of the octagon ? Three consecutive integers form the lengths of a right-angled triangle. How many sets of such three consecutive integers Ware possible ? **(a) (b)** Only one Only two Only three Infirutely many Two circles are drawn with the same centre. The circumference of the smaller circle is 44 cm and that of the bigger circle is double the smaller one. What is the area between these two circles ? 154 Square Square Cm 462 Square Cm Square Cm A rectangular red carpet of size 6 ft x 12 ft has a dark red border 6 inches wide. What is the area of the dark red border ? 82. The of a right-angled triangle is k times the shortest side. If the ratio of the other side to hym»tenuse is 4 : 5, then What is the value of k ? **(c) (d)** A 12 m long wire is cut into two pieces. one Of ' which is bent into a circle and the other into a square enclosing the circle. What is the radius Of the circle ? 12

72. [CDS (I) Paper 20 Q76] The corners of a square of side •a' are cut away 80. so as to form a regular octagon. What is the side Of the octagon ? **(a) (b)** Only one Only two Only three Infirutely many Two circles are drawn with the same centre. The circumference of the smaller circle is 44 cm and that of the bigger circle is double the smaller one. What is the area between these two circles ? 154 Square Square Cm 462 Square Cm Square Cm A rectangular red carpet of size 6 ft x 12 ft has a dark red border 6 inches wide. What is the area of the dark red border ? 82. The of a right-angled triangle is k times the shortest side. If the ratio of the other side to hym»tenuse is 4 : 5, then What is the value of k ? **(c) (d)** A 12 m long wire is cut into two pieces. one Of ' which is bent into a circle and the other into a square enclosing the circle. What is the radius Of the circle ? 12

73. [CDS (I) Paper 21 Q3] Unit in is 7. of the square of two is always divisi- by **(a)** be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? **(b)** 80 **(c)** 60 36 • **(d)** 24 14. WEch Of

74. [CDS (I) Paper 21 Q2] 3. Unit in is 7. of the square of two is always divisi- by **(a)** be 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? **(b)** 80 **(c)** 60 36 • **(d)** 24 14. WEch Of

75. [CDS (I) Paper 22 Q23] If the area of a square is 2401++ 196r2+4, then what is its side length ? **(a)** 49x2 -f- 3x + 2 **(b)** 49x2 _ 3x + 2 **(c)** 49++2 **(d)** 59++2

76. [CDS (I) Paper 22 Q36] In a pi-diagram there are three sectors. If the ratio Of the angles Of the sectors is 1 : 2 : 3, then what is the angle of the largest sector ? **(a)** 2000 **(b)** 1800 **(c)** 1500 **(d)** 1200

77. [CDS (I) Paper 22 Q65] A rectangle is 48 cm long and 14 cm wide. If the diagonal makes an angle 9 with the longer side, then what is (sec0+ cosec9) equal to ? 775 168 168 375 84 84 **(a)** 2sine **(b)** 2cos0 61. What is **(c)** cosec x + cot x 63. If 0 lies in the first quadrant and 63 then what is the value of cot 6 = 16 **(d)**

78. [CDS (I) Paper 22 Q72] If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - PIXI-T-MTK **(a)** +1 **(b)** + 3 **(c)** I **(d)** 266+1

79. [CDS (I) Paper 22 Q75] A square and a rectangle have equal areas. If one side of the rectangle is of length numerically equal to the square of the length of the side of the square then the other side of the rectangle is 25 25 **(a)** square root of the side of the **(b)** half the side of the square **(c)** of unit length **(d)** double the side of the square

80. [CDS (I) Paper 22 Q77] Suppose a region is formed by removing a sector of 200 from a circular region of radius 30 feet. What is the area of this new region ? **(a)** 1507t square feet **(b)** 550z square feet **(c)** square feet **(d)** 850% square feet A - PLKI-T-MTK

81. [CDS (I) Paper 22 Q1] Area of APDA is equal to that of APCQ **(a)** — AB **(b)** a AB2 **(c)** 1, 2,3 82. 83. 2 **(d)** 4

82. [CDS (I) Paper 22 Q2] Area of AQAB is equal to twice that of APCQ Select the correct answer using the code given below : **(a)** 1 only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither 1 nor 2

83. [CDS (I) Paper 22 Q86] What is the area of the largest square plate cut from a circular disk Of radius one unit ? **(a)** 4 square units **(b)** 245 square units **(c)** r square units **(d)** 2 square units

84. [CDS (I) Paper 22 Q92] 66 units 60 units In the given figure, what is the area of the shaded region ? **(a) (b) (c) (d)** — 6) sq units 3(4z —30) sq units sq units —z) sq units -31 -31

85. [CDS (I) Paper 22 Q99] 6 cm What is the approximate area of the shaded region in the figure given ? 35 **(a)** 15•3 cm2 **(b)** 25•5 cm2 **(c)** 284 cm2 **(d)** cm2

86. [CDS (I) Paper 22 Q3] AB+ BC+CD+DA> AC+BD Which of the above statements are correct ? **(a)** 1 and 2 only **(b)** 2 and 3 only **(c)** 1 and 3 only **(d)** 1, 2 and 3 A - PIXI-T-MTK In the given figure, there are three semi circles ABC, AEF and CDE The dis- tance between and C is 28 units and F is the mid-point of AC. What is the total area of the three semi circles ? **(a)** 924 square units **(b)** 824 square units **(c)** 624 square units **(d)** 462 square units

87. [CDS (I) Paper 23 Q78] 74. In a triangle, value Of all the angles are integers (in degree measure). Which one Of the following cannot the proportion of their **(a) (b)** The length of a rectangle is by and breadth is decreased by 10%. Then the area Of the new rectangle is **(a) (b)** (e) em2 186 cm2 150 cm2 180 cm2 **(a) (b) (c)** neither nor by decreased by I % by IOS diagonals Of a rhombus differ by 2 units and its perimeter exceeds the Sum Of the diagonals by 6 units. What is the of the rhombus ? **(b) (d)** 48 units 36 square units 24 units 12 square units What is the area of a right-angled triangle, if the radius of the circumcircle is 5 em and altitude drawn to the hypotenuæ is 4 cm ? The Surfree areas of two spheres are in the ratio I :

88. [CDS (I) Paper 23 Q2] AB.2PQ Select the correct answer using the code given **(a) (b) (c) (d)** only 2 only Both I and 2 Neither I nor 2 Four circular coins of equal radius are placed their centres coinciding with frur vertices of a square. Each coin touches two other coins. If the uncovered area of the square is 42 em 2, then what is the t•adius of each coin ? (e) **(d)** 7 cm IO cm 14 em DZOL-T-LKM be

89. [CDS (I) Paper 24 Q73] A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55 cm, then What is the area of the sector? **(a)** 577•5 cm2 **(b)** 612•5 cm 2 **(c)** 705•5 cm 2 **(d)** 725•5 cm 2

90. [CDS (I) Paper 25 Q30] A persm wishes to fence 375 m2 rectangular garden. He has 65 m of barbed wire and is able to fence only three sides of the garden. What is the perimeter of the garden ? **(a)** 80 m **(b)** 84 m **(c)** 90 m **(d)** 100m

91. [CDS (I) Paper 25 Q63] Let p be the area of a square X and q be the area of the square formed on the diagonal of the square X. What is the value of — ? **(a)** sin 9 **(b)** cose **(c)** tan 0 **(d)** cot O 59. If A, B and C are interior angles of a triangle ABC, then what is equal to ? 2 A+B+C

92. [CDS (I) Paper 25 Q64] The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? **(a)** 200 cm **(b)** 120 cm **(c)** IOOcm **(d)** 90 cm A - NYRG-T-MTK 2 sin39— sino

93. [CDS (I) Paper 26 Q18] What is the least value of n if 194480+nzm4, where m and n are natural numbers? If then What is the value Of ? What is the area Of the region enclosed by three identical circles (each of radius 4 cm) touching each other? **(a)** — square cm **(b)** 16vTä— — square cm **(c)** (160 — 8r) square cm **(d)** — square cm

94. [CDS (I) Paper 26 Q7] If the volume Of the cone is 528 cm3 , then What is the height of the cone? (Take —J **(c)** 10•5 cm **(d)** 140 cm A cylindrical pipe has inner diameter Of 14 cm. Water flows through it at a rate of 154 litres per minute. What is the speed Of water in km/hr? wake 2?) A spherical ball Of lead 6 cm in diameter is melted and recast into three spherical balls. The diameters of the balls are 2 cm, 4 cm and x cm. Which one of the following is correct? **(b)** 52 cm < x cm < 54 cm **(c)** cm < x cm < 56 cm **(d)** 56 cm < x cm < cm A rectangle of length IO units and breadth 8 units is split into two squares each of area x square units and two rectangles each of area y units. Consider the following statements : I. y is alway s greater than x. 2. y can be 15 square units. Which Of the above statements is/ are **(a)** I only **(b)** 2 only Both I and 2 **(d)** Neither I nor 2 GHSA-T-MTK/45A 19 19

95. [CDS (I) Paper 26 Q68] A square sheet is formed by joining n identical Square sheets of same size. If the length of the diagonal of the bigger square sheet formed is m, then What is the side length Of a smaller square sheet? **(c) (a)** 100 The radius and height Of a right circular cone are in the ratio 3 7. If the volume Of the cone is 528 cm3 , then What is the height of the cone? **(b)** 10•5 cm **(c)** 140 cm A square sheet is forrned by joining n identical Square sheets of same size. If the length of the diagonal of the bigger square sheet formed is m, then What is the side length Of a smaller square sheet? **(d)** A cylindrical pipe has inner diameter Of 14 cm. Water flows through it at a rate of 154 litres per minute. What is the speed Of water in km/hr? wake 2?

96. [CDS (I) Paper 26 Q73] ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB— DC 4 cm, then what is the value of ABX DC? **(a)** 84 square cm **(b)** 96 square cm **(c)** 100 square cm **(d)** 108 square cm

97. [CDS (I) Paper 26 Q75] If the perimeter of an isosceles triangle is 4 (2 cm, then What is its area in square cm? **(b)** 12 (d' 24 **(a)** 12 **(b)** 943 cm **(c)** 18 cm **(d)** 27 cm 72. A farmland is in the shape Of a rhombus. The perimeter of the land is 100 m and the length of one of the diagonals is 40 m. The land is divided into four equal parts. What is the area of each 75. 76. 77. 21 If the perimeter of an isosceles right triangle is 4

98. [CDS (I) Paper 26 Q76] 77. 21 The 8 16 diagonal of a square is 120 cm and the area of an equilateral triangle is 64" square cm. Which of the following • statements is/ are correct? **(a)** 12 **(b)** 943 cm **(c)** 18 cm **(d)** 27 cm 72. A farmland is in the shape Of a rhombus. The perimeter of the land is 100 m and the length of one of the diagonals is 40 m. The land is divided into four equal parts. What is the area of each 75. 76. 77. 21 If the perimeter of an isosceles right triangle is 4

99. [CDS (I) Paper 26 Q77] 21 If the perimeter of an isosceles right triangle is 4 (2 cm, then What is its area in square cm? The 8 16 24 diagonal of a square is 120 cm and the area of an equilateral triangle is 64" Square Cm. Which Of the following • statements is/ are correct? part? **(a)** 150 **(b)** 225 **(c)** 300 **(d)** 450

100. [CDS (I) Paper 27 Q87] A square copper plate of side 16 cm weighs 128 gm. A circular disc of diameter 14 cm is cut off from the plate. What is the weight Of 22 the remaining part ? (z z A rectangular metal sheet is of length 24 cm and breadth 18 em. From each Of its corners a square of side x cm is cut off and an open box is made of the remaining sheet. If the volume of the box is 640 cubic em, then what is the value Of x ? **(a) (b) (c) (d)** 48 gm 49 gm 50 gm 51 gm **(c) (d)** 2 3 4 6 The area Of a rhombus is 96 square em and one of its diagonals is of length 12 cm. What is the perimeter of the rhombus ? Let x, y, z be the length, breadth, height of a cuboid. If its volume is 400 cubic cm and total surface area is 340 square em, then What is x-i + y-l + El equal to ? 17 **(a)** 20 40 10 **(d)** FDGT-T-EMT **(a) (b) (c) (d)** 36 cm 40 cm 44 cm 48 cm A right-angled triangle ABC is inscribed in a circle of radius IO cm. The altitude drawn to the hypotenuse AC is of length 8 cm. If AB x cm and BC y cm, then what is the value of xy ? **(a) (b)** 60 80 160

101. [CDS (I) Paper 27 Q89] 90. (31 -A) **(a) (b) (c) (d)** 2-5 em 36 cm A and a rectangle have same perimeter. They differ in areas by I cm. The length Of the rectangle exceeds its breadth by **(a) (b) (c) (d)** 2 cm 4 cm

102. [CDS (I) Paper 27 Q93] A closed box is built of wood Of uniform thickness. Its external dimensions are 12 em, IO cm and 8 cm. If the inner surface area is 376 square em, then what is the thickness of the wood ? **(a) (b)** cm X, Y and Z are three equilateral triangles. The sum Of the areas Of X and Y is equal to the area Of Z. If the side lengths of X and Y are 6 cm and 8 cm respectively, then What is the side length Of Z ? **(a) (b) (c) (d)** Cm IO cm 10•5 cm What is the time taken by a person to cover one round of a circular park of diameter 210 m 22 if he walks at a speed of 6 km,'hr ? (x = **(a) (b) (c) (d)** minutes 5' 5 minutes 4 •4 minutes 3 •3 minutes Two rectangles are Of same area equal to

103. **[CDS (I) Paper 28 Q12]** Ifthe sum Of the squares Of the two numbers is 4896, then what is the mean Of the two ? **(a) (b) (c) (d)** 24 36 48 96

104. **[CDS (I) Paper 28 Q62]** 2h z 3r 63. **(a) (b) (c)** 48 square cm 50 square cm 52 square cm 54 square em TWO times the total surface area of a solid right circular cylinder is three times its curved surface area. If h is the height and r is the radius Of the base Of the cylinder, then which one of the following is correct ? **(a) (b) (d)** A floor Of a big hall has dimensions 30 m 60 cm and 23 m 40 em. It is to be paved With square tiles of same size. What is the minimum number of tiles required ? **(a) (c)** 30 36 169 221

105. **[CDS (I) Paper 28 Q66]** 67. How long will a man take to walk around the kxnmdary of a square field of area 25 hectares at the rate Of 5 km./hr ? **(a) (b) (c) (d)** 36 minutes 30 minutes 24 minutes 18 minutes Let x be the area of a square inscribed in a circle of radius r and y be the area Of an equilateral triangle inscribed in the same circle. Which one of the following is correct ? **(a)** 9x2 16y2 **(b)** 64y2 **(c)** 36x2 **(d)** 16x2 z 21y2 If the length of a rectangle is increased by 66 — %, then by what percent should the width of the rectangle be decreased in order to maintain the same area ? **(a) (b) (c) (d)** 45% 35% What is the maximum area that can be covered by three non-intersecting circles drawn inside a rectangle of sides 8 em and 12 cm ? **(a) (b) (c) (d)** 167t square cm 187t Square cm 207t square cm 247t square cm JNBY-D-MTK (23- le

106. **[CDS (I) Paper 28 Q68]** 69. 70. a-A) ABCD is a square field With AB = x. A vertical pole OP of height 2x stands at the centre O Of the square field. If Z APO = B, then what is cot O equal to ? **(b) (c) (d)** A solid iron ball is melted and 64 smaller solid balls Of equal size are made using the entire volume Of iron. What is the ratio of the surface area Of the larger ball to the sum of the surface areas Of all the smaller balls ? **(a) (c)** (Y25 075 A triangle ABC has been divided into four smaller triangles p, Q, R, S whose perimeters are 16 cm, 12 cm, 4 cm and 12 em respectively. P, R and S contain the vertices A, B and C respectively. What is the perimeter Of the triangle ABC ? **(a) (b) (c) (d)** 18 cm 20 cm 22 cm 24 cm

107. **[CDS (I) Paper 28 Q64]** How long will a man take to walk around the 68. kxnmdary of a square field of area 25 hectares at the rate Of 5 km./hr ? **(a) (b) (c) (d)** 36 minutes 30 minutes 24 minutes 18 minutes Let x be the area of a square inscribed in a circle of radius r and y be the area Of an equilateral triangle inscribed in the same circle. Which one of the following is correct ? 69.

108. **[CDS (I) Paper 28 Q65]** 66. 67. (23-A) **(a) (b) (c) (d)** 36 minutes 30 minutes 24 minutes 18 minutes Let x be the area of a square inscribed in a circle of radius r and y be the area Of an equilateral triangle inscribed in the same circle. Which one of the following is correct ? 69. **(a)** 9x2 16y2 **(b)** 64y2 **(c)** 36x2 **(d)** 16x2 z 21y2 If the length of a rectangle is increased by 66 — %, then by what percent should the width of the rectangle be decreased in order to maintain the same area ? ABCD is a square field With AB = x. A vertical pole OP of height 2x stands at the centre O Of the square field. If Z APO = B, then what is cot O equal to ? **(b) (c) (d)** A solid iron ball is melted and 64 smaller solid balls Of equal size are made using the entire volume Of iron. What is the ratio of the surface area Of the larger ball to the sum of the surface areas Of all the smaller balls ? **(a) (c)** (Y25 075 **(a) (b) (c) (d)** 45% 35%

109. **[CDS (I) Paper 28 Q70]** What is the maximum area that covered by three non-intersecting drawn inside a rectangle Of sides 8 12 cm ? can be circles em and A triangle ABC has been divided into four smaller triangles p, Q, R, S whose perimeters are 16 cm, 12 cm, 4 cm and 12 em respectively. P, R and S contain the vertices A, B and C respectively. What is the perimeter Of the triangle ABC ? **(a) (b) (c) (d)** 167t square cm 187t Square cm 207t square cm 247t square cm **(a) (b) (c) (d)** 18 cm 20 cm 22 cm 24 cm JNBY-D-MTK

110. **[CDS (I) Paper 28 Q89]** 90. The volume Of a cuboid is 3600 cubic cm. The areas of two adjacent faces are tha 225 square cm and 144 square em. What is the area Of the Other adjacent face ? **(a) (b) (c) (d)** 400 square Cm 360 41are cm 320 square cm 300 cm The perimeter and the area Of a right-angled triangle are 36 cm and 54 square cm respectively. What is the length Of the hypotenuse ? **(a) (b) (c) (d)** Let 12 cm 14 em 15 cm 16 cm X = (x I, 2, 3, ...241. Let S be a subset of X such that the sum Of no two elements of S is

111. **[CDS (I) Paper 28 Q100]** What is the maximum possible number of elements in S ? **(a)** 10 **(b)** 11 12 **(d)** 13 The perimeter of a sector of a circle Of radius 52 cm is 164 em. What is the area Of the sector ? **(a) (b) (c) (d)** 15•6 square em 15 square cm 144 square cm 14•1 square cm JNBY-D-MTK (31 —A . Consider the following for the next three (03) items that follow : In the triangle ABC, AB = 6 em, BC 8 em and AC z IO cm. The perpendicular dropped from B meets the side AC at D. A circle Of radius BD (with centre B) cuts AB and BC at p and Q respe-ctively as shcmrn in the figure.

112. **[CDS (I) Paper 28 Q92]** 52 cm is 164 em. What is the area Of the sector ? What is the length Of QC ? **(a) (b) (c) (d)** 3•6 cm 3•2 cm **(a) (b) (c) (d)** 15•6 square em 15 square cm 144 square cm 14•1 square cm If Z ABD = e, then What is sin equal to ? **(a) (b) (c) (d)** JNBY-D-MTK

113. **[CDS (I) Paper 28 Q95]** 96. What is the area Of the smaller Square ? **(a) (b) (c) (d)** 50 (3 —) square cm 25 (3 — 2) square cm 25 (3 + 2) square cm 50 (3 — 2) square cm What is the area of the shaded region ? **(b) (c) (d)** (96 — 25%) square cm (92 — 251) square cm (96 — 16%) square cm (92 — 16%) square em Consider the following for the next two (02) items that follow : In the following figure, a rectangle ABCD is inscribed in a circle Of radius r. Given ZDAE and Z ACD

114. [CDS Paper 30 Q23] 24. then What is pa + 3bp equal to ? **(a) (c) (d)** 2a A plank Of wood 4•25 m long and 34 m wide is to be cut into square pieces of equal size. How many square Of largest size can be cut from the plank, if no wastage is allowed ? **(a) (b) (c) (d)** 45 90 400 500 What is the HCF Of x' — 13x2y2 **(a) (b) (c)** — 4x2y — 4xy2 — 5y3 and — 125ya ? x2 + 5xy • 25y2 - 300y4,

115. [CDS Paper 30 Q36] 37. then what is the value Of a x b x c x d x e ? **(a) (b) (c)** 720 480 360 60 A cube whose edge is 14 cm long has on each of its faces a circle of 7 cm radius painted yellow. What is the total area of unpainted surface ? (Take = —) **(a) (b) (c) (d)** 126 square em 189 square cm 252 square cm 315 square em SRSU-T-EMT (130

116. [CDS Paper 30 Q42] 43. (15—A) A sum of money invest«l at simple interest triples itself in 8 years and becomes n times in 20 years. What is the value Of n ? **(c)** 5 6 75 9 **(a) (b) (c) (d)** 40 square em 42 square cm 45 square cm 48 square cm

117. [CDS Paper 30 Q49] 50. (17-A) **(a) (b) (d)** 8 x < 12 12 < x < 16 16 < x < 18 18 x < 20 In a quarter circle of radius R, a circle of radius r is inscribed. What is the ratio Of R **(a) (c) (d)** In a quadrilateral ABCD, AB = BC and CD = DA; AC and BD are diagonals such that AC = 6 em and BD 12 cm. What is the area of the quadrilateral ? 53. **(a) (b) (c) (d)** 24 square cm 30 square cm 36 square cm 40 square cm SRSU-T-EMT

118. [CDS Paper 30 Q72] What are the values of r and r is an integer. R is an integer. Con, that p, Q, R, S are the mid-lN)ints of sides AB, BC, CD, DA respectively of a quadrilateral ABCD. Question : Statement-I : Statement-II : SRSU-T-EMT What is the difference in the 4 a dist area Of the quadrilateral ABCD and the area Of the quadrilateral PQRS ? Area Of the quadrilateral ABCD is 100 square unit. Area Of the quadrilateral PQRS is 50 unit. 73. r the following for the next two (02) items hat follow : [n a pie-diagram (With radius 7 em), the central of the sectors are in the ratio 2 : 3 : 7 : 5 : l. Take a If P is the area of the smallest and Q is the area of the largest sector, then what is p + Q equal to ? **(a) (b) (c)** — square cm 3 77 — square em square em 6 616 72. square cm 9 If p is the perimeter of the smallest sectnr, then what is the value of 9p ? **(a)** 142 cm **(b)** 148 cm **(c)** 156 cm **(d)** 221 cm Consider the following for the next three (03) items that follow : Two trains A and B started from stations P and Q towards each other. Train A started at 7 p.m. at a speed Of 60 km."hr and train B started at 4 a.m. (next day) at a speed of 90 km/hr. The distance between the two stations P and Q is 800 km.

119. [CDS Paper 30 Q76] 77. 78. -A) What is the area Of the square ? **(a) (b) (c) (d)** 5:28 a.m. 5:44 a.m. 4:56 a.m. 6:24 a.m. If the lengths Of the two trains A and B are 400 m and 500 m respectively, then what is the time taken by them to cross each other ?

120. [CDS Paper 30 Q79] 80. What is the area Of the shaded region ? **(a) (b) (c)** 38•5 41are cm 48 square em 52' 5 square cm 666 square cm 81. 82. **(d)**

121. [CDS Paper 30 Q85] What is the difference between number of people mming from R having age group A and those coming from Q having age group D ? **(b) (c) (d)** 6,000 8,000 12,000 18,000 Consider the following for the next two (02) items that : Consider two identical rectangles ABCD and BEDF as shown in the figure given below. Let AB = I cm and BC 2 em. c **(a)** 6,000 **(b)** 8,000 **(c)** 12,000 **(d)** 18,000 these migrating people for each State. s Consider the following for the next two

122. [CDS Paper 30 Q86] 87. -A) What is the area of the overlapping region ? **(a) (b) (c) (d)** — square cm — square cm — square cm — square cm What is the area Of the non-overlapping region ? **(a) (c) (d)** — square cm 11 — square cm 4 — square em — square cm Consider the following for the next three (03) items 85• that follow : The following Pie-Chart-l shows the people migrating to Delhi from different Indian States (P, Q and R are three different States and S is the combined group of other States) and Pie-Chart•ll indicates the different age groups A, B, C and D of (29-A) What is the difference between number of people coming from R having age group A and those coming from Q having age group D ? **(a)** 6,000 **(b)** 8,000 **(c)** 12,000 **(d)** 18,000 these migrating people for each State. s Consider the following for the next two (02) items that : Consider two identical rectangles ABCD and BEDF as shown in the figure given belmv. Let AB = I cm and BC 2 em. 15% c 11 c What is the area of the overlapping region ?

123. [CDS Paper 30 Q83] 84. If the people coming from a particular State belonging to S are and in number, then What is the total number of migrating 86. people belonging to the age group B ? **(b) (d)** 1-2 lac 125 lac 1-30 lac 1-50 lae 87. **(a) (b) (c) (d)** — square cm — square cm — square cm — square cm What is the maximum Of differences between the number Of people coming from different groups P, Q, R and S ? What is the area Of the non-overlapping region ? **(a) (b) (c) (d)** I •6 lac lac 2•4 lac 2•6 lac **(a) (c) (d)** — square cm 11 — square cm 4 — square em — square cm SRSU-T.EMT

124. [CDS Paper 30 Q88] 89. 90. **(a) (b) (c)** IO cm 12 cm 13 cm 14 cm What is the radius of the circle with centre at 01 ? **(d)** 3-6 What is the radius of the circle with centre at 02 ? **(b)** 1+2-6 22 - 4*fiö 9 22 - 2Jiö 9 SRSU-T-EMT Consider the following for the next three (03) items that follow : Consider two identical semicircles and one circle inscribed in a rectangle of length IO cm as shown in the figure given below. and Jä 0 What is the area Of triangle EOF ? **(a)** 12•5.Ä squ are cm **(b)** 6 •25 square cm **(c)** 12•5 square cm **(d)** 6•25 square cm What is the area of trapezium AEFB ? **(a)** 30 square cm **(b)** 25 square em **(c)** 20 square em **(d)** 18-75 square cm What is the area of the shaded region ? **(a) (c) (d)** 14•75 square cm 14•25 square cm 7 •225 square cm 7 •625 square cm Consider the following for the next three (03) items Consider the following for the next three (03) items that follow : that follow : ABC is a right-angled triangle with Z ABC

125. **[CDS Paper 30 Q90]** What is the radius of the circle with centre at 0 What is the area Of triangle EOF ? **(a)** 12•5.Ä squ are cm **(b)** 6 •25 square cm **(c)** 12•5 square cm **(d)** 6•25 square cm What is the area of trapezium AEFB ? **(a)** 30 square cm **(b)** 25 square em **(c)** 20 square em **(d)** 18-75 square cm What is the area of the shaded region ? 02 ? **(b)** 1+2-6 22 - 4*fiö 9 22 - 2Jiö 9 SRSU-T-EMT 93. (31-A) **(a) (c) (d)** 14•75 square cm 14•25 square cm 7 •225 square cm 7 •625 square cm

126. **[CDS Paper 30 Q97]** What is the area of the shaded region ? **(a) (b) (c)** 2 unit 4 unit 6 unit 8 unit **(d)**

127. **[CDS Paper 30 Q96]** 97. -A) What is the sum of the areas Of the two circles ? **(b) (d)** 171 square unit 16• 757 square unit 16-5K square unit 162511 square unit Which one Of the following is correct in respect Of angle O ? 450<0 < 600 600 < 9<900 Consider the following for the next two (02) items Consider the following for the next three (03) items (33 -A) that follow : that follow : Consider a circle Of area square unit and an TWO circles with centres at 01 and 02 touching each equilateral triangle ABC as shown in the figure Other are placed inside a rectangle Of sides 9 cm and given below. 8 cm as shown in the figure given below. 01 c c **(a) (b) (c)** 2 unit 4 unit 6 unit 8 unit **(d)**

128. **[CDS Paper 30 Q98]** What is the area of the shaded region ? (35-A) 240 240 120 240 — 1211 — n O square unit 41are unit square unit unit **(a) (b) (c)** 100. What is the perimeter of the shaded region ? **(d)**

129. **[CDS Paper 30 Q100]** What is the ratio Of the area Of the shaded region to that Of the non-shaded region ? **(a) (b) (c)** 100. What is the perimeter of the shaded region ? **(d)**

130. **[CDS Paper 30 Q99]** 240 240 120 240 — 1211 — n O square unit 41are unit square unit unit What is the ratio Of the area Of the shaded region to that Of the non-shaded region ? **(a) (b) (c)** 100. What is the perimeter of the shaded region ? **(d)**

131. **[CDS (I) Paper 31 Q48]** A square is drawn inside a square of side 14 cm in such a way that the corners of the inner square coincide with the mid points of the sides of the outer square. What is the area lying between the two squares ? **(a)** 98 square cm **(b)** 56 square cm **(c)** 49 square cm **(d)** 24•5 square cm

132. **[CDS (I) Paper 31 Q50]** What is the area of a triangle having sides 4, 4 and 6 units ? **(a)** 3-6 square unit **(b)** 8 square unit **(c)** 7 square unit **(d)** 70 square unit 17 Consider the following for the next three (03) items that follow : Let ABC be a triangle right-angled at B. Let p be the point on BC such that BP = PC. If 10 cm, ZBAP=450 and ZCAP=O tana + tanß use tan(a + D) = 1 — tana tan

133. **[CDS (I) Paper 31 Q60]** What is the area of triangle OMN? **(a)** 36 square cm **(b)** 40 square cm 21 21 **(c)** 45 square cm **(d)** 48 square cm

134. **[CDS (I) Paper 31 Q72]** What is the maximum area of a rectangle, in square cm, whose perimeter is 400 cm ? **(a)** 100 **(b)** 200 **(c)** 1000 **(d)** 10,000

135. **[CDS (I) Paper 31 Q82]** There are n concentric squares. The area of the innermost square is 1 unit and the distance between corres- ponding comers of any two consecu- tive squares is 1 unit. Consider the following statements : I. The diagonal of the nth square is **(a)** 81. Three amounts x, y, z are such that y is the compound interest on x; and z is the compound interest on y. The rate of interest per annum and the time period in years are same. Which one of the following is correct ? **(b)** Y2 = zx **(c)** z2 — xy 82. There are n concentric squares. The area of the innermost square is 1 unit and the distance between corres- ponding comers of any two consecu- tive squares is 1 unit. Consider the following statements : I. The diagonal of the nth square is 11. The area included between nth square and **(d)** I only

136. **[CDS (I) Paper 31 Q11]** The area included between nth square and (n — l)th square is independent of n 35 Which of the statements given above is/ are correct ? **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

137. **[CDS (I) Paper 31 Q83]** In a rectangle ABCD, AC is one of the diagonals. If AC+AB=3AD and AC— AD 4 units, then what is the area of the triangle ? **(a)** 24 square unit **(b)** 36 square unit **(c)** 48 square unit **(d)** 72 square unit

138. **[CDS (I) Paper 31 Q85]** In a triangle ABC, 21 cm, BC—20cm and CA = 13 cm. A per- pendicular CD is drawn upon the longest side. What is the area of the triangle BCD ? **(a)** 96 square cm **(b)** 84 square cm **(c)** 80 square cm **(d)** 72 square cm

139. **[CDS (I) Paper 31 Q92]** The section of a solid right circular cone by a plane containing vertex and perpendicular to base is an equilateral triangle of side 14 cm. What is the volume of the cone ? = 39 **(a)** 1078-6 cubic cm **(b)** 1078 cubic cm **(c)** 539,6 cubic cm 539 **(d)** cubic cm 93. •mree identical cones each with base radius 3 cm are placed on their bases so that each is touching the other two. There will be one and only circle that would pass through each of the vertices of the cones. What is the area of the circle ? **(a)** 3 It square cm **(b)** 67t square cm **(c)** square cm **(d)** 12r square cm

140. **[CDS (I) Paper 32 Q40]** Which one Of the following does not refer to the bow (and arrow) as a weapon of war in ancient India ? **(a) (b) (c) (d)** Sharnga Kunta Kodanda Karmuka What is the military-device referred to by the term manjaniq ? **(a) (b) (c) (d)** A device for loading field-guns With gunlx•wder Preparing cavalry for rapid charge A bolting device for strengthening the gates of forts A device for throwing stones (mangonel) In medieval India, the process Of indigo extraction underaent great imprcwement as a result Of the : **(a) (b) (c) (d)** use of fertilizer enhancing the quality of indigo crop. introduction of mechanical crushing tools for indigo stems. use of gypsum and lime mortars in making the inner surfaces Of indigo vats water-proo f. employment Of slave labour in quick desilting Of indigo crucibles. Which one of the following statements about

141. **[CDS (I) Paper 34 Q11]** Q can be a perfect square. Q is positive for all real values of x. 9 Select the answer using the code given below : **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11

142. **[CDS (I) Paper 34 Q62]** A square and a rectangle have the same perimeter p and their areas differ by q umts. What is the square of the difference between the length and breadth of the rectangle ? 25 25 **(c)** 2-5q **(d)** 4q **(a)** 120 **(b)** 180 **(c)** 240 **(d)** 300 61. The diameter of a copper solid sphere is 6 cm. The sphere is melted and recast into a wire. If the diameter of the wire is cm, then what is the length of the wire ?

143. **[CDS (I) Paper 34 Q66]** A metallic sheet of rectangular shape is 48 cm x 36 cm. From each corner a square of side x cm is cut off and an open box is made of the remaining sheet. If the box holds 5•12 litre of water, then what is the value of x ? **(a)** 6 cm **(b)** 7-5 cm **(c)** 8 cm **(d)** 8-5 cm 27

144. **[CDS (I) Paper 35 Q60]** On a circular metal plate Of uniform thickness, 16 holes, each Of diameter 2 em, are made. If the plate thereby has lost one-ninth Of its original weight, then what is the diameter of the plate ? **(a) (b) (c) (d)** 12 cm 15 cm 21 cm 24 cm A square Of side length 4 cm has its corners cut away in such a manner so as to form a regular octagon. What is the length of the side of the octagon ? (Take 1-41) **(a) (b)** (C) **(d)** 1•00 cm 1-64 em I •68 cm 1•72cm A tall cylindrical reservoir kept vertically is 20 m in diameter. Water is poured into it at the rate of 264 ma hour. What is the rate at which the water level rises in the reservoir per minute ? (Take **(a) (b) (c) (d)** cm cm 14 cm

145. **[CDS (I) Paper 35 Q69]** ABC is a triangle such that Z ABC = 1206. If BD is the bisector of Z B that meets AC at D, then what is the ratio Of the area Of A ABD to the area of CBD ? **(a)** AB:BC **(b)** AB2 Consider the following statements : 11. The areas Of two similar triangles are in the ratio Of the squares of the corresponding medians. The areas of two similar triangles are in the ratio of the squares Of the corresponding heights. **(c) (d)**

146. **[CDS (I) Paper 35 Q73]** 74. (27—A) How many square metres Of canvas (approximately) will be required to make a conical tent 3 m high so that a man 2 m tall may stand any-where within a radius of I m from its centre without stooping ? **(c) (d)** 36 40 m2 44 m 48 rn2 PQRS is a square. M is a point on PS such that PM : MS = 2 : I and N is a point on SR such that SN : NR I : **(a) (b)** 36 40 m2 44 m 48 rn2 **(c) (d)**

147. **[CDS (I) Paper 35 Q2]** If the area of triangle MQN is IO Square units, then what is the perimeter of the ? **(a) (b)** 36 40 m2 44 m 48 rn2 **(c) (d)**

148. **[CDS (I) Paper 35 Q77]** A square Of maximum area is inscribed in an equilateral triangle. If the side of the triangle is equal to (6 + 4) em, then what is the area of the square ? **(c) (d)** 36 40 m2 44 m 48 rn2 **(a) (b) (c)** 36 cm2 27 cm2 21 em 18 cm2 PQRS is a square. M is a point on PS such that PM : MS = 2 : I and N is a point on SR such that SN : NR I :

CHAPTER 12

Total Archive Questions: 156 Questions

MENSURATION 3D: SURFACE AREAS & VOLUMES

Typical Exam Weightage: 8 - 12 Questions (8 - 12 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q50] Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? **(a)** 300 cml **(b)** 350 cm2 **(c)** 375 cm2 **(d)** 400 cm2

2. [CDS (I) Paper 1 Q54] 55. What is the area of a right angled isosceles triangle whose hypotenuse is ? **(a)** 12 cml **(b)** 18 cm2 **(c)** 24 cm2 **(d)** 36 cml If A is the area of a triangle in cm2, whose sides are 9 cm, IO and 11 cm, then which one Of the following is correct ? **(a)** A < 40 cm2 **(b)** 40 cm2 < A < 45 cm2 **(c)** 45 cma < A < 50 cma **(d)** A > 50 cm2 A roller Of diameter 70 cm and length 2 m is rolling on the yound. What is the area covered by the roller in 50 revolutions ? **(a)** 180 m2 **(b)** 200 mi **(c)** 220 **(d)** 240 ma A cylindrical rod of length h is melted and cast into a cone of base radius twice that of the cylinder. What is the height Of the cone ? **(a)** 3h/4 **(b)** 411/3 **(c)** 2h

3. [CDS (I) Paper 1 Q57] 58. 59. it, 14 A lead pencil is in the shape Of a cylinder. The pencil is 21 cm long with radius 0-4 cm and its lead is of radius cm. What is the volume of wood in the pencil ? **(a)** 9 cml **(c)** cm3 **(d)** 10•1 cm' What is the angle (in radian) included between the hands of a Clock, when the time is IO minutes past 5. ? **(a) (b) (c) (d)** 17m'36 19Tt/36 51t/9 W12 Y In the figure given above LABD LPQD = LCDQ - 2 AB x, PQ z and CD = y, then which one of the following is correct ? **(a)** z x 2 z

4. [CDS (I) Paper 1 Q59] 14 A lead pencil is in the shape Of a cylinder. The pencil is 21 cm long with radius 0-4 cm and its lead is of radius cm. What is the volume of wood in the pencil ? **(a)** 9 cml **(c)** cm3 **(d)** 10•1 cm' What is the angle (in radian) included between the hands of a Clock, when the time is IO minutes past 5. ? **(a) (b) (c) (d)** 17m'36 19Tt/36 51t/9 W12 Y In the figure given above LABD LPQD = LCDQ - 2 AB x, PQ z and CD = y, then which one of the following is correct ?

5. [CDS (I) Paper 2 Q55] A bucket which is in the form Of a frustum of a cone, has radii 3 and 5 units and vertical • height 6 units. HOW much water Can the bucket hold ? 56. pa bc pb = Ca pc ab p2 = ab 57. ely) **(a) (b) (c) (d)** 60 68 73 75

6. [CDS (I) Paper 2 Q58] 14 -A) **(a) (b) (c) (d)** 337t cubic units 4511 Cubic units 48n cubic units None of the above ABC is triangle right-angled at C. If p is the length Of the perpendicular from C to AB and a, b, c are the sides, then which one Of the following is correct ? **(a) (b) (c) (d)** A person rides a bicycle round a circular path of radius 50 m. The radius of the wheel Of the bicycle is 50 cm. The cycle comes to the starting point for the first time in I hour. What is the number Of revolutions of the wheel in 15 minutes ? **(b) (c) (d)** 20 25 30 35 A cylindrical can Of internal diameter 24 cm contains water. A solid sphere Of radius 6 cm is completely immersed in water in the cylinder. The water level increases by **(a) (b) (d)** 0•25 cm 0-5 cm 52. pa bc pb = Ca pc ab p2 = ab

7. [CDS (I) Paper 2 Q57] 58. A bucket which is in the form Of a frustum of a cone, has radii 3 and 5 units and vertical • height 6 units. HOW much water Can the bucket hold ? **(a) (b) (c) (d)** 60 68 73 75 **(a) (b) (c) (d)** 337t cubic units 4511 Cubic units 48n cubic units None of the above PQRS is a diameter of a circle of radius 6 cm as shown in the figure above. The lengths PQ, QR and RS are equal. Semi-circles are drawn on PQ and QS as diameters. What is the perimeter of the shaded region ? ABC is triangle right-angled at C. If p is the length Of the perpendicular from C to AB and a, b, c are the sides, then which one Of the following is correct ? **(a) (b) (c) (d)** A person rides a bicycle round a circular path of radius 50 m. The radius of the wheel Of the bicycle is 50 cm. The cycle comes to the starting point for the first time in I hour. What is the number Of revolutions of the wheel in 15 minutes ? **(a) (b) (c) (d)** 127 147t 167t 1811 **(b) (c) (d)** 20 25 30 35 Hcrw many titres Of water (approximately) can a hemispherical container of radius 21 cm hold ? A cylindrical can Of internal diameter 24 cm contains water. A solid sphere Of radius 6 cm is completely immersed in water in the cylinder. The water level increases by **(c)** 194 388 194 388 **(a) (b) (d)** 0•25 cm 0-5 cm P-DETB.J.FN

8. [CDS (I) Paper 2 Q61] 62. If a sphere Of radius IO em is intersected by a plane at a distance 8 cm from its centre, what is the radius Of the curve of intersection Of the plane and the sphere ? **(a) (b) (c) (d)** From a cylindrical log whose height is equal to its diameter, the greatest possible sphere has been taken Out. What is the fraction of the original log which is eut away ? **(a) (b) (c) (d)** 1/2 2/3 . In the triangle ABC, AB 2 cm, BC = 3 cm and AC 4 cm. D is the middle point of AC. If a square is constructed on the side BD, what is thc area Of the square ? **(c) (d)** 4-5 Cm cm2 625 etn2 Nonc Of the above A solid cone Of height 8 Crn and base radius 6 cm is melted and recast into identical cones, each Of height' 2 cm and radius I cm. What is thc number Of cones formed ? **(a) (c) (d)** 36 72 144 180 P-DETB-J-FN t 16

9. [CDS (I) Paper 3 Q57] The volume of a cone is equal to that of sphere. If the diameter of base of cone is equal to the diameter of the sphere, what is the ratio of height of cone to the diameter of sphere ? **(a)** 1911-21 **(b)** 1921-31 **(c)** 1951-61 **(d)** 1961-71 55. The middle points of ihe parallel sides AB and CD of a parallelogram ABCD are p and Q respectiveiy. If AQ and CP divide the diagonal BD into three parts BX, XY and YD, then which one of the following is correct ?

10. **[CDS (I) Paper 3 Q1]** If the surface area of a cube is equal to the surface area of this parallelopiped; then what is the ratio of the volume of the cube to the volume of the parallelopiped ? **(a)** Rs. 500 @ 10%, Rs. 800 @ 12% **(b)** Rs. 700 @ 10%, Rs.600 @ **(c)** Rs. 800 @ 10%, Rs. 400 @ **(d)** @ 10%, ROOO @ 68. A man is watching from the top of a tower a boat sl*eding away from the tower. The boat makes an angle Of depression of 450 with the man's eye when at a distance of 60m from the bottom of tower. After 5 seconds, the angle of depression becomes 300. What is the approximate speed of the boat assuming that it is running in still water ? 70. The marks of the students of a class who appeared for a test in English are represented in the following frequency table : 11-20 21-30 31-40 41-50 51.60 What is/are the modal class

11. **[CDS (I) Paper 3 Q74]** What is volume of the frustum of a cone with height h and radii r, , ? **(a)** 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? **(b)** 9th **(c)** 15m **(d)** l •5m

12. **[CDS (I) Paper 3 Q75]** A cylinder having base of circumference 60 cm is rolling without sliding at a rate of 5 rounds per second. How much dis- tance will the cylinder roll in 5 seconds ? **(a)** 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ? **(b)** 9th **(c)** 15m **(d)** l •5m

13. **[CDS (I) Paper 3 Q79]** A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(l cm3 of the metal weighs 9 g) ? **(a)** 54 g **(b)** 366 g **(c)** 122 g **(d)** 1087t g

14. **[CDS (I) Paper 3 Q80]** A sphere and a cube have sarne surface area. What is the ratio of the square of volume of the sphere to the square of volume of the cube ? **(a)** 45 **(b)** 225 then what is the value Of tan 6 ? one **(c) (d)**

15. **[CDS (I) Paper 4 Q63]** What is the maximum length of a rod that Can be placed inside a box having the hape Of a cuboid Of length 30 cm, breadth 24 cm and height 18 em? **(a)** 30 cm **(c)** 24 Crn 1843 **(d)** Q-DETB-K-FN / 26Å 64. is ide 66. a? rod 67. of 20 The volume of a sphere is 8 times that of another sphere. What is the ratio of their surface areas? 6S. A rectangular area Of 6 square metres is to be painted on a 3m x 4 m board leaving a border Of uniform Width on all sides. What should be the width Of the border? **(a)** 0•25 m A solid cylinder of height g m has its curved surface area equal to one-third of the total surface area. What is the radius of the base? **(b)** 18m **(c)** 27 m **(d)** 30 m A figure ig formed by revolving a rectangular sheet Of dimensions 7 cm x 4 Crn about its length. What is the volume Of the figure thus formed? **(c)** 352 cm 296 cm 176 cm 616 Crn

16. **[CDS (I) Paper 4 Q67]** 20 The volume of a sphere is 8 times that of another sphere. What is the ratio of their surface areas? 6S. A rectangular area Of 6 square metres is to be painted on a 3m x 4 m board leaving a border Of uniform Width on all sides. What should be the width Of the border? **(a)** 0•25 m A solid cylinder of height g m has its curved surface area equal to one-third of the total surface area. What is the radius of the base? **(b)** 18m **(c)** 27 m **(d)** 30 m A figure ig formed by revolving a rectangular sheet Of dimensions 7 cm x 4 Crn about its length. What is the volume Of the figure thus formed? **(c)** 352 cm 296 cm 176 cm 616 Crn

17. **[CDS (I) Paper 4 Q68]** The diagonals Of the three faces Of a cuboid are x, y; z respectively. What is the volume ef the cuboid? e 22 **(b)** (y2.z2_x 2 _ 22) **(c) (d)** None of the above **(a) (b) (c)** None of the above 69. Half of a large cylindrical tank open at the top is fillpd with water and identical heavy *pherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius Of a ball, What is the rpaximum number of balls that can pe dropped? **(d)** 12

18. **[CDS (I) Paper 4 Q72]** If three cubic biscuits having edges m, m and 0-5 m respectively are melted and formed into a single cubic biscuit, then What is the total surface area Of the cubic biscuit? **(a)** rn2 **(b)** 1-56 rn2 **(c)** 1-84 m2 **(d)** 2•16 m 2

19. **[CDS (I) Paper 4 Q97]** What is the total surface area of, the given frustum? **(a)** + Crn 2 **(c)** 44) cm 2 **(d)** cm2 of For the next three (3) items which follow : Let ABCD be a quadrilateral. Let the diagonals AC and BD meet at O. Let the perpendicular drawn from A to CD meet CD at E. Further, .40: AB = 30 cm, CD 40 cm and the area Of the quadrilateral ABCD is 1050 2 of **(a)** 30 **(b)** 45

20. **[CDS (I) Paper 5 Q12]** What is the locus Of the centre of the front wheel Of the bicycle whose diameter is d ? The centroid Of a triangle ABC is 8 Cm from the vertex A. What is the length Of the median of the triangle through A ? **(a) (b) (c)** A line parallel to the path Of the wheel Of the bicycle at a height d cm A line parallel to the path Of the wheel Of the bicycle at a height d/2 cm A circle Of radius d/2 A circle of radius d cm 13. **(a) (b) (d)** The 20 cm 16 cm 12 cm IO Cm equation whose roots are twice the roots of the equation x2 — 2x + 4 O is **(a) (c)** x TO pass an examination, a candidate needs 40% marks. All questions carry equal marks. A candidate just passed by getting IO answers correct by attempting 15 Of the total questions. HOW many questiong are - 2x +4 20 -2X+16.o -4X+8=o 4X+ 16-0 there in the examination ? 14. of radius 15. circle Of The outer and inner diameters a circular pipe are 6 cm and 4 cm respectively. If its length is IO em, then what is the total surface area in square centimeters ? **(a) (b) (c)** 25 ao 40 45 **(a) (c) (d)** 55 150 None of the above Assume the Earth to be a sphere R. What is the radius Of the latitude 400 S ? A cylindrical rod Of iron whose radius is one-fourth of its height is melted and cast into spherical balls Of the same radius as that Of the cylinder. What is the number of spherical balls ? **(a) (b) (c) (d)** R cos 400 R sin 800 R sin 400 R tan 400 **(a) (b) (c) (d)** 2 3 4 P-PQFA.L.Rz

21. [CDS (I) Paper 5 Q70] 71. If x 27x - then what is x equgl to ? **(a) (b) (c) (d)** The 4 5 6 7 two diagonals Of a rhombus are of lengths 55 Cm and 48 cm. If p is the perpendicular height Of the rhombus, then which One of the following is correct ? **(a) (b) (c)** 36 cm < p < 37 cm 35 cm < p 36 cm 34 cm < p 35 cm 33 cm < p < 34 cm A toy is in the form Of a cone mounted on a hemisphere such that the diameter Of the base of the cone is equal to that of the hemisphere. If thc diameter of the base Of the cone is 6 cm and its height is 4 cm, what is the surface area Of the toy in cm ? (Take 3-14) **(b)** 93-62 103-62 113•62 115-50 perimeter of a triangular field is 240 m. If two of its sides are 78 m and 50 m, then what is the length of the perpendicular on the side Of length 50 m from the opposite vertex ? **(a)** 43 m 52•2 m 672 m 70 m P-PQFA.L-Rz 20 - A of

22. [CDS (I) Paper 5 Q91] The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? **(a)** 15 cm **(b)** 20 cm **(c)** 25 cm **(d)** 30 cm

23. [CDS (I) Paper 5 Q94] A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? **(a)** 22 m **(b)** 20 m **(c)** 18 m **(d)** None Of the above

24. [CDS (I) Paper 6 Q72] If the diameter of a sphere is doubled, how does its surface•area change ? **(a)** It increases two times. **(b)** It increases three times. **(c)** It increases four times, **(d)** It increases eight times.

25. [CDS (I) Paper 6 Q75] A sphere is inscribed in a cubical box such that the sphere is tangent to all six faces of the box. What is the ratio of the volume Of the cubical box to the volume of sphere ? **(b)** 3671 **(c)** 471/3 **(d)** 6/1t **(a)** 286 **(b)** 296 73. The areas of two circles are in the ratio I : 2. If the two circles are bent in the form Of squares, what is the ratio of their areas ? The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? **(c)** 3 units **(d)** 5 units

26. [CDS (I) Paper 6 Q80] What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? **(a)** 256 **(b)** 512 **(c)** 768 **(d)** 1024

27. [CDS (I) Paper 7 Q8] 9. IO. Which One among the following statements is not correct ? **(a)** For size ofreadymade garments, mode is me best suitable measure **(b)** For average rate of increase when the rate Of population gröwth is given, geometric mean is best suitable **(c)** For average rate of speed when different distances are covered by different rates ofspæd, harmonic mean is best suitable For average level of intelligence of students in a class, arithmetic mean is the best suitable Ifthe surface area of a sphere is 616 square cm, What is its volume ? **(a)** 4312/3 cubic Cm **(b)** 4102/3 cubic cm **(c)** 1257 cubic cm **(d)** 1023 cubic cm What are the dimensions (length, breadth and height respsctively) of a cuboid with Volume 720 cubic cm, surface area 484 square cm and the area Of the base 72 square cm ? **(a)** 9, 8 and O cm **(b)** 12,6 and I O Crn **(c)** 18, 4 and 10cm **(d)** 30,2and12cm

28. [CDS (I) Paper 7 Q2] Every whole number is a natural number Which Of the statements given above Ware correct ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 Consider thetfölllowing statements : If two straight lines intersect, then I. vertically opposite angles are equal. 2. vertically opposite angles are supplementary. 3. adjacent angles' are Complementary. Which of the statements given above is/are correct ? **(a)** 3 only **(b)** I only **(c)** land3 **(d)** 2and3 A large solid metallic cylinder whose radius and height are equal to each Other iöto be melted and 48 identical solid balls are to be recast from the liquid metal so formed. What is the ratio of the radius of a ball to the radius of the cylinder ? **(a)** 1:16 **(b)** 1:12

29. [CDS (I) Paper 7 Q24] 25. n-,e number 58129745812974 iS divisible **(a)** 11 **(d)** None Ofthe above What is the HCF of 36(3x• + 5x' - N), 9(6x' + -2x), 54(27x' -x) ? **(a)** 9x(x +) **(b)** 9x(3x- 1) **(c)** 18x(3x-1) **(d)** 18x(X+l) What is the length of the uniform wire of diameter 04 cm that can be drawn from a solid sphere of radius 9 cm ? **(a)** 243m **(b)** 240m **(c)** 60-75m **(d)** 60 m What will be the cost to plaster the inner surface of a well 14 m deep and 4 m in diameter at the rate of Rs. 25 per square meter ? **(a)** Rs. 4000 **(b)** Rs. 4200 **(c)** Rs. 4400

30. [CDS (I) Paper 7 Q46] A cardboard sheet in the form of a circular sector of radius 30 cm and central angle 1440 is folded to make a cone. What is the radius of the cone ? **(a)** 12 cm **(b)** 18 cm **(c)** 21 cm **(d)** None Of the above

31. [CDS (I) Paper 7 Q69] 70. What is the value of k that (2x'— 1) may be a factor — (k — l)x3 +•kx2— 6x + I ? **(c)** 12 **(d)** 13 If 196x4 = x6, then x3 is equal to which one of the following ? **(a)** x'/ 14 **(b)** 14xa **(c)** x2/14 sides of a parallelogram are 12 cm and 8 cm long and one of the diagonals is 10 cm long: If d is the length of other diagonal, then which one of the following is correct ? **(a)** d < 8 cm **(b)** 8 d < Ocm **(c)** 12 cm **(d)** 12 cm Ihe base o! a triangular wall is 7 times its height. If the cost of painting the wall at Rs. 350 per 100 square meter is Rs. 1225, then what is the base length ? **(a)** 50m **(b)** 70 m **(c)** 75 m **(d)** 100m 20 For the next TWO (02) questions that follow : The-areas of the ends of a frustum Of a pyramid are P and Q where p < Q and H is its thickness.

32. [CDS (I) Paper 7 Q73] 74. 20 What is the difference in radii of the ends of the frustum ? **(b) (d)** None of the above What is the volume ofthe frustum ? **(a)** 12 **(b)** 13 If 196x4 = x6, then x3 is equal to which one of the following ? **(c)** x'/ 14 **(d)** 14xa

33. [CDS (I) Paper 8 Q76] 77. 78. o In the figure given above, LOM is a straight line. What is x equal to ? **(a)** 45 **(b)** 60 **(c)** 70 **(d)** 80 What is the volume of the largest sphere that can be carved out of a cube of edge **(a)** 97t cubic cm **(b)** 67t cubic cm **(c)** 4.51t cubic cm **(d)** 37t cubic cm If the ratio of the diameters of two spheres is 3 : 5, then what is the ratio of their surface areas ? **(a)** 9:25 **(b)** 9: 10 **(d)** 27: 125

34. **[CDS (I) Paper 8 Q85]** 86. The volume Of a cube is numerically equal to sum of its edges. What is the total surface area in square units ? **(a)** 12 **(b)** 36 **(c)** 72 **(d)** 144 Two poles of heights 6 m and I I m stand on a plane ground. Ifthe distance between their feet is 12 m, what is the distance between their tops ? **(a)** 13m **(b)** 17m **(c)** 18m **(d)** 23 m A right circular metal cone (solid) is 8 cm high and the radius is 2 cm. It is melted and recast into a sphere. What is the radius Of the sphere ? **(a)** 2 cm **(d)** 5 cm The Sum ofthe perpendiculars drawn from an interior point Of an equilateral triangle is 20 cm. What is the length of side of the triangle ? **(a) (b) (c) (d)** 40 206 cm 20

35. **[CDS (I) Paper 8 Q89]** 90. Let the largest possible right circular cone and largest possible sphere be fitted into two cubes of same length. Let C and S denote the volume of cone and volume of sphere respectively, then which one of the following is correct ? The radius of the base of a right circular cone is increased by 15% keeping the height fixed. The volume of the cone will be increased by **(c)** 32.25% **(d)** 34.75% If the circumference Of a circle is equal to the perimeter of square, then which one of the following is correct ? **(a)** Area of circle = Area Of square **(b)** Area of circle 2 Area of square **(c)** Area of circle > Area of square **(d)** Area Of circle < Area Of square 10 circular plates each of thickness 3 cm each are placed one above the other and a hemisphere of radius 6 cm is placed on the top just to cover the cylindrical solid. What is the volume of the solid so formed ? **(a)** 2647t cubic cm **(b)** 2527t cubic cm **(c)** 23611 cubic cm **(d)** None of the above 9 26 O

36. **[CDS (I) Paper 9 Q64]** 65. A solid spherical ball Of iron Of radius 4 i melted to form spheres OT radius 2 cm each The number Of spheres 30 formed **(d)** 16 From a mlid wooden right eirculer cylinder. right circular cone whose radius and heig are same as the radius and height Of cylinder respectively, is carved Out. What i the ratio of the volume of the utilized wood that the wut-ed wasd ? The arc AB of the circle with centre at 'O' and 61 radius IO em has lengi.h 16 em. What is the area of the sector bounded by the radii OA, 0B and the ? **(a)** 40Æ Square em **(b)** 40 square cm (e) SO square cm **(d)** 20a square cm height Of cylinder ig 15 cm. The lateral are. is 660 square em. Its volume is **(a)** 1156 cubic cm 1215 cubic cm 1230 cubic em **(d)** 2310 cubic cm A bucket is Of height 25 em. Its top and bottom redii are 20 em and IO cm respectively. Its capaåty in litres is **(a)** 1757,'3 17-5m **(d)** 25't O-DETA.N.FMN t 18 -A 68. volume Of a right circular Of height 3 em and slant height 5 cm ig 49•3 cm 508 cubic 52 cubie 5S cubic cm A cylindrical tube at both ends is mA-de Of metal. The internal diemeter Of tube ig 6 cm and length of the tube ig IO cm. If the thickness of the metal used is 1 cm. then the outer curved surßee area the is **(a)** 140n equ ere em **(b)** square cm **(c)** 70r •quare cm **(d)** None Of the above The ratio Of area to diamet« a whose volume is etA'ic cm is **(a) (d)** 6 None of Lhe volume Of the material Of a Bheil With 808 em **(b)** eubie (C) 816 cubic **(d)** 824 cubic cm The breadth Of a are resrctively 14 m, 12 m and 13 m. Itie length of the greateøt rod that can be put in it is 22-31 m 22-56 m 20 m 19-5 m

37. **[CDS (I) Paper 9 Q68]** The volume of a right circular cone of height 3 and slant height 5 cm ig **(a)** 49•3 cubic em **(b)** 508 cubi c cm **(c)** 52 cubic cm **(d)** 5S cubic gm A cylindrical tube Opon at both ends is made Of metal. The internal di9meter Of the tube is 6 Crn ana length of the tube is IO cm. If the thickness OF the metal used is 1 cm, then the Outer curved surface area Of thc tube is **(a)** 140n squere em **(b)** 1465% square cm **(c)** 70n square em **(d)** None Of the above The ratio Of surfiee area to diameter Of a spheav Whose volume is 361 cubie em is **(a)** Srt **(d)** None Of thc above volume of the material Of a hemispherieal shell with outer and radii era and cm respectively ig approximately 808 cubic em **(b)** 800 eubic cm (C) 816 cubic gm **(d)** 824 cubic cm The length, breadth and height Of a are respectively 14 m, 12 m and 13 m. The length of the greatest rod that be put in it is **(d)** 22-56 m 20 m 19-5 m **(d)** 25't O-DETA.N.FMN t 18 -A)

38. **[CDS (I) Paper 10 Q47]** Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for **(a)** A only **(b)** B only **(c)** both A and B **(d)** neither A nor B

39. **[CDS (I) Paper 10 Q58]** A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the cylinder is 24 m. The height of the cylindrical portion is 1 m, while the vertex of the cone is 16 m above the ground. What is the area of the curved surface for conical portion? **(a) (b)** 13 cm2 15 cm 15 Crn < x < 17 17 cm2 < x < 19 cm 19 cm < x < 2 lcm 58. A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the cylinder is 24 m. The height of the cylindrical portion is 1 m, while the vertex of the cone is 16 m above the ground _ What is the area of the curved surface for conical portion? **(c) (d)**

40. **[CDS (I) Paper 10 Q61]** 62. 63 64. -The area of an isosceles triangle ABC with AB = AC and altitude AD = 3 cm is 12 square cm- What is its perimeter? **(a)** 18 cm **(b)** 16 cm **(c)** 14 cm **(d)** 12 cm If the total surface area Of a cube is 6 square unit, then what is the volume of the cube? **(a) (b) (d)** . The 1 cubic unit 2 cubic unit 4 cubic unit 6 cubic unit diameter of the moon is approximately one-fourth of the diameter of the earth. What is thc ratio (approximate) of their volumes? **(a)** 1:16 **(b)** 1: 64 **(d)** 1: 128 A hospital room is to accommodate 56 patients. It should be done in such a way that every patient gets 20 m 2 of floor and m 3 of space. If the length of the room is 14 m, the breadth and the height of the room are respectively **(a) (d)** 84 m, m B-BCRB-M-NBU / 48A

41. **[CDS (I) Paper 11 Q40]** 10, 41. (IO-A) Out of 105 students taking an examination for English and Mathematics, 80 students pass in English, 75 students pass in Mathematics.and IO students fail in both the subjects. How many students pass in only one subject ? **(a) (b) (c) (d)** 26 30 35 45 At What point of time after g o'clock, do the hour-hand and the minute-hand of a clcrk occur at right angles for the first time ? **(a) (b) (c) (d)** 9 0'élock 4 hours 37 — minutes 3 hours 30 — minutes 11 9 hours 32 — minutes If 64 identical small spheres are made Out of a big sphere Of diameter 8 em, what is the surface area Of each small sphere ? **(b)** (C) 8Æ Cm 3.ABQA-N-N8U

42. **[CDS (I) Paper 11 Q48]** 49. 12 -A) How many circular of diameter d Can be taken out of a square plate of side 2d With minimum loss of material ? **(a)** 8 6 4 2 The volume of a hollow cube is 216x3. The surface area of the largest sphere which can be enclosed in it is **(a) (b) (c)** 18xx2 271tx2 72Jtx2 What is the diameter of the largest circle lying on the surface of a sphere of surface area 616 square cm ? **(a) (d)** 14 cm 35 cm What is the total area of three equilateral triangles inscribed in a semicircle of radius 2 em ? **(a) (c)** 12 cm 4

43. **[CDS (I) Paper 11 Q67]** 68. The area of a sector Of a circle Of radius 36 cm is 72K cm2_ The length of the corresponding arc of the sector is **(a) (c) (d)** 311 cm 41 cm A cone of radius r cm and height h cm is divided into two parts by drawing a plane through the middle point of its height and parallel to the base. What is the ratio of the volume Of the original cone to the volurne of 7' the smaller cone ? **(b) (c) (d)** Which one of the following is a Pythagorean triple in Which one Side differs from the hypotenuse by two units ? **(a)** (2n I, 4n, 2n2 + 2n) **(b)** (2n, 4n, 4 1) **(c)** (2n2, 2n, 2n * I) **(d)** (2n, n 2—1, n + I) where n is a positive real number. A cube has each edge 2 cm and a cuboid is I Cm long, 2 em wide and 3 cm high. The paint in a certain container is sufficient to paint an area equal to 54 cm . Which one of the following is correct ? **(a) (c) (d)** Both cube and cuboid can be painted Only cube can be painted Only cuboid can be painted Neither cube nor cuboid can be painted 34BQA-N-NBU (18-A m 69. of 70,

44. **[CDS (I) Paper 11 Q71]** 72. ABC is a triangle right-angled at A where AB = 6 cm and AC 8 cm.' Semicircles are drawn (Outside the triangle) On AB, AC and BC as diameters which enclose areas x, y and z square units respectively. What is x + y — z equal to ? **(a) (b)** 48 cm 32 em2 None of the above The three sides Of a triangle are 15, 25 and x units. Which One of the following is correct ? **(a) (b) (c)** 40 IOS xc40 IO S x < 40 10 < «40 TWO light rods AB = a b, CD = a — b are symmetrically lying on a horizontal plane. There are •kept intact by two strings AC and BD. The perpendicular distance between the rods is length of AC is given by **(b) (d)** Let PQRS be a rectangle such that PQ v'5QR.Whatis z PRSequa1to?- 450 300 **(d)** 150 A cone of radius r cm and height h cm is divided into two parts by drawing a plane through the middle point of its height and parallel to the base. What is the ratio of the volume Of the original cone to the volurne of the smaller cone ? **(b) (c) (d)** Which one of the following is a Pythagorean triple in Which one Side differs from the hypotenuse by two units ? **(a)** (2n I, 4n, 2n2 + 2n) **(b)** (2n, 4n, 4 1) **(c)** (2n2, 2n, 2n * I) **(d)** (2n, n 2—1, n + I) where n is a positive real number. A cube has each edge 2 cm and a cuboid is I Cm long, 2 em wide and 3 cm high. The paint in a certain container is sufficient to paint an area equal to 54 cm . Which one of the following is correct ? **(a)** Both cube and cuboid c an be painted **(b)** Only cube can be painted **(c)** Only cuboid can be painte d **(d)** Neither cu be nor cuboid can be painted 34BQA-N-NBU

45. **[CDS (I) Paper 11 Q75]** The area of a rectangle lies E*twæn 40 cm and 45 cm . If one of the sides is 5 em, then its diagonal lies between **(a) (c) (d)** 8 cm and IO cm 9 and 11 cm IO cm and 12 em ll.cm and 13 cm A cylinder is surmounted by a cone at One end, a hemisphere at the Other end. The radius is cm, the height of the cylinder Lis 6-5 cm and the total height or the structure is 12•8 cm. The volume V Of the structure lies between **(b) (c) (d)** 370 cm3 and 380 em 380 cm3 and 390 cm 390 cm and 400 em None Of the above In a triangle ABC, AB BC = CA. The ratio of the radius Of the Circumcircle to Of the **(a) (b) (c)** None of the above B-A3QA-N-NB'J

46. **[CDS (I) Paper 12 Q18]** The diameter Of a metallic sphere is 6 cm. The sphere is melted and drawn into a Wire Of uniform circular cross-section. If the length Of the wire is 36 m, then what is its radius equal to? **(a) (b) (c) (d)** 001 cm 0•001 Cm B-ABQB-N-LZS/ IA less the is h of and lgth

47. **[CDS (I) Paper 12 Q2]** The surface area Of sphere C is equal to the sum Of areas of spheres A and B The volume Of sphere D is equal to the sum of volumes of spheres A, B and C Which Of the above statements is / are correct? **(a) (b) (c) (d)** I only 2 only Both 1 and 2 Neither 1 nor 2 B;-ABQB-N-LZS/ IA

48. **[CDS (I) Paper 12 Q66]** What is the number of pairs of perpendicular planes in a cuboid? **(d)** 12 None Of the above **(a)** sin x **(b)** cosx 62. What is **(c)** sin x + cos x **(d)** sin x — cos x 63. Consider the following statements • l. sin > sin I 2. COS < COS Which Of the above Stater-nents 66. What is the number of pairs of perpendicular planes in a cuboid?

49. **[CDS (I) Paper 12 Q81]** 82. What is the ratio Of AC to AB? What is the between two side 2 cm? @ cm **(b)** 2qrä cm **(c)** 40 cm maximum distance points of a cube of The areas of the three adjacent faces Of a cuboidal box are x, 4x and 9x square unit. What is the volume Of the box? **(b)** (C) **(d)** 6x2 cubic unit 6x3/2 cubic unit 3x3/2 cubic unit 2x3/2 cubic unit **(a)** 2 only above statements **(b)** AC/5 **(c)** AC/3 **(d)** AC/4

50. **[CDS (I) Paper 12 Q83]** 84. A cylinder circumscribes a sphere. What is the ratio of volume Of the sphere to that Of the cylinder? If for a triangle the radius of the circumcircle is double the radius Of the inscribed circle, then which one of the following is correct? @ The triangle is right angled **(b)** The triangle is isosceles **(c)** The triangle is equilateral **(d)** None Of the ab ove For the next two (2) items that follow A toy is in the form of a cone mounted on the hemisphere With the same radius. The diameter Of the base of the conical portion is 12 Crn and its height is 8 cm. 8S. What is the total area Of the **(a)** 132n cm 2 **(b)** 1121 cm2 **(d)** 661 cm 2

51. **[CDS (I) Paper 12 Q87]** What is the volume Of the double cone so formed? **(a)** 3124 **(b)** 3424 cm3 (C) 3768 C m **(d)** 3924 cm3

52. **[CDS (I) Paper 12 Q88]** What is the surface area of the double cone so formed? **(c)** 1310-4 cm 2 **(d)** 13188 cm2 **(a)** The triangle is isosceles **(b)** The triangle is equilateral **(c)** None Of the ab ove For the next three **(d)** 3124

53. **[CDS (I) Paper 12 Q2]** The volume of the cone generated when the triangle is made to revolve its longer leg is same as the volume Of the cone generated when the triangle is made to revolve about its shorter leg. The sum of the volume of the cone generated when the triangle is made to revolve about its longer leg and the volume Of the cone generated when the triangle is made to revolve about its shorter leg is equal to the volume Of the double cone generated when the triangle is made to revolve about its hypotenuse. Which Of the above statements is / are correct? **(b) (c)** only 2 only Both I and 2 Neither 1 nor 2 **(a)** The triangle is isosceles **(b)** The triangle is equilateral **(c)** None Of the ab ove For the next three **(d)** 3124

54. **[CDS (I) Paper 13 Q46]** If three metallic spheres Of radii 6 8 cm and IO cm are melted to form a single sphere, then the diameter Of the new sphere will be **(a)** 12 Cm **(b)** 24 cm **(c)** 30 cm **(d)** 36 Crn

55. **[CDS (I) Paper 13 Q79]** 80. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume Of the cone. The ratio Of the height Of the cone to the radius Of its base is **(d)** Water flows through a cylindrical pipe of internal diameter 7 cm at the rate Of 5 m/ s. The time, in minutes, the pipe would take to fill an empty rectangular tank 4m x 3m 31m is **(b) (d)** 28 24 20 12 **(a) (b) (c)** . Two area Of a rhombus With side cm and one diagonal IO cm 140 square cm 130 square cm 120 square cm I I O square cm circles, each of radius r, with 79. 80. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume Of the cone. The ratio Of the height Of the cone to the radius Of its base is **(d)** Water flows through a cylindrical pipe of internal diameter 7 cm at the rate Of 5 m/ s. The time, in minutes, the pipe would take to fill an empty rectangular tank 4m x 3m 31m is centres p and Q, are such that each circle passes through the centre Of the other circle. Then the area common to the circles is less than one-third of the sum Of the areas Of the two circles by 3 A spher and a cube have same surface area. The ratio of square of their volumes is

56. **[CDS (I) Paper 13 Q85]** A rectangular paper of 44 cm long and 6 cm wide is rollcd to form a cylinder of height equal to width of the paper. The radius Of the base Of the cylinder so rolled is **(a) (b)** 83. 86. If a **(c)** 264 **(d)** 300

57. **[CDS (I) Paper 13 Q84]** A rectangular block Of wood having dimensions 3 m x 2m x •75m has to be painted On all its faces. The layer of paint must be O I mm thick. Paint comes in cubical boxes having their edges equal to IO cm. The minimum number Of boxes of paint to be purchased is **(c) (d)** 3 2 The diagonals of three faces Of a cuboid are 13, and 20 linear units. Then the total surface area of the cuboid is 14 cm —b = 4 and a2 + b2 —40, where a and b are positive integers, then a 3 + b 6 is equal to 280 **(a) (b)** 83. 86. If a **(c)** 264 **(d)** 300

58. **[CDS (I) Paper 14 Q59]** Consider the following statements . l. If the height of a cylinder is doubled, the area of the curved surface is doubled. **(a)** 15-5 km/hour **(b)** 15•84 kn/hour **(c)** 16 km/hour **(d)** 16-36 km/hour 56. 'lhere are 437 fruit plants in an orchard planted in rows. distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants.. niere is I m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of Rs. 100 per metre ?

59. **[CDS (I) Paper 14 Q2]** If the radius of a hemispherical solid is doubled, its total surface area becomes fourfold. Which of the above statements is/are correct ? **(a) (b) (c) (d)** 20 I only 2 only Both 1 and 2 Neither 1 nor 2

60. **[CDS (I) Paper 14 Q60]** A large water tank has the shape of a cube. If 128 mJ of water is pumped out, the water level goes down by 2 m. "men the maximum capacity of the tank is **(a)** 512 m3 **(b)** 480 ms **(c)** 324 **(d)** 256 my

61. **[CDS (I) Paper 14 Q62]** From the solid gold in the form of a cube of side length 1 cm, spherical solid balls each having the surface area cma are to be made. Assuming that there is no loss of the material in the process Of making the balls, the maximum number Of balls made will be B-ABVB-O-GPU - A 22 **(a) (b)** 480 **(c)** 324 m 256 my **(d)** 63. Water is filled in a container in such a manner that its volume doubles every 5 minutes. If it takes 30 minutes for the container to be full, in how much time will it be one-fourth full ?

62. **[CDS (I) Paper 14 Q63]** Water is filled in a container in such a manner that its volume doubles every 5 minutes. If it takes 30 minutes for the container to be full, in how much time will it be one-fourth full ? **(a) (b) (c) (d)** 7 •5 minutes 15 minutes 20 minutes 17•5 minutes

63. [CDS (I) Paper 14 Q64] 30 metallic cylinders of same size are melted and cast in the form of cones having the same radius and height as those of the cylinders. Consider the following statements : Statement I Statement 11 A maximum Of 90 will be obtained. The Curved surface of the cylinder can be flattened in the shape of a rectangle but the curved surface of the cone when flattened has the shape of triangle, Which one of the following is correct in respect of the above ? **(a) (b) (c) (d)** 22 Both Statement I and Statement 11 are correct and Statement 11 is the correct explanation Of Statement I Both Statement I and Statement 11 are COITect and Statement 11 is not the correct' explanation of Statement I Statement ment 11 is Statement Statement I is correct but State- not Correct I is not correct but 11 is correct

64. [CDS (I) Paper 14 Q74] The areas of three mutually perpendi- cular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to **(a)** will be 75. Let **(b)** 52% **(c)** 540/0 **(d)** 98 cm 108 cm2

65. [CDS (I) Paper 14 Q75] Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 6 2 **(a)** will be 75. Let **(b)** 52% **(c)** 540/0 **(d)** 98 cm 108 cm2

66. [CDS (I) Paper 14 Q77] Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is **(a)** 4445 cm **(b)** 44-16 cm **(c)** 4745 cm **(d)** 476 cm

67. [CDS (I) Paper 14 Q79] How many spherical bullets each of 4 cm in diameter can be made out of a cube of lead whose edge is 44 cm ? **(a)** 4445 cm **(b)** 44-16 cm **(c)** 4745 cm **(d)** 476 cm 78. The area of a trapezium is 336cm2. If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is

68. [CDS (I) Paper 16 Q43] The ratio of the curved surface area to the total surface area of a right circular cylinder is I : **(a)** 12:45 hours **(b)** 5:00 p.m. **(c)** 11:45 a.m. **(d)** 11:50 a.m. 38. A train is travelling at 48 kmfhour completely crosses another train having half its length and travelling in opposite direction at 42 km/hour in 12 s. It also passes a railway platform in 45 s. What is the length of the platform ?

69. [CDS (I) Paper 16 Q2] If the total surface area is 616 cm2, what is the volume of the cylinder ? **(a)** 539 cm3 **(b)** 616 cm3 **(c)** 1078 cm3 **(d)** 1232 cm3

70. [CDS (I) Paper 16 Q44] A cubic metre of copper weighing 9000 kg is rolled into a squale bar 9 m long. An exact cube is cut off from the bar. How much does the cube weigh ? **(a)** 12:45 hours **(b)** 5:00 p.m. **(c)** 11:45 a.m. **(d)** 11:50 a.m. 38. A train is travelling at 48 kmfhour completely crosses another train having half its length and travelling in opposite direction at 42 km/hour in 12 s. It also passes a railway platform in 45 s. What is the length of the platform ?

71. [CDS (I) Paper 16 Q50] A building is in the form of a cylinder surmounted hy a hemispherical dome on the diameter Of the cylinder. The height of the building is three times the radius of the base of the cylinder. The building contains 67-L rn3 of air. What is the height Of the building ? Sl. The radius of the base and the height Of a solid right circular cylinder are in the ratio 2 : 3 and its volume is 1617 cm3. What is the total surface area Of the cylinder ? **(a) (b) (c) (d)** 462 cm2 616 cm2 770 cm2 786 cm2

72. [CDS (I) Paper 16 Q55] A drinking glass Of height 24 cm is in the shape of frustum of a cone and diameters of its bottom and top circular ends are 4 cm and 18 cm respectively. If we take capacity Of the glass as xx cm , then what is the value Of x ? **(a)** 824 **(b)** 1236 **(c)** 1628 **(d)** 2472

73. [CDS (I) Paper 16 Q58] The height of a cone is 60 cm. A small cone is cut Off at the top by a plane parallel to the base and its volume is -L the volume of original cone. What is the height from the base at which the section is made ? **(b)** 20 cm **(c)** 30 cm **(d)** 45 cm **(a)** Rs. 8 **(b)** Rs. 9 **(c)** Rs. 13 **(d)** Rs. 14 53. If a quadrilateral has an Inscribed then the sum of a pair Of opposite sides equals

74. [CDS (I) Paper 16 Q60] What is the volume of a sphere of radius 3 cm ? **(a)** 367t cm3 **(b)** 187t cm3 (C) 97t crn3 **(d)** cm3

75. [CDS (I) Paper 16 Q63] The diameter Of the front wheel of an engine is cm and that of rear wheel is 2y cm. To cover the same distance, what is the number of times the rear wheel 'revolves when the front wheel revolves n times ? SDC-S-TRD - A 18 xy For the next three (03) items that follow : A tent of a circus is made of canvas and is in the form of right circular cylinder and right circular cone above it. The height and diameter Of the cylindrical part of the tent are 5 m and 126 m respectively. The total height of the tent is 21 m. **(a)** 367t cm3 **(b)** 187t cm3 **(c)** cm3 61. The sides of a triangle are given by c + a2 and **(d)** 2 2 3

76. [CDS (I) Paper 16 Q64] What is the slant height of the cone ? **(a)** 60 m **(b)** 65 m **(c)** 68 m **(d)** 70m

77. [CDS (I) Paper 16 Q65] What is the curved surface area of the cylinder ? **(a)** 1980m' **(b)** 2010 nf **(c)** 2100 rn2 **(d)** 2240 m2

78. [CDS (I) Paper 17 Q84] 85. If the radius Of a right circular cone is increased by p% without increasing its height, then what is the percentage increase in the volume Of the cone ? **(b)** 2p2 100 **(d)** p 2+_p_ 100 A copper Wire when bent in the form Of a square encloses an area Of 121 cm2. If the same Wire is bent in the form Of a circle, it encloses an area equal to **(a) (b) (c) (d)** 121 cm2 144 em2 154 cru 168 cm ABC is a triangle and D is a point on the side BC. If BC = 12 cm, BD=9cm and z ADC - ZBAC, then the length of AC is equal to **(a) (b) (c) (d)** 6 cm If the surface area of a sphere is reduced to one-ninth Of the area, its radius reduces to **(a) (b) (c) (d)** BLS-S-FDE One-fourth One-third One-fifth One-ninth (24. is 86. f a 87. and 88. to (24-A) In a trapezium ABCD, AB is parallel to CD and the diagonals intersect each other at O. What is the ratio of OA to OC equal to ? **(a) (b) (c) (d)** Ratio of 0B to OD Ratio of BC to CD Ratio of AD to AB Ratio of AC to BD Ice-cream, completely filled in a cylinder Of diameter 35 cm and height 32 em, is to be served by completely filling identical disposable cones of diameter 4 cm and height 7 cm. 'Ihe maximum number of persons that can be served in this way is **(a) (b) (c) (d)** 950 1000 1050 1100 The radius of a circle is increased so that its circumference increases by 15%. The area Of the circle will increase by **(a) (b) (c) (d)** 3325% If the radius Of a right circular cone is

79. [CDS (I) Paper 18 Q72] 'A cylinder of height 2x is circumscribed by a sphere of radius 2x such that the circular ends of the cylinder are two small circles on the sphere. What is the ratio of the curved surface area of the cylinder t.o the surface area of the sphere ? **(b) (c)** A cylindrical vessel 60 cm in diameter is partially filled with water. A sphere 30 cm in diameter is gently dropped into the vessel and is compl&ly immersed. To what further height will the water in the cylinder rise ? **(a) (b) (c) (d)** 20 cm 15 cm 5 cm The vertical angle of a right circular cone is 2 and the slant height is r cm. What is the volume of the cone in cubic cm ? **(a) (b) (c) (d)** STS-M-TpT TtrS 91tra 3 3Ttr3 (22-A a

80. [CDS (I) Paper 18 Q75] The radii of the frustum of a right circular cone are in the ratio 2 : **(a) (b) (c) (d)** STS-M-TpT 70. 'A cylinder of height 2x is circumscribed by a 73. sphere of radius 2x such that the circular ends of the cylinder are two small circles on the sphere. What is the ratio of the curved surface area of the cylinder t.o the surface area of the sphere ? 71. 72.

81. [CDS (I) Paper 18 Q1] What is the ratio of the volume of the frustum of the cone to that of the whole cone ? **(a) (b) (c) (d)** 128 From a solid cylinder whose height is 8 cm and of base radius 6 em, a conical cavity of height 8 cm and of base radius 6 cm is formed by hollowing out. What is the inner surface area of the cavity ? **(a) (b) (c) (d)** 67t square cm 8Tt square cm 107t square cm 607t square cm A tent has been constructed which is in the form of a right circular cylinder surmounted by a right circular cone whose axis coincides with the axis of the cylinder. If the radius of the base of the cylinder is 50 m, the height of the cylinder is 10 m and the total height of the tent is 15 m, then what is the capacity of the tent in cubic metres ? **(a) (b) (d)** 375007t 875007t 3 265007t 3 250007t **(a) (b) (c) (d)** STS-M-TpT

82. [CDS (I) Paper 18 Q70] 'A cylinder of height 2x is circumscribed by a 73. sphere of radius 2x such that the circular ends of the cylinder are two small circles on the sphere. What is the ratio of the curved surface area of the cylinder t.o the surface area of the sphere ? **(a) (b) (c) (d)** STS-M-TpT 70. 'A cylinder of height 2x is circumscribed by a 73. sphere of radius 2x such that the circular ends of the cylinder are two small circles on the sphere. What is the ratio of the curved surface area of the cylinder t.o the surface area of the sphere ? 71. 72.

83. [CDS (I) Paper 18 Q71] 72. (22-A) The radii of the frustum of a right circular cone are in the ratio 2 : **(a) (b) (c) (d)** STS-M-TpT 70. 'A cylinder of height 2x is circumscribed by a 73. sphere of radius 2x such that the circular ends of the cylinder are two small circles on the sphere. What is the ratio of the curved surface area of the cylinder t.o the surface area of the sphere ? 71. 72.

84. [CDS (I) Paper 18 Q74] 75. From a solid cylinder whose height is 8 cm and of base radius 6 em, a conical cavity of height 8 cm and of base radius 6 cm is formed by hollowing out. What is the inner surface area of the cavity ? **(a) (b) (c) (d)** 67t square cm 8Tt square cm 107t square cm 607t square cm **(a) (b) (c) (d)** 20 cm 15 cm 5 cm The vertical angle of a right circular cone is 2 and the slant height is r cm. What is the volume of the cone in cubic cm ? A tent has been constructed which is in the form of a right circular cylinder surmounted by a right circular cone whose axis coincides with the axis of the cylinder. If the radius of the base of the cylinder is 50 m, the height of the cylinder is 10 m and the total height of the tent is 15 m, then what is the capacity of the tent in cubic metres ? TtrS 91tra 3 3Ttr3 **(a) (b) (d)** 375007t 875007t 3 265007t 3 250007t

85. [CDS (I) Paper 18 Q76] TWO rectangular Sheets Of sizes 27t x 47t and x 57t are available. A hollow right circular cylinder can be formed by joining a pair of parallel sides of any sheet. What is the maximum possible volume of the circular cylinder that can be formed this way ? **(a) (b) (c)** STS-M-TpT 76. TWO rectangular sheets Of sizes 27t x 47t and 79. x 57t are available. A hollow right circular cylinder can be formed by joining a pair of parallel sides of any sheet. What is the maximum possible volume of the circular cylinder that can be formed this way ? 77. 78. PQ = 2XY PX=QY PQ- 3XY **(d)**

86. [CDS (I) Paper 19 Q67] 68. ABC is a triangle right angled at C With BC a and AC b. If p is the length of the perpendicular from C on AB, then which one of the following is correct ? **(a) (b) (c) (d)** 2a2 b2 p2 (a2 + b2) a2 2p2 (a2 + b2) The radius and slant height of a right circular cone are 5 cm and 13 cm respectively. What is the volume Of the ? **(a) (b) (c) (d)** I cm3 50m cm3 169-a cma Two equal circular regions of greatest possible area are cut off from a given circular sheet of area A. What is the remaining area of the **(a) (b) (c)** Ala 3M5 2N5 If the ratio of the radius of the base of a right circular cone to its slant height is I : 3, what is the ratio of the total surface area to the curved surface area ? **(a) (c) (d) (b) (c)** SPSS-T-XHI

87. [CDS (I) Paper 19 Q69] 70. 71. (22—A A right circular cone is sliced into a smaller cone and a frustum Of a cone by a plane perpendicular to its axis. The volume of the smaller cone and the frustum of the cone are in the ratio 64 ; **(a) (b) (c)** 16:9 64 : 61 81 : 64 In a room whose floor is a square Of side IO m, an equilateral triangular table Of side 2 m is placed. Four book-shelves of size 4 m x I m x 9 m are also placed in the room. If half Of the rest Of the area in the room is to be carpeted at the rate Of 100 per square metre, What is the cost of carpeting **(d)** 'tag 3 3'ta 3

88. [CDS (I) Paper 19 Q74] 75. What is the volume Of a cone of maximum volume cut out from a cube Of edge 2a such that their bases are on the same plane ? **(a)** 'tag 3 3'ta 3 **(d)** The radii Of two circles are 4-5 cm and cm respectively. The distance between the centres of the circles is IO em. What is the length of the transverse common tangent ? **(a) (b) (c)** 5 cm There are as many square centimetres in the surface area of a sphere as there are cubic centimetres in its volume. What is the radius of the sphere ? **(a) (b) (c) (d)** 3 cm 2 cm The length Of line segment AB is 2 cm. It is divided into two parts at a point C such that AC2 AB x CB. What is the length of CB ? **(a) (b)** 3 — em (e) **(d)** SPSS-T-xHl

89. [CDS (I) Paper 19 Q100] What Will be the total cost of tiles for the pathway ? **(c) (d)** 1,200 e 1,100 1,000 900 A cube Of maximum volume (each corner touching the surface from inside) is cut from a sphere. What is the ratio of the volume Of the cube to that of the sphere ? **(d)** 4:31 **(a)** 4:31 SPSS-T-XHI 83. What is the largest power of IO that divides 87. the product I x 2 x 3 x 4 ..x23x24x25? 84. 85. 86. **(b) (c)** 2 4 5 None of the above Walls **(d)**

90. [CDS (I) Paper 19 Q89] 90. In a triangle ABC, a, b and c are the lengths Of the sides and p, q and r are the lengths of its Which one of the following is correct ? **(d)** Il(p + q + r) > IO (a + b + c) Which one Of the following is correct ? **(a)** 4:31 SPSS-T-XHI 83. What is the largest power of IO that divides 87. the product I x 2 x 3 x 4 ..x23x24x25? 84. 85. 86. **(b) (c)** 2 4 5 None of the above Walls **(d)**

91. [CDS (I) Paper 19 Q78] 79. There is a path of width 5 m around a circular plot of land whose area is 144m m2. The total area of the circular plot including the path surrounding it is **(a) (b) (c) (d)** 3497t m2 2897t m2 2097t m2 1497t m2 The lateral surface area Of a cone is 462 cm2. Its slant height is 35 cm. The radius of the base of the cone is **(a) (b) (c) (d)** cm cm cm A semi-circular plate is rolled up to form a conical surface. The angle between the generator and the axis of the cone is **(a) (b) (c) (d)** 450 300 150 A solid right cylinder is Of height em. Tf lateral surface area is half its total surface area, then the radius Of its base is **(a) (b) (c) (d)** 2/'t cm GTHY-T-XMT (25- lar 80. lth

92. [CDS (I) Paper 19 Q81] If the diagonal of a cube is of length l, then the total surface area of the cube is The lateral surface area Of a cone is 462 cm2. Its slant height is 35 cm. The radius of the base of the cone is **(a) (c) (a) (b) (c) (d)** 312 2 12 cm cm cm

93. [CDS (I) Paper 19 Q82] A semi-circular plate is rolled up to form a conical surface. The angle between the generator and the axis of the cone is **(a) (b) (c) (d)** 450 300 150

94. [CDS (I) Paper 20 Q6] Priya's age was cube of an integral number (different from l) four years ago and square Of an integral number after four years. HOW long should she wait so that her age becomes of a number in the previous year and cube of a number in the next year ? **(a) (b) (d)** 5 6 9 11 **(a) (b) (c) (d)** 7 years 12 years 14 years 21 years OFGT-T-HTM

95. [CDS (I) Paper 20 Q2] The perimeter Of a triangle is greater than the sum of its three medians. In any triangle ABC, if D is any JY'int on BC, then AB+BC+CA > 2M). Which Of above ? **(a) (b)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 **(c) (d)** 21: 16 14:11

96. [CDS (I) Paper 20 Q91] 92. 98, -A) What is the ratio Of the volume Of the sphere to that of the cone ? **(b)** (e) 60 . 3vfä . What is the ratio Of the volume Of the cube to that of the cylinder ? **(b)** 21: 16 14:11 **(d)** 45: 32 Consider the following statements I. The surface area of the sphere is times the curved surface area of the **(a) (b)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 **(c) (d)** 21: 16 14:11

97. [CDS (I) Paper 20 Q98] What is the ratio Of the volume Of the sphere to that of the cone ? **(a) (b)** I and 2 only 2 and 3 only I and 3 only 1, 2 and 3 **(c) (d)** 21: 16 14:11

98. [CDS (I) Paper 20 Q100] If S is the sum Of the cubes of the dimensions Of the cuboid and V is its volume, then what is **(a)** What is the area Of triangle ADC ? 15 Cm 20 em cm **(b) (c) (d)** 126 cm2 124 cm2 122 cm2 120 cm2 What is the area Of quadrilateral ABCD ? Consider the following for the next two

99. [CDS (I) Paper 21 Q26] A circulu vasel Of volume V is Nual to its heÜ, is Ømpl&ly with water. A havy ofmaximum volume is completely in üw vasel. What of ranains in ? 3 4 27. parallel line y z Ze cut by two tansv«æls m nd n. m cuts l_i1B x, y, z at P, Q, R rapeüvely; Traxsva•al cuts the x, y, z L, M, N tively. If cm' QR—9 cm nd — 106 cm, lagth oflM? O) 3-5 **(d)** 45 of a of a circle of 4 is 256 is radim of of **(a)** 17 12 32 19 21 24. N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ? **(b)** 2975 **(c)** 3094 **(d)** 3101

100. **[CDS (I) Paper 21 Q27]** 11 Of volume V is qual to its heÜ, is Ømpl&ly filled with water. A havy ofmaximum volume is completely in üw vasel. What of ranains in ? 3 4 parallel line y z cut by two tansv«æls m nd n. m cuts the l_i1B x, y, z at P, Q, R rap&üvely; cuts the x, y, z L, M, N tively. If cm+ QR—9 cm nd — 10•5 cm, "h oflM? O) 3-5 **(d)** 45 of a of a circle of 4 is 256 is radim mware of of C — SDHY T-MTH **(a)** 17 12 32 19 21 24. N is by 17, is equal to 182. ne b«wen the quotiai md is 175. is åe value of N ? **(b)** 2975 **(c)** 3094 **(d)** 3101

101. **[CDS (I) Paper 21 Q41]** The Of thre of a cuboid x, y md z. If V is volume of the cuboid, mwof the 6110wing ? **(c)** C - SDHY-T-MTH **(a)** 33 cm **(b)** 44 **(c)** 66 cm 37. The sides AD, BC of a ü•apdum ABCD prallel md the di.N1s AC BD at O. If the of Gangle AOB is 3 cm2 the Of tiangle BDC is 8 then what is åe of aianøe AOD ? **(d)** 8 cm2

102. **[CDS (I) Paper 21 Q53]** A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of **(a)** 40m cm' **(b)** 4500 cm3 **(c)** 5000 cm3 **(d)** cm3

103. **[CDS (I) Paper 21 Q61]** The volume of a bani"hae is 155232 Wha is the Of Ole **(a)** 40 cm **(b)** 42 **(c)** 38 **(d)** 36 cm A bucket is in tlr Of a Cone. The diam&r. Of üw top of the 6 12 cm respætively. If the —t of bucket is 7 üw of the bucket ? 535 cm3 **(b)** 462 **(c)** 234 cm3 **(d)** 166 C - SDHY-T-MTH A solid Of heÜ 10cm and 6cm is melted to make Wo cona in åe ratio Of volume 1 : 2 md of same height u IO em. WhM is Ore in nat surfæe ? **(c)** 75% **(d)** 100%

104. **[CDS (I) Paper 22 Q74]** The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? **(a)** litre **(b)** litre , **(c)** — litre 10 **(d)** — litre

105. **[CDS (I) Paper 22 Q88]** A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct ? **(b)** p 24 **(c)** 2p = q **(a)** 7-5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ? **(b)** 4 square units **(c)** 245 square units **(d)** r square units

106. **[CDS (I) Paper 23 Q89]** 90. l31-A) The radii of the nat circular faces Of a bucket are and 2x. If the height of the bucket is 3K, is the capacity of the bucket ? 22 (Assume = **(b) (c) (d)** Ilx8 22x3 44xa 55xa If p, q, r, s and t represent length, breadth, height, surface area and volume Of a cuboid respectively, then What is — + — + — equal **(a) (b)** 2t 87. (31 -A) ABC is a triangle at C. Let P be on Q point on BC. Which Of Correct ? AQ2+BP2-AB2+PQ2

107. **[CDS (I) Paper 23 Q90]** Four circular coins of equal radius are placed their centres coinciding with frur vertices of a square. Each coin touches two other coins. If the uncovered area of the square is 42 em 2, then what is the t•adius of each coin ? If p, q, r, s and t represent length, breadth, height, surface area and volume Of a cuboid respectively, then What is — + — + — equal **(a) (b) (c)** Ilx8 22x3 44xa 55xa **(d)**

108. **[CDS (I) Paper 24 Q61]** A cone and a hemisphere have equal bases and equal volumes. What is the ratio of the height of the cone to the radius of the hemisphere? **(a) (b)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 66. 63. The ratio of the radius of base to the height of a cylinder is 2 : 3. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? **(c)** 242 cm **(d)** 385 cm 2

109. **[CDS (I) Paper 24 Q62]** A solid sphere of diameter 60 mm is melted to stretch into a wire of length 144 cm. What is the diameter Of the **(a) (b)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 66. 63. The ratio of the radius of base to the height of a cylinder is 2 : 3. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? **(c)** 242 cm **(d)** 385 cm 2

110. **[CDS (I) Paper 24 Q3]** If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = —l **(a)** 242 cm **(b)** 385 cm 2 **(c)** 462 cm 2 **(d)** 770 cm

111. **[CDS (I) Paper 24 Q2]** The pipe is made of 99 crn3 of metal. If R is the outer radius and r is the inner radius Of the pipe, then What is (R equal to? 22 (Take XVWS-T-MTK/SOA A metal solid cube of edge 24 cm is melted and made into three small cubes. If the edges of two small cubes are 12 cm and 16 cm, then What is the surface area of the third small cube? **(b) (d)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 **(a) (b)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 66. 63. The ratio of the radius of base to the height of a cylinder is 2 : 3. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? **(c)** 242 cm **(d)** 385 cm 2

112. **[CDS (I) Paper 24 Q67]** A metal solid cube of side 22 cm is melted to make a cone Of height 21 cm. What is the radius of the base of the cone? (Take 16•5 cm **(c)** 22 cm **(d)** 275 cm I P.T.O. **(a) (b)** 1200 cm2 1800 cm 2 2400 cm 3600 cm2 66. 63. The ratio of the radius of base to the height of a cylinder is 2 : 3. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? **(c)** 242 cm **(d)** 385 cm 2

113. **[CDS (I) Paper 24 Q68]** A cone Of height 24 cm has a curved surface area 550 cm **(a)** 28 m **(b)** 44 m **(c)** 66 m 72. A sphere of diameter 6 cm is dropped into a cylindrical vessel partly filled With water. The radius Of the vessel is 6 cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised? 73. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55 cm, then What is the area of the sector? **(d)** 577•5 cm2

114. **[CDS (I) Paper 24 Q69]** A rectangular is 44 cm long and 22 cm Wide. Ikt x the volume of the largest cylinder formed by rolling the paper along its length and y be the volume of the largest cylinder formed by rolling the paper along its width. What 3:2 70. is the ratio of x to y? (Take A hollow spherical shell is made up of a metal Of density 3 g /cm3. If the internal and external radii are 5 cm and 6 cm respectively, then what is the mass Of the shell? crake n 7 1144 g 1024 g 840 g 570 g XVWS-T-MTK/SOA **(a)** 28 m **(b)** 44 m **(c)** 66 m 72. A sphere of diameter 6 cm is dropped into a cylindrical vessel partly filled With water. The radius Of the vessel is 6 cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised? 73. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55 cm, then What is the area of the sector? **(d)** 577•5 cm2

115. **[CDS (I) Paper 24 Q71]** A cloth of 3 m width is used to make a conical tent 12 m in diameter with a slant height Of 7 m. What i' the length Of the cloth? (Take E) **(b)** 28 m **(c)** 44 m **(d)** 66 m **(a)** 28 m **(b)** 44 m **(c)** 66 m 72. A sphere of diameter 6 cm is dropped into a cylindrical vessel partly filled With water. The radius Of the vessel is 6 cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised? 73. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55 cm, then What is the area of the sector? **(d)** 577•5 cm2

116. **[CDS (I) Paper 24 Q72]** A sphere of diameter 6 cm is dropped into a cylindrical vessel partly filled With water. The radius Of the vessel is 6 cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised? **(a)** 28 m **(b)** 44 m **(c)** 66 m 72. A sphere of diameter 6 cm is dropped into a cylindrical vessel partly filled With water. The radius Of the vessel is 6 cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised? 73. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55 cm, then What is the area of the sector? **(d)** 577•5 cm2

117. **[CDS (I) Paper 24 Q80]** The surface area Of a cube is equal to that Of a sphere. If x is the volume or the and y is the volume of the sphere, then What is x2:y2 equal to? 1 P.T.O. If the perimeter of a semicircular park is 360 m, then what is its area? (Take **(a)** 3850 m **(b)** 7700 m 15400 m 76. In a trapezium ABCD, AB is parallel to DC. The diagonals AC and BD intersect at P. If AP : PC 4 : Bx — 4 **(c)** 11:23 **(d)** 23:25

118. **[CDS (I) Paper 25 Q69]** A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take **(a)** 1540 cm3 **(b)** 1078cm3 **(c)** 1048 cm3 **(d)** 770 cm3

119. **[CDS (I) Paper 25 Q70]** 3 cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? **(a)** 343 cm2 **(b)** 350 cm2 **(c)** 686 cm2 **(d)** 700 cm2

120. **[CDS (I) Paper 25 Q71]** A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) **(a)** 1330 cm2 **(b)** 1306 cm2 **(c)** 1296 cm2 **(d)** 1256 cm2

121. **[CDS (I) Paper 25 Q72]** How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) **(a)** 420 **(b)** 210 **(c)** 200 **(d)** 168 A - NYRG-T-MTK

122. **[CDS (I) Paper 25 Q73]** A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? **(a)** 63•5 cm **(b)** 65 cm **(c)** 73-5 cm **(d)** 75 cm

123. **[CDS (I) Paper 25 Q75]** A solid rcxi consists of a cylinder of height 20 cm and radius 7 cm. It is surmounted by another solid cylinder of height IO cm and radius cm. If 1 cubic metre of rod weighs 10000 kg, what is the mass of the rod ? (take 4) **(a)** 34-65 kg **(b)** 31•56kg **(c)** 3•465 kg **(d)** 3-156kg Consider the following for the next three (03) items that follow : A solid consisting of a right circular cone of radius x and height 28 standing on a hemisphere of radius x (take 21 21 76. nie volume of the solid is equal to that of a **(a)** sphere of radius x **(b)** sphere Of diameter x **(c)** cylinder of radius x **(d)** cylinder of radius x

124. **[CDS (I) Paper 25 Q77]** What is the approximate total surface area of the solid ? **(b)** 12-5x2 **(c)** 13-3* **(d)** 15-1 P **(a)** 63•5 cm **(b)** 65 cm **(c)** 73-5 cm **(d)** 75 cm 74. ABC is a triangle right-angled at B with AB 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surmce area of the cone so formed ?

125. **[CDS (I) Paper 25 Q78]** The solid is placed in a right circular cylinder full of water such that it touches the bottom. If the internal radius of cylinder is x and height is 3x, what is the approximate volume of water left in the cylinder ? **(a)** 504 **(b)** 5•09x3 **(c)** 5-14x3 **(d)** 5-24+ Consider the following for the next two (02) items that follow : A chord of a circle of radius cm subtends an angle of 1200 at the centre. (take = and = 1•732)

126. **[CDS (I) Paper 26 Q63]** 17 The cm 6•5 cm surface area Of a cube Of length x is equal to the surface area Of a sphere Of radius y Consider the following statements : **(a)** cos29 **(b)** 2 sin B **(c)** 2cOsB 59. If O, where then what is 'the value Of 12cot2 B + 3cosece? 61. Considér the following statements : 2. If two chords AB and AC Of a circle are equal, then the centre Of the circle lies on the angle bisector Of angle CAB. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC BD. **(d)**

127. **[CDS (I) Paper 26 Q69]** A rectangular sheet is of length x and breadth y If p is the volume Of the cylinder by rolling the Sheet along its breadth and q is the volume of the cylinder formed by rolling the sheet along its length, and q = 2 p, then Which one Of the following is correct? **(c)** x 2y **(d)** 3x 4y **(a)** 100 The radius and height Of a right circular cone are in the ratio 3 7. If the volume Of the cone is 528 cm3 , then What is the height of the cone? **(b)** 10•5 cm **(c)** 140 cm A square sheet is forrned by joining n identical Square sheets of same size. If the length of the diagonal of the bigger square sheet formed is m, then What is the side length Of a smaller square sheet? **(d)** A cylindrical pipe has inner diameter Of 14 cm. Water flows through it at a rate of 154 litres per minute. What is the speed Of water in km/hr? wake 2?

128. **[CDS (I) Paper 26 Q85]** The length, breadth and height Of a cuboid are in the ratio 27: 8: L The cuboid is melted and recast into a cube. If p is the surface area Of the cuboid and q is the surface area Of the cutr, then what is p / q equal to? 247 216 251 108 505 **(d)** 216 1 P.T.O. **(a)** 2x-90' **(b)** 2x-18 **(c)** 18m-2x ABC is a triangle right angled at B with AC-2BC If ZA-x, then what is ZC equal to? 78. 80. Let D, E and F be the midpoints Of the sides BC, CA and AB respectively Of a triangle ABC Triangle DEF is congruent to which of the following 1. AEF 3. EDC Select the correct answer using the code given below. 81. 84. In a triangle ABC, AB— 16 cm , 12 cm and AD is the bisector Of ZA If BD —4 cm, then what is CD equal to? **(d)** 2-5 cm 82. An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct?

129. **[CDS (I) Paper 27 Q70]** 16 and breadth is — times its height. If H is the height Of the rm:jm and L is the longest rod that can be placed in the room, then which one Of the following is correct ? 121, = 29H 9L = 25H 7L z 23K **(c)** 13H (25-A) FDGT-T-EMT Consider the following for the next three (03) items that follow : A conical vessel of radius 12 cm and height 16 em is filled with water. A sphere is lowered into water and its size is such that it touches the sides of the vessel and it is just immersed. What is the radius Of the sphere ? **(a) (b) (c)** 5 em 900 — 72. Hmv much water will remain in the vessel after the overflow ? **(d)**

130. **[CDS (I) Paper 27 Q72]** Hmv much water will remain in the vessel after the overflow ? **(a) (b) (c)** 2887t mL • 360r mL 480m mL 500't mL 21 The length Of a room is — times its breadth 16 and breadth is — times its height. If H is the height Of the rm:jm and L is the longest rod 73. that can be placed in the room, then which one Of the following is correct ? 121, = 29H **(c)** FDGT-T-EMT is the ratio of lateral surface area Of the vessel to the surface area Of the sphere ? 4 **(a) (c) (d)**

131. **[CDS (I) Paper 27 Q82]** 83. (29-A) **(a) (b) (c) (d)** IO cm 12 cm 18 em 20 cm A rectangular metal sheet is of length 24 cm and breadth 18 em. From each Of its corners a square of side x cm is cut off and an open box is made of the remaining sheet. If the volume of the box is 640 cubic em, then what is the value Of x ? **(c) (d)** 2 3 4 6 Let x, y, z be the length, breadth, height of a cuboid. If its volume is 400 cubic cm and total surface area is 340 square em, then What is x-i + y-l + El equal to ? 17 **(a)** 20 40 10 **(d)** FDGT-T-EMT Two parallel sides of a trapezium are 29 and 21 cm. Non-parallel sides are equal and each is of length 8-5 cm. What is the area Of the trapezium ? **(a) (b) (c) (d)** 187 •5 square em 227 •5 square em 375 square em 455 square cm A square copper plate of side 16 cm weighs 128 gm. A circular disc of diameter 14 cm is cut off from the plate. What is the weight Of 22 the remaining part ? (z z **(a) (b) (c) (d)** 48 gm 49 gm 50 gm 51 gm The area Of a rhombus is 96 square em and one of its diagonals is of length 12 cm. What is the perimeter of the rhombus ? **(a) (b) (c) (d)** 36 cm 40 cm 44 cm 48 cm A right-angled triangle ABC is inscribed in a circle of radius IO cm. The altitude drawn to the hypotenuse AC is of length 8 cm. If AB x cm and BC y cm, then what is the value of xy ? **(a) (b)** 60 80 160 A) The height Of a cone is 30 cm. A small cone is 84• 81. cut off at the top by a plane parallel to its base. If its volume is — Of the volume of the 27 given cone, then what is the height of the frustum of the cone ?

132. **[CDS (I) Paper 28 Q58]** The surface area of a cube is increased by **(a) (b) (c) (d)** 16 < p < 18 14 cpc 16 12 < p < 14 62.

133. **[CDS (I) Paper 28 Q59]** 60. (21 -A **(a) (b) (c) (d)** 16 < p < 18 14 cpc 16 12 < p < 14 A solid cube is cut into two cuboids Of equal volume. What is the ratio of total surface area of the given cube to that Of one Of the cuboids ? **(b) (c) (d)** The length of a diagonal of a cuboid is 11 cm.

134. **[CDS (I) Paper 28 Q68]** The surface area is 240 cm. What is the sum of its length, breadth and height ? **(a) (b) (c) (d)** 16 cm 17 cm 18 cm 19 cm JNBY-D-MTK

135. **[CDS (I) Paper 28 Q86]** A square sheet Of side length 44 em is rolled along one of its sides to form a cylinder by making opposite edges just to touch each other. What is the volume Of the cylinder ? (Take = **(a) (b) (c)** 60 72 75 80 **(d)**

136. **[CDS (I) Paper 29 Q41]** 42. If r is the radius of S and R is the radius of S2, then which one of the following is correct ? R=3r **(b) (c)** If m is the area of the circle S and n is the area of semi-cir•cle Sl, then which one of the following is correct ? **(a) (b) (c) (d)** Consider the following for the next two (02) Consider the following for the next two (02) 3R = 4r 9m 4n 7m = 3n (15-A) items that follow : items that follow A right conical cap just covers two spheres placed one the other on a table such that it touches A line segment AB is bisected at C and semi-circles both the spheres. Let r be the radius Of the smaller Sl, and are drawn respectively on AB, AC and sphere and R be the radius of the bigger sphere. CB as diameters such that they all lie on same side Let 29 be the vertical angle Of the cone. of AB. A circle S is drawn touching internally S, and externally S2 and Ss.

137. **[CDS (I) Paper 29 Q53]** What is the radius Of the sphere passing 54. through the corners Of the cuboid with edges 8 em, 12 cm and 24 em ? **(a) (b) (c) (d)** 10-5 cm 14 cm 21 cm 28 cm A lamp shade is in the shape Of a part Of a cone and its top and bottom ends are circles whose circumferences are respectively 30 em and 40 cm. The perpendicular distance between the ends is 6 cm. If the cone were to be completed, then hmv far would its vertex be from the top end ? **(a) (b) (c) (d)** 20 cm 18 cm 12 em 9 cm A sum Of money at 20% rate of compound interest per annum becomes more than 100 times in n years. What is the least value 3=0-477) **(a) (d)** 23 24 25 26 JNUP-T-EMT (21 -A

138. **[CDS (I) Paper 29 Q51]** What is the radius Of the sphere passing 54. through the corners Of the cuboid with edges 8 em, 12 cm and 24 em ? **(a) (b) (c) (d)** 10-5 cm 14 cm 21 cm 28 cm 55. A lamp shade is in the shape Of a part Of a cone and its top and bottom ends are circles whose circumferences are respectively 30 em and 40 cm. The perpendicular distance between the ends is 6 cm. If the cone were to be completed, then hmv far would its vertex be from the top end ?

139. **[CDS (I) Paper 29 Q65]** 8 cm and IO cm are melted together and recast as a solid sphere. What is the diminution of the surface area as compared to the sum of the surface areas of the three spheres ? **(c)** A solid sphere Of radius 3 cm is melted to form a hollow cylinder of height 4 cm and external diameter IO cm. What is the thickness of the cylinder ? **(a)** 67. 203: 117 203 : 157 172 : 91 165: 88 **(b) (c) (d)** — ya

140. **[CDS (I) Paper 31 Q87]** A cone, a hemisphere and a" cylinder stand on equal base of radius r and have the same height. If the sum of volumes of cone, the hemisphere and the cylinder is equal to volume of a R3 sphere of radius R, then what is —Ü equal to ? **(a)** 96 square cm **(b)** 84 square cm **(c)** 80 square cm **(d)** 72 square cm 86. There are two containers A and B. In container A. the ratio of milk and water is I : 3 and in container B, the ratio of milk and water is m : n. If the mixture in the containers A and B are mixed in the ratio 2 : 3 to get 20 litres of a mixture having milk and water in the ratio 3 : 7, then what is the value of 2 4 4 37 87. A cone, a hemisphere and a" cylinder stand on equal base of radius r and have the same height. If the sum of volumes of cone, the hemisphere and the cylinder is equal to volume of a R3 sphere of radius R, then what is equal to ?

141. **[CDS (I) Paper 32 Q4]** 5. 6. -A) Which of the following statements with reference to Surface inversion Of temperature Ware correct ? I. It cause s instability in the lower layers Of the atmosphere. **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

142. **[CDS (I) Paper 32 Q30]** A metallic sphere Of mass I kg and volume **(a) (b) (c) (d)** Brahmacharya Samyag-darshana Sunrita Asteya

143. **[CDS (I) Paper 32 Q27]** A metallic sphere Of mass I kg and volume **(a) (b) (c) (d)** Brahmacharya Samyag-darshana Sunrita Asteya

144. **[CDS (I) Paper 33 Q62]** The sum of the height and the radius Of a right circular cylinder is 2 1 cm, and the radius is less than the height. If the curved surface area of the cylinder is 616 cm2, then What is the volume Of the cylinder? (Take It 22 / **(a)** 1078 cm **(b)** 1617 cm **(c)** 1927 cm 3 **(d)** 2156 cma ANFB-T-EMT/45A

145. **[CDS (I) Paper 33 Q64]** The length, breadth and height Of a cuboid are consecutive integers. If the volume of the cuboid is 336 cubic units, then what is the total surface area Of the cuboid? **(a)** 288 square units **(b)** 292 square units **(c)** 296 square units **(d)** Cannot be determined due to insuffcient data

146. **[CDS (I) Paper 33 Q77]** 78. A conical tent has an angle Of 600 at the vertex. If the curved surface area is 100 m 2 , then what is the volume of the tent? 250.0 5000 10000 10000 A right circular cone and a hemisphere have equal base and equal volume. What is the ratio Of the height of the cone to the radius Of the hemisphere? ANFB-T-EMT/4SA **(a) (b) (c)** — cm None of the above 80. A right circular cone and a hemisphere have equal base and equal volume. What is the ratio Of the height of the cone to the radius Of the hemisphere? ANFB-T-EMT/4SA Let the area of the largest possible square inscribed in a circle of unit radius be x. Let the area of the largest possible circle inscribed in a square of unit side length be y What is the relation between x and y? **(d)** NX 2y

147. **[CDS (I) Paper 34 Q47]** Let x be the length of a diagonal of a face of a cube and y be the length of a diagonal of the cube. If x + y = (5 + 2-16) units, then what is the total surface area of the cube ? 19 **(a)** 220 m **(b)** 225 m **(c)** 230m **(d)** 235 m 46. In a cyclic quadrilateral ABCD, the diagonal AC bisects the angle C. Which of the following staternents is/are correct ? 11. The diagonal BD is parallel to the tangent of the circle at A Select the answer using the code given below :

148. **[CDS (I) Paper 34 Q58]** A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? 7 **(a)** — cm **(b)** — cm 14 **(c)** — cm 16 **(d)** — cm

149. **[CDS (I) Paper 34 Q61]** The diameter of a copper solid sphere is 6 cm. The sphere is melted and recast into a wire. If the diameter of the wire is cm, then what is the length of the wire ? **(a)** 5-24 m **(b)** 5-36 m **(c)** 5•76 m **(d)** 596 m

150. **[CDS (I) Paper 34 Q69]** The volume of a cube is 8 cm3 and is equal to the volume of a cuboid of length x, breadth y and height z, where x, y and z are natural numbers (x > y > z). Let n be the number of such cuboids with different dimen- sions. What is the value of n ? **(d)** 27 More than 3 A - KPRS-T-MTE **(a)** I only **(b)** 11 only **(c)** Both I and 11 **(d)** Neither I nor 11 66. A metallic sheet of rectangular shape is 48 cm x 36 cm. From each corner a square of side x cm is cut off and an open box is made of the remaining sheet. If the box holds 5•12 litre of water, then what is the value of x ?

151. **[CDS (I) Paper 34 Q70]** Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? **(a)** 60 square units **(b)** 62 square units **(c)** 64 square units **(d)** Cannot be determined due to insufficient data

152. **[CDS (I) Paper 35 Q55]** 56. 57. (19-A) A resenoir is in the form Of a cuboid. Its length is 30 m. If 24 kL Of water is removed from the reservoir, the water level goes down by 20 em. What is the width of the reservoir ? **(a) (b) (c)** 3-5 m **(d)**

153. **[CDS (I) Paper 35 Q58]** 59. On a circular metal plate Of uniform thickness, 16 holes, each Of diameter 2 em, are made. If the plate thereby has lost one-ninth Of its original weight, then what is the diameter of the plate ? **(b) (c) (d)** 3-5 m **(a) (b) (c) (d)** 12 cm 15 cm 21 cm 24 cm A wrell is dug With a diameter of m and 16 m depth. The earth soil so excavated is spread in the form of a right circular cone of radius 7 m. What is the height of the cone ? A square Of side length 4 cm has its corners cut away in such a manner so as to form a regular octagon. What is the length of the side of the octagon ? (Take 1-41) **(a) (b) (c) (d)** 1•5 m **(a) (b)** (C) **(d)** 1•00 cm 1-64 em I •68 cm 1•72cm

154. **[CDS (I) Paper 35 Q67]** 69. A thin metallic sheet, 1-92 m2 in area, is cut into two equal pieces. One piece is used to make a hollow cube of volume P (in m 8) and the other is used to make a hollow cuboid Of volume Q (in m") with d' in the ratio 4 : 2 I _ If no sheet is wasted, which one of the following is correct ? **(a)** AB:BC **(b)** AB2 Consider the following statements : 11. The areas Of two similar triangles are in the ratio Of the squares of the corresponding medians. The areas of two similar triangles are in the ratio of the squares Of the corresponding heights. **(c) (d)**

155. **[CDS (I) Paper 35 Q11]** It is possible to have a perimeter equal to 72 cm. Which of the statements given above Ware correct ? For **(a) (b) (c) (d)** I only 11 only Both I and 11 Neither I nor 11 N number Of cubes each Of side length equal to IO cm are joined end to end in a row. If the total surface area of the resulting cuboid is 3800 em 2, then What is the value Of N ? 10 **(d)** 12 In a quadrilateral ABCD, Z ABC 900 and ACD

156. **[CDS (I) Paper 35 Q83]** If x is the perimeter, in em, Of the triangle, then which one of the following is correct ? **(a) (b) (c) (d)** I only 11 only Both I and 11 Neither I nor 11 **(a) (b) (c) (d)** 69 cm < x < 65 em 65 cm x 67 cm 67 < x < 69 cm 69 cm < x < 71 cm N number Of cubes each Of side length equal to IO cm are joined end to end in a row. If the total surface area of the resulting cuboid is 3800 em 2, then What is the value Of N ? 10 **(d)** 12 In a quadrilateral ABCD, Z ABC 900 and ACD

CHAPTER 13

STATISTICS & DATA INTERPRETATION

Total Archive Questions: 122 Questions

Typical Exam Weightage: 8 - 12 Questions (8 - 12 Marks)

UPSC Examiner Scope: Elementary Mathematics — Objective Multiple Choice Items. Each question carries 1.0 mark with 1/3rd negative marking for incorrect responses.

1. [CDS (I) Paper 1 Q92] 93. The arithmetic mean of a Set of 10 numbers is **(a)** 20 **(b)** 25 **(c)** 40 **(d)** 45 Consider the following types of data 95. 96. 97. Data on percentage distribution Of area of land in acres owned by households in two districts of a particular State are as follows : Land holding District-A District-B 001—0-99 2-5—749 7-5—12-49 12-5—19-99 20-0—29-99 18•35 47-12 19-34 7-21 2-36 13•53 21-84 39-32 12-15 743 5•73 2. 3. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ?
2. [CDS (I) Paper 1 Q20] If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? **(a)** 20 **(b)** 25 **(c)** 40 **(d)** 45 Consider the following types of data
3. [CDS (I) Paper 1 Q2] 3. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? **(a)** I and 2 only **(b)** 2 Only **(c)** I and 3 only **(d)** 1 , 2 and 3
4. [CDS (I) Paper 1 Q94] The mean Of 25 observations is **(a)** 20 **(b)** 25 **(c)** 40 **(d)** 45 Consider the following types of data 95. 96. 97. Data on percentage distribution Of area of land in acres owned by households in two districts of a particular State are as follows : Land holding District-A District-B 001—0-99 2-5—749 7-5—12-49 12-5—19-99 20-0—29-99 18•35 47-12 19-34 7-21 2-36 13•53 21-84 39-32 12-15 743 5•73 2. 3. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ?
5. [CDS (I) Paper 1 Q36] The mean of first 13 observations is 32 and that of last 13 observations is **(a)** 20 **(b)** 25 **(c)** 40 **(d)** 45 Consider the following types of data 95. 96. 97. Data on percentage distribution Of area of land in acres owned by households in two districts of a particular State are as follows : Land holding District-A District-B 001—0-99 2-5—749 7-5—12-49 12-5—19-99 20-0—29-99 18•35 47-12 19-34 7-21 2-36 13•53 21-84 39-32 12-15 743 5•73 2. 3. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ?
6. [CDS (I) Paper 2 Q18] What is the first term ? **(a)** 2 3 -2 -3 If 27 x (81)2n.a _ equal to ? 3m — O, then what is **(a) (b) (c)** 2n + 5 5n + 6 811+3 8n + 15 Consider the following pairs Of numbers : (8, 12) (9, 11) (6, 24) Which pairs Of numbers have the harmonic mean ? **(a) (c) (d)** I and 2 only 2 and 3 only I and 3 Only I, 2 and 3 P.DETB-J.FN 26 97. first
7. [CDS (I) Paper 2 Q5] What is the least possible value of the arithmetic mean of these integers ? **(a) (c) (d)** 2 3 4 No such least possible value exists Which One of the following can be obtained from a histogram ? **(a) (b) (c) (d)** Mean Median Mode None of the above Examples of data are given below •
8. [CDS (I) Paper 2 Q100] What is the geometric mean of the observations 125, 729, 1331 ? **(a) (b) (c) (d)** 495 1485 2221 None Of the above
9. [CDS (I) Paper 3 Q7] What is the mean of the ages of minor children ? **(a)** 3 years **(b)** 4 years **(c)** 5 years **(d)** 6 years
10. [CDS (I) Paper 3 Q42] Consider the following data : The xithmctic mean Of the above distribution is **(a)** tan2t50 + tan2200 + tan2250 . + tan275Ø **(b)** cos2150 + cos2200 + cos2250 cos2750 **(c)** cot2150 — cot220e + cot2250 — + cot2750 **(d)** sec2150 + sec2200 + sec2250 + sec2750 42. Consider the following data : 5 9 2 The arithmetic mean of the above distribution is 296. What is the missing frequency ? 43. If a is an acute angle and then what is tan equal to ?
11. [CDS (I) Paper 3 Q50] A new frequency distribution is con- structed by doubling each frequency of the original distribution keeping the other entries intact. The following measures are computed for both the tables : I. Arithmetic mean Median **(a)** 12 49. ABCD is a square. p. Q, R, S are points on the sidesAB, BC, CD, DA respectively such that CR=DS. What is ZSPQ equal ? **(b)** 300 **(c)** 450 **(d)** 600
12. [CDS (I) Paper 3 Q3] Harmonic mean Which of the following statements with reference to above is correct ? **(a)** CorTesponding values of I only are equal in both the but ions **(b)** Corresponding values of I only are equal in both the butions **(c)** Corresponding values of 2 only are equal in both the butions and 2 distri- and 3 distri - and 3 distri- **(d)** Corresponding values of , 2 and 3 are equal in both the distributions T-DETA-K-FN - A 14

13. [CDS (I) Paper 3 Q70] The marks of the students of a class who appeared for a test in English are represented in the following frequency table : 11-20 21-30 31-40 41-50 51.60 What is/are the modal class(es) ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

14. [CDS (I) Paper 3 Q83] Frequency polygon Frequency curve Frequency distribution All of the above D E Three straight lines X, Y and Z are parallel and the angles are as shown in the figure above. What is ZAFB equal to ? **(a)** 200 **(b)** 150 **(c)** 300 **(d)** 100 T.DETA-K-FN - A 24 **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

15. [CDS (I) Paper 3 Q97] IA there three simultaneous linear equations in two unknowns. What can be the number of solutions (if they do exist) ? **(a)** One or infinite **(b)** Only one **(c)** Exactly two **(d)** Exactly three (For the next three questions to follow) : The arithmetic mean, geometric mean and median of 06 positive numbers a, a, b, b, c, c where a < b < c are 4, 2, 2 respectively.

16. [CDS (I) Paper 3 Q100] What is the mode ? **(a)** IO cm 5 cm **(b)** cm **(c)** 56 94. When the roots of the quadratic equation + bx + c 0 are negative of recipro- Cals of each other, then which one of the 2 2 97. IA there three simultaneous linear equations in two unknowns. What can be the number of solutions **(d)** One or infinite

17. [CDS (I) Paper 4 Q80] The imean weight of 150 students in a certain class is 60 kg. The mean weight of the boys from the class is 70 kg, While that Of girls is 55 kg. What is the number Of girls in the **(b)** hoo **(d)** do **(a)** hoo 83. 84. 26 Consider the following statements in respect of histogram 2. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. **(b)** do 81. In ant examination, 40% of the candidates wrote their answers in Hindi and the others in English. The laverage marks Of the candidates written in Hindi is 74 and the average marks of the candidates written in English is 77. What is the average marks Of all the candidates? **(c)** 758 **(d)** 75-8

18. [CDS (I) Paper 4 Q82] Consider •the following frequency distributién • Class 0-10 0-20 0-30 0-40 0-50 Frequency 3 What is the above frequency distributioÖ known as? **(a)** Cumulative distribution in more than type **(b)** Cumulåtive distribution in less than type **(c)** Continuous frequency distri- bu tion **(d)** None Of above Q-DETB-K-FN / 26A 26

19. [CDS (I) Paper 4 Q83] 84. 26 Consider the following statements in respect of histogram **(a)** hoo 83. 84. 26 Consider the following statements in respect of histogram 2. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. **(b)** do 81. In ant examination, 40% of the candidates wrote their answers in Hindi and the others in English. The laverage marks Of the candidates written in Hindi is 74 and the average marks of the candidates written in English is 77. What is the average marks Of all the candidates? **(c)** 758 **(d)** 75-8

20. [CDS (I) Paper 4 Q2] Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. Which . of the statements given above is/ are correct? **(a)** only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 What is the median Of the values • 11, 7, 6, 9, 12, 15, **(b)** 11 **(c)** 12 **(d)** 15

21. [CDS (I) Paper 4 Q85] Let (Xl, YI), (x2, Y2), (xn, yn) are n pairs Of positive numbers. The arithmetic mean and geometric mean Of any set Of positive numbers (q, c2, cn) are denoted by M(Ci), G (Ci) respectively. Consider the following : **(a)** hoo 83. 84. 26 Consider the following statements in respect of histogram 2. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. **(b)** do 81. In ant examination, 40% of the candidates wrote their answers in Hindi and the others in English. The laverage marks Of the candidates written in Hindi is 74 and the average marks of the candidates written in English is 77. What is the average marks Of all the candidates? **(c)** 758 **(d)** 75-8

22. [CDS (I) Paper 4 Q2] If each number is reduced by 2, then the geometric mean Of the set may not always exist. If each number is increased by 2, then the ieometric mean of the set is inereased by **(a)** + Crn 2 **(b)** 44 **(c)** cm2 Which of the above statements is are correct? **(d)**

23. [CDS (I) Paper 5 Q2] What is the ratio between the times taken by the cars to cover the same distance ? **(a) (b) (c) (d)** The arithmetic mean Of IO numbers was computed as

24. [CDS (I) Paper 5 Q76] It was later discovered that a number 8 was wrongly read as 3 during the computation. What should be the correct mean ? **(a) (b) (c) (d)** 112 108 96 84

25. [CDS (I) Paper 5 Q63] For the next 02 (two) items to follow : The following table gives the frequency distribution Of life length in hours of 100 electric bulbs having median life 20 hours Life of bulbs (in hours) Number Of bulbs 8-13 13 - 18 18-29 40 23-28 28 - 33 10 33 - 38 -2 What is the missing frequency x ? **(a) (b) (c) (d)** 61. 62. 63. For the next 02

26. **[CDS (I) Paper 5 Q74]** 75. Consider the following statements in respect Of a histogram . I. The hisüjgram consists Of vertical rectangular bars With a common base such that there is no gap between consecutive bars. **(a) (b) (c) (d)** The 4 5 6 7 two diagonals Of a rhombus are of lengths 55 Cm and 48 cm. If p is the perpendicular height Of the rhombus, then which One of the following is correct ? Consider the following statements in respect Of a histogram . I. The hisüjgram consists Of vertical rectangular bars With a common base such that there is no gap between consecutive bars. 2. The height Of the rectangle is determined by the frequency Of the class it represents. Which Of the statements given above Ware correct ?

27. **[CDS (I) Paper 6 Q1]** The number is **(a)** 31 **(b)** 47 **(c)** 49 **(d)** 53 Let the observations at hand be arranged in increasing. order. Which one of the following measures will not be affected when the smallest and the largest observations are removed ? **(a)** Mean **(b)** Median **(c)** Mode **(d)** Standard deviation If sin and cos 0 are the roots Of the equation ax2 — bx + c = O, then which one of the following is correct ? **(a)** a2 + b2 + 2ae 0 **(b)** a2 — b2+ 2ac = 0 **(c)** a2 + c2 + 2ab — O **(d)** a2 — b2 — 2ac — O A unit radian is approximately equal to **(a) (b) (c) (d)** 570 570 570 570 14 (Contd e)

28. **[CDS (I) Paper 6 Q58]** 59. The arithmetic mean and geometric mean of two numbers are 14 and 12 respectively. What is the harmonic mean of the numbers ? **(a)** 10 **(b)** 13 **(c)** 32/3 **(d)** 7217 Let A and B be two points. What is the locus of the point P such that angle APB = 900 ? **(a)** the line AB itself **(b)** the point P itself **(c)** the circumference of the circle with AB as diameter **(d)** the line perpendicular to AB and bisecting AB 76 ladies complete a job in 33 days. Due to some reason some ladies did not join the work and therefore it was completed in 44 days. The number Of ladies who did not report for the work is **(a)** 19 **(b)** 18 **(c)** 19 **(d)** 20 If 3x3 — 2x2y — 13xy2 + IOy3 is divided by x — 2y, then what is the remainder ? **(c)**

29. **[CDS (I) Paper 6 Q67]** X is as given below : x frequency **(a)** 36 **(b)** 44 **(c)** 52 64. **(d)** 150

30. **[CDS (I) Paper 6 Q68]** 69. 20 : 27 9:10 27 : 20 The frequency distribution of a variate 4 14 22 6 30 What is the cumulative frequency of variate **(c)** 45 Consider the following statements in respect of the set S = {l, 2, 3, (n + l)f2 is the median of the numbers in S. 2. n is the mode of the numbers in S. Which Of the above statements is/are correct ? **(a) (b) (c) (d)** 1 only 2 only Both I and 2 Neither nor 2 The radii of two cylinders are in the ratio 2 : 3 and their curved surface areas are in the ratio 5 :

31. **[CDS (I) Paper 6 Q69]** The frequency distribution of a variate X is as given below : **(a)** 36 **(b)** 44 **(c)** 52 64. **(d)** 150

32. **[CDS (I) Paper 7 Q16]** 17. The geometric mean of vitiates 32, 4, 8, 2 is **(a)** 16 In the triangle ABC, the base BC is trisected at D and E. The line through D, parallel to AB, meets AC at F and the line through E parallel to AC meets AB at G Let EG and DF intersect at H. What is the ratio of the sum Of the area of parallelogramÅGEF and the area of the triangle DHE to the area Of the triangle ABC ? **(b)** 1/6 Ifx — 2 is a factor + kx2 what is the value of k ? — 2K —24 then AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number ofpoints which are at a distance of I cm from AB and 5 cm from C ? 18. 19. 20. 21. Ihe angle of elevation of the tip of a tower from a point On the ground is 450. Moving 21 m directly towards the base ofthe tower, the angle ofelevation changes to 600. What is the height Of the tower, to the nearest meter ? **(c)** 48 m **(d)** . 49 m

33. **[CDS (I) Paper 8 Q97]** 98. Which one of the following relations for the numtxrs IO, 7, 8, 5, 6, 8, 5, 8, 6 is correct ? **(a)** Mean Median **(b)** Mean = Mode **(c)** Mean > Median **(d)** Mean > Mode What is the harmonic mean of IO, 20, 25, 40 and 50 ? **(a)** 25 **(b)** 30 **(c)** 26.1 **(d)** 21.3 The mean weight of 150 students in a class is 60 kg. The mean weight of boys is 70 kg and that of girls is 55 kg. What is the number of boys in the class ? **(a)** 50 **(b)** 60 **(c)** 75 **(d)** 100 The mean of 100 values is

34. **[CDS (I) Paper 8 Q45]** If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : **(a)** 45 **(b)** 48 **(c)** 51 **(d)** 55 28

35. **[CDS (I) Paper 8 Q100]** Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? **(a)** Mean Median **(b)** Mean Mode **(c)** Mean > Median **(d)** Mean > Mode What is the harmonic mean 25, 40 and 50 ?

36. **[CDS (I) Paper 9 Q93]** 1 The mean observation 1 if CA > CB CA2 of 7 observations is **(a) (b)** equal to equal to r greater than 1 less than I •us 91. and tio 92. ABCD is AD and the following statements I. me La ngent Of a circle ig a line the t meets the circle in one and Only one g. The Of circle at the end p t of the diameter perrgdiculaT t. the Which Of the above ? 1 only **(c)** 2 only Booa and 2 Neither I nor 2 triangle Z C 900 ana CD is perpendicular to A.B. If ABxCD-CAxCE, then cna IS equal t. In the figure given above. EF is parallel to y equal 460 **(d)** 93. 1 The mean observation 1 if CA > CB CA2 of 7 observations is 7. If each is increased by 2, then the new In the figure given above, Z PQR and QL is a median, PQ 5 cm and QR 12 an. QL is equal to G em

37. **[CDS (I) Paper 9 Q4]** Number of o n branchee tree. the Correct answer using the code **(b) (d)** I only and 4 3 and 4 weight Of Of 15 boys and IO girls is S8•4 kg. If average weight Of is 40 kg, then what is the average weight Of the girls ? 366 kg 35 kg 36 856 kg : ' Consi&r the following for the next three question. that a di8t.ribütion having class intervals O- 10, 10-20. 20-30 SO-40S respective frequencies are x, x 8, x — 2 and x —4 and the arithmetic mean Of the 96. dstribution is 17•8. value Of x ie **(a) (b)** I only and 4 3 and 4 **(c)** o: 10 10—20 20 -so 30-40 weight Of Of 15 boys and IO girls is S8•4 kg. If average weight Of is 40 kg, then what is the average weight Of the girls ? 366 kg 35 kg 36 856 kg : ' Consi&r the following for the next three question. that a di8t.ribütion having class intervals O- 10, 10-20. 20-30 SO-40S respective frequencies are x, x 8, x — 2 and x —4 and the arithmetic mean Of the Consider the statements I. A freq uenCy ution condenses the data and reveals ite important features. 2. A frequency distribution is an representation of originpl Which of the above statements Ware corect ? **(d)** only 2 onbr Both and 2 Neither nor 2 96. dstribution is 17•8. value Of x ie

38. **[CDS (I) Paper 10 Q98]** If m is the mean of p, q, r, s, t, u, v, then Ä.vhat is equal to? **(c) (d)** None of the above For the next two (2) items thatfoUow : The median of the following distribution is 14 •4 and the total frequency is 20 Class interval 0-6 6-12 12-18 18-24 24-30 Frequency 5 **(a)** tan e **(b)** cote A spherical balloon of radius r subtends angle 60' at the eye of an observer. If the angle of elevation of its centre is 600 and h is the height 95. 96. **(c)** 30 **(d)** 35 of the which

39. **[CDS (I) Paper 11 Q2]** The mode is both 2 and **(a) (b) (c) (d)** 45 square cm

40. **[CDS (I) Paper 11 Q7]** Which of the above statements is/are correct ? **(a)** I only **(b)** 2 only **(c)** Both 1 and 2 **(d)** Neither I nor 2 Consider the following Statements w•taining to a frequency polygon of a frequency distribution Of a continuous variable with seven class intervals of equal width I. The original frequency distribution can be reconstructed from the frequency polygon. 2. frequency polygon touches the x-axis in its extreme right and extreme left. Which of the above statements Ware correct ? **(a) (c)** I only 2 only Both I and 2 Neither I nor 2

41. **[CDS (I) Paper 11 Q96]** The mean Of the following distribution is **(a) (b) (c) (d)** 1 only 2 only Both I and 2 Neither I nor 2

42. **[CDS (I) Paper 11 Q18]** Class interval 11 - 13 13 -15 6 15-17 9 17-19 19 - 21 21 - 23 23 - 25 What is the value off ? **(a) (b) (c) (d)** 1 only 2 only Both I and 2 Neither I nor 2

43. **[CDS (I) Paper 11 Q2]** The arithmetic mean Of two unequal positive numbers is always greater than their geometric mean. The geometric mean Of two unequal positive numbers is always greater than their harmonic mean. Which of the above statements Ware correct ? **(a) (b) (c) (d)** 1 only 2 only Both I and 2 Neither I nor 2 Consider the following statements in respect of a discrete æt of numbers : I. The arithmetic mean uses all the data and is always uniquely defined.

44. **[CDS (I) Paper 11 Q100]** The geometric mean Of (h, x2, Xs, xn) is X and the geometric mean cf (YI, Y2, h, yn)ig Y. Which of the following Ware correef ? The geometric mean Of xnyn)isXY. geometric mean Of x X 2 Xs xn is — Select the correct answer using the code given **(a) (b) (c) (d)** 26-A) I only 2 only Both I and 2 Neither I nor 2

45. **[CDS (I) Paper 12 Q96]** Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range **(a)** 18 **(b)** 20 **(c)** 24 Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range 98. When we take class intervals On the x-axis and corresponding frequencies on the y-axis, and draw rectangles with the areas proportional to the frequencies Of the respective class intervals, the graph so Obtained is called 2. The cumulative frequency polygon gives an equivalent representation Of frequency distribution table. The cumulative frequency polygon is a closed polygon with one horimntal and one vertical side. The have nan-negative Which of the above is/ are correct? 1 only **(d)** 2 only Both 1 and 2

46. **[CDS (I) Paper 12 Q2]** The cumulative frequency polygon gives an equivalent representation Of frequency distribution table. The cumulative frequency polygon is a closed polygon with one horimntal and one vertical side. The other sides have nan-negative slope. Which of the above statements is/ are correct? 1 only **(b)** 2 only Both 1 and 2 **(d)** Neither I nor 2 **(a)** 18 **(b)** 20 **(c)** 24 Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range 98. When we take class intervals On the x-axis and corresponding frequencies on the y-axis, and draw rectangles with the areas proportional to the frequencies Of the respective class intervals, the graph so Obtained is called 2. The cumulative frequency polygon gives an equivalent representation Of frequency distribution table. The cumulative frequency polygon is a closed polygon with one horimntal and one vertical side. The have nan-negative Which of the above is/ are correct? 1 only **(d)** 2 only Both 1 and 2

47. **[CDS (I) Paper 12 Q100]** Ten observations 6, 14, 15, 17, x +1, 2x— 13, 30, 32, 34, 43 are written in ascending order. The median Of the data is **(a)** 18 **(b)** 20 **(c)** 24 Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range 98. When we take class intervals On the x-axis and corresponding frequencies on the y-axis, and draw rectangles with the areas proportional to the frequencies Of the respective class intervals, the graph so Obtained is called 2. The cumulative frequency polygon gives an equivalent representation Of frequency distribution table. The cumulative frequency polygon is a closed polygon with one horimntal and one vertical side. The have nan-negative Which of the above is/ are correct? 1 only **(d)** 2 only Both 1 and 2

48. **[CDS (I) Paper 12 Q98]** When we take class intervals On the x-axis and corresponding frequencies on the y-axis, and draw rectangles with the areas proportional to the frequencies Of the respective class intervals, the graph so Obtained is called **(a)** bar diagram **(b)** frequency curve **(c)** ogwe None of the above If Xi's are the midpoints Of the class 99. intervals Of grouped data, 's are the corresponding frequencies and is the mean, then what is — i) equal to? **(b)** 18 **(c)** 20 **(d)** 24 **(a)** 0 96. **(b)** 18 **(c)** 20 **(d)** 24 Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range

49. **[CDS (I) Paper 13 Q41]** 42 If G and are the arithmetic, geometric and harmonic means between a and b respectively. then Which One of the following relations is correct? **(b)** . The G is the geometric between A and H A is the arithmetic between G and H H is the harmonic between A and G None of the above geometric mean Of m ean mean mean three positive numbers a, b, c is 3 and the geometric mean Of another three positive numbers d, e, f is **(a)** 100 If G and are the arithmetic, geometric and harmonic means between a and b respectively. then Which One of the following relations is correct? 43. There are five parties A, B, C, D and E in an election. Out of total I OCIOOO votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allocated for party E in the pie Ch art? **(b)** . The G is the geometric between A and H A is the arithmetic between G and H H is the harmonic between A and G None of the above geometric mean Of m ean mean mean three **(c) (d)** 44. Out in 4S. If 540 60S 72 or 532 savings accounts held a post office, 218 accounts have deposits over 10,000 each, Further, in 302 accounts, the first or sole depositors are men, of which the deposits exceed 10,000 in 102 accounts. In how many accounts the first or sole depositors are women and the deposits are up to e 10,000 only? positive numbers a, b, c is 3 and the geometric mean Of another three positive numbers d, e, f is 4. Also, at least three elements in the set {a, b, c, d, e, f} are distinct. Which One Of the following inequalities gives the best information about M, the arithmetic mean Of the six numbers?

50. **[CDS (I) Paper 13 Q4]** Also, at least three elements in the set {a, b, c, d, e, f} are distinct. Which One Of the following inequalities gives the best information about M, the arithmetic mean Of the six numbers? **(c)** M > 3-5 M 23.5 It is not possible to set any precise lower limit for M B -EFUA-O-PDW / 52A **(a)** 100 If G and are the arithmetic, geometric and harmonic means between a and b respectively. then Which One of the following relations is correct? 43. There are five parties A, B, C, D and E in an election. Out of total I OCIOOO votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allocated for party E in the pie Ch art? **(b)** . The G is the geometric between A and H A is the arithmetic between G and H H is the harmonic between A and G None of the above geometric mean Of m ean mean mean three **(c) (d)** 44. Out in 4S. If 540 60S 72 or 532 savings accounts held a post office, 218 accounts have deposits over 10,000 each, Further, in 302 accounts, the first or sole depositors are men, of which the deposits exceed 10,000 in 102 accounts. In how many accounts the first or sole depositors are women and the deposits are up to e 10,000 only? positive numbers a, b, c is 3 and the geometric mean Of another three positive numbers d, e, f is 4. Also, at least three elements in the set {a, b, c, d, e, f} are distinct. Which One Of the following inequalities gives the best information about M, the arithmetic mean Of the six numbers?

51. **[CDS (I) Paper 13 Q97]** What is the mean Of the distribution? **(a)** I only **(b)** 2 only Both I and 2 **(c)** Neither I nor 2 Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 Cm is tangent to the inner circle. The radius Of the inner circle is **(d)** B-EFUA-O-PDW/52A 94. The complement angle of 80' is

52. **[CDS (I) Paper 14 Q99]** If each of n numbers x, — i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 **(d) (a)** 24 **(b)** a —25 **(c)** 28 **(d)** a— 1 25

53. **[CDS (I) Paper 14 Q100]** The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to **(d)** 385 7 14 35 **(a)** 24 **(b)** a —25 **(c)** 28 **(d)** a— 1 25

54. **[CDS (I) Paper 15 Q91]** The election result in Which six parties contested was depicted by a pie chart. Party A had an angle 135E on this pie chart. If it secured 21960 votes, how many valid votes in total were cast? **(a)** 51240 **(b)** 58560 **(c)** 78320 **(d)** 87840

55. **[CDS (I) Paper 15 Q92]** The mean and median of 5 observations are 9 and 8 respectively. If I is subtracted from each observation, then new mean and the new median the respectively be will 8 and 7 **(b)** 9 and 7 8 and 9 Cannot be determined duc to insufficient data **(a) (b) (c) (d)** 400, — 502, y- 40', y 5x-2y= 10 z- 1050 z-II

56. **[CDS (I) Paper 15 Q93]** The age distribution of 40 children is as follows : Age (in years) 7-8 9-10 10-11 No. of childre n 4 Consider the following statements in respect of the above frequency distribution : **(a) (b) (c) (d)** 400, — 502, y- 40', y 5x-2y= 10 z- 1050 z-II

57. **[CDS (I) Paper 16 Q40]** If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? **(a)** 60 **(b)** 50 **(c)** 45 **(d)** 40 SDC-S-TRD - A 12

58. **[CDS (I) Paper 16 Q88]** The median of set Of 9 distinct observations is If each Of the largest 4 observations of the set is increased by 2, then the median Of the new set **(a)** is increased by 2 **(b)** is decreased by 2 **(c)** is two times the original median **(d)** remains the same as that of original set

59. **[CDS (I) Paper 16 Q90]** The mean of 20 observations is **(a)** x-1-25 **(b)** x-o.5 **(c)** 1-25 88. The median of set Of 9 distinct observations is If each Of the largest 4 observations of the set is increased by 2, then the median Of the new set **(d)** is increased by 2

60. **[CDS (I) Paper 16 Q17]** On checking it was found that the two observations were wrongly copied as 3 and **(a)** x-1-25 **(b)** x-o.5 **(c)** 1-25 88. The median of set Of 9 distinct observations is If each Of the largest 4 observations of the set is increased by 2, then the median Of the new set **(d)** is increased by 2

61. **[CDS (I) Paper 16 Q6]** If wrong observations are replaced by correct values 8 and 9, then what is the correct mean ? SDC-S-TRD - A 26 **(a)** 17-4 **(b)** 16•6 15-8 **(c)** i4-2 **(d)**

62. **[CDS (I) Paper 16 Q95]** Suppose the class interval 10—15 has frequency 30, then what is the frequency density of this class interval ? **(a)** I only **(b)** 2 only **(c)** Both I and 2 **(d)** Neither I nor 2 95. Suppose the class interval 10—15 has frequency 30, then what is the frequency density of this class interval ? 98. 99. 100. 28 W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ?

63. **[CDS (I) Paper 17 Q51]** The mean Of 5 numbers is **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

64. **[CDS (I) Paper 17 Q15]** 'If one more number is included, the mean of the 6 numbers becomes **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

65. **[CDS (I) Paper 17 Q53]** 20 — 40 40-60 60-80 80 — 100 (14 —A) **(a) (b) (c) (d)** 24 25 26 27 The mean marks Obtained by 300 students in a subject are

66. **[CDS (I) Paper 17 Q60]** The mean Of top 100 students was found to be 80 and the mean Of last 100 students was found to be **(a) (b) (c) (d)** I only 2 only Both I and 2 Neither I nor 2

67. **[CDS (I) Paper 17 Q50]** The mean marks Of the remaining 100 students are **(a) (b) (c) (d)** 70 65 60 50 Consider the following distribution • Freq uency 17 28 32 If the mean of the above distribution is 50, What is the value Of f ? **(a) (b) (c) (d)** 24 34 56 96

68. **[CDS (I) Paper 17 Q58]** Consider the following frequency distribution x Frequency Cumulative frequency Odd **(a)** The I argest slice has angle 1506 **(b)** The smallest slice has angle 700 **(c)** The largest slice has angle 1800 **(d)** The smallest slice has angle 900 In an asymmetrical distribution, if the mean and median Of the distribution are 270 and 220 respectively, then the mode of the data is

69. **[CDS (I) Paper 18 Q91]** 92. of 93. to 28-A) Frequency density Of a class is computed by the ratio **(a) (b) (c) (d)** Class frequency to the class width Class frequency to total frequency Class frequency to total number of classes Cumulative frequency up to that class to total frequency A small company pays each of its 5 category workers 20,000, each of its 3 category 'B' workers 25,000 and a category 'A' worker 65,000. The number Of workers earning less **(b) (c) (d)** 8 5 4 3 A man travelled 12 km at a speed of 4 km/hr and further 10 km at a speed of 5 km/hr. What was his average speed ? **(a) (b) (c) (d)** km/hr 4•5 kmfhr 5 •O km/hr 2 •5 kmmr The number of girls studying Statistics is

70. **[CDS (I) Paper 18 Q88]** 89. 90. (28-A) Frequency density Of a class is computed by the ratio **(a) (b) (c) (d)** 58-8 73-5 78•7 80•6 **(a) (b) (c) (d)** Class frequency to the class width Class frequency to total frequency Class frequency to total number of classes Cumulative frequency up to that class to total frequency In which subjects is the difference between the number of boys and girls equal ? **(a) (b) (c) (d)** Mathematics and Chemistry Statistics and Chemistry

71. **[CDS (I) Paper 18 Q98]** 99. rect are I is 11 is 100. pre (30 —A) The geometric mean of x and y is 6 and the geometric mean of x, y and z is Then the value Of z is **(c)** 12 6 The total number of live births in a specific locality during different months of a specific year was obtained from the omce Of the Birth Registrar. This set of data may be called **(a) (b) (c) (d)** Primary data Secondary data Recorded data Countable data The heights (in em) Of 5 students are 150, 165, 161, 144 and

72. **[CDS (I) Paper 18 Q99]** 100. The geometric mean of x and y is 6 and the geometric mean of x, y and z is Then the value Of z is **(c)** 12 6 **(a) (c) (d)** 16,000 and 9,000 8,000 and 4,500 e 25,600 and 8,100 e 4,000 and e 3,000 The total number of live births in a specific locality during different months of a specific year was obtained from the omce Of the Birth Registrar. This set of data may be called Consider the following statements : Statement I: The value of a random variable having the highest frequency is mode. Statement 11 : Mode is unique. Which one of the following is correct in respect of the above statements ? **(a) (b) (c) (d)** Primary data Secondary data Recorded data Countable data **(a) (b) (c) (d)** Both Statement I and Statement LI are true and Statement 11 is the correct explanation of Statement I Both Statement I and Statement 11 are true but Statement 11 is not the correct explanation of Statement I Statement I is true but Statement 11 is Statement I is false but Statement 11 is The heights (in em) Of 5 students are 150, 165, 161, 144 and

73. **[CDS (I) Paper 19 Q54]** IO-A) The arithmetic mean Of two numbers is IO and their geometric mean is **(a) (b) (c)** 15, 5 12, 8 16, 4 18, 2 **(d)** 13,500 e 18,000 e 19,750 20,250 53. 54. The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10•5 and the arithmetic mean of the last 6 observations is 11•5. What is the sixth observation ? A Pie Chart is drawn for the following data : Sector Agriculture and Rural Develo t Irrigati on Energy Industry and Minerals Transport and Communication Social Services Percen tage 12-9 12-5 272 15-4 15-9 16•1

74. **[CDS (I) Paper 19 Q8]** What are the two numbers ? **(a) (b) (c)** 15, 5 12, 8 16, 4 18, 2 **(d)** 13,500 e 18,000 e 19,750 20,250 53. 54. The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10•5 and the arithmetic mean of the last 6 observations is 11•5. What is the sixth observation ? A Pie Chart is drawn for the following data : Sector Agriculture and Rural Develo t Irrigati on Energy Industry and Minerals Transport and Communication Social Services Percen tage 12-9 12-5 272 15-4 15-9 16•1

75. **[CDS (I) Paper 19 Q11]** The arithmetic mean of the first 6 observations is 10•5 and the arithmetic mean of the last 6 observations is 11•5. What is the sixth observation ? **(a) (b) (c) (d)** 100 10, 5 11•0 11-5 What is sin' O — cos4 0 equal to for any real **(a) (b) (c) (d)** I—2sin20 2 cos2 0 I I — 2 cos2 A, B, C, D and E start a partnership firm.

76. **[CDS (I) Paper 19 Q52]** 50. Capital contributed by A is three times that contributed by D. E contributes half Of A's Contribution, B contributes one-third of E's contribution and C contributes two-third of A's contribution. [f the difference between the combined shares Of A, D and E and the combined shares Of B and C in the total profit of the firm is 13,500, what is the combined share of B, C and E ? (The shares are supposed to be proportional to the contri bu tions) 51. (16-A) The arithmetic mean Of two numbers is IO and their geometric mean is **(a) (b) (c)** 15, 5 12, 8 16, 4 18, 2 **(d)** 13,500 e 18,000 e 19,750 20,250 53. 54. The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10•5 and the arithmetic mean of the last 6 observations is 11•5. What is the sixth observation ? A Pie Chart is drawn for the following data : Sector Agriculture and Rural Develo t Irrigati on Energy Industry and Minerals Transport and Communication Social Services Percen tage 12-9 12-5 272 15-4 15-9 16•1

77. **[CDS (I) Paper 19 Q53]** 54. The arithmetic mean of 11 observations is **(a) (b) (c)** 15, 5 12, 8 16, 4 18, 2 **(d)** 13,500 e 18,000 e 19,750 20,250 53. 54. The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10•5 and the arithmetic mean of the last 6 observations is 11•5. What is the sixth observation ? A Pie Chart is drawn for the following data : Sector Agriculture and Rural Develo t Irrigati on Energy Industry and Minerals Transport and Communication Social Services Percen tage 12-9 12-5 272 15-4 15-9 16•1

78. **[CDS (I) Paper 19 Q70]** If the average marks of both the sections combined are 67, then the ratio of number of students of section A to that of section B is **(a) (b) (c) (d)** The median of 19 observations is

79. **[CDS (I) Paper 19 Q30]** Two more observations are made and the values of these are 8 and **(a) (b) (c) (d)** Nominal scale Ordinal scale Interval scale Ratio scale The average marks of section A are 65 and that of section B are 70. If the average marks of both the sections combined are 67, then the ratio of number of students of section A to that of section B is

80. **[CDS (I) Paper 19 Q32]** What is the median of the 21 observations ? **(a) (b) (c)** 32 30 20 Cannot be determined insuffcient data due to As the nun-Iber of observations and claggeg increases, the shape of a frequency polygon **(a) (b) (c) (d)** Tends to become jagged Tends to become in&easingly smooth Stays the same Varies only if data become more reliable GTHY-T-XMT (19-

81. **[CDS (I) Paper 19 Q59]** 60. Of to 19-A) Let il and (where >) be the means of two sets comprising nr and nQ (where < m) observations respectively. If is the mean when they are pooled, then which one of the following is correct ? **(a) (b) (c) (d)** Consider the following statements : Statement I : Median can be computed even when the end intervals of a frequency distribution are open. Statement : Median is a positional average. Which one of the following is correct in respect of the above statements ? **(a) (b) (c) (d)** Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I Both Statement I and Statement 11 are true and Statement 11 is not the correct explanation of Statement I Statement I is true but Statement 11 is Statement I is false but Statement 11 is Data on ratings of hotels in a city is measured

82. **[CDS (I) Paper 19 Q60]** Let il and (where >) be the means of two sets comprising nr and nQ (where < m) observations respectively. If is the mean when they are pooled, then which one of the following is correct ? **(a) (b) (c) (d)** Consider the following statements : Statement I : Median can be computed even when the end intervals of a frequency distribution are open. Statement Median is a positional average. Which one of the following is correct in respect of the above statements ? **(a) (b) (c)** 32 30 20 Cannot be determined insuffcient data due to As the nun-Iber of observations and claggeg increases, the shape of a frequency polygon **(a) (b) (c) (d)** Tends to become jagged Tends to become in&easingly smooth Stays the same Varies only if data become more reliable **(a) (b) (c) (d)** Both Statement I and Statement 11 are true and Statement 11 is the correct explanation of Statement I Both Statement I and Statement 11 are true and Statement 11 is not the correct explanation of Statement I Statement I is true but Statement 11 is Statement I is false but Statement 11 is GTHY-T-XMT

83. **[CDS (I) Paper 20 Q62]** 19—A) C%nsider the following frequency distribution : What is the median for the distribution ? **(a) (b)** 61. , then What Consider the distribution : following grouped 10 frequency What is the value Of sin2 250 + sin2 65' ? **(c) (d)** 2 4 What is the value Of sing e 0 3 sin2 0 cos2 0 — I ? If the mean of the above data is 25•2, then what is the value Of p ?

84. **[CDS (I) Paper 20 Q5]** Item of expenditure Food Clothing Education Miscellaneous . 1,660 If the above data is represented by a percentage bar diagram of height 15 em, then what are the lengths of the two segments of the bar diagram corresponding to education and miscellaneous respectively ? **(a) (b) (c) (d)** 44 55 66 If the above data is represented by a percentage bar diagram of height 15 em, then what are the lengths of the two segments of the bar diagram corresponding to education and miscellaneous respectively ? Diagrammatic representation Of data ineludes which of the following ? I.

85. [CDS (I) Paper 20 Q70] 74. 71. **(a) (d)** The Median Mean Geometric mean Mode Graphical location Graphical location Ogive following pairs relate to frequency distribution Of a discrete variable and its frequency polygon. Which one Of the following pairs is not correctly matched ? **(a) (b) (c)** Base line Of the — polygon Ordinates of the — vertices of the Absci of the — vertices of the Area of the X-axis Class frequencies Class marks of the fre quency Total freq uency Of the distribution In a rectangle, length is three times its breadth. If the length and the breadth of the rectangle are increased by and , its DFGT-T-HTM by (23 -A

86. [CDS (I) Paper 22 Q12] If x is the mean Of the numbers in the first pair and y is the mean of the numbers in the second pair, then what is the mean of x and y ? **(a)** 60 **(b)** 72 **(c)** 75 **(d)** 78

87. [CDS (I) Paper 22 Q45] For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? **(b)** S A - PIXI-T-MTK **(a)** A, B and C **(b)** A, B and D **(c)** A, E and F 41. The number of three digit numbers **(d)** Six

88. [CDS (I) Paper 23 Q7] What is the median of marks of all the fifteen candidates ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

89. [CDS (I) Paper 23 Q95] 33-A) When the class intervals have equal width, the height of a rectangle in a histogram **(a) (b) (c) (d)** Width Of the class Lower class limit Upper class limit Frequency of the class The ages Of 7 family members are 2, 5, 12, 18, 38, 40 and 60 years respectively. After 5 years a new member aged x years is added. If the mean age of the family now goes up by years, then what is the value Of x ? **(a) (b) (c) (d)** 2 4 mean weight Of 100 Students in a is 46 kg. The mean weight of boys is 50 kg and that of girls is 40 kg. The of boys exceeds the number of girls by **(a) (b) (c)** 10 15 20 25

90. [CDS (I) Paper 23 Q92] 93. (33-A) When the class intervals have equal width, the height of a rectangle in a histogram **(a) (d) (a) (b) (c) (d)** Width Of the class Lower class limit Upper class limit Frequency of the class 66 7 75

91. [CDS (I) Paper 23 Q97] 98. What is the algebraic sum of the deviations from the mean Of a Of 25, 65, 73, 75, 83, 76, 17, 15, 7, 14? **(a) (b) (c) (d)** 2 The mean Of five observations x, x + 2, x 4, x + 6, x 8 is m. What is the mean Of the first three observations ? **(a) (b) (c) (d)** DZOL-T-LKM (35 ns

92. [CDS (I) Paper 23 Q100] The harmonic mean and the geometric mean of two numbers are IO and 12 respectively. What is their arithmetic mean ? **(a) (b) (c) (d) (a)** 1 **(b)** 2 **(c)** 3 **(d)** 4

93. [CDS (I) Paper 24 Q94] What is the arithmetic mean of the first ten composite numbers? 102 **(d)** 112 **(a)** In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? **(b)** and E **(c)** A, C and D only **(d)** C D and E

94. [CDS (I) Paper 24 Q96] Let p be the mean of m Observations and q the mean of n observations, Where p q If the combined mean of (m observations is then which one Of the following is correct? Directions : For the next four (4) items, consider the following data with regard to different types (I, 11, Ill, [V, V) of multivitamin tablets produced in a company (in lakhs) : Year 2000 2001 2002 2003 2004 2005 160 200 135 240 180 210 150 35 95 110 150 111 44 120 85 100 160 95 80 95 92 100 85 120 115 110 **(a)** Type 11 **(b)** Type 111 **(c)** Type **(d)** Type V XVWS-T-MTK,'SOA 98. In Which one Of the following pairs of years, the difference in total number Of tablets produced between them is m inimum?

95. [CDS (I) Paper 25 Q100] What is the arithmetic mean of p and Q? **(d)** Cannot be determined due to in- sufficient data A - NYRG-T-MTK Consider the following for the next four (04) üems that follow : The following Bar-Chart gives the pro- duction of cars by three different com- panies X, Y and Z in different years : 2014 2015 2016 2017 2018 YEARS **(a)** 100% **(b)** 150% **(c)** 200% 96. Consider the production of each company separately. In how many instances was the percentage increase in production of cars over the previous year's production greater than 20% ? 97. In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? **(d)** 2015

96. [CDS (I) Paper 26 Q30] The arithmetic. mean and the geometr•ic mean of two positive numbers p and q (p > q) are A and G respectively. Which one Of the following is correct? **(d)** A2=G 1 P.T.O. **(a)** 12 **(b)** 10 **(c)** 21 26. 28. If -2 then what is the value Of as + b3? 2n.2 + m and q •2 — m Let p 2 **(d)**

97. [CDS (I) Paper 27 Q50] A frequency distribution table is given below : q 25 10 5 Total frequency. is 200 and mean Of the distribution is 1•46. What is the value of p ? 70 72 76 78 What is the value of q ? **(b) (c) (d)** 32 34 36 38 What is the medign Of the distribution ? **(a) (c)** FDGT-T-EMT (19-A

98. [CDS (I) Paper 27 Q70] What is the mean of the values ? **(a)** 38 **(b)** 40 **(c)** 42 **(d)** 44 FDGT-T-EMT GI. 62. 63. 64.

99. [CDS (I) Paper 27 Q67] The data of different natural numbers 4, 7, 10, 5, 22, 23, 30, 50 are in ascending order. How many possible values are there for the median of the data for various values Of x ? **(a) (b) (c)** Only one value Only two values Only three values Five values If M is the mean Of the first 100 even natural numbers, then what is the mean Of the first 100 Odd natural numbers ? **(c) (d)** Mean marks of 50 students were found to 78•4. But later it was detected that 95 was misread as 59 and 25 was misread as

100. [CDS (I) Paper 27 Q52] What is the difference between correct mean and incorrect mean ? **(a) (c) (d)** 0•04 012 0-18 If the average marks Of the students of all four schools are 58, then - how many students appeared from School-IV ? **(a)** 38 **(b)** 40 **(c)** 42 **(d)** 44 FDGT-T-EMT GI.

101. [CDS (I) Paper 28 Q81] 82. **(a) (b) (c)** 60 72 75 80 **(d)**

102. [CDS (I) Paper 28 Q83] What is the mean Of the distribution ? **(a) (b) (c)** 60 72 75 80 **(d)**

103. [CDS (I) Paper 29 Q8] 9. 2-4, IO. If , then What is the value of 2-b2 ? 375 The sum of deviations of n numbers from IO and 20 are a, b respectively. If then what is the mean of these n numbers ? **(b) (c) (d)** 12 14 16 18 **(a)** Onle one Only two **(b)** Only three **(c) (d)** None Ifx= 97 +56"

104. [CDS (I) Paper 29 Q11] 14. If the median Of observations 12, 1, 8, 54, 61, 28, 45, 35, 21, 17 is M, then What is the value Of 2M + 5 ? **(a)** Onle one Only two **(b)** Only three **(c) (d)** None Ifx= 97 +56"

105. [CDS (I) Paper 29 Q87] 84. length Of the perpendicular from B to AC. If BC IO cm and AC = 12 em, then what is the value Of p ? 5Jfi **(a)** 15 **(b)** 16 **(c)** 18 JNUP-T-EMT In a triangle ABC, angle B = 900 and p is the 87. 84. length Of the perpendicular from B to AC. If BC IO cm and AC = 12 em, then what is the value Of p ? 5Jfi **(d)** 1 oJii

106. [CDS Paper 30 Q3] The difference between new median and old median is I. The difference between new mean and Old mean is less than 0-1. The difference between new mode and old mode is zero. Which of the statements given above are 19. correct ? **(a) (b) (c) (d)** I and 2 only 2 and 3 only I and 3 only I, 2 and 3 What is the digit at the 100th place Of number (225)40 ? **(b) (c)** 6 5 4 If a, b, c, d are natural numbers, then how many possible remainders are there when Ia + 2b + 3c + 4d is divided by IO ? **(a) (b) (c)** 4 5 SRSU-T-EMT

107. [CDS (I) Paper 31 Q21] The arithmetic mean of n numbers is M. If the sum of first (n — 1) terms is k, then what is the nth number ? **(b)** nM — k **(c)** n(M — k) **(a)** nM — k **(b)** n **(c)** 81 **(d)** 105

108. [CDS (I) Paper 31 Q22] What is the geometric mean of 3, 9, 27, 81, 243, 729, 2187? **(a)** 81 **(b)** 105 **(c)** 144 **(d)** 243

109. [CDS (I) Paper 31 Q41] The mean weight of 150 students in a class is 60 kg. The mean weight of boys in the class is 70 kg and that of girls is 55 kg. What is the ratio of number of boys to number of girls ? **(a)** IO mimin **(b)** 5 mimin **(c)** 10 cm/sec **(d)** 5 cm/sec 43. A spherical wooden ball of radius r is to be divided into eight identical parts by cutting by planes passing through the same diameter. What is the surface area of each final piece ? 3 2 2m•2 3 4%r2 A trolley with two wheels one metre 44. apan is moved clockwise on the circular track around a ground with radius 50 m

110. [CDS (I) Paper 33 Q91] Which measure of central tendency is least affected by the presence Of extreme Observations in the data? **(b) (c)** Arithmetic mean Harmonic mean Geometric mean Median **(a)** 500 **(b)** 300 **(c)** 60 92. **(d)**

111. [CDS (I) Paper 33 Q93] The Of arithmetic mean geometric mean harmonic mean mode frequency distribution 100 candidates in a of marks particular examination is as follows : Number Of Candidates More than 10 More than 20 More than 30 More than 40 What are 100 60 40 the average marks Of the candidates? **(a)** 500 **(b)** 300 **(c)** 60 92. **(d)**

112. [CDS (I) Paper 33 Q94] The arithmetic mean Of 200 observations is **(a)** 500 **(b)** 300 **(c)** 60 92. **(d)**

113. [CDS (I) Paper 33 Q60] If 5 is multiplied to each observation, then what Will be the new arithmetic mean? **(a)** 500 **(b)** 300 **(c)** 60 **(d)** 40

114. [CDS (I) Paper 33 Q95] A distribution consists of 3 components With frequencies 45, 40 and 55 having their means 2, 2 • 5 and 2 respectively. What is the mean of the combined dis tribu tion? **(a)** 2 14 **(b)** 225 **(c)** 2 37 **(d)** 2 50

115. [CDS (I) Paper 33 Q96] Which one Of the following is a average ? **(a) (b) (c) (d)** Arithmetic mean Median Mode Geometric mean P.T.O.

116. [CDS (I) Paper 34 Q37] All possible groups of 3 distinct numbers from among A, B, C, D and E are formed. Ifthe aggregate of sums of numbers of each group is 120, then what is the arithmetic mean of A, B, C, D and E? **(c)** 10 **(d)** 20 **(a)** 100 **(b)** 81 **(c)** 64 **(d)** 49 35. Two numbers P and Q are such that . P : Q —x: y. If 1 is added to both the numbers, the ratio becomes 7 : IO. If 3 is added to both the numbers, the ratio becomes 3 : 4. What is

117. [CDS (I) Paper 34 Q38] The arithmetic mean of I observa- tions is **(a)** 100 **(b)** 81 **(c)** 64 **(d)** 49 35. Two numbers P and Q are such that . P : Q —x: y. If 1 is added to both the numbers, the ratio becomes 7 : IO. If 3 is added to both the numbers, the ratio becomes 3 : 4. What is

118. [CDS (I) Paper 34 Q50] If 10 is added to each observation then what will be the new arithmetic mean ? **(a)** 50 **(b)** 55 **(c)** 60 **(d)** 65 15 A - KPRS-T-MTE

119. [CDS (I) Paper 34 Q54] What is the arithmetic mean ? **(a)** 3-25 **(b)** 3-50 **(c)** 3-75 **(d)** 4-25

120. [CDS (I) Paper 35 Q93] 94. (33-A) [f the mean of a set of 7 observations is I O and the mean Of a set Of 3 observations is 5, then what is the combined mean ? **(a) (b)** 8-5 6-5 5-5 **(c)** The average weekly wages of male employees in a company is 4,200 and that Of females is 3,200. If the average weekly wage of all employees is 4,000, then What is the ratio of male to female employees ? **(d)**

121. **[CDS (I) Paper 35 Q97]** 15 10 What is the mode Of the above distribution ? **(a) (c) (d)** B83 25•00 24, 79 If the mean Of 15 observations, namely Xl, x2, x3, x15 is 2•5, then what is the value of Eli5-1 **(b) (c)** 400 525 625 825

122. **[CDS (I) Paper 35 Q99]** 0-10 10—20 30-40 , -55 , = 20 (35-A) 12 14 10 16 18 6 8 10 9 What is the arithmetic mean of the above distribution ? **(b)** 1369 13-39 13-19 1309 Consider the following incomplete frequency distribution with a total frequency Of 100 : Class Frequency 10 20 If the mean Of the frequency distribution is 25, then what are the missing frequencies ? **(a) (b) (c) (d)** x, 15 Y xz50 Y x 220, y 45 BFVS-T-TME